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Gjerrit Meinsma
Arjan van der Schaft

A Course on Optimal Control

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Gjerrit Meinsma · Arjan van der Schaft

A Course on Optimal Control

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Preface

This book reflects a long history of teaching *optimal control* for students in mathematics and engineering at the universities of Twente and Groningen, the Netherlands. In fact, the book has grown out of lecture notes that were tested, adapted, and expanded over many years of teaching.

The present book provides a self-contained treatment of what the undersigned consider to be the core topics of optimal control for finite-dimensional deterministic dynamical systems. The style of writing aims at carefully guiding the students through the mathematical developments, and emphasizes motivational examples; either of a mathematical nature or motivated by applications of optimal control.

Chapter 1 covers the basics of the classical *calculus of variations*, including second-order conditions and integral constraints. This directly motivates the introduction of the *minimum principle* (more commonly known as the *maximum principle*) in Chapter 2. Although the presentation of the book aims at minimizing generalities, the treatment of the minimum principle as given in Chapter 2 is self-contained, and suited for a basic course on optimal control. Chapter 3 continues with the *dynamic programming* approach to optimal control, culminating in the Hamilton-Jacobi-Bellman equation. The connection with the minimum principle is discussed, as well as the relation between infinite horizon optimal control and Lyapunov functions. In Chapter 4, the theory of Chapters 2 and 3 is applied, and specialized, to *linear* control systems with *quadratic* cost criteria (LQ-optimal control). This includes a succinct but detailed treatment of Riccati differential equations and algebraic Riccati equations. The chapter is concluded with a section on controller design based on LQ-optimal control.

In our experience, the material of Chapters 1–4 provides a good coverage of an introductory, but mathematically self-contained, course on optimal control for final year BSc (or beginning MSc) students in (applied) mathematics and beginning MSc students in engineering. We have taught the course for such an audience for many years as an 8-week course of 4 lecture hours and 2 tutorial hours per week (5 ECTS in the European credit system). Of course, the contents of the course can still be adapted. For example, in some of the editions of the course we did not cover Section 1.7 on integral constraints, but instead we paid more attention to Lyapunov stability theory as detailed in Appendix B.

Required background for the course is linear algebra and calculus, basic knowledge of differential equations, and (rudimentary) acquaintance with control systems. Some mathematical background is summarized in Appendix A for easy recollection and for bringing students to the same mathematical level. Appendix B goes further: it does not only recall some of the basics of differential equations, but also provides a rather detailed treatment of Lyapunov stability theory including LaSalle's invariance principle, as occasionally used in Chapters 3 and 4 of the book.

Chapter 5 of the book is of a different nature. It is *not* considered to be part of the basic material for a course on optimal control. Instead, it provides brief outlooks to a number of (somewhat arbitrarily chosen) topics that are related to optimal control, in order to raise interest of students. As such Chapter 5 is written differently from Chapters 1–4. In particular the treatment of the covered topics is not always self-contained.

At the end of each of Chapters 1–4, as well as of Appendix B, there is a rich collection of exercises, including a number of instructive examples of applications of optimal control. Solutions to the odd-numbered exercises are provided.

Main contributors to the first versions of the lecture notes (developed since the 1990s) were Hans Zwart, Jan Willem Polderman (both University of Twente), and Henk Nijmeijer (University of Twente, currently Eindhoven University of Technology). We thank them for their initial contributions.

In 2006–2008 Arjan van der Schaft (University of Groningen) made a number of substantial revisions and modifications to the then available lecture notes. In the period 2010–2018 Gjerrit Meinsma (University of Twente) rewrote much of the material, and added more theory, examples, and illustrations. Finally, in 2021–2023 the book took its final shape. We thank the students and teaching assistants for providing us with constant feedback and encouragement over the years.

We have profitably used many books and papers in the writing of this book. Some of these books are listed in the References at the end. In particular some of our examples and exercises are based on those in Bryson and Ho (1975) and Seierstad and Sydsaeter (1987). We thank Leonid Mirkin (Technion, Haifa) for Example 4.6.2.

Unavoidably, there will be remaining typos and errors in the book. Mentioning of them to us is highly welcomed. A list of errata will be maintained at the website https://people.utwente.nl/g.meinsma?tab=projects. This website can also be reached by scanning the QR code below.

Enschede, The Netherlands — Gjerrit Meinsma
Groningen, The Netherlands — Arjan van der Schaft
May 2023

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Notation and Conventions

While most of the notation used in this book is standard, there are a few conventions we would like to emphasize.

Notation for vectors and functions of time. We frequently switch from functions

$$\mathrm{x} : \mathbb{R} \to \mathbb{R}^n$$

to vectors

$$x \in \mathbb{R}^n$$

and back to functions, and this can be confusing upon first reading. To highlight the difference we typeset functions of time usually in an upright math font, e.g., x, instead of the standard italic math font, e.g., x. This convention is used in differential equations

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{x}(t) = a\mathrm{x}(t), \quad \mathrm{x}(0) = x_0,$$

and solutions of them, e.g., $\mathrm{x}(t) = \mathrm{e}^{at}x_0$. But it is used mainly to avoid possibly ambiguous expressions. For instance whenever we use $V(x)$ we mean that V is a function of $x \in \mathbb{R}^n$ and not of the whole time function $\mathrm{x} : \mathbb{R} \to \mathbb{R}^n$. We still use the italic math font for functions of time if they only play a minor role, such as $a(t)$ in

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{x}(t) = a(t)\mathrm{x}(t) + \mathrm{u}(t).$$

In equations like these $a : \mathbb{R} \to \mathbb{R}$ is typically just a given function.

Notation for derivatives. The material in this book requires partial derivatives, total derivatives, and derivatives with respect to vectors. There is no universally accepted notation for this. In this book, we use the following notation.

For functions of a single real variable, $g : \mathbb{R} \to \mathbb{R}^k$, we denote its derivative at $t \in \mathbb{R}$ as

$$\dot{g}(t) \quad \text{or} \quad g'(t) \quad \text{or} \quad \frac{\mathrm{d}}{\mathrm{d}t}g(t) \quad \text{or} \quad \frac{\mathrm{d}g(t)}{\mathrm{d}t}.$$

Now if, for example, $F : \mathbb{R}^3 \to \mathbb{R}$ and $\mathrm{x} : \mathbb{R} \to \mathbb{R}$, then

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}F(t, \mathrm{x}(t), \dot{\mathrm{x}}(t))$$

means the *total* derivative with respect to t of the *partial* derivative $\frac{\partial}{\partial v}F(t, x, v)$ evaluated at $(t, x, v) = (t, \mathrm{x}(t), \dot{\mathrm{x}}(t))$. For instance, if $F(t, x, v) = t\,v^3$ then

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}F(t,\mathrm{x}(t),\dot{\mathrm{x}}(t)) = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\left(t\,\dot{\mathrm{x}}^3(t)\right)}{\partial \dot{x}} = \frac{\mathrm{d}\left(3t\,\dot{\mathrm{x}}^2(t)\right)}{\mathrm{d}t} = 3\,\dot{\mathrm{x}}^2(t) + 6t\,\dot{\mathrm{x}}(t)\,\ddot{\mathrm{x}}(t).$$

Notation for differentiation with respect to a column or a row vector. For functions $f:\mathbb{R}^n \to \mathbb{R}$ we think of its gradient at some $\bar{x} \in \mathbb{R}^n$ as a *column* vector, and we denote it as $\dfrac{\partial f(\bar{x})}{\partial x}$, so

$$\frac{\partial f(\bar{x})}{\partial x} = \begin{bmatrix} \dfrac{\partial f(\bar{x})}{\partial x_1} \\ \dfrac{\partial f(\bar{x})}{\partial x_2} \\ \vdots \\ \dfrac{\partial f(\bar{x})}{\partial x_n} \end{bmatrix} \in \mathbb{R}^n.$$

In fact, throughout we think of $\mathbb{R}^n$ as the linear space of n-dimensional *column* vectors. The above is a derivative with respect to a column vector $x \in \mathbb{R}^n$, and the outcome (the gradient) is then a column vector as well. By the same logic, if we differentiate f with respect to a *row* vector $x^{\mathrm{T}} \in \mathbb{R}^{1\times n}$—mind the transpose—then we mean the gradient seen as a row vector,

$$\frac{\partial f(\bar{x})}{\partial x^{\mathrm{T}}} = \begin{bmatrix} \dfrac{\partial f(\bar{x})}{\partial x_1} & \dfrac{\partial f(\bar{x})}{\partial x_2} & \cdots & \dfrac{\partial f(\bar{x})}{\partial x_n} \end{bmatrix} \in \mathbb{R}^{1\times n}.$$

The previous two conventions are also combined: if $F:\mathbb{R}\times\mathbb{R}^n\times\mathbb{R}^n \to \mathbb{R}$ and $\mathrm{x}:\mathbb{R}\to\mathbb{R}^n$, then

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}F(t,\mathrm{x}(t),\dot{\mathrm{x}}(t))$$

means the total derivative with respect to t of the column vector with n entries of $\frac{\partial}{\partial \dot{x}}F(t,\mathrm{x}(t),\dot{\mathrm{x}}(t))$. Then $\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}F(t,\mathrm{x}(t),\dot{\mathrm{x}}(t))$ is a column vector with n entries as well.

Occasionally we also need second-order derivatives with respect to vectors x, such as Hessian matrices. Their notation is discussed in Appendix A.2.



Chapter 1

Calculus of Variations

1.1 Introduction

Optimal control theory is deeply rooted in the classical mathematical subject referred to as the *calculus of variations*; the name of which seems to go back to the famous mathematician Leonhard Euler (1707–1783). Calculus of variations deals with minimization of expressions of the form

$$\int_0^T F(t, \mathrm{x}(t), \dot{\mathrm{x}}(t))\, \mathrm{d}t$$

over all functions

$$\mathrm{x} : [0, T] \to \mathbb{R}^n.$$

Here $F : \mathbb{R} \times \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ is some given function. Recall that $\dot{\mathrm{x}}(t)$ denotes the derivative of x with respect to its argument, i.e., $\dot{\mathrm{x}}(t) = \frac{\mathrm{d}\mathrm{x}(t)}{\mathrm{d}t}$. In contrast to basic optimization—where we optimize over a finite number of variables—we minimize in principle over all (sufficiently smooth) functions $\mathrm{x} : [0, T] \to \mathbb{R}^n$. Many fruitful applications of the calculus of variations have been developed in physics, in particular, in connection with Hamilton's principle of least action. Also in other sciences such as economics, biology, and chemistry, the calculus of variations has led to many useful applications.

We start with some motivating examples. The first example is the celebrated *brachistochrone problem*. This problem was introduced in 1696 by Johann Bernoulli and it was one of the first problems of this type[1]. When Bernoulli formulated the problem it immediately attracted a lot of attention and several mathematicians submitted solutions to this problem, including Leibniz, Newton, l'Hôpital, and Johann Bernoulli's older brother Jakob.

[1] Newton's minimal resistance problem can also be seen as a calculus of variations problem, and it predates the brachistochrone problem by almost 10 years.

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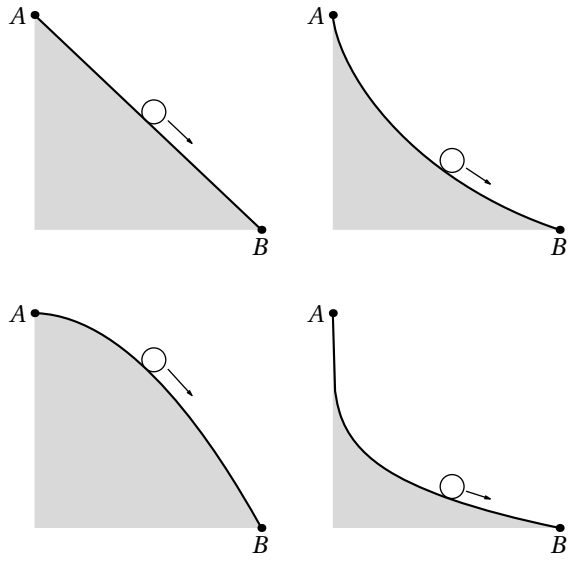


FIGURE 1.1: Four paths from A to B. Which is fastest? See Example 1.1.1.

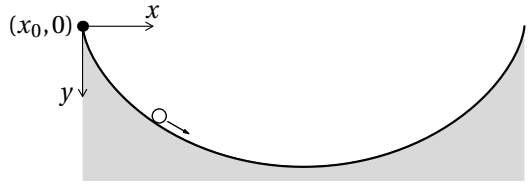


FIGURE 1.2: In this case a positive y means a negative altitude. See Example 1.1.1.

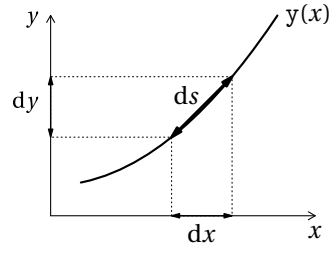


FIGURE 1.3: $\mathrm{d}s = \sqrt{1 + \dot{y}^2(x)}\,\mathrm{d}x$. Here $\dot{y}(x) := \frac{\mathrm{d}y(x)}{\mathrm{d}x}$.

Example 1.1.1 (Brachistochrone). A present-day formulation[2] of the *brachistochrone* problem is as follows. Consider two points $A = (x_0, y_0)$ and $B = (x_1, y_1)$ in $\mathbb{R}^2$. The problem is to find a path (a function) $\mathrm{y} : [x_0, x_1] \to \mathbb{R}$ through A and B such that a point mass released with zero speed at A and that glides along this path without friction reaches B in minimal time. It is assumed that the gravitational acceleration g is constant. Figure 1.1 depicts four possible such paths and, actually, one of them is the optimal solution. Can you guess which one? This is a hard problem, and we will solve this problem step by step in a series of examples (Examples 1.2.4 and 1.3.1). First we set it up as a calculus of variations problem. It is convenient to take the vertical displacement y to increase when going down (i.e., y points downwards, in the same direction as the gravitational force, see Fig. 1.2). Also, without loss of generality we take $(x_0, y_0) = (x_0, 0)$. That is, we release the point mass at zero altitude. As the mass moves friction free we have that kinetic plus potential energy is constant, i.e.,

$$\frac{1}{2}mv^2 - mgy = c$$

for some constant c. Here v is the speed of the mass. We release the mass at zero altitude and with zero speed, so $c = 0$. Hence the speed v follows uniquely from y as

$$v = \sqrt{2gy}.$$

By the Pythagorean theorem, an infinitesimal horizontal displacement $\mathrm{d}x$ corresponds to a displacement along the curve $\mathrm{y}(x)$ of $\mathrm{d}s := \sqrt{1 + \dot{\mathrm{y}}^2(x)}\,\mathrm{d}x$, see Fig. 1.3. The amount of time that this takes is

$$\mathrm{d}t = \frac{\mathrm{d}s}{v} = \sqrt{\frac{1 + \dot{\mathrm{y}}^2(x)}{2g\mathrm{y}(x)}}\mathrm{d}x.$$

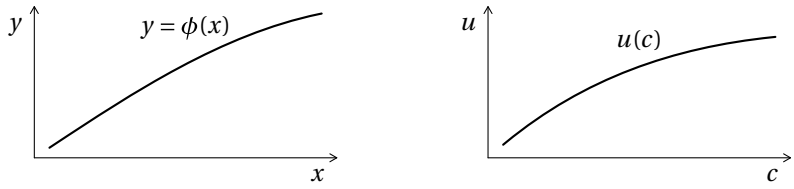
This way the time T needed to travel from $(x_0, 0)$ to (x_1, y_1) can be seen as an integral over x,

$$T = \int_0^T 1\,\mathrm{d}t = \int_{x_0}^{x_1} \sqrt{\frac{1 + \dot{\mathrm{y}}^2(x)}{2g\mathrm{y}(x)}}\,\mathrm{d}x. \tag{1.1}$$

Thus the brachistochrone problem is to minimize the integral (1.1) over all functions $\mathrm{y} : [x_0, x_1] \to \mathbb{R}$ subject to $\mathrm{y}(x_0) = y_0 = 0$ and $\mathrm{y}(x_1) = y_1$. □

Example 1.1.2 (Optimal economic investment). This example is based on an example from Seierstad and Sydsaeter (1987). We consider a simple model of

[2] For more on the history of the brachistochrone problem and subsequent developments see H.J. Sussmann and J.C. Willems. 300 years of optimal control: from the brachistochrone to the maximum principle. *IEEE Control Systems Magazine*, 17:32–44, 1997.



(a) (b)

FIGURE 1.4: Left: national product y is concave in capital stock x. Right: utility functions u are strictly concave in consumption c. See Example 1.1.2.

an economy of some country. We distinguish its *capital stock* $\mathtt{x}(t)$ in, say, euros, which is a measure of the physical capital in the country at time t. We also need the net *national product* $\mathtt{y}(t)$ in euros per unit time, which is the value of all that is produced at time t per unit time in the country. The derivative $\dot{\mathtt{x}}(t)$ of capital stock with respect to t is the increase in physical capital, and it is called *investment*. Therefore, what is left for consumption (euros per unit time) at time t is the difference between national product and investment,

$$\mathtt{c}(t) := \mathtt{y}(t) - \dot{\mathtt{x}}(t).$$

This is called *consumption*. It is assumed that the national product y follows from capital stock x, so at all times t

$$\mathtt{y}(t) = \phi(\mathtt{x}(t)) \tag{1.2}$$

for some function ϕ (which we assume to be twice continuously differentiable). It is a standard assumption that ϕ is strictly increasing and concave, that is, $\phi'(x) > 0$ and $\phi''(x) \le 0$ for all $x > 0$, see Fig. 1.4(a). This captures the not unreasonable assumption that the national product increases with increasing capital stock, but that the *rate* of this increase reduces as x gets larger.

Suppose the economy at $t = 0$ has a certain initial capital stock

$$\mathtt{x}(0) = x_0. \tag{1.3}$$

Then, given an arbitrary investment function $\dot{\mathtt{x}}(t)$, all variables in our model are determined since

$$\mathtt{x}(t) = x_0 + \int_0^t \dot{\mathtt{x}}(\tau)\,\mathrm{d}\tau. \tag{1.4}$$

The question now is: what is a good investment function $\dot{\mathtt{x}}(t)$? A way to answer this question is as follows. Suppose we have a *utility function* $u(c)$ that models the enjoyment of consuming c. Standard assumptions on utility functions are that they are strictly increasing, strictly concave, and twice continuously

differentiable, so $u'(c) > 0, u''(c) < 0$ for all $c > 0$, see Fig. 1.4(b). This is just to say that additional enjoyment of additional consumption flattens at high levels of consumption.

An investment function $\dot{\mathrm{x}}(t)$ is now considered optimal if it maximizes the integrated utility $\int_0^T u(\mathrm{c}(t))\,\mathrm{e}^{-\alpha t}\,\mathrm{d}t$, that is, if it maximizes

$$\int_0^T u(\mathrm{c}(t))\,\mathrm{e}^{-\alpha t}\,\mathrm{d}t = \int_0^T u\big(\phi(\mathrm{x}(t)) - \dot{\mathrm{x}}(t)\big)\,\mathrm{e}^{-\alpha t}\,\mathrm{d}t \tag{1.5}$$

over all investment functions $\dot{\mathrm{x}}(t)$ or, equivalently, over all functions $\mathrm{x}(t)$ satisfying (1.3). The term $\mathrm{e}^{-\alpha t}$ is a so-called *discount factor* (and α is a *discount rate*, assumed positive). This is included to express that the importance of the future utility $u(\mathrm{c}(t))$ is considered to be declining with t further in the future. The optimization problem is of the same type as before apart from the fact that we are maximizing instead of minimizing. Clearly, *maximizing* the integrated utility (1.5) is equivalent to *minimizing* its negation

$$\int_0^T -u(\mathrm{c}(t))\,\mathrm{e}^{-\alpha t}\,\mathrm{d}t = \int_0^T -u\big(\phi(\mathrm{x}(t)) - \dot{\mathrm{x}}(t)\big)\,\mathrm{e}^{-\alpha t}\,\mathrm{d}t.$$

The end time of the planning period is denoted as T, and we will assume in addition that

$$\mathrm{x}(T) = x_T \tag{1.6}$$

for some given desired capital stock x_T. This type of model for optimal economic growth was initiated by F.P. Ramsey in 1928. □

Example 1.1.3 (Cheese production). A cheesemaker is to deliver an amount of x_T kilos of cheese at a delivery time T. The cheesemaker wants to find a production schedule for completing the order with minimal costs. Let $\mathrm{x}(t)$ denote the amount of cheese at time t. We assume that both producing and storing cheese is costly. The total cost might be modeled as

$$\int_0^T \alpha\dot{\mathrm{x}}^2(t) + \beta\mathrm{x}(t)\,\mathrm{d}t, \tag{1.7}$$

where $\beta\mathrm{x}(t)$ models the storage cost per unit time and $\alpha\dot{\mathrm{x}}^2(t)$ models the production cost per unit time. The constants α, β are positive numbers. The objective of the cheesemaker is to determine a production profile $\mathrm{x}(t)$ that minimizes the above cost, subject to the conditions

$$\mathrm{x}(0) = 0, \qquad \mathrm{x}(T) = x_T, \qquad \dot{\mathrm{x}}(t) \geq 0. \tag{1.8}$$

□

Example 1.1.4 (Shortest path). What is the shortest path between two points (x_0, y_0) and (x_1, y_1) in $\mathbb{R}^2$? Of course we know the answer but let us anyway formulate this problem in more detail.

Clearly the path is characterized by a function $\mathrm{y} : [x_0, x_1] \to \mathbb{R}$. As explained in Example 1.1.1, the length $\mathrm{d}s$ of an infinitesimal part of the path follows from an infinitesimal part $\mathrm{d}x$ as $\mathrm{d}s = \sqrt{1 + \dot{\mathrm{y}}^2(x)}\,\mathrm{d}x$, see Fig. 1.3. So the total length of the path is

$$\int_{x_0}^{x_1} \sqrt{1 + \dot{\mathrm{y}}^2(x)}\,\mathrm{d}x. \tag{1.9}$$

This has to be minimized subject to

$$\mathrm{y}(x_0) = y_0, \qquad \mathrm{y}(x_1) = y_1. \tag{1.10}$$

Note that this problem is different from the brachistochrone problem. □

With the exception of the final example, the optimal solution—if one exists at all—is not easy to find.

1.2 Euler-Lagrange Equation

The examples given in the preceding section are instances of what is called *the simplest problem in the calculus of variations*:

Definition 1.2.1 (Simplest problem in the calculus of variations). Given a final time $T > 0$ and a function $F : [0, T] \times \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, and $x_0, x_T \in \mathbb{R}^n$, the *simplest problem in the calculus of variations* is to minimize the cost J defined as

$$J(\mathrm{x}) = \int_0^T F(t, \mathrm{x}(t), \dot{\mathrm{x}}(t))\,\mathrm{d}t \tag{1.11}$$

over all functions $\mathrm{x} : [0, T] \to \mathbb{R}^n$ that satisfy the boundary conditions

$$\mathrm{x}(0) = x_0, \qquad \mathrm{x}(T) = x_T. \tag{1.12}$$

□

The function J is called the *cost (function)* or *cost criterion*, and the integrand F of this cost is called the *running cost* or the *Lagrangian*. For $n = 1$ the problem is visualized in Fig. 1.5: given the two points $(0, x_0)$ and (T, x_T) each smooth function x that connects the two points determines a cost $J(\mathrm{x})$ as defined in (1.11), and the problem is to find the function x that minimizes this cost.

The calculus of variations problem can be regarded as an infinite-dimensional version of the basic optimization problem of finding a $z_* \in \mathbb{R}^n$ that minimizes a function $K : \mathbb{R}^n \to \mathbb{R}$. The difference is that the function K is replaced by an integral expression J, while vectors $z \in \mathbb{R}^n$ are replaced by functions $\mathrm{x} : [0, T] \to \mathbb{R}^n$.

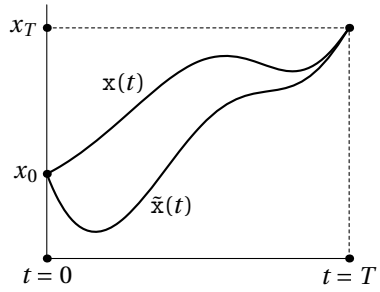


FIGURE 1.5: Two functions $\mathrm{x}, \tilde{\mathrm{x}} : [0, T] \to \mathbb{R}$ that satisfy the boundary conditions (1.12).

Mathematically, Definition 1.2.1 is not complete. We have to be more precise about the class of functions x over which we want to minimize the cost (1.11). A minimal requirement is that x is differentiable. Also, optimization problems usually require some degree of smoothness on the cost function, and this imposes further restrictions on x as well as on F. Most of the times we assume that $F(t, x, \dot{x})$ and $\mathrm{x}(t)$ are either once or twice continuously differentiable in all their arguments. This is abbreviated to C^1 (for *once continuously differentiable*) and C^2 (for *twice continuously differentiable*).

We next derive a differential equation that every solution to the simplest problem in the calculus of variations must satisfy. This differential equation is the generalization of the well-known first-order condition in basic optimization that the gradient vector $\frac{\partial K(z_*)}{\partial z}$ must be equal to zero for every $z_* \in \mathbb{R}^n$ that minimizes a differentiable function $K : \mathbb{R}^n \to \mathbb{R}$.

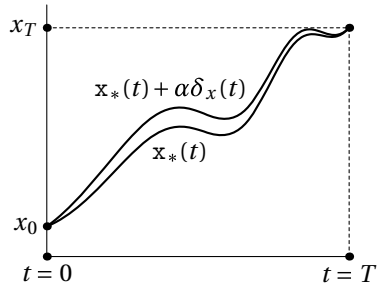


FIGURE 1.6: A function $\mathrm{x}_*(t)$ and a possible perturbed function $\mathrm{x}_*(t) + \alpha\delta_x(t)$. At $t = 0$ and $t = T$ the perturbation $\alpha\delta_x(t)$ is zero. See the proof of Theorem 1.2.2.

Theorem 1.2.2 (Euler-Lagrange equation—necessary first-order condition for optimality). Suppose that F is C^1. Necessary for a C^1 function x_* to minimize (1.11) subject to (1.12) is that it satisfies the differential equation

$$\left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\right) F(t, \mathrm{x}_*(t), \dot{\mathrm{x}}_*(t)) = 0 \qquad \text{for all } t \in [0, T]. \tag{1.13}$$

(Recall page ix for an explanation of the notation.)

Proof. Suppose x_* is a C^1 solution to the simplest problem in the calculus of variations, and let $\delta_x : [0,T] \to \mathbb{R}^n$ be an arbitrary C^1 function on $[0,T]$ that vanishes at the boundaries,

$$\delta_x(0) = \delta_x(T) = 0. \tag{1.14}$$

We use it to form a variation of the optimal solution

$$\mathrm{x}(t) = \mathrm{x}_*(t) + \alpha\delta_x(t),$$

in which $\alpha \in \mathbb{R}$. Notice that this x for every $\alpha \in \mathbb{R}$ satisfies the boundary conditions $\mathrm{x}(0) = \mathrm{x}_*(0) = x_0$ and $\mathrm{x}(T) = \mathrm{x}_*(T) = x_T$, see Fig. 1.6. Since x_* is a minimizing solution for our problem we have that

$$J(\mathrm{x}_*) \le J(\mathrm{x}_* + \alpha\delta_x) \qquad \text{for all } \alpha \in \mathbb{R}. \tag{1.15}$$

For every fixed function δ_x the cost $J(\mathrm{x}_* + \alpha\delta_x)$ is a function of the scalar variable α,

$$\bar{J}(\alpha) := J(\mathrm{x}_* + \alpha\delta_x), \qquad \alpha \in \mathbb{R}.$$

The minimality condition (1.15) thus implies that $\bar{J}(0) \le \bar{J}(\alpha)$ for all $\alpha \in \mathbb{R}$. Given that x_*, δ_x and F are all assumed C^1, it follows that $\bar{J}(\alpha)$ is differentiable as a function of α, and so the above implies that $\bar{J}'(0) = 0$. This derivative is[3]

$$\begin{aligned}\bar{J}'(0) &= \left[\frac{\mathrm{d}}{\mathrm{d}\alpha}\int_0^T F(t, \mathrm{x}_*(t) + \alpha\delta_x(t), \dot{\mathrm{x}}_*(t) + \alpha\dot{\delta}_x(t))\,\mathrm{d}t\right]_{\alpha=0} \\ &= \int_0^T \frac{\partial F(t, \mathrm{x}_*(t), \dot{\mathrm{x}}_*(t))}{\partial x^{\mathrm{T}}}\delta_x(t) + \frac{\partial F(t, \mathrm{x}_*(t), \dot{\mathrm{x}}_*(t))}{\partial \dot{x}^{\mathrm{T}}}\dot{\delta}_x(t)\,\mathrm{d}t. \end{aligned} \tag{1.16}$$

In the rest of the proof we assume that F and x_* and δ_x are C^2. (The case when they are only C^1 is slightly more involved; this is covered in Exercise 1.7.) Integration by parts of the second term in (1.16) yields[4]

$$\begin{aligned}&\int_0^T \frac{\partial F(t, \mathrm{x}_*(t), \dot{\mathrm{x}}_*(t))}{\partial \dot{x}^{\mathrm{T}}}\dot{\delta}_x(t)\,\mathrm{d}t \\ &\quad= \frac{\partial F(t, \mathrm{x}_*(t), \dot{\mathrm{x}}_*(t))}{\partial \dot{x}^{\mathrm{T}}}\delta_x(t)\Big|_0^T - \int_0^T \left(\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial F(t, \mathrm{x}_*(t), \dot{\mathrm{x}}_*(t))}{\partial \dot{x}^{\mathrm{T}}}\right)\delta_x(t)\,\mathrm{d}t. \end{aligned} \tag{1.17}$$

[3] Leibniz' integral rule says that $\frac{\mathrm{d}}{\mathrm{d}\alpha}\int G(\alpha,t)\mathrm{d}t = \int \frac{\partial G(\alpha,t)}{\partial \alpha}\mathrm{d}t$ if $G(\alpha,t)$ and $\frac{\partial G(\alpha,t)}{\partial \alpha}$ are continuous in t and α. Here they are continuous because F and δ_x are assumed C^1.

[4] The *integration by parts* rule holds if $\frac{\partial}{\partial \dot{x}^{\mathrm{T}}}F(t, \mathrm{x}_*(t), \dot{\mathrm{x}}_*(t))$ and $\delta_x(t)$ are C^1 with respect to time. This holds if $F, \mathrm{x}_*, \delta_x$ are C^2 in all their arguments.

Plugging (1.17) into (1.16) and using that $\bar{J}'(0) = 0$ we find that

$$0 = \frac{\partial F(t, \mathrm{x}_*(t), \dot{\mathrm{x}}_*(t))}{\partial \dot{x}^{\mathrm{T}}} \delta_x(t) \Big|_0^T + \int_0^T \left[\left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial}{\partial \dot{x}} \right) F(t, \mathrm{x}_*(t), \dot{\mathrm{x}}_*(t)) \right]^{\mathrm{T}} \delta_x(t) \, \mathrm{d}t. \tag{1.18}$$

The first term on the right-hand side is actually zero because of the boundary conditions (1.14). Hence we have

$$0 = \int_0^T \left[\left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial}{\partial \dot{x}} \right) F(t, \mathrm{x}_*(t), \dot{\mathrm{x}}_*(t)) \right]^{\mathrm{T}} \delta_x(t) \, \mathrm{d}t. \tag{1.19}$$

So far the perturbation δ_x in our derivation was some fixed function. However since δ_x can be arbitrarily chosen, the equality (1.19) must hold for every C^2 perturbation δ_x that satisfies (1.14). But this implies, via the result presented next (Lemma 1.2.3), that the term in between the square brackets in (1.19) is zero for all $t \in [0, T]$, i.e., that (1.13) holds. ∎

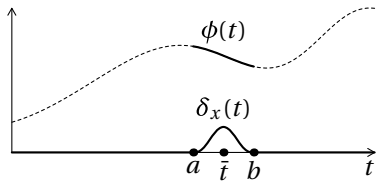


FIGURE 1.7: The function $\delta_x(t)$ defined in (1.21).

Lemma 1.2.3 (Fundamental lemma (or Lagrange's lemma)). A continuous function $\phi : [0, T] \to \mathbb{R}^n$ has the property that

$$\int_0^T \phi^{\mathrm{T}}(t) \delta_x(t) \, \mathrm{d}t = 0 \tag{1.20}$$

for every C^2 function $\delta_x : [0, T] \to \mathbb{R}^n$ satisfying (1.14) iff $\phi(t) = 0$ for all $t \in [0, T]$.

Proof. We prove it for $n = 1$. Figure 1.7 explains it all: suppose that ϕ is not the zero function, i.e., that $\phi(\bar{t})$ is nonzero for some $\bar{t} \in [0, T]$. For example, $\phi(\bar{t}) > 0$. Then, by continuity, $\phi(t)$ is positive on some interval $[a, b]$ around $\bar{t}$ (with $0 \le a < b \le T$). In order to provide a formal proof consider the function δ_x defined as

$$\delta_x(t) = \begin{cases} ((t-a)(b-t))^3 & t \in [a, b], \\ 0 & \text{elsewhere}, \end{cases} \tag{1.21}$$

see Figure 1.7. Clearly this δ_x fulfills the requirements of (1.14), but it violates (1.20) because both ϕ and δ_x are positive on $[a, b]$, and hence the integral in (1.20) is positive as well. A similar argument works for $\phi(\bar{t}) < 0$. The assumption that $\phi(\bar{t}) \neq 0$ at some $\bar{t} \in [0, T]$ hence is wrong. ∎

Theorem 1.2.2 was derived independently by Euler and Lagrange, and in honor of its inventors Equation (1.13) is nowadays called the *Euler-Lagrange equation* (or the *Euler equation*).

We want to stress that the Euler-Lagrange equation is only a *necessary* condition for optimality. All it guarantees is that a "small" perturbation of $\mathtt{x}_*$ results in a "very small" change in cost. To put it more mathematically, solutions $\mathtt{x}_*$ of the Euler-Lagrange equation are precisely those functions for which for every allowable function δ_x and $\alpha \in \mathbb{R}$ we have

$$J(\mathtt{x}_* + \alpha\delta_x) = J(\mathtt{x}_*) + \mathfrak{o}(\alpha),$$

with $\mathfrak{o}$ some *little-o* function[5]. Such solutions $\mathtt{x}_*$ are referred to as *stationary* solutions. They might be minimizing $J(\mathtt{x})$, or maximizing $J(\mathtt{x})$, or neither.

Interestingly, the Euler-Lagrange equation does not depend on the initial or final values x_0, x_T. More on this in § 1.5.

Example 1.2.4 (Brachistochrone; Example 1.1.1 continued). The Euler-Lagrange equation for the brachistochrone problem, see (1.1), reads

$$\left(\frac{\partial}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x}\frac{\partial}{\partial \dot{y}}\right)\sqrt{\frac{1+\dot{\mathtt{y}}^2(x)}{2g\mathtt{y}(x)}} = 0, \tag{1.22}$$

with the boundary conditions $\mathtt{y}(x_0) = y_0$ and $\mathtt{y}(x_1) = y_1$. One may expand (1.22) but in this form the problem is still rather complicated, and defying an explicit solution. In the following section, we use a more sophisticated approach. □

Example 1.2.5 (Shortest path; Example 1.1.4 continued). The Euler-Lagrange equation for the shortest path problem described by (1.9) and (1.10) is

$$0 = \left(\frac{\partial}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x}\frac{\partial}{\partial \dot{y}}\right)\sqrt{1+\dot{\mathtt{y}}^2(x)}, \tag{1.23}$$

with boundary conditions $\mathtt{y}(x_0) = y_0$ and $\mathtt{y}(x_1) = y_1$. Since $\frac{\partial}{\partial y}\sqrt{1+\dot{\mathtt{y}}^2(x)}$ is zero, we obtain from (1.23) that

$$0 = \frac{\mathrm{d}}{\mathrm{d}x}\frac{\partial}{\partial \dot{y}}\sqrt{1+\dot{\mathtt{y}}^2(x)} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\dot{\mathtt{y}}(x)}{\sqrt{1+\dot{\mathtt{y}}^2(x)}}\right) = \ddot{\mathtt{y}}(x)(1+\dot{\mathtt{y}}^2(x))^{-3/2}. \tag{1.24}$$

Clearly, the solution of (1.24) is given by the differential equation

$$\ddot{\mathtt{y}}(x) = 0,$$

which is another way of saying that $\mathtt{y}(x)$ is a straight line. In light of the boundary conditions $\mathtt{y}(x_0) = y_0$ and $\mathtt{y}(x_1) = y_1$, it has the unique solution

$$\mathtt{y}_*(x) = y_0 + \tfrac{y_1-y_0}{x_1-x_0}(x-x_0).$$

[5] A *little-o* function $\mathfrak{o}: \mathbb{R}^m \to \mathbb{R}^k$ is any function with the property that $\lim_{y\to 0} \frac{\|\mathfrak{o}(y)\|}{\|y\|} = 0$.

This solution is not surprising. It is of course *the* solution, although formally we may not yet draw this conclusion because the theory presented so far only claims that solutions of (1.24) are stationary solutions, not necessarily optimal solutions. Optimality is proved later (Example 1.6.8). □

Example 1.2.6 (Economic investment; Example 1.1.2 continued). For the problem of Example 1.1.2 the Euler-Lagrange equation (1.13) takes the form

$$\left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\right)\left(u\big(\phi(\mathrm{x}(t)) - \dot{\mathrm{x}}(t)\big)\,\mathrm{e}^{-\alpha t}\right) = 0,$$

which is the same as

$$u'\big(\phi(\mathrm{x}(t)) - \dot{\mathrm{x}}(t)\big)\phi'(\mathrm{x}(t))\,\mathrm{e}^{-\alpha t} - \frac{\mathrm{d}}{\mathrm{d}t}\left(-u'\big(\phi(\mathrm{x}(t)) - \dot{\mathrm{x}}(t)\big)\,\mathrm{e}^{-\alpha t}\right) = 0, \tag{1.25}$$

where u' and ϕ' denote the usual derivatives of functions of one variable. Taking the time derivative in (1.25) yields

$$\begin{aligned} &u'\big(\phi(\mathrm{x}(t)) - \dot{\mathrm{x}}(t)\big)\phi'(\mathrm{x}(t))\,\mathrm{e}^{-\alpha t} \\ &\quad + u''\big(\phi(\mathrm{x}(t)) - \dot{\mathrm{x}}(t)\big)\big(\phi'(\mathrm{x}(t))\dot{\mathrm{x}}(t) - \ddot{\mathrm{x}}(t)\big)\,\mathrm{e}^{-\alpha t} - u'\big(\phi(\mathrm{x}(t) - \dot{\mathrm{x}}(t))\big)\,\mathrm{e}^{-\alpha t} = 0. \end{aligned}$$

Dividing by $\mathrm{e}^{-\alpha t}$ (and omitting the time argument) we obtain

$$u'(\phi(\mathrm{x}) - \dot{\mathrm{x}})\phi'(\mathrm{x}) + u''(\phi(\mathrm{x}) - \dot{\mathrm{x}})(\phi'(\mathrm{x})\dot{\mathrm{x}} - \ddot{\mathrm{x}}) - u'(\phi(\mathrm{x}) - \dot{\mathrm{x}}) = 0.$$

This, together with the boundary conditions (1.3) and (1.6), has to be solved for the unknown function $\mathrm{x}(t)$, or—see also (1.4)—for the unknown investment function $\dot{\mathrm{x}}(t)$. This can be done once the utility function $u(c)$ and the consumption function $\phi(x)$ are specified. □

Example 1.2.7 (Cheese production; Example 1.1.3 continued). Corresponding to the criterion to be minimized, (1.7), we find the Euler-Lagrange equation

$$0 = \left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\right)(\alpha\dot{\mathrm{x}}^2(t) + \beta\mathrm{x}(t)) = \beta - \frac{\mathrm{d}}{\mathrm{d}t}\big(2\alpha\dot{\mathrm{x}}(t)\big) = \beta - 2\alpha\ddot{\mathrm{x}}(t).$$

So $\ddot{\mathrm{x}}(t) = \frac{\beta}{2\alpha}$, that is,

$$\mathrm{x}(t) = \frac{\beta}{4\alpha}t^2 + \dot{x}_0\,t + x_0. \tag{1.26}$$

The constants x_0 and $\dot{x}_0$ follow from the boundary conditions $\mathrm{x}(0) = 0$ and $\mathrm{x}(T) = x_T$, i.e., $x_0 = 0$ and $\dot{x}_0 = x_T/T - \beta T/(4\alpha)$. Of course, it still remains to be seen whether the $\mathrm{x}(t)$ defined in (1.26) is indeed minimizing (1.7). Notice that the extra constraint, $\dot{\mathrm{x}}(t) \geq 0$, from (1.8) puts a further restriction on the total amount of x_T and the final time T. □

All examples so far considered scalar-valued functions x, but the theory holds for general vector-valued functions. Here is an example.

Example 1.2.8 (Two-dimensional problem). Consider minimization of the integral

$$J(\mathrm{x}_1, \mathrm{x}_2) := \int_0^{\frac{\pi}{2}} \dot{\mathrm{x}}_1^2(t) + \dot{\mathrm{x}}_2^2(t) - 2\mathrm{x}_1(t)\mathrm{x}_2(t)\,\mathrm{d}t$$

over all functions $\mathrm{x}_1, \mathrm{x}_2 : [0, T] \to \mathbb{R}$ subject to the boundary conditions

$$\mathrm{x}_1(0) = 0,\ \mathrm{x}_2(0) = 0, \qquad \mathrm{x}_1(\tfrac{\pi}{2}) = 1,\ \mathrm{x}_2(\tfrac{\pi}{2}) = 1.$$

Since the minimization is over a vector $\mathrm{x} = \left[\begin{smallmatrix}\mathrm{x}_1\\ \mathrm{x}_2\end{smallmatrix}\right]$ of two components, the Euler-Lagrange equation is given by a two-dimensional system of differential equations

$$\begin{bmatrix}-2\mathrm{x}_2(t)\\ -2\mathrm{x}_1(t)\end{bmatrix} - \frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix}2\dot{\mathrm{x}}_1(t)\\ 2\dot{\mathrm{x}}_2(t)\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix},$$

that is, $\ddot{\mathrm{x}}_1(t) = -\mathrm{x}_2(t)$ and $\ddot{\mathrm{x}}_2(t) = -\mathrm{x}_1(t)$. This yields the fourth-order differential equations for each of the components, $\frac{\mathrm{d}^4}{\mathrm{d}t^4}\mathrm{x}_1(t) = \mathrm{x}_1(t)$ and $\frac{\mathrm{d}^4}{\mathrm{d}t^4}\mathrm{x}_2(t) = \mathrm{x}_2(t)$. These are linear differential equations with constant coefficients, and they can be solved with standard methods (see Appendix A.4). The general solution is

$$\begin{aligned}\mathrm{x}_1(t) &= a\,\mathrm{e}^t + b\,\mathrm{e}^{-t} + c\cos(t) + d\sin(t),\\ \mathrm{x}_2(t) = -\ddot{\mathrm{x}}_1(t) &= -a\,\mathrm{e}^t - b\,\mathrm{e}^{-t} + c\cos(t) + d\sin(t),\end{aligned}$$

with $a, b, c, d \in \mathbb{R}$. The given boundary conditions are satisfied iff $a = b = c = 0$ and $d = 1$, that is,

$$\mathrm{x}_{*1}(t) = \mathrm{x}_{*2}(t) = \sin(t).$$

□

1.3 Beltrami Identity

In many applications, the running cost $F(t, x, \dot{x})$ does not depend on t and thus has the form

$$F(x, \dot{x}).$$

Obviously the partial derivative $\frac{\partial F(x,\dot{x})}{\partial t}$ is zero now. An interesting consequence is that then

$$F(\mathrm{x}(t), \dot{\mathrm{x}}(t)) - \dot{\mathrm{x}}^{\mathrm{T}}(t)\frac{\partial F(\mathrm{x}(t), \dot{\mathrm{x}}(t))}{\partial \dot{x}}$$

is constant over time for every solution x of the Euler-Lagrange equation. To see this, we differentiate the above expression with respect to time (and for ease of notation we momentarily write $\mathrm{x}(t)$ simply as x):

$$\begin{aligned}
&\frac{\mathrm{d}}{\mathrm{d}t}\left(F(\mathrm{x},\dot{\mathrm{x}}) - \dot{\mathrm{x}}^{\mathrm{T}}\frac{\partial F(\mathrm{x},\dot{\mathrm{x}})}{\partial \dot{x}}\right)\\
&\quad= \frac{\mathrm{d}}{\mathrm{d}t}F(\mathrm{x},\dot{\mathrm{x}}) - \frac{\mathrm{d}}{\mathrm{d}t}\Big(\dot{\mathrm{x}}^{\mathrm{T}}\frac{\partial F(\mathrm{x},\dot{\mathrm{x}})}{\partial \dot{x}}\Big)\\
&\quad= \left(\dot{\mathrm{x}}^{\mathrm{T}}\frac{\partial F(\mathrm{x},\dot{\mathrm{x}})}{\partial x} + \ddot{\mathrm{x}}^{\mathrm{T}}\frac{\partial F(\mathrm{x},\dot{\mathrm{x}})}{\partial \dot{x}}\right) - \left(\ddot{\mathrm{x}}^{\mathrm{T}}\frac{\partial F(\mathrm{x},\dot{\mathrm{x}})}{\partial \dot{x}} + \dot{\mathrm{x}}^{\mathrm{T}}\Big(\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial F(\mathrm{x},\dot{\mathrm{x}})}{\partial \dot{x}}\Big)\right)\\
&\quad= \dot{\mathrm{x}}^{\mathrm{T}}\left(\frac{\partial F(\mathrm{x},\dot{\mathrm{x}})}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial F(\mathrm{x},\dot{\mathrm{x}})}{\partial \dot{x}}\right).
\end{aligned} \tag{1.27}$$

This is zero for every solution x of the Euler-Lagrange equation. Hence every stationary solution x_* has the property that

$$F(\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t)) - \dot{\mathrm{x}}_*^{\mathrm{T}}(t)\frac{\partial F(\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))}{\partial \dot{x}} = C \qquad \forall\, t\in[0,T]$$

for some integration constant C. This identity is known as the *Beltrami identity*. We illustrate the usefulness of this identity by explicitly solving the brachistochrone problem. It is good to realize, though, that the Beltrami identity is not equivalent to the Euler-Lagrange equation. Indeed, every constant function $\mathrm{x}(t)$ satisfies the Beltrami identity. The Beltrami identity and the Euler-Lagrange equation *are* equivalent for scalar functions $\mathrm{x}:[0,T]\to\mathbb{R}$ if $\dot{\mathrm{x}}(t)$ is nonzero for almost all t, as can be seen from (1.27).

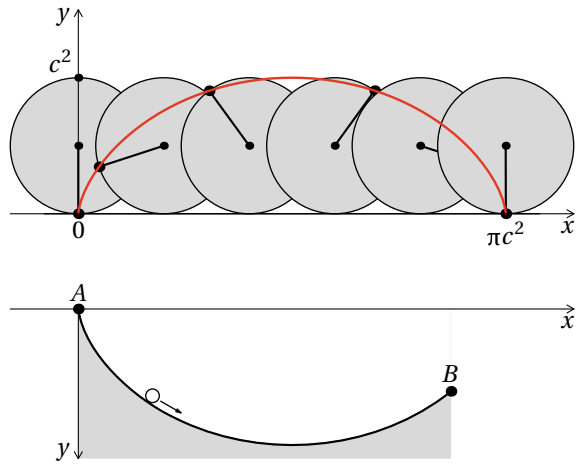


FIGURE 1.8: Top: shown in red is the cycloid $\mathrm{x}(\phi) = \frac{c^2}{2}(\phi - \sin(\phi))$, $\mathrm{y}(\phi) = \frac{c^2}{2}(1-\cos(\phi))$ for $\phi\in[0,2\pi]$. It is the curve that a point on a rolling disk of radius $c^2/2$ traces out. Bottom: a downwards facing cycloid (solution of the brachistochrone problem). See Example 1.3.1.

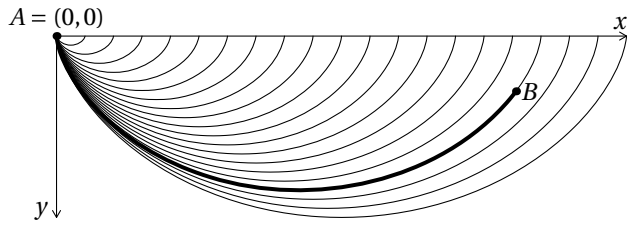


FIGURE 1.9: Cycloids (1.29) for various $c > 0$. Given a B to the right and below $A = (0,0)$ there is a unique cycloid that joins A and B. See Example 1.3.1.

Example 1.3.1 (Brachistochrone; Example 1.1.1 continued). The running cost $F(x, y, \dot{y})$ of the brachistochrone problem is

$$F(y, \dot{y}) = \frac{\sqrt{1+\dot{y}^2}}{\sqrt{2gy}}.$$

It does not depend on x, so Beltrami applies which says that the solution of the brachistochrone problem makes the following function constant (as a function of x):

$$F(\mathrm{y}(x), \dot{\mathrm{y}}(x)) - \dot{\mathrm{y}}(x)\frac{\partial F(\mathrm{y}(x), \dot{\mathrm{y}}(x))}{\partial \dot{y}} = \frac{\sqrt{1+\dot{\mathrm{y}}^2(x)}}{\sqrt{2g\mathrm{y}(x)}} - \frac{\dot{\mathrm{y}}^2(x)}{\sqrt{2g\mathrm{y}(x)(1+\dot{\mathrm{y}}^2(x))}}$$
$$= \frac{1}{\sqrt{2g\mathrm{y}(x)(1+\dot{\mathrm{y}}^2(x))}}.$$

Denote this constant as C. Squaring and inverting both sides gives

$$\mathrm{y}(x)(1+\dot{\mathrm{y}}^2(x)) = c^2, \tag{1.28}$$

where $c^2 = 1/(2gC^2)$. This equation can be solved parametrically by[6]

$$\mathrm{x}(\phi) = \frac{c^2}{2}(\phi - \sin(\phi)), \qquad \mathrm{y}(\phi) = \frac{c^2}{2}(1-\cos(\phi)). \tag{1.29}$$

The curve $(\mathrm{x}(\phi), \mathrm{y}(\phi))$ is known as the *cycloid*. It is the curve that a fixed point on the boundary of a wheel with radius $c^2/2$ traces out while rolling with-

[6]Quick derivation: since the cotangent $\cos(\phi/2)/\sin(\phi/2)$ for $\phi \in [0, 2\pi]$ ranges over all real numbers once (including $\pm\infty$) it follows that any $\mathrm{d}y/\mathrm{d}x$ can uniquely be written as $\mathrm{d}y/\mathrm{d}x = \cos(\phi/2)/\sin(\phi/2)$ with $\phi \in [0, 2\pi]$. Then (1.28) implies that $\mathrm{y}(\phi) = c^2/(1 + \cos^2(\phi/2)/\sin^2(\phi/2)) = c^2\sin^2(\phi/2) = c^2(1-\cos(\phi))/2$ and then $\mathrm{d}x/\mathrm{d}\phi = (\mathrm{d}y/\mathrm{d}\phi)/(\mathrm{d}y/\mathrm{d}x) = [c^2\sin(\phi/2)\cos(\phi/2)]/[\cos(\phi/2)/\sin(\phi/2)] = c^2\sin^2(\phi/2) = c^2(1-\cos(\phi))/2$. Integrating this expression shows that $\mathrm{x}(\phi) = c^2(\phi - \sin(\phi))/2 + d$ where d is some integration constant. This d equals zero because $(x, y) := (0,0)$ is on the curve. (See Exercise 1.4 for more details.)

out slipping on a horizontal line (think of the valve on your bike's wheel), see Fig. 1.8. For the cycloid, the Beltrami identity and the Euler-Lagrange equation are equivalent because $\dot{y}(x)$ is nonzero almost everywhere. Hence all sufficiently smooth stationary solutions of the brachistochrone problem are *precisely* these cycloids.

Varying c in (1.29) generates a family of cycloids, see Fig. 1.9. Given a destination point B to the right and below $A = (0,0)$ there is a unique cycloid that connects A and B, and the solution of the brachistochrone problem is that segment of the cycloid. Notice that for certain final destinations B the curve extends *below* the final destination! □

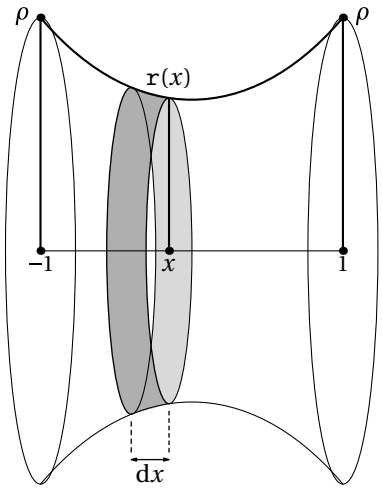


FIGURE 1.10: Given a nonnegative function $\mathtt{r} : [-1,1] \to [0,\infty)$ and its surface of revolution, the infinitesimal $\mathrm{d}x$-strip of this surface has area $2\pi\mathtt{r}(x)\sqrt{1+\dot{\mathtt{r}}^2(x)}\,\mathrm{d}x$. See Example 1.3.2.

Example 1.3.2 (Minimal surface). This is an elaborate example. We want to determine a nonnegative radius $\mathtt{r} : [-1,1] \to [0,\infty)$ for which the surface of revolution about the x-axis,

$$\{(x,y,z) \mid x \in [-1,1], y^2+z^2 = \mathtt{r}^2(x)\},$$

has minimal area, see Fig. 1.10. We assume that the radii at the endpoints are the same and equal to a given $\rho > 0$,

$$\mathtt{r}(-1) = \mathtt{r}(+1) = \rho.$$

The area of the surface of revolution over an infinitesimal $\mathrm{d}x$-strip at x equals $2\pi\mathtt{r}(x)\sqrt{1+\dot{\mathtt{r}}^2(x)}\,\mathrm{d}x$ (see Fig. 1.10) and therefore the total area $J(\mathtt{r})$ of the surface of revolution is

$$J(\mathtt{r}) = \int_{-1}^{1} 2\pi\mathtt{r}(x)\sqrt{1+\dot{\mathtt{r}}^2(x)}\,\mathrm{d}x.$$

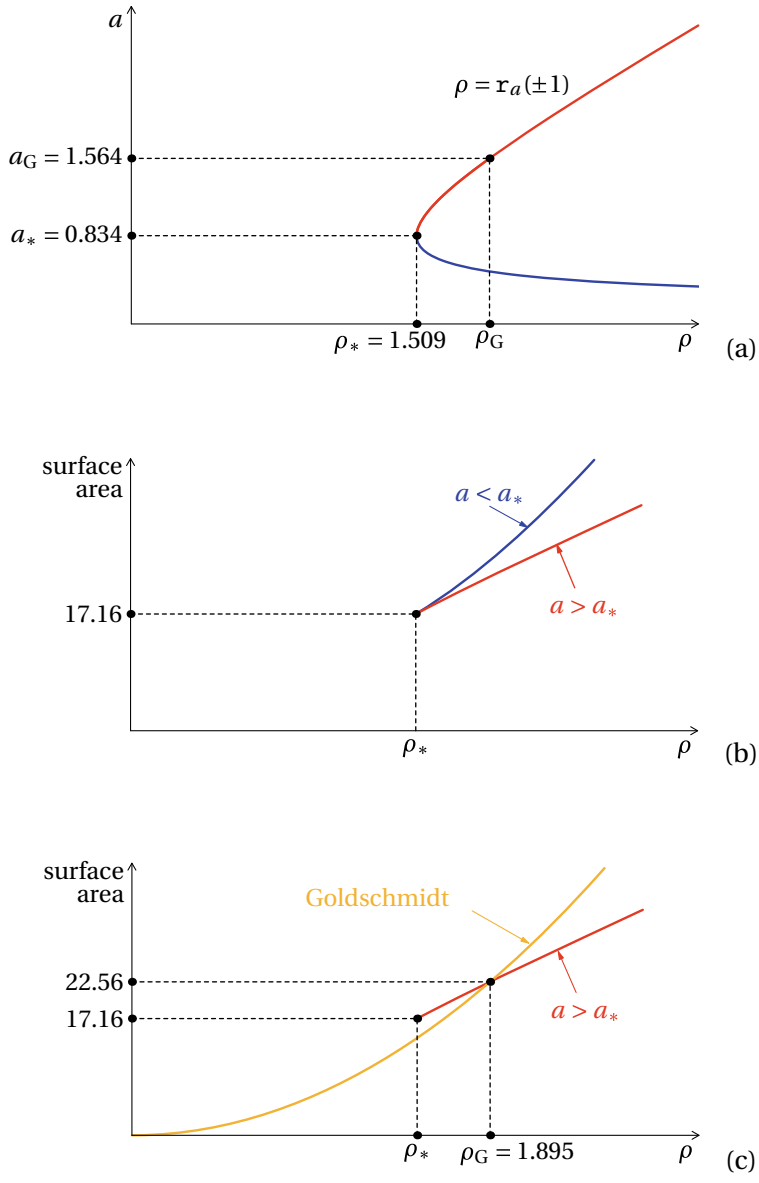


FIGURE 1.11: (a) The endpoint radius $\mathtt{r}_a(\pm 1) := a\cosh(1/a)$ of the catenoid as a function of a. Its minimal value $\mathtt{r}_a(\pm 1)$ is $\rho_* = 1.509$ (attained at $a_* = 0.834$); (b) the area of the catenoid as a function of endpoint radius ρ; (c) the area of the catenoid (in red) and of the Goldschmidt solution (in yellow) as a function of endpoint radius ρ. The two areas are the same at $\rho_G = 1.895$. This ρ_G corresponds to $a_G = 1.564$ (see part (a) of this figure). See Example 1.3.2.

Beltrami applies and it gives us that

$$2\pi \mathtt{r}(x)\sqrt{1+\dot{\mathtt{r}}^2(x)} - \dot{\mathtt{r}}(x)2\pi\mathtt{r}(x)\frac{\dot{\mathtt{r}}(x)}{\sqrt{1+\dot{\mathtt{r}}^2(x)}} = C$$

for some constant C. Multiplying left and right by the nonzero $\sqrt{1+\dot{\mathtt{r}}^2(x)}/(2\pi)$ turns this into

$$\mathtt{r}(x)(1+\dot{\mathtt{r}}^2(x)) - \mathtt{r}(x)\dot{\mathtt{r}}^2(x) = \frac{C}{2\pi}\sqrt{1+\dot{\mathtt{r}}^2(x)},$$

that is,

$$\mathtt{r}(x) = \frac{C}{2\pi}\sqrt{1+\dot{\mathtt{r}}^2(x)}.$$

Since the radius $\mathtt{r}(x)$ is nonnegative we have that $C \geq 0$, and thus $a := C/(2\pi)$ is nonnegative as well. Squaring left- and right-hand side we end up with

$$\mathtt{r}^2(x) = a^2(1+\dot{\mathtt{r}}^2(x)). \tag{1.30}$$

The nonnegative even solutions of this differential equation are[7]

$$\mathtt{r}_a(x) := a\cosh(x/a), \qquad a \geq 0. \tag{1.31}$$

Figure 1.10 shows an example of such a hyperbolic cosine. The two-dimensional surface of revolution of a hyperbolic cosine is called *catenoid*. From the shape of hyperbolic cosines, it will be clear that for every $a > 0$ the derivative $\dot{\mathtt{r}}(x)$ is nonzero almost everywhere, and so the Beltrami identity and Euler-Lagrange equation are equivalent.

But are such hyperbolic cosines optimal solutions? Not necessarily, and Figure 1.11(a) confirms this. It depicts the endpoint radius ρ of the hyperbolic cosine solution

$$\mathtt{r}_a(\pm 1) = a\cosh(1/a)$$

as a function of a (notice the flipped axes in Figure 1.11(a)). The figure demonstrates that the endpoint radius has a minimum, and the minimum is $\rho_* = 1.509$, and it is attained at $a_* = 0.834$. So if we choose an endpoint radius ρ less than ρ_* then none of these hyperbolic cosines $\mathtt{r}_a$ is the solution to our problem! Also, if $\rho > \rho_*$ then apparently there are *two* hyperbolic cosines that meet the endpoint condition, $\mathtt{r}_a(\pm 1) = \rho$, and at most one of them is the optimal solution. It can be shown that the area of the catenoid is

$$J(\mathtt{r}_a) = 2\pi a^2(\tfrac{1}{a} + \sinh(\tfrac{1}{a})\cosh(\tfrac{1}{a})).$$

[7] This hyperbolic cosine solution can be derived using separation of variables (see Appendix A.3). However, there is a technicality in this derivation that is often overlooked, see Exercise 1.6, but we need not worry about that now.

It is interesting to plot this against $\mathtt{r}_a(\pm 1) = a\cosh(1/a)$. This is done in Fig. 1.11(b). The blue curve is for $a < a_*$, and the red curve is for $a > a_*$. The plot reveals that for a given $\mathtt{r}_a(\pm 1) > \rho_*$ the area of the catenoid is the smallest for the largest of the two a's. Thus we need to only consider $a \geq a_*$.

Now the case that $\rho < \rho_*$. Then no hyperbolic cosine meets the endpoint condition. What does it mean? It means that no *smooth function* $\mathtt{r}(x)$ exists that is stationary and satisfies $\mathtt{r}(\pm 1) < \rho_*$. A deeper analysis shows that the only other stationary surface of revolution is the so-called *Goldschmidt solution*, see Fig. 1.12. The Goldschmidt solution consists of the two disks with radius ρ at respective centers $(x, y, z) = (\pm 1, 0, 0)$, and the line of radius zero, $\{(x, y, z) \mid x \in (-1, 1), y = z = 0\}$, that connects the two disks. The area of the Goldschmidt solution is the sum of the areas of the two disks at the endpoints, $2 \times \pi\rho^2$. (The line does not contribute to the area.) This set can *not* be written as the surface of revolution of a graph $(x, \mathtt{r}(x))$ of a function $\mathtt{r}$, thus it is not a surprise that it does not show up in our analysis.

It can be shown that a *global* optimal solution exists, and since it must be stationary it is either the Goldschmidt solution or the catenoid for an appropriate $a \geq a_*$. If $\rho < \rho_*$ then clearly the Goldschmidt solution is the only stationary solution, hence is optimal. For the other case, $\rho > \rho_*$, something odd occurs: Fig. 1.11(c) gives us the area of the surface of revolution of the Goldschmidt solution as well as that of the catenoid. We see that there is an endpoint radius, $\rho_G = 1.895$, at which the Goldschmidt and catenoid solutions have the same area. This point is attained at $a_G = 1.564$. For $\rho > \rho_G$ the catenoid (for the corresponding $a > a_G$) has the smallest area, hence is optimal, but for $\rho < \rho_G$ it is the Goldschmidt solution that is globally optimal. The conclusion is that the optimal shape depends *discontinuously* on the endpoint radius ρ! □

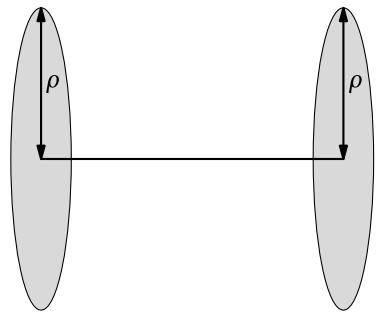


FIGURE 1.12: The Goldschmidt solution is the union of disks around the two endpoints, combined with a line that connects the centers of the two disks. See Example 1.3.2.

Example 1.3.3 (Lagrangian mechanics). Consider the one-dimensional motion of a mass m attached to a linear spring with spring constant k, see Fig. 1.13. Denote the extension of the spring caused by the mass by $q \in \mathbb{R}$. Remarkably

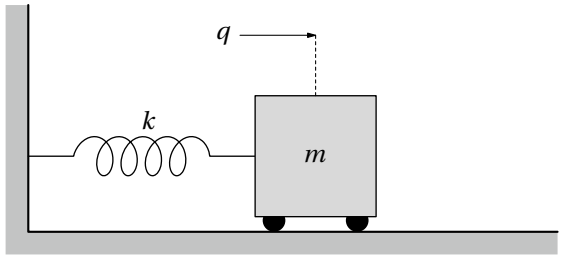


FIGURE 1.13: A mass m attached to a linear spring with spring constant k. See Example 1.3.3.

enough, the dynamics of the mass is given by the Euler-Lagrange equation corresponding to

$$F(q,\dot{q}) := \tfrac{1}{2}m\dot{q}^2 - \tfrac{1}{2}kq^2,$$

that is, the *difference* of the kinetic energy $\frac{1}{2}m\dot{q}^2$ of the mass and the potential energy $\frac{1}{2}kq^2$ of the spring. Indeed, the Euler-Lagrange equation corresponding to this $F(q,\dot{q})$ is

$$0 = \Big(\frac{\partial}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{q}}\Big)\big(\tfrac{1}{2}m\dot{\mathrm{q}}^2(t) - \tfrac{1}{2}k\mathrm{q}^2(t)\big) = -k\mathrm{q}(t) - \frac{\mathrm{d}}{\mathrm{d}t}\big(m\dot{\mathrm{q}}(t)\big) = -k\mathrm{q}(t) - m\ddot{\mathrm{q}}(t),$$

which can be recognized as Newton's law (mass times acceleration, $m\ddot{\mathrm{q}}(t)$, equals the force impressed on the mass by the spring, $-k\mathrm{q}(t)$). Hence according to Beltrami the quantity

$$\begin{aligned}\frac{\partial F(\mathrm{q}(t),\dot{\mathrm{q}}(t))}{\partial \dot{q}}\dot{\mathrm{q}}(t) - F(\mathrm{q}(t),\dot{\mathrm{q}}(t)) &= m\dot{\mathrm{q}}^2(t) - \Big(\tfrac{1}{2}m\dot{\mathrm{q}}^2(t) - \tfrac{1}{2}k\mathrm{q}^2(t)\Big)\\ &= \tfrac{1}{2}m\dot{\mathrm{q}}^2(t) + \tfrac{1}{2}k\mathrm{q}^2(t)\end{aligned}$$

is constant over time. This quantity is nothing else than the *total energy*, that is, kinetic *plus* potential energy. Thus the Beltrami identity is in this case the well-known *conservation of energy* of a mechanical system with conservative forces (in this case the spring force).

In general, in classical mechanics the difference of the kinetic and potential energy $F(\mathrm{q}(t),\dot{\mathrm{q}}(t))$ is referred to as the *Lagrangian*, while the integral

$$\int_0^T F(\mathrm{q}(t),\dot{\mathrm{q}}(t))\,\mathrm{d}t$$

is referred to as the *action integral*. The stationary property of the action integral is known as *Hamilton's principle*; see, e.g., Lanczos (1986) for the close connection between the calculus of variations and classical mechanics. □

1.4 Higher-Order Euler-Lagrange Equation

The Euler-Lagrange equation can directly be extended to the case that the integral $J(\mathrm{x})$ depends on *higher-order* derivatives of x. Let us state explicitly the second-order case.

Proposition 1.4.1 (Higher-order Euler-Lagrange equation). Consider the problem of minimizing

$$J(\mathrm{x}) := \int_0^T F(t, \mathrm{x}(t), \dot{\mathrm{x}}(t), \ddot{\mathrm{x}}(t))\,\mathrm{d}t \tag{1.32}$$

over all C^2 functions $\mathrm{x} : [0,T] \to \mathbb{R}^n$ that satisfy the boundary conditions

$$\begin{aligned} \mathrm{x}(0) &= x_0, \qquad & \mathrm{x}(T) &= x_T, \\ \dot{\mathrm{x}}(0) &= x_0^d, \qquad & \dot{\mathrm{x}}(T) &= x_T^d, \end{aligned} \tag{1.33}$$

for given $x_0, x_0^d, x_T, x_T^d \in \mathbb{R}^n$. Suppose F is C^2. A necessary condition that a C^2 function x_* minimizes (1.32) and satisfies (1.33) is that x_* is a solution of the differential equation

$$\left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}} + \frac{\mathrm{d}^2}{\mathrm{d}t^2}\frac{\partial}{\partial \ddot{x}}\right) F(t, \mathrm{x}_*(t), \dot{\mathrm{x}}_*(t), \ddot{\mathrm{x}}_*(t)) = 0 \qquad \forall\, t \in [0,T]. \tag{1.34}$$

Proof. We prove it for the case that F and x_* are C^3. (If they are only C^2 then one can use the lemma of du Bois-Reymond as explained for the standard problem in Exercise 1.7.) Define $\bar{J}(\alpha) = J(\mathrm{x}_* + \alpha\delta_x)$ where $\delta_x : [0,T] \to \mathbb{R}^n$ is a C^3 perturbation that satisfies the boundary conditions $\delta_x(0) = \delta_x(T) = 0$ and $\dot{\delta}_x(0) = \dot{\delta}(T) = 0$. Then, as before, the derivative $\bar{J}'(0)$ is zero. Analogously to (1.16) we compute $\bar{J}'(0)$. For ease of exposition we momentarily omit all time arguments in $\mathrm{x}_*(t)$ and $\delta_x(t)$ and, sometimes, F:

$$\begin{aligned} 0 = \bar{J}'(0) &= \left[\frac{\mathrm{d}}{\mathrm{d}\alpha}\int_0^T F(t, \mathrm{x}_* + \alpha\delta_x, \dot{\mathrm{x}}_* + \alpha\dot{\delta}_x, \ddot{\mathrm{x}}_* + \alpha\ddot{\delta}_x)\,\mathrm{d}t\right]_{\alpha=0} \\ &= \int_0^T \frac{\partial F}{\partial x^{\mathrm{T}}}\delta_x + \frac{\partial F}{\partial \dot{x}^{\mathrm{T}}}\dot{\delta}_x + \frac{\partial F}{\partial \ddot{x}^{\mathrm{T}}}\ddot{\delta}_x\,\mathrm{d}t. \end{aligned} \tag{1.35}$$

Integration by parts of the second term of the integrand yields

$$\int_0^T \frac{\partial F}{\partial \dot{x}^{\mathrm{T}}}\dot{\delta}_x\,\mathrm{d}t = \underbrace{\left[\frac{\partial F}{\partial \dot{x}^{\mathrm{T}}}\delta_x\right]_0^T}_{=0} - \int_0^T \left(\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial F}{\partial \dot{x}^{\mathrm{T}}}\right)\delta_x\,\mathrm{d}t = -\int_0^T \left(\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial F}{\partial \dot{x}^{\mathrm{T}}}\right)\delta_x\,\mathrm{d}t.$$

The last equality follows from the boundary condition that $\delta_x(0) = \delta_x(T) = 0$. Integration by parts of the third term in (1.35) similarly gives

$$\int_0^T \frac{\partial F}{\partial \ddot{x}^{\mathrm{T}}}\ddot{\delta}_x\,\mathrm{d}t = \underbrace{\left[\frac{\partial F}{\partial \ddot{x}^{\mathrm{T}}}\dot{\delta}_x\right]_0^T}_{=0} - \int_0^T \left(\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial F}{\partial \ddot{x}^{\mathrm{T}}}\right)\dot{\delta}_x\,\mathrm{d}t = -\int_0^T \left(\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial F}{\partial \ddot{x}^{\mathrm{T}}}\right)\dot{\delta}_x\,\mathrm{d}t, \tag{1.36}$$

where now the second equality is the result of the boundary conditions that $\dot{\delta}_x(0) = \dot{\delta}_x(T) = 0$. In fact, we can apply integration by parts again on the final term of (1.36) to obtain

$$\int_0^T \frac{\partial F}{\partial \ddot{x}^{\mathrm{T}}} \ddot{\delta}_x \,\mathrm{d}t = -\int_0^T \left(\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial F}{\partial \ddot{x}^{\mathrm{T}}}\right)\dot{\delta}_x \,\mathrm{d}t = \underbrace{\left[-\left(\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial F}{\partial \ddot{x}^{\mathrm{T}}}\right)\delta_x\right]_0^T}_{=0} + \int_0^T \left(\frac{\mathrm{d}^2}{\mathrm{d}t^2}\frac{\partial F}{\partial \ddot{x}^{\mathrm{T}}}\right)\delta_x \,\mathrm{d}t.$$

Thus (1.35) equals

$$0 = \bar{J}'(0) = \int_0^T \left(\frac{\partial F}{\partial x^{\mathrm{T}}} - \left(\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial F}{\partial \dot{x}^{\mathrm{T}}}\right) + \left(\frac{\mathrm{d}^2}{\mathrm{d}t^2}\frac{\partial F}{\partial \ddot{x}^{\mathrm{T}}}\right)\right)\delta_x \,\mathrm{d}t.$$

As before, Lemma 1.2.3 now yields (1.34). ∎

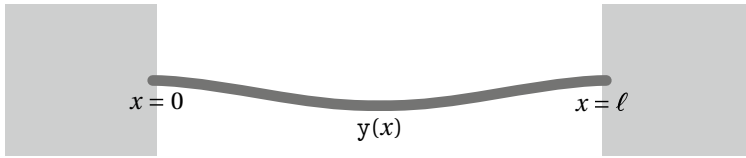


FIGURE 1.14: Elastic bar. See Example 1.4.2.

Example 1.4.2 (Elastic bar). Consider an elastic bar clamped at its two ends, see Fig. 1.14. The bar bends under the influence of gravity. The horizontal and vertical positions we denote by x and y, respectively. The shape of the bar is modeled with the function $\mathrm{y}(x)$. We assume the bar has a uniform cross section (independent of x). If the curvature of the elastic bar is not too large then the potential energy due to elastic forces can be considered, up to first order, to be proportional to the square of the second derivative,

$$V_1 := \frac{k}{2}\int_0^\ell \left(\frac{\mathrm{d}^2\mathrm{y}(x)}{\mathrm{d}x^2}\right)^2 \mathrm{d}x,$$

where k is a constant depending on the elasticity of the bar. Furthermore, the potential energy due to gravity is given by

$$V_2 := \int_0^\ell g\rho(x)\mathrm{y}(x)\,\mathrm{d}x.$$

Here, $\rho(x)$ is the mass density of the bar at x, and, again, we assume that the curvature is small. The total potential energy thus is

$$\int_0^\ell \frac{k}{2}\left(\frac{\mathrm{d}^2\mathrm{y}(x)}{\mathrm{d}x^2}\right)^2 + g\rho(x)\mathrm{y}(x)\,\mathrm{d}x.$$

The minimal potential energy solution satisfies the Euler-Lagrange equation (1.34), and this gives the fourth-order differential equation

$$k\frac{\mathrm{d}^4\mathrm{y}(x)}{\mathrm{d}x^4} = -g\rho(x) \qquad \forall x \in [0,\ell].$$

If $\rho(x)$ is constant then $\mathrm{y}(x)$ is a polynomial of degree 4. Figure 1.14 depicts a solution for constant ρ and boundary conditions $\mathrm{y}(0) = \mathrm{y}(\ell) = 0$ and $\dot{\mathrm{y}}(0) = \dot{\mathrm{y}}(\ell) = 0$. In this case, the solution is $\mathrm{y}(x) = -\frac{g\rho}{4!k}\big(x(x-\ell)\big)^2$. □

1.5 Relaxed Boundary Conditions

In the problems considered so far, the initial $\mathrm{x}(0)$ and final $\mathrm{x}(T)$ were *fixed*. A useful extension is obtained by removing some of these conditions. This means that we allow *more* functions x to optimize over, and, consequently, we expect that the Euler-Lagrange equation still holds for the optimal solution. To get an idea we first look at an example.

Suppose x has three components and that the first component of $\mathrm{x}(0)$ and the last component of $\mathrm{x}(T)$ are free to choose:

$$\mathrm{x}(0) = \begin{bmatrix} \text{free} \\ \text{fixed} \\ \text{fixed} \end{bmatrix}, \qquad \mathrm{x}(T) = \begin{bmatrix} \text{fixed} \\ \text{fixed} \\ \text{free} \end{bmatrix}. \tag{1.37}$$

In the proof of Theorem 1.2.2 we found the following necessary first-order condition for optimality (Eqn. (1.18)):

$$\frac{\partial F(t,\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))}{\partial \dot{x}^{\mathrm{T}}}\delta_x(t)\bigg|_0^T + \int_0^T \left(\left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\right)F(t,\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))\right)^{\mathrm{T}}\delta_x(t)\,\mathrm{d}t = 0. \tag{1.38}$$

This equality needs to hold for every possible perturbation δ_x. In particular, it needs to hold for every perturbation δ_x that is zero at $t = 0$ and $t = T$. For these special perturbations, the first-order condition (1.38) reduces to that of the standard problem, i.e., that

$$\int_0^T \left(\left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\right)F(t,\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))\right)^{\mathrm{T}}\delta_x(t)\,\mathrm{d}t = 0$$

for all such special δ_x. It proves that also for relaxed boundary conditions the Euler-Lagrange equation holds (as was to be expected). Knowing this, the first-order condition (1.38) simplifies to

$$\frac{\partial F(t,\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))}{\partial \dot{x}^{\mathrm{T}}}\delta_x(t)\bigg|_0^T = 0. \tag{1.39}$$

When is this equal to zero for every allowable perturbation? Since the perturbed $\mathrm{x}_*(t) + \alpha\delta_x(t)$ for our example must obey the boundary condition (1.37) it follows that the allowable perturbations are exactly those that satisfy

$$\delta_x(0) = \begin{bmatrix} \text{free} \\ 0 \\ 0 \end{bmatrix}, \qquad \delta_x(T) = \begin{bmatrix} 0 \\ 0 \\ \text{free} \end{bmatrix}.$$

Clearly, the first-order condition (1.39) holds for all such δ_x iff

$$\frac{\partial F(0,\mathrm{x}(0),\dot{\mathrm{x}}(0))}{\partial \dot{x}} = \begin{bmatrix} 0 \\ \text{free} \\ \text{free} \end{bmatrix}, \qquad \frac{\partial F(T,\mathrm{x}(T),\dot{\mathrm{x}}(T))}{\partial \dot{x}} = \begin{bmatrix} \text{free} \\ \text{free} \\ 0 \end{bmatrix}.$$

This example demonstrates that *to every initial or final entry of* x *that is free to choose there corresponds a condition on the derivative of* F *with respect to that component of* $\dot{\mathrm{x}}$. Incidentally, by allowing functions x with free entries at initial and/or final time, it can now make sense to include an initial- and/or final cost to the cost function:

$$J(\mathrm{x}) = \int_0^T F(t,\mathrm{x}(t),\dot{\mathrm{x}}(t))\,\mathrm{d}t + G(\mathrm{x}(0)) + K(\mathrm{x}(T)). \tag{1.40}$$

Here $G(\mathrm{x}(0))$ denotes an *initial cost*, and $K(\mathrm{x}(T))$ a *final cost* (also known as *terminal cost*). The addition of these two costs does not complicate matters much, as detailed in the next proposition.

Proposition 1.5.1 (Relaxed boundary conditions). Let $T > 0$, and suppose $F : [0,T] \times \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ is C^1, and that $K, G : \mathbb{R}^n \to \mathbb{R}$ are C^1. Let $\mathbb{I}_0, \mathbb{I}_T$ be subsets of $\{1,\dots,n\}$, and consider the functions $\mathrm{x} : [0,T] \to \mathbb{R}^n$ whose initial $\mathrm{x}(0)$ and final $\mathrm{x}(T)$ are fixed except for the components

$$\mathrm{x}_i(0) = \text{free} \quad \forall i \in \mathbb{I}_0 \qquad \text{and} \qquad \mathrm{x}_j(T) = \text{free} \quad \forall j \in \mathbb{I}_T.$$

Among these functions, a C^1 function x_* is a stationary solution of the cost (1.40) iff it satisfies the Euler-Lagrange equation (1.13) together with

$$\frac{\partial F(0,\mathrm{x}_*(0),\dot{\mathrm{x}}_*(0))}{\partial \dot{x}_i} - \frac{\partial G(\mathrm{x}_*(0))}{\partial x_i} = 0 \qquad \forall i \in \mathbb{I}_0, \tag{1.41}$$

$$\frac{\partial F(T,\mathrm{x}_*(T),\dot{\mathrm{x}}_*(T))}{\partial \dot{x}_j} + \frac{\partial K(\mathrm{x}_*(T))}{\partial x_j} = 0 \qquad \forall j \in \mathbb{I}_T. \tag{1.42}$$

Proof. See Exercise 1.10. ∎

This general result is needed in the next chapter when we tackle the optimal control problem. A common special case is the *free endpoint* problem, which is when $\mathrm{x}(0)$ is completely fixed and $\mathrm{x}(T)$ is completely free. In the terminology of Proposition 1.5.1 this means $\mathbb{I}_0 = \emptyset$ and $\mathbb{I}_T = \{1,\dots,n\}$. In this case Proposition 1.5.1 simplifies as follows.

Corollary 1.5.2 (Free endpoint). Let $T > 0$, $x_0 \in \mathbb{R}^n$, and suppose both $F : [0,T] \times \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ and $K : \mathbb{R}^n \to \mathbb{R}$ are C^1. Necessary for a C^1 function $\mathrm{x}_* : [0,T] \to \mathbb{R}^n$ to minimize

$$J(\mathrm{x}) = \int_0^T F(t,\mathrm{x}(t),\dot{\mathrm{x}}(t))\,\mathrm{d}t + K(\mathrm{x}(T))$$

over all functions with $\mathrm{x}(0) = x_0$ is that it satisfies the Euler-Lagrange equation (1.13) together with the free endpoint boundary condition

$$\frac{\partial F(T, \mathrm{x}_*(T), \dot{\mathrm{x}}_*(T))}{\partial \dot{x}} + \frac{\partial K(\mathrm{x}_*(T))}{\partial x} = 0 \in \mathbb{R}^n. \tag{1.43}$$

□

Example 1.5.3 (Quadratic cost with fixed and free endpoint). Let $\alpha \in \mathbb{R}$, and consider minimization of

$$\int_{-1}^{1} \alpha^2 \mathrm{x}^2(t) + \dot{\mathrm{x}}^2(t)\,\mathrm{d}t \tag{1.44}$$

over all functions $\mathrm{x} : [-1,1] \to \mathbb{R}$. First we solve the standard problem, so where both $\mathrm{x}(0)$ and $\mathrm{x}(T)$ are fixed. For instance, assume that

$$\mathrm{x}(-1) = 1, \qquad \mathrm{x}(1) = 1. \tag{1.45}$$

The running cost $\alpha^2 \mathrm{x}^2(t) + \dot{\mathrm{x}}^2(t)$ is a sum of two squares, so with minimization we would like both terms small. But one depends on the other. The parameter α models a trade-off between small $\dot{\mathrm{x}}^2(t)$ and small $\mathrm{x}^2(t)$. Whatever α is, the optimal solution x needs to satisfy the Euler-Lagrange equation,

$$0 = \left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\right)\left(\alpha^2 \mathrm{x}^2(t) + \dot{\mathrm{x}}^2(t)\right) = 2\alpha^2 \mathrm{x}(t) - \frac{\mathrm{d}}{\mathrm{d}t}(2\dot{\mathrm{x}}(t)) = 2\alpha^2 \mathrm{x}(t) - 2\ddot{\mathrm{x}}(t).$$

Therefore

$$\ddot{\mathrm{x}}(t) = \alpha^2 \mathrm{x}(t).$$

This differential equation can be solved using characteristic equations (do this yourself, see Appendix A.4), and the general solution is

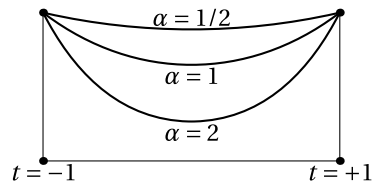
$$\mathrm{x}(t) = c\,\mathrm{e}^{\alpha t} + d\,\mathrm{e}^{-\alpha t} \tag{1.46}$$

with c, d two arbitrary constants. The two constants follow from the two boundary conditions (1.45):

$$1 = \mathrm{x}(-1) = c\,\mathrm{e}^{-\alpha} + d\,\mathrm{e}^{+\alpha},$$
$$1 = \mathrm{x}(1) = c\,\mathrm{e}^{+\alpha} + d\,\mathrm{e}^{-\alpha}.$$

The solution is $c = d = 1/(\mathrm{e}^{\alpha} + \mathrm{e}^{-\alpha})$. That c equals d is expected because of the symmetry of the boundary conditions. We see that there is exactly one function x that satisfies the Euler-Lagrange equation and that meets the boundary conditions:

$$\mathrm{x}_*(t) = \frac{\mathrm{e}^{\alpha t} + \mathrm{e}^{-\alpha t}}{\mathrm{e}^{\alpha} + \mathrm{e}^{-\alpha}}$$



For $\alpha = 0$ the solution is a constant, $\mathrm{x}_*(t) = 1$, which, in hindsight, is not a surprise because for $\alpha = 0$ the running cost is just $F(t, \mathrm{x}(t), \dot{\mathrm{x}}(t)) = \dot{\mathrm{x}}^2(t)$ and then clearly a zero derivative (a constant $\mathrm{x}(t)$) is optimal. For large values of α, on the other hand, the term $\mathrm{x}^2(t)$ is penalized strongly in the running cost, $\dot{\mathrm{x}}^2(t) + \alpha^2\mathrm{x}^2(t)$, so then it pays to take $\mathrm{x}(t)$ close to zero, even if that is at the expense of some increase of $\dot{\mathrm{x}}^2(t)$. Indeed this is what happens.

Consider next the free endpoint problem with

$$\mathrm{x}(-1) = 1 \text{ but where } \mathrm{x}(1) \text{ is free.}$$

We stick to the same cost function (1.44). In the terminology of (1.40) this means we take the initial and final costs equal to zero, $G(x) = K(x) = 0$. Hence $\frac{\partial K(\mathrm{x}(T))}{\partial x} = 0$, and the free endpoint boundary condition (1.43) thus becomes

$$0 = \frac{\partial F(T, \mathrm{x}(T), \dot{\mathrm{x}}(T))}{\partial \dot{x}} + \frac{\partial K(\mathrm{x}(T))}{\partial x} = \frac{\partial \alpha^2\mathrm{x}^2(1) + \dot{\mathrm{x}}^2(1)}{\partial \dot{x}} + 0 = 2\dot{\mathrm{x}}(1).$$

The parameters c, d in (1.46) now follow from the initial condition $\mathrm{x}(-1) = 1$ and the above boundary condition $0 = \dot{\mathrm{x}}(1)$:

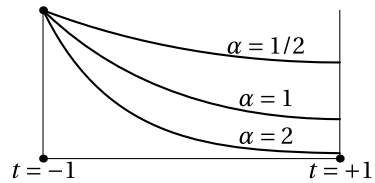
$$\begin{aligned} 1 &= \mathrm{x}(-1) = c\,\mathrm{e}^{-\alpha} + d\,\mathrm{e}^{+\alpha}, \\ 0 &= \dot{\mathrm{x}}(1) \quad = c\alpha\,\mathrm{e}^{+\alpha} - d\alpha\,\mathrm{e}^{-\alpha}. \end{aligned}$$

The solution is

$$c = \frac{\mathrm{e}^{-\alpha}}{\mathrm{e}^{2\alpha} + \mathrm{e}^{-2\alpha}}, \qquad d = \frac{\mathrm{e}^{+\alpha}}{\mathrm{e}^{2\alpha} + \mathrm{e}^{-2\alpha}},$$

(check it for yourself). We see that also in this case the first-order conditions together with the boundary condition have a unique solution,

$$\mathrm{x}_*^{\text{free}}(t) = \frac{\mathrm{e}^{\alpha(t-1)} + \mathrm{e}^{-\alpha(t-1)}}{\mathrm{e}^{2\alpha} + \mathrm{e}^{-2\alpha}}.$$



The free endpoint condition is that the derivative of x is zero at the final time. Again we see that the solution approaches zero fast if α is large. □

1.6 Second-Order Conditions for Minimality

The Euler-Lagrange equation was derived from the condition that minimizing solutions x_* are necessarily stationary solutions, i.e., solutions for which

$$J(\mathrm{x}_* + \alpha\delta_x) = J(\mathrm{x}_*) + \mathfrak{o}(\alpha)$$

for every fixed admissible perturbation function δ_x and all scalars α. But not all stationary solutions are minimizing solutions. To be minimizing the above term "$\mathfrak{o}(\alpha)$" needs to be nonnegative in a neighborhood of $\alpha = 0$. In this section we analyze this problem. We derive a *necessary* condition and a *sufficient* condition for stationary solutions to be minimizing. These conditions are *second-order conditions* and they require a second-order Taylor series expansion of $F(t,x,y)$ for fixed t around $(x,y) \in \mathbb{R}^n \times \mathbb{R}^n$:

$$\begin{aligned} F(t,x+\delta_x,y+\delta_y) = F(t,x,y) &+ \begin{bmatrix} \dfrac{\partial F(t,x,y)}{\partial x^{\mathrm{T}}} & \dfrac{\partial F(t,x,y)}{\partial y^{\mathrm{T}}} \end{bmatrix} \begin{bmatrix} \delta_x \\ \delta_y \end{bmatrix} \\ &+ \frac{1}{2} \begin{bmatrix} \delta_x^{\mathrm{T}} & \delta_y^{\mathrm{T}} \end{bmatrix} \underbrace{\begin{bmatrix} \dfrac{\partial^2 F(t,x,y)}{\partial x \partial x^{\mathrm{T}}} & \dfrac{\partial^2 F(t,x,y)}{\partial x \partial y^{\mathrm{T}}} \\ \dfrac{\partial^2 F(t,x,y)}{\partial y \partial x^{\mathrm{T}}} & \dfrac{\partial^2 F(t,x,y)}{\partial y \partial y^{\mathrm{T}}} \end{bmatrix}}_{\text{Hessian of } F} \begin{bmatrix} \delta_x \\ \delta_y \end{bmatrix} \\ &+ \mathfrak{o}\Big(\Big\| \begin{bmatrix} \delta_x \\ \delta_y \end{bmatrix} \Big\|^2 \Big). \end{aligned} \tag{1.47}$$

(The role of the transpose is explained on page x. More details about this notation can be found in Appendix A.2.) We assume that $F(t,x,y)$ is C^2 so the above Taylor series is valid, and the $2n \times 2n$ Hessian of F exists and is symmetric.

Theorem 1.6.1 (Legendre condition—second-order necessary condition). Consider the simplest problem in the calculus of variations, and suppose that F is C^2. Let x_* be a C^2 solution of the Euler-Lagrange equation (1.13) and which satisfies the boundary conditions (1.12). Necessary for x_* to be minimizing is that

$$\frac{\partial^2 F(t, \mathrm{x}_*(t), \dot{\mathrm{x}}_*(t))}{\partial \dot{x} \partial \dot{x}^{\mathrm{T}}} \geq 0 \qquad \forall\, t \in [0,T]. \tag{1.48}$$

Proof. For ease of notation we prove it for the case that x has one component. Similar to the proof of Theorem 1.2.2, let δ_x be a C^2-perturbation on $[0,T]$ that satisfies the boundary condition (1.14). Let $\alpha \in \mathbb{R}$ and define $\bar{J}(\alpha)$ as

$$\bar{J}(\alpha) := J(\mathrm{x}_* + \alpha \delta_x).$$

By construction we have that every solution x_* of the Euler-Lagrange equation achieves $\bar{J}'(0) = 0$. For simplicity of notation we omit time arguments in what follows. With the help of (1.47) we find that

$$\begin{aligned} \bar{J}''(0) &= \int_0^T \begin{bmatrix} \delta_x & \dot{\delta}_x \end{bmatrix} \underbrace{\begin{bmatrix} \frac{\partial^2 F(t,\mathrm{x}_*,\dot{\mathrm{x}}_*)}{\partial x^2} & \frac{\partial^2 F(t,\mathrm{x}_*,\dot{\mathrm{x}}_*)}{\partial x \partial \dot{x}} \\ \frac{\partial^2 F(t,\mathrm{x}_*,\dot{\mathrm{x}}_*)}{\partial x \partial \dot{x}} & \frac{\partial^2 F(t,\mathrm{x}_*,\dot{\mathrm{x}}_*)}{\partial \dot{x}^2} \end{bmatrix}}_{\text{Hessian}} \begin{bmatrix} \delta_x \\ \dot{\delta}_x \end{bmatrix} \mathrm{d}t \\ &= \int_0^T \frac{\partial^2 F}{\partial x^2} \delta_x^2 + 2 \frac{\partial^2 F}{\partial x \partial \dot{x}} \delta_x \dot{\delta}_x + \frac{\partial^2 F}{\partial \dot{x}^2} \dot{\delta}_x^2 \, \mathrm{d}t. \end{aligned} \tag{1.49}$$

If $\mathtt{x}_*$ is optimal then this has to be nonnegative for every allowable δ_x. This does not necessarily mean that the Hessian is positive semi-definite because δ_x and $\dot{\delta}_x$ are related. Indeed, using integration by parts, the cross term can be rewritten as

$$\int_0^T 2\tfrac{\partial^2 F}{\partial x \partial \dot{x}} \delta_x \dot{\delta}_x \,\mathrm{d}t = \int_0^T \tfrac{\partial^2 F}{\partial x \partial \dot{x}} (\tfrac{\mathrm{d}}{\mathrm{d}t} \delta_x^2) \,\mathrm{d}t = \underbrace{\tfrac{\partial^2 F}{\partial x \partial \dot{x}} \delta_x^2 \Big|_0^T}_{0} - \int_0^T (\tfrac{\mathrm{d}}{\mathrm{d}t} \tfrac{\partial^2 F}{\partial x \partial \dot{x}}) \delta_x^2 \,\mathrm{d}t.$$

Therefore

$$\bar{J}''(0) = \int_0^T \big(\tfrac{\partial^2 F}{\partial x^2} - \tfrac{\mathrm{d}}{\mathrm{d}t} \tfrac{\partial^2 F}{\partial x \partial \dot{x}}\big) \delta_x^2 + \tfrac{\partial^2 F}{\partial \dot{x}^2} \dot{\delta}_x^2 \,\mathrm{d}t. \tag{1.50}$$

If $\mathtt{x}_*$ is optimal then $\bar{J}''(0) \geq 0$ for every allowable perturbation δ_x. Lemma 1.6.2 (presented next) applied to (1.50) shows that this implies that $\frac{\partial^2 F(t, \mathtt{x}_*(t), \dot{\mathtt{x}}_*(t))}{\partial \dot{x}^2}$ is nonnegative for all time, i.e., that (1.48) holds. ∎

The above proof uses the following lemma.

Lemma 1.6.2 (Technical lemma). Let ϕ and ψ be continuous functions from $[0, T]$ to $\mathbb{R}$, and suppose that

$$\int_0^T \phi(t)\delta_x^2(t) + \psi(t)\dot{\delta}_x^2(t) \,\mathrm{d}t \geq 0 \tag{1.51}$$

for every C^2 function $\delta_x : [0, T] \to \mathbb{R}$ with $\delta_x(0) = \delta_x(T) = 0$. Then

$$\psi(t) \geq 0 \qquad \forall\, t \in [0, T].$$

Proof. Suppose, on the contrary, that $\psi(\bar{t}) < 0$ for some $\bar{t} \in [0, T]$. Then for every $\epsilon > 0$ we can construct a possibly small interval $[a, b]$ about $\bar{t}$ in $[0, T]$ and a C^2 function δ_x on $[0, T]$ that is zero for $t \notin [a, b]$ and that satisfies

$$\int_a^b \delta_x^2(t) \,\mathrm{d}t < \epsilon \quad \text{and} \quad \int_a^b \dot{\delta}_x^2(t) \,\mathrm{d}t > 1.$$

This may be clear from Figure 1.15. Such a δ_x satisfies all the conditions of the lemma but renders the integral in (1.51) negative for small enough $\epsilon > 0$. That is a contradiction, and so the assumption that $\psi(\bar{t}) < 0$ is wrong. ∎

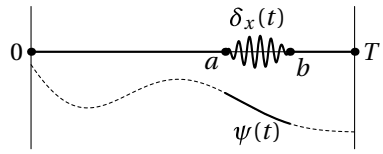


FIGURE 1.15: About the construction of a $\delta_x(t)$ that violates (1.51). See the proof of Lemma 1.6.2.

This second-order condition (1.48) is known as the *Legendre condition*. Notice that the inequality (1.48) means that $\frac{\partial^2 F(t,\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))}{\partial \dot{x} \partial \dot{x}^\intercal}$ (which is an $n \times n$ matrix if x has n components) is a symmetric positive semi-definite matrix at every moment in time.

Example 1.6.3 (Example 1.1.3 continued). The running cost of Example 1.1.3 is

$$F(t,x,\dot{x}) = \alpha \dot{x}^2 + \beta x,$$

and so the second derivative with respect to $\dot{x}$ is $\frac{\partial^2 F(t,x,\dot{x})}{\partial \dot{x}^2} = 2\alpha$. It is given that $\alpha > 0$, hence the Legendre condition,

$$\frac{\partial^2 F(t,\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))}{\partial \dot{x}^2} \geq 0 \qquad \forall\, t \in [0,T],$$

trivially holds for the solution x_* of the Euler-Lagrange equation. □

Example 1.6.4 (Example 1.5.3 continued). The running cost of Example 1.5.3 is $F(t,x,\dot{x}) = \alpha^2 x^2 + \dot{x}^2$. Therefore $\partial^2 F(t,\mathrm{x}(t),\dot{\mathrm{x}}(t))/\partial \dot{x}^2 = 2 \geq 0$ for all functions x and all t. This holds in particular for x_*, so the Legendre condition holds. □

Example 1.6.5 (Optimal investment, Example 1.1.2 continued). The running cost F for the optimal investment application of Example 1.1.2 is

$$F(t,x,\dot{x}) = -u\big(\phi(x) - \dot{x}\big)\,\mathrm{e}^{-\alpha t}.$$

This is derived from (1.5), but we added a minus sign because the application is about maximization, not minimization. Now

$$\frac{\partial^2 F(t,x,\dot{x})}{\partial \dot{x}^2} = -u''\big(\phi(x) - \dot{x}\big)\,\mathrm{e}^{-\alpha t},$$

and this is nonnegative for every $t, x, \dot{x}$ since the utility function u is assumed to be concave, i.e., $u''(c) \leq 0$ for all $c > 0$. So, apart from the standard economic interpretation that utility functions are concave, this assumption is also crucial for the optimization problem to have a solution. □

In the preceding examples, the Legendre condition was easy to verify because the second derivative of F with respect to $\dot{x}$ turned out to be trivially nonnegative for all $x, \dot{x}$ and all time, and not just for the optimal $\mathrm{x}_*(t), \dot{\mathrm{x}}_*(t)$.

The Euler-Lagrange condition together with the Legendre condition is necessary but is still *not sufficient* for minimality. This is illustrated by the next example.

Example 1.6.6 (Stationary solution, but not a minimizer). The Euler-Lagrange equation for the minimization of

$$\int_0^1 \left(\frac{\dot{\mathrm{x}}(t)}{2\pi}\right)^2 - \mathrm{x}^2(t)\,\mathrm{d}t \tag{1.52}$$

is the differential equation $(2\pi)^2 \mathrm{x}(t) + \ddot{\mathrm{x}}(t) = 0$. Assuming the boundary conditions

$$\mathrm{x}(0) = \mathrm{x}(1) = 0,$$

it is easy to see that the stationary solutions are

$$\mathrm{x}_*(t) = A\sin(2\pi t), \qquad A \in \mathbb{R}.$$

Each such solution x_* satisfies the Legendre condition (1.48) since

$$\frac{\partial^2 F(t, \mathrm{x}_*(t), \dot{\mathrm{x}}_*(t))}{\partial \dot{x}^2} = \frac{2}{(2\pi)^2} > 0.$$

Also, each such x_* renders the integral in (1.52) equal to zero. There are however many other functions x that satisfy $\mathrm{x}(0) = \mathrm{x}(1) = 0$ but for which the integral (1.52) takes a negative value. For example $\mathrm{x}(t) = -t^2 + t$. By scaling this last function with a constant we can make the cost as negative as we desire. Thus in this example there is no optimal solution x_*. □

A closer look at the proof of Theorem 1.6.1 actually provides us with an elegant *sufficient* condition for optimality, in fact for *global* optimality. If the Hessian of F, defined earlier as

$$H(t,x,y) := \begin{bmatrix} \dfrac{\partial^2 F(t,x,y)}{\partial x \partial x^{\mathrm{T}}} & \dfrac{\partial^2 F(t,x,y)}{\partial x \partial y^{\mathrm{T}}} \\ \dfrac{\partial^2 F(t,x,y)}{\partial y \partial x^{\mathrm{T}}} & \dfrac{\partial^2 F(t,x,y)}{\partial y \partial y^{\mathrm{T}}} \end{bmatrix}, \tag{1.53}$$

for each t is positive semi-definite for all $x \in \mathbb{R}^n$ and all $y \in \mathbb{R}^n$, then at each t the running cost $F(t,x,\dot{x})$ is convex in $x, \dot{x}$ (see Appendix A.7). For convex functions it is known that stationarity implies global optimality:

Theorem 1.6.7 (Convexity—global optimal solutions). Consider the simplest problem in the calculus of variations, and suppose that F is C^2. If the Hessian (1.53) is positive semi-definite[8] for all $x, y \in \mathbb{R}^n$ and all $t \in [0,T]$ then every C^1 solution x_* of the Euler-Lagrange equation that meets the boundary conditions is a global optimal solution.

If the Hessian is positive definite for all $x, y \in \mathbb{R}^n$ and all $t \in [0,T]$ then this x_* is the *unique* optimal solution.

Proof. Suppose that the Hessian is positive semi-definite. Let x_*, x be two functions that satisfy the boundary conditions, and suppose x_* satisfies the Euler-Lagrange equation. Define the function $\delta = \mathrm{x} - \mathrm{x}_*$ and $\bar{J}(\alpha) = J(\mathrm{x}_* + \alpha\delta)$. This way $\bar{J}(0) = J(\mathrm{x}_*)$ while $\bar{J}(1) = J(\mathrm{x})$. We need to prove that $\bar{J}(1) \geq \bar{J}(0)$.

[8] The relation between positive semi-definite Hessians and convexity is explained in Appendix A.7.

As before, we have that $\bar{J}'(0)$ is zero by the fact that x_* satisfies the Euler-Lagrange equation.

The second derivative of $\bar{J}(\alpha)$ with respect to α is (omitting time arguments)

$$\bar{J}''(\alpha) = \int_0^T \begin{bmatrix} \delta^{\mathrm{T}} & \dot{\delta}^{\mathrm{T}} \end{bmatrix} H(t, \mathrm{x}_* + \alpha\delta, \dot{\mathrm{x}}_* + \alpha\dot{\delta}) \begin{bmatrix} \delta \\ \dot{\delta} \end{bmatrix} \mathrm{d}t.$$

Since $H(t,x,y)$ is positive semi-definite for all $x, y \in \mathbb{R}^n$ and all t, we see that $\bar{J}''(\alpha) \geq 0$ for all $\alpha \in \mathbb{R}$. Therefore for every $\beta \geq 0$ there holds

$$\bar{J}'(\beta) = \bar{J}'(0) + \int_0^\beta \bar{J}''(\alpha)\,\mathrm{d}\alpha \geq \bar{J}'(0) = 0.$$

But then $\bar{J}(1) = \bar{J}(0) + \int_0^1 \bar{J}'(\beta)\,\mathrm{d}\beta \geq \bar{J}(0)$, which is what we had to prove.

Next suppose that $H(t,x,y)$ is positive definite and that $\mathrm{x} \neq \mathrm{x}_*$. Then $\delta := \mathrm{x} - \mathrm{x}_*$ is not the zero function and so by positive definiteness of $H(t,x,y)$ we have $J''(\alpha) > 0$ for every $\alpha \in [0,1]$. Then $J(\mathrm{x}) = \bar{J}(1) > \bar{J}(0) = J(\mathrm{x}_*)$. ∎

This result produces a lot, but also requires a lot. Indeed the convexity assumption fails in many cases of interest. Here are a couple examples where the convexity assumption *is* satisfied.

Example 1.6.8 (Shortest path; Example 1.2.5 continued). In the notation of the shortest path Example 1.1.4 we have $F(x,y,\dot{y}) = \sqrt{1+\dot{y}^2}$, and so we find that

$$\frac{\partial F(x,y,\dot{y})}{\partial \dot{y}} = \frac{\dot{y}}{(1+\dot{y}^2)^{1/2}},$$

and

$$\frac{\partial^2 F(x,y,\dot{y})}{\partial \dot{y}^2} = \frac{1}{(1+\dot{y}^2)^{3/2}}.$$

Clearly, this second derivative is positive for all $y, \dot{y} \in \mathbb{R}$. This implies that the solution y_* found in Example 1.2.5—namely, the straight line through the points (x_0, y_0) and (x_1, y_1)—satisfies the Legendre condition.

The Hessian (1.53) is

$$H(x,y,\dot{y}) = \begin{bmatrix} 0 & 0 \\ 0 & \frac{1}{(1+\dot{y}^2)^{3/2}} \end{bmatrix} \geq 0.$$

It is positive semi-definite, and, hence, the straight-line solution y_* is globally optimal. □

Example 1.6.9 (Quadratic cost; Example 1.5.3 continued). For the quadratic cost

$$J(\mathrm{x}) := \int_{-1}^1 \alpha^2 \mathrm{x}^2(t) + \dot{\mathrm{x}}^2(t)\,\mathrm{d}t,$$

as used in Example 1.5.3, the Hessian is constant,

$$H(t,x,\dot{x}) = \begin{bmatrix} 2\alpha^2 & 0 \\ 0 & 2 \end{bmatrix}.$$

This Hessian is positive definite for every $\alpha \neq 0$ and, hence, the solution x_* of the Euler-Lagrange equation found in Example 1.5.3 is the unique optimal solution of the problem. For $\alpha = 0$, the Hessian is positive *semi*-definite, so Theorem 1.6.7 guarantees that x_* is optimal, but possibly not unique. (Actually, for $\alpha = 0$ the solution x_* found in Example 1.5.3 *is* the unique differentiable optimal solution because it achieves a zero cost, $J(\mathrm{x}_*) = 0$, and for all other differentiable x the cost is positive). □

The Legendre condition (1.48) is only one of several necessary conditions for optimality. Additional necessary conditions go under the names of Weierstrass and Jacobi. Actually, the necessary condition of Weierstrass follows nicely from the dynamic programming approach as explained in Chapter 3, Exercise 3.10 (p. 114).

One can pose many different types of problems in the calculus of variations by giving different boundary conditions, for instance, involving $\dot{\mathrm{x}}(T)$, or by imposing further constraints on the required solution. An example of the latter we saw in (1.8) where $\dot{\mathrm{x}}(t)$ needs to be nonnegative for all time. Also, in Exercise 1.18, we explain what to do if $\mathrm{x}(T)$ needs to satisfy an *in*equality. Another variation is considered in the next section.

1.7 Integral Constraints

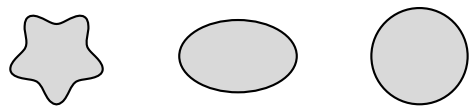


FIGURE 1.16: Three areas enclosed by ropes of the same length. See § 1.7.

An interesting extension is when the function x that is to minimize the cost

$$J(\mathrm{x}) := \int_0^T F(t, \mathrm{x}(t), \dot{\mathrm{x}}(t))\,\mathrm{d}t$$

is not free to choose, but is subject to an *integral constraint*

$$C(\mathrm{x}) := \int_0^T M(t, \mathrm{x}(t), \dot{\mathrm{x}}(t))\,\mathrm{d}t = c_0.$$

The standard example of this type is Queen Dido's isoperimetric problem. This is the problem of determining an area as large as possible that is enclosed by a rope of a given length. Intuition tells us that the optimal area is a disk (the

right-most option in Fig. 1.16). To put it more mathematically, in this problem we have to find a function $\mathrm{x} : [0, T] \to \mathbb{R}$ with given boundary values $\mathrm{x}(0) = x_0, \mathrm{x}(T) = x_T$, that maximizes the area

$$J(\mathrm{x}) = \int_0^T \mathrm{x}(t)\,\mathrm{d}t$$

subject to the constraint that

$$\int_0^T \sqrt{1 + \dot{\mathrm{x}}^2(t)}\,\mathrm{d}t = \ell$$

for a given ℓ.

How to solve such constrained minimization problems? A quick-and-dirty argument goes as follows: from calculus it is known that the solution of a minimization problem of some function $J(\mathrm{x})$ subject to the constraint $C(\mathrm{x}) - c_0 = 0$ is a stationary solution of the augmented function $\mathfrak{J}$ defined as

$$\mathfrak{J}(\mathrm{x}, \mu) := J(\mathrm{x}) + \mu(C(\mathrm{x}) - c_0) = \int_0^T F(t, \mathrm{x}(t), \dot{\mathrm{x}}(t)) + \mu M(t, \mathrm{x}(t), \dot{\mathrm{x}}(t))\,\mathrm{d}t \; - \mu c_0$$

for some *Lagrange multiplier*[9] $\mu \in \mathbb{R}$. The stationary solutions (x_*, μ_*) of $\mathfrak{J}(\mathrm{x}, \mu)$ must satisfy the Euler-Lagrange equation,

$$\left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\right)(F(t, \mathrm{x}_*(t), \dot{\mathrm{x}}_*(t)) + \mu_* M(t, \mathrm{x}_*(t), \dot{\mathrm{x}}_*(t)) = 0.$$

Below we formally prove that this argument is essentially correct. This may sound a bit vague, but it does put us on the right track. The theorem presented next is motivated by the above, but the proof is given from scratch. The proof assumes knowledge of the *inverse function theorem*.

Theorem 1.7.1 (Euler-Lagrange for integral-constrained minimization). Let c_0 be some constant. Suppose that F and M are C^1 in all of its components, and that x_* is a minimizer of

$$\int_0^T F(t, \mathrm{x}(t), \dot{\mathrm{x}}(t))\,\mathrm{d}t$$

subject to boundary conditions $\mathrm{x}(0) = x_0, \mathrm{x}(T) = x_T$ and integral constraint

$$\int_0^T M(t, \mathrm{x}(t), \dot{\mathrm{x}}(t))\,\mathrm{d}t = c_0,$$

and that x_* is C^2. Then either there is a Lagrange multiplier $\mu_* \in \mathbb{R}$ such that

$$\left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\right)\left(F(t, \mathrm{x}_*(t), \dot{\mathrm{x}}_*(t)) + \mu_* M(t, \mathrm{x}_*(t), \dot{\mathrm{x}}_*(t))\right) = 0 \tag{1.54}$$

[9] Lagrange multipliers are usually denoted as λ. We use μ in order to avoid a confusion in the next chapter.

for all $t \in [0,T]$, or M satisfies the Euler-Lagrange equation itself,

$$\left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\right) M(t, \mathtt{x}_*(t), \dot{\mathtt{x}}_*(t)) = 0 \qquad \forall\, t \in [0,T]. \tag{1.55}$$

Proof. This is not an easy proof. Suppose $\mathtt{x}_*$ solves the constrained minimization problem, and fix two C^2 functions δ_x, ϵ_x that vanish at the boundaries,

$$\delta_x(0) = 0 = \epsilon_x(0), \qquad \delta_x(T) = 0 = \epsilon_x(T).$$

Define $J(\mathtt{x}) = \int_0^T F(t, \mathtt{x}(t), \dot{\mathtt{x}}(t))\,\mathrm{d}t$ and $C(\mathtt{x}) = \int_0^T M(t, \mathtt{x}(t), \dot{\mathtt{x}}(t))\,\mathrm{d}t$ and consider the mapping that sends two real numbers (α, β) to the two real numbers

$$\begin{bmatrix} \bar{J}(\alpha,\beta) \\ \bar{C}(\alpha,\beta) \end{bmatrix} := \begin{bmatrix} J(\mathtt{x}_* + \alpha\delta_x + \beta\epsilon_x) \\ C(\mathtt{x}_* + \alpha\delta_x + \beta\epsilon_x) \end{bmatrix}.$$

The mapping from (α,β) to $(\bar{J}(\alpha,\beta), \bar{C}(\alpha,\beta))$ is C^1. So if the Jacobian at $(\alpha,\beta) = (0,0)$,

$$D := \left.\begin{bmatrix} \dfrac{\partial \bar{J}(\alpha,\beta)}{\partial \alpha} & \dfrac{\partial \bar{J}(\alpha,\beta)}{\partial \beta} \\ \dfrac{\partial \bar{C}(\alpha,\beta)}{\partial \alpha} & \dfrac{\partial \bar{C}(\alpha,\beta)}{\partial \beta} \end{bmatrix}\right|_{(\alpha=0,\beta=0)} \tag{1.56}$$

of this mapping is nonsingular then by the inverse function theorem there is a neighborhood of $(\alpha,\beta) = (0,0)$ on which the mapping is invertible. In particular, we then can find small enough α, β such that $\bar{C}(\alpha,\beta) = \bar{C}(0,0) = c_0$—hence satisfying the integral constraint—but rendering a cost $\bar{J}(\alpha,\beta)$ smaller than $\bar{J}(0,0) = J(\mathtt{x}_*)$. This contradicts that $\mathtt{x}_*$ is minimizing. Conclusion: at an optimal $\mathtt{x}_*$ the Jacobian (1.56) is singular for all allowable perturbation functions δ_x, ϵ_x.

We rewrite the Jacobian (1.56) in terms of F and M. To this end define the functions $\mathtt{f}$ and $\mathtt{m}$ as

$$\mathtt{f}(t) = \left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\right) F(t, \mathtt{x}_*(t), \dot{\mathtt{x}}_*(t)),$$
$$\mathtt{m}(t) = \left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\right) M(t, \mathtt{x}_*(t), \dot{\mathtt{x}}_*(t)).$$

This way the Jacobian (1.56) becomes (verify this for yourself)

$$D = \begin{bmatrix} \int_0^T \mathtt{f}(t)\delta_x(t)\mathrm{d}t & \int_0^T \mathtt{f}(t)\epsilon_x(t)\mathrm{d}t \\ \int_0^T \mathtt{m}(t)\delta_x(t)\mathrm{d}t & \int_0^T \mathtt{m}(t)\epsilon_x(t)\mathrm{d}t \end{bmatrix}. \tag{1.57}$$

If $\mathtt{m}(t) = 0$ for all t then (1.55) holds and the proof is complete. Remains to consider the case that $\mathtt{m}(t_0) \neq 0$ for at least one t_0. Suppose, to obtain a contraction, that given such a t_0 there is a t for which

$$\begin{bmatrix} \mathtt{f}(t_0) & \mathtt{f}(t) \\ \mathtt{m}(t_0) & \mathtt{m}(t) \end{bmatrix} \tag{1.58}$$

is nonsingular. Now take δ_x to have support around t_0 and ϵ_x to have support around t. Then by nonsingularity of (1.58) also (1.57) is nonsingular if the support is taken small enough. However nonsingularity of the Jacobian is impossible by the fact that $\mathtt{x}_*$ solves the minimization problem. Therefore we conclude that (1.58) is singular at every t. This means that

$$\mathtt{f}(t_0)\mathtt{m}(t) - \mathtt{f}(t)\mathtt{m}(t_0) = 0 \qquad \forall\, t.$$

In other words $\mathtt{f}(t) + \mu_* \mathtt{m}(t) = 0$ for all t if we take $\mu_* = -\mathtt{f}(t_0)/\mathtt{m}(t_0)$. ∎

The theorem says that the solution $\mathtt{x}_*$ satisfies either (1.54) or (1.55). The first of these two is called the *normal* case, and the second the *abnormal* case. Notice that the abnormal case completely neglects the running cost F. The next example indicates that we usually have the normal case.

Example 1.7.2 (Normal and abnormal Euler-Lagrange equation). Consider minimizing $\int_0^1 \mathtt{x}(t)\,\mathrm{d}t$ subject to the boundary conditions $\mathtt{x}(0) = 0, \mathtt{x}(1) = 1$ and integral constraint

$$\int_0^1 \dot{\mathtt{x}}^2(t)\,\mathrm{d}t = C \tag{1.59}$$

for some given C. The (normal) Euler-Lagrange equation (1.54) becomes

$$0 = \left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\right)(\mathtt{x}_*(t) + \mu \dot{\mathtt{x}}_*^2(t)) = 1 - \frac{\mathrm{d}}{\mathrm{d}t}\big(2\mu \dot{\mathtt{x}}_*(t)\big) = 1 - 2\mu \ddot{\mathtt{x}}_*(t).$$

The general solution of this equation is $\mathtt{x}_*(t) = \frac{1}{4\mu}t^2 + bt + c$. The constants b, c are determined by the boundary conditions $\mathtt{x}(0) = 0, \mathtt{x}(1) = 1$, leading to

$$\mathtt{x}_*(t) = \tfrac{1}{4\mu}t^2 + (1 - \tfrac{1}{4\mu})t.$$

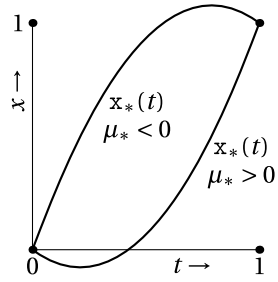
With this form the integral constraint (1.59) becomes

$$C = \int_0^1 \dot{\mathtt{x}}_*^2(t)\,\mathrm{d}t = \int_0^1 (\tfrac{1}{2\mu}t + 1 - \tfrac{1}{4\mu})^2\,\mathrm{d}t = 1 + \frac{1}{48\mu^2}. \tag{1.60}$$

If $C < 1$ then clearly no solution μ exists, and it is not hard to see that then no smooth function with $\mathtt{x}(0) = 0$ and $\mathtt{x}(1) = 1$ exists that meets the integral constraint (see Exercise 1.21). For $C > 1$ there are two μ's that satisfy (1.60):

$$\mu_* = \frac{\pm 1}{\sqrt{48(C-1)}},$$

and the resulting two functions $\mathtt{x}_*$ (for $C = 2$) then are

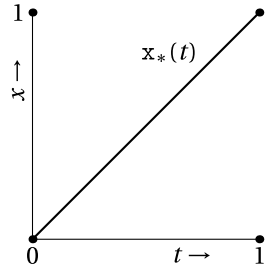


Clearly, out of these two, the cost $J(\mathrm{x}_*) := \int_0^1 \mathrm{x}_*(t)\,\mathrm{d}t$ is minimal for the positive solution μ_*.

In the abnormal case, (1.55), we have that

$$0 = \left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\right)\dot{\mathrm{x}}_*^2(t) = -2\ddot{\mathrm{x}}_*(t).$$

Hence $\mathrm{x}_*(t) = bt + c$ for some b, c. Given the boundary conditions $\mathrm{x}(0) = 0, \mathrm{x}(1) = 1$ it is immediate that this allows for only one solution: $\mathrm{x}_*(t) = t$:



Now $\dot{\mathrm{x}}_*(t) = 1$, and the constant C in the integral constraint necessarily equals $C = \int_0^1 \dot{\mathrm{x}}_*^2(t)\,\mathrm{d}t = 1$. This corresponds to $\mu = \infty$. In this case the integral constraint together with the boundary conditions is tight. There are, so to say, no degrees of freedom left to shape the function. In particular, there is no feasible variation, $\mathrm{x} = \mathrm{x}_* + \alpha\delta_x$, and since the standard Euler-Lagrange equation was derived from such a variation, it is no surprise that the standard Euler-Lagrange equation does not apply in this case. □

1.8 Exercises

1.1 Determine all solutions $\mathrm{x} : [0, T] \to \mathbb{R}$ of the Euler-Lagrange equation for the cost $J(\mathrm{x}) = \int_0^T F(t, \mathrm{x}(t), \dot{\mathrm{x}}(t))\,\mathrm{d}t$ with

(a) $F(t, x, \dot{x}) = \dot{x}^2 - \alpha^2 x^2$.

(b) $F(t, x, \dot{x}) = \dot{x}^2 + 2x$.

(c) $F(t, x, \dot{x}) = \dot{x}^2 + 4t\dot{x}$.

(d) $F(t, x, \dot{x}) = \dot{x}^2 + x\dot{x} + x^2$.

(e) $F(t, x, \dot{x}) = x^2 + 2tx\dot{x}$ (this one is curious).

1.2 Consider minimization of

$$\int_0^1 \dot{\mathrm{x}}^2(t) + 12t\mathrm{x}(t)\,\mathrm{d}t$$

over all functions $\mathrm{x} : [0, 1] \to \mathbb{R}$ that satisfy the boundary conditions $\mathrm{x}(0) = 0$, $\mathrm{x}(1) = 1$.

(a) Determine the Euler-Lagrange equation for this problem.

(b) Determine the solution x_* of the Euler-Lagrange equation and that satisfies the boundary conditions.

1.3 *Trivial running cost.* Consider minimization of

$$J(\mathrm{x}) := \int_0^T F(t, \mathrm{x}(t), \dot{\mathrm{x}}(t))\,\mathrm{d}t$$

over all functions $\mathrm{x} : [0,T] \to \mathbb{R}$ with given boundary conditions $\mathrm{x}(0) = x_0, \mathrm{x}(T) = x_T$. Assume that the running cost has the particular form,

$$F(t, \mathrm{x}(t), \dot{\mathrm{x}}(t)) = \frac{\mathrm{d}}{\mathrm{d}t} G(t, \mathrm{x}(t))$$

for some C^2 function $G(t,x)$.

(a) Derive the Euler-Lagrange equation for this problem.

(b) Show that every differentiable function $\mathrm{x} : [0,T] \to \mathbb{R}$ satisfies the Euler-Lagrange equation.

(c) Explain this remarkable phenomenon by expressing $J(\mathrm{x})$ in terms of the function G and boundary values x_0, x_T.

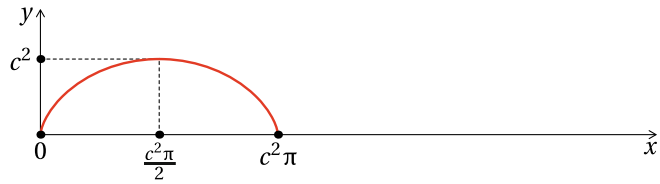
1.4 *Technical problem: the lack of Lipschitz continuity in the Beltrami identity for the brachistochrone problem, and how to circumvent it.* The footnote of Example 1.3.1 derives the cycloid equations (1.29) from

$$c^2 = \mathrm{y}(x)(1 + \dot{\mathrm{y}}^2(x)), \qquad \mathrm{y}(0) = 0. \tag{1.61}$$

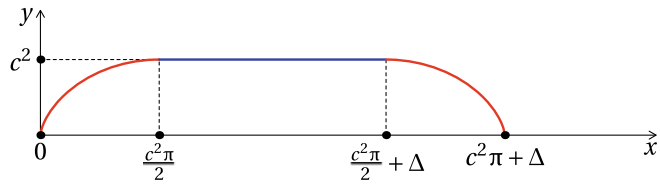
The derivation was quick, and this exercise shows that it was a bit dirty as well.

(a) Let $\mathrm{x}(\phi), \mathrm{y}(\phi)$ be the cycloid solution (1.29). Use the identity $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}\phi}{\mathrm{d}x/\mathrm{d}\phi}$ to show that they satisfy (1.61).

(b) The curve of this cycloid solution for $\phi \in [0, 2\pi]$ is



From this solution we construct a new solution by inserting in the middle a constant part of some length $\Delta \geq 0$:



Argue that for every $\Delta \geq 0$ also this new function satisfies the Beltrami identity (1.61) for all $x \in (0, c^2\pi + \Delta)$.

(c) This is not what the footnote of Example 1.3.1 says. What goes wrong in this footnote?

(d) This new function $\mathrm{y}(x)$ is constant over the interval $[\frac{c^2\pi}{2}, \frac{c^2\pi}{2} + \Delta]$. Show that a constant function $\mathrm{y}(x)$ does not satisfy the Euler-Lagrange equation of the brachistochrone problem.

(e) It can be shown that $\mathrm{y}(x)$ solves (1.61) iff it is of this new form for some $\Delta \geq 0$ (possibly $\Delta = \infty$). Argue that the only function that satisfies the Euler-Lagrange equation with $\mathrm{y}(0) = 0$ is the cycloid solution (1.29).

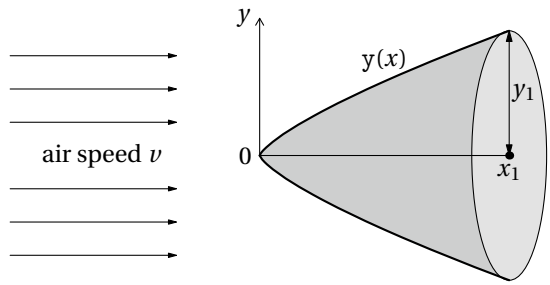


FIGURE 1.17: Solid of least resistance. See Exercise 1.5.

1.5 *A simplified Newton's minimal resistance problem.* Consider a solid of revolution with diameter $\mathrm{y}(x)$ as shown in Fig. 1.17. At $x = 0$ the diameter is 0, and at $x = x_1$ it is $y_1 > 0$. If the air flows with a constant speed v, then the total air resistance (force) can be modeled as

$$4\pi\rho v^2 \int_0^{x_1} \frac{\mathrm{y}(x)\dot{\mathrm{y}}^3(x)}{1+\dot{\mathrm{y}}^2(x)}\,\mathrm{d}x.$$

Here ρ is the air density. The question is: given $\mathrm{y}(0) = 0$ and $\mathrm{y}(x_1) = y_1 > 0$, for which function $\mathrm{y} : [0, x_1] \to \mathbb{R}$ is the resistance minimal? Now we are going to cheat! To make the problem *a lot* easier we discard the quadratic term in the denominator of the running cost, that is, we consider instead the cost function

$$J(\mathrm{y}) := 4\pi\rho v^2 \int_0^{x_1} \mathrm{y}(x)\dot{\mathrm{y}}^3(x)\,\mathrm{d}x.$$

Given the boundary conditions $\mathrm{y}(0) = 0$ and $\mathrm{y}(x_1) = y_1 > 0$, show that

$$\mathrm{y}(x) = \left(\frac{x}{x_1}\right)^{3/4} y_1$$

is a solution of the Beltrami identity with the given boundary conditions. (This function y is depicted in Fig. 1.17.)

1.6 *Technical problem: the lack of Lipschitz continuity in the minimal-surface problem, and how to circumvent it.* In Example 1.3.2 we claimed that $\mathtt{r}_a(x) := a\cosh(x/a)$ is the only positive even solution of (1.30). That is not completely correct. In this exercise we see that the differential equation (1.30), as derived from the Beltrami identity, has more solutions, but that $\mathtt{r}_a(x)$ is the only even solution that satisfies the Euler-Lagrange equation. We assume that $a > 0$.

(a) Show that the function

$$f(r) := \sqrt{r^2/a^2 - 1} \tag{1.62}$$

is not Lipschitz continuous at $r = a$ (see Appendix B.1). Hence we can expect multiple solutions of the differential equation $\frac{\mathrm{d}\mathtt{r}(x)}{\mathrm{d}x} = \sqrt{\mathtt{r}^2(x)/a^2 - 1}$ if $\mathtt{r}(x) = a$.

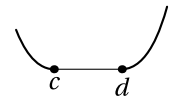
(b) Show that (1.30) can be separated as

$$\frac{\mathrm{d}\mathtt{r}(x)}{\sqrt{\mathtt{r}^2(x)/a^2 - 1}} = \mathrm{d}x.$$

(c) If $\mathtt{r}(x_0) > a$, show that $\mathtt{r}(x) = a\cosh((x-c)/a)$ around $x = x_0$ for some c.

(d) Argue that $\mathtt{r}(x)$ is a solution of (1.30) iff it is pieced together from a hyperbolic cosine, a constant, and a hyperbolic cosine again, as in

$$\mathtt{r}(x) = \begin{cases} a\cosh(\frac{x-c}{a}) & \text{if } x < c \\ a & \text{if } x \in [c,d] \\ a\cosh(\frac{x-d}{a}) & \text{if } x > d. \end{cases}$$



Here $c \le d$. (Notice that for $x \in [c,d]$ the value of $\mathtt{r}(x)$ equals a, so at that point the function f as defined in (1.62) is not Lipschitz continuous.)

(e) If $c < d$ then on the strip $[c,d]$ the function $\mathtt{r}(x)$ is a constant (equal to $a > 0$). Show that this $\mathtt{r}(x)$ does not satisfy the Euler-Lagrange equation. (Recall that the Beltrami identity may have more solutions than the Euler-Lagrange equation.)

(f) Verify that $\mathtt{r}_a(x) := a\cosh(x/a)$ is the only function that satisfies the Euler-Lagrange equation of the minimal-surface problem (Example 1.3.2) and that has the symmetry property that $\mathtt{r}(-1) = \mathtt{r}(+1)$.

1.7 *Lemma of du Bois-Reymond.* The proof of Theorem 1.2.2 at some point assumes that both $\mathtt{x}_*$ and F are C^2. The lemma of du Bois-Reymond that we explore in this exercise shows that the result also holds if $\mathtt{x}_*$ and F are merely C^1. Throughout this exercise we assume that $\mathtt{x}_*$ and F are C^1.

(a) *Lemma of du Bois-Reymond.* Let $f : [0,T] \to \mathbb{R}$ be a continuous function, and suppose that $\int_0^T f(t)\phi(t)\,\mathrm{d}t = 0$ for all continuous functions $\phi : [0,T] \to \mathbb{R}$ for which $\int_0^T \phi(t)\,\mathrm{d}t = 0$. Show that $f(t)$ is constant on $[0,T]$.
[Hint: If f is not constant then $a, b \in [0,T]$ exist for which $f(a) \neq f(b)$. Then construct a ϕ for which $\int_0^T f(t)\phi(t)\,\mathrm{d}t \neq 0$.]

(b) We showed in the proof of Theorem 1.2.2 that C^1 optimal solutions x_* satisfy

$$\int_0^T \frac{\partial F(t,\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))}{\partial x^{\mathrm{T}}}\delta_x(t) + \frac{\partial F(t,\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))}{\partial \dot{x}^{\mathrm{T}}}\dot{\delta}_x(t)\,\mathrm{d}t = 0 \quad (1.63)$$

for all $t \in [0,T]$ and all C^1 functions $\delta_x : [0,T] \to \mathbb{R}^n$ with $\delta_x(0) = \delta_x(T) = 0$. In the proof of Theorem 1.2.2, we performed integration by parts on the second term of the integral in (1.63). Now, instead, we perform integration by parts on the *first* term in (1.63). Use that to show that (1.63) holds iff

$$\int_0^T \left(-\int_0^t \frac{\partial F(\tau,\mathrm{x}_*(\tau),\dot{\mathrm{x}}_*(\tau))}{\partial x^{\mathrm{T}}}\,\mathrm{d}\tau + \frac{\partial F(t,\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))}{\partial \dot{x}^{\mathrm{T}}} \right) \dot{\delta}_x(t)\,\mathrm{d}t = 0$$

for all C^1 functions $\delta_x : [0,T] \to \mathbb{R}^n$ with $\delta_x(0) = \delta_x(T) = 0$.

(c) Use the lemma of du Bois-Reymond to show that C^1 optimal solutions x_* satisfy

$$\frac{\partial F(t,\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))}{\partial \dot{x}} = c + \int_0^t \frac{\partial F(\tau,\mathrm{x}_*(\tau),\dot{\mathrm{x}}_*(\tau))}{\partial x}\,\mathrm{d}\tau \qquad \forall\, t \in [0,T]$$

for some constant $c \in \mathbb{R}^n$. [Hint: $\int_0^T \dot{\delta}_x(t)\,\mathrm{d}t = 0$.]

(d) Show that for C^1 optimal solutions x_* the expression

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial F(t,\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))}{\partial \dot{x}}$$

is well defined and continuous at every $t \in [0,T]$.

(e) Show that C^1 optimal solutions x_* satisfy the Euler-Lagrange equation (1.13).

1.8 *Free endpoint.* Minimize

$$\mathrm{x}^2(1) + \int_0^1 \dot{\mathrm{x}}^2(t)\,\mathrm{d}t$$

over all functions x subject to $\mathrm{x}(0) = 1$ and free endpoint $\mathrm{x}(1)$.

1.9 *Free endpoint.* Consider minimization of

$$J(\mathrm{x}) = \int_0^1 \dot{\mathrm{x}}^2(t) - 2\mathrm{x}(t)\dot{\mathrm{x}}(t) - \dot{\mathrm{x}}(t)\,\mathrm{d}t$$

with initial condition $\mathrm{x}(0) = 1$ and free endpoint $\mathrm{x}(1)$.

(a) Show that no function x exists that satisfies the Euler-Lagrange equation with $\mathrm{x}(0) = 1$ and the free endpoint boundary condition (1.43).

(b) Conclude that there is no C^1 function x that minimizes $J(\mathrm{x})$ subject to $\mathrm{x}(0) = 1$ with free endpoint.

(c) Determine all functions x that satisfy the Euler-Lagrange equation and such that $\mathrm{x}(0) = 1$. Then compute $J(\mathrm{x})$ explicitly and conclude, once more, that the free endpoint problem has no solution.

1.10 *Relaxed boundary conditions.* In this exercise we prove Proposition 1.5.1.

(a) For $G(x) = K(x) = 0$ the first-order conditions are that (1.38) holds for all possible perturbations. Adapt this equation for the case that $G(x)$ and $K(x)$ are arbitrary C^1 functions.

(b) Prove that this equality implies that the Euler-Lagrange equation holds.

(c) Finish the proof of Proposition 1.5.1.

1.11 Show that the minimal surface example (Example 1.3.2) satisfies the Legendre second-order necessary condition of Theorem 1.6.1.

1.12 *Smoothness assumptions in Legendre's necessary condition.* Theorem 1.6.1 assumes that F is C^2, but looking at the proof it seems we need F to be C^3 (see Eqn. (1.50)). However, C^2 is sufficient: argue that the integral in (1.49) is nonnegative for all allowable δ_x only if the Legendre condition holds. [Hint: Formulate a lemma similar to Lemma 1.6.2.]

1.13 Show that the minimization problem in Example 1.2.8 satisfies the Legendre condition. [Hint: The condition now involves a 2×2 matrix.]

1.14 *The optimal solar challenge.* A solar vehicle receives power from solar radiation. This power $p(x,t)$ depends on position x (due to clouds) and on time t (due to moving clouds and the sun's angle of inclination). Driving at some speed $\dot{x}$ also consumes power. Denote this power loss by $f(\dot{x})$. This assumes that it is a function of speed alone, which is reasonable if we do not change speed aggressively and if friction depends only on speed. Driving at higher speed requires more energy per meter than driving at lower speed. This means that f is convex, in fact

$$f(\dot{x}) \geq 0, \quad f'(\dot{x}) > 0, \quad f''(\dot{x}) > 0.$$

Suppose the solar team starts at

$$\mathrm{x}(0) = 0,$$

and at time T it wants to be at some position $\mathrm{x}(T) = x_T$, and, of course, all that using minimal net energy

$$\int_0^T f(\dot{\mathrm{x}}(t)) - p(\mathrm{x}(t), t)\,\mathrm{d}t.$$

(a) Derive the Euler-Lagrange equation for this problem.

(b) Argue from the Euler-Lagrange equation that we should speed up if we drive into a cloud.

(c) Is Legendre's second-order condition satisfied?

(d) From now on assume that

$$f(\dot{x}) = \dot{x}^2$$

(this is actually quite reasonable, modulo scaling) and that $p(x,t)$ does not depend on time,

$$p(x,t) = q(x),$$

i.e., that the sun's angle does not change much over our time window $[0,T]$ and that clouds are not moving. Use the Beltrami identity to express $\dot{\mathrm{x}}(t)$ in terms of $q(\mathrm{x}(t))$ and the initial speed $\dot{\mathrm{x}}(0)$ and initial $q(0)$.

(e) Argue once again (but now using the explicit relation of the previous part) that we should speed up if we drive into a cloud.

(f) *(A computer might be useful for this part.)* Continue with $f(\dot{x}) = \dot{x}^2$ and $p(x,t) = q(x)$. Suppose that up to position $x = 20$ the sky is clear but that from $x = 20$ onwards heavy clouds limit the power input:

$$q(x) = \begin{cases} 100 & x < 20, \\ 4 & x > 20. \end{cases}$$

Determine the optimal speed $\dot{\mathrm{x}}_*(t), t \in [0,7]$ that brings us from $\mathrm{x}(0) = 0$ to $\mathrm{x}(7) = 90$.

1.15 Consider minimization of

$$\int_0^1 \dot{\mathrm{x}}^2(t) - \mathrm{x}(t)\ \mathrm{d}t$$

over all functions $\mathrm{x} : [0,1] \to \mathbb{R}$ that satisfy the boundary conditions $\mathrm{x}(0) = 0$, $\mathrm{x}(1) = 1$.

(a) Determine the Euler-Lagrange equation for this problem.

(b) Determine the solution x_* of the Euler-Lagrange equation and that satisfies the boundary conditions.

(c) Does the x_* found in (b) satisfy Legendre's second-order condition?

(d) Is the convexity condition (Theorem 1.6.7) satisfied?

(e) Show that the solution x_* found in (b) is globally optimal.

1.16 *Convex quadratic cost.* Consider minimization of the quadratic cost

$$J(\mathrm{x}) = \int_0^1 \dot{\mathrm{x}}^2(t) + \mathrm{x}^2(t) + 2t\mathrm{x}(t)\,\mathrm{d}t$$

with boundary conditions

$$\mathrm{x}(0) = 0, \quad \mathrm{x}(1) = 1$$

over all functions $\mathrm{x} : [0,1] \to \mathbb{R}$.

(a) Determine the Euler-Lagrange equation for this problem.

(b) Determine the function x_* that satisfies the Euler-Lagrange equation and the given boundary conditions.

(c) Does x_* satisfy Legendre's second-order condition?

(d) Show that

$$J(\mathrm{x}_* + \delta_x) = J(\mathrm{x}_*) + \int_0^1 \delta_x^2(t) + \dot{\delta}_x^2(t)\,\mathrm{d}t$$

for every continuously differentiable function δ_x with $\delta_x(0) = \delta_x(1) = 0$, and conclude that x_* is globally optimal.

(e) Is the convexity condition (Theorem 1.6.7) satisfied?

1.17 *Smoothness.* This exercise is from Liberzon (2012). It shows that smooth running costs F may result in non-smooth optimal solutions x_*. Consider minimization of

$$J(\mathrm{x}) = \int_{-1}^1 (1 - \dot{\mathrm{x}}(t))^2 \mathrm{x}^2(t)\,\mathrm{d}t$$

subject to the boundary conditions

$$\mathrm{x}(-1) = 0, \qquad \mathrm{x}(1) = 1.$$

(a) Show that $J(\mathrm{x}) \geq 0$ for every function x.

(b) Determine a continuous optimal solution x_* and argue that it is unique. (Hint: $J(\mathrm{x}_*) = 0$ and do not use Euler-Lagrange or Beltrami.)

(c) Argue that there is no continuously differentiable optimal solution x_*.

1.18 *Inequalities.* The calculus of variations problems considered in this chapter all assume that the entries of $\mathrm{x}(0)$ and $\mathrm{x}(T)$ are either fixed or completely free. But what if we demand an *inequality*? Consider, as an example, the calculus of variations problem with standard cost $\int_0^T F(t,\mathrm{x}(t),\dot{\mathrm{x}}(t))\,\mathrm{d}t$ and standard initial condition, $\mathrm{x}(0) = x_0$, but whose final condition is an inequality,

$$\mathrm{x}(T) \geq x_T.$$

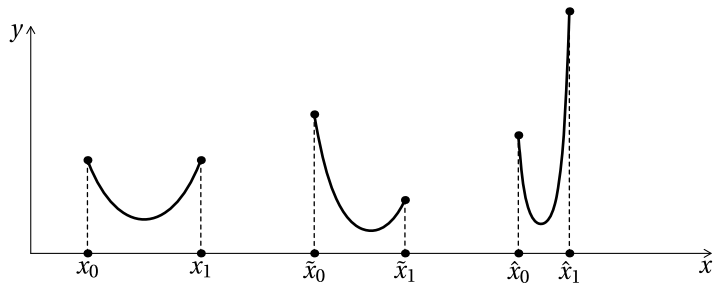
Assume sufficient smoothness of all functions involved.

(a) Show that optimal solutions x_* must obey the Euler-Lagrange equation, and the inequality

$$\frac{\partial F(\mathrm{x}_*(T),\dot{\mathrm{x}}_*(T),T)}{\partial \dot{x}} \geq 0.$$

(b) Verify this statement for the cost $\int_0^1 (\mathrm{x}(t) - \dot{\mathrm{x}}(t))^2\,\mathrm{d}t$ with $\mathrm{x}(0) = 1, \mathrm{x}(1) \geq x_T$, and distinguish the cases $x_T \leq \mathrm{e}$ and $x_T > \mathrm{e}$.

1.19 *The hanging cable.* Every hanging cable eventually comes to a halt in a position of minimal energy, such as these three:



What is the shape of this minimal energy position? When hanging still it has no kinetic energy, it only has potential energy. If the cable is very flexible then the potential energy is only due to its height y. We assume that the cable is very thin, does not stretch and that it has a constant mass per unit length. In a constant gravitational field with gravitational acceleration g the potential energy $J(\mathrm{y})$ equals

$$J(\mathrm{y}) = \int_{x_0}^{x_1} \rho g \mathrm{y}(x) \sqrt{1 + \dot{\mathrm{y}}^2(x)}\,\mathrm{d}x,$$

with ρ the mass per unit length of the cable. We want to minimize the potential energy over all functions $\mathrm{y} : [x_0, x_1] \to \mathbb{R}$, subject to $\mathrm{y}(x_0) = y_0, \mathrm{y}(x_1) = y_1$ and such that the length of the cable is ℓ. The length of the cable can be expressed as

$$\int_{x_0}^{x_1} \sqrt{1 + \dot{\mathrm{y}}^2(x)}\,\mathrm{d}x = \ell.$$

To solve this problem we use Theorem 1.7.1.

(a) Consider first the normal case, and the associated Euler-Lagrange equation (1.54). Analyze the Beltrami identity of this case to show that the minimal energy solution y_* satisfies

$$\mathrm{y}_*(x) + \mu_* \tfrac{1}{\rho g} = a\sqrt{1+\dot{\mathrm{y}}^2(x)}$$

for some constant a and Lagrange multiplier μ_*. (Hint: We considered a similar problem in Example 1.3.2.) It can be shown that the general solution of the above differential equation is $\mathrm{y}_*(x) = a\cosh(\frac{x-b}{a}) - \mu_* \frac{1}{\rho g}$ with $b \in \mathbb{R}$.

(b) Show that the minimal energy solution y_* (if it exists) is of the form

$$\mathrm{y}_*(x) = \begin{cases} a\cosh(\frac{x-b}{a}) - \mu_* \frac{1}{\rho g} & \text{in the normal case (Eqn. (1.54))} \\ cx+d & \text{in the abnormal case (Eqn. (1.55))} \end{cases}$$

for certain constants $a, b, c, d \in \mathbb{R}$ and Lagrange multiplier $\mu_* \in \mathbb{R}$.

(c) Describe in terms of ℓ and x_0, x_1, y_0, y_1 when we have the normal case, the abnormal case, or no solution at all.

1.20 *Integral constraint.* Minimize $\int_0^1 \dot{\mathrm{x}}^2(t)\,\mathrm{d}t$ subject to $\mathrm{x}(0) = \mathrm{x}(\pi) = 0$ and $\int_0^1 \mathrm{x}^2(t)\,\mathrm{d}t = 1$.

1.21 Consider Example 1.7.2. Prove that for $C < 1$ there is no smooth function that satisfies the boundary conditions and integral constraint.

1.22 *Discrete calculus of variations.* A discrete version of the simplest problem in the calculus of variations (Definition 1.2.1) can be formulated as follows. Consider a final time T, a function $F : \{0,1,\dots,T-1\} \times \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, denoted as $F(t, x^1, x^2)$, and fixed $x_0, x_T \in \mathbb{R}^n$. Consider the problem of minimizing

$$\sum_{t=0}^{T-1} F(t, \mathrm{x}(t), \mathrm{x}(t+1))$$

over all sequences $\mathrm{x}(0), \mathrm{x}(1), \mathrm{x}(2), \dots, \mathrm{x}(T-1), \mathrm{x}(T)$ with $\mathrm{x}(0) = x_0, \mathrm{x}(T) = x_T$ (fixed initial and final conditions). In order to derive a discrete version of the Euler-Lagrange equation for this problem we proceed as follows. Let

$$(\mathrm{x}_*(0), \mathrm{x}_*(1), \dots, \mathrm{x}_*(T-1), \mathrm{x}_*(T))$$

be a minimizing sequence with $\mathrm{x}_*(0) = x_0, \mathrm{x}_*(T) = x_T$, and consider variations

$$(\mathrm{x}_*(0), \mathrm{x}_*(1), \dots, \mathrm{x}_*(T-1), \mathrm{x}_*(T)) \;+\; (\delta_x(0), \delta_x(1), \dots, \delta_x(T-1), \delta_x(T))$$

with $\delta_x(0) = \delta_x(T) = 0$.

(a) Show that this implies that

$$\sum_{t=0}^{T-1} \frac{\partial F(t, \mathrm{x}_*(t), \mathrm{x}_*(t+1))}{\partial x^{1\mathrm{T}}} \delta_x(t) + \sum_{t=0}^{T-1} \frac{\partial F(t, \mathrm{x}_*(t), \mathrm{x}_*(t+1))}{\partial x^{2\mathrm{T}}} \delta_x(t+1) = 0$$

for all $\delta_x(t)$ with $\delta_x(0) = \delta_x(T) = 0$.

(b) Rearrange this equation (partly changing the summation index) so as to obtain the equivalent condition

$$\sum_{t=1}^{T-1} \left[\frac{\partial F(t, \mathrm{x}_*(t), \mathrm{x}_*(t+1))}{\partial x^{1\mathrm{T}}} + \frac{\partial F(t-1, \mathrm{x}_*(t-1), \mathrm{x}_*(t))}{\partial x^{2\mathrm{T}}} \right] \delta_x(t) = 0,$$

and show that this implies

$$\frac{\partial F(t, \mathrm{x}_*(t), \mathrm{x}_*(t+1))}{\partial x^1} + \frac{\partial F(t-1, \mathrm{x}_*(t-1), \mathrm{x}_*(t))}{\partial x^2} = 0$$

for all $t = 1, \ldots, T-1$. This system of equations can be called the *discrete Euler-Lagrange equation*.

(c) Extend this to the minimization of (with $S(\mathrm{x}(T))$ some final cost)

$$\sum_{t=0}^{T-1} F(t, \mathrm{x}(t), \mathrm{x}(t+1)) + S(\mathrm{x}(T))$$

over all sequences $\mathrm{x}(0), \mathrm{x}(1), \ldots, \mathrm{x}(T)$ with $\mathrm{x}(0) = x_0$.

(d) Show how this could be used for obtaining numerical schemes solving the ordinary Euler-Lagrange equation (1.13). For example, given a running cost $\tilde{F}(t, \mathrm{x}(t), \dot{\mathrm{x}}(t)), t \in [0, T]$, replace $\dot{\mathrm{x}}(t)$ by its approximation $\mathrm{x}(t+1) - \mathrm{x}(t)$ so as to obtain the discretized running cost

$$F(t, \mathrm{x}(t), \mathrm{x}(t+1)) := \tilde{F}(t, \mathrm{x}(t), \mathrm{x}(t+1) - \mathrm{x}(t)).$$

Write out the discrete Euler-Lagrange equation in this case.



Chapter 2

Minimum Principle

2.1 Optimal Control

In the solar challenge problem (Exercise 1.14) we assumed that we could choose the speed $\dot{\mathtt{x}}$ of the car at will, but in reality, the speed is limited by the dynamics of the car. For instance, the acceleration of the car is bounded. In this chapter, we take such dynamical constraints into account. We assume that the state $\mathtt{x} : [0, T] \to \mathbb{R}^n$ satisfies a system of differential equations with initial state

$$\dot{\mathtt{x}}(t) = f(\mathtt{x}(t), \mathtt{u}(t)), \qquad \mathtt{x}(0) = x_0, \tag{2.1}$$

and that we can not choose $\mathtt{x}$ directly but only can choose $\mathtt{u}$, which is known as the *input* of the system. Furthermore, the input is restricted to take values in some given set $\mathbb{U} \subseteq \mathbb{R}^m$, that is,

$$\mathtt{u} : [0, T] \to \mathbb{U}. \tag{2.2}$$

For instance, in a car-parking problem, the input $\mathtt{u}$ might be the throttle opening and this takes values in between $u = 0$ (fully closed) and $u = 1$ (fully open), so then $\mathbb{U} = [0, 1]$. For a given $\mathbb{U}$ and given (2.1), the *optimal control problem* is to determine an input $\mathtt{u} : [0, T] \to \mathbb{U}$ that minimizes a given cost function of the form

$$J(\mathtt{u}) := \int_0^T L(\mathtt{x}(t), \mathtt{u}(t))\,\mathrm{d}t + K(\mathtt{x}(T)). \tag{2.3}$$

Here, $K : \mathbb{R}^n \to \mathbb{R}$ and $L : \mathbb{R}^n \times \mathbb{U} \to \mathbb{R}$. The part $K(\mathtt{x}(T))$ is called the *terminal cost* or *final cost*, and $L(\mathtt{x}(t), \mathtt{u}(t))$ is commonly called the *running cost*. The optimal $\mathtt{u}$ is referred to as the *optimal input* or *optimal control*, and it is often denoted with a star, i.e., $\mathtt{u}_*$.

A variation of the optimal control problem is to fix the final state $\mathtt{x}(T)$ to a given x_T. Clearly, in this case, there is no need for a final cost $K(\mathtt{x}(T))$ in that every allowable input results in the same final cost. In this case, the optimal

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G. Meinsma and A. van der Schaft, *A Course on Optimal Control*, Springer
Undergraduate Texts in Mathematics and Technology,
https://doi.org/10.1007/978-3-031-36655-0_2

control problem is to find, for a given system (2.1) and x_T and $\mathbb{U}$, an input $\mathrm{u} : [0,T] \to \mathbb{U}$ that minimizes

$$\int_0^T L(\mathrm{x}(t),\mathrm{u}(t))\,\mathrm{d}t$$

subject to $\mathrm{x}(T) = x_T$. Later in this chapter, we consider the optimal control problem where also the final time T is variable, and where the cost is to be minimized over all allowable inputs u as well as all $T \geq 0$.

As is clear from the definition, optimal control problems are more general than calculus of variations problems. This is also reflected by the fact that optimal controls u_* may even be discontinuous as a function of time, as illustrated by the next example.

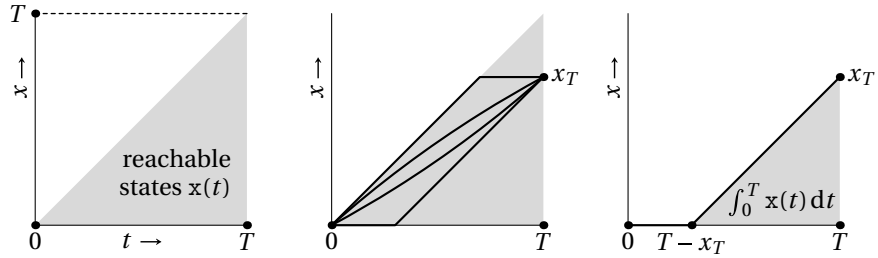


FIGURE 2.1: Reachable states and candidate optimal states for the optimal control problem of Example 2.1.1.

Example 2.1.1 (A bang-bang control example). Let

$$\dot{\mathrm{x}}(t) = \mathrm{u}(t), \qquad \mathrm{x}(0) = 0, \qquad \mathbb{U} = [0,1].$$

We want to determine the control $\mathrm{u} : [0,T] \to \mathbb{U}$ that minimizes

$$J(\mathrm{u}) := \int_0^T \mathrm{x}(t)\,\mathrm{d}t - \mathrm{x}(T) \qquad \text{for some given } T \geq 1.$$

An ad hoc "solution by picture" goes as follows. Since $\dot{\mathrm{x}}(t) = \mathrm{u}(t) \in [0,1]$ it follows that the final state $x_T := \mathrm{x}(T)$ is an element of $[\mathrm{x}(0), \mathrm{x}(0)+T] = [0,T]$, see Fig. 2.1(a). There are usually many ways to reach a given final state x_T by choice of u. Some possible state trajectories x, all reaching the same x_T, are shown in Fig. 2.1(b). Now, for any *fixed* $x_T \in [0,T]$, it is clear that $J(\mathrm{u})$ is minimal iff $\int_0^T \mathrm{x}(t)\,\mathrm{d}t$ is minimal, but this integral is the area under the curve. So for the fixed x_T of Fig. 2.1(b), the optimal state is the one shown Fig. 2.1(c), and then the area under the curve is $\int_0^T \mathrm{x}(t)\,\mathrm{d}t = \frac{1}{2}x_T^2$. Hence, the minimal cost for a fixed x_T is

$$\tfrac{1}{2}x_T^2 - x_T.$$

This cost equals $\frac{1}{2}(x_T-1)^2-\frac{1}{2}$ and, therefore, $x_T=1$ achieves the smallest possible cost over all x_T. Now we solved the problem: optimal is $x_T=1$, and the optimal state $\mathrm{x}_*(t)$ is zero for all $t\le T-x_T=T-1$, and increases with $\dot{\mathrm{x}}_*(t)=+1$ for all $t>T-1$. Therefore we conclude that the optimal control u_* is

$$\mathrm{u}_*(t)=\begin{cases}0 & \text{if } t<T-1,\\ 1 & \text{if } t>T-1.\end{cases} \tag{2.4}$$

In particular, *the optimal control is discontinuous as a function of time.* □

The derivation in this example is ad hoc. We want a theory that can deal with optimal control problems systematically, including problems whose solution is discontinuous. To develop this theory we first assume that all functions involved are sufficiently smooth, and that $\mathbb{U}=\mathbb{R}^m$. Combined with the classic method of Lagrange multipliers we can then employ the theory of calculus of variations, and this provides first-order conditions that optimal controls must satisfy. This is derived in § 2.3. Motivated by these first-order conditions, we then formulate and prove the truly fabulous *minimum principle* of Pontryagin (§ 2.5). This result shocked the scientific community when it was presented in the late fifties of the previous century. The minimum principle is very general, and it provides necessary conditions for a control to be optimal, even if the optimal control is discontinuous. In many applications, these conditions are numerically tractable and allow us to construct the optimal control, assuming one exists. But be warned: the proof of the minimum principle is involved.

2.2 Quick Summary of the Classic Lagrange Multiplier Method

Optimal control problems are minimization problems subject to dynamical *constraints*. The classic way of dealing with constraints is to introduce *Lagrange multipliers*. This short section provides a quick summary of this method; more details can be found in Appendix A.8.

Consider minimizing a function $J:\mathbb{R}^n\to\mathbb{R}$ over a constrained set of $\mathbb{R}^n$ defined as the zero set of some function $G:\mathbb{R}^n\to\mathbb{R}^k$:

$$\begin{cases}\min_{z\in\mathbb{R}^n} J(z)\\ \text{subject to } G(z)=0.\end{cases} \tag{2.5}$$

The method of *Lagrange multipliers* can help to find minimizers. In short, the idea is to associate with this constrained problem in z an *unconstrained* problem in (z,λ) with cost function

$$\mathfrak{J}(z,\lambda):=\lambda^{\mathrm{T}}G(z)+J(z).$$

This function $\mathfrak{J}:\mathbb{R}^n\times\mathbb{R}^k\to\mathbb{R}$ is sometimes called the *augmented cost function*[1],

[1] In optimization, the function $\mathfrak{J}$ is often called "Lagrangian," but in the calculus of variations, as well as in classical mechanics, that terminology is normally reserved for $F(t,x,\dot{x})$, see Chapter 1.

and the components of the vector λ are known as *Lagrange multipliers*. Assuming $\mathfrak{J}$ is sufficiently smooth, a pair (z_*, λ_*) is a stationary solution of the *unconstrained* cost $\mathfrak{J}(z, \lambda)$ over all z *and* λ iff both gradients vanish,

$$\frac{\partial \mathfrak{J}(z_*, \lambda_*)}{\partial z} = 0, \qquad \frac{\partial \mathfrak{J}(z_*, \lambda_*)}{\partial \lambda} = 0. \tag{2.6}$$

The gradient of $\mathfrak{J}(z, \lambda)$ with respect to λ is $G^{\mathrm{T}}(z)$. Hence, stationary solutions (z_*, λ_*) of $\mathfrak{J}(z, \lambda)$ necessarily satisfy $G(z_*) = 0$, and, therefore, $\mathfrak{J}(z_*, \lambda_*) = J(z_*)$. In fact, under mild assumptions, the unconstrained first-order conditions (2.6) are equivalent to the first-order conditions of the constrained minimization problem (2.5), see Appendix A.8 for details.

For the optimal control problem, we take a similar approach, however with the complication that we are not dealing with a minimization over a finite number of variables $z \in \mathbb{R}^n$, but over uncountably many functions $\mathtt{u}, \mathtt{x}$, and the constraints are the dynamical constraints $\dot{\mathtt{x}}(t) = f(\mathtt{x}(t), \mathtt{u}(t))$, and these need to be satisfied for all $t \in [0, T]$.

2.3 First-order Conditions for Unbounded and Smooth Controls

We return to the optimal control problem of minimizing a cost

$$J(\mathtt{u}) := \int_0^T L(\mathtt{x}(t), \mathtt{u}(t))\, \mathrm{d}t + K(\mathtt{x}(T)),$$

subject to

$$\dot{\mathtt{x}}(t) = f(\mathtt{x}(t), \mathtt{u}(t)), \qquad \mathtt{x}(0) = x_0. \tag{2.7}$$

In this section, we do not restrict the inputs, i.e.,

$$\mathbb{U} = \mathbb{R}^m,$$

and we further assume for now that all functions involved are sufficiently smooth.

The optimal control problem can be regarded as a *constrained* optimization problem, with (2.7) being the dynamical constraint. This observation provides a clue to its solution: introduce Lagrange multiplier *functions* $\mathtt{p} : [0, T] \to \mathbb{R}^n$ corresponding to these dynamical constraints. Analogous to the classic Lagrange multiplier method, we introduce an *augmented running cost* $\mathfrak{L} : \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{U} \times \mathbb{R}^n \to \mathbb{R}$, defined as

$$\mathfrak{L}(x, \dot{x}, u, p) = p^{\mathrm{T}}(f(x, u) - \dot{x}) + L(x, u), \tag{2.8}$$

and analyze the first-order conditions for the corresponding cost. That is, we want to know which conditions are satisfied by stationary solutions

$$\mathtt{q}_* := (\mathtt{x}_*, \mathtt{p}_*, \mathtt{u}_*)$$

of the unconstrained problem with cost

$$\mathfrak{J}(\mathtt{q}) := \int_0^T \mathfrak{L}(\mathtt{x}(t), \dot{\mathtt{x}}(t), \mathtt{u}(t), \mathtt{p}(t))\, \mathrm{d}t + K(\mathtt{x}(T)). \tag{2.9}$$

Before we delve into the resulting Euler-Lagrange equation, it is interesting to first see what the Beltrami identity gives us. Indeed, the $\mathfrak{L}$ defined in (2.8) is of the form $\mathfrak{L}(q, \dot{q})$ and so does not depend on time. As a result, the Beltrami identity holds, which says that the function

$$\mathfrak{L}(\mathtt{q}(t), \dot{\mathtt{q}}(t)) - \dot{\mathtt{q}}^{\mathrm{T}}(t) \frac{\partial \mathfrak{L}(\mathtt{q}(t), \dot{\mathtt{q}}(t))}{\partial \dot{q}}$$

is constant over time for every stationary solution $\mathtt{q}$ of the cost (2.9). For our $\mathfrak{L}$ we have

$$\begin{aligned}
&\mathfrak{L}(q, \dot{q}) - \dot{q}^{\mathrm{T}} \frac{\partial \mathfrak{L}(q, \dot{q})}{\partial \dot{q}} \\
&\qquad = \mathfrak{L}(q, \dot{q}) - \left(\dot{x}^{\mathrm{T}} \frac{\partial \mathfrak{L}(q, \dot{q})}{\partial \dot{x}} + \dot{p}^{\mathrm{T}} \frac{\partial \mathfrak{L}(q, \dot{q})}{\partial \dot{p}} + \dot{u}^{\mathrm{T}} \frac{\partial \mathfrak{L}(q, \dot{q})}{\partial \dot{u}} \right) \\
&\qquad = p^{\mathrm{T}}(f(x,u) - \dot{x}) + L(x,u) - (-\dot{x}^{\mathrm{T}} p + 0 + 0) \\
&\qquad = p^{\mathrm{T}} f(x,u) + L(x,u).
\end{aligned} \tag{2.10}$$

The final function is known as the *(optimal control) Hamiltonian* and it plays a central role in optimal control. First, we use it to formulate the necessary first-order conditions for the augmented problem:

Lemma 2.3.1 (Hamiltonian equations). Let $\mathbb{U} = \mathbb{R}^m$, $x_0 \in \mathbb{R}^n$, and consider $\mathfrak{L}$ as defined in (2.8), and assume $f(x), L(x,u), K(x)$ are C^1. Then sufficiently smooth functions $\mathtt{x}_*, \mathtt{p}_*, \mathtt{u}_*$ are stationary solutions of the cost (2.9) with $\mathtt{x}_*(0) = x_0$, iff they satisfy

$$\dot{\mathtt{x}}_*(t) = \frac{\partial H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t))}{\partial p}, \qquad \mathtt{x}_*(0) = x_0, \tag{2.11a}$$

$$\dot{\mathtt{p}}_*(t) = -\frac{\partial H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t))}{\partial x}, \qquad \mathtt{p}_*(T) = \frac{\partial K(\mathtt{x}_*(T))}{\partial x}, \tag{2.11b}$$

$$0 = \frac{\partial H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t))}{\partial u}. \tag{2.11c}$$

Here $H : \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{U} \to \mathbb{R}$ is the *(optimal control) Hamiltonian* defined as

$$H(x, p, u) = p^{\mathrm{T}} f(x,u) + L(x,u). \tag{2.12}$$

Proof. The triple $(\mathtt{x}_*, \mathtt{p}_*, \mathtt{u}_*)$ is a stationary solution iff it satisfies the Euler-Lagrange equation together with the boundary conditions of Proposition 1.5.1 (p. 23). Define $\mathfrak{L}(q, \dot{q})$ as in (2.8) with $q := (x, p, u)$, and notice that $\mathfrak{L}(q, \dot{q})$ in terms of the Hamiltonian $H(x, p, u)$ is

$$\mathfrak{L}(q, \dot{q}) = H(q) - p^{\mathrm{T}} \dot{x}.$$

For ease of exposition, we momentarily denote $\mathtt{x}(t)$ simply as $\mathtt{x}$, etc. The Euler-Lagrange equation, $0 = (\frac{\partial}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{q}})\mathfrak{L}(\mathtt{q}, \dot{\mathtt{q}})$, is a vector equation, that is to say it holds component-wise. For component $\mathtt{x}$, it says

$$0 = \Big(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\Big)(H(\mathtt{x},\mathtt{p},\mathtt{u}) - \mathtt{p}^{\mathrm{T}}\dot{\mathtt{x}}) = \frac{\partial H(\mathtt{x},\mathtt{p},\mathtt{u})}{\partial x} + \dot{\mathtt{p}}.$$

Hence, $\dot{\mathtt{p}} = -\frac{\partial H(\mathtt{x},\mathtt{p},\mathtt{u})}{\partial x}$. For component $\mathtt{p}$, it says

$$0 = \Big(\frac{\partial}{\partial p} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{p}}\Big)(H(\mathtt{x},\mathtt{p},\mathtt{u}) - \mathtt{p}^{\mathrm{T}}\dot{\mathtt{x}}) = \frac{\partial H(\mathtt{x},\mathtt{p},\mathtt{u})}{\partial p} - \dot{\mathtt{x}}.$$

Hence, $\dot{\mathtt{x}} = \frac{\partial H(\mathtt{x},\mathtt{p},\mathtt{u})}{\partial p}$. For component $\mathtt{u}$, it says

$$0 = \Big(\frac{\partial}{\partial u} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{u}}\Big)(H(\mathtt{x},\mathtt{p},\mathtt{u}) - \mathtt{p}^{\mathrm{T}}\dot{\mathtt{x}}) = \frac{\partial H(\mathtt{x},\mathtt{p},\mathtt{u})}{\partial u}.$$

The free final point (also known as free *endpoint*) conditions (1.42) become $0 = \frac{\partial \mathfrak{L}(\mathtt{q}(T),\dot{\mathtt{q}}(T))}{\partial \dot{q}} + \frac{\partial K(\mathtt{x}(T))}{\partial q}$, and per component this is

$$\begin{aligned}
0 &= \frac{\partial \mathfrak{L}(\mathtt{q}(T),\dot{\mathtt{q}}(T))}{\partial \dot{x}} + \frac{\partial K(\mathtt{x}(T))}{\partial x} = -\mathtt{p}(T) + \frac{\partial K(\mathtt{x}(T))}{\partial x},\\
0 &= \frac{\partial \mathfrak{L}(\mathtt{q}(T),\dot{\mathtt{q}}(T))}{\partial \dot{p}} + \frac{\partial K(\mathtt{x}(T))}{\partial p} = 0 + 0,\\
0 &= \frac{\partial \mathfrak{L}(\mathtt{q}(T),\dot{\mathtt{q}}(T))}{\partial \dot{u}} + \frac{\partial K(\mathtt{x}(T))}{\partial u} = 0 + 0.
\end{aligned}$$

The first says that $\mathtt{p}(T) = \frac{\partial K(\mathtt{x}(T))}{\partial x}$, and the other two are void.

Since we have an initial condition on $\mathtt{x}$ but not on $\mathtt{p}$ and $\mathtt{u}$, the free initial-point conditions (1.41) on $\mathtt{q}$ need to hold for the components $\mathtt{p}$ and $\mathtt{u}$ (see Proposition 1.5.1). The initial-point conditions become $0 = \frac{\partial \mathfrak{L}(\mathtt{q}(0),\dot{\mathtt{q}}(0))}{\partial \dot{q}}$, and for the respective components $\mathtt{p}$ and $\mathtt{u}$, this gives

$$0 = \frac{\partial \mathfrak{L}(\mathtt{q}(0),\dot{\mathtt{q}}(0))}{\partial \dot{p}} = 0 \quad \text{and} \quad 0 = \frac{\partial \mathfrak{L}(\mathtt{q}(0),\dot{\mathtt{q}}(0))}{\partial \dot{u}} = 0.$$

These conditions are void. ∎

The differential equations (2.11a, 2.11b) are known as the *Hamiltonian equations*. Note that

$$\frac{\partial H(\mathtt{x}(t),\mathtt{p}(t),\mathtt{u}(t))}{\partial p} = f(\mathtt{x}(t),\mathtt{u}(t)).$$

Therefore, the first Hamiltonian equation (2.11a) is nothing else than the given system equation: $\dot{\mathtt{x}}(t) = f(\mathtt{x}(t),\mathtt{u}(t)),\ \mathtt{x}(0) = x_0$.

The Lagrange multiplier $\mathtt{p}$ is called the *costate* (because mathematically, it lives in a dual space to the (variations) of the state $\mathtt{x}$). In examples it often has

interesting interpretations—shadow prices in economics and contact forces in mechanical systems—in terms of the *sensitivity* of the minimized cost function. This is already illustrated by the condition $\mathtt{p}_*(T) = \frac{\partial K(\mathtt{x}_*(T))}{\partial x}$, which means that $\mathtt{p}_*(T)$ equals the sensitivity of the final time cost with respect to variations in the optimal state at the final time. In Chapter 3 we show that

$$\mathtt{p}_*(0) = \frac{\mathrm{d}J(\mathtt{u}_*)}{\mathrm{d}x_0},$$

where $\mathtt{u}_*$ now means the optimal input depending on the initial state x_0, see § 3.5. A large $\mathtt{p}_*(0)$ hence means that the optimal cost is sensitive to changes in the initial state.

2.4 Towards the Minimum Principle

Based on the previous section, one would conjecture that smooth optimal controls, for $\mathbb{U} = \mathbb{R}^m$, must satisfy the first-order conditions of the augmented problem (Lemma 2.3.1). Specifically, if $\mathtt{u}_*$ is an optimal control, and $\mathtt{x}_*$ is the resulting optimal state, then one would conjecture the existence of a function $\mathtt{p}_*$ that satisfies

$$\dot{\mathtt{p}}_*(t) = -\frac{\partial H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t))}{\partial x}, \qquad \mathtt{p}_*(T) = \frac{\partial K(\mathtt{x}_*(T))}{\partial x},$$

and such that $(\mathtt{x}_*, \mathtt{p}_*, \mathtt{u}_*)$ satisfies (2.11c). We will soon see that that is indeed the case (under some mild smoothness assumption). In fact, it holds in a far more general setting. To motivate this general result, it is instructive to rewrite the Legendre condition of calculus of variations problems in terms of Hamiltonians:

Example 2.4.1 (Legendre condition in terms of Hamiltonians). Consider the calculus of variations problem with free endpoint, and where $F(t, x, \dot{x})$ does not depend on t:

$$\min_{\mathtt{x}:[0,T]\to\mathbb{R}^n} \int_0^T F(\mathtt{x}(t), \dot{\mathtt{x}}(t))\,\mathrm{d}t \quad \text{subject to } \mathtt{x}(0) = x_0.$$

Clearly, this equals the optimal control problem with

$$\dot{\mathtt{x}}(t) = \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0, \qquad \mathbb{U} = \mathbb{R}^n, \qquad L(x, u) = F(x, u).$$

The Hamiltonian in this case is

$$H(x, p, u) = p^{\mathrm{T}} u + F(x, u).$$

The Legendre condition says that optimal solutions of this calculus of variations problem must satisfy

$$\frac{\partial^2 F(\mathtt{x}_*(t), \dot{\mathtt{x}}_*(t))}{\partial \dot{x} \partial \dot{x}^{\mathrm{T}}} \geq 0 \qquad \forall\, t \in [0, T].$$

From the equality $H(x,p,u) = p^{\mathrm{T}}u + F(x,u)$, it follows immediately that the Legendre condition for our problem, in terms of the Hamiltonian, is that

$$\frac{\partial^2 H(\mathtt{x}_*(t),\mathtt{p}(t),\mathtt{u}_*(t))}{\partial u \partial u^{\mathrm{T}}} \geq 0 \qquad \forall\, t \in [0,T], \tag{2.13}$$

for whatever $\mathtt{p}(t)$. □

Condition (2.13) is particularly interesting if we take $\mathtt{p}(t) = \mathtt{p}_*(t)$ because we also have condition (2.11c), which says that

$$\frac{\partial H(\mathtt{x}_*(t),\mathtt{p}_*(t),\mathtt{u}_*(t))}{\partial u} = 0.$$

These two conditions combined suggest that $H(\mathtt{x}_*(t),\mathtt{p}_*(t),u)$ at each point in time is *minimized* by the optimal control $u = \mathtt{u}_*(t)$. Could it be? In the next section, we see that the answer is yes, and that every optimal control problem has this pointwise minimality property!

2.5 Minimum Principle

The following celebrated theorem by Pontryagin and coworkers provides a necessary condition for solutions of the true minimization problem (not just stationary ones), and it can even deal with restricted sets $\mathbb{U}$ and discontinuous controls. The basic feature is that it replaces the first-order optimality condition (2.11c) with a pointwise minimization condition. Here is the famous result. It is very general and it is the central result of this chapter.

Theorem 2.5.1 (Minimum principle). Consider the optimal control problem defined by (2.1, 2.2, 2.3), and assume that $f(x,u)$ and $\partial f(x,u)/\partial x$ and $L(x,u)$ and $\partial L(x,u)/\partial x$ are continuous in x and u, and that $K(x)$ and $\partial K(x)/\partial x$ are continuous in x.

Suppose $\mathtt{u}_* : [0,T] \to \mathbb{U}$ is a solution of the optimal control problem, and assume it is piecewise continuous[2], and let $\mathtt{x}_* : [0,T] \to \mathbb{R}^n$ be the resulting optimal state. Then there is a unique function $\mathtt{p}_* : [0,T] \to \mathbb{R}^n$ that satisfies

$$\dot{\mathtt{x}}_*(t) = \frac{\partial H(\mathtt{x}_*(t),\mathtt{p}_*(t),\mathtt{u}_*(t))}{\partial p}, \qquad \mathtt{x}_*(0) = x_0, \tag{2.14a}$$

$$\dot{\mathtt{p}}_*(t) = -\frac{\partial H(\mathtt{x}_*(t),\mathtt{p}_*(t),\mathtt{u}_*(t))}{\partial x}, \qquad \mathtt{p}_*(T) = \frac{\partial K(\mathtt{x}_*(T))}{\partial x}, \tag{2.14b}$$

and along the solution $(\mathtt{x}_*(t),\mathtt{p}_*(t))$, the input $\mathtt{u}_*(t)$ minimizes the Hamiltonian,

$$H(\mathtt{x}_*(t),\mathtt{p}_*(t),\mathtt{u}_*(t)) = \min_{u\in\mathbb{U}} H(\mathtt{x}_*(t),\mathtt{p}_*(t),u), \tag{2.15}$$

at every $t \in [0,T]$ where $\mathtt{u}_*(t)$ is continuous.

[2] A function $\mathtt{u}$ is *piecewise continuous* on $[0,T]$ if it is continuous everywhere except for finitely many instances $t_i \in (0,T)$, and that $\lim_{t\uparrow t_i} \mathtt{u}(t)$ and $\lim_{t\downarrow t_i} \mathtt{u}(t)$ exist for all points of discontinuity $t_i \in (0,T)$, and that also $\lim_{t\downarrow 0} \mathtt{u}(t)$ and $\lim_{t\uparrow T} \mathtt{u}(t)$ exist.

Proof. (This proof requires a couple of easy-to-believe but technical results regarding continuity of solutions of differential equations. Upon first reading these can be discarded, but for a full understanding one should have a look at Appendix B.)

Let $\mathtt{u}_*$ be an optimal input, and let $\mathtt{x}_*$ be the corresponding optimal state. First notice that the costate equations are linear in the costate:

$$\dot{\mathtt{p}}_*(t) = A(t)\mathtt{p}_*(t) + b(t), \qquad \mathtt{p}_*(T) = \partial K(\mathtt{x}_*(T))/\partial x$$

for $A(t) := -\partial f(\mathtt{x}_*(t), \mathtt{u}_*(t))/\partial x^{\mathrm{T}}$ and $b(t) := -\partial L(\mathtt{x}_*(t), \mathtt{u}_*(t))/\partial x$. By assumption, both $A(t)$ and $b(t)$ are piecewise continuous, and so the solution $\mathtt{p}_*(t)$ exists for all $t \in [0, T]$, is continuous and is unique.

Now assume, to obtain a contradiction, that at some time $\bar{t} \in (0, T)$ where the input is continuous, a $\hat{u} \in \mathbb{U}$ exists that achieves a smaller value of the Hamiltonian $H(\mathtt{x}_*(\bar{t}), \mathtt{p}_*(\bar{t}), \hat{u})$ than $\mathtt{u}_*(\bar{t})$ does. That is, c defined as

$$c = H(\mathtt{x}_*(\bar{t}), \mathtt{p}_*(\bar{t}), \hat{u}) - H(\mathtt{x}_*(\bar{t}), \mathtt{p}_*(\bar{t}), \mathtt{u}_*(\bar{t}))$$

is negative. Then, by continuity, for some small enough $\epsilon > 0$ the function defined as

$$\bar{\mathtt{u}}(t) = \begin{cases} \hat{u} & \text{if } t \in [\bar{t}, \bar{t} + \epsilon], \\ \mathtt{u}_*(t) & \text{elsewhere} \end{cases}$$

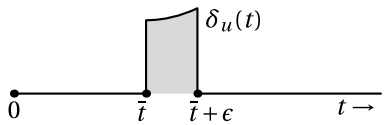
achieves a smaller (or equal) value of the Hamiltonian for all time, and

$$\int_0^T H(\mathtt{x}_*(t), \mathtt{p}_*(t), \bar{\mathtt{u}}(t)) - H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t))\,\mathrm{d}t = c\epsilon + o(\epsilon).$$

Now write $\bar{\mathtt{u}}(t)$ as a perturbation of the optimal input,

$$\bar{\mathtt{u}}(t) = \mathtt{u}_*(t) + \delta_u(t).$$

The so defined perturbation $\delta_u(t) = \bar{\mathtt{u}}(t) - \mathtt{u}_*(t)$ has a support of ϵ. Its graph might look like



In the rest of the proof we fix this perturbation and we consider only very small and positive ϵ. Such perturbations are called "needle" perturbations.

By perturbing the input, $\bar{\mathtt{u}} = \mathtt{u}_* + \delta_u$, the solution of $\dot{\mathtt{x}}(t) = f(\mathtt{x}(t), \mathtt{u}_*(t) + \delta_u(t))$ for $t > \bar{t}$ perturbs as well. Denote the perturbed state as $\mathtt{x}(t) = \mathtt{x}_*(t) + \delta_x(t)$. The perturbation $\delta_x(t)$ is probably not a needle but at each $t > \bar{t}$ it is of

order[3] ϵ. To avoid clutter, we now drop all time arguments, that is, $\mathtt{x}(t)$ is simply denoted as $\mathtt{x}$, etc. The derivative of δ_x with respect to time satisfies

$$\dot{\delta}_x = (\dot{\mathtt{x}}_* + \dot{\delta}_x) - \dot{\mathtt{x}}_* = f(\mathtt{x}_* + \delta_x, \mathtt{u}_* + \delta_u) - f(\mathtt{x}_*, \mathtt{u}_*). \tag{2.16}$$

This expression we soon need.

Let Δ be the change in cost, $\Delta := J(\mathtt{u}_* + \delta_u) - J(\mathtt{u}_*)$. We have

$$\begin{aligned}\Delta &= J(\mathtt{u}_* + \delta_u) - J(\mathtt{u}_*) \\ &= K(\mathtt{x}_*(T) + \delta_x(T)) - K(\mathtt{x}_*(T)) + \int_0^T L(\mathtt{x}_* + \delta_x, \mathtt{u}_* + \delta_u) - L(\mathtt{x}_*, \mathtt{u}_*)\, \mathrm{d}t \\ &= \frac{\partial K(\mathtt{x}_*(T))}{\partial x^{\mathrm{T}}}\delta_x(T) + \int_0^T L(\mathtt{x}_* + \delta_x, \mathtt{u}_* + \delta_u) - L(\mathtt{x}_*, \mathtt{u}_*)\, \mathrm{d}t + \mathfrak{o}(\epsilon).\end{aligned}$$

Next use that $L(x,u) = -p^{\mathrm{T}} f(x,u) + H(x,p,u)$, and substitute p for the optimal costate $\mathtt{p}_*$:

$$\begin{aligned}\Delta = \mathtt{p}_*^{\mathrm{T}}(T)\delta_x(T) &+ \int_0^T -\mathtt{p}_*^{\mathrm{T}}[f(\mathtt{x}_* + \delta_x, \mathtt{u}_* + \delta_u) - f(\mathtt{x}_*, \mathtt{u}_*)]\, \mathrm{d}t \\ &+ \int_0^T H(\mathtt{x}_* + \delta_x, \mathtt{p}_*, \mathtt{u}_* + \delta_u) - H(\mathtt{x}_*, \mathtt{p}_*, \mathtt{u}_*)\, \mathrm{d}t + \mathfrak{o}(\epsilon).\end{aligned}$$

The term in between square brackets according to (2.16) is $\dot{\delta}_x$, so

$$\begin{aligned}\Delta = \mathtt{p}_*^{\mathrm{T}}(T)\delta_x(T) &+ \int_0^T -\mathtt{p}_*^{\mathrm{T}}\dot{\delta}_x + H(\mathtt{x}_* + \delta_x, \mathtt{p}_*, \mathtt{u}_* + \delta_u) - H(\mathtt{x}_*, \mathtt{p}_*, \mathtt{u}_* + \delta_u)\, \mathrm{d}t \\ &+ \int_0^T H(\mathtt{x}_*, \mathtt{p}_*, \mathtt{u}_* + \delta_u) - H(\mathtt{x}_*, \mathtt{p}_*, \mathtt{u}_*)\, \mathrm{d}t + \mathfrak{o}(\epsilon).\end{aligned}$$

Here, we also subtracted and added a term $H(\mathtt{x}_*, \mathtt{p}_*, \mathtt{u}_* + \delta_u)$. The reason is that now the difference of the first two Hamiltonian terms can be recognized as an approximate partial derivative with respect to x, and the difference of the final two Hamiltonian terms is what we considered earlier (it equals $c\epsilon + \mathfrak{o}(\epsilon)$). So:

$$\Delta = \mathtt{p}_*^{\mathrm{T}}(T)\delta_x(T) + \int_0^T -\mathtt{p}_*^{\mathrm{T}}\dot{\delta}_x + \frac{\partial H(\mathtt{x}_*, \mathtt{p}_*, \mathtt{u}_* + \delta_u)}{\partial x^{\mathrm{T}}}\delta_x\, \mathrm{d}t + c\epsilon + \mathfrak{o}(\epsilon).$$

Notice that the partial derivative $\frac{\partial H(\mathtt{x}_*, \mathtt{p}_*, \mathtt{u}_* + \delta_u)}{\partial x}$ equals $-\dot{\mathtt{p}}_* = \frac{\partial H(\mathtt{x}_*, \mathtt{p}_*, \mathtt{u}_*)}{\partial x}$ everywhere except for ϵ units of time (for $t \in [\bar{t}, \bar{t} + \epsilon]$). This, combined with the fact that δ_x at each moment in time is also of order ϵ, allows us to conclude that

$$\Delta = \mathtt{p}_*^{\mathrm{T}}(T)\delta_x(T) + \int_0^T -\mathtt{p}_*^{\mathrm{T}}\dot{\delta}_x - \dot{\mathtt{p}}_*^{\mathrm{T}}\delta_x\, \mathrm{d}t + c\epsilon + \mathfrak{o}(\epsilon).$$

[3] For $t \le \bar{t}$, we have $\delta_x(t) = 0$. For $t \in [\bar{t}, \bar{t} + \epsilon]$, we have $\|\delta_x(t)\| = \|\mathtt{x}(t) - \mathtt{x}_*(t)\| = \|\mathtt{x}(t) - \mathtt{x}(\bar{t}) - (\mathtt{x}_*(t) - \mathtt{x}(\bar{t}))\| \le \|\mathtt{x}(t) - \mathtt{x}(\bar{t})\| + \|\mathtt{x}_*(t) - \mathtt{x}(\bar{t})\| = \|(t - \bar{t})(f(\mathtt{x}(\bar{t}), \mathtt{u}(\bar{t})))\| + \|(t - \bar{t})(f(\mathtt{x}(\bar{t}), \mathtt{u}(\bar{t})))\| + \mathfrak{o}(t - \bar{t}) \le M\epsilon$ for some $M > 0$ and all small enough $\epsilon > 0$. So at $t = \bar{t} + \epsilon$ the solutions $\mathtt{x}(t)$ and $\mathtt{x}_*(t)$ differ, in norm, at most $M\epsilon$. Now for $t > \bar{t} + \epsilon$, apply Lemma B.1.6 with $g(t) = 0$.

The integrand $-\mathtt{p}_*^{\mathrm{T}}\dot{\delta}_x - \dot{\mathtt{p}}_*^{\mathrm{T}}\delta_x$ we recognize as the total derivative of $-\mathtt{p}_*^{\mathrm{T}}\delta_x$ with respect to time. Now it is better to add the time dependence again:

$$\begin{aligned}\Delta &= \mathtt{p}_*^{\mathrm{T}}(T)\delta_x(T) + \left[-\mathtt{p}_*^{\mathrm{T}}(t)\delta_x(t)\right]_0^T + c\epsilon + \mathfrak{o}(\epsilon)\\ &= \mathtt{p}_*^{\mathrm{T}}(0)\delta_x(0) + c\epsilon + \mathfrak{o}(\epsilon)\\ &= c\epsilon + \mathfrak{o}(\epsilon).\end{aligned}$$

Here we used that $\delta_x(0) = 0$. This is because of the initial condition $\mathtt{x}(0) = x_0$. Since $c < 0$ we see that Δ is negative for small enough ϵ. But that would mean that $\bar{\mathtt{u}}$ for small enough ϵ achieves a smaller cost than optimal. Not possible. Hence, the assumption that $\mathtt{u}_*(t)$ does not minimize the Hamiltonian at every t where $\mathtt{u}_*(t)$ is continuous, is wrong. ■

The theory of the minimum principle was developed during the 1950s in the former Soviet Union by a group of mathematicians led by Lev Pontryagin, and in honor of him it is called *Pontryagin's minimum principle*. Actually, Pontryagin followed the classical mechanics sign convention, $p^{\mathrm{T}}f(x,u)$ *minus* $L(x,u)$. Hence, the principle is better known as *Pontryagin's maximum principle*.

The principle *assumes the existence* of an optimal control $\mathtt{u}_*$, and then guarantees that $\mathtt{u}_*$ minimizes the Hamiltonian at each moment in time. In practical situations, this pointwise minimization is used to *determine* the optimal control, tacitly assuming an optimal control exists. Hence, one could say that the principle provides *necessary* conditions for optimality. In Section 2.8, we discuss under which conditions these conditions are *sufficient* as well; see also Chapter 3 for the alternative approach offered by dynamic programming.

Example 2.5.2 (Simple problem). Consider the system and cost

$$\dot{\mathtt{x}}(t) = \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0, \qquad J(\mathtt{u}) = \int_0^1 \mathtt{x}(t)\,\mathrm{d}t.$$

So we have that $f(x,u) = u$, $K(x) = 0$, and $L(x,u) = x$. As input set we take

$$\mathbb{U} = [-1,1].$$

The Hamiltonian for this problem becomes

$$H(x,p,u) = pu + x,$$

and the equation for the costate hence is

$$\dot{\mathtt{p}}_*(t) = -\frac{\partial H(\mathtt{x}_*(t),\mathtt{p}_*(t),\mathtt{u}_*(t))}{\partial x} = -1, \qquad \mathtt{p}_*(1) = \frac{\partial K(\mathtt{x}_*(T))}{\partial x} = 0.$$

Clearly, this means that the costate equals

$$\mathtt{p}_*(t) = 1 - t.$$

The optimal input $\mathtt{u}_*(t)$—assuming it exists—at each $t \in [0,1]$ minimizes the Hamiltonian $\mathtt{p}_*(t)\mathtt{u}(t) + \mathtt{x}_*(t)$. Since $\mathtt{p}_*(t) = 1 - t > 0$ for all $t \in [0,1)$ the value of the optimal input is the minimal value in $\mathbb{U}$,

$$\mathtt{u}_*(t) = -1 \qquad \forall\, t \in [0,1].$$

This makes perfect sense: to minimize $\int_0^1 \mathtt{x}(t)\,\mathrm{d}t$, we want $\mathtt{x}(t)$ to go down as fast as possible, which, given the system dynamics $\dot{\mathtt{x}}(t) = \mathtt{u}(t)$, means taking $\mathtt{u}(t)$ as small (negative) as possible. □

Example 2.5.3 (Switching inputs). We consider again the integrator system

$$\dot{\mathtt{x}}(t) = \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0, \qquad \mathbb{U} = [-1,1],$$

but now we add a final cost $-\frac{1}{2}\mathtt{x}(1)$ to the cost function,

$$J(\mathtt{u}) = \int_0^1 \mathtt{x}(t)\,\mathrm{d}t - \frac{1}{2}\mathtt{x}(1).$$

In this case it is not obvious what to do with $\mathtt{u}(t)$ because the faster $\mathtt{x}(t)$ goes down the larger the final cost $-\frac{1}{2}\mathtt{x}(1)$ is going to be. So possibly $\mathtt{u}(t) = -1$ is no longer optimal. In fact, we will see that it is not. We have $H(x,p,u) = pu + x$ and $K(x) = -\frac{1}{2}x$, and so the costate equation now is

$$\dot{\mathtt{p}}_*(t) = -1, \qquad \mathtt{p}_*(1) = \frac{\partial K(\mathtt{x}_*(1))}{\partial x} = -\frac{1}{2}.$$

Hence,

$$\mathtt{p}_*(t) = \frac{1}{2} - t.$$

The costate is positive for $0 \le t < \frac{1}{2}$ but negative for $\frac{1}{2} < t \le 1$. The optimal control minimizes the Hamiltonian $\mathtt{p}_*(t)\mathtt{u}_*(t) + \mathtt{x}_*(t)$, and, because of the sign change in $\mathtt{p}_*(t)$ at $t = 1/2$, we see that the optimal input switches sign at $t = 1/2$:

$$\mathtt{u}_*(t) = \begin{cases} -1 & \text{if } 0 \le t < \frac{1}{2} \\ +1 & \text{if } \frac{1}{2} < t \le 1 \end{cases}.$$

Assuming an optimal control exists, it must be this one. Apparently, it is now optimal to move $\mathtt{x}(t)$ down as fast as possible over the first half of the time interval and then back up as fast as possible over the second half. □

Example 2.5.4 (Linear system with quadratic cost). Consider the system with cost

$$\dot{\mathtt{x}}(t) = \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0, \qquad J(\mathtt{u}) = \int_0^1 \mathtt{x}^2(t) + \mathtt{u}^2(t)\,\mathrm{d}t.$$

Now we allow every $\mathtt{u}(t) \in \mathbb{R}$, that is, $\mathbb{U} = \mathbb{R}$. Notice that this is the same cost function as in Example 1.5.3 (for $\alpha = 1$) because $\mathtt{u}(t) = \dot{\mathtt{x}}(t)$. The associated Hamiltonian is

$$H(x, p, u) = pu + x^2 + u^2.$$

Since $H(x, p, u)$ is quadratic in u, and $\mathbb{U} = \mathbb{R}$, the minimizing u is the one at which the gradient of $H(x, p, u)$ with respect to u is zero. This yields

$$\mathtt{u}_*(t) = -\tfrac{1}{2}\mathtt{p}_*(t), \tag{2.17}$$

and then the Hamiltonian equations (2.14) become

$$\begin{aligned} \dot{\mathtt{x}}_*(t) &= -\tfrac{1}{2}\mathtt{p}_*(t), & \mathtt{x}_*(0) &= x_0, \\ \dot{\mathtt{p}}_*(t) &= -2\mathtt{x}_*(t), & \mathtt{p}_*(1) &= 0. \end{aligned}$$

This implies that $\ddot{\mathtt{p}}_*(t) = \mathtt{p}_*(t)$. The general solution of this second-order differential equation is

$$\mathtt{p}_*(t) = c_1 \mathrm{e}^t + c_2 \mathrm{e}^{-t},$$

and since $\mathtt{x}_*(t) = -\frac{1}{2}\dot{\mathtt{p}}_*(t)$, we find that

$$\mathtt{x}_*(t) = -\tfrac{1}{2}c_1 \mathrm{e}^t + \tfrac{1}{2}c_2 \mathrm{e}^{-t}.$$

The two constants c_1, c_2, follow uniquely from the two boundary conditions $\mathtt{x}_*(0) = x_0$ and $\mathtt{p}_*(1) = 0$, and it gives (verify this yourself)

$$\begin{aligned} \mathtt{x}_*(t) &= \frac{x_0}{\mathrm{e} + \mathrm{e}^{-1}} \left(\mathrm{e}^{1-t} + \mathrm{e}^{t-1}\right), \\ \mathtt{p}_*(t) &= \frac{2x_0}{\mathrm{e} + \mathrm{e}^{-1}} \left(\mathrm{e}^{1-t} - \mathrm{e}^{t-1}\right). \end{aligned}$$

The optimal control $\mathtt{u}_*(t)$ now follows from (2.17). □

Example 2.5.5 (Optimal reinvestment). Let $\mathtt{x}(t)$ be the rate of production of, say, gold of some mining company. At each moment in time a fraction $\mathtt{u}(t) \in [0, 1]$ of the produced gold is used for reinvestment so as to increase the production rate. This is modeled as

$$\dot{\mathtt{x}}(t) = \alpha \mathtt{u}(t)\mathtt{x}(t), \qquad \mathtt{x}(0) = x_0, \qquad \mathtt{u}(t) \in [0, 1],$$

where α is some positive parameter that models the success of investment. Since we reinvest $\mathtt{u}(t)\mathtt{x}(t)$, the net production rate is $(1 - \mathtt{u}(t))\mathtt{x}(t)$. After T units of time we want to net total production $\int_0^T (1 - \mathtt{u}(t))\mathtt{x}(t)\,\mathrm{d}t$ to be as large as possible. In our setup (with minimization), it means that we want the quantity with the opposite sign

$$J(\mathtt{u}) := \int_0^T (\mathtt{u}(t) - 1)\mathtt{x}(t)\,\mathrm{d}t$$

to be as small as possible. The Hamiltonian in this case is

$$H(x,p,u) = p\alpha ux + (u-1)x = ux(1+\alpha p) - x, \tag{2.18}$$

and the Hamiltonian equations become

$$\begin{aligned} \dot{\mathtt{x}}(t) &= \alpha \mathtt{u}(t)\mathtt{x}(t), & \mathtt{x}(0) &= x_0, \\ \dot{\mathtt{p}}(t) &= (1-\mathtt{u}(t)) - \mathtt{p}(t)\alpha \mathtt{u}(t), & \mathtt{p}(T) &= 0. \end{aligned}$$

These differential equations are, in their present form, still hard to solve. However, the Hamiltonian (2.18) is linear in u, so the minimizer u of the Hamiltonian (2.18) depends solely on the sign of $x(1+\alpha p)$. In fact, since the production rate $\mathtt{x}(t)$ is inherently positive (because $\mathtt{x}(0) = x_0 > 0$ and $\dot{\mathtt{x}}(t) = \alpha \mathtt{u}(t)\mathtt{x}(t) \geq 0$), the Hamiltonian at each moment in time is minimal for

$$\mathtt{u}_*(t) = \begin{cases} 0 & \text{if } 1+\alpha \mathtt{p}_*(t) > 0, \\ 1 & \text{if } 1+\alpha \mathtt{p}_*(t) < 0. \end{cases}$$

The value of the costate $\mathtt{p}_*(t)$ where this $\mathtt{u}_*(t)$ switches is $\mathtt{p}_*(t) = -1/\alpha$, see Fig. 2.2 (left). Now at $t = T$ we have $\mathtt{p}_*(T) = 0$, so near the final time T we have $\mathtt{u}_*(t) = 0$ (invest nothing, sell all), and then the Hamiltonian dynamics reduces to $\dot{\mathtt{x}}_*(t) = 0$ and

$$\dot{\mathtt{p}}_*(t) = 1 \text{ near } t = T, \text{ and } \mathtt{p}_*(T) = 0.$$

That is, $\mathtt{p}_*(t) = t - T$ near $t = T$, see Fig. 2.2. Solving backwards in time, starting at $t = T$, we see that the costate reduces linearly, until at time

$$t_{\mathrm{s}} := T - 1/\alpha$$

it reaches the level $\mathtt{p}_*(t_{\mathrm{s}}) = -1/\alpha < 0$ at which point $\mathtt{u}_*(t)$ switches sign. Since $\dot{\mathtt{p}}(t) > 0$ for every input, the value of $\mathtt{p}_*(t)$ is *less* than $-1/\alpha$ for all $t < t_{\mathrm{s}}$, which implies that $\mathtt{u}_*(t) = 1$ all $t < t_{\mathrm{s}}$. For this case, the Hamiltonian dynamics simplify to

$$\dot{\mathtt{x}}_*(t) = \alpha \mathtt{x}_*(t), \qquad \dot{\mathtt{p}}_*(t) = -\alpha \mathtt{p}_*(t) \qquad \text{if } t < t_{\mathrm{s}}.$$

Both $\mathtt{x}_*(t)$ and $\mathtt{p}_*(t)$ now have exponential solutions. The combination of before-and-after-switch is shown in Fig. 2.2. This settles $\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t)$ for all $t \in [0,T]$.

Notice that if $t_{\mathrm{s}} < 0$ then on the time window $[0,T]$ no switch takes place. It is then optimal to invest nothing and sell everything throughout $[0,T]$. This happens if $\alpha < 1/T$, and the interpretation is that α is then too small to benefit from investment. If, on the other hand, $\alpha > 1/T$ then $t_{\mathrm{s}} > 0$ and then investment is beneficial and the above shows that it is optimal to first invest everything, and in the final $1/\alpha$ time units to sell everything. Of course, this model is a simplification of reality. □

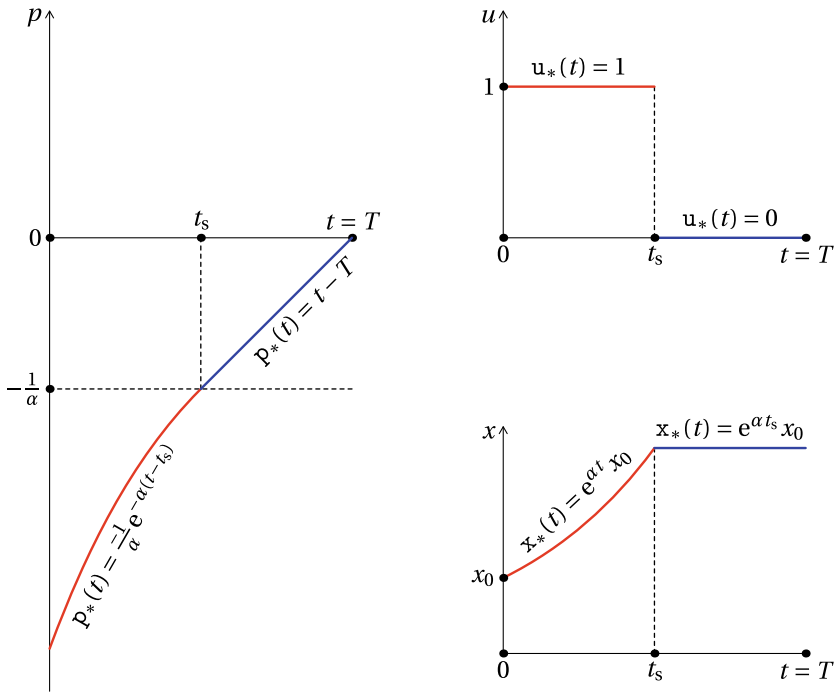


FIGURE 2.2: Optimal costate $\mathtt{p}_*(t)$, optimal input $\mathtt{u}_*(t)$ and optimal state $\mathtt{x}_*(t)$. See Example 2.5.5.

The Hamiltonian was derived from the Beltrami identity (see Eqn. (2.10)). Hence, we could expect that $H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t))$ is constant as a function of time. For the unconstrained inputs ($\mathbb{U} = \mathbb{R}^m$) and smooth enough solutions, this may easily be verified directly from the first-order equations for optimality expressed in Lemma 2.3.1. Indeed, if $(\mathtt{x}_*, \mathtt{p}_*, \mathtt{u}_*)$ are a smooth triple satisfying (2.11), then a direct computation yields (and for the sake of exposition, we momentarily drop here the arguments of H and other functions)

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t)) &= \frac{\partial H}{\partial x^{\mathrm{T}}} \dot{\mathtt{x}}_* + \frac{\partial H}{\partial p^{\mathrm{T}}} \dot{\mathtt{p}}_* + \frac{\partial H}{\partial u^{\mathrm{T}}} \dot{\mathtt{u}}_* \\
&= \frac{\partial H}{\partial x^{\mathrm{T}}} \frac{\partial H}{\partial p} + \frac{\partial H}{\partial p^{\mathrm{T}}} (-\frac{\partial H}{\partial x}) + \frac{\partial H}{\partial u^{\mathrm{T}}} \dot{\mathtt{u}}_* \\
&= \frac{\partial H}{\partial u^{\mathrm{T}}} \dot{\mathtt{u}}_* \\
&= 0.
\end{aligned} \tag{2.19}$$

The final equality follows from (2.11c). Actually, the constancy of the Hamiltonian $H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t))$ also holds for restricted input sets $\mathbb{U}$ (such as $\mathbb{U} = [0, 1]$, etc.). This is remarkable because in such cases the input quite often is not even continuous.

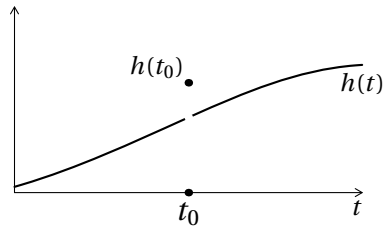


FIGURE 2.3: Suppose a function $h : \mathbb{R} \to \mathbb{R}$ is not continuous at some t_0, but that the limit from the left and from the right at t_0 are the same: $\lim_{t \uparrow t_0} h(t) = \lim_{s \downarrow t_0} h(s)$. Then the discontinuity of h at t_0 is said to be *removable*. See the proof of Theorem 2.5.6.

Theorem 2.5.6 (Constancy of the Hamiltonian). Let all the assumptions of Theorem 2.5.1 be satisfied. Suppose, in addition, that $\partial f(x,u)/\partial u$ and $\partial L(x,u)/\partial u$ are continuous, and that $\mathtt{u}_*$ is C^1 at all but finitely many $t \in [0,T]$. Then a constant H_* exists such that

$$H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t)) = H_*$$

at every t where $\mathtt{u}_*(t)$ is continuous.

Proof. Clearly, $H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t))$ as a function of time is continuous wherever $\mathtt{u}_*(t)$ is continuous. So, the Hamiltonian has finitely many discontinuities because $\mathtt{u}_*(t)$ has finitely many discontinuities. We first prove that all discontinuities in $H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t))$ are *removable*. This notion is illustrated in Fig. 2.3. It means that by suitably redefining the function at the discontinuities the function becomes continuous everywhere. Let $t_0 \in (0,T)$ be a point of discontinuity of $\mathtt{u}_*(t)$, and realize that $\mathtt{x}_*(t)$ and $\mathtt{p}_*(t)$ *are* continuous at t_0. Because of the minimality property of the Hamiltonian, we have that $H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t)) \le H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(s))$ at every t where $\mathtt{u}_*(t)$ is continuous, and at every $s \in [0,T]$. This also holds in the limit $t \uparrow t_0$ and $s \downarrow t_0$:

$$\lim_{t \uparrow t_0} H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t)) \le \lim_{t \uparrow t_0, s \downarrow t_0} H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(s)) \tag{2.20}$$

$$= \lim_{s \downarrow t_0} H(\mathtt{x}_*(s), \mathtt{p}_*(s), \mathtt{u}_*(s)). \tag{2.21}$$

The final equality is because $\mathtt{x}_*(t), \mathtt{p}_*(t)$ are continuous functions, also at t_0. Again by the minimality property, the Hamiltonian in (2.21) satisfies $H(\mathtt{x}_*(s), \mathtt{p}_*(s), \mathtt{u}_*(s)) \le H(\mathtt{x}_*(s), \mathtt{p}_*(s), \mathtt{u}_*(\tau))$ at every s where $\mathtt{u}_*(s)$ is continuous, and at every $\tau \in [0,T]$. Thus, we also have

$$\lim_{s \downarrow t_0} H(\mathtt{x}_*(s), \mathtt{p}_*(s), \mathtt{u}_*(s)) \le \lim_{s \downarrow t_0, \tau \uparrow t_0} H(\mathtt{x}_*(s), \mathtt{p}_*(s), \mathtt{u}_*(\tau)) \tag{2.22}$$

$$= \lim_{\tau \uparrow t_0} H(\mathtt{x}_*(\tau), \mathtt{p}_*(\tau), \mathtt{u}_*(\tau)). \tag{2.23}$$

The last identity is again by continuity of $\mathtt{x}_*, \mathtt{p}_*$ at t_0. Finally, notice that (2.23) equals the first limit in (2.20). Therefore, all the above limits in (2.20)–(2.23)

are the same. In particular, the limits in (2.21) and (2.23) are the same. It means that the possible discontinuity at t_0 is removable. This shows that $H(\mathrm{x}_*(t), \mathrm{p}_*(t), \mathrm{u}_*(t))$ equals some continuous function $H_*(t)$ at all but finitely many t.

Now let t be a time at which u_* is C^1. Since x_* and p_* are C^1 at t, the equality (2.19) gives

$$q := \frac{\mathrm{d}H(\mathrm{x}_*(t), \mathrm{p}_*(t), \mathrm{u}_*(t))}{\mathrm{d}t} = \frac{\partial H(\mathrm{x}_*(t), \mathrm{p}_*(t), \mathrm{u}_*(t))}{\partial u^{\mathrm{T}}} \dot{\mathrm{u}}_*(t). \tag{2.24}$$

This q is zero by the fact that $\mathrm{u}_*(t)$ minimizes the Hamiltonian. To see this more clearly, realize that this q also enters the following Taylor series:

$$H(\mathrm{x}_*(t), \mathrm{p}_*(t), \mathrm{u}_*(t+\epsilon)) = H(\mathrm{x}_*(t), \mathrm{p}_*(t), \mathrm{u}_*(t)) + q\epsilon + \mathfrak{o}(\epsilon).$$

Since $\mathrm{u}_*(t+\epsilon)$ minimizes $H(\mathrm{x}_*(t), \mathrm{p}_*(t), \mathrm{u}_*(t+\epsilon))$ for $\epsilon = 0$, the above Taylor series reveals that q must indeed be zero. Hence, (2.24) is zero at every t where u_* is C^1.

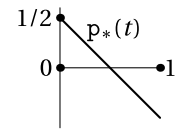
Summarizing: $H(\mathrm{x}_*(t), \mathrm{p}_*(t), \mathrm{u}_*(t))$ equals some continuous function $H_*(t)$ at all but finitely many t, and it has zero derivative at all but finitely many t. Hence, also the continuous function $H_*(t)$ has zero derivative at all but finitely many t. But that means that $H_*(t)$ is constant, $H_*(t) = H_*$. The function $H(\mathrm{x}_*(t), \mathrm{p}_*(t), \mathrm{u}_*(t))$ is continuous at every t where $\mathrm{u}_*(t)$ is continuous, so it equals this constant H_* at all such t. ∎

We note that Theorem 2.5.6 can be proved under weaker smoothness assumptions. However, the above assumptions hold in all practical cases, and avoid further technicalities in the proof.

The following example illustrates the constancy property of the Hamiltonian for a case where the optimal input is not even continuous.

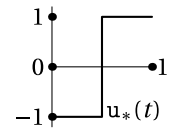
Example 2.5.7 (Example 2.5.3 continued). In Example 2.5.3, we considered $\dot{\mathrm{x}}(t) = \mathrm{u}(t)$ with initial condition $\mathrm{x}(0) = x_0$ and cost $J(\mathrm{u}) = \int_0^1 \mathrm{x}(t)\,\mathrm{d}t - \frac{1}{2}\mathrm{x}(1)$. We found that the optimal costate trajectory equals

$$\mathrm{p}_*(t) = \frac{1}{2} - t,$$



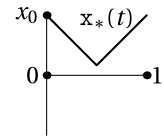
and that the optimal input switches halfway,

$$\mathrm{u}_*(t) = \begin{cases} -1 & \text{if } 0 \le t < \frac{1}{2}, \\ 1 & \text{if } \frac{1}{2} \le t \le 1. \end{cases}$$



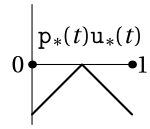
Therefore, the description of the optimal state trajectory also switches halfway: from $\dot{\mathrm{x}}(t) = \mathrm{u}_*(t)$ it follows that

$$\mathrm{x}_*(t) = \begin{cases} x_0 - t & \text{for } 0 \le t \le \frac{1}{2}, \\ x_0 - 1 + t & \text{for } \frac{1}{2} \le t \le 1. \end{cases}$$



Based on this, it seems unlikely that the Hamiltonian along the optimal solution is constant, but realize that $\mathrm{p}_*(t)\mathrm{u}_*(t)$ equals

$$\mathrm{p}_*(t)\mathrm{u}_*(t) = \begin{cases} -(\frac{1}{2} - t) & \text{if } 0 \le t < \frac{1}{2}, \\ (\frac{1}{2} - t) & \text{if } \frac{1}{2} \le t \le 1, \end{cases}$$



and, therefore, that the Hamiltonian is constant as a function of time,

$$\begin{aligned} H(\mathrm{x}_*(t), \mathrm{p}_*(t), \mathrm{u}_*(t)) &= \mathrm{p}_*(t)\mathrm{u}_*(t) + \mathrm{x}_*(t) \\ &= \begin{cases} -(\frac{1}{2} - t) + (x_0 - t) & \text{if } t < 1/2 \\ (\frac{1}{2} - t) + (x_0 - 1 + t) & \text{if } t \ge 1/2 \end{cases} \\ &= x_0 - \tfrac{1}{2} \quad \text{for all } t \in [0,1]. \end{aligned}$$

□

2.6 Optimal Control with Final Constraints

So far in this chapter the final state $\mathrm{x}(T)$ was not constrained. In quite a few applications, however, there *are* constraints on the final state $\mathrm{x}(T)$. (Note furthermore that in Chapter 1, calculus of variations, we actually started with a fully constrained $\mathrm{x}(T)$.) In the car parking application, for instance, we obviously want the speed of the car to equal zero at the final time. Let r denote the number of components of the final state that are constrained. Without loss of generality, we assume these to be the *first* r components. So consider the system with initial *and* final conditions

$$\dot{\mathrm{x}}(t) = f(\mathrm{x}(t), \mathrm{u}(t)), \qquad \mathrm{x}(0) = x_0, \qquad \mathrm{x}_i(T) = \hat{x}_i, \quad i = 1, \ldots, r. \tag{2.25}$$

Keep in mind that no conditions are imposed on the remaining final state components $\mathrm{x}_{r+1}(T), \ldots, \mathrm{x}_n(T)$. As before we take a cost of the form

$$J(\mathrm{u}) = \int_0^T L(\mathrm{x}(t), \mathrm{u}(t))\,\mathrm{d}t + K(\mathrm{x}(T)). \tag{2.26}$$

Lemma 2.3.1 (the first-order conditions for $\mathbb{U} = \mathbb{R}^m$) can be generalized to this case as follows. In the proof of this lemma, the conditions on the final costate

$$\mathrm{p}_*(T) = \frac{\partial K(\mathrm{x}_*(T))}{\partial x}$$

were derived from the free endpoint condition (1.42), but in Proposition 1.5.1 we saw that these conditions are absent if the final state is constrained. With that in mind, it will be no surprise that fixing the first r components of the final state, $\mathtt{x}_i(T)$, $i = 1, \dots, r$, implies that the conditions on the corresponding first r components of the final costate are absent, i.e., only the remaining components of $\mathtt{p}_*(T)$ are constrained:

$$\mathtt{p}_{*i}(T) = \frac{\partial K(\mathtt{x}_*(T))}{\partial x_i}, \qquad i = r+1, \dots, n.$$

That is indeed the case. However, there is a catch: the first-order conditions were derived using a *perturbation* of the solution $\mathtt{x}_*$, but if we have constraints on the both initial state and final state, then it may happen that nonzero perturbations *do not exist*. An example is

$$\dot{\mathtt{x}}(t) = \mathtt{u}^2(t), \qquad \mathtt{x}(0) = 0, \qquad \mathtt{x}(1) = 0.$$

Clearly, in this case, there is only one feasible control and one feasible state: the zero function. (This system will be the starting point of Example 2.6.2.) We have seen similar difficulties in the calculus of variations problems subject to integral constraints, with its "normal" and "abnormal" Euler-Lagrange equation (have a look at § 1.7, in particular, Example 1.7.2). Also now we make a distinction between a normal and an abnormal case, but the proof of the resulting theorem is involved, and it would take too long to explain the details here. The interested reader might want to consult the excellent book (Liberzon, 2012). We just provide the solution. It involves the *modified Hamiltonian* defined as

$$H_\lambda(x, p, u) = p^{\mathrm{T}} f(x, u) + \lambda L(x, u).$$

It is the Hamiltonian but with an extra parameter λ, and this parameter is either zero or one,

$$\lambda \in \{0, 1\}.$$

Observe that $H_1(x, p, u)$ is the "normal" Hamiltonian, while $H_0(x, p, u)$ completely neglects the running cost $L(x, u)$. Optimal control problems where we need H_0 are referred to as "abnormal" problems, indicating that they are not likely to happen. With this modified Hamiltonian, the minimum principle (Theorem 2.5.1) generalizes as follows.

Theorem 2.6.1 (Minimum principle for constrained final state). Consider (2.25) with standard cost (2.26), and assume that $f(x, u)$ and $\partial f(x, u)/\partial x$ and $L(x, u)$ and $\partial L(x, u)/\partial x$ are all continuous in x and u, and that $K(x)$ and $\partial K(x)/\partial x$ are continuous in x.

Suppose $\mathtt{u}_* : [0, T] \to \mathbb{U}$ is a solution of the optimal control problem, and assume it is piecewise continuous, and let $\mathtt{x}_* : [0, T] \to \mathbb{R}^n$ be the resulting opti-

mal state. Then there is a function $\mathtt{p}_* : [0,T] \to \mathbb{R}^n$ and a constant $\lambda \in \{0,1\}$ such that $(\lambda, \mathtt{p}_*(t)) \neq (0,0)$ for all $t \in [0,T]$, and

$$\dot{\mathtt{x}}_*(t) = \frac{\partial H_\lambda(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t))}{\partial p}, \qquad \mathtt{x}_*(0) = x_0, \quad \mathtt{x}_{*i}(T) = \hat{x}_i, \ i = 1, \dots, r, \tag{2.27a}$$

$$\dot{\mathtt{p}}_*(t) = -\frac{\partial H_\lambda(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t))}{\partial x}, \quad \mathtt{p}_{*i}(T) = \frac{\partial K(\mathtt{x}_*(T))}{\partial x_i}, \ i = r+1, \dots, n, \tag{2.27b}$$

and along the solution $\mathtt{x}_*(t), \mathtt{p}_*(t)$, the input $\mathtt{u}_*(t)$ minimizes the modified Hamiltonian,

$$H_\lambda(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t)) = \min_{u \in \mathbb{U}} H_\lambda(\mathtt{x}_*(t), \mathtt{p}_*(t), u), \tag{2.28}$$

at every $t \in [0,T]$ where $\mathtt{u}_*(t)$ is continuous. □

Example 2.6.2 (Singular optimal control—an abnormal case). Consider the system with given initial and final states

$$\dot{\mathtt{x}}(t) = \mathtt{u}^2(t), \qquad \mathtt{x}(0) = 0, \qquad \mathtt{x}(1) = 0,$$

and with $\mathbb{U} = \mathbb{R}$ and cost

$$J(\mathtt{u}) = \int_0^1 \mathtt{u}(t)\, \mathrm{d}t.$$

As mentioned before, the only feasible control is the zero function. So the minimal cost is 0, and $\mathtt{x}_*(t) = \mathtt{u}_*(t) = 0$ for all time. The modified Hamiltonian is

$$H_\lambda(x, p, u) = pu^2 + \lambda u, \qquad \lambda \in \{0,1\}.$$

If we try to solve the normal Hamiltonian equations (2.27a, 2.28) (so for $\lambda = 1$), we find that the costate is constant and that $\mathtt{u}_*(t)$ at every t minimizes $\mathtt{p}_*(t)\mathtt{u}^2(t) + \mathtt{u}(t)$. But the true optimal control is $\mathtt{u}_*(t) = 0$ and this does not minimize $\mathtt{p}_*(t)\mathtt{u}^2(t) + \mathtt{u}(t)$.

If we take $\lambda = 0$ (the abnormal case), then the Hamiltonian simplifies to $H_0(x, p, u) = pu^2$. This again implies that the costate is constant, $\mathtt{p}_*(t) = \hat{p}$. The input $\mathtt{u}(t)$ that minimizes the Hamiltonian $\hat{p}\mathtt{u}^2(t)$ now is either not defined (if $\hat{p} < 0$), or is non-unique (if $\hat{p} = 0$), or equals zero (if $\hat{p} > 0$). This last case (the zero input) is the true optimal input. □

One more abnormal case is discussed in Exercise 2.15. All other examples in this chapter are normal.

Example 2.6.3 (Shortest path—a normal case). In the previous chapter (Example 1.1.4 and Example 1.2.5), we solved the (trivial) shortest path problem by

formulating it as an example of the simplest problem in the calculus of variations. We now formulate it as an optimal control problem with final condition. Let $\mathtt{x} : [0, T] \to \mathbb{R}$ be a function through the points $\mathtt{x}(0) = a$ and $\mathtt{x}(T) = b$, and assume $T > 0$. The length of the curve of the function is

$$\tilde{J}(\mathtt{x}) := \int_0^T \sqrt{1 + \dot{\mathtt{x}}^2(t)}\, \mathrm{d}t.$$

We want to minimize $\tilde{J}(\mathtt{x})$. This can be seen as an optimal control problem for the system

$$\dot{\mathtt{x}}(t) = \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0, \qquad \mathtt{x}(T) = x_T,$$

with cost

$$J(\mathtt{u}) = \int_0^T \sqrt{1 + \mathtt{u}^2(t)}\, \mathrm{d}t.$$

Its normal Hamiltonian is

$$H_1(x, p, u) = pu + \sqrt{1 + u^2}.$$

If we apply Theorem 2.6.1, we find that $\mathtt{p}_*(t)$ is constant. We denote this constant as $\hat{p}$. Since $\mathtt{u}_*(t)$ minimizes the Hamiltonian, we necessarily have that

$$0 = \frac{\partial H(\mathtt{x}_*(t), \hat{p}, \mathtt{u}_*(t))}{\partial u} = \hat{p} + \frac{\mathtt{u}_*(t)}{\sqrt{1 + \mathtt{u}_*^2(t)}}, \tag{2.29}$$

whenever $\mathtt{u}_*(t)$ is finite. After some rearrangements, this yields the following candidates for the optimal input (verify this yourself):

$$\mathtt{u}_*(t) = \begin{cases} -\infty & \text{if } \hat{p} \geq 1 \\ \frac{-\hat{p}}{\sqrt{1-\hat{p}^2}} & \text{if } -1 < \hat{p} < 1 \\ \infty & \text{if } \hat{p} \leq -1 \end{cases}.$$

We can strike off the first and the last candidates, because they clearly fail to achieve the final condition $\mathtt{x}(T) = x_T$. The second candidate says that $\mathtt{u}_*(t)$ is some constant. But for a constant input $\hat{u} := \mathtt{u}_*(t)$, the solution $\mathtt{x}(t)$ of the differential equation is $\mathtt{x}(t) = \hat{u}t + x_0$, which is a straight line. From the initial and final conditions, it follows that $\hat{u} = (x_T - x_0)/T$. Hence, as expected,

$$\mathtt{x}_*(t) = x_0 + \frac{x_T - x_0}{T}\, t, \qquad \mathtt{u}_*(t) = \frac{x_T - x_0}{T}.$$

The constant costate then follows from (2.29),

$$\mathtt{p}_*(t) = \hat{p} = \frac{-\mathtt{u}_*(t)}{\sqrt{1 + \mathtt{u}_*^2(t)}} = \frac{-(x_T - x_0)}{\sqrt{T^2 + (x_T - x_0)^2}}.$$

It is interesting to compare this with the optimal cost (the minimal length of the curve)

$$J(\mathrm{u}_*) = \sqrt{T^2 + (x_T - x_0)^2}.$$

We see that $\mathrm{p}_*(0)$ equals $\frac{\mathrm{d}J(\mathrm{u}_*)}{\mathrm{d}x_0}$. That is, $\mathrm{p}_*(0)$ expresses how strongly the optimal cost changes if x_0 changes. We return to this sensitivity property of the costate in § 3.5. □

Example 2.6.4 (Integrator system with fixed initial and final states). Consider the system with bounded derivative,

$$\dot{\mathrm{x}}(t) = \mathrm{u}(t), \qquad \mathrm{u}(t) \in [-1, 1],$$

and with cost

$$J(\mathrm{u}) = \int_0^T \mathrm{x}(t)\,\mathrm{d}t.$$

In Example 2.5.2, we analyzed the same system and cost (for $T = 1$), but now we fix both the initial and final states,

$$\mathrm{x}(0) = \mathrm{x}(T) = 0.$$

To minimize the cost we want $\mathrm{x}(t)$ as small (negative) as possible, yet it needs to start at zero, $\mathrm{x}(0) = 0$, and needs to end at zero, $\mathrm{x}(T) = 0$. The normal Hamiltonian is

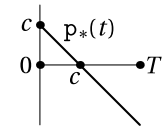
$$H_1(x, p, u) = pu + x,$$

and therefore the costate equations become

$$\dot{\mathrm{p}}_*(t) = -1.$$

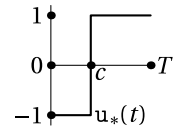
Notice that since the state is fixed at the final time, $\mathrm{x}(T) = 0$, there is no condition on the costate at the final time. So all we know, for now, about the costate is that its derivative is -1, i.e.,

$$\mathrm{p}_*(t) = c - t$$



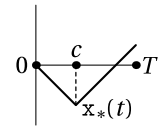
for some as yet unknown constant c. Given this $\mathrm{p}_*(t) = c - t$, the minimizer $\mathrm{u}_*(t)$ of the Hamiltonian is

$$\mathrm{u}_*(t) = \begin{cases} -1 & \text{if } t < c, \\ +1 & \text{if } t > c. \end{cases}$$



This function switches sign (from negative to positive) at $t = c$, and, as a result, the state $\mathrm{x}_*(t)$ is piecewise linear. First, it goes down, and from $t = c$ on it goes up,

$$\mathrm{x}_*(t) = \begin{cases} -t & \text{if } t < c, \\ -2c + t & \text{if } t > c. \end{cases}$$



It will be clear that the only value of c for which $\mathrm{x}_*(T)$ is zero, is

$$c = T/2.$$

This completely settles the optimal control problem. In the first half, $[0, T/2]$, we have $\dot{\mathrm{x}}_*(t) = -1$ and, in the second half, $[T/2, T]$, we have $\dot{\mathrm{x}}_*(t) = +1$. The optimal cost is $J(\mathrm{u}_*) = \int_0^T \mathrm{x}_*(t)\,\mathrm{d}t = -T^2/4$. □

2.7 Free Final Time

So far, the final time T in the optimal control problem was fixed. Now we extend the optimal control problem by minimizing the cost over all inputs as well as over all final times $T \geq 0$. As before we assume a cost of the form

$$J_T(\mathrm{u}) := \int_0^T L(\mathrm{x}(t), \mathrm{u}(t))\,\mathrm{d}t + K(\mathrm{x}(T)).$$

Since we now have one extra degree of freedom, we expect that the minimum principle still holds but with one extra condition. This turns out to be true, and the extra condition is quite elegant:

Theorem 2.7.1 (Minimum principle with free final time). Consider the system (2.25) with cost (2.26), and assume that $f(x,u)$ and $\partial f(x,u)/\partial x$ and $L(x,u)$ and $\partial L(x,u)/\partial x$ are continuous in x and u, and that $K(x)$ and $\partial K(x)/\partial x$ are continuous in x.

Suppose (u_*, T_*) is a solution of the optimal control problem with free final time, and that u_* is piecewise continuous on $[0, T_*]$, and that $0 \leq T_* < \infty$. Then all conditions of Theorem 2.6.1 hold (with $T = T_*$), and, in addition,

$$H_\lambda(\mathrm{x}_*(T_*), \mathrm{p}_*(T_*), \mathrm{u}_*(T_*)) = 0. \tag{2.30}$$

Proof. We prove it only for the normal case ($\lambda = 1$). If the pair (u_*, T_*) is optimal, then u_* is also optimal for the *fixed* final time $T = T_*$; hence, all conditions of Theorem 2.6.1 hold.

Since u_* is assumed to be piecewise continuous, the limit $u_{T_*} := \lim_{t \uparrow T_*} \mathrm{u}_*(t)$ exists. The given u_* is defined on $[0, T_*]$, and we now extend its definition by letting $\mathrm{u}_*(t) = u_{T_*}$ for all $t \geq T_*$. That way u_* is continuous at $T = T_*$, and the

cost becomes differentiable with respect to T at $T = T_*$. By the fact that $T = T_*$ is time-optimal, we have that

$$\frac{\mathrm{d}J(\mathtt{u}_*, T_*)}{\mathrm{d}T} = 0.$$

This derivative equals

$$\begin{aligned}\frac{\mathrm{d}J(\mathtt{u}_*, T_*)}{\mathrm{d}T} &= \frac{\partial K(\mathtt{x}_*(T_*))}{\partial x^{\mathrm{T}}}\dot{\mathtt{x}}_*(T_*) + L(\mathtt{x}_*(T_*), \mathtt{u}_*(T_*)) \\ &= \mathtt{p}_*^{\mathrm{T}}(T_*) f(\mathtt{x}_*(T_*), \mathtt{u}_*(T_*)) + L(\mathtt{x}_*(T_*), \mathtt{u}_*(T_*)) \\ &= H_1(\mathtt{x}_*(T_*), \mathtt{p}_*(T_*), \mathtt{u}_*(T_*)).\end{aligned}$$

■

The remarks about λ made in the previous section also apply to this situation. Also the constancy property of the Hamiltonian (Theorem 2.5.6) remains. This is interesting because it shows that for final time problems the Hamiltonian $H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t))$ is actually zero for all time!

An important special case is when $L(x,u) = 1$ and $K(x) = 0$. Then the cost function equals $J_T(\mathtt{u}) = \int_0^T 1\,\mathrm{d}t = T$, that is, the task of the control input is to realize given boundary conditions in minimal time. This only makes sense if we have both initial and final conditions. Such problems are known as *time-optimal control* problems. A classic time-optimal control problem is the *problem of Zermelo*.

Example 2.7.2 (Zermelo). Consider a boat on a river as depicted in Fig. 2.4. The problem of Zermelo is to steer the boat in minimal time from a given point on one side of the river to a given point on the other side. The coordinates of the boat are denoted by x_1, x_2. The boat starts at $(\mathtt{x}_1(0), \mathtt{x}_2(0)) = (0,0)$, and the destination point is $(\mathtt{x}_1(T), \mathtt{x}_2(T)) = (a,b)$ for some given a, b, where b is the width of the river. The complicating factor is the flow velocity of the water in the river. We assume the flow velocity to be parallel to the river banks, and to be of the form $w(x_2)$, where x_2 is the distance to one of the river banks, see Fig. 2.4. We assume that the speed of the boat with respect to the water is constant and equal to 1. The control u of the boat is the angle between the boat's principal axis and the x_1-axis, see Fig. 2.4. This leads to the following equations of motion:

$$\begin{aligned}&\dot{\mathtt{x}}_1(t) = \cos(\mathtt{u}(t)) + w(\mathtt{x}_2(t)), && \mathtt{x}_1(0) = 0, \quad \mathtt{x}_1(T) = a, \\ &\dot{\mathtt{x}}_2(t) = \sin(\mathtt{u}(t)), && \mathtt{x}_2(0) = 0, \quad \mathtt{x}_2(T) = b. \end{aligned} \tag{2.31}$$

As we want to cross the river in minimal time, the cost to be minimized is

$$J_T(\mathtt{u}) = \int_0^T 1\,\mathrm{d}t = T.$$

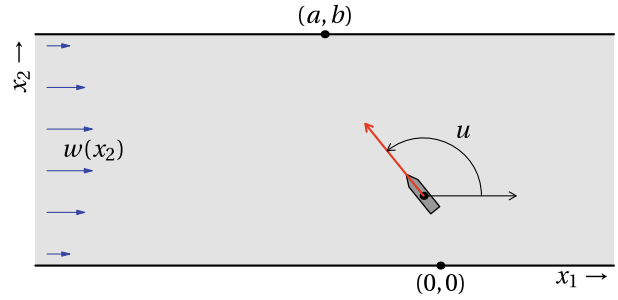


FIGURE 2.4: The problem of Zermelo. See Example 2.7.2.

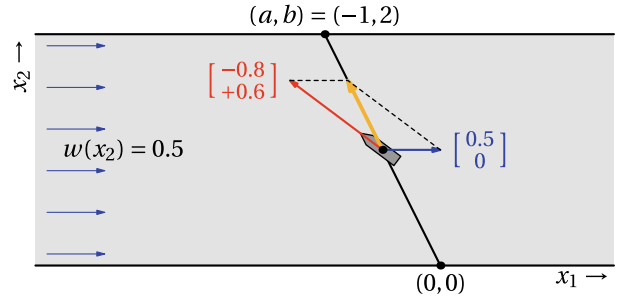


FIGURE 2.5: The problem of Zermelo. We assume that the speed of the boat with respect to the water is $v = 1$ and that the flow velocity is $w(x_2) = 0.5$, and $(a, b) = (-1, 2)$. Then optimal is to take u constant such that $\begin{bmatrix}\cos(u)\\ \sin(u)\end{bmatrix} = \begin{bmatrix}-0.8\\ +0.6\end{bmatrix}$ (shown in red), for then the sum of $\begin{bmatrix}\cos(u)\\ \sin(u)\end{bmatrix}$ and $\begin{bmatrix}0.5\\ 0\end{bmatrix}$ (shown in blue) is $\begin{bmatrix}-0.3\\ 0.6\end{bmatrix}$ (shown in yellow), and this direction brings the boat to $(a, b) = (-1, 2)$ as required. It takes $T = b/0.6 = 3\frac{1}{3}$ units of time. See Example 2.7.2.

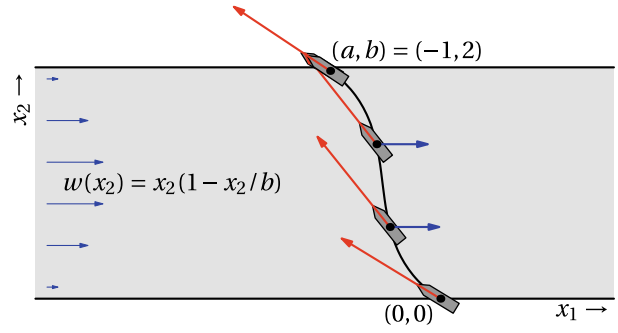


FIGURE 2.6: The problem of Zermelo. We assume that the speed of the boat with respect to the water is $v = 1$ and that the flow velocity is $w(x_2) = x_2(1 - x_2/b)$ with $(a, b) = (-1, 2)$. The optimal trajectory $\mathtt{x}_{*1}(t), \mathtt{x}_{*2}(t)$ as shown here was computed by iterating over u_0 in (2.36). The optimal value is $u_0 = 2.5937$, and the optimal (minimal) time is $T = 2.7570$. The optimal angle $\mathtt{u}_*(t)$ of the boat does not vary much with time. See Example 2.7.2.

With this cost, the normal Hamiltonian is given by

$$H_1(x,p,u) = p_1\cos(u) + p_1 w(x_2) + p_2\sin(u) + 1.$$

The optimal control $\mathtt{u}_*$ minimizes the Hamiltonian, so we need $\frac{\partial H_1(\mathtt{x}_*,\mathtt{p}_*,\mathtt{u}_*)}{\partial u}$ to be equal to zero for all time. This gives

$$-\mathtt{p}_{*1}\sin(\mathtt{u}_*) + \mathtt{p}_{*2}\cos(\mathtt{u}_*) = 0. \tag{2.32}$$

(To avoid clutter, we drop the argument t in most of the equations.) Also, the free final time condition (2.30) has to hold, and this means that

$$\mathtt{p}_{*1}\cos(\mathtt{u}_*) + p_{*1} w(\mathtt{x}_{*2}) + \mathtt{p}_{*2}\sin(\mathtt{u}_*) + 1 = 0 \tag{2.33}$$

for all time. Equations (2.32) and (2.33) are two *linear* equations in $\mathtt{p}_{*1}, \mathtt{p}_{*2}$, so is easy to solve as

$$\begin{bmatrix}\mathtt{p}_{*1}\\ \mathtt{p}_{*2}\end{bmatrix} = \frac{-1}{\cos(\mathtt{u}_*)w(\mathtt{x}_{*2})+1}\begin{bmatrix}\cos(\mathtt{u}_*)\\ \sin(\mathtt{u}_*)\end{bmatrix}. \tag{2.34}$$

Incidentally, since we assumed the speed of the boat to be 1, we can not expect to be able to reach every destination point (a,b) if the flow velocity $w(x_2)$ exceeds 1 everywhere. We assume from now on that $|w(x_2)| < 1$ for every $x_2 \in [0,b]$ even though this assumption is stronger than needed for the problem to have a solution. With this assumption, the inverse in (2.34) is guaranteed to exist.

We have not yet exploited the costate equations. From the Hamiltonian, we readily get the costate equations

$$\begin{aligned}\dot{\mathtt{p}}_{*1} &= 0,\\ \dot{\mathtt{p}}_{*2} &= -\mathtt{p}_{*1} w'(\mathtt{x}_{*2}),\end{aligned} \tag{2.35}$$

in which w' is the derivative of w. Notice that we do not have final conditions on the costate. Interestingly, $\dot{\mathtt{p}}_{*1}$ is zero. Using the formula for $\mathtt{p}_{*1}$ in (2.34), we find that this derivative is

$$\dot{\mathtt{p}}_{*1} = \frac{\sin(\mathtt{u}_*)(\dot{\mathtt{u}}_* + \cos^2(\mathtt{u}_*)w'(\mathtt{x}_{*2}))}{(\cos(\mathtt{u}_*)w(\mathtt{x}_{*2})+1)^2}.$$

(Verify this yourself.) This needs to be zero for all time, so either $\sin(\mathtt{u}_*)$ is zero or $\dot{\mathtt{u}}_* = -\cos^2(\mathtt{u}_*)w'(\mathtt{x}_{*2})$. Likewise it can be shown that the costate equation for $\mathtt{p}_{*2}$ holds iff $\dot{\mathtt{u}}_* = -\cos^2(\mathtt{u}_*)w'(\mathtt{x}_{*2})$ or $\cos(\mathtt{u}_*) + w(\mathtt{x}_{*2}) = 0$. Since $\sin(\mathtt{u}_*)$ and $\cos(\mathtt{u}_*) + w(\mathtt{x}_{*2})$ cannot be zero simultaneously (because we assumed $|w(x_2)| < 1$), we conclude that both costate equations hold iff

$$\dot{\mathtt{u}}_* = -\cos^2(\mathtt{u}_*)w'(\mathtt{x}_{*2}).$$

Summarizing, we have the following three coupled differential equations:

$$\dot{\mathtt{x}}_{*1}(t) = \cos(\mathtt{u}_*(t)) + w(\mathtt{x}_{*2}(t)), \qquad \mathtt{x}_{*1}(0) = 0, \tag{2.36a}$$

$$\dot{\mathtt{x}}_{*2}(t) = \sin(\mathtt{u}_*(t)), \qquad \mathtt{x}_{*2}(0) = 0, \tag{2.36b}$$

$$\dot{\mathtt{u}}_*(t) = -\cos^2(\mathtt{u}_*(t))\, w'(\mathtt{x}_{*2}(t)), \qquad \mathtt{u}_*(0) = u_0, \tag{2.36c}$$

and its solution by construction makes the costate defined in (2.34) satisfy the Hamiltonian equations and makes the Hamiltonian equal to zero for all time. The game is now to determine the initial condition u_0 of the control for which $(\mathtt{x}_1(T), \mathtt{x}_2(T))$ equals (a, b) for some $T > 0$. Without further assumptions on the flow velocity $w(x_2)$, there does not seem to be an easy answer to this problem. For the special case of a constant flow velocity, $w(x_2) = w_0$, however, we see that $\mathtt{u}_*(t)$ is constant, and then $(\mathtt{x}_{*1}(t), \mathtt{x}_{*2}(t))$ is a straight line. A particular instance is shown in Fig. 2.5. A more realistic scenario is when the flow velocity $w(x_2)$ is small near the banks. One such example is depicted in Fig. 2.6. The solution shown in this figure was determined numerically (by iterating over u_0). □

Example 2.7.3 (Minimal time car parking problem). This is an elegant and classic application. We want to steer a car into a parking spot, and we want to do it in minimal time. To keep things manageable, we assume that we can steer the car in one dimension only (like a cart on a rail). The position of the car is denoted as x_1 and its speed as x_2. The acceleration u is bounded, specifically, $\mathtt{u}(t) \in [-1, 1]$ for all t. The equations thus are

$$\dot{\mathtt{x}}_1(t) = \mathtt{x}_2(t), \qquad \mathtt{x}_1(0) = x_{01},$$

$$\dot{\mathtt{x}}_2(t) = \mathtt{u}(t), \qquad \mathtt{x}_2(0) = x_{02}, \qquad \mathtt{u}(t) \in [-1, 1].$$

The parking spot we take to be $x_1 = 0$, and the time we reach the parking spot we denote by T, and, of course, at that moment our speed should become zero. So the final conditions are

$$\mathtt{x}_1(T) = 0, \qquad \mathtt{x}_2(T) = 0.$$

We want to achieve this in minimal time, thus we take as cost $J_T(\mathtt{u}) = \int_0^T 1 \, \mathrm{d}t$.

The normal Hamiltonian for this problem is

$$H_1(x, p, u) = p_1 x_2 + p_2 u + 1.$$

From the Hamiltonian, the costate equations follow as

$$\dot{\mathtt{p}}_1(t) = 0,$$

$$\dot{\mathtt{p}}_2(t) = -\mathtt{p}_1(t).$$

Since both components of the final state $\mathtt{x}(T)$ are fixed, the final conditions on both components of the costate are absent. Therefore, in principle, every constant $\mathtt{p}_1(t) = -a$ is allowed and, consequently, every linear function

$$\mathtt{p}_2(t) = at + b.$$

We can not have $a = b = 0$ because that contradicts the fact that the Hamiltonian $\mathtt{p}_1(t)\mathtt{x}_2(t) + \mathtt{p}_2(t)\mathtt{u}(t) + 1$ is zero along optimal solutions (Theorem 2.7.1). As a result, the second costate entry, $\mathtt{p}_2(t)$, is not the zero function. This, in turn, implies that $\mathtt{p}_2(t)$ switches sign at most once. Why is this important? Well, the optimal $\mathtt{u}_*(t)$ minimizes the Hamiltonian, $\mathtt{p}_1(t)\mathtt{x}_2(t) + \mathtt{p}_2(t)\mathtt{u}(t) + 1$, and since $\mathtt{u}_*(t) \in [-1, 1]$ this yields

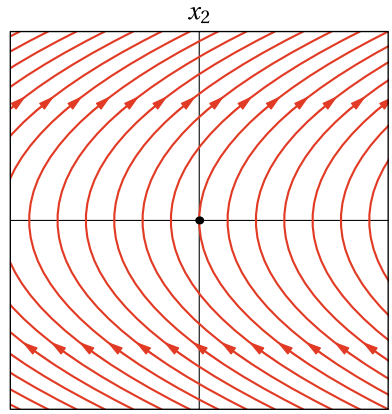
$$\mathtt{u}_*(t) = -\operatorname{sgn}(\mathtt{p}_2(t)).$$

This is well defined because $\mathtt{p}_2(t)$ is nontrivial. In fact, as $\mathtt{p}_2(t)$ switches sign at most once, also

$\mathtt{u}_*(t)$ switches sign at most once.

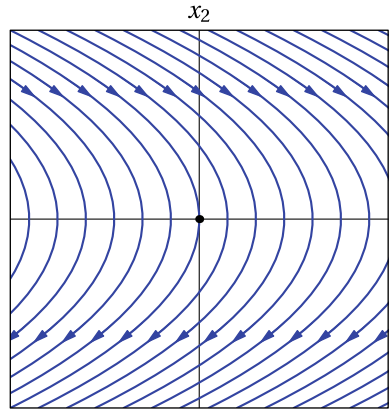
Let t_s be the time of switching. Then the input for $t > t_s$ by definition does not switch any more and so is either $+1$ throughout or -1 throughout. Now for $\mathtt{u}(t) = +1$, the system equations become $\dot{\mathtt{x}}_2(t) = 1, \dot{\mathtt{x}}_1(t) = \mathtt{x}_2(t)$, so $\mathtt{x}_2(t) = t + c$ and $\mathtt{x}_1(t) = \frac{1}{2}(t + c)^2 + d = \frac{1}{2}\mathtt{x}_2^2(t) + d$. Hence, the trajectories $(\mathtt{x}_1(t), \mathtt{x}_2(t))$ are shifted parabolas, shown here in red:

$\mathtt{u}(t) = +1:$

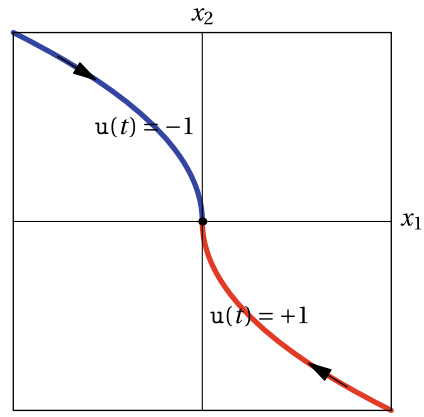


(The arrows indicate the direction of the trajectory as time increases.) Likewise, if $\mathtt{u}(t) = -1$, then all possible $(\mathtt{x}_1(t), \mathtt{x}_2(t))$ are the shifted "reversed" parabolas, shown here in blue:

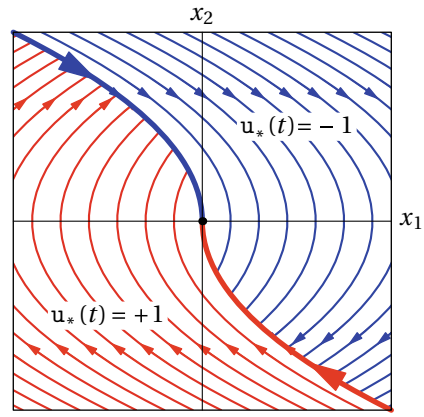
$\mathtt{u}(t) = -1:$



Since on $(t_s, T]$ the input does not change, and since we demand that $\mathrm{x}(T) = (0,0)$, it must be that on $(t_s, T]$ the state either follows this red or blue parabola:



After all, these two are the only two trajectories that end up at the desired final state $\mathrm{x}(T) = (0,0)$. Before the moment of switching, the input $\mathrm{u}(t)$ had the opposite sign. For instance, if after the switch we have $\mathrm{u}(t) = +1$ (the red trajectory), then before the switch we have $\mathrm{u}(t) = -1$, i.e., any of the blue parabolas. These have to end up at the above red parabola at $t = t_s$. Inspection shows that the possible trajectories are any of these:



This solves the problem for every initial state $(\mathrm{x}_1(0), \mathrm{x}_2(0))$. If before the switch the trajectory follows a blue parabola, then, when it reaches the thick red parabola, the inputs switches sign, and the trajectory continues along the thick red parabola, ending up at $(0,0)$. Likewise, if it first follows a red parabola then, when it reaches the thick blue parabola, the input switches sign, and the trajectory continues along the thick blue parabola, ending up at $(0,0)$. □

2.8 Convexity and the Minimum Principle

The minimum principle *assumes* the existence of an optimal control, and then derives some conditions for it: (2.14) and (2.15). These conditions are necessary for optimality, but in general not sufficient (see Exercise 2.16). If, however, the problem has certain convexity properties then the necessary conditions *are* sufficient. That is what the following theorem is about. It requires some knowledge of convex sets and functions as discussed in Appendix A.7.

Theorem 2.8.1 (Mangasarian). Consider the optimal control problem defined by (2.1), (2.2), (2.3), and assume that $f(x,u)$, $L(x,u)$, and $K(x)$ are C^1. Suppose $(\mathtt{x}_*, \mathtt{p}_*, \mathtt{u}_*)$ are piecewise continuous functions that satisfy (2.14) and (2.15), and that

- $\mathbb{U}$ is a convex set,
- $H(x, \mathtt{p}_*(t), u)$ for every $t \in [0,T]$ is convex in $(x,u) \in (\mathbb{R}^n, \mathbb{U})$,
- $K(x)$ is convex in $x \in \mathbb{R}^n$.

Then $(\mathtt{x}_*, \mathtt{p}_*, \mathtt{u}_*)$ is an optimal triple; in particular, $\mathtt{u}_*$ is an optimal control.

Proof. In order not to digress too much, we allow ourselves here some "proofs by picture". Details can be found in Appendix A.7.

Convexity of the Hamiltonian in (x,u) means that

$$H(x, \mathtt{p}_*(t), u) \geq H(\bar{x}, \mathtt{p}_*(t), \bar{u}) + \frac{\partial H(\bar{x}, \mathtt{p}_*(t), \bar{u})}{\partial x^{\mathrm{T}}}(x - \bar{x}) + \frac{\partial H(\bar{x}, \mathtt{p}_*(t), \bar{u})}{\partial u^{\mathrm{T}}}(u - \bar{u})$$

for all $x, \bar{x} \in \mathbb{R}^n$, $u, \bar{u} \in \mathbb{U}$. This is illustrated in Fig. 2.7 (left). By convexity of $\mathbb{U}$ and the fact that $\mathtt{u}_*(t)$ minimizes the Hamiltonian for almost all times, we have that

$$\frac{\partial H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t))}{\partial u^{\mathrm{T}}}(u - \mathtt{u}_*(t)) \geq 0 \quad \forall u \in \mathbb{U}$$

for almost all times. This property is illustrated in Fig. 2.7 (right). The above two inequalities, combined with the fact that $\dot{\mathtt{p}}_*(t) = -\frac{\partial H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t))}{\partial x}$, shows that

$$H(x, \mathtt{p}_*(t), u) \geq H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t)) - \dot{\mathtt{p}}_*^{\mathrm{T}}(t)(x - \mathtt{x}_*(t))$$

for all $x \in \mathbb{R}^n$, $u \in \mathbb{U}$, for almost all times. For simplicity, we now assume that the terminal cost is absent, $K = 0$. (Exercise 2.19 considers nonzero K.) The final inequality gives us that

$$
\begin{aligned}
J(x_0,\mathtt{u}) - J(x_0,\mathtt{u}_*) &= \int_0^T L(\mathtt{x},\mathtt{u})\,\mathrm{d}t - L(\mathtt{x}_*,\mathtt{u}_*)\,\mathrm{d}t \\
&= \int_0^T \left(H(\mathtt{x},\mathtt{p}_*,\mathtt{u}) - \mathtt{p}_*^{\mathrm{T}}\dot{\mathtt{x}}\right) - \left(H(\mathtt{x}_*,\mathtt{p}_*,\mathtt{u}_*) - \mathtt{p}_*^{\mathrm{T}}\dot{\mathtt{x}}_*\right)\mathrm{d}t \\
&= \int_0^T H(\mathtt{x},\mathtt{p}_*,\mathtt{u}) - H(\mathtt{x}_*,\mathtt{p}_*,\mathtt{u}_*) - \mathtt{p}_*^{\mathrm{T}}(\dot{\mathtt{x}} - \dot{\mathtt{x}}_*)\,\mathrm{d}t \\
&\geq \int_0^T -\dot{\mathtt{p}}_*^{\mathrm{T}}(\mathtt{x} - \mathtt{x}_*) - \mathtt{p}_*^{\mathrm{T}}(\dot{\mathtt{x}} - \dot{\mathtt{x}}_*)\,\mathrm{d}t \\
&= \left[-\mathtt{p}_*^{\mathrm{T}}(t)(\mathtt{x}(t) - \mathtt{x}_*(t))\right]_0^T = 0.
\end{aligned}
$$

(In the last equality, we used that $\mathtt{p}_*(T) = 0$ and that $\mathtt{x}(0) = \mathtt{x}_*(0) = x_0$.) Therefore $J(x_0,\mathtt{u}) \geq J(x_0,\mathtt{u}_*)$. So $\mathtt{u}_*$ is optimal. ■

Many of the examples considered in the chapter satisfy the above convexity properties, see Exercises 2.20 and 2.21 for illustrations.

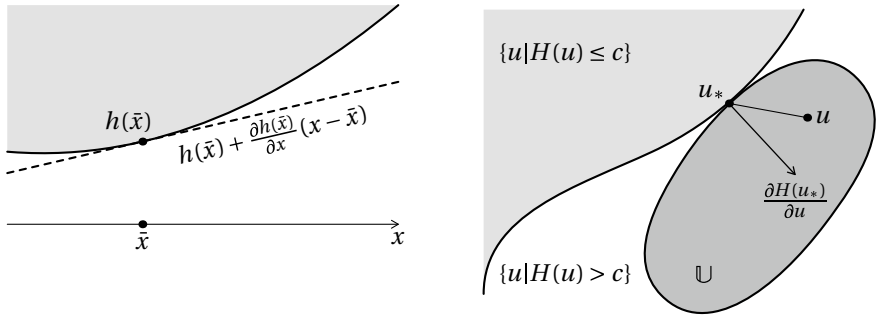


FIGURE 2.7: Left: a C^1 function $h : \mathbb{R} \to \mathbb{R}$ is convex iff $h(x) \geq h(\bar{x}) + \frac{\partial h(\bar{x})}{\partial x}(x - \bar{x})\,\forall \bar{x}, x \in \mathbb{R}$. Right: Suppose $H : \mathbb{R}^2 \to \mathbb{R}$ is C^1 and that $\mathbb{U} \subseteq \mathbb{R}^2$. If $H(u_*) = \min_{u\in\mathbb{U}} H(u)$ for some $u_* \in \mathbb{U}$, and $\mathbb{U}$ is convex, then $\frac{\partial H(u_*)}{\partial u^{\mathrm{T}}}(u - u_*) \geq 0\,\forall u \in \mathbb{U}$. This is used in the proof of Theorem 2.8.1. Appendix A.7 has more details.

2.9 Exercises

2.1 Consider the scalar system

$$\dot{\mathtt{x}}(t) = \mathtt{x}(t)\mathtt{u}(t), \qquad \mathtt{x}(0) = x_0 = 1,$$

with $\mathbb{U} = \mathbb{R}$ and cost function $J(\mathtt{u}) = 2\mathtt{x}(T) + \int_0^T \mathtt{x}^2(t) + \mathtt{u}^2(t)\,\mathrm{d}t$.

(a) Determine the Hamiltonian $H(x,p,u)$ and the differential equation for the costate.

(b) Determine the optimal input $\mathtt{u}_*(t)$ as a function of $\mathtt{x}_*(t), \mathtt{p}_*(t)$.

(c) Show that $H(\mathtt{x}_*(T), \mathtt{p}_*(T), \mathtt{u}_*(T))$ is zero.

(d) Show that $\mathtt{p}_*(t) = 2$ (the constant function 2) satisfies the costate equation.

(e) Express the optimal $\mathtt{u}_*(t)$ as a function of $\mathtt{x}_*(t)$, and then determine $\mathtt{x}_*(t)$. [Hint: see Example B.1.5.]

In the next chapter, we analyze this problem from another perspective (Exercise 3.3).

2.2 Consider the system

$$\dot{\mathtt{x}}(t) = \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0,$$

and cost function $J(\mathtt{u}) = \int_0^T \mathtt{x}^2(t)\,\mathrm{d}t$. We want to minimize this cost over all $\mathtt{u} : [0,T] \to [0,1]$.

(a) Give the Hamiltonian and the differential equation for the costate.

(b) Argue from the Hamiltonian that $\mathtt{u}_*(t)$ most likely assumes just one or two values.

(c) Argue that if $x_0 > 0$, then $\mathtt{x}_*(t) > 0$ for all $t \in [0,T]$.

(d) Prove that $\mathtt{p}_*(t)$ under the conditions stated in (c) has at most one sign change. What does this mean for $\mathtt{u}_*(t)$?

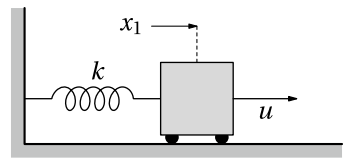
(e) Solve the optimization problem for $x_0 > 0$. Also give the solution for $\mathtt{p}_*(t)$.

(f) Determine an optimal input for the case that $x_0 < 0$, and verify that it satisfies the Hamiltonian equations (2.14), (2.15). [Hint: directly formulate an optimal input $\mathtt{u}_*$ without using the minimum principle.]

2.3 Consider the scalar system $\dot{\mathtt{x}}(t) = \mathtt{x}(t) + \mathtt{u}(t)$ with initial condition $\mathtt{x}(0) = x_0$. Determine the input $\mathtt{u} : [0,1] \to [0,4]$ that minimizes $\int_0^1 \frac{1}{2}\mathtt{u}^2(t) - 2\mathtt{u}(t) - 2\mathtt{x}(t)\,\mathrm{d}t$ over all functions $\mathtt{u} : [0,1] \to [0,4]$.

2.4 *Optimal potential energy.* Consider

$$\dot{\mathtt{x}}_1(t) = \mathtt{x}_2(t),$$
$$\dot{\mathtt{x}}_2(t) = -k\mathtt{x}_1(t) + \mathtt{u}(t).$$



This is the standard linear model for a point mass of mass $m = 1$, displacement from the equilibrium x_1, velocity x_2, and which is subject to an external force u and a spring force with spring constant $k > 0$.

We assume that $|\mathtt{u}(t)| \le 1$ for all t.

(a) Show that without external force $\mathtt{u}$, the mass follows a harmonic motion with period $T = 2\pi/\sqrt{k}$.

The system has to be controlled in such a way that after a time $T = 2\pi/\sqrt{k}$ the potential energy $\frac{1}{2}k\mathtt{x}_1^2(T)$ of the mass is maximal.

(b) Formulate the associated optimal control problem. [Hint: use $L(x,u) = 0$.]

(c) Derive the equations for the costate, including final conditions. Show that the optimal control only depends on the second component $\mathtt{p}_2$ of the costate. How are the control and $\mathtt{p}_2$ connected?

(d) Derive, by elimination of the first component $\mathtt{p}_1$, a differential equation for $\mathtt{p}_2$, including final condition. Determine all possible solutions $\mathtt{p}_2$, and, from this, derive all possible optimal controls.

(e) Determine an optimal displacement $\mathtt{x}_{*1}(t)$ as a function of t for the case that $\mathtt{x}_{*1}(0) = \mathtt{x}_{*2}(0) = 0$. What is the maximal potential energy at the final time in this case?

2.5 *Initial and final conditions.* Consider $\dot{\mathtt{x}}(t) = \mathtt{x}(t) + \mathtt{u}(t)$ with an initial and a final condition, $\mathtt{x}(0) = 1, \mathtt{x}(3) = 0$, and $\mathbb{U} = \mathbb{R}$. As cost we take $\int_0^3 \frac{1}{4}\mathtt{u}^4(t)\,\mathrm{d}t$. Determine the optimal state $\mathtt{x}_*(t)$ and optimal control $\mathtt{u}_*(t)$.

2.6 *Initial and final conditions.* Let $\mathtt{u}, \mathtt{y} : [0,T] \to \mathbb{R}$. Consider the second-order differential equation

$$\ddot{\mathtt{y}}(t) + \mathtt{y}(t) = \mathtt{u}(t), \qquad \mathtt{y}(0) = y_0, \quad \dot{\mathtt{y}}(0) = \dot{y}_0,$$

with cost

$$J(\mathtt{u}) = \frac{1}{2}\int_0^T \mathtt{u}^2(t)\,\mathrm{d}t.$$

Determine the optimal control $\mathtt{u}_*$ that drives the system from the initial state $\mathtt{y}(0) = y_0$, $\dot{\mathtt{y}}(0) = \dot{y}_0$ to the final state $\mathtt{y}(T) = \dot{\mathtt{y}}(T) = 0$.

2.7 *Maximal distance.* We want to move a mass in T seconds, beginning and ending with zero speed, using bounded acceleration. With x_1 its position and x_2 its speed, a model for this problem is

$$\begin{aligned} \dot{\mathtt{x}}_1(t) &= \mathtt{x}_2(t), & \mathtt{x}_1(0) &= 0, \\ \dot{\mathtt{x}}_2(t) &= \mathtt{u}(t), & \mathtt{x}_2(0) &= 0, \qquad \mathtt{x}_2(T) = 0. \end{aligned}$$

Here $\mathtt{u}$ is the acceleration which we take to be bounded in magnitude by one, that is, $\mathtt{u}(t) \in [-1,1]$ for all t. We want to *maximize* the traveled distance $\mathtt{x}_{*1}(T)$.

(a) Determine the Hamiltonian $H(x,p,u)$.

(b) Determine the Hamiltonian equations in $\mathtt{x}(t)$ and $\mathtt{p}(t)$ as used in the minimum principle, including all initial and final conditions.

(c) Determine the general solution of the costate $\mathtt{p}(t)$ for $t \in [0, T]$.

(d) Determine the optimal input $\mathtt{u}_*(t)$ for all $t \in [0, T]$ and compute the maximal distance $\mathtt{x}_{*1}(T)$.

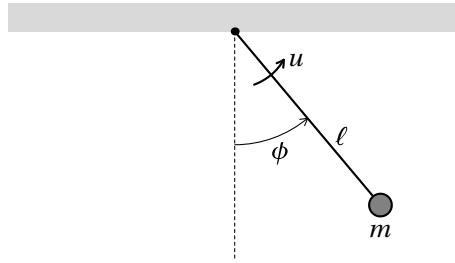


FIGURE 2.8: A pendulum with a torque u. See Exercise 2.8.

2.8 *Control of pendula via torques.* Consider a mass m hanging from a ceiling on a thin massless rod of length ℓ, see Fig. 2.8. We can control the pendulum with a torque u exerted around the suspension point. The differential equation describing the pendulum without damping is

$$m\ell^2\ddot{\phi}(t) + gm\ell\sin(\phi(t)) = \mathtt{u}(t),$$

where ϕ is the angle with respect to the stable equilibrium state (the vertical hanging position). The objective is to minimize the cost

$$J(\mathtt{u}) := m\ell^2\dot{\phi}^2(T) - 2mg\ell\cos(\phi(T)) + \int_0^T \dot{\phi}^2(t) + \mathtt{u}^2(t)\,\mathrm{d}t.$$

It is convenient to use $\mathtt{x}_1 := \phi$ and $\mathtt{x}_2 := \dot{\phi}$.

(a) Determine the state differential equation $\dot{\mathtt{x}}(t) = f(\mathtt{x}(t), \mathtt{u}(t))$.

(b) Determine the Hamiltonian $H(x, p, u)$ and the differential equation for the costate, including final conditions.

(c) Calculate $\int_0^T \dot{\phi}(t)\mathtt{u}(t)\,\mathrm{d}t$. What do you see?

(d) Express the optimal control in terms of ϕ and/or $\dot{\phi}$.

2.9 *Optimal capacitor charging.* Consider the RC-circuit of Fig. 2.9. We want to determine at any moment in time the voltage $\mathtt{u}(t)$ of the voltage source that charges the capacitor in T seconds from zero voltage, $\mathtt{x}(0) = 0$, to a certain desired voltage, $\mathtt{x}(T) = x_{\text{desired}}$, with minimal dissipation of energy through the resistor. The voltage $\mathtt{v}(t)$ across the resistor is given by Kirchhoff's voltage law as $\mathtt{v}(t) = \mathtt{u}(t) - \mathtt{x}(t)$. Hence the current $\mathtt{i}(t)$ through the resistor with resistance R equals $\mathtt{i}(t) = \frac{1}{R}(\mathtt{u}(t) - \mathtt{x}(t))$,

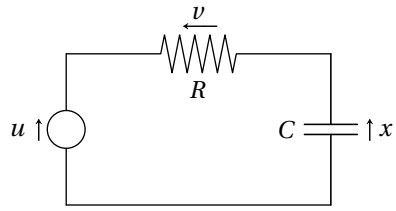


FIGURE 2.9: An RC-circuit with resistance R and capacitance C. The voltage source is denoted by u, and the voltage across the capacitor is denoted by x. See Exercise 2.9.

and thus the dynamics of the charge $\mathtt{q}(t)$ at the capacitor is given as $\dot{\mathtt{q}}(t) = \mathtt{i}(t) = \frac{1}{R}(\mathtt{u}(t) - \mathtt{x}(t))$. For a linear capacitor with capacitance C the charge $\mathtt{q}(t)$ equals $C\mathtt{x}(t)$. This leads to the model

$$\dot{\mathtt{x}}(t) = -\frac{1}{RC}\mathtt{x}(t) + \frac{1}{RC}\mathtt{u}(t). \tag{2.37}$$

Furthermore, the power dissipated in the resistor is given as $\mathtt{v}(t)\mathtt{i}(t) = \frac{1}{R}\mathtt{v}^2(t)$, and, hence, the total energy loss is

$$J(\mathtt{u}) = \int_0^T \frac{1}{R}\big(\mathtt{u}(t) - \mathtt{x}(t)\big)^2 \,\mathrm{d}t.$$

For the rest of this exercise we take $R = 1$ (ohm), and $C = 1$ (farad), and $x_{\text{desired}} = 1$ (volt). We further assume that $\mathtt{x}(0) = 0$, and that $\mathbb{U} = \mathbb{R}$.

(a) Determine the solution $(\mathtt{x}_*, \mathtt{p}_*, \mathtt{u}_*)$ of the normal Hamiltonian equations (2.27a), (2.28) explicitly as functions of time.

(b) Are the convexity assumptions of Theorem 2.8.1 satisfied?

(c) Determine the minimal cost $J(\mathtt{u}_*)$. It turns out that the minimal cost decreases as T grows. Explain why this makes sense.

2.10 *Soft landing on the Moon.* We consider the problem of optimal and safe landing on the Moon. The situation is depicted in Fig. 2.10. We assume the lunar ship only moves in the vertical direction. Its position relative to the Moon's surface is denoted by $\mathtt{y}(t)$, and its mass is denoted by $\mathtt{m}(t)$. The ship can generate an upwards force by thrusting out gasses downwards (in the direction of the Moon). We assume it does so with a constant velocity c, but that it can control the rate $-\dot{\mathtt{m}}(t)$ at which it expels the mass. This results in an upwards force of $-c\dot{\mathtt{m}}(t)$. The gravitational pull on the lunar ship is $-g\mathtt{m}(t)$. (On the Moon the gravitational acceleration g is 1.624 m/s^2.) The altitude $\mathtt{y}(t)$ of the ship satisfies the differential equation

$$\mathtt{m}(t)\ddot{\mathtt{y}}(t) = -g\mathtt{m}(t) - c\dot{\mathtt{m}}(t).$$

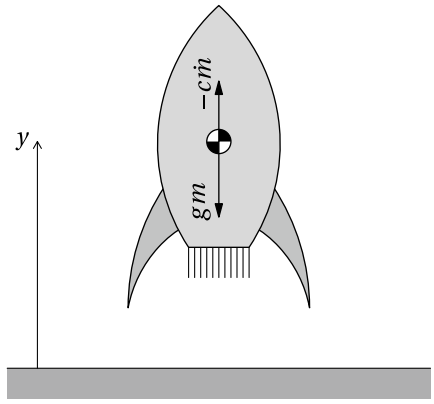


FIGURE 2.10: Soft landing on the Moon. See Exercise 2.10.

As mention earlier, we can control the rate of expel

$$\mathtt{u}(t) := -\dot{\mathtt{m}}(t).$$

Clearly, $\mathtt{u}(t)$ is bounded and nonnegative. Specifically we assume that $\mathtt{u}(t) \in \mathbb{U} := [0,1]$. The total expelled mass over $[0,T]$ equals

$$J(\mathtt{u}) = \int_0^T \mathtt{u}(t)\,\mathrm{d}t.$$

The objective is to determine the control $\mathtt{u}$ and final time $T > 0$ that minimizes the total expelled mass, $J(\mathtt{u})$, while achieving a safe, soft landing on the Moon at final time T. The latter means that $\mathtt{y}(T) = 0$ and $\dot{\mathtt{y}}(T) = 0$. With the state variables $\mathtt{x}_1 := \mathtt{y}, \mathtt{x}_2 := \dot{\mathtt{y}}, \mathtt{x}_3 := \mathtt{m}$ we can rewrite the differential equations as

$$\begin{aligned}
&\dot{\mathtt{x}}_1(t) = \mathtt{x}_2(t), && \mathtt{x}_1(0) = y_0, \quad \mathtt{x}_1(T) = 0,\\
&\dot{\mathtt{x}}_2(t) = c\frac{\mathtt{u}(t)}{\mathtt{x}_3(t)} - g, && \mathtt{x}_2(0) = \dot{y}_0, \quad \mathtt{x}_2(T) = 0,\\
&\dot{\mathtt{x}}_3(t) = -\mathtt{u}(t), && \mathtt{x}_3(0) = m_0.
\end{aligned}$$

(a) Explain in words why in this application we need $\mathtt{x}_1(t) \geq 0, \mathtt{x}_3(t) > 0$ for all $t \in [0,T]$, and that $\dot{\mathtt{x}}_2(T) \geq 0$.

(b) Determine the Hamiltonian and the differential equation for the costate $\mathtt{p} = (\mathtt{p}_1, \mathtt{p}_2, \mathtt{p}_3)$, including possible final conditions.

(c) Show that $\mathtt{z}(t) := \frac{\partial H(\mathtt{x}(t),\mathtt{p}(t),\mathtt{u}(t))}{\partial u}$ equals

$$\mathtt{z}(t) = c\frac{\mathtt{p}_2(t)}{\mathtt{x}_3(t)} - \mathtt{p}_3(t) + 1,$$

and show that

$$\dot{\mathtt{z}}(t) = \frac{-c\mathtt{p}_1(T)}{\mathtt{x}_3(t)}.$$

Also, use the fact $H(\mathrm{x}_*(t), \mathrm{p}_*(t), \mathrm{u}_*(t))$ is the zero function for time-optimal problems to show that $\mathrm{z}_*(t)$ cannot be the zero function.

(d) Conclude from (c) that $\mathrm{u}_*(t)$ is of the form

$$\mathrm{u}_*(t) = \begin{cases} 0 & \text{if } t < t_\mathrm{s} \\ 1 & \text{if } t > t_\mathrm{s} \end{cases}$$

for some $t_\mathrm{s} < T$. Thus it is optimal to thrust out gasses only during the final stage of descent, and then to do so at maximal rate.

2.11 *Initial and final conditions.* Consider the system

$$\dot{\mathrm{x}}(t) = \mathrm{x}(t)(1 - \mathrm{u}(t)), \qquad \mathrm{x}(0) = 1, \quad \mathrm{x}(1) = \tfrac{1}{2}\mathrm{e}$$

with cost

$$J(\mathrm{u}) = \int_0^1 -\ln\bigl(\mathrm{x}(t)\mathrm{u}(t)\bigr)\,\mathrm{d}t.$$

Since $\mathrm{x}(0) > 0$ we have that $\mathrm{x}(t) \geq 0$ for all t. For a well-defined cost we hence need $\mathrm{u}(t) \in [0, \infty)$ but for the moment we allow any $\mathrm{u}(t) \in \mathbb{R}$ and later verify that the optimal $\mathrm{u}_*(t)$ is in fact positive for all $t \in [0, 1]$.

(a) Determine the Hamiltonian.

(b) Determine the Hamiltonian equations (2.11).

(c) Show that $\mathrm{u}(t) = -1/(\mathrm{p}(t)\mathrm{x}(t))$ is the candidate optimal control.

(d) Substitute this $\mathrm{u}(t) = -1/(\mathrm{p}(t)\mathrm{x}(t))$ into the Hamiltonian equations and solve for $\mathrm{p}_*(t)$ and then $\mathrm{x}_*(t)$ and subsequently $\mathrm{u}_*(t)$.

(e) Is $\mathrm{u}_*(t) > 0$ for all $t \in [0, 1]$?

2.12 *Running cost depending on time.* Consider the second-order system with mixed initial and final conditions

$$\begin{aligned} \dot{\mathrm{x}}_1(t) &= \mathrm{u}(t), & \mathrm{x}_1(0) &= 0, \quad \mathrm{x}_1(1) = 1, \\ \dot{\mathrm{x}}_2(t) &= 1, & \mathrm{x}_2(0) &= 0, \end{aligned}$$

and with cost

$$J(\mathrm{u}) = \int_0^1 \mathrm{u}^2(t) + 12\mathrm{x}_2(t)\mathrm{x}_1(t)\,\mathrm{d}t.$$

(Notice that $\mathrm{x}_2(t) = t$. This is a simple strategy to allow running costs that depend on t, here $L(\mathrm{x}(t), \mathrm{u}(t)) = \mathrm{u}^2(t) + 12t\mathrm{x}_1(t)$.)

We assume that the input is not restricted, i.e., $\mathbb{U} = \mathbb{R}$.

(a) Determine the Hamiltonian for this problem.

(b) Determine the differential equations for state $\mathtt{x}(t)$ and costate $\mathtt{p}(t)$, including the boundary conditions.

(c) Express the candidate minimizing $\mathtt{u}_*(t)$ as a function of $\mathtt{x}(t), \mathtt{p}(t)$.

(d) Solve the equations for $\mathtt{x}_*, \mathtt{p}_*, \mathtt{u}_*$ (that is, determine $\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t)$ as explicit functions of time $t \in [0,1]$).

2.13 *Two-sector economy.* Consider an economy consisting of two sectors where sector 1 produces investment goods and sector 2 produces consumption goods. Let $\mathtt{x}_i(t), i = 1,2$, denote the production rate in the i-th sector at time t, and let $\mathtt{u}(t)$ be the fraction of investments allocated to sector 1. Suppose the dynamics of the $\mathtt{x}_i(t)$ are given by

$$\dot{\mathtt{x}}_1(t) = a\mathtt{u}(t)\mathtt{x}_1(t),$$
$$\dot{\mathtt{x}}_2(t) = a(1-\mathtt{u}(t))\mathtt{x}_1(t),$$

where a is a positive constant. Hence, the increase in production per unit of time in each sector is assumed to be proportional to the investment allocated to the sector. By definition we have

$$0 \le \mathtt{u}(t) \le 1 \quad \text{for all } t \in [0,T],$$

where $[0,T]$ denotes the planning period. As optimal control problem we may consider the problem of maximizing the total consumption in the given planning period $[0,T]$, thus our problem is to *maximize*

$$\tilde{J}(\mathtt{u}) := \int_0^T \mathtt{x}_2(t)\,\mathrm{d}t$$

subject to

$$\mathtt{x}_1(0) = x_{01}, \qquad \mathtt{x}_1(T) = \text{free},$$
$$\mathtt{x}_2(0) = x_{02}, \qquad \mathtt{x}_2(T) = \text{free},$$

in which $x_{01} > 0, x_{02} \ge 0$.

(a) Argue that $\mathtt{x}_1(t) > 0$ for all time.

(b) Determine an optimal input using the minimum principle. [Hint: it may help to realize that $\dot{\mathtt{p}}_1(T) - \dot{\mathtt{p}}_2(T) < 0$.]

2.14 Consider the second-order system with mixed initial and final conditions

$$\dot{\mathtt{x}}_1(t) = \mathtt{u}(t), \qquad \mathtt{x}_1(0) = 0, \qquad \mathtt{x}_1(1) = 2,$$
$$\dot{\mathtt{x}}_2(t) = 1, \qquad \mathtt{x}_2(0) = 0,$$

and with cost

$$J(\mathtt{u}) = \int_0^1 \mathtt{u}^2(t) + 4\mathtt{x}_2(t)\mathtt{u}(t)\;\mathrm{d}t.$$

The input $\mathtt{u} : [0,1] \to \mathbb{R}$ is not restricted, i.e., $\mathtt{u}(t)$ can take on any real value.

(a) Determine the Hamiltonian for this problem.

(b) Determine the differential equations for the costate $\mathtt{p}(t)$, including the boundary conditions.

(c) Express the candidate minimizing $\mathtt{u}_*(t)$ as a function of $\mathtt{x}_*(t), \mathtt{p}_*(t)$.

(d) Solve the equations for $\mathtt{x}_*, \mathtt{p}_*, \mathtt{u}_*$ (that is, determine $\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t)$ as explicit functions of time $t \in [0,1]$).

2.15 *Integral constraints.* Let us return to the calculus of variations problem of minimizing

$$\int_0^T F(\mathtt{x}(t), \dot{\mathtt{x}}(t))\,\mathrm{d}t$$

over all functions $\mathtt{x} : [0,T] \to \mathbb{R}^n$ that satisfy an *integral constraint*

$$\int_0^T M(\mathtt{x}(t), \dot{\mathtt{x}}(t))\,\mathrm{d}t = c_0.$$

Theorem 1.7.1 (p. 32) says that the optimal solution satisfies either (1.54) for some $\mu_* \in \mathbb{R}$, or satisfies (1.55). This problem can also be cast as an optimal control problem with a final condition, and then Theorem 2.6.1 gives us the same two conditions (depending on whether the Hamiltonian is normal or abnormal):

(a) Let $\dot{\mathtt{x}}(t) = \mathtt{u}(t)$ and define $\dot{\mathtt{z}}_{n+1}(t) = M(\mathtt{x}(t), \mathtt{u}(t))$ and $\mathtt{z} := (\mathtt{x}, \mathtt{z}_{n+1})$. Formulate the above calculus of variations problem as an optimal control problem with a final condition on state $\mathtt{z}$ and with $\mathbb{U} = \mathbb{R}^n$. (I.e., express $f(z), L(z,u), K(z)$ in terms of F, M, c_0.)

(b) Since $\mathtt{z} := (\mathtt{x}, \mathtt{z}_{n+1})$ has $n+1$ components, also the corresponding costate $\mathtt{p}$ has $n+1$ components. Show that $\mathtt{p}_{n+1}(t)$ is constant for the normal Hamiltonian $H_1(z,p,u)$ as well as the abnormal Hamiltonian $H_0(z,p,u)$.

(c) For the normal Hamiltonian $H_1(z,p,u)$, show that the existence of a solution of the Hamiltonian equations (2.27a) and (2.28) imply that (1.54) holds for $\mu_* = \mathtt{p}_{n+1}$.

(d) For the abnormal Hamiltonian $H_0(z,p,u)$, show that the existence of a solution of the Hamiltonian equations (2.27a) and (2.28) with $\mathtt{p}_{n+1} \neq 0$ implies that (1.55) holds.

2.16 *The minimum principle is not sufficient for optimality.* The minimum principle is necessary for optimality, but it is not sufficient. That is to say, if we are able to solve the Hamiltonian equations (2.14) (including the pointwise minimization (2.15)) then it is not guaranteed that the so found input is optimal. Here is an example: consider the system and cost

$$\dot{\mathtt{x}}(t) = \mathtt{u}(t), \qquad \mathtt{x}(0) = 1, \qquad J(\mathtt{u}) = \int_0^{2\pi} -\tfrac{1}{2}\mathtt{x}^2(t) + \tfrac{1}{2}\mathtt{u}^2(t)\,\mathrm{d}t.$$

We allow every input, so $\mathbb{U} = \mathbb{R}$.

(a) Solve the equations (2.14), (2.15).

(b) Compute $J(\mathtt{u}_*)$ for the input $\mathtt{u}_*$ found in the previous part.

(c) Find an input $\mathtt{u}$ for which $J(\mathtt{u})$ is less than $J(\mathtt{u}_*)$. (Just guess a simple one; many inputs $\mathtt{u}$ will do the job.)

2.17 *Time-optimal control.* Consider the optimal control problem of Example 2.1.1.

(a) Solve this problem using the minimum principle.

(b) For every $T \geq 0$ determine $H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(t))$.

(c) For every $T \geq 0$ determine the optimal cost $J(\mathtt{u}_*)$.

(d) Now suppose we also optimize the cost over all final times $T \geq 0$. Which T's are optimal, and does this agree with Theorem 2.7.1?

2.18 *Calculus of variations.* Consider the calculus of variations problem of Example 2.4.1.

(a) Use (2.11c) to express $\mathtt{p}_*(t)$ explicitly in terms of $L, \mathtt{x}_*, \mathtt{u}_*, t$ and derivatives.

(b) Given the above form of $\mathtt{p}_*(t)$, and (2.11a), show that the costate equation (2.11b) is equivalent to the Euler-Lagrange equation.

2.19 *Convexity.* The proof of Theorem 2.8.1 assumes that the terminal cost is absent, i.e., $K(x) = 0$. Now consider more general $K(x)$. Assume $K(x)$ and $\frac{\partial K(x)}{\partial x}$ are continuous, and that $K(x)$ is a convex function.

(a) Adapt the proof of Theorem 2.8.1 so that it also works for nonzero convex $K(x)$. [Hint: have a look at Lemma A.7.1 (p. 198).]

(b) Theorem 2.8.1 considers the standard (free endpoint) optimal control problem of Theorem 2.5.1. Show that Theorem 2.8.1 remains valid for the case that the final state is constrained as in Theorem 2.6.1.

2.20 *Convexity.* Does Example 2.5.3 satisfy the convexity assumptions of Theorem 2.8.1?

2.21 *Convexity.* Does Example 2.5.4 satisfy the convexity assumptions of Theorem 2.8.1?



Chapter 3

Dynamic Programming

3.1 Introduction

The minimum principle was developed in the Soviet Union in the late fifties of the previous century. At about the same time Richard Bellman in the USA developed an entirely different approach to optimal control, called *dynamic programming*. In this chapter, we deal with dynamic programming. As in the previous chapter, we assume that the state satisfies a system of differential equations

$$\dot{\mathrm{x}}(t) = f(\mathrm{x}(t), \mathrm{u}(t)), \qquad \mathrm{x}(0) = x_0, \tag{3.1a}$$

in which $\mathrm{x} : [0, T] \to \mathbb{R}^n$, and $x_0 \in \mathbb{R}^n$, and that the input u at each moment in time takes values in a given subset $\mathbb{U}$ of $\mathbb{R}^m$,

$$\mathrm{u} : [0, T] \to \mathbb{U}. \tag{3.1b}$$

As before, we associate with system (3.1a) a cost over a finite time horizon $[0, T]$ of the form

$$J_{[0,T]}(x_0, \mathrm{u}) := \int_0^T L(\mathrm{x}(t), \mathrm{u}(t))\, \mathrm{d}t + K(\mathrm{x}(T)). \tag{3.1c}$$

This cost depends on the input u, but in dynamic programming it is convenient to also emphasize the dependence of this cost on x_0 and the time interval $[0, T]$. The final time T and the functions $K : \mathbb{R}^n \to \mathbb{R}$ and $L : \mathbb{R}^n \times \mathbb{U} \to \mathbb{R}$ are assumed as given.

The crux of dynamic programming is to associate with this single cost over time horizon $[0, T]$ a whole family of costs over subsets of this time horizon,

$$J_{[\tau,T]}(x, \mathrm{u}) := \int_\tau^T L(\mathrm{x}(t), \mathrm{u}(t))\, \mathrm{d}t + K(\mathrm{x}(T)), \tag{3.2}$$

for each initial time $\tau \in [0, T]$ and for each initial state $\mathrm{x}(\tau) = x$, and then to establish a dynamic relation between the family of optimal costs (hence the

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G. Meinsma and A. van der Schaft, *A Course on Optimal Control*, Springer Undergraduate Texts in Mathematics and Technology,
https://doi.org/10.1007/978-3-031-36655-0_3

name *dynamic* programming). On the one hand, this complicates the problem because many optimal control problems need to be considered. On the other hand, if this dynamic relation can be solved, then it turns out to produce *sufficient* conditions for optimality.

3.2 Principle of Optimality

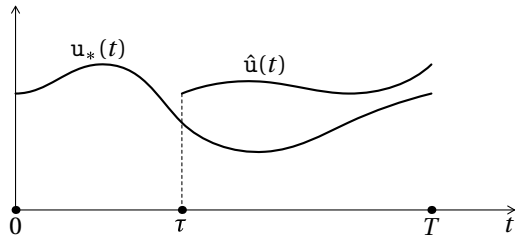


FIGURE 3.1: Principle of optimality, see § 3.2.

The *principle of optimality* is a simple yet powerful result in optimal control. Roughly speaking, it says that *every tail of an optimal control is optimal*. We formalize this result. Figure 3.1 should be instructive here. It depicts an optimal control u_* on $[0,T]$ and an alternative control $\hat{u}$ on a restricted time window $[\tau,T]$ for some $\tau \in [0,T]$. The optimal control u_* steers the state from $x(0) = x_0$ to some value $x_*(\tau)$ at time τ. Is it now possible that the alternative control $\hat{u}$ achieves a smaller cost-to-go $J_{[\tau,T]}(x_*(\tau),u)$ over the remaining window $[\tau,T]$ than u_*? That is, is it possible that

$$J_{[\tau,T]}(x_*(\tau),\hat{u}) < J_{[\tau,T]}(x_*(\tau),u_*)?$$

No, because if it would, then the new input $\tilde{u}$ constructed from u_* over the initial interval $[0,\tau]$, and from $\hat{u}$ over the remaining $[\tau,T]$, would improve on u_* over the entire interval:

$$\begin{aligned} J_{[0,T]}(x_0,\tilde{u}) &= \int_0^\tau L(x(t),\tilde{u}(t))\,dt + J_{[\tau,T]}(x(\tau),\tilde{u}) \\ &= \int_0^\tau L(x_*(t),u_*(t))\,dt + J_{[\tau,T]}(x_*(\tau),\hat{u}) \\ &< \int_0^\tau L(x_*(t),u_*(t))\,dt + J_{[\tau,T]}(x_*(\tau),u_*) = J_{[0,T]}(x_0,u_*), \end{aligned}$$

and this contradicts the assumed optimality of u_*. So we conclude: if $u_* : [0,T] \to \mathbb{U}$ is optimal for $J_{[0,T]}(x_0,u)$ then for every $\tau \in [0,T]$ this u_* restricted to $[\tau,T]$ is optimal for $J_{[\tau,T]}(x_*(\tau),u)$ as well. This is the *principle of optimality*. It will be of great help in the analysis to come. Notice that this principle hinges on the causality property of the system that $x(t)$ for $t < \tau$ does not depend on $u(t')$ for $t' > \tau$. Also, the additivity property of the cost is crucial, which is that the "cost-to-go" $J_{[\tau,T]}(x_*(\tau),u)$ adds to "cost-so-far" $\int_0^\tau L(x_*(t),u(t))\,dt$.

3.3 Discrete-Time Dynamic Programming

The main idea of dynamic programming and the reason for its popularity is explained best for systems that evolve over *discrete* time—as opposed to the systems that evolve over continuous time, which we normally consider in this book. Thus, for the time being, consider a discrete-time system

$$\mathtt{x}(t+1) = f(\mathtt{x}(t), \mathtt{u}(t)), \qquad \mathtt{x}(0) = x_0, \tag{3.3}$$

on some discrete finite time horizon

$$t \in \{0, 1, \ldots, T-1\},$$

with x_0 given, and T a given positive integer. We want to find a control sequence $\mathtt{u}_* = (\mathtt{u}_*(0), \mathtt{u}_*(1), \ldots, \mathtt{u}_*(T-1))$, called *optimal control (sequence)*, and resulting state sequence $(\mathtt{x}_*(0), \mathtt{x}_*(1), \ldots, \mathtt{x}_*(T))$, that minimizes a cost of the form

$$J_{[0,T]}(x_0, \mathtt{u}) = \sum_{t=0}^{T-1} L(\mathtt{x}(t), \mathtt{u}(t)) \; + \; K(\mathtt{x}(T)). \tag{3.4}$$

Incidentally, in discrete-time systems there is no need to restrict the state space $\mathbb{X}$ to some set on which derivatives are defined, like our default $\mathbb{R}^n$. Indeed, the state space in applications is often a finite set. The same is true for the input set $\mathbb{U}$. In what follows, the number of elements of a set $\mathbb{X}$ is denoted as $|\mathbb{X}|$.

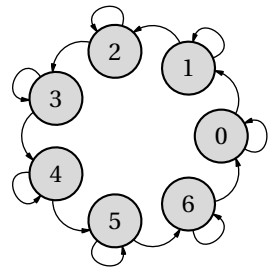


FIGURE 3.2: A discrete-time system with 7 states. See Example 3.3.1.

Example 3.3.1 (Naive optimization). Suppose the state space $\mathbb{X}$ consists of the 7 integer elements

$$\mathbb{X} = \{0, 1, 2, \ldots, 6\}.$$

Align the states in a circle (see Fig. 3.2), and suppose that at each moment in time, the state can either move one step counter-clockwise, or stay where it is. Thus, at each moment in time, we have a choice of two. The input space $\mathbb{U}$ hence has two elements. If we take

$$\mathbb{U} = \{0, 1\}$$

then the transition from one state to the next is modeled by the discrete system

$$\mathtt{x}(t+1) = \mathtt{x}(t) + \mathtt{u}(t), \qquad \mathtt{u}(t) \in \mathbb{U}, \qquad t \in \{0, 1, \ldots, T-1\}$$

(counting modulo 7, so $6+1=0$). Each transition from one state $\mathtt{x}(t)$ to the next $\mathtt{x}(t+1)$ is assumed to cost a certain amount $L(\mathtt{x}(t), \mathtt{u}(t))$, and the final state $\mathtt{x}(T)$ costs an additional $K(\mathtt{x}(T))$. The total cost hence is (3.4). The naive approach to determine the optimal control $(\mathtt{u}(0), \ldots, \mathtt{u}(T-1))$ and resulting optimal state sequence $(\mathtt{x}(1), \ldots, \mathtt{x}(T))$ is to just explore them all and pick the best. As we can move in two different ways each at moment in time, this naive approach requires 2^T sequences $(\mathtt{x}(1), \ldots, \mathtt{x}(T))$ to explore. Since each sequence has length T the evaluation of the cost for each sequence is (roughly) linear in T, and, therefore, the total number of operations required in this naive approach is of order

$$T \times 2^T.$$

□

It is not hard to see that for arbitrary systems (3.3) the total number of operations that the naive approach requires is of order

$$T \times |\mathbb{U}|^T.$$

Thus the total number of operations is exponential in T.

In general, in dynamic programming we solve the minimization *backwards* in time. This may at first sight seem to complicate the analysis, but it allows us to exploit the principle of optimality. The following example explains it all.

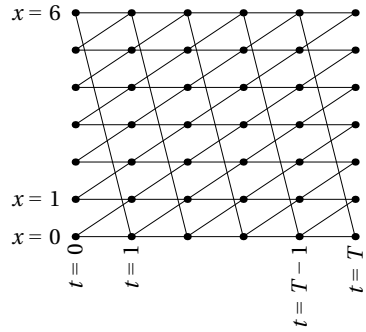
Example 3.3.2 (Dynamic programming). Continue with the system of Example 3.3.1:

$$\mathtt{x}(t+1) = \mathtt{x}(t) + \mathtt{u}(t), \qquad \mathtt{u}(t) \in \{0, 1\}, \qquad t \in \{0, 1, \ldots, T-1\}$$

with $\mathtt{x}(t) \in \mathbb{X} := \{0, 1, \ldots, 6\}$, and, to make it more explicit, assume that the final cost is x^2 and that each counter-clockwise move costs 1, i.e.,

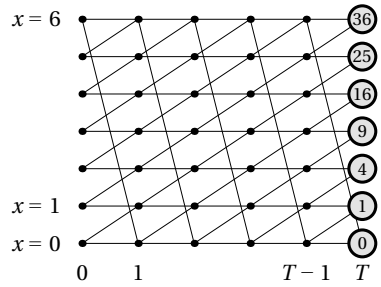
$$K(x) = x^2 \qquad \text{and} \qquad L(x, u) = u \in \mathbb{U} := \{0, 1\}.$$

This system over a given time horizon we now visualize as



(Here we took $T = 5$.) The horizontal axis represents the time, $t = 0, 1, \ldots, T$, and the vertical axis represents the states, $x = 0, 1, \ldots, 6$. Vertices (dots) denote pairs (t, x), and lines (edges) between vertices represent possible transitions. For instance, the line connecting $(t, x) = (0, 6)$ with $(t, x) = (1, 0)$ says that we can move from $x = 6$ to $x = 0$ in one time step.

Let us first figure out the cost at the final time T. Since we do not know in which final state, $\mathtt{x}(T)$, we end up, we have to determine this cost for every element of the state space. This cost we denote as $\mathcal{V}(x, T)$, and clearly this is simply the final cost, so $\mathcal{V}(x, T) = K(x) = x^2$:



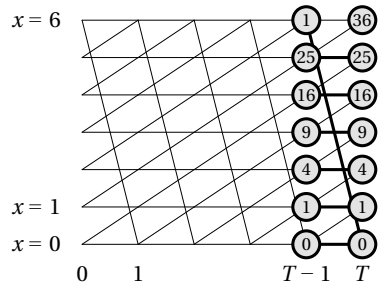
Now that $\mathcal{V}(x, T)$ is known, consider the optimal *cost-to-go* from $t = T - 1$ onwards. This optimal cost-to-go we denote by $\mathcal{V}(x, T-1)$ and it is defined as the minimal cost from $t = T - 1$ onwards if at time $t = T - 1$ we are at state $\mathtt{x}(T-1) = x$. It satisfies

$$\mathcal{V}(x, T-1) = \min_{\mathtt{u}(T-1) \in \{0,1\}} \big(L(x, \mathtt{u}(T-1)) + K(\mathtt{x}(T)) \big),$$

because $L(x, \mathtt{u}(T-1))$ is the cost of the transition if we apply input $\mathtt{u}(T-1)$, and $K(\mathtt{x}(T))$ is the final cost. Since $\mathtt{x}(T) = f(x, \mathtt{u}(T-1))$, this final cost equals $\mathcal{V}(f(x, \mathtt{u}(T-1)), T)$, so we can also write

$$\mathcal{V}(x, T-1) = \min_{u \in \{0,1\}} \big(L(x, u) + \mathcal{V}(f(x, u), T) \big).$$

With $\mathcal{V}(x, T)$ already established for all x, this minimization requires $|\mathbb{U}| = |\{1, 2\}| = 2$ inputs to explore at each state, and, hence, the total number of operations that this requires is of order $|\mathbb{X}| \times |\mathbb{U}|$. The so determined $\mathcal{V}(x, T-1)$, together with $\mathcal{V}(x, T)$, are shown here:



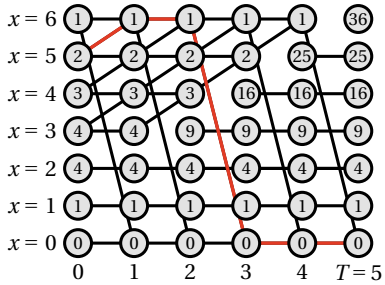
Along the way we also determined for each state $\mathtt{x}(T-1)$ an optimal control $\mathtt{u}_*(T-1)$, indicated in the figure by the thick edges. Notice that none of the states $\mathtt{x}(T-1)$ switch to $\mathtt{x}(T)=6$. We can continue in this fashion and determine backwards in time—for $t=T-2$, then $t=T-3$, etc., till $t=0$—the optimal cost-to-go from t onwards for any state $\mathtt{x}(t)=x$. At this stage, we exploit the principle of optimality: since every tail of an optimal control is optimal, the optimal cost-to-go $\mathcal{V}(x,t)$, defined as the optimal cost from t onwards starting at $\mathtt{x}(t)=x$, satisfies the equation:

$$\mathcal{V}(x,t)=\min_{u\in\{0,1\}}\big(L(x,u)+\mathcal{V}(f(x,u),t+1)\big).$$

This equation expresses that the optimal cost from t onwards, starting at $\mathtt{x}(t)=x$, is the cost of the transition, $L(x,u)$, plus the optimal cost from $t+1$ onwards. Once $\mathcal{V}(x,t+1)$ is known for all x, this easily gives us $\mathcal{V}(x,t)$ for all x. For

$$T=5$$

we end up this way with the following complete solution:



This solves the optimal control problem for every initial state x_0. For some initial states the optimal control sequence, $\mathtt{u}_*=(\mathtt{u}(0),\mathtt{u}(1),\mathtt{u}(2),\mathtt{u}(3),\mathtt{u}(4))$, is actually not unique. For instance, the control sequence shown here in red, $\mathtt{u}_*=(1,0,1,0,0)$, is one of several optimal controls for $x_0=5$. The optimal cost-to-go $\mathcal{V}(x,t)$ of course *is* unique. □

In general, in dynamic programming, we compute the optimal cost-to-go $\mathcal{V}(x,t)$ via the recursion

$$\mathcal{V}(x,t)=\min_{u\in\mathbb{U}}\big(L(x,u)+\mathcal{V}(f(x,u),t+1)\big),\qquad t\in\{0,1,\dots,T-1\},\tag{3.5}$$

starting at the final time where $\mathcal{V}(x,T)=K(x)$ for all states, and then subsequently going backwards in time, $t=T-1,t=T-2,\dots$, until we reach $t=0$. In this way, the optimal control problem is split into T ordinary minimization problems. To determine the final cost $\mathcal{V}(x,T)=K(x)$ for all $x\in\mathbb{X}$ requires order $|\mathbb{X}|$ operations. Then determining $\mathcal{V}(x,T-1)$ for all $x\in\mathbb{X}$ requires $|\mathbb{X}|$ times the

number of inputs $|\mathbb{U}|$ to explore, etc., and so the total number of operations over all $t \in \{0, 1, \ldots, T-1\}$ is of order

$$T \times |\mathbb{U}| \times |\mathbb{X}|.$$

If the number of states is modest or if T is large, then this typically outperforms the naive approach (which requires order $T \times |\mathbb{U}|^T$ operations). Equation (3.5) is called *Bellman's equation of dynamic programming*.

In continuous time the same basic idea survives, except for the results regarding computational complexity. Note that, in the continuous time case, the optimization is over a set of input functions on the time interval $[0, T]$, which is an infinite-dimensional space. Furthermore, it is clear that, contrary to the discrete-time case, we will not be able to split the problem into a series of finite-dimensional minimization problems.

3.4 Hamilton-Jacobi-Bellman Equation

We return to the continuous time. In dynamic programming in continuous time we minimize *all* costs $J_{[\tau,T]}(x, \mathtt{u})$—for all $\tau \in [0, T]$ and all states $\mathtt{x}(\tau) = x$—and not just the one cost $J_{[0,T]}(x_0, \mathtt{u})$ that we are asked to minimize. To tackle this problem, we will again exploit the principle of optimality, and we will again need the notion of *optimal cost-to-go*, also known as the *value function*.

Definition 3.4.1 (Value function/optimal cost-to-go). Consider the optimal control problem (3.1). The *value function* $\mathcal{V} : \mathbb{R}^n \times [0, T] \to \mathbb{R}$ at state x and time τ is defined as the *optimal cost-to-go* over time horizon $[\tau, T]$ with initial state $\mathtt{x}(\tau) = x$, that is,

$$\mathcal{V}(x, \tau) = \inf_{\mathtt{u}:[\tau,T]\to\mathbb{U}} J_{[\tau,T]}(x, \mathtt{u}), \tag{3.6}$$

with $J_{[\tau,T]}$ as defined in (3.2). □

In most cases of interest the infimum in (3.6) is attained by some $\mathtt{u}_*$, in which case the infimum (3.6) is a minimum. In general, though, a minimizer need not exist, while the infimum does exist (but it might be $\pm\infty$).

Example 3.4.2 (Integrator with linear cost). Consider once again the optimal control problem of Example 2.5.2:

$$\dot{\mathtt{x}}(t) = \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0, \tag{3.7}$$

with bounded inputs

$$\mathbb{U} = [-1, 1],$$

and with cost

$$J_{[0,1]}(x_0, \mathtt{u}) = \int_0^1 \mathtt{x}(t)\, \mathrm{d}t.$$

From the fact that $\dot{\mathtt{x}}(t) = \mathtt{u}(t) \in [-1,1]$, it is immediate that the optimal control is $\mathtt{u}_*(t) = -1$ and, hence, $\mathtt{x}_*(t) = x_0 - t$. Therefore, the value function at $\tau = 0$ is

$$\mathcal{V}(x_0,0) = J_{[0,1]}(x_0,\mathtt{u}_*) = \int_0^1 x_0 - t \,\mathrm{d}t = x_0 - 1/2.$$

Next, we determine the value function at the other time instances. It is easy to see that $\mathtt{u}_*(t) = -1$ is optimal for $J_{[\tau,1]}(x,\mathtt{u})$ for every $\tau > 0$ and every $\mathtt{x}(\tau) = x$. Hence, in this case, $\mathtt{x}_*(t) = x - (t-\tau)$ and

$$\mathcal{V}(x,\tau) = \int_\tau^1 x - (t-\tau) \,\mathrm{d}t = \Big[xt - \tfrac{1}{2}(t-\tau)^2\Big]_\tau^1 = x(1-\tau) - \tfrac{1}{2}(1-\tau)^2.$$

As expected, the value function is zero at the final time $\tau = 1$. It is *not* necessarily monotonic in τ, see Fig. 3.3. Indeed, for $x = 1/2$, the value function is zero at $\tau = 0$ and at $\tau = 1$, yet it is positive in between. □

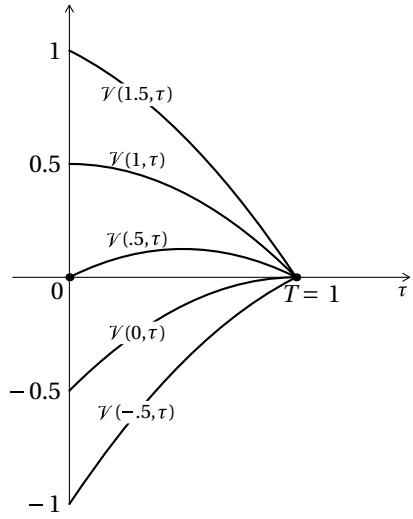


FIGURE 3.3: The value function $\mathcal{V}(x,\tau)$ of the problem of Example 3.4.2 for various x as a function of $\tau \in [0,1]$.

Now it is time to derive, or rather motivate, the continuous-time version of Bellman's equation of dynamic programming (3.5). For any input $\mathtt{u}$—optimal or not—the cost-to-go from τ onwards equals the cost over $[\tau, \tau+\epsilon]$ plus the cost over the remaining $[\tau+\epsilon, T]$, that is

$$J_{[\tau,T]}(x,\mathtt{u}) = \int_\tau^{\tau+\epsilon} L(\mathtt{x}(t),\mathtt{u}(t))\,\mathrm{d}t + J_{[\tau+\epsilon,T]}(\mathtt{x}(\tau+\epsilon),\mathtt{u}) \tag{3.8}$$

with initial state $\mathtt{x}(\tau) = x$. The value function is defined as the infimum of this cost over all inputs. Suppose the infimum is attained by some input, i.e., that the

infimum is a minimum. Taking the minimum over all $\mathtt{u}$ of the left- and right-hand sides of (3.8) shows that

$$\mathcal{V}(x,\tau) = \min_{\mathtt{u}:[\tau,T]\to\mathbb{U}} \Big(\int_\tau^{\tau+\epsilon} L(\mathtt{x}(t),\mathtt{u}(t))\,\mathrm{d}t + J_{[\tau+\epsilon,T]}(\mathtt{x}(\tau+\epsilon),\mathtt{u}) \Big).$$

By the principle of optimality, any optimal control over $[\tau,T]$ is optimal for $J_{[\tau+\epsilon,T]}(\mathtt{x}(\tau+\epsilon),\mathtt{u})$ as well. The right-hand side of the above equality can thus be simplified to

$$\mathcal{V}(x,\tau) = \min_{\mathtt{u}:[\tau,\tau+\epsilon]\to\mathbb{U}} \Big(\int_\tau^{\tau+\epsilon} L(\mathtt{x}(t),\mathtt{u}(t))\,\mathrm{d}t + \mathcal{V}(\mathtt{x}(\tau+\epsilon),\tau+\epsilon) \Big)$$

with initial state $\mathtt{x}(\tau) = x$. Notice that, in this last equation, we need only optimize over inputs defined on the time window $[\tau,\tau+\epsilon]$ because the optimization over the remaining time window $[\tau+\epsilon,T]$ is incorporated in the value function $\mathcal{V}(\mathtt{x}(\tau+\epsilon),\tau+\epsilon)$. For further analysis, it is beneficial to move the term $\mathcal{V}(x,\tau)$ to the right-hand side and to scale the equation by ϵ,

$$0 = \min_{\mathtt{u}:[\tau,\tau+\epsilon]\to\mathbb{U}} \frac{\int_\tau^{\tau+\epsilon} L(\mathtt{x}(t),\mathtt{u}(t))\,\mathrm{d}t + \mathcal{V}(\mathtt{x}(\tau+\epsilon),\tau+\epsilon) - \mathcal{V}(x,\tau)}{\epsilon}. \tag{3.9}$$

In this form we can take the limit $\epsilon \to 0$. It is plausible that *functions* $\mathtt{u}:[\tau,\tau+\epsilon]\to\mathbb{U}$ in the limit can be identified with *constants* $u\in\mathbb{U}$, and that the difference between the two value functions in (3.9) converges for $\epsilon\to 0$ to the total derivative of $\mathcal{V}(\mathtt{x}(\tau),\tau)$ with respect to τ. Thus,

$$0 = \min_{u\in\mathbb{U}} \left(L(\mathtt{x}(\tau),u) + \frac{\mathrm{d}\mathcal{V}(\mathtt{x}(\tau),\tau)}{\mathrm{d}\tau} \right) \tag{3.10}$$

for all $\tau\in[0,T]$ and all $\mathtt{x}(\tau)=x\in\mathbb{R}^n$. Incidentally, this identity is reminiscent of the cost-to-go (B.14) as explained in Section B.5 of Appendix B. The total derivative of $\mathcal{V}(\mathtt{x}(\tau),\tau)$ with respect to τ is

$$\frac{\mathrm{d}\mathcal{V}(\mathtt{x}(\tau),\tau)}{\mathrm{d}\tau} = \frac{\partial\mathcal{V}(\mathtt{x}(\tau),\tau)}{\partial x^{\mathrm{T}}} f(\mathtt{x}(\tau),\mathtt{u}(\tau)) + \frac{\partial\mathcal{V}(\mathtt{x}(\tau),\tau)}{\partial\tau}.$$

Inserting this into (3.10), and using $u=\mathtt{u}(\tau), x=\mathtt{x}(\tau)$, we arrive at the partial differential equation:

$$0 = \min_{u\in\mathbb{U}} \left(L(x,u) + \frac{\partial\mathcal{V}(x,\tau)}{\partial x^{\mathrm{T}}} f(x,u) + \frac{\partial\mathcal{V}(x,\tau)}{\partial\tau} \right)$$

for all $\tau\in[0,T]$ and all $x\in\mathbb{R}^n$. The partial derivative of $\mathcal{V}(x,\tau)$ with respect to τ does not depend on u and so does not contribute to the minimization. This, finally, brings us to the famous equation

$$\frac{\partial\mathcal{V}(x,\tau)}{\partial\tau} + \min_{u\in\mathbb{U}} \left(\frac{\partial\mathcal{V}(x,\tau)}{\partial x^{\mathrm{T}}} f(x,u) + L(x,u) \right) = 0. \tag{3.11}$$

This equation is known as the *Hamilton-Jacobi-Bellman* equation—or just *HJB* equation—because it extends the Hamilton-Jacobi equation from classical mechanics (see Lanczos (1986)).

What did we do so far? We made it *plausible* that the relation between the value functions at neighboring points in state x and time τ is the partial differential equation (3.11). We need to stress here the word "plausible", because we have "derived" (3.11) only under several technical assumptions including existence of an optimal control, existence of a value function, and existence of some limits. However, we can turn the analysis around, and show that (3.11) provides *sufficient* conditions for optimality. This is the following theorem, and it is the central result of this chapter. In this formulation, the time τ is called t again, and the solution of the partial differential equation we denote by V, and not $\mathcal{V}$, because the solution V of the partial differential equation is not always the value function (although in most cases it is).

Theorem 3.4.3 (Hamilton-Jacobi-Bellman equations). Consider the optimal control problem (3.1). Suppose $V : \mathbb{R}^n \times [0,T] \to \mathbb{R}$ is a continuously differentiable function that satisfies the partial differential equation

$$\frac{\partial V(x,t)}{\partial t} + \min_{u \in \mathbb{U}} \left(\frac{\partial V(x,t)}{\partial x^{\mathrm{T}}} f(x,u) + L(x,u) \right) = 0 \tag{3.12a}$$

for all $x \in \mathbb{R}^n$ and all $t \in [0,T]$, and that satisfies the final time condition

$$V(x,T) = K(x) \tag{3.12b}$$

for all $x \in \mathbb{R}^n$. Then

1. $V(x,\tau)$ is a lower bound of the cost over $[\tau,T]$ starting at $\mathtt{x}(\tau) = x$, that is,

   $$J_{[\tau,T]}(x,\mathtt{u}) \geq V(x,\tau) \quad \text{for every input } \mathtt{u}.$$

2. Suppose $\mathtt{u}_* : [0,T] \to \mathbb{U}$ is such that the solution $\mathtt{x}$ of $\dot{\mathtt{x}}(t) = f(\mathtt{x}(t), \mathtt{u}_*(t))$ with $\mathtt{x}(0) = x_0$ is well defined on $[0,T]$, and that at almost every $t \in [0,T]$ the vector $\mathtt{u}_*(t)$ minimizes

   $$\frac{\partial V(\mathtt{x}(t),t)}{\partial x^{\mathrm{T}}} f(\mathtt{x}(t),u) + L(\mathtt{x}(t),u)$$

   over all $u \in \mathbb{U}$. Then $\mathtt{u}_*$ is a solution of the optimal control problem and the optimal cost is

   $$J_{[0,T]}(x_0,\mathtt{u}_*) = V(x_0,0). \tag{3.13}$$

   Furthermore, in this case, $V(x_0,0)$ equals the value function $\mathcal{V}(x_0,0)$. (Note however that for other states and times, the solution $V(x,t)$ may differ from the value function $\mathcal{V}(x,t)$.)

3. Suppose the minimization problem in (3.12a) for each $x \in \mathbb{R}^n$ and each $t \in [0,T]$ has a (possibly non-unique) solution u. Denote one such solution as $\mathtt{u}(x,t)$. If for every $x \in \mathbb{R}^n$ and every $\tau \in [0,T]$ the solution $\mathtt{x}$ of $\dot{\mathtt{x}}(t) = f(\mathtt{x}(t), \mathtt{u}(\mathtt{x}(t),t))$ with initial condition $\mathtt{x}(\tau) = x$ is well defined for all $t \in [\tau, T]$, then V equals the value function,

$$V(x,t) = \mathcal{V}(x,t) \qquad \forall x \in \mathbb{R}^n, t \in [0,T],$$

and $\mathtt{u}_*(t) := \mathtt{u}(\mathtt{x}(t),t)$ is an optimal control for $J_{[\tau,T]}(x,\mathtt{u})$ for every $x \in \mathbb{R}^n, \tau \in [0,T]$.

Proof.

1. Let $\mathtt{x}(\tau) = x$. We have that

$$\begin{aligned}
J_{[\tau,T]}(x,\mathtt{u}) &= \int_\tau^T L(\mathtt{x}(t),\mathtt{u}(t))\,\mathrm{d}t + K(\mathtt{x}(T)) \\
&= \int_\tau^T \frac{\partial V(\mathtt{x}(t),t)}{\partial x^{\mathrm{T}}} f(\mathtt{x}(t),\mathtt{u}(t)) + L(\mathtt{x}(t),\mathtt{u}(t))\,\mathrm{d}t \\
&\qquad - \int_\tau^T \frac{\partial V(\mathtt{x}(t),t)}{\partial x^{\mathrm{T}}} f(\mathtt{x}(t),\mathtt{u}(t))\,\mathrm{d}t + K(\mathtt{x}(T)) \\
&\geq \int_\tau^T \min_{u\in\mathbb{U}} \left(\frac{\partial V(\mathtt{x}(t),t)}{\partial x^{\mathrm{T}}} f(\mathtt{x}(t),u) + L(\mathtt{x}(t),u) \right) \mathrm{d}t \qquad (3.14) \\
&\qquad - \int_\tau^T \frac{\partial V(\mathtt{x}(t),t)}{\partial x^{\mathrm{T}}} f(\mathtt{x}(t),\mathtt{u}(t))\,\mathrm{d}t + K(\mathtt{x}(T)) \\
&= \int_\tau^T -\frac{\partial V(\mathtt{x}(t),t)}{\partial t} - \frac{\partial V(\mathtt{x}(t),t)}{\partial x^{\mathrm{T}}} f(\mathtt{x}(t),\mathtt{u}(t))\,\mathrm{d}t + K(\mathtt{x}(T)) \\
&= -\int_\tau^T \frac{\mathrm{d}V(\mathtt{x}(t),t)}{\mathrm{d}t}\,\mathrm{d}t + V(\mathtt{x}(T),T) \\
&= -\big[V(\mathtt{x}(t),t)\big]_\tau^T + V(\mathtt{x}(T),T) = V(\mathtt{x}(\tau),\tau) = V(x,\tau).
\end{aligned}$$

2. By assumption, $\mathtt{x}(t)$ is well defined for all $t \in [0,T]$. Let $x = x_0$ and $\tau = 0$. For the input $\mathtt{u}_*$, the inequality in (3.14) is an equality. Hence, $J_{[0,T]}(x_0,\mathtt{u}_*) = V(x_0,0)$, and we already established that no control achieves a smaller cost.

3. Similar to part 2: by assumption for every $\tau \in [0,T]$ and every $\mathtt{x}(\tau) = x \in \mathbb{R}^n$ the solution $\mathtt{x}$ is well defined for all $t \in [\tau,T]$. For the input $\mathtt{u}_*$, the inequality in (3.14) is an equality. Hence, the optimal cost equals $J_{[\tau,T]}(x,\mathtt{u}_*) = V(x,\tau)$ and it is attained by $\mathtt{u}_*$. Since this holds for every $x \in \mathbb{R}^n$ and every $\tau \in [0,T]$ the function $V(x,\tau)$ equals the value function $\mathcal{V}(x,\tau)$ at every $x \in \mathbb{R}^n$ and every $\tau \in [0,T]$.

■

The reasoning in the proof of this theorem (especially Part 1) is very similar to the one used by Caratheodory in his approach to the calculus of variations[1]. This approach was called the "royal road of the calculus of variations[2]".

Parts 2 and 3 are technical but this is needed because the input found by solving the minimization problem (3.12a) *pointwise* (for each x and each t) does not always give us an input $\mathtt{u}(\mathtt{x}(t),t)$ for which $\mathtt{x}(t)$ is well defined for all $t \in [0,T]$, see Exercise 3.3(c) and Exercise 3.7(e). Such cases are ruled out in parts 2 and 3. In most applications this problem does not occur, and then the above says that the so determined input is the optimal control and that $V(x,t)$ is the value function $\mathcal{V}(x,t)$.

Theorem 3.4.3 provides a *sufficient* condition for optimality: *if* we can solve the Hamilton-Jacobi-Bellman equations (3.12) and *if* the conditions of Theorem 3.4.3 are satisfied, then it is guaranteed that $\mathtt{u}_*$ is an optimal control. Recall, on the other hand, that the conditions formulated in the minimum principle (Theorem 2.5.1) are *necessary* for optimality. So in a sense, dynamic programming and the minimum principle complement each other.

Another difference between the two methods is that an optimal control $\mathtt{u}_*$ derived from the minimum principle is given as a function of state $\mathtt{x}_*$ and costate $\mathtt{p}_*$, which, after solving the Hamiltonian equations, gives us $\mathtt{u}_*(t)$ as a function of time, while in dynamic programming the optimal control is given in *state feedback* form, $\mathtt{u}(x,t)$. Applying the feedback $\mathtt{u}(x,t)$ to the system gives, what is called, the *closed-loop system*[3]

$$\dot{\mathtt{x}}_*(t) = f(\mathtt{x}_*(t), \mathtt{u}(\mathtt{x}_*(t),t)), \qquad \mathtt{x}_*(0) = x_0,$$

and its solution (if it exists) determines $\mathtt{x}_*(t)$ and the optimal control $\mathtt{u}_*(t) := \mathtt{u}(\mathtt{x}_*(t),t)$. In applications, the state feedback form is often preferred, because its implementation is way more robust. For example, if the evolution of the state is affected by disturbances, then the optimal control as a function of time, $\mathtt{u}_*(t)$, derived from the undisturbed case can easily be very different from the true optimal control, whereas the optimal control given in state feedback form, $\mathtt{u}(\mathtt{x}(t),t)$, will automatically keep track of possible disturbances in the system dynamics. Most of the following examples exhibit this feedback property.

Example 3.4.4 (Integrator with quadratic cost). Consider

$$\dot{\mathtt{x}}(t) = \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0$$

[1] C. Carathéodory. *Variationsrechnung und partielle Differentialgleichungen erster Ordnung*. B.G. Teubner, Leipzig, 1935.

[2] "Königsweg der Variationsrechnung" in H. Boerner, Caratheodorys Eingang zur variationsrechnung, Jahresbericht der Deutschen Mathematiker Vereinigung, 56 (1953), 31—58; see H.J. Pesch, Caratheodory's royal road of the Calculus of Variations: Missed exits to the Maximum Principle of Optimal Control Theory, AIMS.

[3] Controlling a system with an input $\mathtt{u}$ (optimal or not) that depends on $\mathtt{x}$ is known as *closed-loop control*, and the resulting system is known as the *closed-loop system*. Controlling the system with a given time function $\mathtt{u}(t)$ is called *open-loop* control.

with cost

$$J_{[0,T]}(x_0, \mathtt{u}) = \mathtt{x}^2(T) + \int_0^T r\mathtt{u}^2(t)\,\mathrm{d}t$$

for some $r > 0$. We allow every input, that is, $\mathbb{U} = \mathbb{R}$. Then the HJB equations (3.12) become

$$\frac{\partial V(x,t)}{\partial t} + \min_{u\in\mathbb{R}} \left(\frac{\partial V(x,t)}{\partial x} u + r u^2 \right) = 0, \qquad V(x,T) = x^2. \tag{3.15}$$

Since the term to be minimized is quadratic in u (and $r > 0$), the optimal u is where the derivative of $\frac{\partial V(x,t)}{\partial x} u + r u^2$ with respect to u is zero. This u depends on x and t,

$$\mathtt{u}(x,t) = -\frac{1}{2r} \frac{\partial V(x,t)}{\partial x}, \tag{3.16}$$

and thereby reduces the HJB equations (3.15) to

$$\frac{\partial V(x,t)}{\partial t} - \frac{1}{4r} \left(\frac{\partial V(x,t)}{\partial x} \right)^2 = 0, \qquad V(x,T) = x^2.$$

Motivated by the boundary condition we try a $V(x,t)$ that is quadratic in x for all time, so of the form $V(x,t) = x^2 P(t)$. (Granted, this is a magic step because at this point it is not clear that a quadratic form works.) This way the HJB equations simplify to

$$x^2 \dot{P}(t) - \frac{1}{4r} (2xP(t))^2 = 0, \qquad x^2 P(t) = x^2.$$

It has a common quadratic term x^2. Canceling this quadratic term x^2 gives

$$\dot{P}(t) = P^2(t)/r, \qquad P(T) = 1.$$

This is an ordinary differential equation and its solution can be found with separation of variables. The solution is

$$P(t) = \frac{r}{r + T - t}.$$

It is well defined throughout $t \in [0, T]$ and, therefore,

$$V(x,t) = x^2 \frac{r}{r+T-t} \tag{3.17}$$

is a solution of the HJB equations (3.15). Now that $V(x,t)$ is known we can compute the optimal input (3.16). It is expressed in feedback form, i.e., depending on $\mathtt{x}(t)$ (as well as on t),

$$\mathtt{u}_*(t) = \mathtt{u}(\mathtt{x}(t), t) = -\frac{1}{2r} \frac{\partial V(\mathtt{x}(t), t)}{\partial x} = -\frac{2\mathtt{x}(t)P(t)}{2r} = -\frac{\mathtt{x}(t)}{r+T-t}.$$

The optimal state $\mathtt{x}_*$ therefore satisfies the closed-loop differential equation

$$\dot{\mathtt{x}}_*(t) = \mathtt{u}_*(t) = -\frac{\mathtt{x}_*(t)}{r+T-t}.$$

This is a linear differential equation which has a well-defined solution $\mathtt{x}_*(t)$ for all $t \in [0,T]$ and all initial states, and, hence, also the above $\mathtt{u}_*(t)$ is well defined for all $t \in [0,T]$. This, finally, allows us to conclude that (3.17) *is* the value function, that the above $\mathtt{u}_*$ *is* the optimal input, and that the optimal cost *is* $J_{[0,T]}(x_0, \mathtt{u}_*) = V(x_0,0) = x_0^2/(1+T/r)$. □

The next example is a minor variation of the previous example.

Example 3.4.5 (Quadratic control). Consider the linear system

$$\dot{\mathtt{x}}(t) = \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0$$

with $\mathbb{U} = \mathbb{R}$ and cost

$$J_{[0,T]}(x_0, \mathtt{u}) = \int_0^T \mathtt{x}^2(t) + \rho^2 \mathtt{u}^2(t)\,\mathrm{d}t$$

for some $\rho > 0$. For this problem, the HJB equations (3.12) are

$$\frac{\partial V(x,t)}{\partial t} + \min_{u\in\mathbb{R}} \left(\frac{\partial V(x,t)}{\partial x} u + x^2 + \rho^2 u^2 \right) = 0, \qquad V(x,T) = 0.$$

The term to be minimized is quadratic in u. Hence, it is minimal only if the derivative with respect to u is zero. This gives

$$u = -\frac{1}{2\rho^2} \frac{\partial V(x,t)}{\partial x}.$$

So, we can rewrite the HJB equations as

$$\frac{\partial V(x,t)}{\partial t} + x^2 - \frac{1}{4\rho^2} \left(\frac{\partial V(x,t)}{\partial x} \right)^2 = 0, \qquad V(x,T) = 0. \tag{3.18}$$

This is a nonlinear partial differential equation, and this might be complicated. But it has an interesting physical dimension[4] property, which implies that

$$V(x,t) = P(t)x^2.$$

[4] Outside the scope of this book, but still: let $[x]$ denote the dimension of a quantity x. For example, $[t]$ = time. From $\dot{\mathtt{x}} = \mathtt{u}$, it follows that $[u] = [x][t]^{-1}$. Also, the expression $x^2 + \rho^2 u^2$ implies that $\rho^2 u^2$ has the same dimension as x^2. Hence, $[\rho] = [t]$, and then $[V] = [J] = [x]^2[t]$. This suggests that $V(x,t) = x^2 P(t)$. In fact, application of the *Buckingham π-theorem* (not part of this course) shows that $V(x,t)$ *must* have the form $V(x,t) = x^2 \rho G((t-T)/\rho)$ for some dimensionless function $G : \mathbb{R} \to \mathbb{R}$.

This form turns (3.18) into

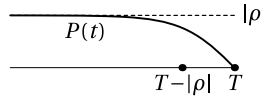
$$\dot{P}(t)x^2 + x^2 - \frac{1}{4\rho^2}(2P(t)x)^2 = 0, \qquad P(T) = 0.$$

It has a common factor x^2. Division by x^2 yields the ordinary differential equation

$$\dot{P}(t) + 1 - \frac{1}{\rho^2}P^2(t) = 0, \qquad P(T) = 0.$$

This type of differential equation is discussed at length in the next chapter. The solution is

$$P(t) = \rho \frac{e^{(T-t)/\rho} - e^{(t-T)/\rho}}{e^{(T-t)/\rho} + e^{(t-T)/\rho}}.$$



(3.19)

So for this $P(t)$, the function $V(x,t) := P(t)x^2$ solves the HJB equations (3.18). The candidate optimal control thus takes the form $\mathtt{u}(\mathtt{x}(t),t) = -\frac{1}{2\rho^2}\frac{\partial V(\mathtt{x}(t),t)}{\partial x} = -\frac{1}{\rho^2}P(t)\mathtt{x}(t)$, and the candidate optimal state satisfies the linear time-varying differential equation

$$\dot{\mathtt{x}}_*(t) = \mathtt{u}(\mathtt{x}_*(t),t) = -\frac{1}{\rho^2}P(t)\mathtt{x}_*(t), \qquad \mathtt{x}_*(0) = x_0.$$

Since $P(t)$ is well defined and bounded, it is clear that the solution $\mathtt{x}_*(t)$ is well defined for all $t \in [0,T]$. In fact the solution is

$$\mathtt{x}_*(t) = e^{-\frac{1}{\rho^2}\int_0^t P(\tau)\,d\tau}x_0.$$

Having a well-defined solution for all $t \in [0,T]$ allows us to conclude that $\mathtt{x}_*(t)$ *is* the optimal state, that

$$\mathtt{u}_*(t) := -\frac{1}{\rho^2}P(t)\mathtt{x}_*(t)$$

is the optimal control, and that $V(x_0,0) = P(0)x_0^2$ *is* the optimal cost. □

Example 3.4.6 (Quartic control). This is an uncommon application, but interesting. We again consider the integrator system $\dot{\mathtt{x}}(t) = \mathtt{u}(t)$, $\mathtt{x}(0) = x_0$, but now with the cost equal to a sum of quartics

$$J_{[0,T]}(x_0,\mathtt{u}) = \int_0^T \mathtt{x}^4(t) + \mathtt{u}^4(t)\,dt.$$

Again we assume that the input is not restricted: $\mathbb{U} = \mathbb{R}$. The HJB equations become

$$\frac{\partial V(x,t)}{\partial t} + \min_{u\in\mathbb{R}}\left(\frac{\partial V(x,t)}{\partial x}u + x^4 + u^4\right) = 0, \qquad V(x,T) = 0.$$

Encouraged by the previous example, we try a $V(x,t)$ of the form

$$V(x,t) = x^4 P(t). \tag{3.20}$$

(We will soon see that this form works.) Substitution of this form in the HJB equations yields

$$x^4 \dot{P}(t) + \min_{u\in\mathbb{R}} \left(4x^3 P(t)u + x^4 + u^4\right) = 0, \qquad x^4 P(T) = 0.$$

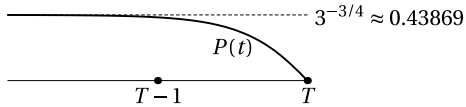
The minimizing u is $u = -\sqrt[3]{P(t)}x$. This can be obtained by setting the gradient of $4x^3P(t)u + x^4 + u^4$ with respect to u equal to zero (verify this yourself). This reduces the HJB equations to

$$x^4 \dot{P}(t) - 4x^4 P^{4/3}(t) + x^4 + x^4 P^{4/3}(t) = 0, \qquad x^4 P(T) = 0.$$

Canceling the common factor x^4 leaves us with

$$\dot{P}(t) = 3P^{4/3}(t) - 1, \qquad P(T) = 0. \tag{3.21}$$

The equation here is a simple first-order differential equation, except that no closed-form solution appears to be known. The graph of the solution (obtained numerically) is



Clearly, $P(t)$ is well defined and bounded on $(-\infty, T]$. This shows that the HJB equations have a solution of the quartic form (3.20). As $t \to -\infty$ the solution $P(t)$ converges to the equilibrium solution where $0 = 3P^{4/3} - 1$, i.e., $P = 3^{-3/4} \approx 0.43869$. For now the function $V(x,t) = x^4 P(t)$ is just a *candidate* value function. The resulting candidate optimal control

$$\mathtt{u}_*(t) = -\sqrt[3]{P(t)}\,\mathtt{x}(t) \tag{3.22}$$

is linear in $\mathtt{x}(t)$, and thus the optimal closed-loop system is linear as well,

$$\dot{\mathtt{x}}_*(t) = -\sqrt[3]{P(t)}\,\mathtt{x}_*(t), \qquad \mathtt{x}_*(0) = x_0.$$

Since $P(t)$ is bounded, also $-\sqrt[3]{P(t)}$ is bounded. Therefore the closed-loop system has a well-defined solution $\mathtt{x}_*(t)$ for every initial condition x_0 and all $t \in [0,T]$. We thus conclude that $V(x,t) = x^4 P(t)$ is the value function, that (3.22) is the optimal control and that $x_0^4 P(0)$ is the optimal cost. □

In the above examples, the functions V all turned out to be true value functions: $V = \mathcal{V}$. We need to stress that examples exist where this is not the case, see Exercise 3.3(c). The next example is one where $\mathbb{U}$ is bounded (while again $V = \mathcal{V}$).

Example 3.4.7 (Example 3.4.2 extended). We consider the system of Example 3.4.2, that is, $\dot{\mathtt{x}}(t) = \mathtt{u}(t)$, $\mathtt{x}(0) = x_0$, with the input taking values in $\mathbb{U} = [-1,1]$. The cost, however, we extend with a final cost,

$$J_{[0,T]}(x_0, \mathtt{u}) = \int_0^T \mathtt{x}(t)\,\mathrm{d}t - \alpha \mathtt{x}(T).$$

We assume that $\alpha > 0$. The HJB equations (3.12) become

$$\frac{\partial V(x,t)}{\partial t} + \min_{u\in[-1,1]} \left(\frac{\partial V(x,t)}{\partial x} u + x \right) = 0, \qquad V(x,T) = -\alpha x.$$

The function to be minimized, $\frac{\partial V(x,t)}{\partial x} u + x$, is linear in u. So the minimum is attained at one of the boundaries of $[-1,1]$. One way to proceed would be to analyze the HJB equations for the two cases $u = \pm 1$. But the equations are partial differential equations and these are often very hard to solve. We take another route: in Example 3.4.2 we analyzed a similar problem and ended up with a value function of the form

$$V(x,t) = xP(t) + Q(t)$$

for certain functions $P(t), Q(t)$. We will demonstrate that this form also works for the present problem. The HJB equations for this form simplify to

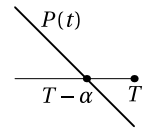
$$x\dot{P}(t) + \dot{Q}(t) + \min_{u\in[-1,1]} (P(t)u + x) = 0, \qquad xP(T) + Q(T) = -\alpha x.$$

This has to hold for all x and all t so the HJB equations hold iff

$$\dot{P}(t) = -1, \quad \dot{Q}(t) = -\min_{u\in[-1,1]} P(t)u, \qquad P(T) = -\alpha, \quad Q(T) = 0. \tag{3.23}$$

This settles $P(t)$:

$$P(t) = T - \alpha - t.$$

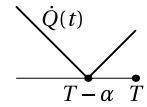


Thus, $P(t)$ is positive for $t < T-\alpha$ and negative for $t > T-\alpha$. The $u \in [-1,1]$ that minimizes $P(t)u + x$ hence is

$$\mathtt{u}_*(t) = \begin{cases} -1 & \text{if } t < T-\alpha \\ +1 & \text{if } t > T-\alpha \end{cases}. \tag{3.24}$$

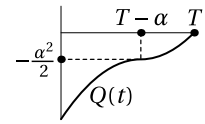
This, in turn, specializes the differential equation for $Q(t)$ as given in (3.23) to

$$\dot{Q}(t) = \begin{cases} +(T-\alpha-t) & \text{if } t < T-\alpha \\ -(T-\alpha-t) & \text{if } t > T-\alpha \end{cases}.$$



Since $Q(T) = 0$, it follows that

$$Q(t) = \begin{cases} -\frac{1}{2}(T-\alpha-t)^2 - \frac{\alpha^2}{2} & \text{if } t < T-\alpha \\ +\frac{1}{2}(T-\alpha-t)^2 - \frac{\alpha^2}{2} & \text{if } t > T-\alpha \end{cases}.$$



This function is continuously differentiable. Now all conditions of (3.23) are met and, therefore, $V(x,t) := xP(t) + Q(t)$ satisfies the HJB equations. Along the way, we also determined the candidate optimal input: (3.24). Clearly, for this input, the solution $\mathtt{x}_*(t)$ of the closed loop $\dot{\mathtt{x}}_*(t) = \mathtt{u}_*(t), \mathtt{x}_*(0) = x_0$ is well defined for all $t \in [0,T]$. Hence, (3.24) is the optimal input, the above $V(x,t)$ is the value function, and $V(x_0,0) = x_0P(0) + Q(0)$ is the optimal cost.

Does it agree with the minimum principle? The Hamiltonian is $H(x,p,u) = pu + x$, and so the Hamiltonian equation for the costate is $\dot{\mathtt{p}}_*(t) = -1, \mathtt{p}_*(T) = -\alpha$. Clearly, this means that $\mathtt{p}_*(t) = T-\alpha-t$. Now the input $\mathtt{u}_*(t)$ that minimizes the Hamiltonian

$$H(\mathtt{x}_*(t), \mathtt{p}_*(t), u) = (T-\alpha-t)u + \mathtt{x}_*(t)$$

agrees with what we found earlier: (3.24). But, of course, the fundamental difference is that the minimum principle *assumes* the existence of an optimal control, whereas satisfaction of the HJB equations proves that the control *is* optimal. □

The examples might give the impression that dynamic programming is superior to the minimum principle. In applications, however, it is often the other way around. The thing is that the equations needed in the minimum principle (i.e., the Hamiltonian equations) are *ordinary* differential equations, and numerical routines exist that are quite efficient in solving these equations. The HJB equations, in contrast, consist of a *partial* differential equation and a boundary condition. Furthermore, the standard HJB theory requires a higher degree of smoothness than the minimum principle.

For restricted sets such as $\mathbb{U} = [0,1]$, the value function often is continuously differentiable everywhere (see Example 3.4.7 and Exercise 3.7) but there are also cases where it is not even differentiable everywhere (let alone continuously differentiable). Here is an example:

Example 3.4.8 (Non-smooth value functions). Let

$$\dot{\mathtt{x}}(t) = -\mathtt{x}(t)\mathtt{u}(t), \qquad \mathtt{x}(0) = x_0, \qquad \mathbb{U} = [0,1],$$

and take $J_{[0,T]}(x_0, \mathtt{u}) = \mathtt{x}(T)$. So, we have to make $\mathtt{x}(T)$ as small (negative) as possible. From the differential equation we can deduce that one optimal input as a function of state x and time t is

$$\mathtt{u}(x,t) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases},$$

and then the value function follows as

$$\mathcal{V}(x,t) = \mathtt{x}(T) = \begin{cases} \mathrm{e}^{t-T}\, x & \text{if } x > 0 \\ x & \text{if } x \leq 0 \end{cases}.$$

This value function is *not* continuously differentiable with respect to x at $x = 0$, and, therefore, the standard theory does not apply. It does satisfy the HJB equation (3.12a) at all x where it is continuously differentiable (at all $x \neq 0$):

$$\frac{\partial \mathcal{V}(x,t)}{\partial t} + \min_{u \in [0,1]} \left(\frac{\partial \mathcal{V}(x,t)}{\partial x}(-xu) \right) = \begin{cases} \mathrm{e}^{t-T}\, x + \mathrm{e}^{t-T}(-x) = 0 & \text{if } x > 0 \\ 0 + 0 = 0 & \text{if } x < 0 \end{cases}.$$

□

3.5 Connection with the Minimum Principle

In Chapter 2, we claimed that the initial costate $\mathtt{p}_*(0)$ measures the sensitivity of the optimal cost with respect to changes in the initial state (end of § 2.3, see also Example 2.6.3). This connection can now be proved. In fact, it is a by-product of a more general connection between the solution of the HJB equations (assuming it equals the value function) and the costate of the minimum principle. First of all, we note that the HJB equation (3.12a) can be expressed in terms of the Hamiltonian $H(x,p,u) := p^{\mathrm{T}} f(x,u) + L(x,u)$ as

$$\frac{\partial V(x,t)}{\partial t} + \min_{u \in \mathbb{U}} H\left(x, \frac{\partial V(x,t)}{\partial x}, u\right) = 0,$$

(whence the name Hamilton-Jacobi). This suggests that the costate is closely related to $\partial V(x,t)/\partial x$. In fact, since we know that $\mathtt{p}_*(T) = \partial K(\mathtt{x}_*(T))/\partial x = \partial V(\mathtt{x}_*(T), T)/\partial x$ we conjecture that

$$\mathtt{p}_*(t) = \frac{\partial V(\mathtt{x}_*(t), t)}{\partial x} \qquad \text{for all } t \in [0,T].$$

Under mild assumptions that is indeed the case. To avoid technicalities, we derive this connection only for $\mathbb{U} = \mathbb{R}^m$ and value functions that are C^2.

Theorem 3.5.1 (Connection between costate and value function). Assume $f(x,u)$, $L(x,u)$, $K(x)$ are all C^1. Let $\mathbb{U} = \mathbb{R}^m$ and suppose there is a C^2 function $V : \mathbb{R}^n \times [0,T] \to \mathbb{R}$ that satisfies the HJB equations

$$\frac{\partial V(x,t)}{\partial t} + \min_{u \in \mathbb{U}} H\Big(x, \frac{\partial V(x,t)}{\partial x}, u\Big) = 0, \qquad V(x,T) = K(x)$$

for all $x \in \mathbb{R}^n$ and all $t \in [0,T]$. Denote, for each x,t, one possible minimizer as $\mathtt{u}_*(x,t)$, and assume that all conditions of Theorem 3.4.3 are satisfied. In particular that $V(x,t)$ equals the value function $\mathcal{V}(x,t)$ and that the differential equation $\dot{\mathtt{x}}_*(t) = f(\mathtt{x}_*(t), \mathtt{u}_*(\mathtt{x}_*(t),t))$ has a well-defined solution $\mathtt{x}_*(t)$ for all $t \in [\tau, T]$ and all $\mathtt{x}(\tau) \in \mathbb{R}^n$. Then $\mathtt{p}_*(t)$ defined as

$$\mathtt{p}_*(t) = \frac{\partial V(\mathtt{x}_*(t),t)}{\partial x} \tag{3.25}$$

is the solution of the Hamiltonian costate equation

$$\dot{\mathtt{p}}_*(t) = -\frac{\partial H(\mathtt{x}_*(t), \mathtt{p}_*(t), \mathtt{u}_*(\mathtt{x}_*(t),t))}{\partial x}, \qquad \mathtt{p}_*(T) = \frac{\partial K(\mathtt{x}_*(T))}{\partial x}. \tag{3.26}$$

In particular, $\mathtt{p}_*(0) = \partial V(x_0,0)/\partial x_0$.

Proof. Let $H(x,p,u) = p^{\mathrm{T}} f(x,u) + L(x,u)$. By definition, the minimizing $\mathtt{u}_*(x,t)$ satisfies the HJB equation

$$\frac{\partial V(x,t)}{\partial t} + H\Big(x, \frac{\partial V(x,t)}{\partial x}, \mathtt{u}_*(x,t)\Big) = 0.$$

In the rest of this proof, we drop all function arguments. The partial derivative of the previous expression with respect to (row vector) x^{T} yields

$$\frac{\partial^2 V}{\partial x^{\mathrm{T}} \partial t} + \frac{\partial H}{\partial x^{\mathrm{T}}} + \frac{\partial H}{\partial p^{\mathrm{T}}} \frac{\partial^2 V}{\partial x^{\mathrm{T}} \partial x} + \underbrace{\frac{\partial H}{\partial u^{\mathrm{T}}} \frac{\partial \mathtt{u}_*}{\partial x^{\mathrm{T}}}}_{0} = 0.$$

The underbraced term is zero because $\mathtt{u}_*$ minimizes the Hamiltonian. Using this expression and the fact that $\frac{\partial H}{\partial p} = f$, we find that

$$\frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial V(\mathtt{x}(t),t)}{\partial x} = \frac{\partial^2 V}{\partial t \partial x} + \frac{\partial^2 V}{\partial x \partial x^{\mathrm{T}}} f = \frac{\partial^2 V}{\partial t \partial x} + \frac{\partial^2 V}{\partial x^{\mathrm{T}} \partial x} \frac{\partial H}{\partial p} = -\frac{\partial H}{\partial x}.$$

Because $V(x,T) = K(x)$ for all x, we also have $\frac{\partial V(\mathtt{x}(T),T)}{\partial x} = \frac{\partial K(\mathtt{x}(T))}{\partial x}$. Hence, $\mathtt{p}_*(t) := \frac{\partial V(\mathtt{x}_*(t),t)}{\partial x}$ satisfies the costate equation (3.26) for all time. ∎

Example 3.5.2. We apply Theorem 3.5.1 to the optimal control problem of Example 3.4.5. For simplicity, we take $\rho = T = 1$. Then Example 3.4.5 says that

$$V(x,t) = x^2 P(t) \quad \text{where} \quad P(t) = \frac{\mathrm{e}^{1-t} - \mathrm{e}^{t-1}}{\mathrm{e}^{1-t} + \mathrm{e}^{t-1}}.$$

Using this and the formula for $\mathrm{x}_*(t)$ (determined in Example 2.5.4), we find that

$$\begin{aligned}\mathrm{p}_*(t) = \frac{\partial V(\mathrm{x}_*(t),t)}{\partial x} &= 2\mathrm{x}_*(t)P(t)\\ &= 2\frac{x_0}{\mathrm{e}+\mathrm{e}^{-1}}\left(\mathrm{e}^{1-t}+\mathrm{e}^{t-1}\right)\frac{\mathrm{e}^{1-t}-\mathrm{e}^{t-1}}{\mathrm{e}^{1-t}+\mathrm{e}^{t-1}}\\ &= 2\frac{x_0}{\mathrm{e}+\mathrm{e}^{-1}}\left(\mathrm{e}^{1-t}-\mathrm{e}^{t-1}\right).\end{aligned}$$

This equals the $\mathrm{p}_*(t)$ as determined in Example 2.5.4. □

3.6 Infinite Horizon Optimal Control and Lyapunov Functions

For *infinite* horizon optimal control problems, there is an interesting connection with Lyapunov functions and stabilizing inputs. Given a system of differential equations and a class of inputs

$$\dot{\mathrm{x}}(t) = f(\mathrm{x}(t),\mathrm{u}(t)), \qquad \mathrm{x}(0) = x_0, \qquad \mathrm{u}:[0,\infty)\to\mathbb{U}, \tag{3.27a}$$

the *infinite horizon optimal control problem* is to minimize over all inputs the infinite horizon cost

$$J_{[0,\infty)}(x_0,\mathrm{u}) := \int_0^\infty L(\mathrm{x}(t),\mathrm{u}(t))\,\mathrm{d}t. \tag{3.27b}$$

The only difference with the previous formulation is the cost function. The integral that defines the cost is now over all $t>0$, and the "final" cost $K(\mathrm{x}(\infty))$ has been dropped because in applications we normally send the state to a unique equilibrium $\mathrm{x}(\infty) := \lim_{t\to\infty}\mathrm{x}(t)$, and thus all such controls achieve the same final cost (i.e., the final cost would not affect the optimal control).

As before we define the *value function* as

$$\mathcal{V}(x,\tau) = \inf_{\mathrm{u}:[\tau,\infty)\to\mathbb{U}} J_{[\tau,\infty)}(x,\mathrm{u}) \tag{3.28}$$

in which $J_{[\tau,\infty)}(x,\mathrm{u}) := \int_\tau^\infty L(\mathrm{x}(t),\mathrm{u}(t))\,\mathrm{d}t$ for $\mathrm{x}(\tau)=x$. Because of the infinite horizon, however, the value function no longer depends on τ (see Exercise 3.14(a)), and so we can simply write

$$\mathcal{V}(x) = \inf_{\mathrm{u}:[0,\infty)\to\mathbb{U}} J_{[0,\infty)}(x,\mathrm{u}). \tag{3.29}$$

The derivative of $\mathcal{V}(x)$ with respect to time vanishes, and thus the HJB equation (3.12a) simplifies to

$$\min_{u\in\mathbb{U}}\left(\frac{\partial V(x)}{\partial x^{\mathrm{T}}}f(x,u)+L(x,u)\right)=0. \tag{3.30}$$

As we will soon see, this equation typically has more than one solution V, and, clearly, at most one of them will be the value function $\mathcal{V}$. The next example suggests that the "right" solution gives us a *stabilizing input*[5], and a Lyapunov function for that equilibrium.

Example 3.6.1 (Quartic control—design of optimal stabilizing inputs and Lyapunov function). Consider the infinite horizon optimal control problem with

$$\dot{\mathrm{x}}(t) = \mathrm{u}(t), \quad \mathrm{x}(0) = x_0, \quad \mathbb{U} = \mathbb{R}, \quad J_{[0,\infty)}(x_0, \mathrm{u}) = \int_0^\infty \mathrm{x}^4(t) + \mathrm{u}^4(t)\,\mathrm{d}t.$$

For this problem the infinite horizon HJB equation (3.30) is

$$\min_{u\in\mathbb{R}} \left(\frac{\partial V(x)}{\partial x} u + x^4 + u^4 \right) = 0.$$

The solution u of the above minimization problem is

$$u = -\left(\frac{1}{4} \frac{\partial V(x)}{\partial x} \right)^{1/3} \tag{3.31}$$

(verify this yourself), and then the HJB equation becomes

$$\left(\frac{1}{4} - 1\right)\left(\frac{1}{4}\right)^{1/3} \left(\frac{\partial V(x)}{\partial x} \right)^{4/3} + x^4 = 0.$$

This looks rather ugly but actually it says that

$$\frac{\partial V(x)}{\partial x} = \pm 4(3^{-3/4}) x^3,$$

and, therefore, all possible solutions are

$$V_{\text{all}}(x) = \pm 3^{-3/4} x^4 + d \tag{3.32}$$

with d some integration constant. Which $V_{\text{all}}(x)$ gives us a stabilizing input (3.31)? First of all, we can take $d = 0$ without loss of generality in the sense that it does not affect the control (3.31). Since $L(x, u) = x^4 + u^4 \geq 0$ we know that the value function $\mathcal{V}$ defined in (3.29) is nonnegative; hence we consider the nonnegative option of (3.32), that is,

$$V(x) = 3^{-3/4} x^4.$$

We claim that this V is a Lyapunov function for the equilibrium $\bar{x} = 0$ of the closed-loop system $\dot{\mathrm{x}}(t) = \mathrm{u}(t)$ for the control equal to the candidate optimal control defined in (3.31),

$$\mathrm{u}_*(t) = -\left(\frac{1}{4} \frac{\partial V(\mathrm{x}(t))}{\partial x}\right)^{1/3} = -3^{-1/4} \mathrm{x}(t).$$

[5] A *stabilizing* input is an input that steers the state to a given equilibrium. Better would have been to call it "asymptotically stabilizing" input or "attracting" input, but "stabilizing" is the standard in the literature.

Indeed, $\bar{x} = 0$ is an equilibrium of the closed-loop system, and V clearly is C^1 and positive definite, and *by construction* the HJB equation (3.30) gives us that

$$\dot{V}(\mathtt{x}(t)) = \frac{\partial V(\mathtt{x}(t))}{\partial x^{\mathrm{T}}} f(\mathtt{x}(t), \mathtt{u}_*(t)) = -L(\mathtt{x}(t), \mathtt{u}_*(t)) = -(\mathtt{x}^4(t) + \mathtt{u}_*^4(t)) \quad (3.33)$$

which is < 0 for all $\mathtt{x}(t) \neq 0$. Hence V is a strong Lyapunov function of the closed-loop system with equilibrium $\bar{x} = 0$ and, therefore, it is asymptotically stable at $\bar{x} = 0$ (see Theorem B.3.2). The closed-loop system is even *globally* asymptotically stable because all conditions of Theorem B.3.5 (p. 216) are met. For this reason the control input $\mathtt{u}_*$ is called a *stabilizing* input. In fact it is the input that minimizes the cost $J_{[0,\infty)}(x_0, \mathtt{u})$ over all inputs that stabilize the system! Indeed, for every input $\mathtt{u}$ that steers the state to zero we have the inequality

$$\begin{aligned} J_{[0,\infty)}(x_0, \mathtt{u}) &= \int_0^\infty L(\mathtt{x}(t), \mathtt{u}(t))\,\mathrm{d}t \\ &\geq \int_0^\infty -\frac{\partial V(\mathtt{x}(t))}{\partial x^{\mathrm{T}}} f(\mathtt{x}(t), \mathtt{u}(t))\,\mathrm{d}t \qquad \text{because of (3.30)} \\ &= \int_0^\infty -\dot{V}(\mathtt{x}(t))\,\mathrm{d}t = V(x_0) - \underbrace{V(\mathtt{x}(\infty))}_{0} = V(x_0) = 3^{-3/4} x_0^4, \end{aligned}$$

while in view of (3.33) equality holds if $\mathtt{u} = \mathtt{u}_*$. □

In this example we demonstrated that $\mathtt{u}(t) := -3^{-1/4}\mathtt{x}(t)$ minimizes the cost $J_{[0,\infty)}(x_0, \mathtt{u})$ over all *stabilizing* inputs, not necessarily over all inputs (see, however, Exercise 18). In applications closed-loop stability is such a crucial property that we prefer to consider only stabilizing inputs. We thus define:

Definition 3.6.2 (Optimal control with stability). Given a system $\dot{\mathtt{x}}(t) = f(\mathtt{x}(t), \mathtt{u}(t)), \mathtt{x}(0) = x_0$ and candidate closed-loop equilibrium $\bar{x}$, the infinite horizon *optimal control problem with stability* is to minimize $J_{[0,\infty)}(x_0, \mathtt{u})$ over all inputs $\mathtt{u} : [0,\infty) \to \mathbb{U}$ that *stabilize* the system (meaning the solution $\mathtt{x}(t)$ of $\dot{\mathtt{x}}(t) = f(\mathtt{x}(t), \mathtt{u}(t)), \mathtt{x}(0) = x_0$ is defined for all $t > 0$, and $\lim_{t\to\infty} \mathtt{x}(t) = \bar{x}$). □

The following proposition is a generalization of the previous example. We assume in this result that the equilibrium is the origin, $\bar{x} = 0$.

Proposition 3.6.3 (Optimal control with stability & Lyapunov functions). Consider the optimal control problem (3.27), and assume that $f(x,u)$ and $L(x,u)$ are C^1 and that $f(0,0) = 0, L(0,0) = 0$, and $L(x,u) \geq 0$ for all $x \in \mathbb{R}^n, u \in \mathbb{U}$. Then

1. $\mathcal{V}(0) = 0$, and $\mathcal{V}(x) \geq 0$ for all $x \neq 0$. (Possibly $\mathcal{V}(x) = \infty$.)
2. Suppose V is a C^1 solution of the HJB equation (3.30) and that $V(0) = 0$ and $V(x) > 0$ for all $x \neq 0$. Let $\hat{\mathtt{u}}(x)$ be a minimizer of (3.30), i.e.,

$$\frac{\partial V(x)}{\partial x^{\mathrm{T}}} f(x, \hat{\mathtt{u}}(x)) + L(x, \hat{\mathtt{u}}(x)) = 0 \qquad \forall x \in \mathbb{R}^n.$$

If $f(x,\hat{u}(x))$ is Lipschitz continuous on a neighborhood of $\bar{x}=0$ then the closed-loop system $\dot{x}(t) = f(x(t),\hat{u}(x(t)), x(0) = x_0$ has as a well defined solution $x(t)$ for all x_0 sufficiently close to $\bar{x} = 0$ for all $t > 0$, and the closed-loop system at equilibrium $\bar{x}=0$ is stable.

3. Suppose, in addition, that $L(x,\hat{u}(x)) > 0$ for all $x \neq 0$. Then the closed-loop system is asymptotically stable, and for all x_0 sufficiently close to $\bar{x} = 0$ the input $u_*(t) := \hat{u}(x(t))$ solves the infinite horizon optimal control problem with stability. Moreover the optimal cost then equals $V(x_0)$.

Proof. This proof refers to several definitions and results from Appendix B.

1. Since $L(x,u) \geq 0$ it is immediate that $\mathcal{V}(x) \geq 0$. Also $\mathcal{V}(0) = 0$ because for $x_0 = 0$ the control $u(t) = 0$ achieves $x(t) = 0$ for all time, and, hence, $L(x(t),u(t)) = 0$ for all time.

2. Lipschitz continuity assures existence and uniqueness of the solution $x(t)$ (for x_0 sufficiently close to 0), see Theorem B.1.3. The function V is a Lyapunov function for the equilibrium $\bar{x}=0$ because it is C^1 and positive definite (by assumption), and $\dot{V}(x) \leq 0$ because

$$\dot{V}(x(t)) = \frac{\partial V(x(t))}{\partial x^{\mathrm{T}}} f(x(t),\hat{u}(x(t))) = -L(x(t),\hat{u}(x(t)) \leq 0.$$

Theorem B.3.2 now guarantees that the equilibrium is stable.

3. Then V is a *strong* Lyapunov function, so then the equilibrium is *asymptotically* stable (Theorem B.3.2). As in the previous example we have for every stabilizing input u that

$$\begin{aligned} J_{[0,\infty)}(x_0,u) &= \int_0^\infty L(x(t),u(t))\,\mathrm{d}t \\ &\geq \int_0^\infty -\frac{\partial V(x(t))}{\partial x^{\mathrm{T}}} f(x(t),u(t))\,\mathrm{d}t \qquad \text{because of (3.30)} \\ &= \int_0^\infty -\dot{V}(x(t))\,\mathrm{d}t = V(x_0) - \underbrace{V(x(\infty))}_{0} = V(x_0), \end{aligned}$$

and equality holds if $u(t) = \hat{u}(x(t))$. ∎

3.7 Exercises

3.1 *Maximization.* Consider the system

$$\dot{x}(t) = f(x(t),u(t)), \qquad x(0) = x_0.$$

We want to *maximize* the cost

$$\int_0^T L_0(x(t),u(t))\,\mathrm{d}t + K_0(x(T)).$$

Find a new cost such that the maximization problem becomes a minimization problem, in the sense that an input $\mathtt{u}$ solves the minimization problem iff it solves the maximization problem. Also comment how the associated two costs are related? (Note: this exercise is trivial.)

3.2 *An optimal control problem that has no solution.* Not every optimal control problem is solvable. Consider the system $\dot{\mathtt{x}}(t) = \mathtt{u}(t), x_0 = 1$ with cost

$$J_{[0,T]}(x_0, \mathtt{u}) = \int_0^1 \mathtt{x}^2(t)\,\mathrm{d}t,$$

and $\mathbb{U} = \mathbb{R}$.

(a) Determine the value function (from the definition, not from the HJB equations).

(b) Show that the value function does *not* satisfy the HJB equations.

3.3 *The solution V need not equal the value function $\mathcal{V}$.* Consider again the optimal control problem of Exercise 2.1:

$$\dot{\mathtt{x}}(t) = \mathtt{x}(t)\mathtt{u}(t), \qquad \mathtt{x}(0) = x_0$$

with cost function

$$J_{[0,T]}(x_0, \mathtt{u}) = \int_0^T \mathtt{x}^2(t) + \mathtt{u}^2(t)\,\mathrm{d}t + 2\mathtt{x}(T),$$

and with the input free to choose: $\mathbb{U} = \mathbb{R}$.

(a) Determine a solution $V(x,t)$ of (3.12), and a candidate optimal control $\mathtt{u}_*(\mathtt{x}(t), t)$ (possibly still depending on $\mathtt{x}(t)$).
[Hint: assume that $V(x,t)$ does not depend on t, i.e., that it has the form $V(x,t) = Q(x)$ for some function Q.]

(b) Now let $x_0 = 1$ and $T > 0$. Show that $V(x_0, 0)$ is the optimal cost and determine the optimal control $\mathtt{u}_*(t)$ explicitly as a function of time.
[Hint: have look at Example B.1.5.]

(c) Now let $x_0 = -1$ and $T = 2$. Show that $V(x,t)$ and $\mathtt{u}_*(\mathtt{x}(t), t)$ are *not* the value function and *not* the optimal input! (In other words: what condition of Theorem 3.4.3 fails here?)

3.4 *Direct solution.* Even though dynamic programming and the HJB equations are powerful concepts, we should always aim for simpler approaches. Consider the system

$$\dot{\mathtt{x}}(t) = \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0$$

with cost function $J_{[0,T]}(x_0, \mathtt{u}) = \int_0^T \mathtt{x}^2(t)\,\mathrm{d}t$. The problem is to minimize this with bounded inputs

$$0 \le \mathtt{u}(t) \le 1.$$

(a) Use your common sense to solve the problem for $x_0 \geq 0$.

(b) What is the cost for the optimal control found in (a)?

(c) Use (b) to find a candidate solution $V(x,t)$ of the HJB equations for $x \geq 0$. Verify that this candidate solution satisfies (3.12) for $x > 0$.

(d) Use your common sense to solve the minimization problem for $x_0 < 0$. What are the minimal costs now?

3.5 *Economic application.* The capital $\mathtt{x}(t) \geq 0$ of an economy at any moment t is divided into two parts: $\mathtt{u}(t)\mathtt{x}(t)$ and $(1-\mathtt{u}(t))\mathtt{x}(t)$, with

$$\mathtt{u}(t) \in [0,1].$$

The first part, $\mathtt{u}(t)\mathtt{x}(t)$, is for investments and contributes to the increase of capital according to the formula

$$\dot{\mathtt{x}}(t) = \mathtt{u}(t)\mathtt{x}(t), \qquad \mathtt{x}(0) > 0.$$

The other part, $(1-\mathtt{u}(t))\mathtt{x}(t)$, is for consumption and is evaluated by the "satisfaction"

$$\hat{J}_{[0,3]}(x_0,\mathtt{u}) := -\mathtt{x}(3) + \int_0^3 (1-\mathtt{u}(t))\mathtt{x}(t)\,\mathrm{d}t.$$

We want to *maximize* the satisfaction.

(a) Let V be a function of the form $V(x,t) = Q(t)x$, and with it determine the HJB equations.

(b) Express the candidate optimal $\mathtt{u}_*(t)$ as a function of $Q(t)$ [Hint: $\mathtt{x}(t)$ is always positive].

(c) Determine $Q(t)$ for all $t \in [0,3]$.

(d) Determine the optimal $\mathtt{u}_*(t)$ explicitly as a function of time, and argue that this is the true optimal control (so not just the candidate optimal control).

(e) What is the maximal satisfaction $\hat{J}_{[0,3]}(x_0,\mathtt{u}_*)$?

3.6 *Weird problem.* Consider the system

$$\dot{\mathtt{x}}(t) = \mathtt{x}(t) + \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0, \qquad \mathbb{U} = \mathbb{R}$$

with cost

$$J_{[0,T]}(x_0,\mathtt{u}) = \tfrac{1}{2}\mathtt{x}^2(T) + \int_0^T -\mathtt{x}^2(t) - \mathtt{x}(t)\mathtt{u}(t)\,\mathrm{d}t. \tag{3.34}$$

(a) Solve the HJB equations. [Hint: try the special form $V(x,t) = Q(x)$.]

(b) Determine an optimal input $\mathtt{u}(t)$.

(c) Determine the optimal cost.

(d) Show directly from (3.34) that *every* input results in the same cost, $J_{[0,T]}(x_0, \mathtt{u}) = \frac{1}{2}x_0^2$. [Hint: use that $\mathtt{u}(t) = \dot{\mathtt{x}}(t) - \mathtt{x}(t)$.]

(e) *About the technicalities of Theorem 3.4.3.* This optimal control problem is a good one to illustrate why we have to be so technical in parts 2 and 3 of Theorem 3.4.3. The HJB equation (3.12a) for this problem is that $\min_{u\in\mathbb{R}} 0 = 0$ for every x, t. Clearly this means that every $\mathtt{u}(x,t)$ solves the HJB equation (3.12a), for instance

$$\mathtt{u}(x,t) = \begin{cases} -x + \frac{1}{T-t} & \text{if } 0 \le t < T \\ 0 & \text{if } t = T \end{cases},$$

and

$$\mathtt{u}(x,t) = -x - x^2,$$

and

$$\mathtt{u}(x,t) = \begin{cases} 1 & \text{if } t \text{ is a rational number} \\ 0 & \text{if } t \text{ is an irrational number} \end{cases}.$$

Why are these three choices problematic? (For the second input Example B.1.5 may be useful.)

3.7 *Value function.* Let $T > 0$ and consider the system with bounded input

$$\dot{\mathtt{x}}(t) = \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0, \qquad \mathtt{u}(t) \in [-1, 1],$$

and define the family of costs

$$J_{[\tau,T]}(\mathtt{x}(\tau), \mathtt{u}) = \mathtt{x}^2(T), \qquad \tau \in [0, T].$$

(a) Argue that

$$\mathtt{u}_*(t) := \begin{cases} +1 & \text{if } \mathtt{x}(t) < 0 \\ 0 & \text{if } \mathtt{x}(t) = 0 \\ -1 & \text{if } \mathtt{x}(t) > 0 \end{cases}$$

is an optimal control for $J_{[\tau,T]}(\mathtt{x}(\tau), \mathtt{u})$ for every $\tau \in [0, T]$ and $\mathtt{x}(\tau)$.

(b) Use the above optimal input to determine the value function $\mathcal{V}(x,t)$. (Use the definition of value function, do not use the HJB equations.)

(c) Verify that this $\mathcal{V}(x,t)$ satisfies the HJB equations (3.12).

3.8 *Quartic control.* Consider the problem of Example 3.4.6 on quartic control. Argue that

$$\frac{\partial V(\mathtt{x}_*(t), t)}{\partial x}\mathtt{u}_*(t) + \mathtt{x}_*^4(t) + \mathtt{u}_*^4(t)$$

equals $\mathtt{x}_*^4(T)$ for all $t \in [0, T]$.

3.9 *Exploiting physical dimensions.* Consider the nonlinear system

$$\dot{\mathtt{x}}(t) = \mathtt{x}(t)\mathtt{u}(t), \quad \mathtt{x}(0) = x_0, \quad \mathbb{U} = \mathbb{R},$$

with a quadratic cost

$$J_{[0,T]}(x_0, \mathtt{u}) = \int_0^T \mathtt{x}^2(t) + \rho^2 \mathtt{u}^2(t)\,\mathrm{d}t$$

that depends on a positive number ρ. The HJB equation (3.12a) is a partial differential equation in $V(x,t)$ which is in general hard to solve. However, by exploiting physical dimensions one can sometimes get an idea of the form of $V(x,t)$. With some basic knowledge of dimensional analysis this goes as follows: let $[x]$ be the dimension of a quantity x, for instance $[t]$ is time. The identity $\dot{\mathtt{x}} = \mathtt{x}\mathtt{u}$ implies that $\mathtt{u}$ has dimension $[\mathtt{u}] = [t]^{-1}$. Furthermore, in order for the sum $\mathtt{x}^2 + \rho^2\mathtt{u}^2$ to be well defined, we need that $\mathtt{x}^2$ and $\rho^2\mathtt{u}^2$ have the same dimension. Hence $[\rho] = [x][t]$, and then we have $[V] = [J] = [x]^2[t]$. Thus, both $V/(x\rho)$ and $(t-T)x/\rho$ are *dimensionless*. The *Buckingham* π*-theorem* (not covered in this book) claims that $V/(x\rho)$ is a function of $(t-T)x/\rho$. That is to say, it claims that the value function $V(x,t)$ for this problem must be of the special form

$$V(x,t) = x\rho G\big(\tfrac{(t-T)x}{\rho}\big)$$

for some function $G : \mathbb{R} \to \mathbb{R}$. We can verify that this is indeed correct:

(a) Show that the HJB equations (3.12) for this form reduce to an ordinary differential equation with "initial" condition,

$$G'(z) + 1 - \tfrac{1}{4}\big(G(z) + zG'(z)\big)^2 = 0, \qquad G(0) = 0.$$

An analytic solution of this ODE is difficult to obtain. Figure 3.4 shows the graphs of $G(z)$ and $G(z) + zG'(z)$ obtained numerically. Both $G(z)$ and $G(z) + zG'(z)$ converge to $+2$ as $z \to -\infty$ and to -2 as $z \to \infty$.

(b) Express the optimal control $\mathtt{u}_*(t)$ in terms of $\mathtt{x}(t), \rho, G$ and $z := (t-T)\mathtt{x}(t)\rho$.

(c) Argue, using the graphs, that the closed-loop system $\dot{\mathtt{x}}_*(t) = \mathtt{x}_*(t)\mathtt{u}_*(t)$, $\mathtt{x}_*(0) = x_0$ has a well-defined solution on $[0,T]$, and then conclude that the optimal cost is $V(x_0,0) = x_0\rho G(-Tx_0/\rho)$.

3.10 *Weierstrass necessary condition.* As we saw in Example 2.4.1, the calculus of variations problem with free endpoint equals the optimal control problem with

$$\dot{\mathtt{x}}(t) = \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0, \qquad \mathbb{U} = \mathbb{R}, \qquad L(x,u) = F(x,u).$$

In what follows we assume that the solution $V(x,t)$ of the HJB equations (3.12) exists, and that it is the value function, and that the optimal $\mathtt{x}_*, \mathtt{u}_*$ are sufficiently smooth.

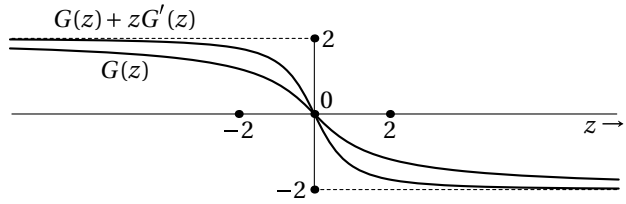


FIGURE 3.4: Graphs of $G(z)$ and $G(z)+zG'(z)$ of the solution of the differential equation $G'(z)+1-\frac{1}{4}\big(G(z)+zG'(z)\big)^2=0$, $G(0)=0$. See Exercise 3.9.

(a) Determine the HJB equations (3.12) for this calculus of variations problem formulated as an optimal control problem.

(b) Show that

$$\frac{\partial V(\mathrm{x}_*(t),t)}{\partial x}+\frac{\partial F(\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))}{\partial \dot{x}}=0.$$

(c) Show that

$$F(\mathrm{x}_*(t),u)-F(\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))-(u-\dot{\mathrm{x}}_*(t))^{\mathrm{T}}\frac{\partial F(\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))}{\partial \dot{x}}\geq 0 \quad (3.35)$$

for all $u\in\mathbb{R}^n$ and all $t\in(0,T)$.

Inequality (3.35) is known as the *Weierstrass necessary condition* for optimality of calculus of variations problems.

3.11 *Optimal capacitor charging.* In Exercise 2.9 we determined a voltage input u that charges a capacitor from zero voltage, $\mathrm{x}(0)=0$, to a desired voltage, $\mathrm{x}(T)=1$, with minimal energy loss. We continue with this problem. As in Exercise 2.9, we take $\mathbb{U}=\mathbb{R}$, and we assume that capacitance and resistance are both equal to one. In that case the relation between x and u is

$$\dot{\mathrm{x}}(t)=-\mathrm{x}(t)+\mathrm{u}(t),\qquad \mathrm{x}(0)=0. \quad (3.36)$$

As cost we take

$$J_{[0,T]}(0,\mathrm{u})=\frac{(\mathrm{x}(T)-1)^2}{\beta}+\int_0^T(\mathrm{u}(t)-\mathrm{x}(t))^2\,\mathrm{d}t. \quad (3.37)$$

Here, as before, the term $\int_0^T(\mathrm{u}(t)-\mathrm{x}(t))^2\,\mathrm{d}t$ is the energy loss, but now we also added a final cost, $(\mathrm{x}(T)-1)^2/\beta$. This final cost depends on a positive tuning parameter β that we can choose. We do not insist on having $\mathrm{x}(T)=1$, but the final cost puts a penalty on the deviation of the final voltage $\mathrm{x}(T)$ from 1 (our desired voltage). For instance, if $\beta\approx 0$ then we expect $\mathrm{x}(T)\approx 1$.

(a) Assume the solution of the HJB equations (3.12) is of the form

$$V(x,t) = (x-1)^2 P(t).$$

Derive an ordinary differential equation for $P(t)$ in terms of $P, \dot{P}, t, T, \beta$ (and nothing else).

(b) Verify that

$$P(t) = \frac{1}{\beta + T - t},$$

and that

$$\mathtt{x}_*(t) = \frac{t}{\beta + T}, \quad \mathtt{u}_*(t) = \frac{t+1}{\beta + T}, \quad \text{and} \quad J_{[0,T]}(0, \mathtt{u}_*) = \frac{1}{\beta + T},$$

respectively, are the optimal state, optimal control, and optimal cost.

(c) Determine $\lim_{\beta \downarrow 0} \mathtt{x}_*(T)$, and explain on the basis of the cost (3.37) why this makes sense.

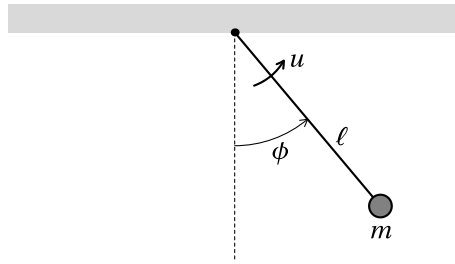


FIGURE 3.5: A pendulum with a torque u. See Exercise 3.12.

3.12 *Optimal stabilization of a pendulum.* Consider a mass m hanging from a ceiling on a thin massless rod of length ℓ, see Fig. 3.5. We can control the pendulum with a torque. The standard mathematical model in the absence of damping is

$$m\ell^2 \ddot{\phi}(t) + g m \ell \sin(\phi(t)) = \mathtt{u}(t),$$

where ϕ is the angle between pendulum and the vertical hanging position, $\mathtt{u}$ is the applied torque, m is the mass of the pendulum, ℓ is the length of the pendulum, and g is the gravitational acceleration.

The objective is to determine a torque $\mathtt{u}$ that stabilizes the pendulum to the vertical hanging equilibrium $\phi = 2k\pi, \dot{\phi} = 0$. This, by definition, means that $\mathtt{u}$ is such that

$$\lim_{t \to \infty} \phi(t) = 2k\pi, \qquad \lim_{t \to \infty} \dot{\phi}(t) = 0.$$

We consider the stabilization "optimal" if the input stabilizes and minimizes

$$J_{[0,\infty)}(x_0, \mathtt{u}) = \int_0^\infty \dot{\phi}^2(t) + \mathtt{u}^2(t)\,\mathrm{d}t$$

over all stabilizing inputs.

(a) Prove that if $\mathtt{u}$ stabilizes the system, then $\int_0^\infty \mathtt{u}(t)\dot{\phi}(t)\,\mathrm{d}t$ only depends on the initial conditions $\phi(0), \dot{\phi}(0)$.
[Hint: there is an explicit anti-derivative of the product $\mathtt{u}\dot{\phi}$.]

(b) Solve the optimal control problem.
[Hint: work out $(\dot{\phi} \pm \mathtt{u})^2$ and use (a).]

(c) Verify that your optimal solution renders the closed-loop asymptotically stable. [Hint: you probably need Lyapunov functions and LaSalle's invariance principle, see § B.4 (p. 218).]

3.13 *Connection between value function and costate.* Consider Example 3.4.8.

(a) Determine the costate directly from the minimum principle.

(b) Argue that for $x_0 = 0$ there are many optimal inputs and many corresponding costates $\mathtt{p}_*(t)$.

(c) Determine the costate via Theorem 3.5.1 for the cases that $x_0 > 0$ and $x_0 < 0$.

3.14 *Infinite horizon.* Consider a system $\dot{\mathtt{x}}(t) = f(\mathtt{x}(t), \mathtt{u}(t))$, $\mathtt{x}(0) = x_0$, and infinite horizon cost

$$J_{[\tau,\infty)}(\mathtt{x}(\tau), \mathtt{u}) = \int_\tau^\infty L(\mathtt{x}(t), \mathtt{u}(t))\,\mathrm{d}t.$$

(a) Argue that the value function $\mathcal{V}(x, \tau)$ defined in (3.28) does not depend on τ.

(b) Suppose $V(x)$ is a continuously differentiable function that solves the HJB equation (3.30). Show that for every input for which $V(\mathtt{x}(\infty)) = 0$ we have that

$$J_{[0,\infty)}(x_0, \mathtt{u}) \geq V(x_0).$$

(c) Consider the integrator $\dot{\mathtt{x}}(t) = \mathtt{u}(t), \mathtt{x}(0) = x_0$, and assume that $\mathtt{u}(t)$ is free to choose (so $\mathtt{u}(t) \in \mathbb{R}$), and that the cost is

$$J_{[0,\infty)}(x_0, \mathtt{u}) = \int_0^\infty \mathtt{x}^2(t) + \mathtt{u}^2(t)\,\mathrm{d}t.$$

There are two continuously differentiable solutions V of the HJB equation (3.30) with the property that $V(0) = 0$. Determine both.

(d) Continue with the system and cost of (c). Find the input $\mathtt{u}_* : [0,\infty) \to \mathbb{R}$ that minimizes $J_{[0,\infty)}(x_0, \mathtt{u})$ over all inputs that steer the state to zero (i.e., such that $\lim_{t\to\infty} \mathtt{x}(t) = 0$).

3.15 *Infinite horizon optimal control.* Determine the input $\mathtt{u} : [0,\infty) \to \mathbb{R}$ that stabilizes the system $\dot{\mathtt{x}}(t) = \mathtt{u}(t), \mathtt{x}(0) = x_0$ (meaning $\lim_{t\to\infty} \mathtt{x}(t) = 0$) and that minimizes $\int_0^\infty \mathtt{x}^4(t) + \mathtt{u}^2(t)\,\mathrm{d}t$ over all inputs that stabilize the system.

3.16 *Infinite horizon optimal control.* Consider the nonlinear system with infinite horizon quadratic cost

$$\dot{\mathtt{x}}(t) = \mathtt{x}(t)\mathtt{u}(t), \quad \mathtt{x}(0) = x_0, \quad \mathbb{U} = \mathbb{R}, \quad J_{[0,T]}(x_0, \mathtt{u}) = \int_0^\infty \mathtt{x}^2(t) + \rho^2 \mathtt{u}^2(t)\,\mathrm{d}t.$$

We assume that $\rho > 0$.

(a) Determine all nonnegative functions V that satisfy (3.30) and such that $V(0) = 0$.

(b) Determine the input $\mathtt{u} : [0,\infty) \to \mathbb{R}$ that stabilizes the system (meaning $\lim_{t\to\infty} \mathtt{x}(t) = 0$) and that minimizes $\int_0^\infty \mathtt{x}^2(t) + \rho^2 \mathtt{u}^2(t)\,\mathrm{d}t$ over all inputs that stabilize the system.

(c) Express the optimal cost $V(x_0)$ in terms of x_0, ρ, and show that this equals $\lim_{T\to\infty} V(x_0, 0)$ where $V(x_0, 0)$ is the value function of the *finite* horizon case as given in Exercise 3.9(c) and Fig. 3.4.

3.17 *Infinite horizon optimal control with and without stability.*

(a) Determine the input $\mathtt{u} : [0,\infty) \to \mathbb{R}$ that stabilizes the system $\dot{\mathtt{x}}(t) = \mathtt{x}(t) + \mathtt{u}(t), \mathtt{x}(0) = 1$ (meaning $\lim_{t\to\infty} \mathtt{x}(t) = 0$) and that minimizes $\int_0^\infty \mathtt{u}^4(t)\,\mathrm{d}t$ over all inputs that stabilize the system.

(b) Show that the optimal input differs if we do not insist on stability. That is, argue that the input (stabilizing or not) that minimizes $\int_0^\infty \mathtt{u}^4(t)\,\mathrm{d}t$ does not stabilize the given system $\dot{\mathtt{x}}(t) = \mathtt{x}(t) + \mathtt{u}(t), \mathtt{x}(0) = 1$.

3.18 *Infinite horizon optimal control with and without stability.* Example 3.6.1 shows that the minimal value of the cost over all *stabilizing* inputs is $3^{-3/4} x_0^4$. Argue that this equals the minimal value of the cost over *all* inputs (stabilizing or not). [Hint: have a look at the finite horizon case as discussed in Example 3.4.6.]

3.19 *Free final time.* Consider the standard optimal control problem (3.1), but now we optimize over all $\mathtt{u} : [0, T] \to \mathbb{U}$ as well as all final times $T \geq 0$. The definition of the value function changes accordingly:

$$\mathcal{V}(x, \tau) = \inf_{T \geq \tau, \mathtt{u}:[\tau,T]\to\mathbb{U}} J_{[\tau,T]}(x, \mathtt{u}), \qquad \tau \geq 0.$$

(a) Show that $\mathcal{V}(x,\tau)$ does not depend on τ. Hence the value function is of the form $\mathcal{V}(x)$.

(b) Assume that the value function $\mathcal{V}(x)$ is well defined and that it is C^1. Let $\mathtt{u}_*(t)$ be an optimal control for a given x_0, and let $\mathtt{x}_*(t)$ be the resulting optimal state, and assume that $L(\mathtt{x}_*(t),\mathtt{u}_*(t))$ is continuous in t. Show that

$$\frac{\mathrm{d}\mathcal{V}(\mathtt{x}_*(t))}{\mathrm{d}t} = \frac{\partial \mathcal{V}(\mathtt{x}_*(t))}{\partial x^{\mathrm{T}}} f(\mathtt{x}_*(t),\mathtt{u}_*(t)) = -L(\mathtt{x}_*(t),\mathtt{u}_*(t)).$$

(c) Let $\mathcal{V}$ and $\mathtt{u}_*$ and $\mathtt{x}_*$ be as in part (b) of this exercise. Show that

$$H(\mathtt{x}_*(t),\mathtt{p}_*(t),\mathtt{u}_*(t)) = 0 \qquad \forall\, t \in [0,T],$$

for $\mathtt{p}_*(t) := \frac{\partial \mathcal{V}(\mathtt{x}_*(t))}{\partial x}$. Which two theorems from Chapter 2 come to mind?



Chapter 4

Linear Quadratic Control

4.1 Linear Systems with Quadratic Costs

Optimal control theory took shape in the late 1950s, among others stimulated by the space programs in the Soviet Union and the USA. At about the same time there was a clear paradigm shift in control theory. Till about 1960 the transfer function was the de-facto standard representation of linear time-invariant dynamical systems with inputs and outputs. This changed in the sixties when Kalman (and others) advocated the use of state space representations. These developments are not unrelated since optimal control theory is based on the representation of dynamical systems by systems of differential equations, i.e., state space models. Furthermore, the introduction of the Kalman-Bucy filter in the early 1960s, again based on state representations and replacing the Wiener filter, contributed to the rise of state space descriptions.

Kalman also introduced and solved the optimal control problem for linear systems with quadratic costs. This has become a standard tool in the control of linear systems, and it also paved the way for the highly influential state space $\mathbb{H}_\infty$ control theory as it emerged in the eighties and nineties (see Chapter 5). In this chapter we study optimal control problems for linear systems with quadratic costs, close to Kalman's original problem. Specifically, we consider the minimization of quadratic costs of the form

$$J_{[0,T]}(x_0, \mathtt{u}) := \int_0^T \mathtt{x}^{\mathrm{T}}(t) Q \mathtt{x}(t) + \mathtt{u}^{\mathrm{T}}(t) R \mathtt{u}(t)\, \mathrm{d}t + \mathtt{x}^{\mathrm{T}}(T) S \mathtt{x}(T) \tag{4.1}$$

over all inputs $\mathtt{u} : [0, T] \to \mathbb{R}^m$ and states $\mathtt{x} : [0, T] \to \mathbb{R}^n$ that are governed by a linear system with given initial state,

$$\dot{\mathtt{x}}(t) = A\mathtt{x}(t) + B\mathtt{u}(t), \qquad \mathtt{x}(0) = x_0. \tag{4.2}$$

This problem is known as the finite horizon *linear quadratic optimal control problem*, or *LQ problem* for short. Later in this chapter we also consider the

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G. Meinsma and A. van der Schaft, *A Course on Optimal Control*, Springer Undergraduate Texts in Mathematics and Technology,
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infinite horizon case, which is when $T = \infty$. No restrictions are imposed on $\mathtt{u}(t)$, that is, at any time $t \in [0, T]$ the input can take any value in $\mathbb{R}^m$.

The matrix B in (4.2) has n rows and m columns, and A is an $n \times n$ matrix. The weighting matrices Q and S in (4.1) are $n \times n$ matrices, and they are assumed to be positive semi-definite, and we assume that R is an $m \times m$ positive definite matrix:

$$S \geq 0, \qquad Q \geq 0, \qquad R > 0. \tag{4.3}$$

This class of linear systems and quadratic costs is broad, yet is simple enough to allow for explicit solutions. Especially for the infinite horizon case there are efficient numerical routines that solve the problem completely, and they can be found in various software packages. The theory is popular because it is a complete and elegant theory, but also because it provides the pragmatic control engineer with a systematic tool to determine stabilizing controllers, and it allows to tune the controller by changing meaningful parameters. As an example, suppose we have the scalar system $\dot{\mathtt{x}}(t) = a\mathtt{x}(t) + b\mathtt{u}(t)$ and that we want to steer the state $\mathtt{x}$ to zero "quickly" by choice of input $\mathtt{u}$, but using only "small" inputs $\mathtt{u}$. LQ control has the potential to solve such problems. The idea is to minimize over all inputs the quadratic cost

$$\int_0^\infty \mathtt{x}^2(t) + \rho^2 \mathtt{u}^2(t)\,\mathrm{d}t. \tag{4.4}$$

Here ρ is a tuning parameter that we choose. If ρ is large then we put a large penalty on the input $\mathtt{u}$ in the cost function, so the input that minimizes the cost is probably going to be "small". Conversely, if ρ is small (close to zero) then inputs $\mathtt{u}$ are "cheap" and then the optimal input is probably going to be "large" and possibly it is able to steer the state $\mathtt{x}$ to zero "quickly". By tuning ρ we can now hope to come up with a good compromise between small $\mathtt{u}$ and small $\mathtt{x}$. In this example we have a single parameter to choose, ρ. In the general case we can choose the matrices Q, R and S; see § 4.6 for a number of examples.

We solve the finite horizon LQ problem in detail, first using Pontryagin's minimum principle and then using dynamic programming. Both methods reveal that the optimal cost is quadratic in the initial state, that is,

$$\min_{\mathtt{u}} J_{[0,T]}(x_0, \mathtt{u}) = x_0^{\mathrm{T}} P x_0 \tag{4.5}$$

for some matrix P. Furthermore, the optimal input can be implemented as a linear time-varying state feedback, $\mathtt{u}_*(t) = -F(t)\mathtt{x}(t)$, for some matrix $F(t)$ depending on time t. Then we tackle the infinite horizon LQ problem. Also in that case the optimal cost turns out to be quadratic in the initial state, and, interestingly, the optimal input $\mathtt{u}_*$ (if it exists) can be implemented as a linear state feedback

$$\mathtt{u}_*(t) = -F\mathtt{x}(t)$$

for some constant matrix F. Note that the linear state feedback form is not imposed on the LQ problem. It will be a *result*.

4.2 Finite Horizon LQ: Minimum Principle

The Hamiltonian (2.12) for system (4.2) and cost (4.1) is $H(x,p,u) = p^{\mathrm{T}}(Ax + Bu) + x^{\mathrm{T}}Qx + u^{\mathrm{T}}Ru$. The computations to come clean up considerably if we express the costate as $2p$, thus our new costate p is half of the standard costate. This way the Hamiltonian becomes

$$H(x,2p,u) = 2p^{\mathrm{T}}(Ax+Bu) + x^{\mathrm{T}}Qx + u^{\mathrm{T}}Ru.$$

Working out the Hamiltonian equations (2.14) for the state $\mathtt{x}$ and the halved costate $\mathtt{p}$, we obtain

$$\begin{aligned} \dot{\mathtt{x}}(t) &= A\mathtt{x}(t) + B\mathtt{u}(t), & \mathtt{x}(0) &= x_0, \\ 2\dot{\mathtt{p}}(t) &= -A^{\mathrm{T}}2\mathtt{p}(t) - 2Q\mathtt{x}(t), & 2\mathtt{p}(T) &= 2S\mathtt{x}(T). \end{aligned} \tag{4.6}$$

According to the minimum principle, the optimal input at each t minimizes the Hamiltonian. Since the Hamiltonian is quadratic in u with positive definite quadratic term (since R is assumed to be positive definite), it is minimal iff its gradient with respect to u is zero. This gradient is

$$\frac{\partial H(x,2p,u)}{\partial u} = B^{\mathrm{T}}2p + 2Ru,$$

and so the input that minimizes the Hamiltonian is

$$\mathtt{u}_*(t) := -R^{-1}B^{\mathrm{T}}\mathtt{p}(t). \tag{4.7}$$

Hence if we can additionally compute $\mathtt{p}(t)$ then this settles the optimal control. Substitution of (4.7) into the Hamiltonian equations (4.6) yields the system of coupled differential equations

$$\begin{bmatrix} \dot{\mathtt{x}}_*(t) \\ \dot{\mathtt{p}}_*(t) \end{bmatrix} = \begin{bmatrix} A & -BR^{-1}B^{\mathrm{T}} \\ -Q & -A^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} \mathtt{x}_*(t) \\ \mathtt{p}_*(t) \end{bmatrix}, \qquad \begin{aligned} \mathtt{x}_*(0) &= x_0 \\ \mathtt{p}_*(T) &= S\mathtt{x}_*(T). \end{aligned} \tag{4.8}$$

The $(2n)\times(2n)$ matrix here is called a *Hamiltonian matrix* and we denote it by $\mathscr{H}$,

$$\mathscr{H} := \begin{bmatrix} A & -BR^{-1}B^{\mathrm{T}} \\ -Q & -A^{\mathrm{T}} \end{bmatrix}. \tag{4.9}$$

Remark 4.2.1. Another way to obtain (4.8), instead of halving the costate as done above, is to halve the cost criterion (4.1). Clearly, this does not change the optimal control $\mathtt{u}_*$, it only scales the cost with a factor $1/2$. This approach leads to the Hamiltonian $p^{\mathrm{T}}(Ax+Bu) + \frac{1}{2}x^{\mathrm{T}}Qx + \frac{1}{2}u^{\mathrm{T}}Ru$ which is half of the expression $H(x,2p,u)$ as considered above, and it also leads to (4.8). □

The coupled differential equations (4.8) form a linear system of differential equations in $\mathtt{x}_*$ and $\mathtt{p}_*$. If we would have had only an initial or only a final condition on $\mathtt{x}_*$ and $\mathtt{p}_*$, then we could easily solve (4.8). Here, though, we have partly an initial condition (on $\mathtt{x}_*$) and partly a final condition (on $\mathtt{p}_*$ and $\mathtt{x}_*$), so it is not immediately clear how to solve the above equation. In fact, at this point it is not even clear that the above differential equation, with its mixed boundary conditions, has a solution at all, or, if it has, that its solution is unique. Later on in this section we will see that it indeed has a unique solution, owing to our assumptions on Q, R and S. This result exploits the following remarkable connection between state and costate and the optimal cost. This connection may come as a surprise but can be understood from the dynamic programming solution presented further on in this chapter.

Lemma 4.2.2 (Optimal cost). For every solution $\mathtt{x}_*, \mathtt{p}_*$ of (4.8), the cost (4.1) for input $\mathtt{u}_* := -R^{-1}B^{\mathrm{T}}\mathtt{p}_*$ equals

$$J_{[0,T]}(x_0, \mathtt{u}_*) = \mathtt{p}_*^{\mathrm{T}}(0)x_0.$$

Proof. Consider first the identity (and here we momentarily skip time arguments)

$$\begin{aligned}\tfrac{\mathrm{d}}{\mathrm{d}t}(\mathtt{p}_*^{\mathrm{T}}\mathtt{x}_*) = \mathtt{p}_*^{\mathrm{T}}\dot{\mathtt{x}}_* + \dot{\mathtt{p}}_*^{\mathrm{T}}\mathtt{x}_* &= \mathtt{p}_*^{\mathrm{T}}(A\mathtt{x} - BR^{-1}B^{\mathrm{T}}\mathtt{p}_*) + (-Q\mathtt{x}_* - A^{\mathrm{T}}\mathtt{p}_*)^{\mathrm{T}}\mathtt{x}_* \\ &= -\mathtt{p}_*^{\mathrm{T}}BR^{-1}B^{\mathrm{T}}\mathtt{p}_* - \mathtt{x}_*^{\mathrm{T}}Q\mathtt{x}_* \\ &= -(\mathtt{u}_*^{\mathrm{T}}R\mathtt{u}_* + \mathtt{x}_*^{\mathrm{T}}Q\mathtt{x}_*).\end{aligned}$$

With this we can express the cost (4.1) as

$$\begin{aligned}J_{[0,T]}(x_0, \mathtt{u}_*) &= \mathtt{x}_*^{\mathrm{T}}(T)S\mathtt{x}_*(T) - \int_0^T \tfrac{\mathrm{d}}{\mathrm{d}t}\big(\mathtt{p}_*^{\mathrm{T}}(t)\mathtt{x}_*(t)\big)\,\mathrm{d}t \\ &= \mathtt{x}_*^{\mathrm{T}}(T)S\mathtt{x}_*(T) - \Big[\mathtt{p}_*^{\mathrm{T}}(t)\mathtt{x}_*(t)\Big]_0^T \\ &= \mathtt{x}_*^{\mathrm{T}}(T)S\mathtt{x}_*(T) - \mathtt{p}_*^{\mathrm{T}}(T)\mathtt{x}_*(T) + \mathtt{p}_*^{\mathrm{T}}(0)\mathtt{x}_*(0) = \mathtt{p}_*^{\mathrm{T}}(0)x_0.\end{aligned}$$

In the final identity we used the final condition $\mathtt{p}_*(T) = S\mathtt{x}_*(T)$. ∎

Example 4.2.3 (First-order system). For the standard integrator system $\dot{\mathtt{x}}(t) = \mathtt{u}(t)$ with quadratic cost

$$J_{[0,T]}(x_0, \mathtt{u}) = \int_0^T \mathtt{x}^2(t) + \mathtt{u}^2(t)\,\mathrm{d}t$$

the Hamiltonian matrix (4.9) is

$$\mathscr{H} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}.$$

This matrix is simple enough to allow for an explicit solution of its matrix exponential,

$$e^{\mathscr{H}t} = \frac{1}{2}\begin{bmatrix} e^t + e^{-t} & -e^t + e^{-t} \\ -e^t + e^{-t} & e^t + e^{-t} \end{bmatrix}.$$

The state and costate thus equal

$$\begin{bmatrix} x_*(t) \\ p_*(t) \end{bmatrix} = \frac{1}{2}\begin{bmatrix} e^t + e^{-t} & -e^t + e^{-t} \\ -e^t + e^{-t} & e^t + e^{-t} \end{bmatrix}\begin{bmatrix} x_0 \\ p_*(0) \end{bmatrix}$$

for an, as yet unknown, initial costate $p_*(0)$. This $p_*(0)$ must be chosen such that $x_*(T), p_*(T)$ match the final condition $p_*(T) = Sx_*(T) = 0x_*(T) = 0$. It is not hard to see that this requires

$$p_*(0) = \frac{e^T - e^{-T}}{e^T + e^{-T}} x_0.$$

This then fully determines the state and costate for all $t \in [0, T]$ as

$$\begin{bmatrix} x_*(t) \\ p_*(t) \end{bmatrix} = \frac{1}{2}\begin{bmatrix} e^t + e^{-t} & -e^t + e^{-t} \\ -e^t + e^{-t} & e^t + e^{-t} \end{bmatrix}\begin{bmatrix} 1 \\ \frac{e^T - e^{-T}}{e^T + e^{-T}} \end{bmatrix} x_0.$$

The initial costate $p_*(0)$ is linear in x_0, and therefore the entire state and costate (x_*, p_*) is linear in x_0. Hence the optimal cost is quadratic in x_0,

$$J_{[0,T]}(x_0, u_*) = p_*(0)x_0 = \frac{e^T - e^{-T}}{e^T + e^{-T}} x_0^2.$$

Furthermore, the optimal input is linear in the costate,

$$u_*(t) = -R^{-1}B^{\mathrm{T}} p_*(t) = -p_*(t),$$

and since the costate is linear in x_0, the optimal input is also linear in x_0. □

In the above example we managed to transform the final condition, $p_*(T) = Sx_*(T)$, into an equivalent initial condition on $p_*(0)$, and this demonstrated that the solution of the Hamiltonian equation exists and is unique. We will shortly see that this always works. The general procedure is as follows. First consider the $(2n) \times (2n)$ matrix exponential of the Hamiltonian matrix $\mathscr{H}$, and split it into four $n \times n$ blocks:

$$\begin{bmatrix} \Sigma_{11}(t) & \Sigma_{12}(t) \\ \Sigma_{21}(t) & \Sigma_{22}(t) \end{bmatrix} := e^{\mathscr{H}t}. \tag{4.10}$$

Now the state-costate solution as a function of (known) x_0 and (unknown) $p_*(0)$ is

$$\begin{bmatrix} x_*(t) \\ p_*(t) \end{bmatrix} = \begin{bmatrix} \Sigma_{11}(t) & \Sigma_{12}(t) \\ \Sigma_{21}(t) & \Sigma_{22}(t) \end{bmatrix}\begin{bmatrix} x_0 \\ p_*(0) \end{bmatrix}.$$

With this expression the final condition, $\mathtt{p}_*(T) = S\mathtt{x}_*(T)$, can be written as

$$\begin{aligned}
0 &= S\mathtt{x}_*(T) - \mathtt{p}_*(T) \\
&= \begin{bmatrix} S & -I \end{bmatrix} \begin{bmatrix} \mathtt{x}_*(T) \\ \mathtt{p}_*(T) \end{bmatrix} \\
&= \begin{bmatrix} S & -I \end{bmatrix} \begin{bmatrix} \Sigma_{11}(T) & \Sigma_{12}(T) \\ \Sigma_{21}(T) & \Sigma_{22}(T) \end{bmatrix} \begin{bmatrix} x_0 \\ \mathtt{p}_*(0) \end{bmatrix} \\
&= \big(S\Sigma_{11}(T) - \Sigma_{21}(T)\big)x_0 + \big(S\Sigma_{12}(T) - \Sigma_{22}(T)\big)\mathtt{p}_*(0). \qquad (4.11)
\end{aligned}$$

This final condition has a unique solution $\mathtt{p}_*(0)$ iff $S\Sigma_{12}(T) - \Sigma_{22}(T)$ is invertible, and then

$$\mathtt{p}_*(0) = Mx_0$$

where M is defined as

$$M = -\big(S\Sigma_{12}(T) - \Sigma_{22}(T)\big)^{-1}\big(S\Sigma_{11}(T) - \Sigma_{21}(T)\big). \qquad (4.12)$$

Hence the question is: does the inverse of $S\Sigma_{12}(T) - \Sigma_{22}(T)$ always exist? The answer is yes:

Theorem 4.2.4 (Existence and uniqueness of solution). Suppose $Q \geq 0, S \geq 0, R > 0$. Then the matrix M in (4.12) is well defined. Hence the linear system with mixed boundary conditions (4.8) has a unique solution $\mathtt{x}_*, \mathtt{p}_*$ on the time interval $[0, T]$, and it is given by

$$\begin{bmatrix} \mathtt{x}_*(t) \\ \mathtt{p}_*(t) \end{bmatrix} = \begin{bmatrix} \Sigma_{11}(t) & \Sigma_{12}(t) \\ \Sigma_{21}(t) & \Sigma_{22}(t) \end{bmatrix} \begin{bmatrix} I \\ M \end{bmatrix} x_0 \qquad \text{for all } t \in [0, T]. \qquad (4.13)$$

Proof. First take $x_0 = 0$, and realize that (4.8) has at least the trivial solution $\mathtt{x}_* = \mathtt{p}_* = 0$. Lemma 4.2.2 showed that for every possible solution $\mathtt{x}_*, \mathtt{p}_*$ we have

$$\mathtt{x}_*^{\mathrm{T}}(T)S\mathtt{x}_*(T) + \int_0^T \mathtt{x}_*^{\mathrm{T}}(t)Q\mathtt{x}_*(t) + \mathtt{u}_*^{\mathrm{T}}(t)R\mathtt{u}_*(t)\,\mathrm{d}t = \mathtt{p}_*^{\mathrm{T}}(0)x_0,$$

and here that is zero because we took $x_0 = 0$. Since all terms on the left-hand side of the above equation are nonnegative, it must be that all these terms are zero. In particular $\mathtt{u}_*^{\mathrm{T}}(t)R\mathtt{u}_*(t) = 0$. Now $R > 0$, so necessarily $\mathtt{u}_*(t) = 0$. This, in turn, implies that $\dot{\mathtt{x}}_*(t) = A\mathtt{x}_*(t) + B\mathtt{u}_*(t) = A\mathtt{x}_*(t)$. Given that $\mathtt{x}_*(0) = x_0 = 0$ we get $\mathtt{x}_*(t) = 0$ for all time and, as a result, $\dot{\mathtt{p}}_*(t) = -Q\mathtt{x}_*(t) - A^{\mathrm{T}}\mathtt{p}_*(t) = -A^{\mathrm{T}}\mathtt{p}_*(t)$ and $\mathtt{p}_*(T) = S\mathtt{x}_*(T) = 0$. This shows that $\mathtt{p}_*(t)$ is zero for all time as well. Conclusion: for $x_0 = 0$ the solution $(\mathtt{x}_*, \mathtt{p}_*)$ of (4.8) exists and is unique. This implies that $S\Sigma_{12}(T) - \Sigma_{22}(T)$ is nonsingular for otherwise there would have existed multiple $\mathtt{p}_*(0)$ that satisfy the boundary condition (4.11). Invertibility of $S\Sigma_{12}(T) - \Sigma_{22}(T)$ in turn shows that the final condition (4.11) has a unique solution, $\mathtt{p}_*(0) = Mx_0$, for *every* x_0. ∎

It gets better: the LQ problem satisfies the convexity conditions of Theorem 2.8.1 if $S \geq 0, Q \geq 0, R > 0$. So solvability of the Hamiltonian equations—which we just proved—is not only *necessary* for optimality, it is also *sufficient*. That is, $\mathtt{u}_*(t) := -R^{-1}B^{\mathrm{T}}\mathtt{p}_*(t)$ is the optimal control. A more direct proof of optimality is discussed in Exercise 4.3. The assumptions that R is positive definite and S and Q positive semi-definite are crucial. Without these assumptions, M might not exist, see Exercise 4.2.

Note that $\mathtt{p}_*(0)$ according to (4.13) is linear in the initial state, $\mathtt{p}_*(0) = Mx_0$. Hence, as follows from Lemma 4.2.2, the optimal cost is *quadratic* in the initial state. There is also an elegant elementary argument why the optimal cost is quadratic in the state, see Exercise 4.5.

Example 4.2.5 (Integrator, see also Example 3.4.4). Consider again the scalar integrator system $\dot{\mathtt{x}}(t) = \mathtt{u}(t), \mathtt{x}(0) = x_0$ and take as cost

$$J_{[0,T]}(x_0, \mathtt{u}) = \mathtt{x}^2(T) + \int_0^T R\mathtt{u}^2(t)\,\mathrm{d}t \tag{4.14}$$

where R is some positive number. Then

$$\mathscr{H} = \begin{bmatrix} 0 & -1/R \\ 0 & 0 \end{bmatrix},$$

and, thus,

$$\mathrm{e}^{\mathscr{H}t} = \begin{bmatrix} 1 & -t/R \\ 0 & 1 \end{bmatrix}.$$

The final condition on $\mathtt{p}_*(T)$ can be transformed into a unique initial condition on $\mathtt{p}_*(0)$. Indeed the final condition is met iff

$$\begin{aligned} 0 &= S\mathtt{x}_*(T) - \mathtt{p}_*(T) \\ &= \begin{bmatrix} S & -1 \end{bmatrix} \mathrm{e}^{\mathscr{H}T} \begin{bmatrix} x_0 \\ \mathtt{p}_*(0) \end{bmatrix} \\ &= \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & -T/R \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_0 \\ \mathtt{p}_*(0) \end{bmatrix} \\ &= x_0 - (T/R+1)\mathtt{p}_*(0). \end{aligned}$$

This is the case iff

$$\mathtt{p}_*(0) = \frac{x_0}{T/R+1}.$$

It is linear in x_0 (as predicted), and the above inverse $(T/R+1)^{-1}$ exists (as predicted) because $T/R \geq 0$ so $T/R+1 \neq 0$. The optimal cost is quadratic in x_0 (predicted as well), in fact,

$$J_{[0,T]}(x_0, \mathtt{u}_*) = \mathtt{p}_*(0)x_0 = \frac{x_0^2}{T/R+1}.$$

Special about this example is that the costate is constant, $\mathtt{p}_*(t) = \mathtt{p}_*(0)$. The optimal control is therefore constant as well,

$$\mathtt{u}_*(t) = -\frac{1}{R}\mathtt{p}_*(t) = -\frac{\mathtt{p}_*(0)}{R} = -\frac{x_0}{T+R}.$$

For $R \gg T$ the optimal control $\mathtt{u}_*(t)$ is small, which is to be expected because for large R the input is penalized strongly in the cost (4.14). If $R \approx 0$ then control is cheap. In this case the control is not necessarily large, $\mathtt{u}_*(t) \approx -x_0/T$, but large enough to steer the final state $\mathtt{x}_*(T)$ to something close to zero, $\mathtt{x}_*(T) = x_0(1 - T/(R+T)) = x_0 R/(T+R) \approx 0$. □

Example 4.2.6 (Second-order system with mixed boundary condition). This is a laborious example. Consider the system with initial condition

$$\begin{bmatrix}\dot{\mathtt{x}}_1(t)\\ \dot{\mathtt{x}}_2(t)\end{bmatrix} = \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}\begin{bmatrix}\mathtt{x}_1(t)\\ \mathtt{x}_2(t)\end{bmatrix} + \begin{bmatrix}0\\ 1\end{bmatrix}\mathtt{u}(t), \qquad \begin{bmatrix}\mathtt{x}_1(0)\\ \mathtt{x}_2(0)\end{bmatrix} = \begin{bmatrix}1\\ 0\end{bmatrix},$$

and with cost

$$J_{[0,3]}(x_0, \mathtt{u}) = \mathtt{x}_1^2(3) + \int_0^3 \mathtt{u}^2(t)\,\mathrm{d}t.$$

The Hamiltonian equations (4.8) then become (verify this yourself)

$$\begin{bmatrix}\dot{\mathtt{x}}_{*1}(t)\\ \dot{\mathtt{x}}_{*2}(t)\\ \dot{\mathtt{p}}_{*1}(t)\\ \dot{\mathtt{p}}_{*2}(t)\end{bmatrix} = \left[\begin{array}{cc|cc} 0 & 1 & 0 & 0\\ 0 & 0 & 0 & -1\\ \hline 0 & 0 & 0 & 0\\ 0 & 0 & -1 & 0\end{array}\right]\begin{bmatrix}\mathtt{x}_{*1}(t)\\ \mathtt{x}_{*2}(t)\\ \mathtt{p}_{*1}(t)\\ \mathtt{p}_{*2}(t)\end{bmatrix} \tag{4.15}$$

with boundary conditions

$$\begin{bmatrix}\mathtt{x}_{*1}(0)\\ \mathtt{x}_{*2}(0)\end{bmatrix} = \begin{bmatrix}1\\ 0\end{bmatrix}, \qquad \begin{bmatrix}\mathtt{p}_{*1}(3)\\ \mathtt{p}_{*2}(3)\end{bmatrix} = \begin{bmatrix}\mathtt{x}_{*1}(3)\\ 0\end{bmatrix}.$$

Now we try to solve (4.15). The differential equation for $\mathtt{p}_{*1}(t)$ simply is $\dot{\mathtt{p}}_{*1}(t) = 0, \mathtt{p}_{*1}(3) = \mathtt{x}_{*1}(3)$, and therefore it has a constant solution,

$$\mathtt{p}_{*1}(t) = \mathtt{x}_{*1}(3). \tag{4.16}$$

The differential equation for $\mathtt{p}_{*2}(t)$ now is

$$\dot{\mathtt{p}}_{*2}(t) = -\mathtt{p}_{*1}(t) = -\mathtt{x}_{*1}(3), \qquad \mathtt{p}_{*2}(3) = 0,$$

so that

$$\mathtt{p}_{*2}(t) = (3-t)\mathtt{x}_{*1}(3). \tag{4.17}$$

With this solution the differential equation for $\mathtt{x}_{*2}(t)$ becomes

$$\dot{\mathtt{x}}_{*2}(t) = -\mathtt{p}_{*2}(t) = (t-3)\mathtt{x}_{*1}(3), \qquad \mathtt{x}_{*2}(0) = 0.$$

This equation, too, is not difficult to solve,

$$\mathrm{x}_{*2}(t) = (\tfrac{1}{2}t^2 - 3t)\mathrm{x}_{*1}(3). \tag{4.18}$$

Finally, we have to solve the differential equation for $\mathrm{x}_{*1}(t)$, given by

$$\dot{\mathrm{x}}_{*1}(t) = \mathrm{x}_{*2}(t) = (\tfrac{1}{2}t^2 - 3t)\mathrm{x}_{*1}(3), \qquad \mathrm{x}_{*1}(0) = 1.$$

Its solution is

$$\mathrm{x}_{*1}(t) = \left(\tfrac{1}{6}t^3 - \tfrac{3}{2}t^2\right)\mathrm{x}_{*1}(3) + 1. \tag{4.19}$$

The only unknown left is $\mathrm{x}_{*1}(3)$. From (4.19) it follows that

$$\mathrm{x}_{*1}(3) = \left(\tfrac{9}{2} - \tfrac{27}{2}\right)\mathrm{x}_{*1}(3) + 1,$$

i.e.,

$$\mathrm{x}_{*1}(3) = \frac{1}{10}. \tag{4.20}$$

Now we have solved the differential equation (4.15), and the solution is given by (4.16)–(4.19), with $\mathrm{x}_{*1}(3)$ equal to 1/10, see (4.20). Hence, the optimal control (4.7) is

$$\mathrm{u}_*(t) = -R^{-1}B^{\mathrm{T}}\mathrm{p}_*(t) = -B^{\mathrm{T}}\mathrm{p}_*(t) = -\begin{bmatrix}0 & 1\end{bmatrix}\mathrm{p}_*(t) = -\mathrm{p}_{*2}(t) = \frac{1}{10}(t-3),$$

and the optimal cost is

$$\begin{bmatrix}\mathrm{p}_{*1}(0)\\ \mathrm{p}_{*2}(0)\end{bmatrix}^{\mathrm{T}}\begin{bmatrix}\mathrm{x}_{*1}(0)\\ \mathrm{x}_{*2}(0)\end{bmatrix} = \begin{bmatrix}1/10\\ 3/10\end{bmatrix}^{\mathrm{T}}\begin{bmatrix}1\\ 0\end{bmatrix} = \frac{1}{10}.$$

□

4.3 Finite Horizon LQ: Dynamic Programming

The LQ problem can be solved with the aid of dynamic programming as well. The equation that has to be solved in dynamic programming is a partial differential equation—the Hamilton-Jacobi-Bellman (HJB) equation—and that is, in general, not an easy task. For LQ it can be done, however.

So consider again a linear system

$$\dot{\mathrm{x}}(t) = A\mathrm{x}(t) + B\mathrm{u}(t), \qquad \mathrm{x}(0) = x_0$$

with a quadratic cost

$$J_{[0,T]}(x_0, \mathrm{u}) = \int_0^T \mathrm{x}^{\mathrm{T}}(t)Q\mathrm{x}(t) + \mathrm{u}^{\mathrm{T}}(t)R\mathrm{u}(t)\,\mathrm{d}t + \mathrm{x}^{\mathrm{T}}(T)S\mathrm{x}(T).$$

Here, as before, S and Q are symmetric $n \times n$ positive semi-definite matrices, and R is an $m \times m$ positive definite matrix: $S \geq 0, Q \geq 0, R > 0$. The HJB equations (3.12) for this problem are

$$\frac{\partial V(x,t)}{\partial t} + \min_{u \in \mathbb{R}^m} \left(\frac{\partial V(x,t)}{\partial x^{\mathrm{T}}} (Ax + Bu) + x^{\mathrm{T}} Q x + u^{\mathrm{T}} R u \right) = 0, \quad V(x,T) = x^{\mathrm{T}} S x.$$

We determine a solution $V(x,t)$ of this equation. Because the optimal cost according to the minimum principle is quadratic, we expect the value function to be quadratic in x as well. Based on this we restrict our $V(x,t)$ to functions of the form

$$V(x,t) = x^{\mathrm{T}} P(t) x$$

with $P(t)$ an $n \times n$ symmetric matrix depending on t. Using this quadratic $V(x,t)$, the above HJB equations become

$$x^{\mathrm{T}} \dot{P}(t) x + \min_{u \in \mathbb{R}^m} \left(2x^{\mathrm{T}} P(t) (Ax + Bu) + x^{\mathrm{T}} Q x + u^{\mathrm{T}} R u \right) = 0, \quad x^{\mathrm{T}} P(T) x = x^{\mathrm{T}} S x.$$

The minimization over u can, like in the previous section, be solved by setting the gradient of $2x^{\mathrm{T}} P(t)(Ax + Bu) + x^{\mathrm{T}} Q x + u^{\mathrm{T}} R u$ with respect to u equal to zero. This gives for each x and each t as input

$$u = -R^{-1} B^{\mathrm{T}} P(t) x$$

(verify this yourself), and thereby the HJB equations reduce to

$$\begin{aligned} x^{\mathrm{T}} \left(\dot{P}(t) + P(t)A + A^{\mathrm{T}} P(t) + Q - P(t) B R^{-1} B^{\mathrm{T}} P(t) \right) x = 0, \\ x^{\mathrm{T}} P(T) x = x^{\mathrm{T}} S x. \end{aligned} \tag{4.21}$$

Here we used the fact that $x^{\mathrm{T}} P(t) A x$ is the same as $x^{\mathrm{T}} A^{\mathrm{T}} P(t) x$. All terms in (4.21) have a factor x^{T} (on the left) and a factor x (on the right), and the matrix inside the brackets is symmetric. Hence (4.21) holds for all $x \in \mathbb{R}^n$ iff the equation in which x^{T} and x are removed holds. Thus

$$\boxed{\dot{P}(t) = -P(t)A - A^{\mathrm{T}} P(t) + P(t) B R^{-1} B^{\mathrm{T}} P(t) - Q, \qquad P(T) = S}. \tag{4.22}$$

This is a nonlinear, $n \times n$-matrix-valued differential equation, called *Riccati differential equation* (or *RDE* for short) because of its (loose) connection with certain quadratic differential equations studied by the Italian nobleman, Count Jacopo Riccati (1676–1754).

The existence of a solution $P(t)$ of this RDE is not straightforward, but *if* there is a continuous solution on $[0,T]$ then the candidate optimal control

$$\mathtt{u}_*(t) = -R^{-1} B^{\mathrm{T}} P(t) \mathtt{x}(t)$$

makes the closed-loop system satisfy

$$\dot{\mathtt{x}}(t) = A\mathtt{x}(t) + B\mathtt{u}_*(t) = (A - BR^{-1}B^{\mathrm{T}}P(t))\mathtt{x}(t), \qquad \mathtt{x}(0) = x_0.$$

This is a linear differential equation, so if $P(t)$ is continuous on $[0,T]$ then it has a unique solution $\mathtt{x}_*(t)$ for every x_0 and all $t \in [0,T]$. Theorem 3.4.3 in that case guarantees that this $\mathtt{u}_*$ *is* the optimal input, and that $V(x,t) = x^{\mathrm{T}}P(t)x$ *is* the value function. In particular $V(x_0,0) = x_0^{\mathrm{T}}P(0)x_0$ is the optimal cost. Thus we proved:

Proposition 4.3.1 (Solution of the finite horizon LQ problem). Let Q,R,S be symmetric, and suppose that $R>0$. *If* the RDE (4.22) has a continuously differentiable solution $P:[0,T]\to\mathbb{R}^{n\times n}$, then the LQ problem (4.1)–(4.2) has a solution for every $x_0\in\mathbb{R}^n$. In particular

$$\mathtt{u}_*(t) := -R^{-1}B^{\mathrm{T}}P(t)\mathtt{x}(t) \tag{4.23}$$

is the optimal input, and the optimal cost is

$$J_{[0,T]}(x_0,\mathtt{u}_*) = x_0^{\mathrm{T}}P(0)x_0,$$

and $V(x,t) := x^{\mathrm{T}}P(t)x$ is its value function. □

Proof. It is an immediate consequence of Theorem 3.4.3 of the previous chapter. But it also has a direct proof. This proof resembles that of Lemma 4.2.2, and for the case that $R=I$ it goes as follows (for ease of exposition we omit here the time arguments):

$$\begin{aligned}
\tfrac{\mathrm{d}}{\mathrm{d}t}(\mathtt{x}^{\mathrm{T}}P\mathtt{x}) &= \dot{\mathtt{x}}^{\mathrm{T}}P\mathtt{x} + \mathtt{x}^{\mathrm{T}}\dot{P}\mathtt{x} + \mathtt{x}^{\mathrm{T}}P\dot{\mathtt{x}}\\
&= (A\mathtt{x}+B\mathtt{u})^{\mathrm{T}}P\mathtt{x} + \mathtt{x}^{\mathrm{T}}(-PA-PA^{\mathrm{T}}+PBB^{\mathrm{T}}P-Q)\mathtt{x} + \mathtt{x}^{\mathrm{T}}P(A\mathtt{x}+B\mathtt{u})\\
&= \mathtt{u}^{\mathrm{T}}B^{\mathrm{T}}P\mathtt{x} + \mathtt{x}^{\mathrm{T}}PB\mathtt{u} + \mathtt{x}^{\mathrm{T}}PBB^{\mathrm{T}}P\mathtt{x} - \mathtt{x}^{\mathrm{T}}Q\mathtt{x}\\
&= (\mathtt{u}+B^{\mathrm{T}}P\mathtt{x})^{\mathrm{T}}(\mathtt{u}+B^{\mathrm{T}}P\mathtt{x}) - \mathtt{x}^{\mathrm{T}}Q\mathtt{x} - \mathtt{u}^{\mathrm{T}}\mathtt{u}.
\end{aligned}$$

(Verify the final identity yourself.) From this it follows that the cost can also be expressed as (where, again, we omit the time arguments),

$$\begin{aligned}
J_{[0,T]}(x_0,\mathtt{u}) &= \mathtt{x}^{\mathrm{T}}(T)S\mathtt{x}(T) + \int_0^T \mathtt{x}^{\mathrm{T}}Q\mathtt{x} + \mathtt{u}^{\mathrm{T}}\mathtt{u}\,\mathrm{d}t\\
&= \mathtt{x}^{\mathrm{T}}(T)S\mathtt{x}(T) + \int_0^T -\tfrac{\mathrm{d}}{\mathrm{d}t}(\mathtt{x}^{\mathrm{T}}P\mathtt{x}) + (\mathtt{u}+B^{\mathrm{T}}P\mathtt{x})^{\mathrm{T}}(\mathtt{u}+B^{\mathrm{T}}P\mathtt{x})\,\mathrm{d}t\\
&= \mathtt{x}^{\mathrm{T}}(T)S\mathtt{x}(T) - \left[\mathtt{x}^{\mathrm{T}}(t)P(t)\mathtt{x}(t)\right]_0^T + \int_0^T (\mathtt{u}+B^{\mathrm{T}}P\mathtt{x})^{\mathrm{T}}(\mathtt{u}+B^{\mathrm{T}}P\mathtt{x})\,\mathrm{d}t\\
&= x_0^{\mathrm{T}}P(0)x_0 + \int_0^T (\mathtt{u}+B^{\mathrm{T}}P\mathtt{x})^{\mathrm{T}}(\mathtt{u}+B^{\mathrm{T}}P\mathtt{x})\,\mathrm{d}t.
\end{aligned}$$

Clearly, the final integral is nonnegative for every $\mathtt{u}$, and it is minimal if we take $\mathtt{u}(t) = -B^{\mathrm{T}}P(t)\mathtt{x}(t)$, and thus the optimal cost is $x_0^{\mathrm{T}}P(0)x_0$. For $R\neq I$ the proof is similar, see Exercise 4.10. ■

Notice that Proposition 4.3.1 assumes symmetry of S and Q, but not that they are positive semi-definite. Also notice that the optimal control (4.23) is of a special form: first we have to determine the solution $P(t)$ of the RDE, but this can be done irrespective of x_0. Once $P(t)$ is determined the optimal control can be implemented as a linear time-varying state feedback (4.23). Thus the "gain" matrix $F(t) := R^{-1}B^{\mathrm{T}}P(t)$ in the optimal feedback can be computed "off-line", i.e., based on the knowledge of the system matrices A, B and the cost criterion matrices Q, R, S only.

Example 4.3.2 (Example 4.2.5 continued). Consider again the integrator system $\dot{\mathrm{x}}(t) = \mathrm{u}(t), \mathrm{x}(0) = x_0$ of Example 4.2.5 with

$$J_{[0,T]}(x_0, \mathrm{u}) = \mathrm{x}^2(T) + \int_0^T R\mathrm{u}^2(t)\,\mathrm{d}t$$

for some $R > 0$. Here $S = 1$ and $Q = 0$, and thus the RDE (4.22) becomes

$$\dot{P}(t) = P^2(t)/R, \qquad P(T) = 1.$$

The solution of this RDE can be found with separation of variables,

$$P(t) = \frac{R}{R+T-t} = \frac{1}{1+(T-t)/R}.$$

Since $t \in [0, T]$ and $R > 0$ we see that $R + T - t > 0$ throughout, and so $P(t)$ is well defined on $[0, T]$. Hence $P(t)x^2$ is the value function with optimal cost

$$J_{[0,T]}(x_0, \mathrm{u}_*) = P(0)x_0^2 = \frac{x_0^2}{1+T/R},$$

and the optimal input is

$$\mathrm{u}_*(t) = -R^{-1}B^{\mathrm{T}}P(t)\mathrm{x}(t) = -\frac{P(t)\mathrm{x}(t)}{R} = -\frac{\mathrm{x}(t)}{R+T-t}. \tag{4.24}$$

In this example the optimal control u_* is given in state feedback form, while in Example 4.2.5 (where we handled the same LQ problem) the control input is given as a function of time. The feedback form is often preferred in applications, but for this particular problem the feedback form (4.24) blurs the fact that the optimal state and optimal control are just linear functions of time, see Example 4.2.5. □

4.4 Riccati Differential Equations

Proposition 4.3.1 *assumes* the existence of a solution $P(t)$ of the RDE, but does not require S and Q to be positive semi-definite. If $S \geq 0, Q \geq 0$ (and $R > 0$) then existence of $P(t)$ can in fact be *guaranteed*. So for standard LQ problems we have a complete solution:

Theorem 4.4.1 (Existence of solution of RDE's). If $S = S^{\mathrm{T}} \geq 0, Q = Q^{\mathrm{T}} \geq 0$ and $R = R^{\mathrm{T}} > 0$, then the RDE (4.22) has a unique continuously differentiable solution $P(t)$ on $[0, T]$, and $P(t)$ is symmetric and positive semi-definite at every $t \in [0, T]$.

Consequently, $\mathtt{u}_*(t) := -R^{-1}B^{\mathrm{T}}P(t)\mathtt{x}(t)$ is a solution of the LQ problem, $V(x, t) := x^{\mathrm{T}}P(t)x$ is its value function, and the optimal cost is $J_{[0,T]}(x_0, \mathtt{u}_*) = x_0^{\mathrm{T}}P(0)x_0$.

Proof. The RDE (4.22) is equivalent to a system of n^2 differential equations in the entries $p_{ij}(t)$, $i, j = 1, \ldots, n$ of $P(t)$. The right-hand side of this equation consists of polynomials in $p_{ij}(t)$, and hence it is continuously differentiable and therefore is locally Lipschitz. We conclude from Theorem B.1.3 (p. 207) that the solution $P(t)$ exists and is unique on an interval $(t_{\mathrm{esc}}, T]$ for some $t_{\mathrm{esc}} < T$. It is easy to see that also $P^{\mathrm{T}}(t)$ is a solution, so, being unique, we have that $P(t)$ is symmetric on $(t_{\mathrm{esc}}, T]$.

Now suppose that it has an escape time $t_{\mathrm{esc}} \in [0, T)$. Theorem B.1.4 (p. 207) then says that as $t \downarrow t_{\mathrm{esc}}$, the norm of the vector with entries $p_{ij}(t), i, j = 1, \ldots, n$, diverges to infinity. This implies that at least one entry, say $p_{ij}(t)$, is unbounded as $t \downarrow t_{\mathrm{esc}}$. We now show that this leads to a contradiction. Let e_i denote the i-th basis vector of $\mathbb{R}^n$. Because $P(t)$ is symmetric, it follows that

$$(e_i + e_j)^{\mathrm{T}}P(t)(e_i + e_j) - (e_i - e_j)^{\mathrm{T}}P(t)(e_i - e_j) = 4p_{ij}(t).$$

Since $p_{ij}(t)$ is unbounded, either $(e_i + e_j)^{\mathrm{T}}P(t)(e_i + e_j)$ or $(e_i - e_j)^{\mathrm{T}}P(t)(e_i - e_j)$ is unbounded (or both are unbounded). Now, choose the initial state z equal to $e_i + e_j$ or $e_i - e_j$, whichever results in an unbounded $z^{\mathrm{T}}P(t)z$ as $t \downarrow t_{\mathrm{esc}}$. From the preceding discussion we know that $z^{\mathrm{T}}P(t)z$ is the value function $\mathcal{V}(z, t)$ for $t \in (t_{\mathrm{esc}}, T]$ but the value function for sure is bounded from above by the cost that we make with the zero input:

$$\forall\, t \in (t_{\mathrm{esc}}, T]: \quad z^{\mathrm{T}}P(t)z = \min_{\mathtt{u}} J_{[t,T]}(z, \mathtt{u}) \leq J_{[t,T]}(z, 0) \leq J_{[t_{\mathrm{esc}},T]}(z, 0) < \infty.$$

Our $z^{\mathrm{T}}P(t)z$ can therefore not escape to $+\infty$ as $t \downarrow t_{\mathrm{esc}}$. Furthermore, it can also not escape to $-\infty$ because $z^{\mathrm{T}}P(t)z = \min_{\mathtt{u}} J_{[t,T]}(z, \mathtt{u}) \geq 0$. This yields a contradiction, so there is no $t_{\mathrm{esc}} \in [0, T)$. Lemma B.1.4 (p. 207) then guarantees that the differential equation has a unique solution $P(t)$ on the entire time interval $[0, T]$.

Now that existence of $P(t)$ is proved, Proposition 4.3.1 tells us that the given $\mathtt{u}_*$ is optimal, and that $x^{\mathrm{T}}P(t)x$ is the value function and $x_0^{\mathrm{T}}P(0)x_0$ is the optimal cost. Finally, we showed at the beginning of the proof that $P(t)$ is symmetric. It is also positive semi-definite because the cost is nonnegative for every x. ∎

In this proof non-negativity of Q, S and positivity of R is crucially used. These assumptions are standard in LQ, but it is interesting to see what happens if one of these assumptions is violated. Then the solution $P(t)$ might escape in finite time. The following example demonstrates this for negative Q.

Example 4.4.2 (Negative Q, finite escape time). Consider the integrator system $\dot{\mathrm{x}}(t) = \mathrm{u}(t)$ with initial state $\mathrm{x}(0) = x_0$ and cost

$$J_{[0,T]}(x_0, \mathrm{u}) = \int_0^T -\mathrm{x}^2(t) + \mathrm{u}^2(t)\, \mathrm{d}t.$$

This is a non-standard LQ problem because $Q = -1 < 0$. The RDE (4.22) simplifies to

$$\dot{P}(t) = P^2(t) + 1, \qquad P(T) = 0.$$

Using separation of variables one can show that its solution is

$$P(t) = \tan(t - T), \qquad t \in (T - \tfrac{\pi}{2}, T].$$

This solution $P(t)$ escapes at

$$t_{\mathrm{esc}} = T - \tfrac{\pi}{2},$$

see Fig. 4.1. If $T < \frac{\pi}{2}$ then there is no escape time in $[0, T]$ and, hence, $P(t) := \tan(t - T)$ is then well defined on the entire horizon $[0, T]$, and, consequently,

$$V(x, t) = x^2 \tan(t - T)$$

is the value function, and

$$\mathrm{u}_*(t) = -R^{-1}B^{\mathrm{T}}P(t)\mathrm{x}(t) = -\tan(t - T)\mathrm{x}(t)$$

is the optimal state feedback.

However, if $T \geq \frac{\pi}{2}$ then the escape time t_{esc} *is* in $[0, T]$, see Fig. 4.1 (right). In this case the optimal cost is unbounded from below. That is, it can be made as close to $-\infty$ as we desire. To see this, take

$$\mathrm{u}_\epsilon(t) = \begin{cases} 0 & t \leq t_{\mathrm{esc}} + \epsilon \\ -\tan(T - t)\mathrm{x}(t) & t > t_{\mathrm{esc}} + \epsilon \end{cases}$$

for some small $\epsilon > 0$. For this input the state $\mathrm{x}(t)$ is constant over $[0, t_{\mathrm{esc}} + \epsilon]$, and continues optimally over $[t_{\mathrm{esc}} + \epsilon, T]$. The cost for this input thus is

$$\begin{aligned} J_{[0,T]}(x_0, \mathrm{u}_\epsilon) &= \int_0^{t_{\mathrm{esc}}+\epsilon} -\mathrm{x}^2(t) + \mathrm{u}^2(t)\, \mathrm{d}t + V(x_0, t_{\mathrm{esc}} + \epsilon) \\ &= -(t_{\mathrm{esc}} + \epsilon)x_0^2 + \tan(-\tfrac{\pi}{2} + \epsilon)x_0^2. \end{aligned}$$

It diverges to $-\infty$ as $\epsilon \downarrow 0$. □

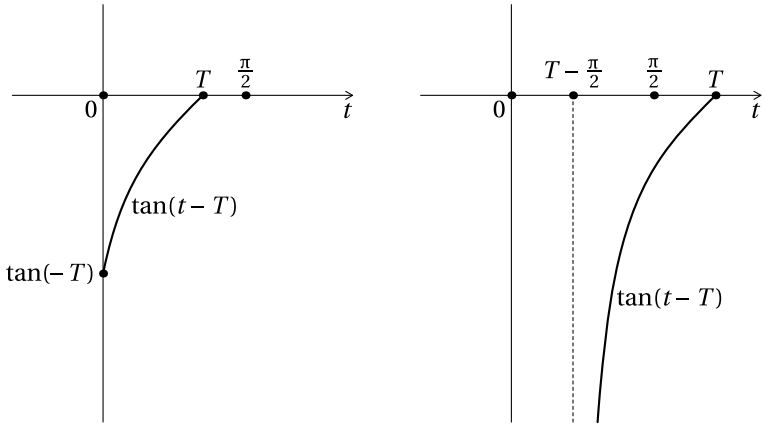


FIGURE 4.1: Graph of $\tan(t-T)$ for $t \in [0,T]$. Left: if $0 < T < \frac{\pi}{2}$. In that case $\tan(t-T)$ is defined for all $t \in [0,T]$. Right: if $T \geq \frac{\pi}{2}$. Then $\tan(t-T)$ is not defined at $T - \frac{\pi}{2} \in [0,T]$. See Example 4.4.2.

Connection between Hamiltonians and RDE's

In Theorem 3.5.1 we established a connection between value functions and standard costates: $\mathtt{p}_*(t) = \frac{\partial V(\mathtt{x}_*(t),t)}{\partial x}$. Given that the standard costate is twice the costate as used in the LQ problem—see the beginning of § 4.2—this connection for the LQ problem becomes

$$2\mathtt{p}_*(t) = \frac{\partial \mathcal{V}(\mathtt{x}_*(t),t)}{\partial x}.$$

For the LQ problem with quadratic value functions $\mathcal{V}(x,t) = x^{\mathrm{T}}P(t)x$ we have $\frac{\partial \mathcal{V}(\mathtt{x}_*(t),t)}{\partial x} = 2P(t)\mathtt{x}_*(t)$. Therefore the connection is

$$\mathtt{p}_*(t) = P(t)\mathtt{x}_*(t). \tag{4.25}$$

Incidentally, this re-proves Lemma 4.2.2 because $\mathtt{p}_*^{\mathrm{T}}(0)x_0 = x_0^{\mathrm{T}}P(0)x_0 = V(x_0,0)$. Equation (4.25) expresses the costate $\mathtt{p}_*(t)$ in terms of the solution $P(t)$ of the RDE, but it can also be used to determine $P(t)$ using the states and costates. This goes as follows. In Theorem 4.2.4 we saw that the optimal $\mathtt{x}_*$ and $\mathtt{p}_*$ follow uniquely from x_0 as

$$\begin{bmatrix} \mathtt{x}_*(t) \\ \mathtt{p}_*(t) \end{bmatrix} = \begin{bmatrix} \Sigma_{11}(t) & \Sigma_{12}(t) \\ \Sigma_{21}(t) & \Sigma_{22}(t) \end{bmatrix} \begin{bmatrix} I \\ M \end{bmatrix} x_0 \tag{4.26}$$

for $M := -(S\Sigma_{12}(T) - \Sigma_{22}(T))^{-1}(S\Sigma_{11}(T) - \Sigma_{21}(T))$. Consider the mapping $x_0 \mapsto \mathtt{x}_*(t)$ given by the upper part of (4.26), i.e., $\mathtt{x}_*(t) = (\Sigma_{11}(t) + \Sigma_{12}(t)M)x_0$. *If* this mapping is invertible at every $t \in [0,T]$ then x_0 follows uniquely from $\mathtt{x}_*(t)$ as $x_0 = (\Sigma_{11}(t) + \Sigma_{12}(t)M)^{-1}\mathtt{x}_*(t)$ and, consequently, $\mathtt{p}_*(t)$ also follows uniquely

from $\mathtt{x}_*(t)$:

$$\begin{aligned}\mathtt{p}_*(t) &= (\Sigma_{21}(t)+\Sigma_{22}(t)M)x_0\\ &= (\Sigma_{21}(t)+\Sigma_{22}(t)M)(\Sigma_{11}(t)+\Sigma_{12}(t)M)^{-1}\mathtt{x}_*(t).\end{aligned}$$

Comparing this with (4.25) suggests the following explicit formula for $P(t)$.

Lemma 4.4.3 (Solution of RDE's using the Hamiltonian). Let S, Q be positive semi-definite $n\times n$ matrices, and $R = R^{\mathrm{T}} > 0$. Then the solution $P(t)$, $t\in[0,T]$, of the RDE

$$\dot{P}(t) = -P(t)A - A^{\mathrm{T}}P(t) + P(t)BR^{-1}B^{\mathrm{T}}P(t) - Q, \qquad P(T) = S,$$

is

$$P(t) = \big(\Sigma_{21}(t)+\Sigma_{22}(t)M\big)\big(\Sigma_{11}(t)+\Sigma_{12}(t)M\big)^{-1}. \tag{4.27}$$

Here $M := -(S\Sigma_{12}(T)-\Sigma_{22}(T))^{-1}(S\Sigma_{11}(T)-\Sigma_{21}(T))$, and Σ_{ij} are $n\times n$ sub-blocks of the matrix exponential $\mathrm{e}^{\mathscr{H}t}$ as defined in (4.10).

Proof. Recall that the solution $P(t)$ of the RDE exists. If $\Sigma_{11}(t)+\Sigma_{12}(t)M$ would have been singular at some $t=\bar{t}$, then any nonzero x_0 in the null space of $\Sigma_{11}(\bar{t})+\Sigma_{12}(\bar{t})M$ renders $\mathtt{x}_*(\bar{t})=0$ while $\mathtt{p}_*(\bar{t})$ is nonzero (because $\Sigma(t) := \mathrm{e}^{\mathscr{H}t}$ is invertible). This contradicts the fact that $\mathtt{p}_*(t) = P(t)\mathtt{x}_*(t)$. Hence $\Sigma_{11}(t)+\Sigma_{12}(t)M$ is invertible for every $t\in[0,T]$ and, consequently, the mapping from $\mathtt{x}_*(t)$ to $\mathtt{p}_*(t)$ follows uniquely from (4.26) and it equals (4.27). ∎

Example 4.4.4. In Example 4.2.3 we tackled the minimization of $\int_0^T \mathtt{x}^2(t)+\mathtt{u}^2(t)\,\mathrm{d}t$ for $\dot{\mathtt{x}}(t)=\mathtt{u}(t)$ using Hamiltonians, and we found that

$$\begin{bmatrix}\Sigma_{11}(t) & \Sigma_{12}(t)\\ \Sigma_{21}(t) & \Sigma_{22}(t)\end{bmatrix} = \frac{1}{2}\begin{bmatrix}\mathrm{e}^t+\mathrm{e}^{-t} & -\mathrm{e}^t+\mathrm{e}^{-t}\\ -\mathrm{e}^t+\mathrm{e}^{-t} & \mathrm{e}^t+\mathrm{e}^{-t}\end{bmatrix}, \qquad M = \frac{\mathrm{e}^T-\mathrm{e}^{-T}}{\mathrm{e}^T+\mathrm{e}^{-T}}.$$

The RDE for this problem is

$$\dot{P}(t) = P^2(t)-1, \qquad P(T)=0.$$

According to (4.27) the solution of this RDE is

$$\begin{aligned}P(t) &= \frac{\Sigma_{21}(t)+\Sigma_{22}(t)M}{\Sigma_{11}(t)+\Sigma_{12}(t)M}\\ &= \frac{-\mathrm{e}^t+\mathrm{e}^{-t}+(\mathrm{e}^t+\mathrm{e}^{-t})M}{\mathrm{e}^t+\mathrm{e}^{-t}+(-\mathrm{e}^t+\mathrm{e}^{-t})M}\\ &= \frac{(-\mathrm{e}^t+\mathrm{e}^{-t})(\mathrm{e}^T+\mathrm{e}^{-T})+(\mathrm{e}^t+\mathrm{e}^{-t})(\mathrm{e}^T-\mathrm{e}^{-T})}{(\mathrm{e}^t+\mathrm{e}^{-t})(\mathrm{e}^T+\mathrm{e}^{-T})+(-\mathrm{e}^t+\mathrm{e}^{-t})(\mathrm{e}^T-\mathrm{e}^{-T})}\\ &= \frac{\mathrm{e}^{T-t}-\mathrm{e}^{-(T-t)}}{\mathrm{e}^{T-t}+\mathrm{e}^{-(T-t)}} = \tanh(T-t).\end{aligned}$$

□

4.5 Infinite Horizon LQ and Algebraic Riccati Equations

Now we turn to the *infinite horizon LQ problem*. This is the problem of minimizing

$$J_{[0,\infty)}(x_0, \mathrm{u}) := \int_0^\infty \mathrm{x}^{\mathrm{T}}(t)Q\mathrm{x}(t) + \mathrm{u}^{\mathrm{T}}(t)R\mathrm{u}(t)\,\mathrm{d}t \tag{4.28}$$

over all $\mathrm{u} : [0,\infty) \to \mathbb{R}^m$ under the dynamical constraint

$$\dot{\mathrm{x}}(t) = A\mathrm{x}(t) + B\mathrm{u}(t), \qquad \mathrm{x}(0) = x_0.$$

As before, we assume that R is positive definite and that Q is positive semi-definite. The terminal cost $\mathrm{x}^{\mathrm{T}}(\infty)S\mathrm{x}(\infty)$ is absent. (For the problems that we have in mind the state converges to zero so the terminal cost would not contribute anyway.)

We first approach the infinite horizon LQ problem as the limit as $T \to \infty$ of the finite horizon LQ problem over the time window $[0, T]$. To make the dependence on T explicit we add a subscript T to the solution of the RDE (4.22), that is,

$$\dot{P}_T(t) = -P_T(t)A - A^{\mathrm{T}}P_T(t) + P_T(t)BR^{-1}B^{\mathrm{T}}P_T(t) - Q, \qquad P_T(T) = 0. \tag{4.29}$$

Example 4.5.1. Consider again the integrator system

$$\dot{\mathrm{x}}(t) = \mathrm{u}(t), \qquad \mathrm{x}(0) = x_0,$$

but still with a *finite* horizon cost

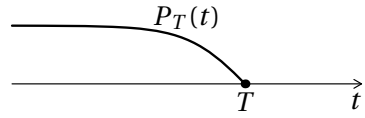
$$J_{[0,T]}(x_0, \mathrm{u}) = \int_0^T \mathrm{x}^2(t) + \mathrm{u}^2(t)\,\mathrm{d}t.$$

The associated RDE (4.29) is

$$\dot{P}_T(t) = P_T^2(t) - 1, \qquad P_T(T) = 0.$$

Its solution was derived in Example 4.4.4,

$$P_T(t) = \tanh(T-t) = \frac{\mathrm{e}^{T-t} - \mathrm{e}^{-(T-t)}}{\mathrm{e}^{T-t} + \mathrm{e}^{-(T-t)}}.$$



Clearly, as T goes to infinity, the solution $P_T(t)$ converges to $P := 1$ and, in particular, it no longer depends on t. It is now tempting to conclude that the constant state feedback $\mathrm{u}_*(t) := -R^{-1}B^{\mathrm{T}}P\mathrm{x}(t) = -\mathrm{x}(t)$ is the optimal solution of the *infinite* horizon LQ problem. It is, as we shall soon see. □

The example suggests that $P_T(t)$ converges to a constant P as the horizon T goes to ∞. It also suggests that $\lim_{T\to\infty} \dot{P}_T(t) = 0$, which in turn suggests that the Riccati *differential* equation in the limit reduces to an *algebraic* equation, known as the *algebraic Riccati equation* (or *ARE* for short) of LQ:

$$\boxed{0 = A^{\mathrm{T}}P + PA - PBR^{-1}B^{\mathrm{T}}P + Q}. \tag{4.30}$$

The following theorem shows that all this is indeed the case. It requires just one extra condition (apart from the standard conditions $Q \geq 0$, $R > 0$): for each x_0 there needs to exist at least one input that renders the cost $J_{[0,\infty)}(x_0, \mathtt{u})$ finite.

Theorem 4.5.2 (Solution of ARE via limit of solution of RDE). Consider $\dot{\mathtt{x}}(t) = A\mathtt{x}(t) + B\mathtt{u}(t), \mathtt{x}(0) = x_0$, and suppose $Q \geq 0, R > 0$, and that for every x_0 an input exists that renders the cost (4.28) finite. Then the solution $P_T(t)$ of (4.29) converges to a matrix independent of t as the final time T goes to infinity. That is, a constant matrix P exists such that

$$\lim_{T\to\infty} P_T(t) = P \qquad \forall\, t > 0.$$

This P is symmetric, positive semi-definite, and it satisfies the ARE (4.30).

Proof. For every fixed x_0 the expression $x_0^{\mathrm{T}} P_T(t) x_0$ is nondecreasing with T because the longer the horizon the higher the cost. Indeed, for every $\epsilon > 0$ and initial $\mathtt{x}(t) = z$ we have

$$\begin{aligned} z^{\mathrm{T}} P_{T+\epsilon}(t) z &= \int_t^{T+\epsilon} \mathtt{x}_*^{\mathrm{T}}(t) Q \mathtt{x}_*(t) + \mathtt{u}_*^{\mathrm{T}}(t) R \mathtt{u}_*(t)\, \mathrm{d}t \\ &\geq \int_t^{T} \mathtt{x}_*^{\mathrm{T}}(t) Q \mathtt{x}_*(t) + \mathtt{u}_*^{\mathrm{T}}(t) R \mathtt{u}_*(t)\, \mathrm{d}t \geq z^{\mathrm{T}} P_T(t) z. \end{aligned}$$

Besides being nondecreasing, it is, for any given z, also bounded from above because by assumption for at least one input $\mathtt{u}_z$ the infinite horizon cost is finite, so that

$$z^{\mathrm{T}} P_T(t) z \leq J_{[t,T]}(z, \mathtt{u}_z) \leq J_{[t,\infty)}(z, \mathtt{u}_z) < \infty.$$

Bounded and nondecreasing implies that $z^{\mathrm{T}} P_T(t) z$ converges as $T \to \infty$. Next we prove that in fact the entire matrix $P_T(t)$ converges as $T \to \infty$. Let e_i be the i-th unit vector in $\mathbb{R}^n$, so $e_i = (0, \ldots, 0, 1, 0, \ldots, 0)^{\mathrm{T}}$, with a 1 on the i-th position. The preceding discussion shows that for each $z = e_i$, the limit

$$p_{ii} := \lim_{T\to\infty} e_i^{\mathrm{T}} P_T(t) e_i$$

exists. The diagonal entries of $P_T(t)$ hence converge. For the off-diagonal entries we use that

$$\begin{aligned} \lim_{T\to\infty} (e_i + e_j)^{\mathrm{T}} P_T(t)(e_i + e_j) &= \lim_{T\to\infty} e_i^{\mathrm{T}} P_T(t) e_i + e_j^{\mathrm{T}} P_T(t) e_j + 2 e_i^{\mathrm{T}} P_T(t) e_j \\ &= p_{ii} + p_{jj} + \lim_{T\to\infty} 2 e_i^{\mathrm{T}} P_T(t) e_j. \end{aligned}$$

The limit on the left-hand side exists, so the limit $p_{ij} := \lim_{T\to\infty} e_i^{\mathrm{T}} P_T(t) e_j$ exists as well. Therefore all entries of $P_T(t)$ converge as $T \to \infty$. The limit is independent of t because $P_T(t) = P_{T-t}(0)$.

Clearly, $P \geq 0$ because it is the limit of $P_T(t) \geq 0$.

Since $P_T(t)$ converges to a constant matrix, also $\dot{P}_T(t) = -P_T(t)A - A^{\mathrm{T}}P_T(t) + P_T(t)BR^{-1}B^{\mathrm{T}}P_T(t) - Q$ converges to a constant matrix as $T \to \infty$. This constant matrix must be zero because $\int_t^{t+1} \dot{P}_T(\tau)\,\mathrm{d}\tau = P_T(t+1) - P_T(t) \to 0$ as $T \to \infty$. ∎

LQ with stability

The classic infinite-horizon LQ problem does not consider asymptotic stability of the closed-loop system. For instance, if we choose as cost $\int_0^\infty \mathrm{u}^2(t)\,\mathrm{d}t$ then optimal is to take $\mathrm{u}_*(t) = 0$, even if it would render the closed-loop system unstable, such as when $\dot{\mathrm{x}}(t) = \mathrm{x}(t) + \mathrm{u}(t)$. In applications closed-loop asymptotic stability is crucial. Classically, closed-loop asymptotic stability is incorporated in LQ by imposing conditions on Q. For example, if $Q = I$ then the cost contains a term $\int_0^\infty \mathrm{x}^{\mathrm{T}}(t)\mathrm{x}(t)\,\mathrm{d}t$, and then the optimal control turns out to necessarily stabilize the system. An alternative approach is to include asymptotic stability in the problem definition. This we explore now.

Definition 4.5.3 (Infinite-horizon LQ problem with stability). Suppose $Q \geq 0, R > 0$, and consider the linear system with given initial state $\dot{\mathrm{x}}(t) = A\mathrm{x}(t) + B\mathrm{u}(t), \mathrm{x}(0) = x_0$. The *LQ problem with stability* is to minimize

$$J_{[0,\infty)}(x_0, \mathrm{u}) := \int_0^\infty \mathrm{x}^{\mathrm{T}}(t)Q\mathrm{x}(t) + \mathrm{u}^{\mathrm{T}}(t)R\mathrm{u}(t)\,\mathrm{d}t \tag{4.31}$$

over all *stabilizing inputs* u, meaning inputs that achieve $\lim_{t\to\infty} \mathrm{x}(t) = 0$. □

The next example shows that in some cases the LQ problem with stability has an easy solution.

Example 4.5.4 (LQ with stability). Consider the problem of Example 4.5.1:

$$\dot{\mathrm{x}}(t) = \mathrm{u}(t), \qquad \mathrm{x}(0) = x_0, \qquad J_{[0,\infty)}(x_0, \mathrm{u}) = \int_0^\infty \mathrm{x}^2(t) + \mathrm{u}^2(t)\,\mathrm{d}t.$$

The running cost, $\mathrm{x}^2 + \mathrm{u}^2$, can also be written as

$$\mathrm{x}^2 + \mathrm{u}^2 = (\mathrm{x} + \mathrm{u})^2 - 2\mathrm{x}\mathrm{u} = (\mathrm{x} + \mathrm{u})^2 - 2\mathrm{x}\dot{\mathrm{x}}.$$

Interestingly, the term $-2\mathrm{x}\dot{\mathrm{x}}$ has an explicit antiderivative, namely $-\mathrm{x}^2$, so

$$\mathrm{x}^2 + \mathrm{u}^2 = \tfrac{\mathrm{d}}{\mathrm{d}t}(-\mathrm{x}^2) + (\mathrm{x} + \mathrm{u})^2.$$

Integrating this over $t \in [0, \infty)$ we see that the cost for stabilizing inputs equals

$$J_{[0,\infty)}(x_0, \mathrm{u}) = x_0^2 + \int_0^\infty (\mathrm{x}(t) + \mathrm{u}(t))^2\,\mathrm{d}t. \tag{4.32}$$

Here we used that $\lim_{t\to\infty} \mathtt{x}(t) = 0$, since $\mathtt{u}$ is assumed to stabilize the system. It is immediate from (4.32) that the cost for every stabilizing input is at least x_0^2, and it equals the minimal value x_0^2 iff

$$\mathtt{u} = -\mathtt{x}.$$

Since the state feedback $\mathtt{u}_* := -\mathtt{x}$ indeed stabilizes (because the closed-loop system becomes $\dot{\mathtt{x}} = -\mathtt{x}$) we conclude that this state feedback is the optimal control, and that the optimal (minimal) cost is

$$J_{[0,\infty)}(x_0, \mathtt{u}_*) = x_0^2.$$

In Example 4.5.1 we *conjectured* that $\mathtt{u}_* := -\mathtt{x}$ is optimal. Now we *know* it is optimal, or at least optimal with respect to all stabilizing inputs. □

In this example, and also in the general finite horizon LQ problem, we have that the optimal cost is quadratic in the initial state, and that the optimal input can be implemented as a state feedback. Inspired by this we expect that every infinite horizon LQ problem has these properties. That is, we conjecture that the optimal cost is of the form

$$x_0^{\mathrm{T}} P x_0$$

for some matrix P, and that the optimal input equals $\mathtt{u}_*(t) := -F\mathtt{x}(t)$ for some matrix F. This leads to the following central result.

Theorem 4.5.5 (Solution of the LQ problem with stability). There is at most one matrix $P \in \mathbb{R}^{n\times n}$ that satisfies the algebraic Riccati equation (ARE)

$$A^{\mathrm{T}}P + PA - PBR^{-1}B^{\mathrm{T}}P + Q = 0 \tag{4.33}$$

with the property that

$$A - BR^{-1}B^{\mathrm{T}}P \quad \text{is asymptotically stable.} \tag{4.34}$$

Such a P is called a *stabilizing solution of the ARE*. In that case P is symmetric, and the linear state feedback

$$\mathtt{u}_*(t) := -R^{-1}B^{\mathrm{T}}P\mathtt{x}(t)$$

is the solution of the LQ problem with stability, and the optimal cost is $x_0^{\mathrm{T}} P x_0$.

Proof. If P satisfies the ARE then it can be verified that $P - P^{\mathrm{T}}$ satisfies

$$(A - BR^{-1}B^{\mathrm{T}}P)^{\mathrm{T}}(P - P^{\mathrm{T}}) + (P - P^{\mathrm{T}})(A - BR^{-1}B^{\mathrm{T}}P) = -Q + Q^{\mathrm{T}} = 0.$$

Using this identity, Corollary B.5.3 (p. 227) shows that for asymptotically stable $A - BR^{-1}B^{\mathrm{T}}P$ we necessarily have $P - P^{\mathrm{T}} = 0$, i.e., stabilizing solutions P of

the ARE are symmetric. To show that there is at most one stabilizing solution we proceed as follows. Suppose P_1, P_2 are two stabilizing solutions of the ARE (hence P_1 and P_2 are symmetric), and let $\mathtt{x}_1, \mathtt{x}_2$ be solutions of the corresponding $\dot{\mathtt{x}}_1 = (A - BR^{-1}B^{\mathrm{T}}P_1)\mathtt{x}_1$ and $\dot{\mathtt{x}}_2 = (A - BR^{-1}B^{\mathrm{T}}P_2)\mathtt{x}_2$. Then

$$\begin{aligned}
&\tfrac{\mathrm{d}}{\mathrm{d}t}(\mathtt{x}_1^{\mathrm{T}}(P_1 - P_2)\mathtt{x}_2) \\
&\quad = \mathtt{x}_1^{\mathrm{T}}\big((A - BR^{-1}B^{\mathrm{T}}P_1)^{\mathrm{T}}(P_1 - P_2) + (P_1 - P_2)(A - BR^{-1}B^{\mathrm{T}}P_2)\big)\mathtt{x}_2 \\
&\quad = \mathtt{x}_1^{\mathrm{T}}\big(A^{\mathrm{T}}P_1 - A^{\mathrm{T}}P_2 - P_1BR^{-1}B^{\mathrm{T}}P_1 + P_1BR^{-1}B^{\mathrm{T}}P_2 \\
&\qquad\qquad + P_1A - P_2A - P_1BR^{-1}B^{\mathrm{T}}P_2 + P_2BR^{-1}B^{\mathrm{T}}P_2\big)\mathtt{x}_2 \\
&\quad = \mathtt{x}_1^{\mathrm{T}}(-Q + Q)\mathtt{x}_2 = 0.
\end{aligned}$$

Hence $\mathtt{x}_1(t)(P_1 - P_2)\mathtt{x}_2(t)$ is constant as a function of time. By asymptotic stability we have that $\lim_{t\to\infty}\mathtt{x}_1(t) = \lim_{t\to\infty}\mathtt{x}_2(t) = 0$. Therefore $\mathtt{x}_1(t)(P_1 - P_2)\mathtt{x}_2(t)$ is, in fact, zero for all time. Since this holds for every initial condition $\mathtt{x}_1(0), \mathtt{x}_2(0)$, we conclude that $P_1 = P_2$.

In the rest of the proof we assume that P is the symmetric stabilizing solution of the ARE. We expect the optimal $\mathtt{u}$ to be a linear state feedback $\mathtt{u} = -F\mathtt{x}$ for some F, so with that in mind define $\mathtt{v} := F\mathtt{x} + \mathtt{u}$. (If our hunch is correct then optimal means $\mathtt{v} = 0$.) Next we write $\mathtt{x}^{\mathrm{T}}Q\mathtt{x} + \mathtt{u}^{\mathrm{T}}R\mathtt{u}$ and $\mathtt{v}^{\mathrm{T}}R\mathtt{v}$ and $\frac{\mathrm{d}}{\mathrm{d}t}(\mathtt{x}^{\mathrm{T}}P\mathtt{x})$ as quadratic expressions in $(\mathtt{x}, \mathtt{u})$:

$$\mathtt{x}^{\mathrm{T}}Q\mathtt{x} + \mathtt{u}^{\mathrm{T}}R\mathtt{u} = \begin{bmatrix}\mathtt{x}^{\mathrm{T}} & \mathtt{u}^{\mathrm{T}}\end{bmatrix}\begin{bmatrix}Q & 0\\ 0 & R\end{bmatrix}\begin{bmatrix}\mathtt{x}\\ \mathtt{u}\end{bmatrix},$$

$$\mathtt{v}^{\mathrm{T}}R\mathtt{v} = (\mathtt{x}^{\mathrm{T}}F^{\mathrm{T}} + \mathtt{u}^{\mathrm{T}})R(F\mathtt{x} + \mathtt{u}) = \begin{bmatrix}\mathtt{x}^{\mathrm{T}} & \mathtt{u}^{\mathrm{T}}\end{bmatrix}\begin{bmatrix}F^{\mathrm{T}}RF & F^{\mathrm{T}}R\\ RF & R\end{bmatrix}\begin{bmatrix}\mathtt{x}\\ \mathtt{u}\end{bmatrix},$$

$$\begin{aligned}
\tfrac{\mathrm{d}}{\mathrm{d}t}(\mathtt{x}^{\mathrm{T}}P\mathtt{x}) = \dot{\mathtt{x}}^{\mathrm{T}}P\mathtt{x} + \mathtt{x}^{\mathrm{T}}P\dot{\mathtt{x}} &= (\mathtt{x}^{\mathrm{T}}A^{\mathrm{T}} + \mathtt{u}^{\mathrm{T}}B^{\mathrm{T}})P\mathtt{x} + \mathtt{x}^{\mathrm{T}}P(A\mathtt{x} + B\mathtt{u}) \\
&= \begin{bmatrix}\mathtt{x}^{\mathrm{T}} & \mathtt{u}^{\mathrm{T}}\end{bmatrix}\begin{bmatrix}A^{\mathrm{T}}P + PA & PB\\ B^{\mathrm{T}}P & 0\end{bmatrix}\begin{bmatrix}\mathtt{x}\\ \mathtt{u}\end{bmatrix}.
\end{aligned}$$

Therefore

$$\mathtt{x}^{\mathrm{T}}Q\mathtt{x} + \mathtt{u}^{\mathrm{T}}R\mathtt{u} - \mathtt{v}^{\mathrm{T}}R\mathtt{v} + \tfrac{\mathrm{d}}{\mathrm{d}t}(\mathtt{x}^{\mathrm{T}}P\mathtt{x}) = \begin{bmatrix}\mathtt{x} & \mathtt{u}\end{bmatrix}^{\mathrm{T}}\begin{bmatrix}A^{\mathrm{T}}P + PA + Q - F^{\mathrm{T}}RF & PB - F^{\mathrm{T}}R\\ B^{\mathrm{T}}P - RF & 0\end{bmatrix}\begin{bmatrix}\mathtt{x}\\ \mathtt{u}\end{bmatrix}.$$

Since P is symmetric and satisfies (4.33), the choice $F := R^{-1}B^{\mathrm{T}}P$ makes the above matrix on the right-hand side equal to zero. So then

$$\mathtt{x}^{\mathrm{T}}Q\mathtt{x} + \mathtt{u}^{\mathrm{T}}R\mathtt{u} = -\tfrac{\mathrm{d}}{\mathrm{d}t}(\mathtt{x}^{\mathrm{T}}P\mathtt{x}) + \mathtt{v}^{\mathrm{T}}R\mathtt{v},$$

and, hence, the cost (4.31) equals

$$J_{[0,\infty)}(x_0, \mathtt{u}) = x_0^{\mathrm{T}}Px_0 + \int_0^\infty \mathtt{v}(t)^{\mathrm{T}}R\mathtt{v}(t)\,\mathrm{d}t,$$

whenever the input stabilizes the system. Given x_0 the above cost is minimal for $\mathtt{v} = 0$, provided it stabilizes. It does: since $\mathtt{v} := \mathtt{u} + F\mathtt{x} = \mathtt{u} + R^{-1}B^{\mathrm{T}}P\mathtt{x}$ we have $\mathtt{v} = 0$ iff $\mathtt{u} = -F\mathtt{x} = -R^{-1}B^{\mathrm{T}}P\mathtt{x}$, and so the closed-loop system is $\dot{\mathtt{x}} = (A - BR^{-1}B^{\mathrm{T}}P)\mathtt{x}$, which, by assumption on P, is asymptotically stable. ∎

The theorem does *not* say that the ARE has a stabilizing solution. It only says that *if* a stabilizing solution P exists, then it is unique and symmetric, and the LQ problem with stability is solved, with $\mathtt{u}_*(t) := -R^{-1}B^{\mathrm{T}}P\mathtt{x}(t)$ being the optimal control. It is not yet clear under what conditions there *exists* a stabilizing solution P of the ARE (4.33). This will be addressed by considering the solution $P_T(t)$ of the finite horizon problem, and showing how under stabilizability and detectability assumptions[1] $\lim_{T\to\infty} P_T(t)$ exists and defines such a solution:

Theorem 4.5.6 (Three ways to solve the LQ problem with stability). Consider the LQ problem with stability as formulated in Definition 4.5.3, and consider the associated ARE (4.33). (In particular assume $Q \geq 0, R > 0$.)

If (A, B) is stabilizable and (Q, A) detectable, then there is a unique stabilizing solution P of the ARE, and this P is symmetric. Consequently, $\mathtt{u}_*(t) := -R^{-1}B^{\mathrm{T}}P\mathtt{x}(t)$ solves the infinite horizon LQ problem with stability, and $x_0^{\mathrm{T}}Px_0$ is the minimal cost. Moreover this unique P can be determined in the following three equivalent ways:

1. P equals $\lim_{T\to\infty} P_T(t)$ where $P_T(t)$ is the solution of RDE (4.29),
2. P is the unique symmetric, positive semi-definite solution of the ARE,
3. P is the unique stabilizing solution of the ARE.

Proof. This proof assumes knowledge of detectability and stabilizability as explained in Appendix A.6. First we prove equivalence of the three ways of computing P, and later we comment on the uniqueness.

(1 $\implies$ 2). Since (A, B) is stabilizable, there is a state feedback $\mathtt{u} = -F\mathtt{x}$ that steers the state to zero exponentially fast for every x_0, and, so, renders the cost finite. Therefore the conditions of Theorem 4.5.2 are met. That is, $P := \lim_{T\to\infty} P_T(t)$ exists and it satisfies the ARE, and it is positive semi-definite.

(2 $\implies$ 3). Assume P is a positive semi-definite solution of the ARE, and let x be an eigenvector of $A - BR^{-1}B^{\mathrm{T}}P$ with eigenvalue λ. We show that $\mathrm{Re}(\lambda) < 0$. The trick is to rewrite the ARE as

$$(A - BR^{-1}B^{\mathrm{T}}P)^{\mathrm{T}}P + P(A - BR^{-1}B^{\mathrm{T}}P) + Q + PBR^{-1}B^{\mathrm{T}}P = 0.$$

Next, postmultiply this equation with the eigenvector x, and premultiply with its complex conjugate transpose x^*:

$$x^*\big((A - BR^{-1}B^{\mathrm{T}}P)^{\mathrm{T}}P + P(A - BR^{-1}B^{\mathrm{T}}P) + Q + PBR^{-1}B^{\mathrm{T}}P\big)x = 0.$$

Since x is an eigenvector of $A - BR^{-1}B^{\mathrm{T}}P$ the above simplifies to a sum of three terms, the last two of which are nonnegative,

$$(\lambda^* + \lambda)(x^*Px) + x^*Qx + x^*PBR^{-1}B^{\mathrm{T}}Px = 0.$$

[1] Stabilizability and detectability are discussed in Appendix A.6.

If $\mathrm{Re}(\lambda) \geq 0$ then $(\lambda^* + \lambda)x^* Px \geq 0$, implying that all the above three terms are in fact zero: $(\lambda^* + \lambda)x^* Px = 0$, $Qx = 0$, and $B^{\mathrm{T}} Px = 0$ (and, consequently, $Ax = \lambda x$). This contradicts detectability. So it cannot be that $\mathrm{Re}(\lambda) \geq 0$. It must be that $A - BR^{-1}B^{\mathrm{T}}P$ is asymptotically stable.

(Uniqueness & 3 $\implies$ 1). Theorem 4.5.5 shows that there is *at most one* stabilizing solution P of the ARE. Now $P := \lim_{T\to\infty} P_T(t)$ is *one* solution that stabilizes (because 1. $\implies$ 2. $\implies$ 3.). Hence the stabilizing solution of the ARE exists and is unique, and it equals the matrices P from Item 1 and 2, which, hence, are unique as well.

Theorem 4.5.5 then guarantees that $\mathtt{u}_* = -R^{-1}B^{\mathrm{T}}P\mathtt{x}$ solves the LQ problem with stability, and that $x_0^{\mathrm{T}} P x_0$ is the optimal cost. ∎

Theorem 4.5.6 shows that we have several ways to determine the solution P that solves the LQ problem with stability, namely (a) $\lim_{T\to\infty} P_T(t)$, (b) the unique symmetric positive semi-definite solution of the ARE, and (c) the unique stabilizing solution of the ARE.

Example 4.5.7 (LQ problem with stability of the integrator system solved in three ways). Consider again the integrator system

$$\dot{\mathtt{x}}(t) = \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0$$

and cost

$$J_{[0,\infty)}(x_0, \mathtt{u}) = \int_0^\infty \mathtt{x}^2(t) + \mathtt{u}^2(t)\,\mathrm{d}t.$$

This system is stabilizable, and $(Q, A) = (1, 0)$ is detectable. We determine the LQ solution P in the three different ways as explained in Theorem 4.5.6:

1. In Example 4.5.1 we handled the finite horizon case of this problem, and we found that $P := \lim_{T\to\infty} P_T(t) = 1$.

2. We could have gone as well for the unique symmetric, positive semi-definite solution of the ARE. The ARE in this case is

   $$-P^2 + 1 = 0,$$

   and, clearly, the only (symmetric) positive semi-definite solution is $P = 1$.

3. The ARE has two solutions, $P = \pm 1$, and Theorem 4.5.6 guarantees that precisely one of them is stabilizing. The solution P is stabilizing if $A - BR^{-1}B^{\mathrm{T}}P = -P$ is less than zero. Clearly this, again, gives $P = 1$.

□

While for low-order systems the 2nd option (that P is positive semi-definite) is often the easiest way to determine P, general numerical recipes usually exploit the 3rd option. This is explained in the final part of this section where we examine the connection with Hamiltonian matrices.

Connection between Hamiltonians and ARE's

For the *finite* horizon LQ problem we established in Lemma 4.4.3 a tight connection between solutions $P(t)$ of RDE's and Hamiltonian matrices $\mathscr{H}$. For the *infinite* horizon case a similar connection exists. This we explore now.

A matrix P satisfies the ARE

$$PA + A^{\mathrm{T}}P - PBR^{-1}B^{\mathrm{T}}P + Q = 0$$

iff

$$-Q - A^{\mathrm{T}}P = P(A - BR^{-1}B^{\mathrm{T}}P),$$

that is, iff

$$\underbrace{\begin{bmatrix} A & -BR^{-1}B^{\mathrm{T}} \\ -Q & -A^{\mathrm{T}} \end{bmatrix}}_{\mathscr{H}} \begin{bmatrix} I \\ P \end{bmatrix} = \begin{bmatrix} I \\ P \end{bmatrix} (A - BR^{-1}B^{\mathrm{T}}P). \tag{4.35}$$

This is interesting because in the case that all matrices here are numbers (and the Hamiltonian matrix $\mathscr{H}$ hence a 2×2 matrix) then it says that $\left[\begin{smallmatrix} I \\ P \end{smallmatrix}\right]$ is an *eigenvector* of the Hamiltonian matrix, and that $A - BR^{-1}B^{\mathrm{T}}P$ is its *eigenvalue*. This connection between P and eigenvectors/eigenvalues of the Hamiltonian matrix $\mathscr{H}$ is the key to most numerical routines for computation of P. This central result is formulated in the following theorem. The subsequent examples show how the result can be used to find P concretely.

Theorem 4.5.8 (Computation of P). Define $\mathscr{H} \in \mathbb{R}^{(2n)\times(2n)}$ as in (4.35), and assume that $Q \geq 0, R > 0$. If (A, B) is stabilizable and (Q, A) detectable, then

1. $\mathscr{H}$ has no imaginary eigenvalues, and it has n asymptotically stable eigenvalues and n unstable eigenvalues. Also, λ is an eigenvalue of $\mathscr{H}$ iff so is $-\lambda$,

2. matrices $V \in \mathbb{R}^{(2n)\times n}$ of rank n exist that satisfy $\mathscr{H}V = V\Lambda$ for some asymptotically stable $\Lambda \in \mathbb{R}^{n\times n}$,

3. for any such $V \in \mathbb{R}^{(2n)\times n}$, if we partition V as $V = \left[\begin{smallmatrix} V_1 \\ V_2 \end{smallmatrix}\right]$ with $V_1, V_2 \in \mathbb{R}^{n\times n}$, then V_1 is invertible,

4. the ARE (4.33) has a unique stabilizing solution P. In fact

   $$P := V_2 V_1^{-1},$$

   is the unique answer, and it is symmetric.

Proof. This proof is involved. We assume familiarity with detectability and stabilizability as explained in Appendix A.6. The proof again exploits the remarkable property that solutions of the associated Hamiltonian system (now with initial conditions, possibly complex-valued),

$$\begin{bmatrix} \dot{\mathrm{x}}(t) \\ \dot{\mathrm{p}}(t) \end{bmatrix} = \begin{bmatrix} A & -BR^{-1}B^{\mathrm{T}} \\ -Q & -A^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} \mathrm{x}(t) \\ \mathrm{p}(t) \end{bmatrix}, \qquad \begin{bmatrix} \mathrm{x}(0) \\ \mathrm{p}(0) \end{bmatrix} = \begin{bmatrix} x_0 \\ p_0 \end{bmatrix} \in \mathbb{C}^{2n} \tag{4.36}$$

satisfy

$$\tfrac{\mathrm{d}}{\mathrm{d}t}(\mathrm{p}^*\mathrm{x}) = -(\mathrm{x}^*Q\mathrm{x} + \mathrm{p}^*BR^{-1}B^{\mathrm{T}}\mathrm{p}), \tag{4.37}$$

(see the proof of Lemma 4.2.2). Note that we consider the system of differential equations over $\mathbb{C}^{2n}$, instead of over $\mathbb{R}^{2n}$, and here p^* means the *complex conjugate* transpose of p. The reason is that eigenvalues and eigenvectors may be complex-valued. Integrating (4.37) over $t \in [0,\infty)$ tells us that

$$\int_0^\infty \mathrm{x}^*(t)Q\mathrm{x}(t) + \mathrm{p}^*(t)BR^{-1}B^{\mathrm{T}}\mathrm{p}(t)\,\mathrm{d}t = p_0^*x_0 - \lim_{t\to\infty} \mathrm{p}^*(t)\mathrm{x}(t), \tag{4.38}$$

provided the limit exists. In what follows we denote by $\left[\begin{smallmatrix} \mathrm{x}(t) \\ \mathrm{p}(t) \end{smallmatrix}\right]$ the solution of (4.36).

1. Suppose $\left[\begin{smallmatrix} x_0 \\ p_0 \end{smallmatrix}\right]$ is an eigenvector of $\mathscr{H}$ with imaginary eigenvalue λ. Then $\left[\begin{smallmatrix} \mathrm{x}(t) \\ \mathrm{p}(t) \end{smallmatrix}\right] = \mathrm{e}^{\lambda t}\left[\begin{smallmatrix} x_0 \\ p_0 \end{smallmatrix}\right]$. Now $\mathrm{p}^*(t)\mathrm{x}(t)$ is constant, hence both sides of (4.37) are zero for all time. So both $\mathrm{x}^*(t)Q\mathrm{x}(t)$ and $B^{\mathrm{T}}\mathrm{p}(t)$ are zero for all time. Inserting this into (4.36) shows that $\lambda x_0 = Ax_0$ and $\lambda p_0 = -A^{\mathrm{T}}p_0$. Thus $\left[\begin{smallmatrix} A-\lambda I \\ Q \end{smallmatrix}\right]x_0 = 0$ and $p_0^*\left[\begin{smallmatrix} A+\lambda I & B \end{smallmatrix}\right] = 0$. Stabilizability and detectability imply that then $x_0 = 0, p_0 = 0$, but $\left[\begin{smallmatrix} x_0 \\ p_0 \end{smallmatrix}\right]$ is an eigenvector, so nonzero. Contradiction, hence $\mathscr{H}$ has no imaginary eigenvalues.

   Exercise 4.19 shows that $r(\lambda) := \det(\lambda I - \mathscr{H})$ equals $r(-\lambda)$. So $\mathscr{H}$ has as many (asymptotically) stable eigenvalues as unstable eigenvalues.

2. Since $\mathscr{H}$ has no imaginary eigenvalues and has n asymptotically stable eigenvalues, linear algebra tells us that a $(2n)\times n$ matrix V exists of rank n such that $\mathscr{H}V = V\Lambda$ with Λ asymptotically stable. (If all n asymptotically stable eigenvalues are distinct then we can simply take $V = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}$ where $v_1,\ldots,v_n$ are eigenvectors corresponding to the asymptotically stable eigenvalues $\lambda_1,\ldots,\lambda_n$ of $\mathscr{H}$, and then Λ is the diagonal matrix with these eigenvalues on the diagonal. If some eigenvalues coincide then one might need a Jordan normal form and use generalized eigenvectors.)

3. Suppose, to obtain a contradiction, that V has rank n but that V_1 is singular. Then the subspace spanned by $\left[\begin{smallmatrix} V_1 \\ V_2 \end{smallmatrix}\right]$ contains an $\left[\begin{smallmatrix} x_0 \\ p_0 \end{smallmatrix}\right]$ with $x_0 = 0, p_0 \neq 0$. The solution $\left[\begin{smallmatrix} \mathrm{x}(t) \\ \mathrm{p}(t) \end{smallmatrix}\right]$ for this initial condition converges to

zero[2]. Hence the integral in (4.38) equals $p_0^* x_0 = 0$. That can only be if $Q\mathrm{x}(t)$ and $B^{\mathrm{T}}\mathrm{p}(t)$ are zero for all time. Equation (4.36) then implies that $\dot{\mathrm{p}}(t) = -A^{\mathrm{T}}\mathrm{p}(t), \mathrm{p}(0) = p_0$. We claim that this contradicts stabilizability. Indeed, since $B^{\mathrm{T}}\mathrm{p}(t) = 0$ for all time, we have

$$\dot{\mathrm{p}}(t) = -(A^{\mathrm{T}} - LB^{\mathrm{T}})\mathrm{p}(t), \qquad \mathrm{p}(0) = p_0 \neq 0 \tag{4.39}$$

for every L. By stabilizability there is an L such that $A - BL^{\mathrm{T}}$ is asymptotically stable. Then all eigenvalues of $-(A^{\mathrm{T}} - LB^{\mathrm{T}})$ are anti-stable, and thus the solution $\mathrm{p}(t)$ of (4.39) diverges. But we know that $\lim_{t\to\infty} \mathrm{p}(t) = 0$. Contradiction, so the assumption that V_1 is singular is wrong.

4. Let $P = V_2 V_1^{-1}$. Since $\mathscr{H}V = V\Lambda$ we have that $\mathscr{H}\left[\begin{smallmatrix} I \\ P \end{smallmatrix}\right] = \left[\begin{smallmatrix} I \\ P \end{smallmatrix}\right] V_1 \Lambda V_1^{-1}$. Also $V_1 \Lambda V_1^{-1}$ is asymptotically stable because it has the same eigenvalues as Λ (assumed asymptotically stable). Hence

$$\begin{bmatrix} A & -BR^{-1}B^{\mathrm{T}} \\ -Q & -A^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} I \\ P \end{bmatrix} = \begin{bmatrix} I \\ P \end{bmatrix} \hat{\Lambda} \tag{4.40}$$

for some asymptotically stable $\hat{\Lambda} \in \mathbb{R}^{n\times n}$. Premultiplying (4.40) from the left with $\begin{bmatrix} -P & I \end{bmatrix}$ shows that

$$\begin{bmatrix} -P & I \end{bmatrix} \begin{bmatrix} A & -BR^{-1}B^{\mathrm{T}} \\ -Q & -A^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} I \\ P \end{bmatrix} = 0.$$

This equation is nothing else than the ARE (verify this for yourself). And P is a stabilizing solution because $A - BR^{-1}B^{\mathrm{T}}P = \hat{\Lambda}$ is asymptotically stable. Uniqueness and symmetry of P we showed earlier (Theorem 4.5.5).

■

Realize that *any* $V \in \mathbb{R}^{(2n)\times n}$ of rank n for which $\mathscr{H}V = V\Lambda$ does the job if Λ is asymptotically stable. That is, even though there are many such V, we always have that V_1 is invertible and that P follows uniquely as $P = V_2 V_1^{-1}$. As already mentioned in the above proof, in case $\mathscr{H}$ has n *distinct* asymptotically stable eigenvalues $\lambda_1, \ldots, \lambda_n$, with eigenvectors $v_1, \ldots, v_n$, then we can take

$$V = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix}$$

for then Λ is diagonal with

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{bmatrix},$$

and this matrix clearly is asymptotically stable.

[2] If $\left[\begin{smallmatrix} x_0 \\ p_0 \end{smallmatrix}\right] = Vz_0$ for some z_0 then $\left[\begin{smallmatrix} \mathrm{x}(t) \\ \mathrm{p}(t) \end{smallmatrix}\right] = V\mathbf{z}(t)$ where $\mathbf{z}(t)$ is the solution of $\dot{\mathbf{z}}(t) = \Lambda\mathbf{z}(t), \mathbf{z}(0) = z_0$. If Λ is asymptotically stable then $\mathbf{z}(t) \to 0$ as $t \to \infty$.

Example 4.5.9 ($n = 1$). Consider once more the integrator system $\dot{\mathrm{x}}(t) = \mathrm{u}(t)$ and cost $\int_0^\infty \mathrm{x}^2(t) + \mathrm{u}^2(t)\,\mathrm{d}t$. That is, $A = 0, B = Q = R = 1$. The Hamiltonian matrix for this case is

$$\mathscr{H} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}.$$

Its characteristic polynomials is $\lambda^2 - 1$, and the eigenvalues are $\lambda_{1,2} = \pm 1$. Its asymptotically stable eigenvalue is $\lambda_{\mathrm{as}} = -1$, and it is easy to verify that v is an eigenvector corresponding to this asymptotically stable eigenvalue iff

$$v := \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} c, \qquad c \neq 0.$$

According to Lemma 4.5.8 the stabilizing solution P of the ARE is

$$P = v_2 v_1^{-1} = \frac{v_2}{v_1} = \frac{c}{c} = 1.$$

This agrees with what we found in Example 4.5.7. As predicted, P does not depend on the choice of eigenvector (the choice of c). Also, the (eigen)value of $A - BR^{-1}B^{\mathrm{T}}P = -1$ as predicted equals the asymptotically stable eigenvalue of the Hamiltonian matrix, $\lambda_{\mathrm{as}} = -1$. The optimal control is $\mathrm{u}_* = -R^{-1}B^{\mathrm{T}}P\mathrm{x} = -\mathrm{x}$.

□

Example 4.5.10 ($n = 2$). Consider the stabilizable system

$$\dot{\mathrm{x}}(t) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \mathrm{x}(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \mathrm{u}(t),$$

with standard cost

$$\int_0^\infty \mathrm{x}_1^2(t) + \mathrm{x}_2^2(t) + \mathrm{u}^2(t)\,\mathrm{d}t.$$

The associated Hamiltonian matrix is (verify this yourself)

$$\mathscr{H} = \left[\begin{array}{cc|cc} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ \hline -1 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 \end{array}\right].$$

Its characteristic polynomial is $\lambda^4 - \lambda^2 + 1$, and the four eigenvalues turn out to be

$$\lambda_{1,2} = -\tfrac{1}{2}\sqrt{3} \pm \tfrac{1}{2}\mathrm{i}, \qquad \lambda_{3,4} = +\tfrac{1}{2}\sqrt{3} \pm \tfrac{1}{2}\mathrm{i}.$$

The first two eigenvalues, $\lambda_{1,2}$, are asymptotically stable so we need eigenvectors corresponding to these two. Not very enlightening manipulation shows that we can take

$$v_{1,2} = \begin{bmatrix} -\lambda_{1,2} \\ -\lambda_{1,2}^2 \\ 1 \\ \lambda_{1,2}^3 \end{bmatrix}.$$

Now $V \in \mathbb{C}^{4\times 2}$ defined as

$$V = \begin{bmatrix} v_1 & v_2 \end{bmatrix} = \begin{bmatrix} -\lambda_1 & -\lambda_2 \\ -\lambda_1^2 & -\lambda_2^2 \\ 1 & 1 \\ \lambda_1^3 & \lambda_2^3 \end{bmatrix}$$

is the V we need. (Note that this matrix is complex; this is not a problem.) With V known, it is easy to compute the stabilizing solution of the ARE,

$$P = V_2 V_1^{-1} = \begin{bmatrix} 1 & 1 \\ \lambda_1^3 & \lambda_2^3 \end{bmatrix} \begin{bmatrix} -\lambda_1 & -\lambda_2 \\ -\lambda_1^2 & -\lambda_2^2 \end{bmatrix}^{-1} = \begin{bmatrix} \sqrt{3} & 1 \\ 1 & \sqrt{3} \end{bmatrix}.$$

The optimal input is $\mathtt{u}_* = -R^{-1}B^{\mathrm{T}}P\mathtt{x} = -p_{21}\mathtt{x}_1 - p_{22}\mathtt{x}_2 = -\mathtt{x}_1 - \sqrt{3}\mathtt{x}_2$. The LQ-optimal closed-loop system is described by

$$\dot{\mathtt{x}}_*(t) = (A - BR^{-1}B^{\mathrm{T}}P)\mathtt{x}_*(t) = \begin{bmatrix} 0 & 1 \\ -1 & \sqrt{3} \end{bmatrix} \mathtt{x}_*(t),$$

and its eigenvalues are $\lambda_{1,2} = -\frac{1}{2}\sqrt{3} \pm \frac{1}{2}\mathrm{i}$ (which, as predicted, are the asymptotically stable eigenvalues of $\mathscr{H}$). □

In the above example the characteristic polynomial $\lambda^4 - \lambda^2 + 1$ is of degree 4, but by letting $\mu = \lambda^2$ it reduces to the polynomial $\mu^2 - \mu + 1$ of degree 2. This works for every Hamiltonian matrix, see Exercise 4.19.

Example 4.5.11. In Example 4.5.10 we found the solution

$$P = \begin{bmatrix} \sqrt{3} & 1 \\ 1 & \sqrt{3} \end{bmatrix}$$

via the eigenvectors of the Hamiltonian, which, by construction, gives us the stabilizing solution of the ARE. This solution is positive semi-definite according to Theorem 4.5.6. Let us verify. Clearly P is symmetric, and since $p_{1,1} = \sqrt{3} > 0$ and $\det(P) = 2 > 0$ it indeed is positive semi-definite (in fact, positive definite).

□

4.6 Controller Design with LQ Optimal Control

In five examples we explore the use of infinite horizon LQ theory for the design of controllers. The first two examples discuss the effect of tuning parameters on the control and cost. The final three examples are about control of cars.

Example 4.6.1 (Tuning the controller). Consider the system with output,

$$\dot{\mathrm{x}}(t) = \mathrm{u}(t), \qquad \mathrm{x}(0) = x_0 = 1,$$
$$\mathrm{y}(t) = 2\mathrm{x}(t),$$

and suppose the task is to steer the output y to zero "quickly" but without using "excessive" inputs u. A way to resolve this problem is by considering the LQ problem with stability with cost

$$\int_0^\infty \mathrm{y}^2(t) + \rho^2 \mathrm{u}^2(t)\,\mathrm{d}t = \int_0^\infty 4\mathrm{x}^2(t) + \rho^2 \mathrm{u}^2(t)\,\mathrm{d}t.$$

This cost includes a *tuning parameter*

$$\rho > 0,$$

which we will choose so as to achieve an acceptable compromise between "small" y and "small" u. For large values of ρ we put a strong penalty on u in the cost function, hence we expect the optimal u_* to be small in that case. Conversely, for ρ close to zero the input is "cheap", and the optimal input in that case is probably going to be "large" and is able to steer to output y to zero quickly.

We have $A = 0, B = 1, R = \rho^2, Q = 4$, thus the ARE (4.33) and optimal input for this problem are

$$4 - \frac{1}{\rho^2} P^2 = 0, \qquad \mathrm{u}_* = -\frac{1}{\rho^2} P\mathrm{x}.$$

Clearly this means $P = \pm 2\rho$. Since $A - BR^{-1}B^{\mathrm{T}}P = \mp 2/\rho$ needs to be negative, we find that

$$P = +2\rho, \qquad \mathrm{u}_* = -\frac{2}{\rho}\mathrm{x},$$

and the optimal closed-loop system is $\dot{\mathrm{x}}_*(t) = -\frac{2}{\rho}\mathrm{x}_*(t)$. In particular we have

$$\mathrm{y}_*(t) = 2\,\mathrm{e}^{-\frac{2}{\rho}t}, \qquad \mathrm{u}_*(t) = -\frac{2}{\rho}\,\mathrm{e}^{-\frac{2}{\rho}t}.$$

Let us consider several different values and ranges of the tuning parameter ρ:

- If $\rho = 1$ then the input u is "as cheap" or "as expensive" as y; they are equally weighted. The closed-loop eigenvalue in that case is $A - BR^{-1}B^{\mathrm{T}}P = -2$, and the optimal u_* and y_* have the same magnitude: $|\mathrm{u}_*(t)| = |\mathrm{y}_*(t)| = 2\,\mathrm{e}^{-2t}$. (See the red graphs of Fig. 4.2.)

- If $0 < \rho \ll 1$ then the control input u is "cheap". The closed-loop system is fast now (the closed-loop eigenvalue is $-2/\rho \ll -2 < 0$), and both u_*, y_* converge to zero quickly, but u_* initially is relatively large (in magnitude): $|u_*(0)| = 2/\rho = |y_*(0)|/\rho \gg |y_*(0)|$. That is to be expected since control is cheap. (See the yellow graphs of Fig. 4.2.)

- Conversely, if $\rho \gg 1$ then the input u is "expensive". The closed-loop system is now slow (the closed-loop eigenvalue is $-2/\rho \approx 0$), and both u_*, y_* converge to zero slowly, although u initially is already small: $u_*(0) = -2/\rho \approx 0$. That is to be expected since control is expensive. (See the blue graphs of Fig. 4.2.)

It is not hard to see that the optimal solutions satisfy

$$\int_0^\infty y_*^2(t)\,dt = \rho \quad \text{and} \quad \int_0^\infty u_*^2(t)\,dt = \frac{1}{\rho}.$$

Hence $\int_0^\infty y_*^2(t)\,dt = 1/\int_0^\infty u_*^2(t)\,dt$. This relation establishes once more that small inputs result in large outputs, and that large inputs result in small outputs. □

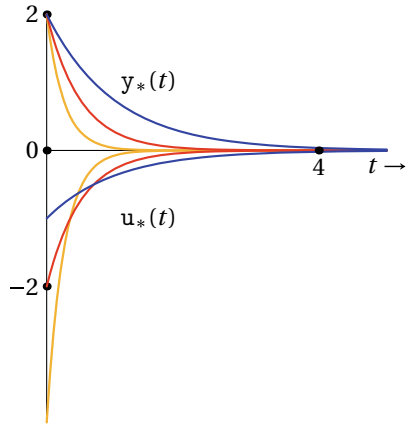


FIGURE 4.2: Graphs of optimal y_* and u_* for $\rho = 1/2$ (yellow), for $\rho = 1$ (red), and $\rho = 2$ (blue). The larger ρ is the slower the system is and the smaller $|u_*(0)|$ is. See Example 4.6.1.

Example 4.6.2 (Two tuning parameters). Consider the third-order system

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & -0.1 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t), \qquad x(0) = x_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},$$

$$y(t) = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} x(t).$$

We want to steer the output y to zero quickly but not too steeply, so $\dot{\mathrm{y}}$ should be small as well, and all that using small u. This requires a cost function with two tuning parameters,

$$\int_0^\infty \sigma^2 \mathrm{y}^2(t) + (1-\sigma^2)\dot{\mathrm{y}}^2(t) + \rho^2 \mathrm{u}^2(t)\,\mathrm{d}t.$$

The parameter $\sigma \in [0,1]$ defines the trade-off between small y and small $\dot{\mathrm{y}}$, and the parameter $\rho > 0$ defines the trade-off between small $(\mathrm{y}, \dot{\mathrm{y}})$ and small u.

Given σ and ρ the LQ solution can be determined numerically using the eigenvalues and eigenvectors of the corresponding Hamiltonian matrix, but we skip the details. In what follows we take as initial state $x_0 = (1,0,0)$. Figure 4.3 shows the response of the optimal u_* and resulting y_* for various combinations of σ and ρ. For $\sigma = 1$ the term $\dot{\mathrm{y}}$ is not included in the cost, so we can expect "steep" behavior in the output. For $\sigma \approx 0$ the output converges slowly to zero. As for ρ, we see that smaller ρ means larger controls u_* and faster convergence to zero of the output y_*.

Assuming we can live with inputs u of at most 2 (in magnitude) then $\rho^2 = 0.2$ is a reasonable choice (the red graphs in Fig. 4.3). Given that, a value of $\sigma^2 = 0.75$ may be a good compromise between overshoot and settling time in the response y_*. For this $\rho^2 = 0.2, \sigma^2 = 0.75$, the optimal control turns out to be

$$\mathrm{u}_* = -R^{-1}B^{\mathrm{T}}P\mathrm{x} = -(1.9365\mathrm{x}_1 + 3.0656\mathrm{x}_2 + 2.6187\mathrm{x}_3),$$

and the eigenvalues of $A - BR^{-1}B^{\mathrm{T}}P$ are -0.7468 and $-0.9859 \pm 1.2732\mathrm{i}$. □

Example 4.6.3 (Control of a car connected to a wall via a spring). Consider a car of mass m connected to a wall via a spring with spring constant k, see Fig. 4.4. The position of the car is denoted by y. Assume the car is controlled by an external force u. Newton's second law says that

$$m\ddot{\mathrm{y}}(t) + k\mathrm{y}(t) = \mathrm{u}(t).$$

To keep matters simple we take $k = 1$ and $m = 1$, so

$$\ddot{\mathrm{y}}(t) + \mathrm{y}(t) = \mathrm{u}(t).$$

For zero input the car continues to oscillate around $y = 0$. The task of the controller is to bring the car quickly to a stand still at position $y = 0$ but without using excessive force u. We propose to take as cost

$$\int_0^\infty \mathrm{y}^2(t) + \tfrac{1}{3}\mathrm{u}^2(t)\,\mathrm{d}t.$$

A state representation of the system is

$$\dot{\mathrm{x}}(t) = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \mathrm{x}(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \mathrm{u}(t),$$
$$\mathrm{y}(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \mathrm{x}(t).$$

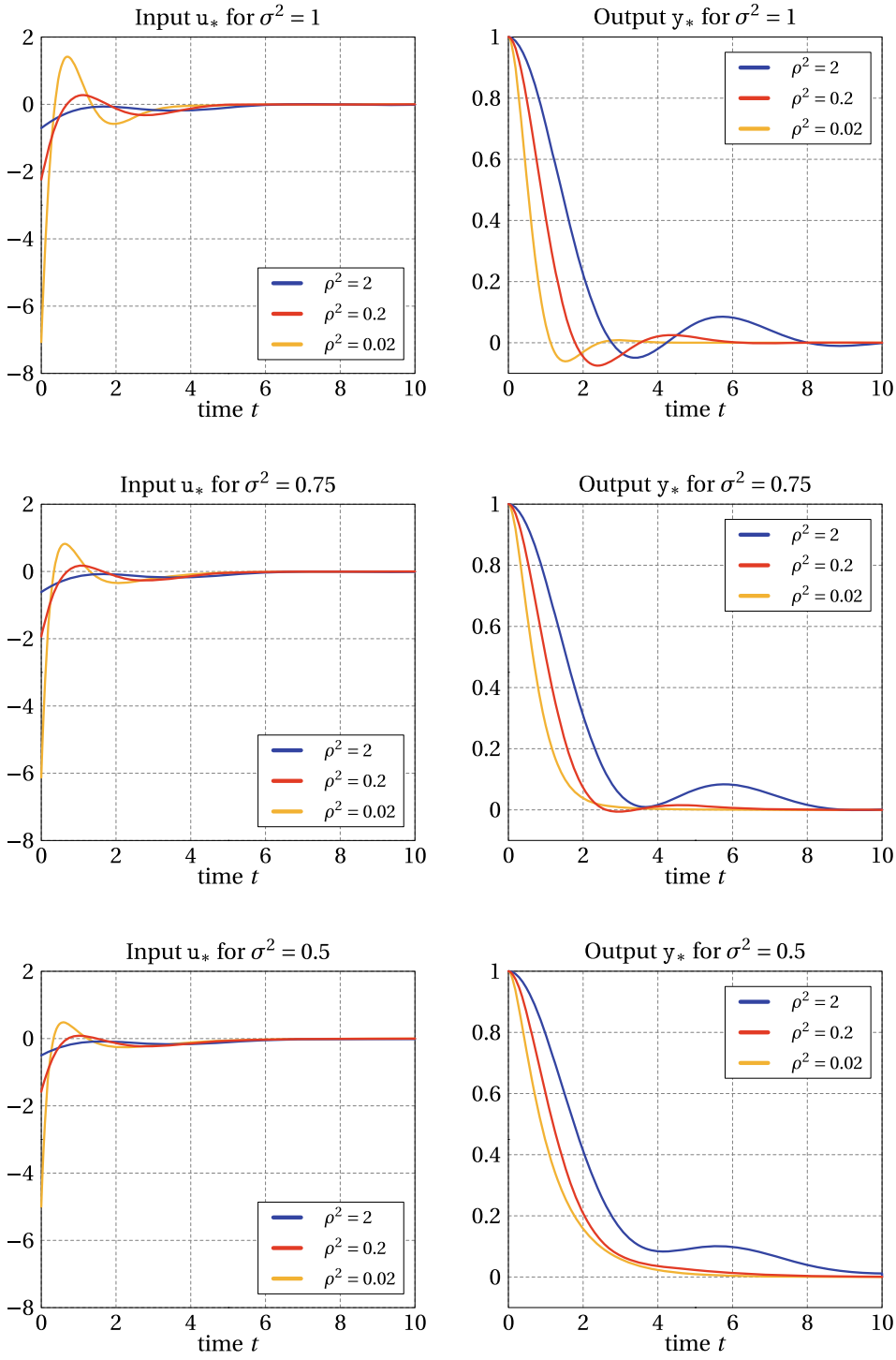


FIGURE 4.3: LQ-optimal responses of u_* (left) and y_* (right) for various combinations of $\sigma \in [0,1]$ and $\rho > 0$. See Example 4.6.2.

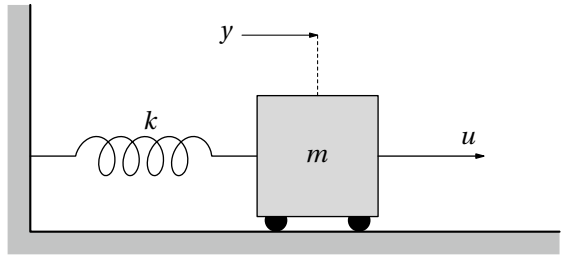


FIGURE 4.4: Car connected to a wall via a spring and with a force control u. See Example 4.6.3.

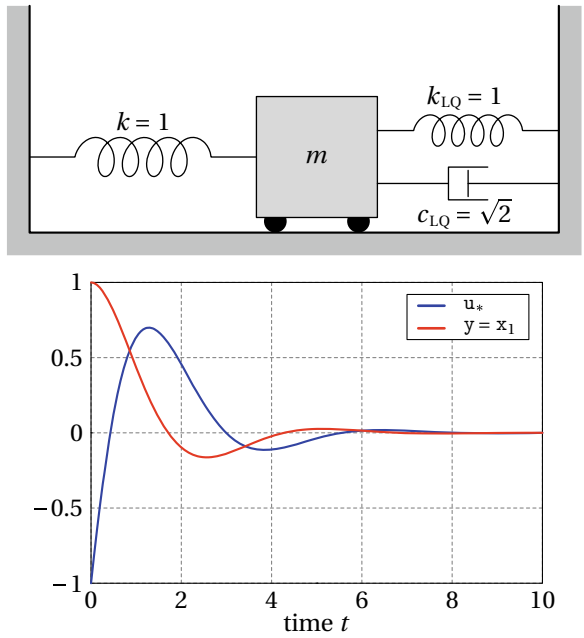


FIGURE 4.5: Top: a car connected to a wall via a spring (on the left). The car is controlled with an LQ-optimal force $\mathtt{u}_*(t) = -k_{\mathrm{LQ}}\mathtt{y}(t) - c_{\mathrm{LQ}}\dot{\mathtt{y}}(t)$ implemented as spring/damper system (on the right). Bottom: responses $\mathtt{u}_*(t)$ and $\mathtt{y}(t) = \mathtt{x}_1(t)$ for initial state $\mathtt{x}(0) = \left[\begin{smallmatrix}1\\0\end{smallmatrix}\right]$. See Example 4.6.3.

Here, the first state component is $\mathrm{x}_1 = \mathrm{y}$ and the second is $\mathrm{x}_2 = \dot{\mathrm{y}}$. This way the cost becomes $\int_0^\infty \mathrm{x}_1^2(t) + \frac{1}{3}\mathrm{u}^2(t)\,\mathrm{d}t$, and the stabilizing solution of the ARE turns out to be[3]

$$P = \frac{1}{3}\begin{bmatrix} 2\sqrt{2} & 1 \\ 1 & \sqrt{2} \end{bmatrix},$$

and the eigenvalues of $A - BR^{-1}B^{\mathrm{T}}P$ are $-\frac{1}{2}\sqrt{2} \pm \sqrt{3/2}\mathrm{i}$, while

$$\mathrm{u}_*(t) = -3B^{\mathrm{T}}P\mathrm{x}(t) = -\begin{bmatrix} 1 & \sqrt{2} \end{bmatrix}\mathrm{x}(t) = -\mathrm{y}(t) - \sqrt{2}\dot{\mathrm{y}}(t).$$

An analog implementation of this control law is a spring with spring constant $k_{\mathrm{LQ}} = 1$ parallel to a damper with friction coefficient $c_{\mathrm{LQ}} = \sqrt{2}$, see Fig. 4.5(top). For $x_0 = \left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right]$ the LQ-optimal input and output converge to zero quickly, although there is some overshoot, see Fig. 4.5 (bottom). □

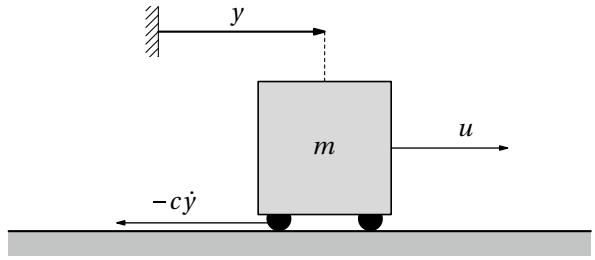


FIGURE 4.6: A car at position y subject to a friction force $-c\dot{y}$ and external force u. See Example 4.6.4.

Example 4.6.4 (Control of a car subject to friction). In the previous example we considered a car connected to a wall via a spring. Now we consider a car that is subject to damping (e.g., linear friction). As in the previous example, m denotes the mass of the car, and y is its position. The input u is an external force, and $-c\dot{y}$ models the friction force, see Fig. 4.6. The model is

$$m\ddot{\mathrm{y}}(t) + c\dot{\mathrm{y}}(t) = \mathrm{u}(t), \qquad c > 0. \tag{4.41}$$

We take the mass equal to $m = 1$, and we leave the friction coefficient c arbitrary (but positive). As state we take $\mathrm{x} := (\mathrm{y}, \dot{\mathrm{y}})$. Then (4.41) becomes

$$\dot{\mathrm{x}}(t) = A\mathrm{x}(t) + B\mathrm{u}(t) \quad \text{with} \quad A = \begin{bmatrix} 0 & 1 \\ 0 & -c \end{bmatrix},\ B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The aim is again to bring the mass to rest but without using excessive control effort. A possible solution is to minimize the cost

$$J_{[0,\infty)}(x_0, \mathrm{u}) = \int_0^\infty \mathrm{y}^2(t) + \rho^2\mathrm{u}^2(t)\,\mathrm{d}t.$$

[3] This is the reason we took $R = \frac{1}{3}$. Other values yield more complicated expressions for P.

Again the parameter $\rho > 0$ defines a trade-off between small y and small u. The matrices Q and R for this cost are

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad R = \rho^2.$$

(It can be verified that (A, B) is stabilizable and (Q, A) is detectable.) The ARE becomes

$$P \begin{bmatrix} 0 & 1 \\ 0 & -c \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & -c \end{bmatrix} P - P \begin{bmatrix} 0 & 0 \\ 0 & \rho^{-2} \end{bmatrix} P + \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

This matrix equation is effectively a set of *three* scalar equations in *three* unknowns. Indeed, the matrix P is symmetric so is characterized by three numbers, $P = \left[\begin{smallmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{smallmatrix}\right]$, and then the above left-hand side is symmetric so it equals zero iff its $(1,1)$-element, $(1,2)$-element and $(2,2)$-element are zero. This gives

$$\begin{aligned} 0 &= 1 - \rho^{-2} p_{12}^2, \\ 0 &= p_{11} - c p_{12} - \rho^{-2} p_{12} p_{22}, \\ 0 &= 2p_{12} - 2c p_{22} - \rho^{-2} p_{22}^2. \end{aligned}$$

From the first equation we find that $p_{12} = \pm\rho$. If $p_{12} = +\rho$ then the third equation gives two possible $p_{22} = \rho^2(-c \pm \sqrt{c^2 + 2/\rho})$. One is positive, the other is negative. We need the positive solution because positive semi-definiteness of P requires $p_{22} \geq 0$. Now that p_{12} and p_{22} are known, the second equation settles p_{11}. This turns out to give

$$P = \rho \begin{bmatrix} \sqrt{c^2 + 2/\rho} & 1 \\ 1 & \rho\big(-c + \sqrt{c^2 + 2/\rho}\big) \end{bmatrix}. \tag{4.42}$$

(Similarly, for $p_{12} = -\rho$ the resulting P turns out not to be positive semi-definite.) Conclusion: the P of (4.42) is the unique positive semi-definite solution P of the ARE. Hence it is the solution we seek. The optimal control is

$$\begin{aligned} \mathrm{u}_*(t) &= -\rho^{-2} B^{\mathrm{T}} P \mathrm{x}(t) \\ &= \begin{bmatrix} -\frac{1}{\rho} & c - \sqrt{c^2 + 2/\rho} \end{bmatrix} \mathrm{x}(t) \\ &= -\tfrac{1}{\rho}\mathrm{y}(t) + \big(c - \sqrt{c^2 + 2/\rho}\big)\dot{\mathrm{y}}(t). \end{aligned}$$

This optimal control is a linear combination of the displacement $\mathrm{y}(t)$ and velocity $\dot{\mathrm{y}}(t)$; similarly to the solution found in Example 4.6.3. These two terms can be interpreted as a spring and friction force in parallel, connected to a wall, see Fig. 4.7. □

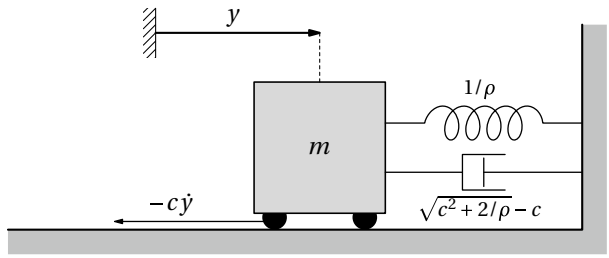


FIGURE 4.7: A car at position y subject to a friction force $-c\dot{y}$. It is optimally controlled with a spring with spring constant $1/\rho$ and a damper with friction coefficient $\sqrt{c^2+2/\rho}-c$. See Example 4.6.4.

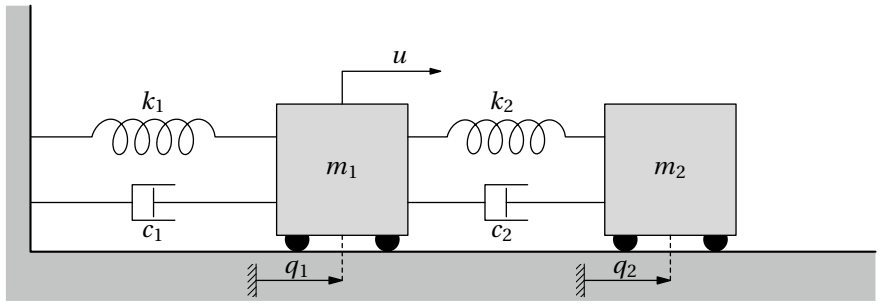


FIGURE 4.8: Two connected cars. The purpose is to control the second car with a force u that acts on the first car. See Example 4.6.5.

Example 4.6.5 (Connected cars). In this example we consider an application of two connected cars. The state dimension in this case is four which is too high to easily determine the solution of the Riccati equation by hand. The solution will be determined numerically.

The two cars are connected to each other with springs and dampers, and with the car on the left connected to a wall, see Fig. 4.8. The two spring constants are denoted k_1 and k_2, and the two friction coefficients c_1 and c_2. The horizontal positions of the two cars relative to the equilibrium positions are denoted q_1 and q_2 respectively, and the two masses are m_1 and m_2. We can control the first car with an additional force u, but we want to control the position q_2 of the second car. This application represents a common situation (for instance in a robotics context) where the control action is physically separated from the part that needs to be controlled.

The standard model for this system is

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{bmatrix} \ddot{\mathtt{q}}_1(t) \\ \ddot{\mathtt{q}}_2(t) \end{bmatrix} + \begin{bmatrix} c_1+c_2 & -c_2 \\ -c_2 & c_2 \end{bmatrix} \begin{bmatrix} \dot{\mathtt{q}}_1(t) \\ \dot{\mathtt{q}}_2(t) \end{bmatrix} + \begin{bmatrix} k_1+k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{bmatrix} \mathtt{q}_1(t) \\ \mathtt{q}_2(t) \end{bmatrix} = \begin{bmatrix} \mathtt{u}(t) \\ 0 \end{bmatrix}.$$

For simplicity we take all masses and spring constants equal to one, $m_1 = m_2 = 1, k_1 = k_2 = 1$, and that the friction coefficients are small and the same: $c_1 = c_2 = 0.1$. Then the linear model in terms of the state $\mathtt{x}$ defined as $\mathtt{x} = (\mathtt{q}_1, \mathtt{q}_2, \dot{\mathtt{q}}_1, \dot{\mathtt{q}}_2)$

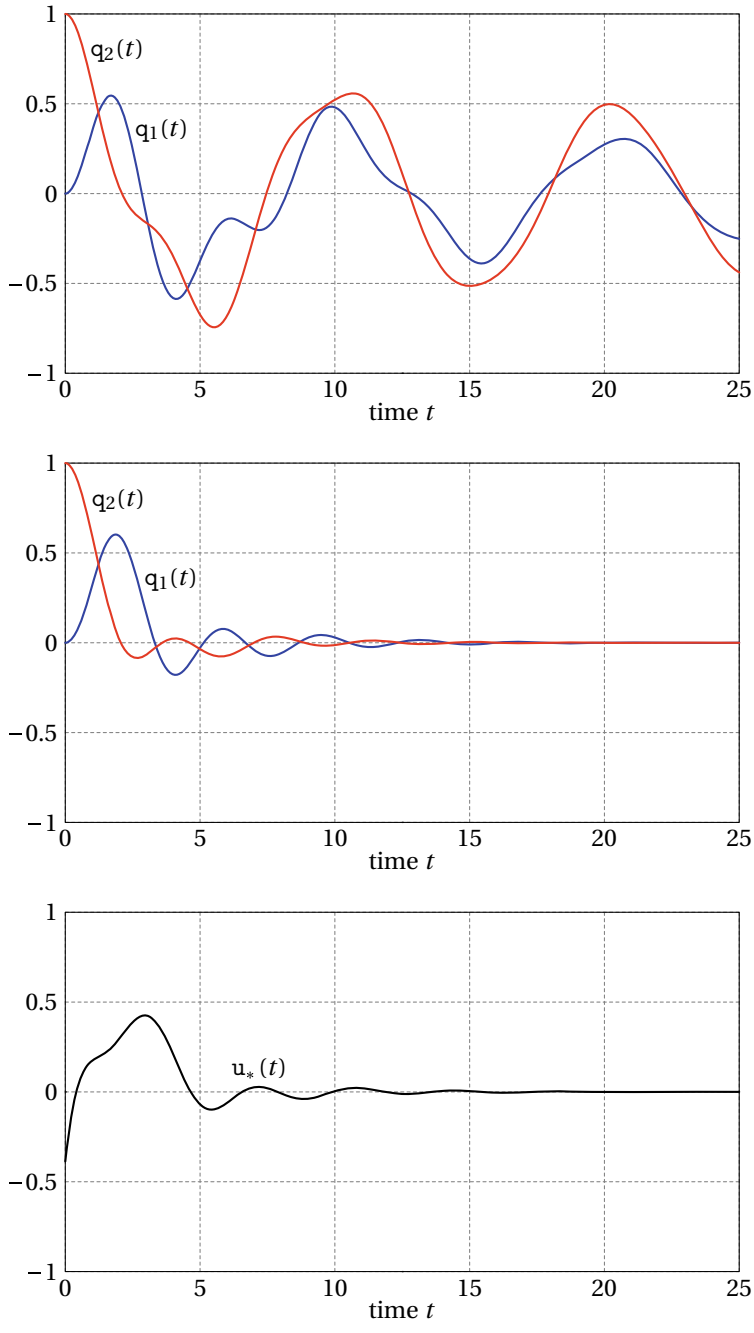


FIGURE 4.9: Top: positions of the uncontrolled cars. Middle: positions of the controlled cars. Bottom: control force u_* for the controlled car. For all cases the initial state is $q_1(0) = 0, q_2(0) = 1, \dot{q}_1(0) = 0, \dot{q}_2(0) = 0$. See Example 4.6.5.

becomes

$$\dot{\mathtt{x}}(t) = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -2 & 1 & -0.2 & 0.1 \\ 1 & -1 & 0.1 & -0.1 \end{bmatrix} \mathtt{x}(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \mathtt{u}(t).$$

As the friction coefficients are small one may expect sizeable oscillations when no control is applied. Indeed, the above A matrix has two eigenvalues close to the imaginary axis (at $-0.0119 \pm \mathrm{i}0.6177$ and $-0.1309 \pm \mathrm{i}1.6127$), and for the initial state $x_0 = (0,1,0,0)$ and $\mathtt{u} = 0$ the positions $\mathtt{q}_1, \mathtt{q}_2$ of the two cars oscillate for a long time, see Fig. 4.9(top).

To control the second car with the force $\mathtt{u}$ we propose the solution of the infinite horizon LQ problem with cost

$$\int_0^\infty \mathtt{q}_2^2(t) + R\mathtt{u}^2(t)\,\mathrm{d}t.$$

The value of R was set, somewhat arbitrarily, to $R = 0.2$. Since A is asymptotically stable we know that (A, B) is stabilizable and (Q, A) is detectable whatever B and Q we have. Therefore the conditions of Theorem 4.5.6 are met, and so we are guaranteed that the stabilizing solution P of the Riccati equation exists and is unique. The solution, obtained numerically, is

$$P = \begin{bmatrix} 0.4126 & 0.2286 & 0.2126 & 0.5381 \\ 0.2286 & 0.9375 & 0.0773 & 0.5624 \\ 0.2126 & 0.0773 & 0.2830 & 0.4430 \\ 0.5381 & 0.5624 & 0.4430 & 1.1607 \end{bmatrix},$$

and the optimal state feedback control $\mathtt{u}_*(t) = -R^{-1}B^{\mathrm{T}}P\mathtt{x}(t)$ follows as

$$\mathtt{u}_*(t) = -\begin{bmatrix} 1.0628 & 0.3867 & 1.4151 & 2.2150 \end{bmatrix} \mathtt{x}(t).$$

Under this control the response for the initial state $x_0 = (0,1,0,0)$ is damped much stronger than without control, see Fig. 4.9(middle). The eigenvalues of the closed loop system $\dot{\mathtt{x}}_*(t) = (A - BR^{-1}B^{\mathrm{T}}P)\mathtt{x}_*(t)$ are $-0.5925 \pm 0.6847\mathrm{i}$ and $-0.2651 \pm 1.7081\mathrm{i}$, and these are considerably further away from the imaginary axis than the eigenvalues of A, and the imaginary parts are almost the same. This confirms the stronger damping in the controlled system.

All this is achieved with a control force $\mathtt{u}_*(t)$ that never exceeds 0.4 in magnitude for this initial state, see Fig. 4.9 (bottom). Notice that the optimal control $\mathtt{u}_*(t)$ starts out negative but turns positive way before $\mathtt{q}_2(t)$ becomes zero for the first time. So apparently it is optimal to initially speed up the first car away from the second car, but only for a very short period of time, and then for the next couple of seconds to move the first car towards the second car.

For the initial state $x_0 = (0,1,0,0)$ the optimal cost $x_0^{\mathrm{T}}Px_0$ follows from the $(2,2)$-element of P: $\int_0^\infty \mathtt{q}_{*2}^2(t) + R\mathtt{u}_*^2(t)\,\mathrm{d}t = 0.9375$. □

4.7 Exercises

4.1 *Hamiltonian matrix.* Let $T > 0$ and consider the system

$$\dot{\mathrm{x}}(t) = 3\mathrm{x}(t) + 2\mathrm{u}(t), \qquad \mathrm{x}(0) = x_0$$

with cost

$$J_{[0,T]}(x_0, \mathrm{u}) = \int_0^T 4\mathrm{x}^2(t) + \mathrm{u}^2(t)\,\mathrm{d}t.$$

(a) Determine the Hamiltonian matrix $\mathscr{H}$.

(b) It can be shown that

$$\mathrm{e}^{\mathscr{H}t} = \frac{1}{5}\begin{bmatrix} 4\mathrm{e}^{5t} + \mathrm{e}^{-5t} & -2\mathrm{e}^{5t} + 2\mathrm{e}^{-5t} \\ -2\mathrm{e}^{5t} + 2\mathrm{e}^{-5t} & \mathrm{e}^{5t} + 4\mathrm{e}^{-5t} \end{bmatrix}.$$

For arbitrary $T > 0$ determine the optimal $\mathrm{x}_*(t), \mathrm{u}_*(t), \mathrm{p}_*(t)$ and the optimal cost.

4.2 *Hamiltonian equations for an LQ problem with negative weight.* Consider the system and cost of Example 4.4.2. Special about the example is that $Q < 0$. This makes it a non-standard LQ problem. In the example we found an optimal control only if $0 \le T < \pi/2$. For $T = \pi/2$ the method failed. In this exercise we use Hamiltonian equations to analyze the case $T = \pi/2$.

(a) Determine the Hamiltonian matrix $\mathscr{H}$ for this problem.

(b) It can be shown that

$$\mathrm{e}^{\mathscr{H}t} = \begin{bmatrix} \cos(t) & -\sin(t) \\ \sin(t) & \cos(t) \end{bmatrix}.$$

Use this to confirm the claim that for $T = \pi/2$ the Hamiltonian equations (4.8) have no solution if $x_0 \neq 0$.

(c) Does Pontryagin's minimum principle allow us to conclude that for $T = \pi/2$ and $x_0 \neq 0$ no optimal control u_* exists?

(d) *A Wirtinger inequality.* Show that $\int_0^{\pi/2} \dot{\mathrm{x}}^2(t)\,\mathrm{d}t \ge \int_0^{\pi/2} \mathrm{x}^2(t)\,\mathrm{d}t$ for all smooth x for which $\mathrm{x}(0) = x_0 := 0$, and show that equality holds iff $\mathrm{x}(t) = A\sin(t)$.

4.3 *A direct proof of why solutions of the Hamiltonian equations in the LQ problem are optimal solutions.* Following Theorem 4.2.4 we argued that the LQ problem satisfies the convexity conditions of Theorem 2.8.1, so satisfaction of the Hamiltonian equations is both necessary and sufficient for optimality. There is also a direct proof of this result. It exploits the

linear/quadratic nature of the LQ problem. To simplify matters a bit we assume in this exercise that

$$R = I, \qquad S = 0,$$

and, as always, $Q \geq 0$.

(a) Show that the finite horizon LQ problem satisfies the convexity assumptions of Theorem 2.8.1. [Hint: Appendix A.7 may be useful.]

(b) Consider a solution $(\mathtt{x}_*, \mathtt{p}_*)$ of (4.8), and define $\mathtt{u}_* = -B^{\mathrm{T}}\mathtt{p}_*$. Now consider an arbitrary input $\mathtt{u}$ and corresponding state $\mathtt{x}$, an define $\mathtt{v} = \mathtt{u} - \mathtt{u}_*$.

i. Show that $\mathtt{z} := \mathtt{x} - \mathtt{x}_*$ satisfies $\dot{\mathtt{z}}(t) = A\mathtt{z}(t) + B\mathtt{v}(t), \mathtt{z}(0) = 0$.

ii. Show that

$$J_{[0,T]}(x_0, \mathtt{u}) - J_{[0,T]}(x_0, \mathtt{u}_*) = \int_0^T \mathtt{z}^{\mathrm{T}} Q\mathtt{z} + \mathtt{v}^{\mathrm{T}}\mathtt{v} + 2\mathtt{z}^{\mathrm{T}} Q\mathtt{x}_* + 2\mathtt{u}_*^{\mathrm{T}}\mathtt{v}\,\mathrm{d}t.$$

(For readability we dropped here the time argument.)

iii. Show that $\frac{\mathrm{d}}{\mathrm{d}t}(\mathtt{p}_*^{\mathrm{T}}(t)\mathtt{z}(t)) = -\mathtt{z}^{\mathrm{T}}(t)Q\mathtt{x}_*(t) - \mathtt{u}_*^{\mathrm{T}}(t)\mathtt{v}(t)$.

iv. Show that

$$J_{[0,T]}(x_0, \mathtt{u}) - J_{[0,T]}(x_0, \mathtt{u}_*) = \int_0^T \mathtt{z}^{\mathrm{T}}(t)Q\mathtt{z}(t) + \mathtt{v}^{\mathrm{T}}(t)\mathtt{v}(t)\,\mathrm{d}t,$$

and argue that $\mathtt{u}_*$ is the optimal control.

4.4 *There are RDE's whose solution is constant.* Let $T > 0$ and consider

$$\dot{\mathtt{x}}(t) = \mathtt{x}(t) + \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0 := 1,$$

with cost

$$J_{[0,T]}(x_0, \mathtt{u}) = 2\mathtt{x}^2(T) + \int_0^T \mathtt{u}^2(t)\,\mathrm{d}t.$$

(a) Determine the RDE.

(b) Solve the RDE. [Hint: the solution happens to be constant.]

(c) Determine the optimal state $\mathtt{x}_*(t)$ and input $\mathtt{u}_*(t)$ explicitly as functions of time.

(d) Verify that $J_{[0,T]}(1, \mathtt{u}_*) = P(0)$.

4.5 *Why LQ-optimal inputs are linear in the state, and costs are quadratic in the state.* In this exercise we prove, using only *elementary* arguments (but not *easy* arguments), that the optimal control in LQ control is *linear* in the state, and that the value function is *quadratic* in the state. Consider $\dot{\mathtt{x}}(t) = A\mathtt{x}(t) + B\mathtt{u}(t)$ with the standard LQ cost over the time window $[t, T]$,

$$J_{[t,T]}(\mathtt{x}(t), \mathtt{u}) = \mathtt{x}^{\mathrm{T}}(T)S\mathtt{x}(T) + \int_t^T \mathtt{x}^{\mathrm{T}}(\tau)Q\mathtt{x}(\tau) + \mathtt{u}^{\mathrm{T}}(\tau)R\mathtt{u}(\tau)\,\mathrm{d}\tau,$$

and let $\mathcal{V}(x, t)$ be its value function.

(a) Exploit the quadratic nature of the cost to prove that for every $\lambda \in \mathbb{R}$, every two $x, z \in \mathbb{R}^n$ and every two inputs $\mathtt{u}, \mathtt{w}$ we have

$$J_{[t,T]}(\lambda x, \lambda \mathtt{u}) = \lambda^2 J_{[t,T]}(x, \mathtt{u}),$$
$$J_{[t,T]}(x+z, \mathtt{u}+\mathtt{w}) + J_{[t,T]}(x-z, \mathtt{u}-\mathtt{w}) = 2J_{[t,T]}(x, \mathtt{u}) + 2J_{[t,T]}(z, \mathtt{w}). \tag{4.43}$$

(The second identity is known as the *parallelogram law*.)

(b) Prove that $\mathcal{V}(\lambda x, t) = \lambda^2 \mathcal{V}(x, t)$, and that input $\lambda \mathtt{u}_*$ is optimal for initial state λx if $\mathtt{u}_*$ is optimal for initial state x.

(c) Conclude that

$$\mathcal{V}(x+z, t) + \mathcal{V}(x-z, t) \leq 2\mathcal{V}(x, t) + 2\mathcal{V}(z, t).$$

[Hint: minimize the right-hand side of (4.43) over all $\mathtt{u}, \mathtt{w}$.]

(d) Likewise conclude that

$$\mathcal{V}(x+z, t) + \mathcal{V}(x-z, t) \geq 2\mathcal{V}(x, t) + 2\mathcal{V}(z, t).$$

[Hint: minimize the left-hand side of (4.43) over all $\mathtt{u}+\mathtt{w}, \mathtt{u}-\mathtt{w}$.]

(e) Suppose $\mathtt{u}_x$ is the optimal input for x, and $\mathtt{w}_z$ is the optimal input for z. Use (a,c,d) to show that

$$J_{[t,T]}(x+z, \mathtt{u}_x + \mathtt{w}_z) - \mathcal{V}(x+z, t) = \mathcal{V}(x-z, t) - J_{[t,T]}(x-z, \mathtt{u}_x - \mathtt{w}_z).$$

(f) Let $\lambda \in \mathbb{R}$. Prove that if $\mathtt{u}_x$ is the optimal input for x, and $\mathtt{w}_z$ is the optimal input for z, then $\mathtt{u}_x + \lambda \mathtt{w}_z$ is the optimal input for $x + \lambda z$. [Hint: both sides of the identity of the previous part are zero. Why?]

(g) The previous part shows that the optimal control $\mathtt{u}_* : [t, T] \to \mathbb{R}^m$ for $J_{[t,T]}(\mathtt{x}(t), \mathtt{u})$ is linear in $\mathtt{x}(t)$. Show that this implies that at each t the optimal control $\mathtt{u}_*(t)$ is linear in $\mathtt{x}(t)$.

(h) Argue that $\mathcal{V}(x, t)$ for each t is quadratic in the state, i.e., that $\mathcal{V}(x, t) = x^{\mathrm{T}} P(t) x$ for some matrix $P(t) \in \mathbb{R}^{n \times n}$. [Hint: it is quadratic iff $\mathcal{V}(x+\lambda z, t) + \mathcal{V}(x-\lambda z, t) = 2\mathcal{V}(x, t) + \lambda^2 \mathcal{V}(z, t)$ for all x, z and scalars λ.]

4.6 *Solution of scalar RDE.* Consider the *scalar* system $\dot{\mathtt{x}}(t) = A\mathtt{x}(t) + B\mathtt{u}(t)$, that is, with $\mathtt{x}$ having just one entry. Then A, B, Q, R are numbers. As always we assume that $Q \geq 0$ and $R > 0$. The RDE in the scalar case may be solved explicitly, as we show in this exercise.

(a) Suppose $B \neq 0$. Show that the RDE (4.22) is of the form

$$\dot{P}(t) = \gamma(P(t) - \alpha)(P(t) - \beta).$$

for some nonzero real numbers α, β, γ and with $\gamma > 0$.

(b) Consider the above scalar differential equation in $P(t)$. Prove that

$$G(t) := \frac{1}{P(t) - \alpha}$$

satisfies

$$\dot{G}(t) = \gamma(\beta - \alpha)G(t) - \gamma.$$

(c) The above assumes that $P(t) \neq \alpha$. Solve the differential equation of (a) directly for the case that $P(\bar{t}) = \alpha$ for some $\bar{t} \in [0, T]$. (Yes, the answer is easy.)

(d) Solve the RDE (4.22) for $A = -1, B = 2, Q = 4, R = 2, S = 0$, and final time $T = 1$.

(e) Solve the RDE (4.22) for $A = -1, B = 2, Q = 4, R = 2, S = 1$, and final time $T = 1$.

(f) Solve the RDE (4.22) for $A = 0, B = 1, Q = 1, R = 1, S = 0$, and final time $T = 1$.

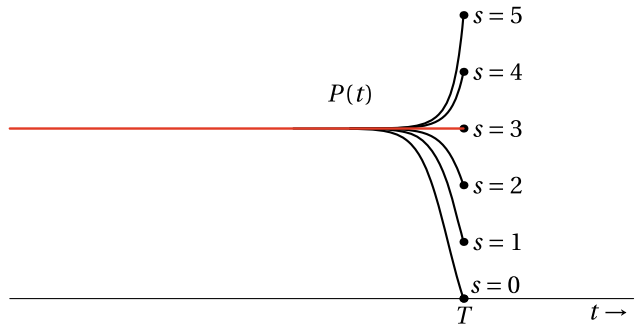


FIGURE 4.10: Graph of $P(t)$ for several values of $P(T) = s$. If $s = 3$ then $P(t)$ is constant (shown in red). See Exercise 4.7.

4.7 *Dependence of $P(t)$ and $\mathrm{x}_*(t)$ on the final cost.* Consider the scalar system

$$\dot{\mathrm{x}}(t) = \mathrm{x}(t) + \mathrm{u}(t), \qquad \mathrm{x}(0) = x_0,$$

and cost

$$J_{[0,T]}(x_0, \mathrm{u}) = s\,\mathrm{x}^2(T) + \int_0^T 3\mathrm{x}^2(t) + \mathrm{u}^2(t)\,\mathrm{d}t.$$

Here s is some nonnegative number.

(a) Determine the associated RDE (4.22), and verify that the solution is given by

$$P(t) = \frac{3 - d\,\mathrm{e}^{4(t-T)}}{1 + d\,\mathrm{e}^{4(t-T)}} \qquad \text{for} \qquad d := \frac{3 - s}{1 + s}.$$

[Hint: Exercise 4.6 is useful.]

(b) Figure 4.10 depicts the graph of $P(t)$ for several $s \geq 0$. The graphs suggest that $P(t)$ is an increasing function if $s > 3$, and a decreasing function if $0 \leq s < 3$. Use the RDE to formally verify this property.

(c) It appears that for $s = 0$ the function $P(t)$ is decreasing. Argue from the value function why it is immediate that $P(t)$ is decreasing if $s = 0$. [Hint: $P(t)$ *decreases* iff for a fixed t the function $P(t)$ *increases* as a function of final time T.]

(d) Figure 4.11 shows graphs of the optimal state $\mathrm{x}_*(t)$ for $T = 1, T = 2$ and various s. The initial condition is $x_0 = 1$ in all cases. The plot on the left considers $T = 1$ and $s = 0, 1, 2, 3, 4, 5$. The plot on the right $T = 2$ and the same $s = 0, 1, 2, 3, 4, 5$. Explain which of the graphs correspond to which value of s, and also explain from the system equation $\dot{\mathrm{x}}(t) = \mathrm{x}(t) + \mathrm{u}(t)$ and cost why it can happen that for some s the optimal $\mathrm{x}_*(t)$ *increases* for t near T.

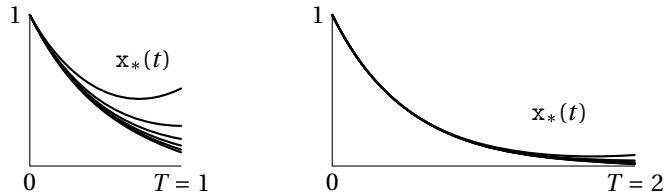


FIGURE 4.11: Graphs of optimal x_* with $x_0 = 1$. Left: for $T = 1$ and $s = 0, 1, 2, 3, 4, 5$. Right: for $T = 2$ and $s = 0, 1, 2, 3, 4, 5$. See Exercise 4.7.

4.8 *State transformation.* Sometimes a transformation of the state variables can facilitate solving the optimal control problem. With $\mathrm{z}(t)$ defined as $\mathrm{z}(t) = E^{-1}\mathrm{x}(t)$, show that the LQ problem for $\dot{\mathrm{x}}(t) = A\mathrm{x}(t) + B\mathrm{u}(t), \mathrm{x}(0) = x_0$ with cost

$$J_{[0,T]}(x_0, \mathrm{u}) = \int_0^T \mathrm{x}^{\mathrm{T}}(t) Q \mathrm{x}(t) + \mathrm{u}^{\mathrm{T}}(t) R \mathrm{u}(t)\, \mathrm{d}t$$

yields the problem

$$\dot{\mathrm{z}}(t) = \tilde{A}\mathrm{z}(t) + \tilde{B}\mathrm{u}(t), \qquad \mathrm{z}(0) = z_0 := E^{-1}x_0$$

with cost

$$\tilde{J}_{[0,T]}(z_0, \mathrm{u}) = \int_0^T \mathrm{z}^{\mathrm{T}}(t) \tilde{Q} \mathrm{z}(t) + \mathrm{u}^{\mathrm{T}}(t) R \mathrm{u}(t)\, \mathrm{d}t,$$

where $\tilde{A} = E^{-1}AE$, $\tilde{B} = E^{-1}B$ and $\tilde{Q} = E^{\mathrm{T}}QE$.

Also, what is the relationship between the value functions for both problems?

4.9 *State transformation.* Consider the infinite horizon LQ problem with stability for the system

$$\dot{\mathtt{x}}(t) = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \mathtt{x}(t) + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0$$

and cost

$$J_{[0,\infty)}(x_0, \mathtt{u}) = \int_0^\infty \mathtt{x}^{\mathrm{T}}(t) \begin{bmatrix} 7 & 2 \\ 2 & 7 \end{bmatrix} \mathtt{x}(t) + \mathtt{u}^{\mathrm{T}}(t)\mathtt{u}(t)\,\mathrm{d}t.$$

(a) Show that the optimal control is $\mathtt{u}_*(t) = -P\mathtt{x}(t)$, where P is the stabilizing solution of the ARE.

(b) To find the solution P, perform the state transformation $\mathtt{z} = E^{-1}\mathtt{x}$ for a suitable matrix E. Choose E such that $E^{-1}AE$ is diagonal, and use it to determine P. [Hint: use Exercise 4.8.]

4.10 *Direct proof of optimality.* The proof of Proposition 4.3.1 assumes that $R = I$. Develop a similar proof for the case that R is an arbitrary $m \times m$ positive definite matrix.

4.11 *Discount factor.* Consider the linear system

$$\dot{\mathtt{x}}(t) = A\mathtt{x}(t) + B\mathtt{u}(t), \qquad \mathtt{x}(0) = x_0$$

with $A \in \mathbb{R}^{n\times n}$ and $B \in \mathbb{R}^{n\times m}$, and suppose that the cost function contains an exponential factor,

$$J_{[0,T]}(x_0, \mathtt{u}) = \int_0^T \mathrm{e}^{-2\alpha t}\big(\mathtt{x}^{\mathrm{T}}(t)Q\mathtt{x}(t) + \mathtt{u}^{\mathrm{T}}(t)R\mathtt{u}(t)\big)\,\mathrm{d}t,$$

for a given constant α. (For $\alpha > 0$ the factor $\mathrm{e}^{-2\alpha t}$ is known as a *discount factor*, rendering running costs further in the future less important.) We also assume that $Q \geq 0$ and $R > 0$.

(a) Write the above cost function and system equations $\dot{\mathtt{x}}(t) = A\mathtt{x}(t) + B\mathtt{u}(t)$ in terms of the new variables $\mathtt{z}(t), \mathtt{v}(t)$ defined as

$$\mathtt{z}(t) = \mathrm{e}^{-\alpha t}\mathtt{x}(t), \quad \mathtt{v}(t) = \mathrm{e}^{-\alpha t}\mathtt{u}(t).$$

(The new version of the cost function and system equations should no longer contain $\mathtt{x}$ and $\mathtt{u}$.)

(b) With the aid of (a) determine the solution $\mathtt{u}_*(t)$ in terms of $\mathtt{x}_*(t)$, of the optimal control problem of the scalar system

$$\dot{\mathtt{x}}(t) = \mathtt{x}(t) + \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0 := 1$$

with cost

$$J_{[0,T]}(x_0, \mathtt{u}) = \int_0^1 \mathrm{e}^{-2t}\big(\mathtt{x}^2(t) + \mathtt{u}^2(t)\big)\,\mathrm{d}t.$$

4.12 *Solving the infinite horizon LQ problem via the eigenvectors of the Hamiltonian.* (This exercise assumes knowledge of basic linear algebra.) Consider the infinite horizon LQ problem of Exercise 4.9.

(a) Verify that the Hamiltonian matrix $\mathscr{H}$ has eigenvalues ± 3. [Hint: perform row operations on $\mathscr{H} + 3I$.]

(b) Determine two linearly independent eigenvectors v_1, v_2 of $\mathscr{H}$ that both have eigenvalue -3, and use these to construct the stabilizing solution P of the ARE.

4.13 Consider the LQ problem of Example 4.6.3.

(a) Verify that P satisfies the ARE, and that $P > 0$.

(b) Verify that

$$A - BR^{-1}B^{\mathrm{T}}P = \begin{bmatrix} 0 & 1 \\ -2 & -\sqrt{2} \end{bmatrix},$$

and show that the closed loop is asymptotically stable.

4.14 *Set-point regulation.* Consider the linear system

$$\dot{\mathtt{x}}(t) = A\mathtt{x}(t) + B\mathtt{u}(t), \qquad \mathtt{x}(0) = x_0$$

with $A \in \mathbb{R}^{n\times n}$ and $B \in \mathbb{R}^{n\times m}$. In applications we often want to steer the state to a "set-point" $\bar{x} \in \mathbb{R}^n$ which is not necessarily zero (think of the heating system in your house). For constant inputs, $\mathtt{u}(t) = \bar{u}$, a set-point $\bar{x}$ is an equilibrium iff

$$0 = A\bar{x} + B\bar{u}.$$

Given such a pair $(\bar{x}, \bar{u})$ we define a cost relative to them as

$$\bar{J}_{[0,\infty)}(x_0, \mathtt{u}) = \int_0^\infty (\mathtt{x}(t) - \bar{x})^{\mathrm{T}} Q(\mathtt{x}(t) - \bar{x}) + (\mathtt{u}(t) - \bar{u})^{\mathrm{T}} R(\mathtt{u}(t) - \bar{u})\,\mathrm{d}t,$$

in which $Q \geq 0$ and $R > 0$.

(a) Show that the transformation

$$\mathtt{z} := \mathtt{x} - \bar{x}, \qquad \mathtt{v} := \mathtt{u} - \bar{u}$$

reduces the above optimal control problem to a standard LQ problem.

(b) Under what general conditions on A, B, Q, R are we guaranteed that the problem of minimizing the above $\bar{J}$ over all inputs that steer $(\mathtt{x}, \mathtt{u})$ to $(\bar{x}, \bar{u})$ has a solution?

(c) What is the general form of the optimal input $\mathtt{u}_*$ in terms of $R, B, P, \mathtt{x}, \bar{x}, \bar{u}$?

(d) Apply the above to the optimal control problem for the scalar system

$$\dot{\mathtt{x}}(t) = -\mathtt{x}(t) + \mathtt{u}(t), \qquad \mathtt{x}(0) = 0$$

with cost

$$\int_0^\infty (\mathtt{x}(t) - 1)^2 + (\mathtt{u}(t) - 1)^2 \, \mathrm{d}t,$$

and compute the optimal control $\mathtt{u}_*$.

4.15 *Infinite horizon LQ.* We consider the same system and cost as in Exercise 4.1, but now with $T = \infty$:

$$\dot{\mathtt{x}}(t) = 3\mathtt{x}(t) + 2\mathtt{u}(t), \quad \mathtt{x}(0) = x_0, \quad J_{[0,\infty)}(x_0, \mathtt{u}) = \int_0^\infty 4\mathtt{x}^2(t) + \mathtt{u}^2(t) \, \mathrm{d}t.$$

(a) Determine the nonnegative solution P of the ARE.

(b) Verify that $A - BR^{-1}B^{\mathrm{T}}P$ is asymptotically stable.

(c) Determine the input that solves the LQ problem with stability, and write it in the form $\mathtt{u}(t) = -F\mathtt{x}(t)$ (that is, find F).

(d) Determine the optimal cost.

(e) Determine the eigenvalues of the Hamiltonian matrix $\mathscr{H}$ without determining the Hamiltonian matrix!

(f) Determine the Hamiltonian matrix and verify that $\left[\begin{smallmatrix} 1 \\ P \end{smallmatrix}\right]$ is an eigenvector of the Hamiltonian matrix.

4.16 *Infinite horizon LQ problems with and without stability.* Consider the system

$$\dot{\mathtt{x}}(t) = \mathtt{x}(t) + \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0, \qquad \mathtt{u}(t) \in \mathbb{R}$$

with the infinite horizon cost

$$J_{[0,\infty)}(x_0, \mathtt{u}) = \int_0^\infty g^2\mathtt{x}^2(t) + \mathtt{u}^2(t) \, \mathrm{d}t.$$

Here g is some nonzero real number.

(a) Determine all solutions P of the ARE.

(b) Which solution P do we need for the infinite horizon LQ problem with stability?

(c) Determine the solutions $\mathtt{u}_*(t), \mathtt{x}_*(t)$ explicitly as a function of time of the infinite horizon LQ problem with stability.

(d) Now take $g = 0$. Argue that the input that minimizes the cost over all *stabilizing* inputs is not the same as the one that minimizes the cost over all inputs (stabilizing or not).

4.17 *Lyapunov equation.* Let $Q \geq 0, R > 0$ and suppose that $B = 0$. Consider Theorem 4.5.6.

(a) Given that $B = 0$, under what conditions on A are the assumptions of Theorem 4.5.6 satisfied?

(b) Determine the ARE for this case.

(c) What result from § B.5 comes to mind?

4.18 *Quadratic cost consisting of two terms.* Consider the system

$$\dot{\mathrm{x}}(t) = \mathrm{u}(t), \qquad \mathrm{x}(0) = x_0, \qquad \mathrm{u}(t) \in \mathbb{R},$$

and the infinite horizon quadratic cost composed of two terms,

$$J_{[0,\infty)}(x_0, \mathrm{u}) = \int_0^1 \mathrm{u}^2(t)\ \mathrm{d}t + \int_1^\infty 4\mathrm{x}^2(t) + \mathrm{u}^2(t)\ \mathrm{d}t.$$

(a) Assume first that $\mathrm{x}(1)$ is given. Minimize $\int_1^\infty 4\mathrm{x}^2(t) + \mathrm{u}^2(t)\ \mathrm{d}t$ over all stabilizing inputs.

(b) Express the optimal cost $J_{[0,\infty)}(x_0, \mathrm{u})$ as $J_{[0,\infty)}(x_0, \mathrm{u}) = \int_0^1 \mathrm{u}^2(t)\ \mathrm{d}t + S\mathrm{x}^2(1)$, that is, determine S.

(c) Solve the optimal control problem, that is, minimize $J_{[0,\infty)}(x_0, \mathrm{u})$ over all stabilizing inputs, and express the optimal input $\mathrm{u}_*(t)$ as a function of $\mathrm{x}(t)$. [Hint: use *separation of variables*, see § A.3.]

4.19 *Properties of the Hamiltonian Matrix.* Let $\mathscr{H}$ be the Hamiltonian matrix as defined in (4.9). Define the characteristic polynomial $r(\lambda)$ as

$$r(\lambda) = \det(\lambda I - \mathscr{H}) = \det \begin{bmatrix} \lambda I - A & BR^{-1}B^{\mathrm{T}} \\ Q & \lambda I + A^{\mathrm{T}} \end{bmatrix}.$$

(a) Argue that

$$r(\lambda) = \det \begin{bmatrix} -Q & -\lambda I - A^{\mathrm{T}} \\ \lambda I - A & BR^{-1}B^{\mathrm{T}} \end{bmatrix}.$$

(b) Show that $r(\lambda) = r(-\lambda)$.

(c) Argue that $r(\lambda)$ is a polynomial in λ^2.

4.20 *Control of a car when the input is either cheap or expensive.* Consider Example 4.6.4. In this exercise we analyze what happens if ρ is either very large or very small (positive but close to zero). Keep in mind that $c > 0$.

(a) Show that

$$\lim_{\rho\to\infty} \frac{1}{\rho} P = \begin{bmatrix} c & 1 \\ 1 & 1/c \end{bmatrix}$$

and

$$\lim_{\rho\to\infty} \rho \mathrm{u}_*(t) = -\mathrm{y}(t) - (1/c)\dot{\mathrm{y}}(t).$$

(b) Assume that $x_0 = \left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right]$. Determine what happens with the optimal control, the optimal cost, and the eigenvalues of the closed-loop system as $\rho \to \infty$?

(c) Argue that for $\rho \approx 0$ (but positive) we have

$$P \approx \begin{bmatrix} \sqrt{2\rho} & \rho \\ \rho & \rho\sqrt{2\rho} \end{bmatrix}$$

and

$$\mathrm{u}_*(t) \approx -(1/\rho)\mathrm{y}(t) - \sqrt{2/\rho}\dot{\mathrm{y}}(t).$$

(d) Assume that $x_0 = \left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right]$. Determine what happens with the optimal control, the optimal cost, and the eigenvalues of the closed-loop system as $\rho \downarrow 0$.



Chapter 5

Glimpses of Related Topics

This chapter provides an outlook on a number of (rather arbitrarily chosen) topics that are related to the main contents of this book. These brief glimpses are meant to raise interest and are written in a style that is different from the rest of the book. Each section, is concluded with a few key references, which offer an entrance to the literature for further study.

5.1 $\mathbb{H}_\infty$ Theory and Robustness

The LQ optimal control problem can be seen as an $\mathbb{L}_2$-*norm* minimization problem. This point of view became widespread in the eighties of the previous century, in part because of its connection with the very popular $\mathbb{H}_\infty$ optimal control problem which emerged in that same decade. The $\mathbb{L}_2$-*norm* of a function $\mathrm{y} : [0,\infty) \to \mathbb{R}^p$ is defined as

$$\|\mathrm{y}\|_{\mathbb{L}_2} = \sqrt{\int_0^\infty \|\mathrm{y}(t)\|^2 \,\mathrm{d}t} \quad \text{where} \quad \|y\| := \sqrt{y^{\mathrm{T}} y},$$

and $\mathbb{L}_2$ is the Hilbert space of functions whose $\mathbb{L}_2$-norm is finite. In this section, we briefly review $\mathbb{L}_2$-norm inequalities, and we make a connection with $\mathbb{H}_\infty$ theory. The starting point is a system that includes an output y,

$$\begin{aligned} \dot{\mathrm{x}}(t) &= A\mathrm{x}(t) + B\mathrm{u}(t), \qquad \mathrm{x}(0) = x_0, \\ \mathrm{y}(t) &= C\mathrm{x}(t). \end{aligned} \tag{5.1}$$

Suppose we want to minimize the indefinite cost

$$-\|\mathrm{y}\|_{\mathbb{L}_2}^2 + \gamma^2 \|\mathrm{u}\|_{\mathbb{L}_2}^2 \tag{5.2}$$

for some given $\gamma > 0$ over all stabilizing inputs. In terms of the state and input, this cost takes the form $\int_0^\infty \mathrm{x}^{\mathrm{T}}(-C^{\mathrm{T}}C)\mathrm{x}(t) + \gamma^2 \mathrm{u}^{\mathrm{T}}(t)\mathrm{u}(t)\,\mathrm{d}t$, and, therefore, the LQ Riccati equation (4.33) and stability condition (4.34) become

$$A^{\mathrm{T}}\tilde{P} + \tilde{P}A - \tfrac{1}{\gamma^2}\tilde{P}BB^{\mathrm{T}}\tilde{P} - C^{\mathrm{T}}C = 0 \text{ and } A - \tfrac{1}{\gamma^2}BB^{\mathrm{T}}\tilde{P} \text{ asymptotically stable.} \tag{5.3}$$

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G. Meinsma and A. van der Schaft, *A Course on Optimal Control*, Springer
Undergraduate Texts in Mathematics and Technology,
https://doi.org/10.1007/978-3-031-36655-0_5

However, since $-C^{\mathrm{T}}C \le 0$ this is not a standard LQ problem, and, as such, the existence of $\tilde{P}$ is not ensured. Assume, for now, that a symmetric solution $\tilde{P}$ of (5.3) *does* exist, and also assume that A is asymptotically stable. Actually, it is customary in this case to express the Riccati equation in terms of its negation, $P := -\tilde{P}$. That way (5.3) becomes

$$A^{\mathrm{T}}P + PA + \tfrac{1}{\gamma^2}PBB^{\mathrm{T}}P + C^{\mathrm{T}}C = 0 \text{ and } A + \tfrac{1}{\gamma^2}BB^{\mathrm{T}}P \text{ asymptotically stable.} \quad (5.4)$$

Exactly as in the proof of Theorem 4.5.5 it now follows that

$$-\|\mathtt{y}(t)\|^2 + \gamma^2\|\mathtt{u}(t)\|^2 = \tfrac{\mathrm{d}}{\mathrm{d}t}\big(\mathtt{x}^{\mathrm{T}}(t)P\mathtt{x}(t)\big) + \gamma^2\|\mathtt{u}(t) - \tfrac{1}{\gamma^2}B^{\mathrm{T}}P\mathtt{x}(t)\|^2. \quad (5.5)$$

In particular, it reveals that every stabilizing input satisfies the equality

$$-\|\mathtt{y}\|_{\mathbb{L}_2}^2 + \gamma^2\|\mathtt{u}\|_{\mathbb{L}_2}^2 = -x_0^{\mathrm{T}}Px_0 + \gamma^2\|\mathtt{u} - \tfrac{1}{\gamma^2}B^{\mathrm{T}}P\mathtt{x}\|_{\mathbb{L}_2}^2. \quad (5.6)$$

Consequently, like in the standard LQ problem, the stabilizing input that minimizes (5.6) is $\mathtt{u}_* = \frac{1}{\gamma^2}B^{\mathrm{T}}P\mathtt{x}$, and the optimal (minimal) cost $-\|\mathtt{y}\|_{\mathbb{L}_2}^2 + \gamma^2\|\mathtt{u}\|_{\mathbb{L}_2}^2$ equals $-x_0^{\mathrm{T}}Px_0$. It also shows that P is positive semi-definite, because for $\mathtt{u} = 0$ (which is stabilizing since we assumed A to be asymptotically stable) the cost is $-\|\mathtt{y}\|_{\mathbb{L}_2}^2 + \gamma^2\|\mathtt{u}\|_{\mathbb{L}_2}^2 = -\|\mathtt{y}\|_{\mathbb{L}_2}^2 \le 0$, so the minimal cost $-x_0^{\mathrm{T}}Px_0$ is less than or equal to zero for every x_0. Knowing this, the identity (5.5) for zero initial state, $x_0 = 0$, gives us the inequality

$$-\|\mathtt{y}\|_{\mathbb{L}_2}^2 + \gamma^2\|\mathtt{u}\|_{\mathbb{L}_2}^2 = \mathtt{x}^{\mathrm{T}}(\infty)P\mathtt{x}(\infty) + \gamma^2\|\mathtt{u} - \tfrac{1}{\gamma^2}B^{\mathrm{T}}P\mathtt{x}\|_{\mathbb{L}_2}^2 \ge 0. \quad (5.7)$$

This is a key observation: *in asymptotically stable systems* (5.1) *that start at rest,* $x_0 = 0$, *the norm* $\|\mathtt{y}\|_{\mathbb{L}_2}$ *never exceeds* $\gamma\|\mathtt{u}\|_{\mathbb{L}_2}$ *if there exists a symmetric* P *that satisfies* (5.4). It is a central result in $\mathbb{H}_\infty$ theory that the existence of such a P is both necessary and sufficient for the norm inequality (5.7) to hold (in a strict sense):

Theorem 5.1.1 (Characterization of the $\mathbb{L}_2$-gain). Let $A \in \mathbb{R}^{n\times n}, B \in \mathbb{R}^{n\times m}, C \in \mathbb{R}^{p\times n}$, and assume A is asymptotically stable. Consider the system with input $\mathtt{u}$ and output $\mathtt{y}$, and zero initial state,

$$\begin{aligned} \dot{\mathtt{x}}(t) &= A\mathtt{x}(t) + B\mathtt{u}(t), \qquad \mathtt{x}(0) = 0, \\ \mathtt{y}(t) &= C\mathtt{x}(t). \end{aligned}$$

For every $\gamma > 0$, the following four conditions are equivalent.

1. $\sup_{\mathtt{u}\in\mathbb{L}_2, \mathtt{u}\neq 0} \frac{\|\mathtt{y}\|_{\mathbb{L}_2}}{\|\mathtt{u}\|_{\mathbb{L}_2}} < \gamma$.
2. The Hamiltonian matrix $\begin{bmatrix} A & \frac{1}{\gamma^2}BB^{\mathrm{T}} \\ -C^{\mathrm{T}}C & -A^{\mathrm{T}} \end{bmatrix}$ has no imaginary eigenvalues.

3. The Riccati equation $A^{\mathrm{T}}P + PA + \frac{1}{\gamma^2}PBB^{\mathrm{T}}P + C^{\mathrm{T}}C = 0$ has a unique solution $P \in \mathbb{R}^{n\times n}$ for which $A + \frac{1}{\gamma^2}BB^{\mathrm{T}}P$ is asymptotically stable, and P is symmetric and positive semi-definite.

4. The Riccati equation $AQ + QA^{\mathrm{T}} + \frac{1}{\gamma^2}QC^{\mathrm{T}}CQ + BB^{\mathrm{T}} = 0$ has a unique solution $Q \in \mathbb{R}^{n\times n}$ for which $A + \frac{1}{\gamma^2}QC^{\mathrm{T}}C$ is asymptotically stable, and Q is symmetric and positive semi-definite.

Proof. The equivalence of the first three is standard and can be found in several books, e.g. (Zhou et al., 1996, Corollary 13.24)[1]. The Riccati equation in Condition 4 we recognize as the Riccati equation of the "transposed" system $\dot{\hat{\mathrm{x}}}(t) = A^{\mathrm{T}}\hat{\mathrm{x}}(t) + C^{\mathrm{T}}\hat{\mathrm{u}}(t), \hat{\mathrm{y}}(t) = B^{\mathrm{T}}\hat{\mathrm{x}}(t)$. Condition 4 is equivalent to Condition 2 because the Hamiltonian matrix $\widehat{\mathscr{H}}$ of the transposed system is similar to the transpose of the Hamiltonian matrix $\mathscr{H}$ of Condition 2:

$$\begin{bmatrix} \frac{-1}{\gamma^2}I & 0 \\ 0 & I \end{bmatrix} \underbrace{\begin{bmatrix} A^{\mathrm{T}} & -C^{\mathrm{T}}C \\ \frac{1}{\gamma^2}BB^{\mathrm{T}} & -A \end{bmatrix}}_{=\mathscr{H}^{\mathrm{T}}} \begin{bmatrix} -\gamma^2 I & 0 \\ 0 & I \end{bmatrix} = \underbrace{\begin{bmatrix} A^{\mathrm{T}} & \frac{1}{\gamma^2}C^{\mathrm{T}}C \\ -BB^{\mathrm{T}} & -A \end{bmatrix}}_{=\widehat{\mathscr{H}}},$$

and, thus, $\widehat{\mathscr{H}}$ and $\mathscr{H}$ have the same eigenvalues. ∎

There are many variations of this theorem, for instance, for systems that include a *direct feedthrough* term, $\mathrm{y}(t) = A\mathrm{x}(t) + D\mathrm{u}(t)$. In general, the expression

$$\sup_{\mathrm{u}\in\mathbb{L}_2, \mathrm{u}\neq 0} \frac{\|\mathrm{y}\|_{\mathbb{L}_2}}{\|\mathrm{u}\|_{\mathbb{L}_2}}$$

is known as the $\mathbb{L}_2$-*gain* of the system. Theorem 5.1.1 shows that the $\mathbb{L}_2$-gain equals the largest $\gamma > 0$ for which the Hamiltonian matrix has imaginary eigenvalues. Thus by iterating over γ, we can calculate the $\mathbb{L}_2$-gain. Also, as $\gamma \to \infty$, the eigenvalues of the Hamiltonian matrix converge to the eigenvalues of A and $-A$. Hence, the $\mathbb{L}_2$-gain is finite whenever A is asymptotically stable. An interesting by-product of Theorem 5.1.1 is that every $\mathbb{L}_2$ input is a stabilizing input if the system is asymptotically stable:

Lemma 5.1.2 (Stability for $\mathbb{L}_2$ inputs). Let $A \in \mathbb{R}^{n\times n}, B \in \mathbb{R}^{n\times m}$ and consider $\dot{\mathrm{x}}(t) = A\mathrm{x}(t) + B\mathrm{u}(t), \mathrm{x}(0) = x_0$. If A is asymptotically stable and $\mathrm{u} \in \mathbb{L}_2$, then

- both x and $\dot{\mathrm{x}}$ are in $\mathbb{L}_2$,
- $\lim_{t\to\infty} \mathrm{x}(t) = 0$.

Proof. Take $\mathrm{y} = \mathrm{x}$, that is, $C = I$. For large enough γ, the Hamiltonian matrix of Theorem 5.1.1 does not have imaginary eigenvalues, so Condition 3 of Theorem 5.1.1 holds for some large enough γ. Thus, given γ large enough, (5.5) says that

$$\gamma^2\|\mathrm{u}\|_{\mathbb{L}_2}^2 + x_0^{\mathrm{T}}Px_0 = \|\mathrm{x}\|_{\mathbb{L}_2}^2 + \mathrm{x}^{\mathrm{T}}(\infty)P\mathrm{x}(\infty) + \gamma^2\|\mathrm{u} - \tfrac{1}{\gamma}B^{\mathrm{T}}P\mathrm{x}\|_{\mathbb{L}_2}^2.$$

[1] See *References and further reading* on page 175.

All terms on the right-hand side are nonnegative, hence if $\mathtt{u} \in \mathbb{L}_2$, then all these terms are finite. In particular, $\mathtt{x} \in \mathbb{L}_2$. Consequently also $\dot{\mathtt{x}} = A\mathtt{x} + B\mathtt{u} \in \mathbb{L}_2$. Now the Cauchy-Schwarz inequality guarantees that $|\mathtt{x}_i^2(b) - \mathtt{x}_i^2(a)| = |\int_a^b 2\dot{\mathtt{x}}_i(t)\mathtt{x}_i(t)\,\mathrm{d}t| \le 2\sqrt{\int_a^b \dot{\mathtt{x}}_i^2(t)\,\mathrm{d}t}\sqrt{\int_a^b \mathtt{x}_i^2(t)\,\mathrm{d}t} \to 0$ as $a, b \to \infty$. So $\mathtt{x}_i^2(t)$ converges as $t \to \infty$. Since $\mathtt{x}_i \in \mathbb{L}_2$, it must in fact converge to zero. ∎

The standard $\mathbb{H}_\infty$ problem

LQ optimal control theory was very successful, and it still is, but it has not been easy to incorporate model uncertainties in this approach. In the late seventies, this omission led Zames[2] to the idea of using $\mathbb{H}_\infty$-*optimization* as an alternative to LQ optimal control. It was the starting point of the wonder years of $\mathbb{H}_\infty$ theory. It attracted the attention of operator theoreticians, and for over a decade there was a very fruitful cooperation between operator theory and systems and control theory. At the end of the eighties the first truly satisfactory solutions of what is called the "standard $\mathbb{H}_\infty$ problem" were obtained.

The $\mathbb{H}_\infty$-norm of a linear time-invariant mapping from $\mathtt{w}$ to $\mathtt{z}$ is, strictly speaking, defined as a property of its *transfer function / matrix* $H_{z/w}(s)$. However, for systems described by

$$\begin{aligned} \dot{\mathtt{x}}(t) &= A\mathtt{x}(t) + B\mathtt{w}(t), \qquad \mathtt{x}(0) = 0, \\ \mathtt{z}(t) &= C\mathtt{x}(t), \end{aligned}$$

the $\mathbb{H}_\infty$-norm of $H_{z/w}$ coincides with its $\mathbb{L}_2$-gain,

$$\|H_{z/w}\|_{\mathbb{H}_\infty} = \sup_{\mathtt{w} \in \mathbb{L}_2, \mathtt{w} \neq 0} \frac{\|\mathtt{z}\|_{\mathbb{L}_2}}{\|\mathtt{w}\|_{\mathbb{L}_2}}.$$

Thus, the $\mathbb{H}_\infty$-norm is an *induced* norm (on a Banach space of bounded linear operators). Hence, we have the well-known *contraction theorem* (aka *small gain theorem*) which for linear mappings $H : \mathbb{L}_2 \to \mathbb{L}_2$ says that $I - H$ is invertible on $\mathbb{L}_2$ if $\|H\|_{\mathbb{H}_\infty} < 1$. This elegant result is very powerful and can be utilized to design controllers for a whole family of systems, or for systems whose models are uncertain in some specific way. The game in $\mathbb{H}_\infty$ optimal control is to exploit the freedom in the design to minimize the $\mathbb{H}_\infty$-norm of a mapping $H_{z/w}$ that we select. In this context, $\mathtt{w}$ is often called the "disturbance" and $\mathtt{z}$ the "error signal". Even though the popularity stems mainly from its ability to deal with *dynamic* uncertainties, we illustrate it only for a problem with *signal* uncertainties:

Example 5.1.3 ($\mathbb{H}_\infty$ filtering). Suppose we have a function $\mathtt{q} : \mathbb{R} \to \mathbb{R}$ that we wish to estimate/reconstruct on the basis of another signal $\mathtt{y} : \mathbb{R} \to \mathbb{R}$ that we can measure. This assumes that $\mathtt{q}$ and $\mathtt{y}$ are in some way related. For instance, $\mathtt{q}$ might be the noise in an airplane as it enters your ear, and $\mathtt{y}$ is the noise

[2] See *References and further reading* on page 175.

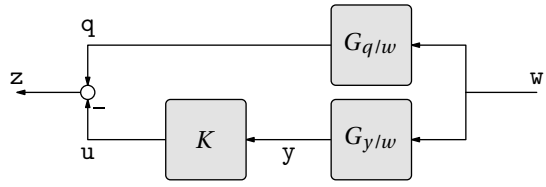


FIGURE 5.1: Filtering configuration. See Example 5.1.3.

picked up somewhere else by a microphone. We model $\mathtt{q}$ and $\mathtt{y}$ as the outputs of systems $G_{q/w}$ and $G_{y/w}$ driven by a common noise source input $\mathtt{w}$, see Fig. 5.1. Let $\mathtt{u}$ be the estimate of $\mathtt{q}$ that we determine based on $\mathtt{y}$:

$$\mathtt{u} = K(\mathtt{y}).$$

In this context, K is usually called a *filter*, and it is the part that we have to design. The configuration of this problem is shown in Fig. 5.1. Ideally, $\mathtt{u}$ equals $\mathtt{q}$ (perfect reconstruction), and then the reconstruction error $\mathtt{z} := \mathtt{q} - \mathtt{u}$ is zero. In practice, that will hardly ever be possible. An option is then to try to minimize the effect of $\mathtt{w}$ on $\mathtt{z}$, for instance, by minimizing the $\mathbb{H}_\infty$-norm of the mapping from $\mathtt{w}$ to $\mathtt{z}$ over all stable causal filters K. The mapping from $\mathtt{w}$ to $\mathtt{z}$ is

$$H_{z/w} := G_{q/w} - KG_{y/w}.$$

Minimizing its $\mathbb{H}_\infty$-norm over all stable causal filters K is a typical $\mathbb{H}_\infty$ optimal control problem. □

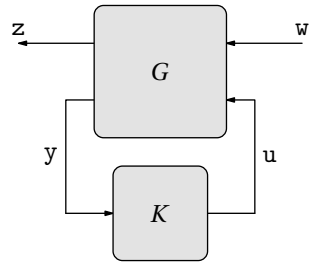


FIGURE 5.2: Configuration for the standard $\mathbb{H}_\infty$ problem.

Having Theorem 5.1.1, it will be no surprise that the theory of $\mathbb{H}_\infty$ optimal control has strong ties with LQ theory and the machinery of Riccati equations. The above $\mathbb{H}_\infty$ filtering problem, and many other $\mathbb{H}_\infty$ problems, are special cases of what is called the "standard $\mathbb{H}_\infty$ (optimal control) problem". In this problem, we are given a system G with two sets of inputs $(\mathtt{w}, \mathtt{u})$ and two sets of outputs $(\mathtt{z}, \mathtt{y})$, described by, say,

$$G: \begin{cases} \dot{\mathtt{x}}(t) = A\mathtt{x}(t) + B_w\mathtt{w}(t) + B_u\mathtt{u}(t), & \mathtt{x}(0) = 0, \\ \mathtt{z}(t) = C_z\mathtt{x}(t) + D_{z/u}\mathtt{u}(t), \\ \mathtt{y}(t) = C_y\mathtt{x}(t) + D_{y/w}\mathtt{w}(t), \end{cases} \tag{5.8}$$

for certain matrices $A, B_w, B_u, C_z, C_y, D_{z/u}, D_{y/w}$. Given this system, the *standard* $\mathbb{H}_\infty$ *problem* is to minimize the $\mathbb{H}_\infty$-norm of the mapping from $\mathtt{w}$ to $\mathtt{z}$ over all stabilizing causal mappings $\mathtt{u} = K(\mathtt{y})$, see Fig. 5.2. The mapping K is usually called *controller*, and it is the part that we have to design. Over the years, many solutions have been put forward, but it is fair to say that the best known solution and best supported in software packages is as follows. It assumes that the state representation (5.8) is such that

- (A, B_u) is stabilizable,
- (C_y, A) is detectable,
- $\begin{bmatrix} A - \mathrm{i}\omega I & B_u \\ C_z & D_{z/u} \end{bmatrix}$ has full column rank for all $\omega \in \mathbb{R}$,
- $\begin{bmatrix} A - \mathrm{i}\omega I & B_w \\ C_y & D_{y/w} \end{bmatrix}$ has full row rank for all $\omega \in \mathbb{R}$,
- $D_{z/u}$ has full column rank, and $D_{y/w}$ has full row rank.

Here is the famous result:

Theorem 5.1.4 (γ-optimal solution of the standard $\mathbb{H}_\infty$ problem). Let $\gamma > 0$. Suppose the above 5 assumptions hold, and that in addition (for reasons of aesthetics only)

$$D_{y/w} \begin{bmatrix} B_w^{\mathrm{T}} & D_{y/w}^{\mathrm{T}} \end{bmatrix} = \begin{bmatrix} 0 & I \end{bmatrix} \quad \text{and} \quad D_{z/u}^{\mathrm{T}} \begin{bmatrix} C_z & D_{z/u} \end{bmatrix} = \begin{bmatrix} 0 & I \end{bmatrix}. \tag{5.9}$$

Then there exists a causal stabilizing controller K for which the $\mathbb{H}_\infty$-norm of the mapping from $\mathtt{w}$ to $\mathtt{z}$ is less than γ iff the following three conditions hold:

1. The Riccati equation $A^{\mathrm{T}}P + PA + P(\frac{1}{\gamma^2} B_w B_w^{\mathrm{T}} - B_u B_u^{\mathrm{T}})P + C_z^{\mathrm{T}} C_z = 0$ has a unique solution P for which $A + (\frac{1}{\gamma^2} B_w B_w^{\mathrm{T}} - B_u B_u^{\mathrm{T}})P$ is asymptotically stable, and P is symmetric and positive semi-definite,
2. The Riccati equation $AQ + QA^{\mathrm{T}} + Q(\frac{1}{\gamma^2} C_z^{\mathrm{T}} C_z - C_y^{\mathrm{T}} C_y)Q + B_w B_w^{\mathrm{T}} = 0$ has a unique solution Q for which $A + Q(\frac{1}{\gamma^2} C_z C_z^{\mathrm{T}} - C_y^{\mathrm{T}} C_y)$ is asymptotically stable, and Q is symmetric and positive semi-definite,
3. All eigenvalues of QP have magnitude less than γ^2.

In that case, one (out of many) causal stabilizing controllers for which the $\mathbb{H}_\infty$-norm of the mapping from $\mathtt{w}$ to $\mathtt{z}$ is less than γ is the mapping $\mathtt{u} = K(\mathtt{y})$ defined by

$$\dot{\hat{\mathtt{x}}} = \big(A + [\tfrac{1}{\gamma^2} B_w B_w^{\mathrm{T}} - B_u B_u^{\mathrm{T}}]P\big)\hat{\mathtt{x}} + \big(I - \tfrac{1}{\gamma^2} QP\big)^{-1} QC_y^{\mathrm{T}}(\mathtt{y} - C_y\hat{\mathtt{x}}), \quad \hat{\mathtt{x}}(0) = 0,$$
$$\mathtt{u} = -B_u^{\mathrm{T}} P\hat{\mathtt{x}}.$$

□

The solutions P and Q of the above two Riccati equations can be constructed from the asymptotically stable eigenvalues and eigenvectors of the corresponding Hamiltonian matrices, much like what we did in the final examples of § 4.5. We need to stress that the additional assumptions (5.9) are for ease of exposition only. Without it, the problem is perfectly solvable but the formulae become unwieldy.

References and further reading

- G. Zames. Feedback and optimal sensitivity: Model reference transformations, multiplicative seminorms, and approximate inverses. *IEEE Trans. Automat. Control*, 26(2): 301–320, 1981.
- S. Skogestad and I. Postlethwaite. *Multivariable Feedback Control. Analysis and Design*. John Wiley and Sons Ltd., Chichester, Sussex, UK, 2nd edition, 2005.
- K. Zhou, J.C. Doyle, and K. Glover. *Robust and Optimal Control*. Prentice Hall: Upper Saddle River, New Jersey, 1996.

5.2 Dissipative Systems

An important approach to the analysis of input-state-output systems

$$\begin{aligned} \dot{\mathtt{x}}(t) &= f(\mathtt{x}(t),\mathtt{u}(t)), \qquad \mathtt{x}(t)\in\mathbb{X}=\mathbb{R}^n, \quad \mathtt{u}(t)\in\mathbb{U}=\mathbb{R}^m, \\ \mathtt{y}(t) &= h(\mathtt{x}(t),\mathtt{u}(t)), \qquad\qquad\qquad\quad \mathtt{y}(t)\in\mathbb{Y}=\mathbb{R}^p, \end{aligned} \tag{5.10}$$

is the theory of *dissipative systems*; as initiated and developed by Willems[3]. In particular, this theory unifies, and generalizes, the classical *passivity* and *small-gain* theorems for feedback interconnections of systems. Perhaps surprisingly, it is intimately related to *optimal control*.

Consider a system (5.10), together with a function $s:\mathbb{U}\times\mathbb{Y}\to\mathbb{R}$, called a *supply rate*. The system is said to be *dissipative* (with respect to the supply rate s) if there exists a nonnegative function $S:\mathbb{X}\to[0,\infty)$ such that

$$S(\mathtt{x}(\tau)) \le S(x_0) + \int_0^\tau s\big(\mathtt{u}(t),\mathtt{y}(t)\big)\,\mathrm{d}t \tag{5.11}$$

for all initial conditions $\mathtt{x}(0)=x_0$, all $\tau\ge 0$, and all $\mathtt{u}:[0,\tau]\to\mathbb{U}$, where $\mathtt{x}(\tau)$ denotes the state at time τ and $\mathtt{y}(t)$ the output at time t resulting from initial condition $\mathtt{x}(0)=x_0$ and input function $\mathtt{u}$. The nonnegative function S is called the *storage* function (corresponding to the supply rate s), and (5.11) is called the *dissipation inequality*. Clearly, if $S(x)$ is satisfies (5.11), then so does the function $S(x)-c$ for any constant c. Hence, if $S(x)$ is a storage function, then so is $S(x)-c$

[3] See *References and further reading* on page 179.

for any c such that $S(x) - c$ is a nonnegative function. However, in many cases, the *non-uniqueness* of storage functions goes much further than this. Two key examples of supply rates $s(u, y)$ are

$$\textit{passivity supply rate}: \quad s(u,y) = y^{\mathrm{T}} u, \qquad (\text{assuming } p = m),$$

and

$$\mathbb{L}_2\textit{-gain supply rate}: \quad s(u,y) = \gamma^2 \|u\|^2 - \|y\|^2, \text{ with } \gamma \geq 0.$$

Here, $\|u\|, \|y\|$ denote the standard Euclidean norms, $\|u\| = \sqrt{u^{\mathrm{T}} u}$, $\|y\| = \sqrt{y^{\mathrm{T}} y}$. The passivity supply rate typically has the interpretation of the supplied *power*, with u, y denoting, for example, generalized forces and velocities (in the mechanical domain), or currents and voltages (in the electrical domain). Then $S(x)$ has the interpretation of the *energy* stored in the system if it is at state x, and (5.11) expresses the property that for all initial conditions x_0 and all input functions the energy stored at any future time instant τ is always less than or equal to the amount of energy stored at time 0 *plus* the total energy that is supplied to the system by its surroundings during the time-interval $[0, \tau]$. That is, the energy of the system can only increase due to supply from outside. Said differently, the system itself cannot create energy; it can only dissipate. Systems that are dissipative with respect to the supply rate $s(u, y) = y^{\mathrm{T}} u$ are also called *passive*.

Dissipativity with respect to the $\mathbb{L}_2$-gain supply rate means that

$$S(\mathrm{x}(\tau)) \leq S(x_0) + \int_0^\tau \gamma^2 \|\mathrm{u}(t)\|^2 \,\mathrm{d}t - \int_0^\tau \|\mathrm{y}(t)\|^2 \,\mathrm{d}t.$$

Since $S(\mathrm{x}(\tau)) \geq 0$ this implies

$$\int_0^\tau \|\mathrm{y}(t)\|^2 \,\mathrm{d}t \leq \int_0^\tau \gamma^2 \|\mathrm{u}(t)\|^2 \,\mathrm{d}t + S(x_0).$$

(This is similar to the infinite horizon case as discussed in § 5.1 for linear systems.) Thus, the $\mathbb{L}_2$-norm of the output on $[0, \tau]$ is bounded by γ times the $\mathbb{L}_2$-norm of the input on $[0, \tau]$, plus a constant (only depending on the initial condition); for all $\tau \geq 0$. The infimal value of γ for which this holds is the $\mathbb{L}_2$-gain of the system; it measures the *amplification* from input to output functions. For linear time-invariant systems, the $\mathbb{L}_2$-gain equals the $\mathbb{H}_\infty$-norm of its transfer matrix; see Section 5.1.

When is the system (5.10) dissipative for a given supply rate s? It turns out that the answer to this question is given by an optimal control problem! Consider the *extended* function $S_a : \mathbb{X} \to [0, \infty]$ ("extended" because the value ∞ is allowed) which is defined by the *free final time optimal control problem*

$$S_a(x_0) = \sup_{\tau, \mathrm{u}:[0,\tau]\to\mathbb{U}} -\int_0^\tau s\big(\mathrm{u}(t), \mathrm{y}(t)\big)\,\mathrm{d}t = -\inf_{\tau, \mathrm{u}:[0,\tau]\to\mathbb{U}} \int_0^\tau s\big(\mathrm{u}(t), \mathrm{y}(t)\big)\,\mathrm{d}t \quad (5.12)$$

for any initial condition x_0, where $\mathtt{y}(t), t \in [0,\tau]$, denotes the output resulting from initial condition $\mathtt{x}(0) = x_0$ and input $\mathtt{u} : [0,\tau] \to \mathbb{U}$. Note that by construction (take $\tau = 0$) $S_a(x_0) \geq 0$ for all $x_0 \in \mathbb{X}$.

Theorem 5.2.1. The system (5.10) is dissipative with respect to the supply rate s iff $S_a(x_0) < \infty$ for all x_0, i.e., $S_a : \mathbb{X} \to [0,\infty)$. Furthermore, if this is the case, then S_a is a storage function, and any other storage function S satisfies $S_a(x_0) \leq S(x_0) - \inf_x S(x)$ for all $x_0 \in \mathbb{X}$. Finally, $\inf_x S_a(x) = 0$.

Proof. Suppose $S_a(x_0) < \infty$ for all x_0, and thus $S_a : \mathbb{X} \to [0,\infty)$. Consider any $\mathtt{u} : [0,\tau] \to \mathbb{U}$ and x_0. Then in general $\mathtt{u}$ will be a *suboptimal* input for the optimal control problem (5.12). Hence,

$$S_a(x_0) \geq -\int_0^\tau s\big(\mathtt{u}(t),\mathtt{y}(t)\big)\,\mathrm{d}t + S_a(\mathtt{x}(\tau)),$$

which is the same as the dissipation inequality (5.11). Thus, (5.10) is dissipative with storage function S_a. Conversely, let (5.10) be dissipative, i.e., there exists nonnegative S satisfying (5.11). Then for any $\mathtt{u} : [0,\tau] \to \mathbb{U}$ and x_0

$$S(x_0) + \int_0^\tau s\big(\mathtt{u}(t),\mathtt{y}(t)\big)\,\mathrm{d}t \geq S(\mathtt{x}(\tau)) \geq 0,$$

and thus $S(x_0) \geq -\int_0^\tau s\big(\mathtt{u}(t),\mathtt{y}(t)\big)\,\mathrm{d}t$, and hence

$$S(x_0) \geq \sup_{\tau,\mathtt{u}:[0,\tau]\to\mathbb{U}} -\int_0^\tau s\big(\mathtt{u}(t),\mathtt{y}(t)\big)\,\mathrm{d}t = S_a(x_0).$$

For any storage function S, the function $\hat{S}(x_0) := S(x_0) - \inf_x S(x), x_0 \in \mathbb{X}$, is a storage function as well. Finally, since $\inf_x \hat{S}(x) = 0$ also $\inf_x S_a(x) = 0$. ∎

In the passivity case, $S_a(x_0)$ has the interpretation of the "maximal" energy that can be extracted from the system being at time 0 at x_0; this quantity should be finite in order that the system is passive. Hence, S_a is called the *available energy*.

Now return to the dissipation inequality (5.11), where we additionally assume that S is *differentiable*. Furthermore, in order to obtain simple formulas, let us restrict attention to systems of input-affine form and without feedthrough term:

$$\begin{aligned} &\dot{\mathtt{x}}(t) = f(\mathtt{x}(t)) + G(\mathtt{x}(t))\mathtt{u}(t), \qquad \mathtt{x}(t) \in \mathbb{X} = \mathbb{R}^n, \quad \mathtt{u}(t) \in \mathbb{U} = \mathbb{R}^m, \\ &\mathtt{y}(t) = h(\mathtt{x}(t)), \qquad\qquad\qquad\qquad\qquad\quad \mathtt{y}(t) \in \mathbb{Y} = \mathbb{R}^p, \end{aligned} \tag{5.13}$$

where $G(x)$ is an $n \times m$ matrix. Bringing $S(x_0)$ to the left-hand side of (5.11), dividing both sides by τ, and letting $\tau \to 0$, turns the dissipation inequality (5.11) into the *differential dissipation inequality*

$$\frac{\partial S(x)}{\partial x^{\mathrm{T}}}\big(f(x) + G(x)u\big) \leq s(u, h(x)) \tag{5.14}$$

for all $x \in \mathbb{X}, u \in \mathbb{U}$. The connection of (5.14) with Lyapunov function theory (Appendix B.3) is clear: if the supply rate s is such that $s(0, y) \leq 0$ for all y (which is, e.g., the case for the passivity and $\mathbb{L}_2$-gain supply rate), then the nonnegative function S satisfies $\frac{\partial S(x)}{\partial x^{\mathrm{T}}} f(x) \leq 0$, and thus is a candidate Lyapunov function for the uncontrolled system $\dot{\mathrm{x}}(t) = f(\mathrm{x}(t))$. In this sense, one could say that the theory of dissipative systems generalizes Lyapunov function theory to systems *with inputs and outputs*.

In case of the $\mathbb{L}_2$-gain supply rate, the optimal control problem (5.12) has an *indefinite* cost criterion. (This was already noted in § 5.1.) Furthermore, the differential dissipation inequality (5.11) amounts to

$$\frac{\partial S(x)}{\partial x^{\mathrm{T}}} \big(f(x) + G(x)u\big) - \gamma^2 \|u\|^2 + \|y\|^2 \leq 0 \tag{5.15}$$

for all x, u and $y = h(x)$. For any x the maximum over all u of the left-hand side of (5.15) is attained at $u = \frac{1}{2\gamma^2} G^{\mathrm{T}}(x) \frac{\partial S(x)}{\partial x}$, and by substitution into (5.15), it follows that (5.15) is satisfied for all x, u iff

$$\frac{\partial S(x)}{\partial x^{\mathrm{T}}} f(x) + \frac{1}{4\gamma^2} \frac{\partial S(x)}{\partial x^{\mathrm{T}}} G(x) G^{\mathrm{T}}(x) \frac{\partial S(x)}{\partial x} + h^{\mathrm{T}}(x) h(x) \leq 0 \tag{5.16}$$

for all $x \in \mathbb{X}$. On the other hand, the Hamilton-Jacobi-Bellman equation (3.12a) (p. 96) for the optimal control problem (5.12) with $s(u, y) = \gamma^2 \|u\|^2 - \|y\|^2$ is readily computed as

$$\frac{\partial S_a(x)}{\partial x^{\mathrm{T}}} f(x) + \frac{1}{4\gamma^2} \frac{\partial S_a(x)}{\partial x^{\mathrm{T}}} G(x) G^{\mathrm{T}}(x) \frac{\partial S_a(x)}{\partial x} + h^{\mathrm{T}}(x) h(x) = 0. \tag{5.17}$$

Hence, we will call (5.16) the Hamilton-Jacobi *inequality*. It thus follows that S_a satisfies the Hamilton-Jacobi equation (5.17), while any other storage function S satisfies the Hamilton-Jacobi inequality (5.16). In general, any infinite horizon optimal control problem of minimizing $\int_0^\infty L(\mathrm{x}(t), \mathrm{u}(t))\, \mathrm{d}t$ over all input functions to a control system $\dot{\mathrm{x}}(t) = f(\mathrm{x}(t), \mathrm{u}(t))$, with $L(x, u)$ an arbitrary running cost, leads to the following inequality involving Bellman's value function $\mathcal{V}$:

$$\frac{\partial \mathcal{V}(x)}{\partial x^{\mathrm{T}}} f(x, u) + L(x, u) \geq 0, \quad x \in \mathbb{X}, u \in \mathbb{U}. \tag{5.18}$$

Thus, (5.18) could be regarded as a *reversed* dissipation inequality.

The optimal control problem (5.12) for the passivity supply rate $s(u, y) = y^{\mathrm{T}} u$ is much harder: by linearity in u, it is *singular*. On the other hand, the differential dissipation inequality (5.14) takes the simple form

$$\frac{\partial S(x)}{\partial x^{\mathrm{T}}} \big(f(x) + G(x)u\big) \leq h^{\mathrm{T}}(x) u$$

for all x, u, or equivalently for all $x \in \mathbb{X}$

$$\frac{\partial S(x)}{\partial x^{\mathrm{T}}} f(x) \leq 0, \qquad h(x) = G^{\mathrm{T}}(x) \frac{\partial S(x)}{\partial x}. \tag{5.19}$$

Let us finally consider (5.19) in case the system (5.13) is a linear system $\dot{\mathrm{x}}(t) = A\mathrm{x}(t) + B\mathrm{u}(t)$, $\mathrm{y}(t) = C\mathrm{x}(t)$, that is, $f(x) = Ax, G(x) = B, h(x) = Cx$. Using the same argumentation as in Exercise 4.5, it follows that S_a is a *quadratic* function of the state; i.e., $S_a(x) = \frac{1}{2}x^{\mathrm{T}}Q_a x$ for some matrix $Q_a = Q_a^{\mathrm{T}} \geq 0$. Restricting anyway to quadratic storage functions $S(x) = \frac{1}{2}x^{\mathrm{T}}Qx$, with $Q = Q^{\mathrm{T}} \geq 0$, the differential dissipation inequality (5.19) takes the form $x^{\mathrm{T}}QAx \leq 0, Cx = B^{\mathrm{T}}Qx$ for all x, that is

$$A^{\mathrm{T}}Q + QA \leq 0, \quad C = B^{\mathrm{T}}Q. \tag{5.20}$$

This is the famous linear matrix inequality (LMI) of the classical Kalman-Yakubovich-Popov lemma. Note that Q_a is the *minimal* solution of (5.20).

References and further reading

- J.C. Willems, Dissipative dynamical systems. Part I: General theory. *Arch. Rat. Mech. Anal.* 1972, *45*, 321–351.
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- A.J. van der Schaft, *L_2-Gain and Passivity Techniques in Nonlinear Control*, 3rd ed.; Springer: Cham, Switzerland, 2017.

5.3 Invariant Lagrangian Subspaces and Riccati

In this section, we will explore some of the *geometry* behind the algebraic Riccati equation, using the geometric theory of *Hamiltonian dynamics*.

Consider a state space $\mathbb{X}$ with elements x; first regarded as a finite-dimensional *abstract linear space*. Let $\mathbb{X}^*$ be its *dual space*, with elements denoted by p. Denote the duality product between $\mathbb{X}$ and $\mathbb{X}^*$ by $\langle p, x\rangle \in \mathbb{R}$, for $x \in \mathbb{X}, p \in \mathbb{X}^*$. After choosing a basis for $\mathbb{X}$, one identifies $\mathbb{X} \cong \mathbb{R}^n$ for some n. Taking the *dual basis* for $\mathbb{X}^*$, thereby also identifying $\mathbb{X}^* \cong \mathbb{R}^n$, the duality product reduces to the vector product $p^{\mathrm{T}}x$. On the product space $\mathbb{X} \times \mathbb{X}^*$, one defines the *symplectic form*

$$\ll (x_1, p_1), (x_2, p_2) \gg := \langle p_1, x_2\rangle - \langle p_2, x_1\rangle, \tag{5.21}$$

which, after choosing a basis for $\mathbb{X}$ and dual basis for $\mathbb{X}^*$, is identified with the matrix

$$\mathscr{J} := \begin{bmatrix} 0 & -I_n \\ I_n & 0 \end{bmatrix}. \tag{5.22}$$

Denoting $z_i := \left[\begin{smallmatrix} x_i \\ p_i \end{smallmatrix}\right], i = 1,2$, the expression (5.21) thus equals $z_1^{\mathrm{T}}\mathscr{J}z_2$. Now consider any differentiable function $H : \mathbb{X} \times \mathbb{X}^* \to \mathbb{R}$, called a *Hamiltonian* function.

Its gradient will be denoted by the $2n$-dimensional vector

$$e_H(x,p) := \begin{bmatrix} \frac{\partial H(x,p)}{\partial x} \\ \frac{\partial H(x,p)}{\partial p} \end{bmatrix}.$$

Furthermore, the *Hamiltonian vector field* v_H on $\mathbb{X} \times \mathbb{X}^*$ is defined as

$$\mathcal{J} v_H(x,p) = e_H(x,p), \quad \text{equivalently,} \quad v_H(x,p) = \begin{bmatrix} \frac{\partial H(x,p)}{\partial p} \\ -\frac{\partial H(x,p)}{\partial x} \end{bmatrix}.$$

In case of a *quadratic* Hamiltonian

$$H(z) = H(x,p) = p^{\mathrm{T}} A x + \frac{1}{2} x^{\mathrm{T}} F x + \frac{1}{2} p^{\mathrm{T}} G p, \qquad F = F^{\mathrm{T}}, G = G^{\mathrm{T}}, \tag{5.23}$$

the Hamiltonian vector field v_H corresponds to the linear Hamiltonian differential equations

$$\begin{bmatrix} \dot{\mathrm{x}}(t) \\ \dot{\mathrm{p}}(t) \end{bmatrix} = \underbrace{\begin{bmatrix} A & G \\ -F & -A^{\mathrm{T}} \end{bmatrix}}_{\mathscr{H}} \begin{bmatrix} \mathrm{x}(t) \\ \mathrm{p}(t) \end{bmatrix}, \tag{5.24}$$

where $\mathscr{H}$ is the *Hamiltonian matrix* corresponding to the Hamiltonian (5.23).

Next, we come to the definition of a *Lagrangian subspace*.

Definition 5.3.1 (Lagrangian subspace). A subspace $\mathscr{L} \subset \mathbb{X} \times \mathbb{X}^*$ is *Lagrangian* if $\mathscr{L} = \mathscr{L}^{\perp\!\!\!\perp}$, where $\mathscr{L}^{\perp\!\!\!\perp}$ is defined as $\mathscr{L}^{\perp\!\!\!\perp} = \{z \in \mathbb{X} \times \mathbb{X}^* \mid z^{\mathrm{T}} \mathcal{J} v = 0 \text{ for all } v \in \mathscr{L}\}$. □

It follows that any Lagrangian subspace $\mathscr{L}$ satisfies $z_1^{\mathrm{T}} \mathcal{J} z_2 = 0$ for all $z_1, z_2 \in \mathscr{L}$, while $\dim \mathscr{L} = \dim \mathbb{X} = n$. Examples of Lagrangian subspaces are $\mathscr{L} = \mathrm{Im} \left[\begin{smallmatrix} I \\ P \end{smallmatrix} \right]$ for P symmetric. Another example is the *generalized stable eigenspace* $\mathscr{N}^-$ of a Hamiltonian matrix $\mathscr{H}$ in case $\mathscr{H}$ does not have purely imaginary eigenvalues. In fact, as already noticed in the previous chapter (Exercise 4.19), if λ is an eigenvalue of a Hamiltonian matrix $\mathscr{H}$, then so is $-\lambda$. Thus, if $\mathscr{H}$ has no purely imaginary eigenvalues, then the number (counting multiplicities) of eigenvalues in the open left half-plane is equal to $n = \dim \mathbb{X}$. Furthermore, for any $z_1, z_2 \in \mathscr{N}^-$, let $\mathrm{z}_1, \mathrm{z}_2$ be the solutions of (5.24) for $\mathrm{z}_1(0) = z_1, \mathrm{z}_2(0) = z_2$. Any Hamiltonian matrix $\mathscr{H}$ as in (5.24) satisfies $\mathscr{H}^T \mathcal{J} + \mathcal{J} \mathscr{H} = 0$, and thus

$$\frac{\mathrm{d}}{\mathrm{d}t} \mathrm{z}_1^{\mathrm{T}}(t) \mathcal{J} \mathrm{z}_2(t) = \dot{\mathrm{z}}_1^{\mathrm{T}}(t) \mathcal{J} \mathrm{z}_2(t) + \mathrm{z}_1^{\mathrm{T}}(t) \mathcal{J} \dot{\mathrm{z}}_2(t) = \mathrm{z}_1^{\mathrm{T}}(t) \big(\mathscr{H}^T \mathcal{J} + \mathcal{J} \mathscr{H} \big) \mathrm{z}_2(t) = 0,$$

implying that $\mathrm{z}_1^{\mathrm{T}}(t) \mathcal{J} \mathrm{z}_2(t)$ is constant as a function of t. Because $\lim_{t\to\infty} \mathrm{z}_1(t) = 0 = \lim_{t\to\infty} \mathrm{z}_2(t)$ this yields $\mathrm{z}_1^{\mathrm{T}}(t) \mathcal{J} \mathrm{z}_2(t) = 0$ for all t, proving that $\mathscr{N}^-$ is indeed a Lagrangian subspace. By time-reversal, the same is shown for the generalized *unstable* eigenspace $\mathscr{N}^+$.

A Lagrangian subspace $\mathscr{L}$ is called *invariant* for the Hamiltonian vector field v_H if $v_H(z) \in \mathscr{L}$ for all $z = (x, p) \in \mathscr{L}$. This means that the Hamiltonian dynamics $\dot{z} = v_H(z)$ leaves $\mathscr{L}$ invariant: starting in $\mathscr{L}$ the solution remains in $\mathscr{L}$. Obviously, both $\mathscr{N}^-$ and $\mathscr{N}^+$ are invariant for (5.24). With respect to *general* invariant Lagrangian subspaces and *general* Hamiltonians, we have the following result.

Proposition 5.3.2. Consider a differentiable Hamiltonian $H : \mathbb{X} \times \mathbb{X}^* \to \mathbb{R}$ and a Lagrangian subspace $\mathscr{L} \subset \mathbb{X} \times \mathbb{X}^*$. Then $\mathscr{L}$ is invariant for the Hamiltonian vector field v_H iff H is zero restricted to $\mathscr{L}$.

Proof. Let $\mathscr{L}$ be such that $v_H(z) \in \mathscr{L}$ for all $z \in \mathscr{L}$. Then for all $v, z \in \mathscr{L}$, $0 = v^{\mathrm{T}} \mathscr{J} v_H(z) = v^{\mathrm{T}} e_H(z)$, which implies that H is constant on $\mathscr{L}$. Thus, since $H(0) = 0$, H is zero on $\mathscr{L}$. Conversely, let H be zero on $\mathscr{L}$, implying that $0 = v^{\mathrm{T}} e_H(z) = v^{\mathrm{T}} \mathscr{J} v_H(z)$ for all $v, z \in \mathscr{L}$. By $\mathscr{L} = \mathscr{L}^{\perp\!\!\!\perp}$ this implies $v_H(z) \in \mathscr{L}$ for all $z \in \mathscr{L}$, and thus $\mathscr{L}$ is invariant for v_H. ∎

Restricting to *quadratic* Hamiltonians H as in (5.23), and taking a basis for $\mathbb{X}$ and dual basis for $\mathbb{X}^*$, this results in the following equations. Any n-dimensional subspace $\mathscr{L} \subset \mathbb{X} \times \mathbb{X}^*$ can be written as

$$\mathscr{L} = \operatorname{Im} \begin{bmatrix} U \\ V \end{bmatrix}, \quad \operatorname{rank} \begin{bmatrix} U \\ V \end{bmatrix} = n, \tag{5.25}$$

for some square matrices U, V. Furthermore, $\mathscr{L}$ given by (5.25) is Lagrangian iff $U^{\mathrm{T}} V = V^{\mathrm{T}} U$. It follows that any $z = \left[\begin{smallmatrix} x \\ p \end{smallmatrix}\right] \in \mathscr{L}$ is given as $x = Uw, p = Vw$ for some $w \in \mathbb{R}^n$. Hence, H given by (5.23) is zero on $\mathscr{L}$ iff $w^{\mathrm{T}} V^{\mathrm{T}} A U w + \frac{1}{2} w^{\mathrm{T}} U^{\mathrm{T}} F U w + \frac{1}{2} w^{\mathrm{T}} V^{\mathrm{T}} G V w = 0$ for all $w \in \mathbb{R}^n$, or equivalently

$$U^{\mathrm{T}} A^{\mathrm{T}} V + V^{\mathrm{T}} A U + U^{\mathrm{T}} F U + V^{\mathrm{T}} G V = 0. \tag{5.26}$$

This will be called the *generalized algebraic Riccati equation*. In case U is *invertible*, the Lagrangian subspace $\mathscr{L}$ given by (5.25) equals

$$\mathscr{L} = \operatorname{Im} \begin{bmatrix} U \\ V \end{bmatrix} = \operatorname{Im} \begin{bmatrix} I \\ V U^{-1} \end{bmatrix}.$$

Hence, H is zero on $\mathscr{L}$ iff $P := V U^{-1}$ satisfies the *standard* algebraic Riccati equation

$$A^{\mathrm{T}} P + P^{\mathrm{T}} A + F + P^{\mathrm{T}} G P = 0.$$

Also, the condition $U^{\mathrm{T}} V = V^{\mathrm{T}} U$ for any Lagrangian subspace (5.25) yields $P = P^{\mathrm{T}}$.

Invertibility of U can be guaranteed for several cases of interest. One case is exploited in Theorem 5.1.1. Another elegant result is the following.

Proposition 5.3.3 (Kučera, 1991). Consider H given by (5.23), and a Lagrangian subspace $\mathscr{L}$ given by (5.25) which is invariant for the Hamiltonian vector field v_H corresponding to the Hamiltonian matrix $\mathscr{H}$ as in (5.24). If (A,G) is *controllable*, then U is invertible. □

Remark 5.3.4. Dually, if (F,A) is observable, then V is invertible. □

The dynamics *restricted* to any invariant Lagrangian subspace $\mathscr{L}$ can be expressed as follows. By $U^{\mathrm{T}}V = V^{\mathrm{T}}U$ and the rank condition on U,V in (5.25), the Lagrangian subspace $\mathscr{L}$ satisfies

$$\mathscr{L} = \operatorname{Im}\begin{bmatrix} U \\ V \end{bmatrix} = \ker\begin{bmatrix} V^{\mathrm{T}} & -U^{\mathrm{T}} \end{bmatrix}.$$

Furthermore, in view of (5.26), premultiplication of

$$\begin{bmatrix} A & G \\ -F & -A^{\mathrm{T}} \end{bmatrix}\begin{bmatrix} U \\ V \end{bmatrix} = \begin{bmatrix} AU+GV \\ -FU-A^{\mathrm{T}}V \end{bmatrix} \tag{5.27}$$

by $\begin{bmatrix} V^{\mathrm{T}} & -U^{\mathrm{T}} \end{bmatrix}$ is seen to be zero. This implies that the subspace spanned by the right-hand side of (5.27) is indeed contained in $\operatorname{Im}\begin{bmatrix} U \\ V \end{bmatrix}$, and thus

$$\begin{bmatrix} A & G \\ -F & -A^{\mathrm{T}} \end{bmatrix}\begin{bmatrix} U \\ V \end{bmatrix} = \begin{bmatrix} U \\ V \end{bmatrix} L$$

for some matrix L describing the dynamics on $\mathscr{L}$. In fact,

$$L = \left(U^{\mathrm{T}}U + V^{\mathrm{T}}V\right)^{-1}\left(U^{\mathrm{T}}AU + U^{\mathrm{T}}GV - V^{\mathrm{T}}A^{\mathrm{T}}V - V^{\mathrm{T}}FU\right).$$

Many of the above geometric considerations for algebraic Riccati equations extend to the *Hamilton-Jacobi equation*

$$H\left(x, \frac{\partial V(x)}{\partial x}\right) = 0, \quad x \in \mathbb{X}, \tag{5.28}$$

with $\mathbb{X}$ a general n-dimensional smooth *manifold*. (Compare with the infinite horizon Hamilton-Jacobi-Bellman equation (3.30).) In this case the *co-tangent bundle* $T^*\mathbb{X}$ is endowed with a symplectic form ω, which in natural cotangent bundle coordinates (x,p) is given by the same matrix expression $\mathscr{J}$ as in (5.22). Furthermore, for any differentiable $V : \mathbb{X} \to \mathbb{R}$, the submanifold

$$\mathscr{L} := \{(x,p) \mid p = \frac{\partial V(x)}{\partial x}\} \tag{5.29}$$

is a *Lagrangian submanifold*, i.e., a submanifold of dimension equal to $\dim \mathbb{X}$ on which ω is zero. Such a Lagrangian submanifold $\mathscr{L}$ is *invariant* for the Hamiltonian vector field v_H on $T^*\mathbb{X}$ for an *arbitrary* differentiable Hamiltonian H iff H is *constant* on $\mathscr{L}$. Hence, if H is zero at some point of $\mathscr{L}$ given by (5.29),

then the Hamilton-Jacobi equation (5.28) corresponds to invariance of $\mathscr{L}$. Furthermore, *tangent spaces* to Lagrangian submanifolds are Lagrangian *subspaces*, while invariance of Lagrangian submanifolds with respect to nonlinear Hamiltonian dynamics implies invariance with respect to their *linearized* Hamiltonian dynamics, see the last reference below.

Finally, the consideration of *invariant* Lagrangian subspaces/manifolds $\mathscr{L}$ can be extended to the *dynamics* of Lagrangian subspaces/manifolds resulting from a Hamiltonian that is *not* constant on $\mathscr{L}$. In the linear case, this corresponds to the Riccati differential equation (RDE), and in the nonlinear case to the time-variant Hamilton-Jacobi equation. In fact, the Lagrangian submanifold at time t is given by $\mathscr{L}(t) = \{(x,p) \mid p = \frac{\partial V(x,t)}{\partial x}\}$ with V a solution of $\frac{\partial V(x,t)}{\partial t} + H(x, \frac{\partial V(x,t)}{\partial x}) = 0$.

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5.4 Model Predictive Control

Model predictive control (MPC) is usually formulated for systems in discrete time (see (3.3) in Chapter 3)

$$\mathtt{x}(t+1) = f(\mathtt{x}(t), \mathtt{u}(t)), \qquad t \in \mathbb{Z} = \{\ldots, -1, 0, 1, \ldots\}. \tag{5.30}$$

Its key idea is the following. Let the state of the system at some discrete time instant $t \in \mathbb{Z}$ be equal to x_t. Then consider for given integer $N > 0$ the optimal control problem of minimizing over all control sequences $\mathtt{u}(t), \ldots, \mathtt{u}(t+N-1)$ the cost criterion

$$J_{[t,t+N-1]}(x_t, \mathtt{u}) = \sum_{k=0}^{N-1} L(\mathtt{x}(t+k), \mathtt{u}(t+k)). \tag{5.31}$$

Here, $L(x,u)$ is the running cost, and $\mathtt{x}(t+k)$ denotes the state at time $t+k$ resulting from initial condition $\mathtt{x}(t) = x_t$ and control sequence $\mathtt{u}(t), \ldots, \mathtt{u}(t+k-1)$. Note that this is the same optimal control problem as considered in § 3.3,

apart from the fact that the minimization is done over the time horizon $t, t+1, \ldots, t+N-1$, instead of $0, 1, \ldots, T-1$, and that there is no terminal cost.

Let $\mathtt{u}_*(t), \mathtt{u}_*(t+1), \ldots, \mathtt{u}_*(t+N-1)$ be the computed optimal control sequence. The basic *difference* between model predictive control and standard optimal control now shows up as follows. Consider just the *first* control value $\mathtt{u}_*(t)$, and apply this to the system. Now consider the *same* optimal control problem, but *shifted* in time by 1. That is, minimize, for the observed initial condition x_{t+1} at time $t+1$, the cost criterion

$$J_{[t+1,t+N]}(x_t, \mathtt{u}) = \sum_{k=0}^{N-1} L(\mathtt{x}(t+1+k), \mathtt{u}(t+1+k)) \tag{5.32}$$

over the *shifted* time horizon $t+1, t+2, \ldots, t+N$. This yields an optimal control sequence $\mathtt{u}_{**}(t+1), \mathtt{u}_{**}(t+2), \ldots, \mathtt{u}_{**}(t+N)$. Then, again, only apply the first value $\mathtt{u}_{**}(t+1)$ of the now obtained optimal control sequence, consider the observed state x_{t+2}, and continue with the next shifted optimal control problem. And so on.

What are the characteristics of this MPC strategy? First, we do not consider one *single* optimal control problem over a fixed horizon, but instead at *every* subsequent time instant, we consider an optimal control problem with a *shifted* time horizon. This is why MPC is also called "receding horizon" optimal control (and in this sense bears similarity with *infinite horizon* optimal control). Second, we do not use the full optimal control sequence computed at each subsequent time instant $t, t+1, \ldots$, but instead only its *first* value; thus yielding for all t the same feedback expression in x_t.

Furthermore, what are the possible advantages of MPC over standard optimal control? Clearly, this very much depends on the aims and on the application context. Model predictive control is often applied in situations where the system model (5.30) is uncertain and incomplete. In such a situation, $f(x_t, \mathtt{u}_*(t))$ resulting from the optimal input $\mathtt{u}_*(t)$ can be rather *different* from the actual, observed, value of x_{t+1} at time $t+1$. Furthermore, such a discrepancy between *computed* values of the next state and the *observed* ones will generally only increase for subsequent time instants $t+2, \ldots$. This is the reason that in MPC only the *first* value of the optimal control sequence is used, and that the initial value of the optimal control problem (5.32) is rather taken to be the *observed* value of x_{t+1}, instead of its computed (or *predicted*) value $f(x_t, \mathtt{u}_*(t))$. Thus, the uncertain system model (5.30) is only used as the "available knowledge" for predicting the future behavior of the system, in order to compute a *current* control value that is optimal with respect to this available knowledge and a cost criterion with time horizon length N.

The computation of the optimal control problems (5.31), (5.32), etc., can be done with the same optimal control techniques (but in discrete time; see § 3.3) as treated in the present book. However, often it is done in a more basic fashion. For example, the optimal control problem (5.31) can be written as a *static*

optimization problem over the vectors $\mathtt{u}(t),\ldots,\mathtt{u}(t+N-1)$, taking into account the system model (5.30). (In fact, this is not so much different from what was done in § 3.3.) This means that in case there are equality and/or inequality *constraints* on the input and state vectors, these can be handled using the standard powerful techniques from static optimization theory.

Contrary to infinite horizon optimal control (§ 4.5), MPC does not give a priori guarantees for (asymptotic) stability. In fact, quite some research has been devoted to enforcing stability in MPC; either by adding suitable terminal cost functions to the optimal control problems (5.31), (5.32), etc., or by supplementing these optimal control problems with terminal constraints; see the references below.

Finally, note that MPC can be also extended to continuous time systems $\dot{\mathtt{x}}(t) = f(\mathtt{x}(t),\mathtt{u}(t))$, by considering for any state x_t at time $t \in \mathbb{Z}$ the optimal control problem of minimizing

$$J_{[t,t+T]}(x_t,\mathtt{u}) = \int_t^{t+T} L\big(\mathtt{x}(\tau),\mathtt{u}(\tau)\big)\,\mathrm{d}\tau$$

for some finite horizon length T, with T a positive integer. Solving this optimal control problem yields an optimal control function $\mathtt{u}_* : [t,t+T] \to \mathbb{U}$, which can be implemented during the restricted time interval $[t,t+1)$. The observed state x_{t+1} at the next discrete time instant $t+1$ then serves as initial condition for a *shifted* optimal control problem on $[t+1,t+1+T]$, and so on, just as in the discrete-time case. (Obviously, all this can be extended to arbitrary time instants $t \in \mathbb{R}$, arbitrary horizon $T \in (0,\infty)$, and an arbitrary implementation time interval $[t,t+\delta]$, $T \geq \delta > 0$.)

References and further reading

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Appendix A

Background Material

This appendix contains summaries of a number of topics that play a role in optimal control. Each section covers one topic, and most of them can be read independently from the other sections. The topics are standard and are covered in one form or another in calculus courses, a course on differential equations, or a first course on systems theory. Nonlinear differential equations and stability of their equilibria are discussed in Appendix B.

A.1 Positive Definite Functions and Matrices

Let $\Omega \subset \mathbb{R}^n$, and suppose Ω is a neighborhood of some $\bar{x} \in \mathbb{R}^n$. A continuously differentiable function $V : \Omega \to \mathbb{R}$ is said to be *positive definite* relative to $\bar{x}$ if

$$V(x) > 0 \ \forall x \in \Omega \backslash \{\bar{x}\} \quad \text{and} \quad V(\bar{x}) = 0.$$

It is *positive semi-definite*—also known as *nonnegative definite*—if

$$V(x) \geq 0 \ \forall x \in \Omega \quad \text{and} \quad V(\bar{x}) = 0.$$

A symmetric matrix $P \in \mathbb{R}^{n \times n}$ is said to be *positive definite* if $V(x) := x^{\mathrm{T}} P x$ is a positive definite function relative to $\bar{x} = 0 \in \mathbb{R}^n$. In this case the neighborhood Ω is irrelevant and we may as well take $\Omega = \mathbb{R}^n$. So a symmetric $P \in \mathbb{R}^{n \times n}$ is positive definite if

$$x^{\mathrm{T}} P x > 0 \ \forall x \in \mathbb{R}^n, \ x \neq 0.$$

It is *positive semi-definite* (or *nonnegative definite*) if

$$x^{\mathrm{T}} P x \geq 0 \ \forall x \in \mathbb{R}^n.$$

The notation $V > 0$ and $P > 0$ means that the function/matrix is positive definite. Interestingly, real symmetric matrices have real eigenvalues only, and there exist simple tests for positive definiteness:

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G. Meinsma and A. van der Schaft, *A Course on Optimal Control*, Springer Undergraduate Texts in Mathematics and Technology,
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Lemma A.1.1 (Tests for positive definiteness). Suppose P is an $n \times n$ real symmetric matrix. The following six statements are equivalent.

1. $P > 0$.
2. All leading principal minors are positive: $\det(P_{1:k,1:k}) > 0$ for all $k \in \{1, 2, \ldots, n\}$. Here $P_{1:k,1:k}$ is the $k \times k$ sub-matrix of P composed of the first k rows and first k columns of P.
3. All eigenvalues of P are real and larger than zero.
4. There is a nonsingular matrix X such that $P = X^{\mathrm{T}}X$.
5. *Cholesky factorization*: $P = X^{\mathrm{T}}X$ for some (unique) upper-triangular matrix X with positive entries on the diagonal.
6. For whatever partition of P,

$$P = \begin{bmatrix} P_{11} & P_{12} \\ P_{12}^{\mathrm{T}} & P_{22} \end{bmatrix}$$

with P_{11} square (hence P_{22} square), we have

$$P_{11} > 0 \quad \text{and} \quad P_{22} - P_{12}^{\mathrm{T}} P_{11}^{-1} P_{12} > 0.$$

(That is, both P_{11} and its so-called *Schur complement* $P_{22} - P_{12}^{\mathrm{T}} P_{11}^{-1} P_{12}$ are positive definite).

□

For positive *semi*-definite matrices similar tests exist, except for the principal minor test which is now more involved:

Lemma A.1.2 (Tests for positive semi-definiteness). Let $P = P^{\mathrm{T}} \in \mathbb{R}^{n \times n}$. The following five statements are equivalent.

1. $P \geq 0$.
2. All principal minors (not just the leading ones) are nonnegative. That is, $\det(P_{\mathbb{I},\mathbb{I}}) \geq 0$ for every subset $\mathbb{I}$ of $\{1, \ldots, n\}$. Here $P_{\mathbb{I},\mathbb{I}}$ is the square matrix composed from all rows $i \in \mathbb{I}$ and columns $j \in \mathbb{I}$.
3. All eigenvalues of P are real and nonnegative.
4. There is a matrix X such that $P = X^{\mathrm{T}}X$.
5. *Cholesky factorization*: $P = X^{\mathrm{T}}X$ for a unique upper-triangular matrix X with nonnegative diagonal entries.

Moreover, if for some partition

$$P = \begin{bmatrix} P_{11} & P_{12} \\ P_{12}^{\mathrm{T}} & P_{22} \end{bmatrix}$$

the matrix P_{11} is square and invertible, then $P \geq 0$ iff $P_{11} > 0$ and $P_{22} - P_{12}^{\mathrm{T}} P_{11}^{-1} P_{12} \geq 0$. □

Example A.1.3. $P = \left[\begin{smallmatrix} 0 & 0 \\ 0 & -1 \end{smallmatrix}\right]$ is not positive semi-definite because the principal minor $\det(P_{2,2}) = -1$ is not nonnegative.

$P = \left[\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right]$ is positive semi-definite because all three principal minors, $\det(0)$, $\det(1)$, $\det(P)$ are nonnegative. □

A.2 A Notation for Partial Derivatives

We introduce a notation for partial derivatives of functions $f : \mathbb{R}^n \to \mathbb{R}^k$.

First the case $k = 1$, so $f : \mathbb{R}^n \to \mathbb{R}$. Then $\frac{\partial f(x)}{\partial x}$ is a vector of partial derivatives of the same dimension as x. For the standard choice of *column* vectors x (with n entries) this means the column vector

$$\frac{\partial f(x)}{\partial x} := \begin{bmatrix} \dfrac{\partial f(x)}{\partial x_1} \\ \dfrac{\partial f(x)}{\partial x_2} \\ \vdots \\ \dfrac{\partial f(x)}{\partial x_n} \end{bmatrix} \in \mathbb{R}^n.$$

With the same logic we get a row vector if we differentiate with respect to a row vector,

$$\frac{\partial f(x)}{\partial x^{\mathrm{T}}} := \begin{bmatrix} \dfrac{\partial f(x)}{\partial x_1} & \dfrac{\partial f(x)}{\partial x_2} & \cdots & \dfrac{\partial f(x)}{\partial x_n} \end{bmatrix} \in \mathbb{R}^{1 \times n}.$$

Now the case $k \geq 1$. If $f(x) \in \mathbb{R}^k$ is itself vectorial (column) then we similarly define

$$\frac{\partial f(x)}{\partial x^{\mathrm{T}}} := \begin{bmatrix} \dfrac{\partial f_1(x)}{\partial x_1} & \dfrac{\partial f_1(x)}{\partial x_2} & \cdots & \dfrac{\partial f_1(x)}{\partial x_n} \\ \vdots & \vdots & \vdots & \vdots \\ \dfrac{\partial f_k(x)}{\partial x_1} & \dfrac{\partial f_k(x)}{\partial x_2} & \cdots & \dfrac{\partial f_k(x)}{\partial x_n} \end{bmatrix} \in \mathbb{R}^{k \times n},$$

and

$$\frac{\partial f^{\mathrm{T}}(x)}{\partial x} := \begin{bmatrix} \frac{\partial f_1(x)}{\partial x_1} & \cdots & \frac{\partial f_k(x)}{\partial x_1} \\ \frac{\partial f_1(x)}{\partial x_2} & \cdots & \frac{\partial f_k(x)}{\partial x_2} \\ \vdots & \vdots & \vdots \\ \frac{\partial f_1(x)}{\partial x_n} & \cdots & \frac{\partial f_k(x)}{\partial x_n} \end{bmatrix} \in \mathbb{R}^{n\times k}.$$

The first is the *Jacobian*, the second is its transpose. Convenient about this notation is that the $n \times n$ *Hessian* of a function $f : \mathbb{R}^n \to \mathbb{R}$ can now compactly be denoted as

$$\begin{aligned} \frac{\partial^2 f(x)}{\partial x \partial x^{\mathrm{T}}} := \frac{\partial}{\partial x}\left(\frac{\partial f(x)}{\partial x^{\mathrm{T}}}\right) &= \frac{\partial}{\partial x}\begin{bmatrix} \frac{\partial f(x)}{\partial x_1} & \frac{\partial f(x)}{\partial x_2} & \cdots & \frac{\partial f(x)}{\partial x_n} \end{bmatrix} \\ &= \begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1^2} & \frac{\partial^2 f(x)}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f(x)}{\partial x_2 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_2^2} & \cdots & \frac{\partial^2 f(x)}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial^2 f(x)}{\partial x_n \partial x_1} & \frac{\partial^2 f(x)}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_n^2} \end{bmatrix}. \end{aligned}$$

Indeed, we first differentiate with respect to the row x^{T}, and subsequently, differentiate the outcome (a row) with respect to the column x, resulting in an $n\times n$ matrix of second-order partial derivatives. If $f(x)$ is twice continuously differentiable then the order in which we determine the second-order derivatives does not matter, so then

$$\frac{\partial^2 f(x)}{\partial x \partial x^{\mathrm{T}}} = \frac{\partial^2 f(x)}{\partial x^{\mathrm{T}} \partial x}.$$

Hence the Hessian is symmetric.

A.3 Separation of Variables

Let $\mathrm{x} : \mathbb{R} \to \mathbb{R}$, and consider the differential equation

$$\dot{\mathrm{x}}(t) = \frac{g(t)}{h(\mathrm{x}(t))}$$

for some given continuous functions $g, h : \mathbb{R} \to \mathbb{R}$ with $h(\mathrm{x}(t)) \neq 0$. Let G, H denote anti-derivatives of g, h. The differential equation is equivalent to

$$h(\mathrm{x}(t))\dot{\mathrm{x}}(t) = g(t),$$

and we see that the left-hand side is the derivative of $H(\mathrm{x}(t))$ with respect to t, and the right-hand side is the derivative of $G(t)$ with respect to t. So it must be that

$$H(\mathrm{x}(t)) = G(t) + c_0$$

for some integration constant c_0. That is

$$\mathrm{x}(t) = H^{-1}(G(t) + c_0).$$

This derivation assumes that H is invertible. The value of c_0 is typically used to match an initial condition $\mathrm{x}(t_0)$.

Example A.3.1. We solve the differential equation

$$\dot{\mathrm{x}}(t) = -\mathrm{x}^2(t), \qquad \mathrm{x}(0) = x_0$$

of Example B.1.5 using separation of variables. We split the solution in two columns; the first column is the example, the second column makes a connection with the general procedure:

$$\begin{array}{rcl|rcl}
\dot{\mathrm{x}}(t) & = & -\mathrm{x}^2(t) & h(\mathrm{x}(t)) & = & 1/\mathrm{x}(t)^2, \quad g(t) = -1 \\
\dfrac{\dot{\mathrm{x}}(t)}{\mathrm{x}^2(t)} & = & -1 & h(\mathrm{x}(t))\dot{\mathrm{x}}(t) & = & g(t) \\
-\dfrac{1}{\mathrm{x}(t)} & = & -t + c_0 & H(\mathrm{x}(t)) & = & G(t) + c_0 \\
\mathrm{x}(t) & = & \dfrac{1}{t - c_0} & \mathrm{x}(t) & = & H^{-1}(G(t) + c_0)
\end{array}$$

In this example the inverse exists as long as $t \neq c_0$. Now $x_0 = \mathrm{x}(0) = -1/c_0$ so c_0 can be expressed in terms of x_0 as $c_0 = -1/x_0$ and the above solution then becomes

$$\mathrm{x}(t) = \frac{1}{t + 1/x_0} = \frac{x_0}{x_0 t + 1}. \tag{A.1}$$

The solution $\mathrm{x}(t)$ escapes at $t = -1/x_0$. (For the escape time we refer to Example B.1.5.) □

Example A.3.2. Suppose that

$$\dot{\mathrm{x}}(t) = a\mathrm{x}(t), \qquad \mathrm{x}(0) = x_0,$$

and that $\mathrm{x}(t) > 0$. Then we may divide by $\mathrm{x}(t)$ to obtain

$$\frac{\dot{\mathrm{x}}(t)}{\mathrm{x}(t)} = a.$$

Integrating both sides and using that $\mathrm{x}(t) > 0$, we find that

$$\ln(\mathrm{x}(t)) = at + c_0.$$

The logarithm is invertible, yielding

$$\mathtt{x}(t) = \mathrm{e}^{at+c_0} = x_0\,\mathrm{e}^{at}.$$

For $\mathtt{x}(t) < 0$ the same solution $x_0\,\mathrm{e}^{at}$ results (verify this yourself), and if $\mathtt{x}(t) = 0$ for some time t then $\mathtt{x}(t) = 0$ for all time, which is also of the form $\mathtt{x}(t) = x_0\,\mathrm{e}^{at}$ (since $x_0 = 0$). In summary, for every $x_0 \in \mathbb{R}$ the solution is $\mathtt{x}(t) = x_0\,\mathrm{e}^{at}$. □

A.4 Linear Constant-Coefficient DE's

On the basis of a few examples we briefly refresh the method of characteristic equations for solving linear differential equations (DE's) with constant coefficients. Several exercises and examples in this book assume familiarity with this method.

To determine all solutions $\mathtt{y} : \mathbb{R} \to \mathbb{R}$ of the homogeneous DE

$$\dddot{\mathtt{y}}(t) - \ddot{\mathtt{y}}(t) - 5\dot{\mathtt{y}}(t) - 3\mathtt{y}(t) = 0 \tag{A.2}$$

we first determine its *characteristic equation*

$$\lambda^3 - \lambda^2 - 5\lambda - 3 = 0.$$

The function $\lambda^3 - \lambda^2 - 5\lambda - 3$ is known as the *characteristic polynomial* of the DE, and in this case it happens to equal $(\lambda+1)^2(\lambda-3)$. Thus the *characteristic roots* of this equation (over the complex numbers) are

$$\lambda = -1 \text{ (of multiplicity two), and } \lambda = +3.$$

To each characteristic root, λ, there corresponds an exponential solution, $\mathtt{y}(t) = \mathrm{e}^{\lambda t}$, of the differential equation, and the general solution $\mathtt{y}$ of (A.2) follows as

$$\mathtt{y}(t) = (c_1 + c_2\,t)\,\mathrm{e}^{-t} + c_3\,\mathrm{e}^{+3t}$$

with c_1, c_2, c_3 arbitrary constants. The number of degrees of freedom per exponential function equals the multiplicity of the corresponding characteristic root.

For *in*homogeneous equations with an exponential term on the right, say,

$$\dddot{\mathtt{y}}(t) - \ddot{\mathtt{y}}(t) - 5\dot{\mathtt{y}}(t) - 3\mathtt{y}(t) = 2\dot{\mathtt{u}}(t) + 3\mathtt{u}(t), \qquad \mathtt{u}(t) = \mathrm{e}^{st} \tag{A.3}$$

one can find a *particular solution*, $\mathtt{y}_{\text{part}}(t)$, of the same exponential form, $\mathtt{y}_{\text{part}}(t) = A\mathrm{e}^{st}$. The constant A follows easily by equating left and right-hand side of (A.3). For this example it gives

$$\mathtt{y}_{\text{part}}(t) = \frac{2s+3}{s^3 - s^2 - 5s - 3}\,\mathrm{e}^{st}.$$

Then the general solution is obtained by adding the general solution of the homogeneous equation to the particular solution,

$$\mathrm{y}(t) = \frac{2s+3}{s^3-s^2-5s-3}\,\mathrm{e}^{st} + (c_1 + c_2\,t)\,\mathrm{e}^{-t} + c_3\,\mathrm{e}^{+3t}.$$

If s equals a characteristic root ($s=-1$ or $s=+3$ in our example) then the above particular solution is invalid due to a division by zero. Then a particular solution exists of the form

$$\mathrm{y}_{\mathrm{part}}(t) = A t^k\,\mathrm{e}^{st}$$

for some constant A and large enough integer k.

If the function $\mathrm{u}(t)$ in (A.3) is polynomial in t, then a polynomial particular solution $\mathrm{y}_{\mathrm{part}}(t) = A_k t^k + \cdots + A_1 t + A_0$ of sufficiently high degree k exists.

A.5 Systems of Linear Time-Invariant DE's

Let $A \in \mathbb{R}^{n\times n}$ and $B \in \mathbb{R}^{n\times m}$. For every $x_0 \in \mathbb{R}^n$ and piecewise continuous $\mathrm{u}:\mathbb{R}\to\mathbb{R}^m$ the solution $\mathrm{x}:\mathbb{R}\to\mathbb{R}^n$ of the system of DE's

$$\dot{\mathrm{x}}(t) = A\mathrm{x}(t) + B\mathrm{u}(t), \qquad \mathrm{x}(0) = x_0$$

follows uniquely as

$$\mathrm{x}(t) = \mathrm{e}^{At}\,x_0 + \int_0^t \mathrm{e}^{A(t-\tau)}\,B\mathrm{u}(\tau)\,\mathrm{d}\tau, \qquad t\in\mathbb{R}. \tag{A.4}$$

Piecewise continuity of u is for technical reasons only. Here e^A is the *matrix exponential*. It exists for square matrices A and can, for instance, be defined in analogy with the Taylor series expansion of e^a as

$$\mathrm{e}^A = \sum_{k=0}^{\infty} \frac{1}{k!}A^k = I + A + \frac{1}{2!}A^2 + \frac{1}{3!}A^3 + \cdots.$$

This series is convergent for every square matrix A. Some characteristic properties of the matrix exponential are:

Lemma A.5.1 (Matrix exponential properties). Let $A, P \in \mathbb{R}^{n\times n}$. Then

1. $\mathrm{e}^0 = I$ for the zero matrix $0 \in \mathbb{R}^{n\times n}$.
2. e^A is invertible, and $(\mathrm{e}^A)^{-1} = \mathrm{e}^{-A}$.
3. If $A = P\Lambda P^{-1}$ for some matrix Λ, then $\mathrm{e}^A = P\,\mathrm{e}^{\Lambda}\,P^{-1}$.
4. Let $t\in\mathbb{R}$. Then $\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{e}^{At} = A\,\mathrm{e}^{At} = \mathrm{e}^{At}\,A$.

□

For the zero signal $\mathtt{u}(t) = 0$, the above equation (A.4) says that the general solution of

$$\dot{\mathtt{x}}(t) = A\mathtt{x}(t), \qquad \mathtt{x}(0) = x_0 \in \mathbb{R}^n$$

is

$$\mathtt{x}(t) = \mathrm{e}^{At} x_0.$$

For diagonal matrices

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \lambda_n \end{bmatrix},$$

the matrix exponential is simply the diagonal matrix of scalar exponentials,

$$\mathrm{e}^{\Lambda t} = \begin{bmatrix} \mathrm{e}^{\lambda_1 t} & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \mathrm{e}^{\lambda_n t} \end{bmatrix}. \tag{A.5}$$

If A is *diagonalizable*—meaning $\mathbb{C}^n$ has a basis $\{v_1, \ldots, v_n\}$ of eigenvectors of A—then the matrix of eigenvectors, $P := \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix}$, is invertible and

$$AP = P\Lambda,$$

with Λ the diagonal matrix of eigenvalues of A. Thus $A = P\Lambda P^{-1}$ which yields

$$\mathrm{e}^{At} = P\,\mathrm{e}^{\Lambda t}\,P^{-1}$$

with $\mathrm{e}^{\Lambda t}$ as in (A.5). This shows that for diagonalizable matrices A, every entry of e^{At} is a linear combination of $\mathrm{e}^{\lambda_i t}$, $i = 1, \ldots, n$. However, not every matrix is diagonalizable. Using Jordan forms it can be shown that:

Lemma A.5.2. Let $A \in \mathbb{R}^{n\times n}$ and denote its eigenvalues as $\lambda_1, \lambda_2, \ldots, \lambda_n$ (possibly some coinciding). Then every entry of e^{At} is a finite linear combination of $t^k\,\mathrm{e}^{\lambda_i t}$ with $k \in \mathbb{N}$ and $i = 1, 2, \ldots, n$. Moreover, the following statements are equivalent.

1. Every entry of e^{At} converges to zero as $t \to \infty$.
2. Every entry of e^{At} converges to zero *exponentially fast* as $t \to \infty$ (meaning for every entry $w(t)$ of e^{At} there is a $\delta > 0$ such that $\lim_{t\to\infty} w(t)\,\mathrm{e}^{\delta t} = 0$).
3. All eigenvalues of A have negative real part: $\mathrm{Re}(\lambda_i) < 0\ \forall i = 1, \ldots, n$.

In that case we say that A is an *asymptotically stable* matrix. □

A.6 Stabilizability and Detectability

Consider the following system of differential equations,

$$\dot{\mathtt{x}}(t) = A\mathtt{x}(t) + B\mathtt{u}(t), \qquad \mathtt{x}(0) = x_0, \qquad t > 0. \tag{A.6}$$

Here $A \in \mathbb{R}^{n\times n}$ and $B \in \mathbb{R}^{n\times m}$. The function $\mathtt{u} : [0,\infty) \to \mathbb{R}^m$ is often called the *(control) input*, and the interpretation is that this $\mathtt{u}$ is for us to choose, and that the state $\mathtt{x} : [0,\infty) \to \mathbb{R}^n$ follows. A natural question is how well the state can be controlled by choice of $\mathtt{u}$:

Definition A.6.1 (Controllability). A system $\dot{\mathtt{x}}(t) = A\mathtt{x}(t) + B\mathtt{u}(t)$ is *controllable* if for every pair of states $x_0, x_1 \in \mathbb{R}^n$, there is a time $T > 0$ and an input $\mathtt{u} : [0,T] \to \mathbb{R}^m$ such that the solution $\mathtt{x}$ with $\mathtt{x}(0) = x_0$ satisfies $\mathtt{x}(T) = x_1$. □

We then say "(A,B) is controllable". Controllability can be tested in many ways. The following four conditions are equivalent:

1. (A,B) is controllable;
2. $\begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix} \in \mathbb{R}^{n\times(mn)}$ has rank n;
3. $\begin{bmatrix} A - sI & B \end{bmatrix}$ has rank n for every $s \in \mathbb{C}$;
4. for every set $\{\lambda_1, \lambda_2, \ldots, \lambda_n\}$ of n points in the complex plane, symmetric with respect to the real axis, there exists a matrix $F \in \mathbb{R}^{m\times n}$ such that the eigenvalues of $A - BF$ are equal to $\{\lambda_1, \lambda_2, \ldots, \lambda_n\}$.

A weaker form is stabilizability:

Definition A.6.2 (Stabilizability). A system $\dot{\mathtt{x}}(t) = A\mathtt{x}(t) + B\mathtt{u}(t)$ is *stabilizable* if for every $\mathtt{x}(0) \in \mathbb{R}^n$ there is a $\mathtt{u} : [0,\infty) \to \mathbb{R}^m$ such that $\lim_{t\to\infty} \mathtt{x}(t) = 0$. □

The following three conditions are equivalent:

1. (A,B) is stabilizable;
2. $\begin{bmatrix} A - sI & B \end{bmatrix}$ has rank n for every $s \in \mathbb{C}$ with $\mathrm{Re}(s) \geq 0$;
3. there is an $F \in \mathbb{R}^{m\times n}$ such that $A - BF$ is asymptotically stable.

The final condition is interesting because it implies that $\mathtt{u}(t) := -F\mathtt{x}(t)$ is a *stabilizing* input for (A.6), irrespective of the initial condition x_0.

Now consider a system with an output $\mathtt{y}$:

$$\begin{aligned} &\dot{\mathtt{x}}(t) = A\mathtt{x}(t), \qquad \mathtt{x}(0) = x_0, \qquad t > 0, \\ &\mathtt{y}(t) = C\mathtt{x}(t). \end{aligned} \tag{A.7}$$

Here A is the same as in (A.6), and C is a (constant) $k \times n$ matrix. The function $\mathtt{y} : [0,\infty) \to \mathbb{R}^k$ is called the *output*, and in applications $\mathtt{y}$ often is the part of the

state that can be measured. It is a natural question to ask how much information the output provides about the state. For example, if we know the output, can we reconstruct the state? For linear systems one might define observability as follows.

Definition A.6.3 (Observability). A system (A.7) is *observable* if a $T > 0$ exists such that x_0 follows uniquely from $\mathrm{y} : [0, T] \to \mathbb{R}^k$. □

We then say "(C, A) is observable". Of course, if x_0 follows uniquely then the state $\mathrm{x}(t) = \mathrm{e}^{At} x_0$ follows uniquely for all time. There are many ways to test for observability. The following five conditions are equivalent:

1. (C, A) is observable;
2. $\begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} \in \mathbb{R}^{(kn)\times n}$ has rank n;
3. $\begin{bmatrix} C \\ A - sI \end{bmatrix}$ has rank n for every $s \in \mathbb{C}$;
4. for every set $\{\lambda_1, \lambda_2, \ldots, \lambda_n\}$ of n points in the complex plane, symmetric with respect to the real axis, there is a matrix $L \in \mathbb{R}^{n\times k}$ such that the eigenvalues of $A - LC$ are equal to $\{\lambda_1, \lambda_2, \ldots, \lambda_n\}$;
5. the "transposed" system $\dot{\tilde{\mathrm{x}}}(t) = A^{\mathrm{T}}\tilde{\mathrm{x}}(t) + C^{\mathrm{T}}\tilde{\mathrm{u}}(t)$ is controllable.

A weaker form of observability is "detectability". A possible definition is as follows (from this definition it is not clear that it is weaker than observability):

Definition A.6.4 (Detectability). A system (A.7) is *detectable* if $\lim_{t\to\infty} \mathrm{y}(t) = 0$ iff $\lim_{t\to\infty} \mathrm{x}(t) = 0$. □

Detectability thus means that a possible instability of $\dot{\mathrm{x}}(t) = A\mathrm{x}(t)$ can be detected from $\mathrm{y}(t)$. The following four statements are equivalent:

1. (C, A) is detectable;
2. $\begin{bmatrix} C \\ A - sI \end{bmatrix}$ has rank n for every $s \in \mathbb{C}$ with $\mathrm{Re}(s) \geq 0$;
3. there is an $L \in \mathbb{R}^{n\times k}$ such that $A - LC$ is asymptotically stable;
4. the "transposed" system $\dot{\tilde{\mathrm{x}}}(t) = A^{\mathrm{T}}\tilde{\mathrm{x}}(t) + C^{\mathrm{T}}\tilde{\mathrm{u}}(t)$ is stabilizable.

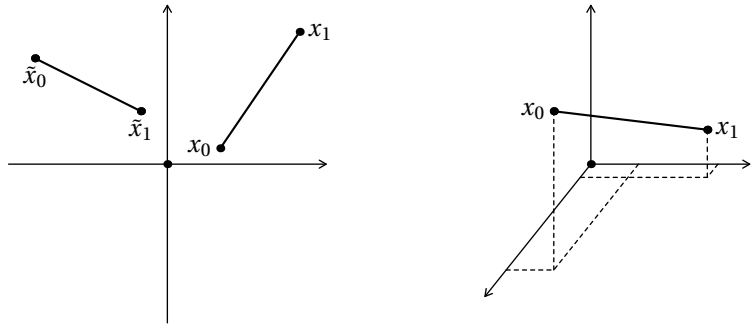


FIGURE A.1: Left: two line segments in $\mathbb{R}^2$. Right: one line segment in $\mathbb{R}^3$. See § A.7.

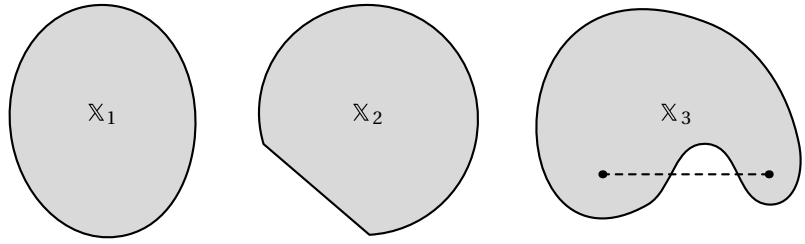


FIGURE A.2: Three subsets of $\mathbb{R}^2$. Sets $\mathbb{X}_1, \mathbb{X}_2$ are convex. The third set, $\mathbb{X}_3$, is not convex because one of its line segments is not contained in $\mathbb{X}_3$. See § A.7.

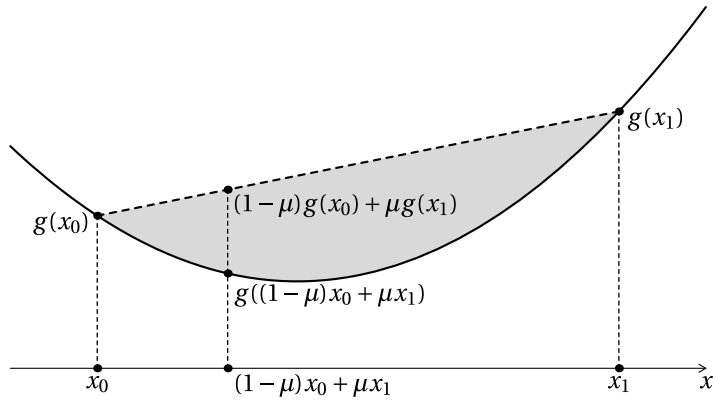


FIGURE A.3: A function $g : \mathbb{R} \to \mathbb{R}$ is convex (on $\mathbb{R}$) if for every $x_0, x_1 \in \mathbb{R}$ and every $\mu \in [0,1]$ we have $g((1-\mu)x_0+\mu x_1) \leq (1-\mu)g(x_0)+\mu g(x_1)$. (In the plot we took $\mu = 1/4$.). See § A.7.

A.7 Convex Sets and Convex Functions

To explain convex functions we first need to know what *line segments* and *convex sets* are. Let $\mathbb{X}$ be a set. A *line segment* with endpoints $x_0, x_1 \in \mathbb{X}$ is the set

$$\{x_0 + \mu(x_1 - x_0) \mid \mu \in \mathbb{R}, 0 \le \mu \le 1\}, \tag{A.8}$$

and the entries of this set are known as the *convex combinations* of x_0 and x_1. For $\mathbb{X} = \mathbb{R}$ the line segments are the closed intervals $[x_0, x_1]$. Figure A.1 depicts a couple of line segments in $\mathbb{R}^2$ and $\mathbb{R}^3$. In order to convey the symmetry in x_0, x_1, line segments are usually denoted as

$$\{(1-\mu)x_0 + \mu x_1 \mid 0 \le \mu \le 1\}.$$

This is the same as (A.8). A set $\mathbb{X}$ is said to be *convex* if it contains all its line segments. That is, if for every $x_0, x_1 \in \mathbb{X}$ also $(1-\mu)x_0 + \mu x_1 \in \mathbb{X}$ for every $\mu \in [0,1]$. Figure A.2 depicts a couple of convex sets in $\mathbb{R}^2$, and also one that is not convex.

Now that convex *sets* are defined, we can define convex *functions*. Such functions are only defined on convex sets. Let $\mathbb{X}$ be a convex set. A function $g : \mathbb{X} \to \mathbb{R}$ is said to be a *convex function* (on $\mathbb{X}$) if for every $x_0, x_1 \in \mathbb{X}$ the graph of the function with endpoints $(x_0, g(x_0))$ and $(x_1, g(x_1))$ is on or below the line segment between these two points. More concrete, it is a convex function if for every $x_0, x_1 \in \mathbb{X}$ we have

$$g((1-\mu)x_0 + \mu x_1) \le (1-\mu)g(x_0) + \mu g(x_1) \quad \forall \mu \in [0,1].$$

This is illustrated in Figure A.3. A function g on a convex set is said to be *concave* if $-g$ is convex. Convex functions enjoy many fantastic properties. We need the following three results. The first of these is illustrated in Fig. A.4.

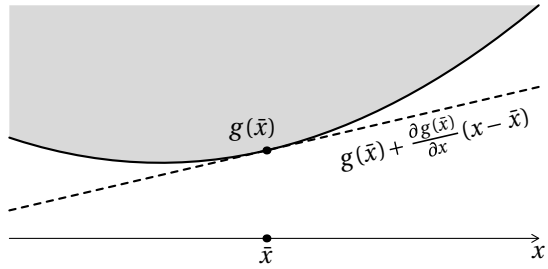


FIGURE A.4: A C^1 function $g : \mathbb{R} \to \mathbb{R}$ is convex iff $g(x) \ge g(\bar{x}) + \frac{\partial g(\bar{x})}{\partial x}(x - \bar{x})$ $\forall \bar{x}, x \in \mathbb{R}$. See Lemma A.7.1.

Lemma A.7.1 (First-order inequality for convexity). Let $g : \mathbb{R}^n \to \mathbb{R}$, and suppose g is C^1. Then g is convex on $\mathbb{R}^n$ iff

$$g(x) \ge g(\bar{x}) + \frac{\partial g(\bar{x})}{\partial x^{\mathrm{T}}}(x - \bar{x}) \qquad \forall \bar{x}, x \in \mathbb{R}^n, \tag{A.9}$$

see Fig. A.4.

Proof. If g is convex then $g(\bar{x}+\mu(x-\bar{x})) \le g(\bar{x})+\mu(g(x)-g(\bar{x}))$ for all $\mu \in [0,1]$. This inequality we can rearrange as

$$g(x)-g(\bar{x}) \ge \frac{g(\bar{x}+\mu(x-\bar{x}))-g(\bar{x})}{\mu}$$

assuming $\mu \in (0,1]$. The right-hand side of the above inequality converges to $\frac{\partial g(\bar{x})}{\partial x^{\mathrm{T}}}(x-\bar{x})$ as $\mu \downarrow 0$. So (A.9) follows.

Conversely, suppose (A.9) holds. Then it also holds for $x_\mu := (1-\mu)\bar{x}+\mu x$ for arbitrary $\mu \in [0,1]$. That is,

$$\begin{aligned} g(\bar{x}) &\ge g(x_\mu)+\frac{\partial g(x_\mu)}{\partial x^{\mathrm{T}}}(\bar{x}-x_\mu) = g(x_\mu)-\frac{\partial g(x_\mu)}{\partial x^{\mathrm{T}}}\mu(x-\bar{x}), \\ g(x) &\ge g(x_\mu)+\frac{\partial g(x_\mu)}{\partial x^{\mathrm{T}}}(x-x_\mu) = g(x_\mu)+\frac{\partial g(x_\mu)}{\partial x^{\mathrm{T}}}(1-\mu)(x-\bar{x}). \end{aligned}$$

Adding the first inequality times $1-\mu$ to the second times μ cancels the derivative and yields $(1-\mu)g(\bar{x})+\mu g(x) \ge g(x_\mu) = g((1-\mu)\bar{x}+\mu x)$. ∎

Lemma A.7.2 (Second-order inequality for convexity). Let $g : \mathbb{R}^n \to \mathbb{R}$, and suppose it is C^2. Then g is convex on $\mathbb{R}^n$ iff the $n \times n$ Hessian $G(x)$ defined as

$$G(x) = \begin{bmatrix} \frac{\partial^2 g(x)}{\partial^2 x_1^2} & \cdots & \frac{\partial^2 g(x)}{\partial x_1 \partial x_n} \\ \vdots & \vdots & \vdots \\ \frac{\partial^2 g(x)}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 g(x)}{\partial^2 x_n^2} \end{bmatrix}$$

is positive semi-definite for every $x \in \mathbb{R}^n$.

Proof. The Hessian is symmetric. By Taylor's formula we have that $g(x) = g(\bar{x}) + \frac{\mathrm{d}g(\bar{x})}{\mathrm{d}x^{\mathrm{T}}}(x-\bar{x}) + \frac{1}{2}(x-\bar{x})^{\mathrm{T}} G(z)(x-\bar{x})$ for some convex combination z of $\bar{x}, x$. Hence $G(z) \ge 0$ implies the inequality of (A.9) which, in turn, implies convexity.

Conversely, if $G(\bar{x})$ is not positive semi-definite for some $\bar{x} \in \mathbb{R}^n$ then a $w \in \mathbb{R}^n$ exists such that $C := w^{\mathrm{T}} G(\bar{x}) w < 0$. Now $g(\bar{x}+\mu w) = g(\bar{x}) + \frac{\mathrm{d}g(\bar{x})}{\mathrm{d}x^{\mathrm{T}}}(\mu w) + \frac{1}{2}\mu^2 C + o(\mu^2) < g(\bar{x}) + \frac{\mathrm{d}g(\bar{x})}{\mathrm{d}x^{\mathrm{T}}}(\mu w)$ for some small enough $\mu > 0$. This contradicts convexity. ∎

Lemma A.7.3. Let $\mathbb{X}$ be a convex subset of $\mathbb{R}^n$, and suppose $f : \mathbb{R}^n \to \mathbb{R}$ is some C^1 function (not necessarily convex). If x_* minimizes $f(x)$ over all $x \in \mathbb{X}$, then

$$\frac{\partial f(x_*)}{\partial x^{\mathrm{T}}}(x-x_*) \ge 0 \quad \forall x \in \mathbb{X}, \tag{A.10}$$

see Fig. A.5. If f in addition is a convex function then x_* minimizes $f(x)$ over all $x \in \mathbb{X}$ *iff* (A.10) holds.

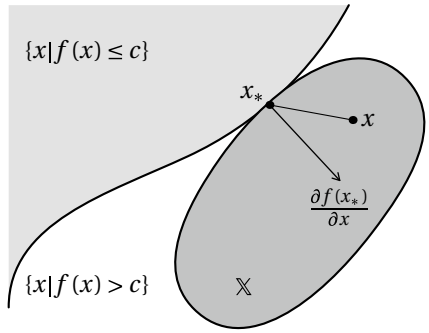


FIGURE A.5: Let $f:\mathbb{R}^2\to\mathbb{R}$, and $\mathbb{X}\subseteq\mathbb{R}^2$. If $\mathbb{X}$ is convex, and x_* minimizes $f(x)$ over all $x\in\mathbb{X}$, then $\frac{\partial f(x_*)}{\partial x^{\mathrm{T}}}(x-x_*)\geq 0\,\forall x\in\mathbb{X}$. See Lemma A.7.3.

Proof. Suppose (A.10) does not hold, i.e., $\frac{\partial f(x_*)}{\partial x^{\mathrm{T}}}(x-x_*)<0$ for some $x\in\mathbb{X}$. Then $f(x_*+\mu(x-x_*))=f(x_*)+\mu\frac{\partial f(x_*)}{\partial x^{\mathrm{T}}}(x-x_*)+o(\mu)<f(x_*)$ for small enough $\mu\in(0,1]$. So x_* does not minimize $f(x)$.

If f is convex then by Lemma A.7.1 we have that $f(x)\geq f(x_*)+\frac{\partial f(x_*)}{\partial x^{\mathrm{T}}}(x-x_*)$. If (A.10) holds then the latter clearly implies $f(x)\geq f(x_*)$. Hence x_* is the solution of the minimization problem. ∎

A.8 Lagrange Multipliers

We recall in this section the notion of Lagrange multipliers for finite-dimensional optimization problems and its connection with first-order conditions of constrained minimization.

Let $J:\mathbb{R}^n\to\mathbb{R}$. Roughly speaking, the first-order condition for unconstrained minimization,

$$\min_{z\in\mathbb{R}^n} J(z),$$

is that no small perturbation $z=z_*+\delta$ of the candidate minimizer z_* decreases $J(z)$. This idea leads to the classic first-order condition that the gradient vector $\partial J(z)/\partial z$ of a differentiable function J must be zero at a minimizer z_*. A similar idea applies to the constrained minimization problem

$$\min_{z\in\mathbb{R}^n} J(z) \quad \text{subject to } G(z)=0, \tag{A.11}$$

where $G:\mathbb{R}^n\to\mathbb{R}^k$. Notice that $G(z)$ has k rows, so $G(z)=0$ is a collection of k constraints. To motivate the general case, consider first the situation of one constraint, $k=1$, and two parameters, $n=2$:

Example A.8.1 (Geometric interpretation of the first-order necessary condition for $k=1$ and $n=2$). An example is depicted in Fig. A.6. Let $J:\mathbb{R}^2\to\mathbb{R}$

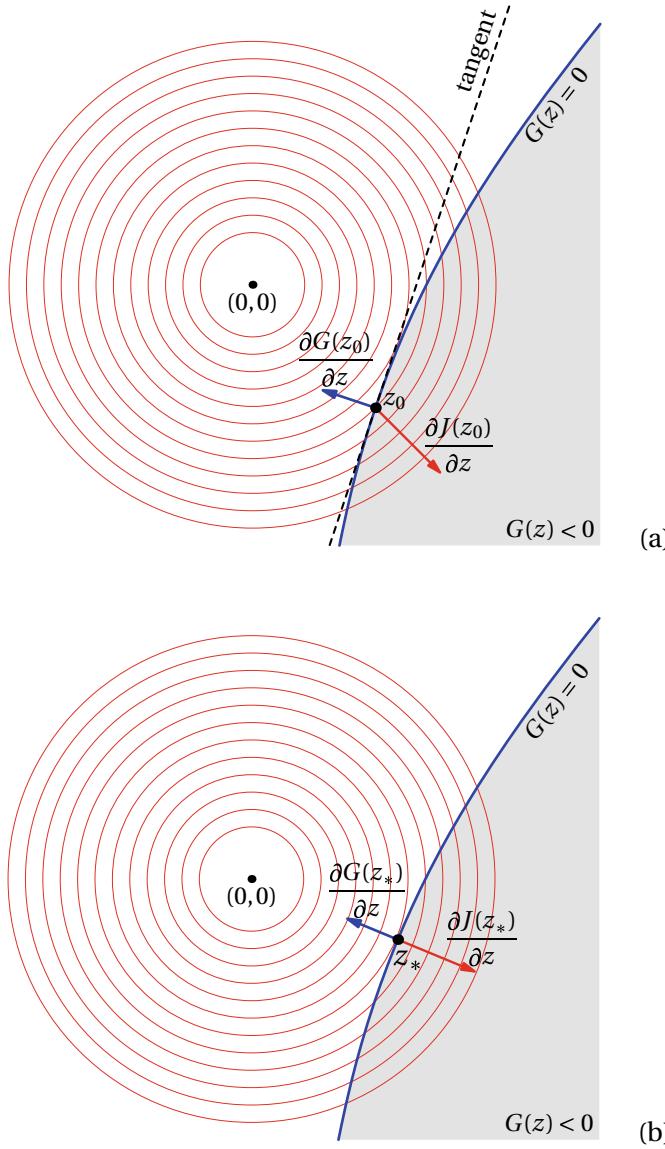


FIGURE A.6: Let $J(z) = z_1^2 + z_2^2$. Its level sets $\{z = (z_1, z_2) | J(z) = c\}$ are circles (shown in red). Suppose the blue curve is where $G(z) = 0$ and the gray region is where $G(z) < 0$. The z_0 in (a) is not a local minimizer of $J(z)$ subject to $G(z) = 0$. The z_* in (b) *is* a local minimizer, and it satisfies the first-order condition that the gradients $\partial J(z)/\partial z$ and $\partial G(z)/\partial z$ are aligned at $z = z_*$. See § A.8.

be a smooth function that we want to minimize over all $z \in \mathbb{R}^2$ subject to the constraint $G(z) = 0$, with $G : \mathbb{R}^2 \to \mathbb{R}$ some differentiable function. Intuition tells us that z_0 in Fig. A.6(a) is not a local minimizer because moving up along the constraint curve brings us to a lower value of $J(z)$. Another way to say this is that the tangent of the constraint curve at z_0 is *not* tangent to the level set of $J(z)$ through z_0. The *first-order* condition for (local) minimality is that every perturbation $\delta \in \mathbb{R}^2$ in the "tangent of the constraint"

$$\{\delta \in \mathbb{R}^2 \mid \frac{\partial G(z_*)}{\partial z^{\mathrm{T}}}\delta = 0\}$$

is also tangent to the level set through z_*, meaning

$$\frac{\partial J(z_*)}{\partial z^{\mathrm{T}}}\delta = 0.$$

The geometric interpretation for $n = 2$ is that the gradients $\frac{\partial J(z)}{\partial z}$ and $\frac{\partial G(z)}{\partial z}$ are aligned at the minimizer z_*, see Fig. A.6(b). □

The findings of the above example generalize as follows: under regularity assumptions[1] one can show that a solution z_* of the general constrained minimization problem (A.11) necessarily has the properties

$$G(z_*) = 0, \tag{A.12a}$$

and for all $\delta \in \mathbb{R}^n$

$$\text{if} \quad \frac{\partial G(z_*)}{\partial z^{\mathrm{T}}}\delta = 0 \quad \text{then} \quad \frac{\partial J(z_*)}{\partial z^{\mathrm{T}}}\delta = 0. \tag{A.12b}$$

Now the brilliant idea of Lagrange is to associate with the constrained minimization problem (A.11) an *augmented* function defined as

$$\mathfrak{J}(z, \lambda) := J(z) + \lambda^{\mathrm{T}} G(z). \tag{A.13}$$

This is a function of $z \in \mathbb{R}^n$ as well as $\lambda \in \mathbb{R}^k$, and the components of λ are known as the *Lagrange multipliers*. The standard *unconstrained* first-order conditions for stationarity of the augmented function $\mathfrak{J}(z, \lambda)$ are that the gradient with respect to both z and λ are zero at (z_*, λ_*):

$$\frac{\partial J(z_*)}{\partial z^{\mathrm{T}}} + \lambda_*^{\mathrm{T}} \frac{\partial G(z_*)}{\partial z^{\mathrm{T}}} = 0 \quad \text{and} \quad G(z_*) = 0. \tag{A.14}$$

The second half of these equations (the first-order conditions with respect to λ) are just the constraint equations themselves. The first half of these equations (the first-order conditions with respect to z) tell us that at a stationary z_* the gradient vector $\frac{\partial J(z)}{\partial z^{\mathrm{T}}}$ is a linear combination of the k rows of $\frac{\partial G(z_*)}{\partial z^{\mathrm{T}}}$, see Fig. A.6(b). The classic result is that the first-order conditions for the *unconstrained* problem in the augmented set of variables (z, λ) are equivalent to the above first-order conditions for the original *constrained* problem in z:

[1] That J and G are continuously differentiable, and $\partial G(z)/\partial z^{\mathrm{T}}$ has full row rank at $z = z_*$.

Lemma A.8.2 (First-order condition). Let $J : \mathbb{R}^n \to \mathbb{R}$, $G : \mathbb{R}^n \to \mathbb{R}^k$, and assume that they are both continuously differentiable. Then $z_* \in \mathbb{R}^n$ satisfies the first-order condition (A.12) iff (A.14) holds for some $\lambda_* \in \mathbb{R}^k$.

Proof. This is an application of the *theorem of alternatives*:

> Given $A \in \mathbb{R}^{k\times n}$ and $\mu \in \mathbb{R}^n$ there is a $\lambda \in \mathbb{R}^k$ such that $\lambda^{\mathrm{T}} A = \mu^{\mathrm{T}}$ iff for every $\delta \in \mathbb{R}^n$ such that $A\delta = 0$ we have $\mu^{\mathrm{T}}\delta = 0$.

(proof: The only-if part is easy: if $\mu^{\mathrm{T}} = \lambda^{\mathrm{T}} A$ and $A\delta = 0$ then $\mu^{\mathrm{T}}\delta = \lambda^{\mathrm{T}} A\delta = 0$. For the if-part we note that the condition that $(A\delta = 0 \implies \mu^{\mathrm{T}}\delta = 0)$ implies that $\ker A \subseteq \ker \mu^{\mathrm{T}}$. This is equivalent to $\operatorname{Im}\mu \subseteq \operatorname{Im} A^{\mathrm{T}}$. Since $\mu \in \operatorname{Im}\mu$ this implies the existence of a $\lambda \in \mathbb{R}^n$ such that $A^{\mathrm{T}}\lambda = \mu$.)

Apply the theorem of alternatives with $A = \partial G(z_*)/\partial z^{\mathrm{T}}$ and $\mu = -\partial J(z_*)/\partial z$.

■

Appendix B

Differential Equations and Lyapunov Functions

This appendix first reviews the existence and uniqueness of solutions $\mathrm{x}_1, \ldots, \mathrm{x}_n : \mathbb{R} \to \mathbb{R}$ of n coupled differential equations

$$\begin{aligned} \dot{\mathrm{x}}_1(t) &= f_1(\mathrm{x}_1(t), \ldots, \mathrm{x}_n(t)), & \mathrm{x}_1(0) &= x_{01}, \\ &\vdots & &\vdots \\ \dot{\mathrm{x}}_n(t) &= f_n(\mathrm{x}_1(t), \ldots, \mathrm{x}_n(t)), & \mathrm{x}_n(0) &= x_{0n}, \quad t \geq 0, \end{aligned} \tag{B.1}$$

and later in this appendix its stability properties are analyzed. Here $x_{01}, \ldots, x_{0n} \in \mathbb{R}$ are given initial conditions, and the $f_i : \mathbb{R}^n \to \mathbb{R}$ are given functions. The vector

$$\mathrm{x} := \begin{bmatrix} \mathrm{x}_1 \\ \vdots \\ \mathrm{x}_n \end{bmatrix} : \mathbb{R} \to \mathbb{R}^n$$

is called the *state*, and

$$x_0 := \begin{bmatrix} x_{01} \\ \vdots \\ x_{0n} \end{bmatrix} \in \mathbb{R}^n,$$

the *initial state*. Define the vector field $f : \mathbb{R}^n \to \mathbb{R}^n$ with components $f_1, \ldots, f_n$. Then (B.1) may be written succinctly as

$$\dot{\mathrm{x}}(t) = f(\mathrm{x}(t)), \qquad \mathrm{x}(0) = x_0, \qquad t \geq 0. \tag{B.2}$$

The solution we normally write as x or $\mathrm{x}(t)$ but sometimes we use $\mathrm{x}(t; x_0)$ if we want to emphasize the dependence on the initial state.

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B.1 Existence and Uniqueness of Solutions

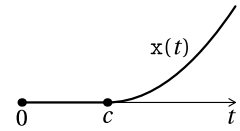
There are differential equations (B.2) whose solution x does not follow uniquely from the initial state:

Example B.1.1 (Non-unique solution). A standard example of a differential equation with a non-unique solution is

$$\dot{\mathrm{x}}(t) = \sqrt{\mathrm{x}(t)}, \qquad \mathrm{x}(0) = 0, \qquad t \geq 0.$$

Clearly the zero function, $\mathrm{x}(t) = 0 \ \forall t$, is one solution, but it is easy to verify that for every $c > 0$ the function

$$\mathrm{x}(t) = \begin{cases} 0 & t \in [0, c] \\ \frac{1}{4}(t-c)^2 & t > c \end{cases}$$



is a solution as well! Weird. It is as if the state x—like Baron Münchhausen—is able to lift itself by pulling on its own hair. □

The vector field in this example is $f(x) = \sqrt{x}$ and it has unbounded derivative around $x = 0$. We will soon see that if $f(x)$ does not increase "too quickly" then uniqueness of solutions x of (B.2) is ensured. A measure for the rate of increase is the *Lipschitz constant*.

Definition B.1.2 (Lipschitz continuity). Let $\Omega \subseteq \mathbb{R}^n$, and let $\|\cdot\|$ be some norm on $\mathbb{R}^n$ (e.g., the standard Euclidean norm). A function $f : \Omega \to \mathbb{R}^n$ is *Lipschitz continuous* on Ω if a *Lipschitz constant* $K \geq 0$ exists such that

$$\|f(x) - f(z)\| \leq K\|x - z\| \tag{B.3}$$

for all $x, z \in \Omega$. It is Lipschitz continuous *at* x_0 if it is Lipschitz continuous on some neighborhood Ω of x_0, and it is *locally Lipschitz continuous* if it is Lipschitz continuous at every $x_0 \in \mathbb{R}^n$. □

Figure B.1 illustrates Lipschitz continuity for $f : \mathbb{R} \to \mathbb{R}$. For the linear $f(x) = kx$, with $k \in \mathbb{R}$, the Lipschitz constant is obviously $K = |k|$, and the solution of the corresponding differential equation

$$\dot{\mathrm{x}}(t) = k\mathrm{x}(t), \qquad \mathrm{x}(0) = x_0$$

is $\mathrm{x}(t) = \mathrm{e}^{kt} x_0$. Given x_0, this solution exists and is unique. The idea is now that for arbitrary Lipschitz continuous $f : \mathbb{R}^n \to \mathbb{R}^n$ the solution of $\dot{\mathrm{x}}(t) = f(\mathrm{x}(t)), \mathrm{x}(0) = x_0$ exists and is unique (one some neighborhood of x_0) and that the solution increases at most exponentially fast as a function of t, with the exponent K equal to the Lipschitz constant (on that neighborhood):

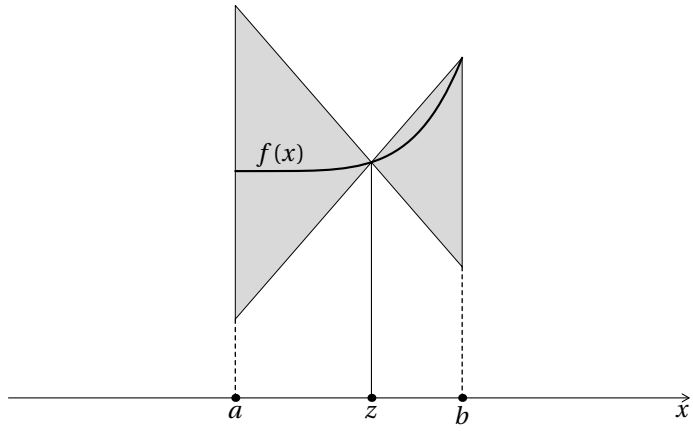


FIGURE B.1: A function $f : \mathbb{R} \to \mathbb{R}$ is Lipschitz continuous on some interval $\Omega = [a, b] \subset \mathbb{R}$ if at each $z \in [a, b]$ the graph $(x, f(x))$ is completely contained in a steep-enough "bow tie" through the point $(z, f(z))$. The slope of the steepest bow tie needed over all $z \in [a, b]$ is a possible Lipschitz constant. See Definition B.1.2.

Theorem B.1.3 (Existence and uniqueness of solution). Let $x_0 \in \mathbb{R}^n$ and $f : \mathbb{R}^n \to \mathbb{R}^n$. If f is Lipschitz continuous at x_0 then, for some $T > 0$, the differential equation (B.2) has a unique solution $\mathrm{x}(t; x_0)$ for all $t \in [0, T)$, and then for every fixed $t \in [0, T)$ the solution $\mathrm{x}(t; x)$ is continuous at $x = x_0$. Specifically, if $\mathrm{x}(t; x_0)$ and $\mathrm{x}(t; z_0)$ are two solutions which for all $t \in [0, T)$ live in some neighborhood Ω, and if f on this neighborhood has a Lipschitz constant K, then

$$\|\mathrm{x}(t; x_0) - \mathrm{x}(t; z_0)\| \le \|x_0 - z_0\| \, \mathrm{e}^{Kt} \qquad \forall\, t \in [0, T).$$

Proof. The proof can be found in many textbooks, e.g., (Khalil, 1996, Thm. 2.2 & Thm. 2.5). ∎

If a *single* Lipschitz constant $K \ge 0$ exists such that (B.3) holds for all $x, z \in \mathbb{R}^n$ then f is said to satisfy a *global Lipschitz condition*.

It follows from the above theorem that the solution $\mathrm{x}(t)$ can be uniquely continued at every t if f is locally Lipschitz continuous. This is such a desirable property that one normally assumes that f is locally Lipschitz continuous. Every continuously differentiable f is locally Lipschitz continuous, so then we can uniquely continue the solution $\mathrm{x}(t)$ at every t. However, the solution may escape in finite time:

Theorem B.1.4 (Escape time). Suppose that $f : \mathbb{R}^n \to \mathbb{R}^n$ is locally Lipschitz continuous. Then for every $\mathrm{x}(0) = x_0$ there is a unique $t(x_0) > 0$ (possibly $t(x_0) = \infty$) such that the solution $\mathrm{x}(t; x_0)$ of (B.2) exists and is unique on the half-open time interval $[0, t(x_0))$ but does not exist for $t > t(x_0)$.

Moreover if $t(x_0) < \infty$ then $\lim_{t \uparrow t(x_0)} \|\mathrm{x}(t; x_0)\| = \infty$.

If f is globally Lipschitz continuous then $t(x_0) = \infty$, i.e., the solution $\mathrm{x}(t; x_0)$ then exists and is unique for all $t \geq 0$.

Proof. See (Khalil 1996, p. 74–75). ∎

This $t(x_0)$—whenever finite—is known as the *escape time*.

Example B.1.5 (Escape time). Consider the scalar differential equation

$$\dot{\mathrm{x}}(t) = -\mathrm{x}^2(t), \qquad \mathrm{x}(0) = x_0.$$

The vector field $f(x) := -x^2$ is locally Lipschitz continuous because it is continuously differentiable. (It is not *globally* Lipschitz continuous however.) Hence for every initial condition x_0 there is a unique solution on some non-empty interval $[0, t(x_0))$ but $t(x_0)$ might be finite. In fact, for this example we can solve the differential equation explicitly (see Appendix A.3), and the solution is

$$\mathrm{x}(t) = \frac{x_0}{tx_0 + 1}.$$

If $x_0 \geq 0$ then $\mathrm{x}(t)$ is well defined for every $t > 0$, so then $t(x_0) = \infty$. If, however, $x_0 < 0$ then the solution escapes at finite time $t(x_0) = -1/x_0$, see Fig. B.2. □

We conclude this section with a result about the continuity of solutions that we need in the proof of the minimum principle (Theorem 2.5.1). Here we take the standard Euclidean norm:

Lemma B.1.6 (Continuity of solutions). Consider the two differential equations in x and in z:

$$\begin{aligned} \dot{\mathrm{x}}(t) &= f(\mathrm{x}(t)), & \mathrm{x}(0) &= x_0, \\ \dot{\mathrm{z}}(t) &= f(\mathrm{z}(t)) + g(t), & \mathrm{z}(0) &= z_0. \end{aligned}$$

Let $T > 0$. If Ω is a set such that $\mathrm{x}(t), \mathrm{z}(t) \in \Omega$ for all $t \in [0, T)$ and if f on Ω has Lipschitz constant K, then

$$\|\mathrm{x}(t) - \mathrm{z}(t)\| \leq \mathrm{e}^{Kt} \left(\|x_0 - z_0\| + \int_0^t \|g(\tau)\| \, \mathrm{d}\tau \right) \qquad \forall\, t \in [0, T).$$

Proof. Let $\Delta(t) = \mathrm{x}(t) - \mathrm{z}(t)$. Then $\dot{\Delta}(t) = f(\mathrm{x}(t)) - f(\mathrm{z}(t)) - g(t)$. By Cauchy-Schwarz we have $|\frac{\mathrm{d}}{\mathrm{d}t}\|\Delta(t)\|| \leq K\|\Delta(t)\| + \|g(t)\|$. From (A.4) it follows that then $\|\Delta(t)\| \leq \mathrm{e}^{Kt}(\|\Delta(0)\| + \int_0^t \|g(\tau)\| \mathrm{d}\tau)$. ∎

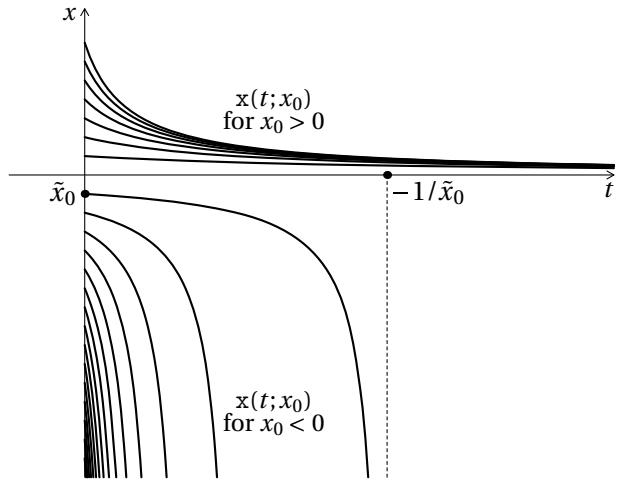


FIGURE B.2: Solutions $\mathrm{x}(t;x_0)$ of $\dot{\mathrm{x}}(t) = -\mathrm{x}^2(t)$ for various initial states $\mathrm{x}(0) = x_0$. If $x_0 < 0$ then the solution escapes at $t = -1/x_0$. See Example B.1.5.

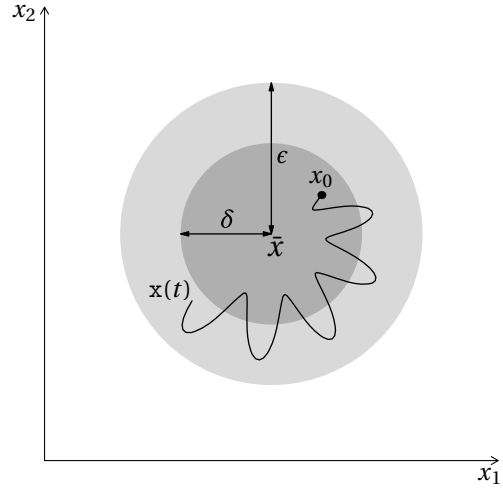


FIGURE B.3: Illustration of stability for systems with two state components, $\mathrm{x} = (\mathrm{x}_1, \mathrm{x}_2)$. See § B.2.

B.2 Definitions of Stability

Asymptotic stability of $\dot{\mathrm{x}}(t) = f(\mathrm{x}(t))$ means, loosely speaking, that solutions $\mathrm{x}(t)$ "come to rest", and stability means that solutions $\mathrm{x}(t)$ remain "close to rest". In order to formalize this, we first have to define the "points of rest". These are the constant solutions, $\mathrm{x}(t) = \bar{x}$, of the differential equation, so solutions $\bar{x}$ of $f(\bar{x}) = 0$.

Definition B.2.1 (Equilibrium). $\bar{x} \in \mathbb{R}^n$ is an *equilibrium (point)* of (B.2) if $f(\bar{x}) = 0$. □

Different possibilities for the behavior of the system near an equilibrium point are described in the following definition (see also Fig. B.3).

Definition B.2.2 (Stability of equilibria). An equilibrium point $\bar{x}$ of a differential equation $\dot{\mathrm{x}}(t) = f(\mathrm{x}(t)), \mathrm{x}(0) = x_0$ is called

1. *stable* if $\forall \epsilon > 0\ \exists \delta > 0$ such that $\|x_0 - \bar{x}\| < \delta$ implies $\|\mathrm{x}(t; x_0) - \bar{x}\| < \epsilon\ \forall\, t \geq 0$.
2. *attractive* if $\exists \delta_1 > 0$ such that $\|x_0 - \bar{x}\| < \delta_1$ implies that $\lim_{t\to\infty} \mathrm{x}(t; x_0) = \bar{x}$.
3. *asymptotically stable* if it is stable and attractive.
4. *globally attractive* if $\lim_{t\to\infty} \mathrm{x}(t; x_0) = \bar{x}$ for every $x_0 \in \mathbb{R}^n$.
5. *globally asymptotically stable* if it is stable and globally attractive.
6. *unstable* if $\bar{x}$ is not stable. This means that $\exists \epsilon > 0$ such that $\forall \delta > 0$ an x_0 and a $t_1 \geq 0$ exists for which $\|x_0 - \bar{x}\| < \delta$ yet $\|\mathrm{x}(t_1; x_0) - \bar{x}\| \geq \epsilon$.

□

In particular, an equilibrium $\bar{x}$ is unstable if every neighborhood of it contains an x_0 that has finite escape time, $t(x_0) < \infty$. Surprisingly, perhaps, we have that attractive equilibria need not be stable, see Exercise B.24. However, it is easy to see that in the case of *linear* dynamics, $\dot{\mathrm{x}}(t) = A\mathrm{x}(t)$, attractivity implies global attractivity and global asymptotic stability. In fact, in the linear case, attractivity, asymptotic stability, global attractivity and global asymptotic stability are all equivalent.

Instead of (in)stability of equilibria, one may also study (in)stability of a specific trajectory $\mathrm{x}(t), t \in \mathbb{R}$, in particular of a *periodic orbit*. We do not explicitly deal with this problem.

There are many ways to analyze stability properties of equilibria. Of particular importance are those methods that do not rely on explicit forms of the solutions $\mathrm{x}(t)$ since explicit forms are in general hard to find. Two methods, both attributed to Lyapunov, that do not require explicit knowledge of $\mathrm{x}(t)$ are *linearization*, also known as *Lyapunov's first method*, and the method of Lyapunov functions, also known as *Lyapunov's second method*. An advantage of the second method over the first is that the first can be proved elegantly with the second. This is why Lyapunov's second method is covered first.

B.3 Lyapunov Functions

Lyapunov's second method mimics the well-known physical property that a system that continually loses energy eventually comes to a halt. Of course, in a mathematical context one may bypass the notion of physical energy, but it is a helpful interpretation.

Suppose we have a function $V : \mathbb{R}^n \to \mathbb{R}$ that does not increase along *any* solution x of the differential equation, i.e., that

$$V(\mathrm{x}(t+h)) \leq V(\mathrm{x}(t)) \qquad \forall h > 0,\ \forall t \tag{B.4}$$

for every solution of $\dot{\mathrm{x}}(t) = f(\mathrm{x}(t))$. If $V(\mathrm{x}(t))$ is differentiable with respect to time t then it is non-increasing for all solutions iff its derivative with respect to time is non-positive everywhere,

$$\dot{V}(\mathrm{x}(t)) \leq 0 \qquad \forall t. \tag{B.5}$$

This condition can be checked for solutions x of $\dot{\mathrm{x}}(t) = f(\mathrm{x}(t))$ without explicitly computing the solutions. Indeed, using the chain rule, we have that

$$\begin{aligned}\dot{V}(\mathrm{x}(t)) = \frac{\mathrm{d}V(\mathrm{x}(t))}{\mathrm{d}t} &= \frac{\partial V(\mathrm{x}(t))}{\partial x_1}\dot{\mathrm{x}}_1(t) + \cdots + \frac{\partial V(\mathrm{x}(t))}{\partial x_n}\dot{\mathrm{x}}_n(t) \\ &= \frac{\partial V(\mathrm{x}(t))}{\partial x_1} f_1(\mathrm{x}(t)) + \cdots + \frac{\partial V(\mathrm{x}(t))}{\partial x_n} f_n(\mathrm{x}(t)),\end{aligned}$$

and, hence, (B.5) holds for all solutions $\mathrm{x}(t)$ iff

$$\frac{\partial V(x)}{\partial x_1} f_1(x) + \cdots + \frac{\partial V(x)}{\partial x_n} f_n(x) \leq 0 \qquad \forall x \in \mathbb{R}^n. \tag{B.6}$$

This final inequality no longer involves time, let alone solutions $\mathrm{x}(t)$ depending on time. This is a key result. We clean up the notation a bit. The left-hand side of (B.6) can be seen as the column vector $f(x)$ premultiplied by the gradient of $V(x)$ seen as a row vector,

$$\frac{\partial V(x)}{\partial x^{\mathrm{T}}} := \begin{bmatrix} \dfrac{\partial V(x)}{\partial x_1} & \dfrac{\partial V(x)}{\partial x_2} & \cdots & \dfrac{\partial V(x)}{\partial x_n} \end{bmatrix}.$$

(Appendix A.2 explains this notation, in particular the role of the transpose.) Thus (B.6) by definition is the same as $\frac{\partial V(x)}{\partial x^{\mathrm{T}}} f(x) \leq 0$. With slight abuse of notation we now use $\dot{V}(x)$ to mean

$$\dot{V}(x) = \frac{\partial V(x)}{\partial x^{\mathrm{T}}} f(x),$$

and we conclude that (B.5) holds for all solutions x of the differential equation iff $\dot{V}(x) \leq 0$ for all $x \in \mathbb{R}^n$. In order to deduce stability from the existence of a non-increasing function $V(\mathrm{x}(t))$ we additionally require that V has a minimum at the equilibrium. Furthermore, we also require a certain degree of differentiability of the function. We formalize these properties in the following definition and theorem. As in Appendix A.1 we define:

Definition B.3.1 (Positive and negative (semi-)definite). Let $\Omega \subseteq \mathbb{R}^n$ and assume it is a neighborhood of some $\bar{x} \in \mathbb{R}^n$. A continuously differentiable function $V : \Omega \to \mathbb{R}$ is *positive definite* on Ω relative to $\bar{x}$ if

$$V(\bar{x}) = 0 \text{ while } V(x) > 0 \text{ for all } x \in \Omega \setminus \{\bar{x}\}.$$

It is *positive semi-definite* if $V(\bar{x}) = 0$ and $V(x) \geq 0$ for all other x. And V is *negative (semi-)definite* if $-V$ is positive (semi-)definite. □

Positive definite implies that V has a unique minimum on Ω, and that the minimum is attained at $\bar{x}$. The assumption that the minimum is zero, $V(\bar{x}) = 0$, is a convenient normalization. Figure B.4 shows an example of each of the four types of "definite" functions.

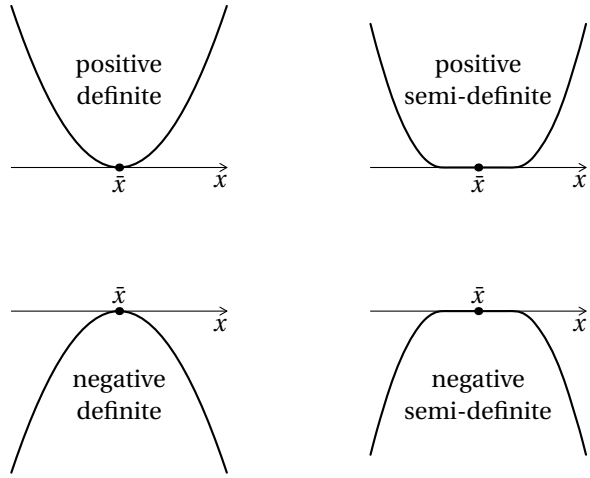


FIGURE B.4: Examples of graphs of positive/negative (semi-)definite functions $V : \mathbb{R} \to \mathbb{R}$. See Definition B.3.1.

The following famous result can now be proved:

Theorem B.3.2 (Lyapunov's second stability theorem). Consider the differential equation $\dot{x}(t) = f(x(t))$ and assume that $f : \mathbb{R}^n \to \mathbb{R}^n$ is locally Lipschitz continuous. Let $\bar{x}$ be an equilibrium of this differential equation. If there is a neighborhood Ω of $\bar{x}$ and a function $V : \Omega \to \mathbb{R}$ such that on Ω

1. V is continuously differentiable,
2. V is positive definite relative to $\bar{x}$,
3. $\dot{V}$ is negative semi-definite relative to $\bar{x}$,

then $\bar{x}$ is a stable equilibrium and we call V a *Lyapunov function*.

If in addition $\dot{V}$ is negative definite on Ω relative to $\bar{x}$ (so not just negative *semi*-definite) then $\bar{x}$ is *asymptotically stable* and we call V a *strong Lyapunov function*.

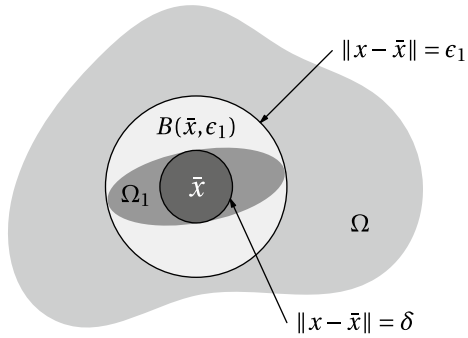


FIGURE B.5: Four inclusions of regions (used in the proof of Theorem B.3.2).

Proof. We denote the open ball with radius r and center $\bar{x}$ by $B(\bar{x},r)$, i.e.,

$$B(\bar{x},r) := \{x \in \mathbb{R}^n \mid \|x-\bar{x}\| < r\}.$$

We first consider the stability property. For every $\epsilon > 0$ we have to find a $\delta > 0$ such that $x_0 \in B(\bar{x},\delta)$ implies $\mathrm{x}(t) \in B(\bar{x},\epsilon)$ for all $t > 0$. To this end we construct a series of inclusions, see Fig. B.5.

Because Ω is a neighborhood of $\bar{x}$, there exists an $\epsilon_1 > 0$ such that $B(\bar{x},\epsilon_1) \subset \Omega$. Without loss of generality we can take it so small that $\epsilon_1 \leq \epsilon$. Because V is continuous on Ω and because the boundary of $B(\bar{x},\epsilon_1)$ is a compact set, the function V has a minimum on the boundary of $B(\bar{x},\epsilon_1)$. We call this minimum α, and realize that $\alpha > 0$. Now define

$$\Omega_1 := \{x \in B(\bar{x},\epsilon_1) \mid V(x) < \alpha\}.$$

This set Ω_1 is open because V is continuous. By definition, Ω_1 is contained in $B(\bar{x},\epsilon_1)$. Clearly, $\bar{x}$ is an element of Ω_1 because $V(\bar{x}) = 0$. Thus, as Ω_1 is open, there exists a $\delta > 0$ such that $B(\bar{x},\delta) \subset \Omega_1$. We prove that this δ satisfies the requirements.

If $x_0 \in B(\bar{x},\delta)$ we find, because $\dot{V}$ is negative semi-definite, that $V(\mathrm{x}(t;x_0)) \leq V(x_0) < \alpha$ for all $t \geq 0$. This means that it is impossible that $\mathrm{x}(t;x_0)$, with initial condition in $B(\bar{x},\delta)$, reaches the boundary of $B(\bar{x},\epsilon_1)$ because on this boundary we have, by definition, that $V(x) \geq \alpha$. Hence, $\|\mathrm{x}(t;x_0)-\bar{x}\| < \epsilon_1 \leq \epsilon$ for all time and the system, therefore, is stable.

Next we prove that the stronger inequality $\dot{V}(x) < 0 \ \forall x \in \Omega\backslash\{\bar{x}\}$ assures *asymptotic* stability. Specifically we prove that for every $x_0 \in B(\bar{x},\delta)$ the solution $\mathrm{x}(t;x_0) \to \bar{x}$ as $t \to \infty$. First note that, because of stability, the state trajectory $\mathrm{x}(t;x_0)$ remains within the bounded set $B(\bar{x},\epsilon_1)$ for all time. Now, to obtain a contradiction, assume that $\mathrm{x}(t;x_0)$ does *not* converge to $\bar{x}$. This implies that there is a $\mu > 0$ and time instances t_k, with $\lim_{k\to\infty} t_k = \infty$, such that

$$\|\mathrm{x}(t_k;x_0)-\bar{x}\| > \mu > 0 \qquad \forall k \in \mathbb{N}.$$

As $\mathrm{x}(t_k;x_0)$ is a bounded sequence, the theorem of Bolzano-Weierstrass guarantees the existence of a subsequence $\mathrm{x}(t_{k_j};x_0)$ that converges to some element x_∞. Clearly $x_\infty \neq \bar{x}$. Since $V(\mathrm{x}(t))$ is non-increasing we have for every $t>0$ that

$$V(\mathrm{x}(t_{k_j};x_0)) \geq V(\mathrm{x}(t_{k_j}+t;x_0)) \geq V(\mathrm{x}(t_{k_{j+m}};x_0)),$$

where m is chosen such that $t_{k_j}+t < t_{k_{j+m}}$. The term in the middle equals $V(\mathrm{x}(t;\mathrm{x}(t_{k_j};x_0)))$. So we also have

$$V(\mathrm{x}(t_{k_j};x_0)) \geq V(\mathrm{x}(t;\mathrm{x}(t_{k_j};x_0))) \geq V(\mathrm{x}(t_{k_{j+m}};x_0)),$$

In the limit $j\to\infty$ the above inequality becomes

$$V(x_\infty) \geq V(\mathrm{x}(t;x_\infty)) \geq V(x_\infty).$$

(Let us be precise here: since the differential equation is locally Lipschitz continuous we have, by Theorem B.1.3, that $\mathrm{x}(t;x)$ depends continuously on x. For that reason we are allowed to say that $\lim_{j\to\infty}\mathrm{x}(t;\mathrm{x}(t_{k_j};x_0)) = \mathrm{x}(t;x_\infty)$.) The above shows that $V(\mathrm{x}(t;x_\infty)) = V(x_\infty)$ for all $t>0$. In particular we see that $V(\mathrm{x}(t;x_\infty))$ is constant. But that would mean that $\dot{V}(x_\infty)=0$, and this violates the fact that $\dot{V}$ is negative definite and $x_\infty\neq\bar{x}$. Therefore the assumption that $\mathrm{x}(t)$ does not converge to $\bar{x}$ is wrong. The system *is* asymptotically stable. ∎

By definition, a Lyapunov function $V(\mathrm{x}(t))$ never increases over time (on Ω), while a strong Lyapunov function $V(\mathrm{x}(t))$ always decreases on Ω unless we are at the equilibrium $\bar{x}$.

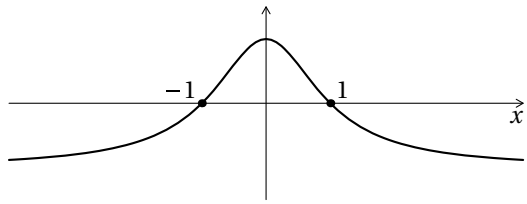


FIGURE B.6: Graph of $\dfrac{1-x^2}{1+x^2}$. See Example B.3.3.

Example B.3.3 (First-order system). The scalar system

$$\dot{\mathrm{x}}(t) = \frac{1-\mathrm{x}^2(t)}{1+\mathrm{x}^2(t)}$$

has two equilibria, $\bar{x}=\pm1$, see Fig. B.6. For equilibrium $\bar{x}=1$ we propose the candidate Lyapunov function

$$V(x) = (x-1)^2.$$

It is positive definite relative to $\bar{x} = 1$ and it is continuously differentiable. On $\Omega = (-1, \infty)$ it is a Lyapunov function because then also the third condition of Theorem B.3.2 holds:

$$\dot{V}(x) = \frac{\partial V(x)}{\partial x} f(x) = 2(x-1)\frac{1-x^2}{1+x^2} = -2\frac{(1-x)^2(1+x)}{1+x^2} \le 0 \quad \forall x \in (-1, \infty).$$

Actually $\dot{V}(x) < 0$ for all $x \in (-1, \infty) \setminus \{1\}$, so it is in fact a *strong* Lyapunov function for equilibrium $\bar{x} = 1$ and, hence, the equilibrium is *asymptotically* stable.

The other equilibrium, $\bar{x} = -1$, is unstable. □

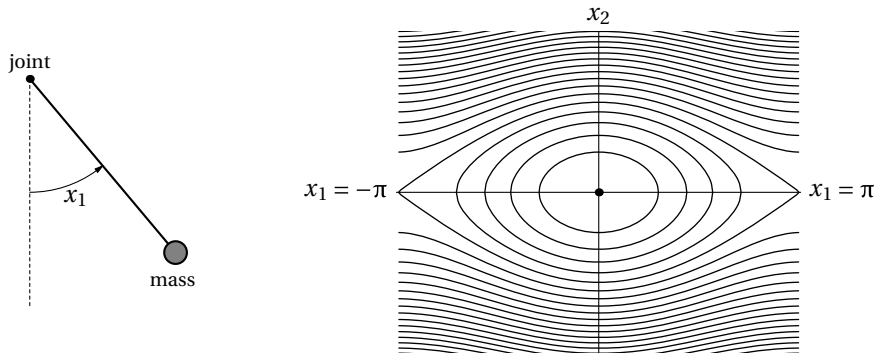


FIGURE B.7: Left: pendulum. Right: level sets of its mechanical energy $V(x)$. See Example B.3.4.

Example B.3.4 (Undamped pendulum). The standard equation of motion of a pendulum without damping is

$$\begin{aligned} \dot{x}_1(t) &= x_2(t), \\ \dot{x}_2(t) &= -\tfrac{g}{\ell}\sin(x_1(t)). \end{aligned}$$

Here x_1 is the angular displacement, x_2 is the angular velocity, g is the gravitational acceleration, and ℓ is the length of the pendulum, see Fig. B.7(left). The mechanical energy of the pendulum with mass m is

$$V(x) = \frac{1}{2}m\ell^2 x_2^2 + mg\ell\big(1 - \cos(x_1)\big).$$

This energy is zero at $(x_1, x_2) = (2k\pi, 0), k \in \mathbb{Z}$ and it is positive elsewhere. To turn this into a Lyapunov function for the hanging position $\bar{x} = (0,0)$ we simply take, say,

$$\Omega = \{x \in \mathbb{R}^2 \mid -2\pi < x_1 < 2\pi\}.$$

This way V on Ω has a unique minimum at equilibrium $\bar{x} = (0,0)$. Hence V is positive definite relative to this $\bar{x}$ for this Ω. Clearly, V is also continuously

differentiable, and $\dot{V}(x)$ equals

$$\dot{V}(x) = \frac{\partial V(x)}{\partial x^{\mathrm{T}}} f(x) = \frac{\partial V(x)}{\partial x_1} f_1(x) + \frac{\partial V(x)}{\partial x_2} f_2(x)$$
$$= mg\ell \sin(x_1) x_2 - m\ell^2 x_2 \frac{g}{\ell} \sin(x_1) = 0.$$

Apparently the mechanical energy is constant over time. Therefore, using Theorem B.3.2, we may draw the conclusion that the system is stable, but not necessarily asymptotically stable. The fact that $V(\mathrm{x}(t))$ is constant actually implies it is *not* asymptotically stable. Indeed if we start at a nonzero state $x_0 \in \Omega$—so with $V(x_0) > 0$—then $V(\mathrm{x}(t)) = V(x_0)$ for all time, and $\mathrm{x}(t)$ thus does not converge to $(0,0)$ as $t \to \infty$. Figure B.7(right) depicts level sets $\{(x_1, x_2) | V(x_1, x_2) = c\}$ of the mechanical energy in the phase plane for several levels $c > 0$. Solutions $\mathrm{x}(t; x_0)$ remain within its level set $\{x | V(x) = V(x_0)\}$. □

For strong Lyapunov functions, Theorem B.3.2 states that $\mathrm{x}(t; x_0) \to \bar{x}$ for initial sates x_0 that are sufficiently close to the equilibrium. At first sight it seems reasonable to expect that the "bigger" the set Ω the "bigger" the region of attraction. Alas, as demonstrated in Exercise B.4, having a strong Lyapunov function on the entire state space $\Omega = \mathbb{R}^n$ does not imply that $\mathrm{x}(t; x_0) \to \bar{x}$ for *all* initial conditions $x_0 \in \mathbb{R}^n$. The question that thus arises is: what is the *region of attraction* of the equilibrium $\bar{x}$ in case it is asymptotically stable, and under which conditions is this region of attraction the entire state space $\mathbb{R}^n$?

The proof of Theorem B.3.2 gives some insight into the region of attraction. In fact, it follows that the region of attraction of $\bar{x}$ includes the largest ball around $\bar{x}$ that is contained in $\Omega_1 := \{x \in B(\bar{x}, \epsilon) \mid V(x) < \alpha\}$, see Fig. B.5. We use this observation to formulate an extra condition on V that guarantees *global* asymptotic stability.

Theorem B.3.5 (Global asymptotic stability). Suppose all conditions of Theorem B.3.2 are met with $\Omega = \mathbb{R}^n$. If $V : \mathbb{R}^n \to \mathbb{R}$ is a strong Lyapunov function with the additional property that

$$V(x) \to \infty \quad \text{as} \quad \|x\| \to \infty, \tag{B.7}$$

then the system is globally asymptotically stable. (Property (B.7) is known as *radial unboundedness*.)

Proof. The proof of Theorem B.3.2 shows that $\mathrm{x}(t) \to \bar{x}$ as $t \to \infty$ whenever $x_0 \in B(\bar{x}, \delta)$, where δ is as indicated in Fig. B.5. Remains to show that every x_0 is in this ball $B(\bar{x}, \delta)$, that is, that δ can be chosen arbitrarily large. We will construct the various regions of Fig. B.5 starting with the smallest and step-by-step working towards the biggest.

Take an arbitrary $x_0 \in \mathbb{R}^n$ and let $\delta := 2\|\bar{x} - x_0\|$. Then by construction we have that $x_0 \in B(\bar{x}, \delta)$. Define $\alpha = \max_{\|x - \bar{x}\| \le \delta} V(x)$. This α is finite and positive. Next let $\Omega_1 = \{x \in \mathbb{R}^n | V(x) < \alpha\}$. This set is bounded because $V(x)$ is radially

unbounded. (This is the reason we require radial unboundedness.) As a result, ϵ_1 defined as $\epsilon_1 = \sup_{x\in\Omega_1} \|x-\bar{x}\|$ is finite. For every $\epsilon > \epsilon_1$ the conditions of Theorem B.3.2 are met, and since $x_0 \in B(\bar{x},\delta)$, the proof of Theorem B.3.2 says that $\mathrm{x}(t) \to \bar{x}$ as $t \to \infty$. This works for every x_0 so the system is globally attractive. Together with asymptotic stability this means it is globally asymptotically stable. ■

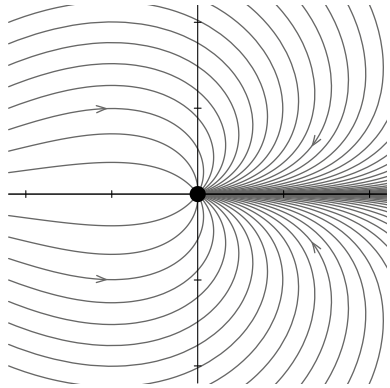


FIGURE B.8: Phase portrait of the system of Example B.3.6. The origin is globally asymptotically stable.

Example B.3.6 (Global asymptotic stability). Consider the system

$$\dot{\mathrm{x}}_1(t) = -\mathrm{x}_1(t) + \mathrm{x}_2^2(t),$$
$$\dot{\mathrm{x}}_2(t) = -\mathrm{x}_2(t)\mathrm{x}_1(t) - \mathrm{x}_2(t).$$

Clearly the origin $(0,0)$ is an equilibrium of this system. We choose

$$V(x) = x_1^2 + x_2^2.$$

This V is radially unbounded and it is a strong Lyapunov function on $\mathbb{R}^2$ because it is positive definite and continuously differentiable and

$$\dot{V}(x) = 2x_1(-x_1 + x_2^2) + 2x_2(-x_2x_1 - x_2) = -2(x_1^2 + x_2^2) < 0 \qquad \forall x \neq 0.$$

Since V is radially unbounded the equilibrium $(0,0)$ is globally asymptotically stable. This also implies that $(0,0)$ is the only equilibrium. Its phase portrait is shown in Fig. B.8. □

Powerful as the theory may be, it does not really tell us how to find a Lyapunov function, assuming one exists. Systematic design of Lyapunov functions is hard, but it does work for linear systems, as discussed in § B.5. In physical systems the construction of Lyapunov functions is often facilitated by the knowledge of existence of conserved quantities, like total energy or total momentum.

B.4 LaSalle's Invariance Principle

Theorem B.3.2 guarantees asymptotic stability when $\dot{V}(x) < 0$ everywhere except at the equilibrium. However, in many cases of interest the natural Lyapunov function does *not* satisfy this condition, while the equilibrium may be asymptotically stable nonetheless. Examples include physical systems whose energy decreases *almost* everywhere but not everywhere. A case in point is the pendulum with damping:

Example B.4.1 (Damped pendulum). The equations of motion of a pendulum subject to damping due to a friction force are

$$\begin{aligned}\dot{\mathrm{x}}_1(t) &= \mathrm{x}_2(t),\\ \dot{\mathrm{x}}_2(t) &= -\frac{g}{\ell}\sin(\mathrm{x}_1(t)) - \frac{d}{m}\mathrm{x}_2(t),\end{aligned}$$

where x_1 is the angular displacement, and x_2 is the angular velocity. The parameter d is a positive friction coefficient. The time-derivative of the mechanical energy $V(x) = \frac{1}{2}m\ell^2 x_2^2 + mg\ell(1-\cos(x_1))$ is

$$\dot{V}(x) = mg\ell\sin(x_1)x_2 - m\ell^2 x_2\frac{g}{\ell}\sin(x_1) - d\ell^2 x_2^2 = -d\ell^2 x_2^2 \le 0.$$

Thus the mechanical energy decreases everywhere except if the angular velocity x_2 is zero. Using Theorem B.3.2 we may draw the conclusion that the system is stable, but not that it is asymptotically stable, because $\dot{V}(x)$ is also zero at points other than the equilibrium (it is zero at every $x = (x_1, 0)$). However from physical considerations we feel that $(0,0)$ *is* asymptotically stable. □

In the above example we would still like to infer asymptotically stability (since we do know from experience that the hanging position of the pendulum with friction is asymptotically stable). If we were to use the theory from the previous section, we would have to find a new Lyapunov function (different from the mechanical energy), but this is not an easy task. In this section we discuss a method that allows to prove asymptotic stability without us having to construct a new Lyapunov function.

From the above pendulum example one might be tempted to conclude that asymptotic stability follows as long as $\dot{V}(x) < 0$ "almost everywhere" in state space. That is not necessarily the case as the following basic example demonstrates.

Example B.4.2 (Simple system). Consider

$$\begin{aligned}\dot{\mathrm{x}}_1(t) &= 0,\\ \dot{\mathrm{x}}_2(t) &= -\mathrm{x}_2(t).\end{aligned}$$

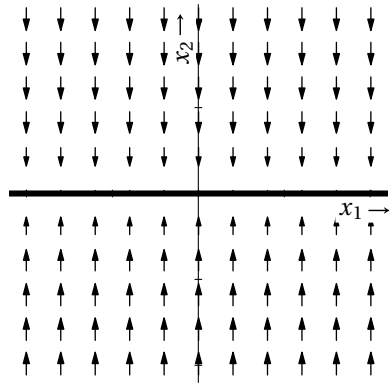


FIGURE B.9: Simple system. All solutions converge to the x_1-axis. See Example B.4.2.

Clearly, x_1 is constant and x_2 converges exponentially fast to zero (see the vector field of Fig. B.9). Now

$$V(x) := x_1^2 + x_2^2$$

is a Lyapunov function for $\bar{x} = (0,0)$ because it is positive definite and continuously differentiable, and

$$\dot{V}(x) = 2x_1\dot{x}_1 + 2x_2\dot{x}_2 = -2x_2^2 \leq 0.$$

The set of states x where $\dot{V}(x) = 0$ is where $x_2 = 0$ (i.e., the x_1-axis) and everywhere else in the plane we have $\dot{V}(x) < 0$. In that sense $\dot{V}(x)$ is negative "almost everywhere". The origin is however *not* asymptotically stable because every $(\bar{x}_1, 0)$ on the x_1-axis is an equilibrium, so no matter how small we take $\delta > 0$, there is always an initial state $x_0 = (\delta/2, 0)$ less than δ away from $(0,0)$ for which the solution $\mathrm{x}(t; x_0)$ is constant, and so does not converge to $(0,0)$. □

We set up a generalized Lyapunov theory that allows to prove that the hanging position in the damped pendulum example (Example B.4.1) is asymptotically stable, and that in Example B.4.2 all solutions converge to the x_1-axis. It requires a bit of terminology.

Definition B.4.3 (Orbit). The *orbit* $\mathcal{O}(x_0)$ with initial condition x_0 is defined as $\mathcal{O}(x_0) = \{y \in \mathbb{R}^n \mid y = \mathrm{x}(t; x_0) \text{ for some } t \geq 0\}$. □

The orbit of x_0 is just the set of states that $\mathrm{x}(t; x_0)$ traces out as t varies over all $t \geq 0$.

Definition B.4.4 (Invariant set). A set $\mathcal{G} \subseteq \mathbb{R}^n$ is called a *(forward) invariant set* for (B.2) if every solution $\mathrm{x}(t; x_0)$ of (B.2) with initial condition x_0 in $\mathcal{G}$, is contained in $\mathcal{G}$ for all $t > 0$. □

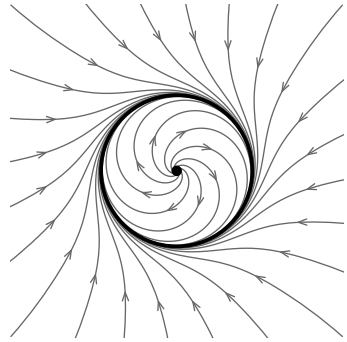


FIGURE B.10: Phase portrait. See Example B.4.6.

So once the state is in an invariant set it never leaves it. Note that every orbit is an example of an invariant set. In particular every equilibrium point is an invariant set.

Example B.4.5. The x_1-axis is an invariant set for the system of Example B.4.2. In fact every element $x = (x_1, 0)$ of this axis is an invariant set because they all are equilibria. The general solution is $\mathrm{x}(t) = (x_{10}, x_{20}\,\mathrm{e}^{-t})$. This shows that for instance also the x_2-axis $\{(0, x_2) | x_2 \in \mathbb{R}\}$ is an invariant set. □

The union of two invariant sets is invariant. In fact, the union of an arbitrary number (finite, infinite, countable, uncountable) of invariant sets is invariant.

Example B.4.6 (Rotation invariant phase portrait). The phase portrait of Fig. B.10 is that of

$$\begin{aligned} \dot{\mathrm{x}}_1(t) &= \mathrm{x}_2(t) + x_1(t)\left(1 - \mathrm{x}_1^2(t) - \mathrm{x}_2^2(t)\right), \\ \dot{\mathrm{x}}_2(t) &= -\mathrm{x}_1(t) + \mathrm{x}_2(t)\left(1 - \mathrm{x}_1^2(t) - \mathrm{x}_2^2(t)\right). \end{aligned} \tag{B.8}$$

Inspired by the rotation-invariant phase portrait (see Fig. B.10) we analyze first how the *squared* radius

$$\mathrm{r}(t) := \mathrm{x}_1^2(t) + \mathrm{x}_2^2(t)$$

changes over time,

$$\begin{aligned} \dot{\mathrm{r}}(t) &= \tfrac{\mathrm{d}}{\mathrm{d}t}\left(\mathrm{x}_1^2(t) + \mathrm{x}_2^2(t)\right) \\ &= 2\mathrm{x}_1(t)\dot{\mathrm{x}}_1(t) + 2\mathrm{x}_2(t)\dot{\mathrm{x}}_2(t) \\ &= 2\mathrm{x}_1(t)\mathrm{x}_2(t) + 2\mathrm{x}_1^2(t)\left(1 - \mathrm{x}_1^2(t) - \mathrm{x}_2^2(t)\right) \\ &\qquad - 2\mathrm{x}_2(t)\mathrm{x}_1(t) + 2\mathrm{x}_2^2(t)\left(1 - \mathrm{x}_1^2(t) - \mathrm{x}_2^2(t)\right) \\ &= 2\left(\mathrm{x}_1^2(t) + \mathrm{x}_2^2(t)\right)\left(1 - \mathrm{x}_1^2(t) - \mathrm{x}_2^2(t)\right). \end{aligned}$$

Therefore

$$\dot{\mathrm{r}}(t) = 2\mathrm{r}(t)(1 - \mathrm{r}(t)). \tag{B.9}$$

If $\mathtt{r}(0) = 1$ then $\mathtt{r}(t)$ is always equal to one, so the unit circle is an invariant set. Furthermore, Eqn. (B.9) shows that if $0 \leq \mathtt{r}(0) < 1$, then $0 \leq \mathtt{r}(t) < 1$ for all time. Hence the open unit disc is also invariant. Using similar arguments, we find that also the complement of the unit disc is invariant. This system has many more invariant sets. □

In the previous example the state does not always converge to a single element, but to a set (e.g., the unit circle in Example B.4.6). We use $\text{dist}(x, \mathcal{G})$ to denote the (infimal) distance between a point $x \in \mathbb{R}^n$ and a set $\mathcal{G} \subseteq \mathbb{R}^n$, thus

$$\text{dist}(x, \mathcal{G}) := \inf_{g \in \mathcal{G}} \|x - g\|,$$

and we say that $\mathtt{x}$ *converges to a set* $\mathcal{G}$ if $\lim_{t\to\infty} \text{dist}(\mathtt{x}(t), \mathcal{G}) = 0$. The following extension of Lyapunov's stability theorem can now be proved.

Theorem B.4.7 (LaSalle's invariance principle). Let $\bar{x}$ be an equilibrium of a locally Lipschitz continuous differential equation $\dot{\mathtt{x}}(t) = f(\mathtt{x}(t))$, and suppose that V is a Lyapunov function for this system on some neighborhood Ω of $\bar{x}$.

Then Ω contains a closed and bounded invariant neighborhood $\mathcal{K}$ of $\bar{x}$, and for every $x_0 \in \mathcal{K}$ the solution $\mathtt{x}(t)$ as $t \to \infty$ converges to the set

$$\mathcal{G} := \{x \in \mathcal{K} \mid \dot{V}(\mathtt{x}(t; x)) = 0 \ \forall\, t \geq 0\}.$$

This set is invariant and non-empty. In particular, if $\mathcal{G} = \{\bar{x}\}$ then $\bar{x}$ is an asymptotically stable equilibrium.

Proof. The construction of $\mathcal{K}$ is very similar to that of Ω_1 in the proof of Theorem B.3.2. Since Ω is a neighborhood of $\bar{x}$ there is, by definition, a small enough ball $B(\bar{x}, \epsilon)$ completely contained in Ω. Let $\alpha = \min_{\|x-\bar{x}\|=\epsilon} V(x)$. This α is larger than zero. Then $\mathcal{K} := \{x \in B(\bar{x}, \epsilon) | V(x) \leq \alpha/2\}$ does the job. Indeed it is bounded, it is closed, and since $\dot{V}(x) \leq 0$ it is also invariant. And, finally, it is a neighborhood of $\bar{x}$.

The set $\mathcal{G}$ is non-empty (it contains $\bar{x}$). Let x be an element of $\mathcal{G}$. Then by invariance of $\mathcal{K}$ for every $t > 0$ the element $y := \mathtt{x}(t; x)$ is in $\mathcal{K}$. Also since $\dot{V}(\mathtt{x}(s; y)) = \dot{V}(\mathtt{x}(t + s; x)) = 0$ this orbit is in $\mathcal{G}$. Hence $\mathcal{G}$ is invariant.

Next let $x_0 \in \mathcal{K}$. Since $\mathcal{K}$ is invariant, the entire orbit $\mathtt{x}(t; x_0)$ is in $\mathcal{K}$ for all time. Now suppose, to obtain a contradiction, that $\mathtt{x}(t)$ does not converge to $\mathcal{G}$. Then, as $\mathtt{x}(t)$ is bounded, there is a sequence t_n of time with $\lim_{n\to\infty} t_n = \infty$ for which $\mathtt{x}(t_n, x_0)$ converges to some $x_\infty \notin \mathcal{G}$. This x_∞ is in $\mathcal{K}$ because $\mathcal{K}$ is closed. We claim that $V(\mathtt{x}(t; x_\infty))$ is constant as a function of time. To see this we need the inequality

$$V(\mathtt{x}(t_n; x_0)) \geq V(\mathtt{x}(t_n + t; x_0)) \geq V(x_\infty) \qquad \forall\, t \geq 0. \tag{B.10}$$

(The first inequality holds because $\dot{V}(x) \leq 0$ and the second inequality follows from $\dot{V}(x) \leq 0$ combined with the fact that $t_n + t < t_{n+k}$ for some large enough k,

so that $V(\mathrm{x}(t_n + t)) \geq V(\mathrm{x}(t_{n+k})) \geq V(x_\infty)$.) Taking the limit $n \to \infty$ turns (B.10) into

$$V(x_\infty) \geq V(\mathrm{x}(t; x_\infty)) \geq V(x_\infty).$$

Hence $V(\mathrm{x}(t; x_\infty))$ is constant for all time, that is, $\dot{V}(\mathrm{x}(t; x_\infty)) = 0$. But then $x_\infty \in \mathscr{G}$ (by definition of $\mathscr{G}$) which is a contradiction. Therefore the assumption that $\mathrm{x}(t)$ does not converge to $\mathscr{G}$ is wrong. ∎

The proof also provides an explicit description of the set $\mathscr{K}$. But if we only want to establish asymptotic stability then we can normally avoid this description. Its existence is enough.

Example B.4.8. Consider the system

$$\begin{aligned} \dot{\mathrm{x}}_1(t) &= \mathrm{x}_2^3(t), \\ \dot{\mathrm{x}}_2(t) &= -\mathrm{x}_1^3(t) - \mathrm{x}_2(t). \end{aligned} \tag{B.11}$$

Clearly, the origin $(0,0)$ is an equilibrium. For this equilibrium we propose the Lyapunov function

$$V(x) = x_1^4 + x_2^4.$$

This function is indeed a Lyapunov function (on $\Omega = \mathbb{R}^2$) because it is continuously differentiable, it is positive definite and

$$\dot{V}(x) = 4x_1^3(x_2^3) + 4x_2^3(-x_1^3 - x_2) = -4x_2^4 \leq 0.$$

This implies that the origin is stable, but not necessarily asymptotically stable. To prove asymptotic stability we use Theorem B.4.7. This theorem says that a bounded, closed invariant neighborhood $\mathscr{K}$ of the equilibrium $(0,0)$ exists, but we need not worry about its form. The set of interest is $\mathscr{G}$. It contains those initial states $x \in \mathscr{K}$ whose solution $\mathrm{x}(t; x)$ satisfies the system equations (B.11) and at the same time is such that $\dot{V}(\mathrm{x}(t)) = 0$ for all time. For our example the latter means

$$\mathrm{x}_2(t) = 0 \quad \forall t.$$

Substituting this into the system equations (B.11) gives

$$\begin{aligned} \dot{\mathrm{x}}_1(t) &= 0, \\ 0 &= -\mathrm{x}_1^3(t) - 0, \qquad \forall t. \end{aligned}$$

Clearly then $\mathrm{x}_1(t) = 0$ for all time as well, and so

$$\mathscr{G} = \{(0,0)\}.$$

LaSalle's invariance principle proves that for every $x_0 \in \mathscr{K}$ the solution $\mathrm{x}(t)$ converges to $(0,0)$ as $t \to \infty$, and, hence, that $(0,0)$ is an asymptotically stable equilibrium of this system. □

Example B.4.9 (Example B.4.1 continued). Consider again the damped pendulum from Example B.4.1,

$$\begin{aligned} \dot{\mathrm{x}}_1(t) &= \mathrm{x}_2(t), \\ \dot{\mathrm{x}}_2(t) &= -\frac{g}{\ell}\sin(\mathrm{x}_1(t)) - \frac{d}{m}\mathrm{x}_2(t). \end{aligned} \tag{B.12}$$

We found that the mechanical energy $V(x) := \frac{1}{2}m\ell^2 x_2^2 + mg\ell(1-\cos(x_1))$ is a Lyapunov function on some small enough neighborhood Ω of the hanging equilibrium $\bar{x} = (0,0)$, and we also found that

$$\dot{V}(x) = -d\ell^2 x_2^2.$$

The equality $\dot{V}(\mathrm{x}(t)) = 0$ hence holds for all time iff $\mathrm{x}_2(t) = 0$ for all time, and the LaSalle set $\mathscr{G}$ therefore is

$$\mathscr{G} = \{x \in \mathscr{K} \mid \mathrm{x}(t;x) \text{ satisfies (B.12) and } \mathrm{x}_2(t;x) = 0 \ \forall t\}.$$

We comment on $\mathscr{K}$ later. Since $\mathrm{x}_2(t) \equiv 0$ the equations (B.12) reduce to $\dot{\mathrm{x}}_1(t) = 0$ and $0 = -g/\ell \sin(\mathrm{x}_1(t))$. This implies that $\mathrm{x}_1(t)$ is constant and $\sin(\mathrm{x}_1(t)) = 0$, so

$$\mathscr{G} = \{x \in \mathscr{K} \mid x_1 = k\pi, k \in \mathbb{Z}, x_2 = 0\}.$$

This set contains at most two physically different solutions: the hanging downwards solution $x = \left[\begin{smallmatrix}0\\0\end{smallmatrix}\right]$ and the standing upwards solution $x = \left[\begin{smallmatrix}\pi\\0\end{smallmatrix}\right]$. To rule out the upwards solution it suffices to take the neighborhood Ω of $\bar{x} = \left[\begin{smallmatrix}0\\0\end{smallmatrix}\right]$ so small that $\left[\begin{smallmatrix}\pi\\0\end{smallmatrix}\right] \notin \Omega$. For example

$$\Omega = \{x \in \mathbb{R}^2 \mid -\pi < x_1 < \pi\}.$$

LaSalle's invariance principle now guarantees the existence of an invariant, closed, bounded neighborhood $\mathscr{K}$ of $\bar{x}$ in Ω. Clearly, this $\mathscr{K}$ does not contain $\left[\begin{smallmatrix}\pi\\0\end{smallmatrix}\right]$ either, so then

$$\mathscr{G} = \{\left[\begin{smallmatrix}0\\0\end{smallmatrix}\right]\},$$

and thus we have asymptotic stability of the hanging position.

Although not strictly needed, it may be interesting to know that by continuity of V we can always take $\mathscr{K}$ equal to the set of states x close enough to $\bar{x} = (0,0)$ and whose energy $V(x)$ is less than or equal to some small enough positive number such as,

$$\mathscr{K} = \{x \in \mathbb{R}^2 \mid -\pi < x_1 < \pi,\ V(x) \le 0.9V(\left[\begin{smallmatrix}\pi\\0\end{smallmatrix}\right])\}.$$

Since the energy does not increase over time it is immediate that this set is invariant. It is also closed and bounded, and it is a neighborhood of $(0,0)$. □

Example B.4.10 (Example B.4.2 continued). Consider again the system

$$\dot{\mathrm{x}}_1(t) = 0,$$
$$\dot{\mathrm{x}}_2(t) = -\mathrm{x}_2(t),$$

with equilibrium $\bar{x} = (0,0)$ and Lyapunov function $V(x) = x_1^2 + x_2^2$. In Example B.4.2 we found that $V(x)$ is a Lyapunov function and that

$$\dot{V}(x) = -2x_2^2 \leq 0.$$

Substitution of $\mathrm{x}_2(t) = 0$ in the system equations reduces the system equations to

$$\dot{\mathrm{x}}_1(t) = 0.$$

Hence $\mathrm{x}_1(t)$ is constant (besides $\mathrm{x}_2(t) = 0$), so

$$\mathscr{G} = \{(x_1, x_2) \in \mathscr{K} \mid x_2 = 0\}.$$

This is a part of the x_1-axis. Now LaSalle's invariance principle says that all states that start in $\mathscr{K}$ converge to the x_1-axis as $t \to \infty$. For $\mathscr{K}$ we can take for instance $\mathscr{K} = \{x \in \mathbb{R}^2 \mid V(x) \leq 1000\}$. □

B.5 Cost-to-Go Lyapunov Functions

It is in general hard to come up with a Lyapunov function for a given $\dot{\mathrm{x}}(t) = f(\mathrm{x}(t))$ and equilibrium point $\bar{x}$. An elegant attempt, with interesting interpretations, goes as follows. Suppose we have to pay an amount

$$L(x) \geq 0$$

per unit time, when we are at state x. As time progresses we move as dictated by the differential equation, and so the cost $L(\mathrm{x}(t))$ typically changes with time. The *cost-to-go* $V(x_0)$ is now defined as the total payment over the infinite future if we start at x_0, that is, it is the integral of $L(\mathrm{x}(t))$ over positive time,

$$V(x_0) := \int_0^\infty L(\mathrm{x}(\tau))\,\mathrm{d}\tau \quad \text{for} \quad \mathrm{x}(0) = x_0. \tag{B.13}$$

If $L(\mathrm{x}(t))$ decreases quick enough as we approach the equilibrium $\bar{x}$ then the cost-to-go may be well defined (finite), and possibly it is going to be continuously differentiable in x_0 as well. These are technical considerations and they might be hard to verify. The interesting property of the cost-to-go $V(\mathrm{x}(t))$ is that it decays as t increases. In fact

$$\dot{V}(x) = -L(x) \tag{B.14}$$

whenever $V(x)$ as defined in (B.13) is convergent. To see this split the cost-to-go into an integral over the first h units of time and an integral over the time beyond h,

$$V(\mathrm{x}(t)) = \int_t^{t+h} L(\mathrm{x}(\tau))\,\mathrm{d}\tau + \int_{t+h}^{\infty} L(\mathrm{x}(\tau))\,\mathrm{d}\tau = \int_t^{t+h} L(\mathrm{x}(\tau))\,\mathrm{d}\tau + V(\mathrm{x}(t+h)).$$

Therefore

$$\dot{V}(\mathrm{x}(t)) = \lim_{h\to 0} \frac{V(\mathrm{x}(t+h)) - V(\mathrm{x}(t))}{h} = \lim_{h\to 0} \frac{-\int_t^{t+h} L(\mathrm{x}(\tau))\,\mathrm{d}\tau}{h} = -L(\mathrm{x}(t))$$

if $L(\mathrm{x}(t))$ is continuous. An interpretation of (B.14) is that the current cost-to-go minus the cost-to-go from tomorrow onwards, is what we pay today. The function $L(x)$ is called the *running cost*. In physical applications $L(x)$ is often the dissipated power, and then $V(x)$ is the total dissipated energy.

Example B.5.1 (Damped pendulum). Consider once more the pendulum as shown in Fig. B.7 (p. 215). Here x_1 is the angular displacement, g is the gravitational acceleration, ℓ is the length of the pendulum, and m its mass. The force balance on the mass is

$$m\ell\ddot{\mathrm{x}}_1(t) = -mg\sin(\mathrm{x}_1(t)) - d\ell\dot{\mathrm{x}}_1(t).$$

The final term, $d\ell\dot{\mathrm{x}}_1(t)$, is a friction force, and d is a positive number known as the friction coefficient. The dissipated power due to this friction equals the friction force times the velocity of the mass, $\ell\dot{\mathrm{x}}_1(t)$. Hence we take this product as our running cost,

$$L(\mathrm{x}_1(t), \dot{\mathrm{x}}_1(t)) = d\ell\dot{\mathrm{x}}_1(t)\ell\dot{\mathrm{x}}_1(t) = d\ell^2\dot{\mathrm{x}}_1^2(t).$$

Notice that $L \geq 0$ because $d > 0$. The cost-to-go is the total dissipated power,

$$\begin{aligned} V(\mathrm{x}_1(0), \dot{\mathrm{x}}_1(0)) &:= \int_0^{\infty} d\ell^2\dot{\mathrm{x}}_1^2(t)\,\mathrm{d}t \\ &= \int_0^{\infty} d\ell^2 \frac{-mg\sin(\mathrm{x}_1(t)) - m\ell\ddot{\mathrm{x}}_1(t)}{d\ell}\dot{\mathrm{x}}_1(t)\,\mathrm{d}t \\ &= \int_0^{\infty} -mg\ell\sin(\mathrm{x}_1(t))\dot{\mathrm{x}}_1(t) - m\ell^2\ddot{\mathrm{x}}_1(t)\dot{\mathrm{x}}_1(t)\,\mathrm{d}t \\ &= \left[mg\ell\cos(\mathrm{x}_1(t)) - m\ell^2\tfrac{1}{2}\dot{\mathrm{x}}_1^2(t)\right]_0^{\infty} \\ &= mg\ell(1-\cos(\mathrm{x}_1(0)) + \tfrac{1}{2}m\ell^2\dot{\mathrm{x}}_1^2(0). \end{aligned} \tag{B.15}$$

In the final equality we used that $\mathrm{x}_1(t)$ and $\dot{\mathrm{x}}_1(t)$ converge to zero as $t \to \infty$. The first term in (B.15) is the potential energy at $t = 0$, and the second term is the kinetic energy at $t = 0$. Thus the cost-to-go V is the mechanical energy. This is the Lyapunov function that we used in Example B.3.4 and Example B.4.1. □

As mentioned earlier, the only obstacle is that the integral (B.13) has to be well defined and continuously differentiable at x_0. If the system dynamics is linear of the form

$$\dot{\mathrm{x}}(t) = A\mathrm{x}(t)$$

then these obstacles can be overcome and we end up with a very useful result. It is a classic result in systems theory. In this result we take the running cost to be quadratic in the state,

$$L(x) = x^{\mathrm{T}} Q x,$$

with $Q \in \mathbb{R}^{n\times n}$ a symmetric positive definite matrix (see Appendix A.1).

Theorem B.5.2 (Lyapunov equation). Let $A \in \mathbb{R}^{n\times n}$ and consider

$$\dot{\mathrm{x}}(t) = A\mathrm{x}(t), \qquad \mathrm{x}(0) = x_0. \tag{B.16}$$

Suppose $Q \in \mathbb{R}^{n\times n}$ is symmetric positive definite, and let

$$V(x_0) := \int_0^\infty \mathrm{x}^{\mathrm{T}}(t) Q \mathrm{x}(t)\, \mathrm{d}t. \tag{B.17}$$

The following four statements are equivalent.

1. $\bar{x} = 0$ is a globally asymptotically stable equilibrium of (B.16).
2. $\bar{x} = 0$ is an asymptotically stable equilibrium of (B.16).
3. $V(x)$ defined in (B.17) exists for every $x \in \mathbb{R}^n$, and it is a strong Lyapunov function for (B.16) with equilibrium $\bar{x} = 0$. In fact $V(x)$ is quadratic, $V(x) = x^{\mathrm{T}} P x$, with $P \in \mathbb{R}^{n\times n}$ the well-defined positive definite matrix

$$P := \int_0^\infty \mathrm{e}^{A^{\mathrm{T}} t} Q \mathrm{e}^{At}\, \mathrm{d}t. \tag{B.18}$$

4. The linear matrix equation (known as the *Lyapunov equation*)

$$A^{\mathrm{T}} P + PA = -Q \tag{B.19}$$

has a unique solution $P \in \mathbb{R}^{n\times n}$, and P is symmetric positive definite.

In that case the P of (B.18) and (B.19) are the same.

Proof. We prove the cycle of implications 1. $\Longrightarrow$ 2. $\Longrightarrow$ 3. $\Longrightarrow$ 4. $\Longrightarrow$ 1.

1. $\Longrightarrow$ 2. Trivial.

2. $\Longrightarrow$ 3. The solution of $\dot{x}(t) = Ax(t)$ is $x(t) = e^{At}x_0$. By asymptotic stability the entire transition matrix converges to zero, $\lim_{t\to\infty} e^{At} = 0 \in \mathbb{R}^{n\times n}$. Now

$$V(x_0) = \int_0^\infty (e^{At}x_0)^T Q(e^{At}x_0)\,dt = \int_0^\infty x_0^T (e^{A^T t} Q e^{At}) x_0\,dt = x_0^T P x_0$$

for $P := \int_0^\infty e^{A^T t} Q e^{At}\,dt$. This P is well defined because e^{At} converges to zero exponentially fast. This P is positive definite because it is the integral of a positive definite matrix.

So $V(x_0)$ is well defined and quadratic and, hence, continuously differentiable. It has a unique minimum at $x_0 = 0$ and, as we showed earlier, $\dot{V}(x) = -L(x) := -x^T Q x \le 0$. Hence V is a Lyapunov function. In fact it is a strong Lyapunov function because $-x^T Q x < 0$ for every $x \neq 0$.

3. $\Longrightarrow$ 4. Since V is a strong Lyapunov function, it follows that A is asymptotically stable. Take P as defined in (B.18). Then

$$A^T P + PA = \int_0^\infty A^T e^{A^T t} Q e^{At} + e^{A^T t} Q e^{At} A\,dt = \left[e^{A^T t} Q e^{At}\right]_0^\infty = -Q.$$

This shows that for every $Q \in \mathbb{R}^{n\times n}$ (symmetric or not) there is a $P \in \mathbb{R}^{n\times n}$ for which $A^T P + PA = -Q$. This means that the linear mapping from $P \in \mathbb{R}^{n\times n}$ to $A^T P + PA \in \mathbb{R}^{n\times n}$ is surjective. It is a standard result from linear algebra that a surjective linear mapping from a finite-dimensional vector space to the same vector space is in fact invertible. Hence, for every $Q \in \mathbb{R}^{n\times n}$ (symmetric or not) the solution P of $A^T P + PA = -Q$ exists and is unique. Our Q is symmetric and positive definite, so the solution P as constructed in (B.18) is symmetric and positive definite.

4. $\Longrightarrow$ 1. Then $V(x) := x^T P x$ satisfies $\dot{V}(x(t)) = \frac{d}{dt} x^T(t) P x(t) = \dot{x}^T(t) P x(t) + x^T(t) P \dot{x}(t) = x^T(t)(A^T P + PA)x(t) = -x^T(t) Q x(t)$. So V is a strong Lyapunov function with $\dot{V}(x) < 0$ for all $x \neq 0$. It is radially unbounded, hence the equilibrium is globally asymptotically stable (Theorem B.3.5).

■

The proof of the above theorem actually shows that for asymptotically stable matrices $A \in \mathbb{R}^{n\times n}$, the Lyapunov equation

$$A^T P + PA = -Q$$

for *every* matrix $Q \in \mathbb{R}^{n\times n}$ (not necessarily symmetric or positive definite) has a unique solution $P \in \mathbb{R}^{n\times n}$. In particular, it immediately yields the following result.

Corollary B.5.3 (Lyapunov equation). If $A \in \mathbb{R}^{n\times n}$ is asymptotically stable, then $A^T P + PA = 0 \in \mathbb{R}^{n\times n}$ iff $P = 0 \in \mathbb{R}^{n\times n}$. □

Example B.5.4 (Scalar Lyapunov equation). The system

$$\dot{x}(t) = -2x(t)$$

is globally asymptotically stable because for $q = 1 > 0$ the Lyapunov equation

$$-2p - 2p = -q = -1$$

has a unique solution, $p = 1/4$, and the solution is positive. Note that Theorem B.5.2 says that we may take any $q > 0$ that we like. Indeed, whatever positive q we take, we have that the solution of the Lyapunov equation is unique and positive: $p = q/4 > 0$. □

It is good to realize that (B.19) is a linear equation in the entries of P, and it is therefore easily solved (it requires a finite number of operations). Also positive definiteness of a matrix can be tested in a finite number of steps (Appendix A.1). Thus asymptotic stability of $\dot{x}(t) = Ax(t)$ can be tested in a finite number of steps.

Example B.5.5 (2×2 Lyapunov equation). Consider

$$\dot{x}(t) = \begin{bmatrix} -1 & 2 \\ 0 & -1 \end{bmatrix} x(t).$$

We take Q equal to the 2×2 identity matrix,

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The candidate solution P of the Lyapunov equation we write as $P = \left[\begin{smallmatrix} \alpha & \beta \\ \beta & \gamma \end{smallmatrix}\right]$. The Lyapunov equation (B.19) then reads

$$\begin{bmatrix} -1 & 0 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} \alpha & \beta \\ \beta & \gamma \end{bmatrix} + \begin{bmatrix} \alpha & \beta \\ \beta & \gamma \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Working out the matrix products on left-hand side leaves us with

$$\begin{bmatrix} -2\alpha & 2\alpha - 2\beta \\ 2\alpha - 2\beta & 4\beta - 2\gamma \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

By symmetry, the upper-right and lower-left entries are identical so the above matrix equation is effectively *three* scalar equations in the three unknowns α, β, γ:

$$\begin{aligned} -2\alpha &= -1, \\ 2\alpha - 2\beta &= 0, \\ 4\beta - 2\gamma &= -1. \end{aligned}$$

This gives $\alpha = \beta = 1/2$ and $\gamma = 3/2$, that is,

$$P = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}.$$

This matrix is positive definite because $P_{11} = \frac{1}{2} > 0$ and $\det(P) = 1 > 0$ (see Appendix A.1). Therefore the differential equation with equilibrium $\bar{x} = (0,0)$ is globally asymptotically stable. □

B.6 Lyapunov's First Method

Through a process called *linearization* we can approximate a nonlinear system around an equilibrium with a linear system. Often the stability properties of the nonlinear system and the so determined linear system are alike. In fact, as we will see, they often share the same Lyapunov function.

We assume that the vector field $f : \mathbb{R}^n \to \mathbb{R}^n$ is differentiable at the given equilibrium $\bar{x}$. This is to say that $f(x)$ is of the form

$$f(\bar{x} + \delta_x) = A\delta_x + \mathfrak{o}(\delta_x) \tag{B.20}$$

with A some $n \times n$ matrix, and $\mathfrak{o} : \mathbb{R}^n \to \mathbb{R}^n$ some *little-o* function which means a function having the property that

$$\lim_{\delta_x \to 0} \frac{\|\mathfrak{o}(\delta_x)\|}{\|\delta_x\|} = 0. \tag{B.21}$$

We think of little-o functions as functions that are "extremely small" around the origin.

To analyze the behavior of the state $\mathrm{x}(t)$ relative to an equilibrium $\bar{x}$, it makes sense to analyze $\delta_x(t)$ defined as the difference between state and equilibrium,

$$\delta_x(t) := \mathrm{x}(t) - \bar{x}.$$

This difference obeys the differential equation

$$\dot{\delta}_x(t) = \dot{\mathrm{x}}(t) - \dot{\bar{x}} = \dot{\mathrm{x}}(t) = f(\mathrm{x}(t)) = f(\bar{x} + \delta_x(t)) = A\delta_x(t) + \mathfrak{o}(\delta_x(t)).$$

The *linearized system* of $\dot{\mathrm{x}}(t) = f(\mathrm{x}(t))$ at equilibrium $\bar{x}$ is defined as the system in which the little-o term, $\mathfrak{o}(\delta_x(t))$, is deleted:

$$\dot{\delta}_x(t) = A\delta_x(t).$$

It constitutes a linear approximation of the original nonlinear system but we expect it to be an accurate approximation as long as $\delta_x(t)$ is "small". The matrix

A equals the *Jacobian* matrix at $\bar{x}$ defined as

$$A = \frac{\partial f(\bar{x})}{\partial x^{\mathrm{T}}} := \begin{bmatrix} \frac{\partial f_1(\bar{x})}{\partial x_1} & \cdots & \frac{\partial f_1(\bar{x})}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_n(\bar{x})}{\partial x_1} & \cdots & \frac{\partial f_n(\bar{x})}{\partial x_n} \end{bmatrix}. \tag{B.22}$$

(See Appendix A.2 for an explanation of this notation.)

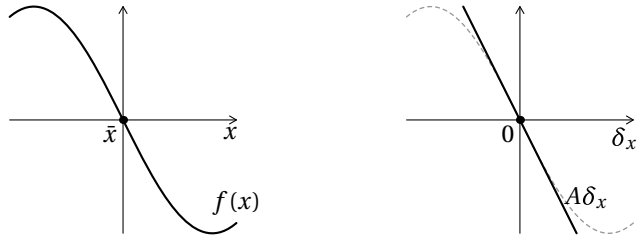


FIGURE B.11: Nonlinear $f(x)$ (left) and its linear approximation $A\delta_x$ (right).

Example B.6.1. Consider the nonlinear differential equation

$$\dot{\mathrm{x}}(t) = -\sin(2\mathrm{x}(t)). \tag{B.23}$$

The function $f(x) = -\sin(2x)$ has many zeros, among which is

$$\bar{x} = 0.$$

The idea of linearization is that around $\bar{x}$ the function $f(x)$ is almost indistinguishable from its tangent with slope

$$A = \frac{\partial f(\bar{x})}{\partial x} = -2\cos(\bar{x}) = -2\cos(0) = -2,$$

see Fig. B.11, and so the solutions $\mathrm{x}(t)$ of (B.23) will probably be quite similar to $\bar{x} + \delta_x(t) = \delta_x(t)$ with $\delta_x(t)$ the solution of the *linear* system

$$\dot{\delta}_x(t) = -2\delta_x(t) \tag{B.24}$$

provided that $\delta_x(t)$ is small. The above linear system (B.24) is the linearized system of (B.23) at equilibrium $\bar{x} = 0$. □

Lyapunov's first method presented next, roughly speaking says that the nonlinear system and the linearized system have the same asymptotic stability properties. The only exception to this rule is if the eigenvalue with the largest real part is on the imaginary axis (so its real part is zero). The proof of this result relies on the fact that every asymptotically stable linear system has a Lyapunov function (namely its cost-to-go) which turns out to be a Lyapunov function for the nonlinear system as well:

Theorem B.6.2 (Lyapunov's first method). Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be a continuously differentiable function and let $\bar{x}$ be an equilibrium of $\dot{x}(t) = f(x(t))$.

1. If all eigenvalues of the Jacobian (B.22) have negative real part, then $\bar{x}$ is an asymptotically stable equilibrium of the nonlinear system.
2. If there is an eigenvalue of the Jacobian (B.22) with positive real part, then $\bar{x}$ is an unstable equilibrium of the nonlinear system.

Proof. (First realize that continuous differentiability of f implies Lipschitz continuity, and so Lyapunov theory is applicable.) Write $f(x)$ as in (B.20). Without loss of generality we assume that $\bar{x} = 0$, and we define A as in (B.22).

1. By the assumptions on the eigenvalues the linearized system $\dot{\delta}_x(t) = A\delta_x(t)$ is asymptotically stable. So Theorem B.5.2 guarantees the existence of a positive definite matrix P that satisfies

   $$A^{\mathrm{T}}P + PA = -I,$$

   and that $V(\delta_x) = \delta_x^{\mathrm{T}} P \delta_x$ is a strong Lyapunov function for the linear system $\dot{\delta}_x(t) = A\delta_x(t)$. We prove that $V(x) := x^{\mathrm{T}}Px$ is also a strong Lyapunov function for $\dot{x}(t) = f(x(t))$ on some neighborhood Ω of $\bar{x} = 0$. Clearly, this V is positive definite and continuously differentiable and positive definite. We have that

   $$\begin{aligned}
   \dot{V}(x) &= \dot{x}^{\mathrm{T}}Px + x^{\mathrm{T}}P\dot{x} \\
   &= f(x)^{\mathrm{T}}Px + x^{\mathrm{T}}Pf(x) \\
   &= [Ax + o(x)]^{\mathrm{T}}Px + x^{\mathrm{T}}P[Ax + o(x)] \\
   &= x^{\mathrm{T}}(A^{\mathrm{T}}P + PA)x + o(x)^{\mathrm{T}}Px + x^{\mathrm{T}}Po(x) \\
   &= -x^{\mathrm{T}}x + 2\,o(x)^{\mathrm{T}}Px \\
   &= -\|x\|^2 + 2\,o(x)^{\mathrm{T}}Px.
   \end{aligned}$$

   The term $2\,o(x)^{\mathrm{T}}Px$ we recognize as the standard inner product of $2\,o(x)$ and Px, so by the Cauchy-Schwarz inequality we can bound it from above by $2\|o(x)\|\|Px\|$, hence

   $$\dot{V}(x) \le -\|x\|^2 + 2\|o(x)\|\|Px\|.$$

   Based on this we now choose Ω as

   $$\Omega := \{x \in \mathbb{R}^n \mid 2\|o(x)\|\|Px\| \le \frac{1}{2}\|x\|^2\}.$$

   From (B.21) it follows that this Ω is a neighborhood of $\bar{x} = 0$. Then, finally, we find that

   $$\dot{V}(x) \le -\frac{1}{2}\|x\|^2 \qquad \forall x \in \Omega.$$

   Therefore $\dot{V}(x) < 0$ for all $x \in \Omega, x \neq \bar{x}$, so V is a strong Lyapunov function for the nonlinear system.

2. See (Khalil, 1996, Thm. 3.7).

■

The two cases of Theorem B.6.2 cover all possible eigenvalue configurations, except when some eigenvalues have zero real part and none have positive real part. In fact, if there are eigenvalues on the imaginary axis then the dynamical behavior crucially depends on the higher-order terms $o(\delta_x)$, which are neglected in the linearization. For example, the three systems

$$\dot{\mathrm{x}}(t) = \mathrm{x}^2(t),$$
$$\dot{\mathrm{x}}(t) = -\mathrm{x}^2(t),$$
$$\dot{\mathrm{x}}(t) = \mathrm{x}^3(t),$$

all have the same linearization at $\bar{x} = 0$, but their dynamical properties are very different. See also Exercise B.5.

Example B.6.3. Consider the system

$$\dot{\mathrm{x}}_1(t) = \mathrm{x}_1(t) + \mathrm{x}_1(t)\mathrm{x}_2^2(t)$$
$$\dot{\mathrm{x}}_2(t) = -\mathrm{x}_2(t) + \mathrm{x}_1^2(t)\mathrm{x}_2(t).$$

The system has equilibrium $\bar{x} := \left[\begin{smallmatrix}0\\0\end{smallmatrix}\right]$, and the Jacobian A at this $\left[\begin{smallmatrix}0\\0\end{smallmatrix}\right]$ is

$$A = \frac{\partial f(\bar{x})}{\partial x^{\mathrm{T}}} = \begin{bmatrix} 1 + x_2^2 & 2x_1x_2 \\ 2x_1x_2 & -1 + x_1^2 \end{bmatrix}\Bigg|_{x=\left[\begin{smallmatrix}0\\0\end{smallmatrix}\right]} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Clearly it has eigenvalues ± 1. In particular it has a positive eigenvalue. Lyapunov's first method hence proves that the system at this equilibrium is unstable. □

B.7 Exercises

B.1 *Equilibria.*

(a) Let $\bar{x}$ be an equilibrium of system (B.2). Show that every continuously differentiable function $V : \mathbb{R}^n \to \mathbb{R}$ satisfies $\dot{V}(\bar{x}) = 0$.

(b) Prove that if a system of the form (B.2) has more than one equilibrium point, then none of these equilibrium points are globally asymptotically stable.

(c) Consider the linear system

$$\dot{\mathrm{x}}(t) = A\mathrm{x}(t),$$

with A an $n \times n$ matrix. Argue that this system either has exactly one equilibrium, or infinitely many equilibria.

B.2 Investigate the stability of the origin for the following two systems (that is, check all six stability types as mentioned in Definition B.2.2). Use a suitable Lyapunov function.

(a)

$$\begin{aligned}\dot{\mathtt{x}}_1(t) &= -\mathtt{x}_1^3(t) - \mathtt{x}_2^2(t),\\ \dot{\mathtt{x}}_2(t) &= \mathtt{x}_1(t)\mathtt{x}_2(t) - \mathtt{x}_2^3(t).\end{aligned}$$

[Hint: take the "standard" $V(x)$.]

(b)

$$\begin{aligned}\dot{\mathtt{x}}_1(t) &= \mathtt{x}_2(t),\\ \dot{\mathtt{x}}_2(t) &= -\mathtt{x}_1^3(t).\end{aligned}$$

[Hint: try $V(x_1,x_2) = x_1^\alpha + cx_2^\beta$ and then determine suitable α,β,c.]

B.3 *Adaptive Control.* The following problem from adaptive control illustrates an extension of the theory of Lyapunov functions to functions that are, strictly speaking, no longer Lyapunov functions. This problem concerns the stabilization of a system of which the parameters are not (completely) known. Consider the scalar system

$$\dot{\mathtt{x}}(t) = a\mathtt{x}(t) + \mathtt{u}(t), \qquad \mathtt{x}(0) = x_0, \tag{B.25}$$

where a is a constant, and where $\mathtt{u} : [0,\infty) \to \mathbb{R}$ is an input that we have to choose in such a way that $\lim_{t\to\infty} \mathtt{x}(t) = 0$. If we know a then $\mathtt{u}(t) = -k\mathtt{x}(t)$, with $k > a$, would solve the problem. However, we assume that a is unknown but that we can measure $\mathtt{x}(t)$. Contemplate the following dynamic state feedback

$$\mathtt{u}(t) = -\mathtt{k}(t)\mathtt{x}(t) \quad \text{where} \quad \dot{\mathtt{k}}(t) = \mathtt{x}^2(t), \quad \mathtt{k}(0) = 0. \tag{B.26}$$

Here, the term $\mathtt{x}^2(t)$ ensures that $\mathtt{k}(t)$ grows if $\mathtt{x}^2(t)$ is not close to zero. The idea is that $\mathtt{k}(t)$ keeps on growing until it is so large that $\mathtt{u}(t) := -\mathtt{k}(t)\mathtt{x}(t)$ stabilizes the system, so until $\mathtt{x}(t)$ is equal to zero.

(a) Write (B.25)–(B.26) as one system with state $(\mathtt{x},\mathtt{k})$ and determine all equilibrium points.

(b) Consider the function $V(x,k) := x^2 + (k-a)^2$. Prove that $\dot{V}(\mathtt{x}(t),\mathtt{k}(t)) = 0$ for all $\mathtt{x},\mathtt{k}$. For which equilibrium point is this a Lyapunov function?

(c) Prove, using the above, that $\mathtt{k}(t)$ is bounded.

(d) Prove, using (B.26), that $\mathtt{k}(t)$ converges as $t \to \infty$.

(e) Prove that $\lim_{t\to\infty} \mathtt{x}(t;x_0) = 0$.

(f) Determine $\lim_{t\to\infty} \mathtt{k}(t)$.

B.4 This exercise is based on an exercise in Khalil (1996) who, in turn, took it from a book by Hahn[1], and it appears that Hahn was inspired by a paper by Barbashin and Krasovskĭ[2]. Consider the system

$$\dot{\mathtt{x}}_1(t) = \frac{-\mathtt{x}_1(t) + \mathtt{x}_2(t)(1+\mathtt{x}_1^2(t))^2}{(1+\mathtt{x}_1^2(t))^2},$$
$$\dot{\mathtt{x}}_2(t) = \frac{-\mathtt{x}_1(t) - \mathtt{x}_2(t)}{(1+\mathtt{x}_1^2(t))^2},$$

and define $V : \mathbb{R}^2 \to \mathbb{R}$ as

$$V(x) = \frac{x_1^2}{1+x_1^2} + x_2^2.$$

(a) Show that $(0,0)$ is the only equilibrium point.

(b) Show that V is a strong Lyapunov function on the entire state space $\Omega = \mathbb{R}^2$.

(c) Show that the level sets $\{x \in \mathbb{R}^2 \mid V(x) = c\}$ of the Lyapunov function are unbounded if $c \geq 1$. Hence the Lyapunov function is not radially unbounded. (Figure B.12 depicts several level sets.)

(d) Figure B.12 also depicts the curve $x_2 = 1/x_1$ and the region to the right of it, so where $x_1 x_2 > 1$. The phase portrait suggests that $\mathtt{x}_1(t)\mathtt{x}_2(t)$ increases if $\mathtt{x}_2(t) = 1/\mathtt{x}_1(t)$. Indeed. Show that

$$\frac{\mathrm{d}\mathtt{x}_1(t)\mathtt{x}_2(t)}{\mathrm{d}t} = \frac{1}{\mathtt{x}_1^2(t)(1+\mathtt{x}_1^2(t))^2} > 0$$

whenever $\mathtt{x}_2(t) = 1/\mathtt{x}_1(t) > 0$.

(e) Use the above to prove that the origin is not globally asymptotically stable.

B.5 *Linearization.* Consider the scalar system

$$\dot{\mathtt{x}}(t) = a\mathtt{x}^3(t)$$

with $a \in \mathbb{R}$.

(a) Prove that the linearization of this system about its equilibrium point is independent of a.

[1] W. Hahn. *Stability of Motion*, volume 138 of *Die Grundlehren der mathematischen Wissenschaften*. Springer-Verlag, New York, 1967.

[2] E.A. Barbashin and N.N. Krasovskĭ. Ob ustoichivosti dvizheniya vtzelom. *Dokl. Akad. Nauk. USSR*, 86(3): 453–456, 1952. (Russian). English title: "On the stability of motion in the large".

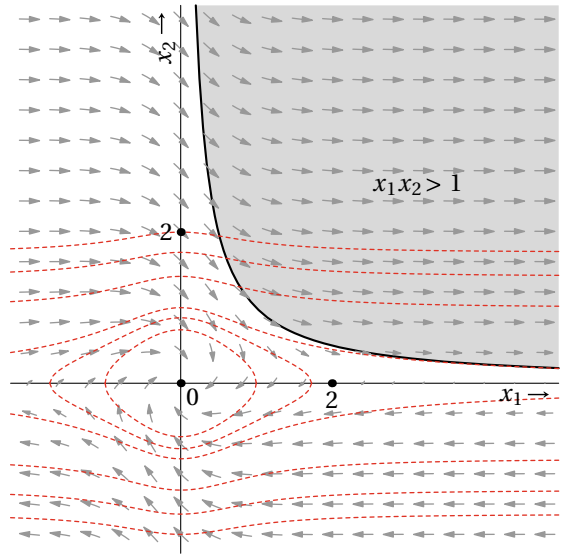


FIGURE B.12: A phase portrait of the system of Exercise B.4. The red dashed lines are level sets of $V(x)$. The boundary of the shaded region $\{(x_1, x_2) \mid x_1, x_2 > 0, x_1 x_2 > 1\}$ is where $x_2 = 1/x_1 > 0$.

(b) Sketch the graph of ax^3 as a function of x, and use it to argue that the equilibrium is
- asymptotically stable if $a < 0$,
- stable if $a = 0$,
- unstable if $a > 0$.

(c) Determine a Lyapunov function for the cases that the system is stable.

(d) Determine a strong Lyapunov function for the cases that the system is asymptotically stable.

B.6 Consider the system

$$\dot{x}_1(t) = -x_1^5(t) - x_2(t),$$
$$\dot{x}_2(t) = x_1(t) - 2x_2^3(t).$$

(a) Determine all points of equilibrium.

(b) Determine a Lyapunov function for the equilibrium $\bar{x} = (0,0)$, and discuss the type of stability that follows from this Lyapunov function (stable? asymptotically stable? globally asymptotically stable?)

B.7 Suppose that

$$\dot{x}_1(t) = x_2(t) - x_1(t),$$
$$\dot{x}_2(t) = -x_1^3(t),$$

and use the candidate Lyapunov function $V(x_1, x_2) = x_1^4 + 2x_2^2$. The equilibrium is $\bar{x} = (0,0)$.

(a) Is this a Lyapunov function?

(b) Is this a strong Lyapunov function?

(c) Investigate the nature of stability of this equilibrium with LaSalle's invariance principle.

B.8 Consider the *Van der Pol equation*

$$\ddot{\mathrm{y}}(t) - \epsilon(1 - \mathrm{y}^2(t))\dot{\mathrm{y}}(t) + \mathrm{y}(t) = 0.$$

This equation occurs in the study of vacuum tubes and then ϵ is positive. However, in this exercise we take $\epsilon < 0$.

(a) Rewrite this equation in the standard form (B.2) with $\mathrm{x}_1 := \mathrm{y}$ and $\mathrm{x}_2 := \dot{\mathrm{y}}$.

(b) Use linearization to show that the origin $(x_1, x_2) = (0,0)$ is an asymptotically stable equilibrium (recall that $\epsilon < 0$).

(c) Determine a neighborhood Ω of the origin for which $V(x_1, x_2) = x_1^2 + x_2^2$ is a Lyapunov function for $\bar{x} = (0,0)$.

(d) Let $V(x_1, x_2)$ and Ω be as in the previous part. Which stability properties can be concluded from LaSalle's invariance principle?

B.9 The well-known *Lotka-Volterra model* describes the interaction between a population of predators (of size x_1) and prey (of size x_2), and is given by the equations

$$\begin{aligned} \dot{\mathrm{x}}_1(t) &= -a\mathrm{x}_1(t) + b\mathrm{x}_1(t)\mathrm{x}_2(t), & \mathrm{x}_1(0) &\geq 0, \\ \dot{\mathrm{x}}_2(t) &= c\mathrm{x}_2(t) - d\mathrm{x}_1(t)\mathrm{x}_2(t), & \mathrm{x}_2(0) &\geq 0. \end{aligned}$$

The first term on the right-hand side of the first equation models that predators become extinct without food, while the second term models that the growth of the number of predators is proportional to the number of prey. Likewise, the term on the right-hand side of the second equation models that without predators the population of prey increases, while its decrease is proportional to the number of predators. For convenience we choose $a = b = c = d = 1$.

(a) Show that, apart from $(0,0)$, the system has a second equilibrium point.

(b) Investigate the stability of both equilibrium points using linearization.

(c) Investigate the stability of the *nonzero* equilibrium point using the function

$$V(x_1, x_2) = x_1 + x_2 - \ln(x_1 x_2) - 2.$$

Here, ln is the natural logarithm.

B.10 The equations of motion of the pendulum with friction are

$$\begin{aligned} \dot{\mathrm{x}}_1(t) &= \mathrm{x}_2(t), \\ \dot{\mathrm{x}}_2(t) &= -\frac{g}{\ell}\sin(\mathrm{x}_1(t)) - \frac{d}{m}\mathrm{x}_2(t). \end{aligned} \tag{B.27}$$

Here x_1 is the angular displacement, x_2 is the angular velocity, g is the gravitational acceleration, ℓ is the length of the pendulum, m is the mass of the pendulum, and d is a friction coefficient. All constants g, ℓ, d, m are positive.

(a) Prove, using Theorem B.6.2, that the origin is an asymptotically stable equilibrium point.

(b) In Example B.4.9 we verified asymptotic stability using LaSalle's invariance principle. Here we want to construct a strong Lyapunov function to show asymptotic stability using Theorem B.3.2: determine a symmetric matrix $P > 0$ such that the function

$$V(x) := x^{\mathrm{T}} P x + g\big(1 - \cos(x_1)\big)$$

is a strong Lyapunov function for (B.27) on some neighborhood Ω of the origin. (This exercise assumes knowledge of Appendix A.1.)

B.11 Consider the system

$$\begin{aligned} \dot{\mathrm{x}}_1(t) &= -2\mathrm{x}_1(t)(\mathrm{x}_1(t) - 1)(2\mathrm{x}_1(t) - 1), \\ \dot{\mathrm{x}}_2(t) &= -2\mathrm{x}_2(t). \end{aligned} \tag{B.28}$$

(a) Determine all equilibrium points of the system (B.28).

(b) Prove that there are two asymptotically stable equilibrium points.

(c) Investigate the stability of the other equilibrium point(s).

B.12 Determine all equilibrium points of

$$\begin{aligned} \dot{\mathrm{x}}_1(t) &= \mathrm{x}_1(t)(1 - \mathrm{x}_2^2(t)), \\ \dot{\mathrm{x}}_2(t) &= \mathrm{x}_2(t)(1 - \mathrm{x}_1^2(t)). \end{aligned}$$

For each of the equilibrium points determine the linearization and the nature of stability of the linearization.

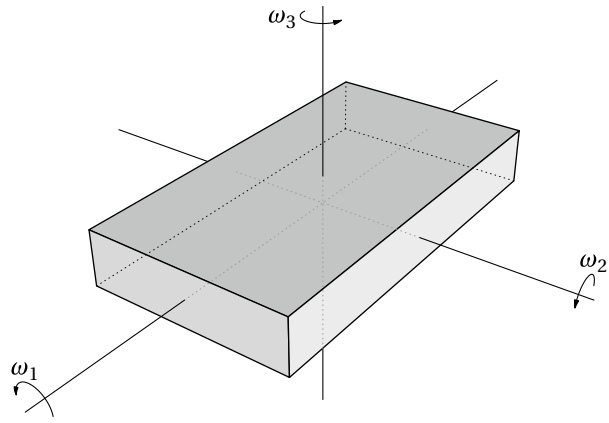


FIGURE B.13: A spinning rigid body. See Exercise B.13.

B.13 The equations of motion of a rigid body spinning around its center of mass are

$$I_1\dot{\omega}_1(t) = (I_2 - I_3)\omega_2(t)\omega_3(t),$$
$$I_2\dot{\omega}_2(t) = (I_3 - I_1)\omega_1(t)\omega_3(t),$$
$$I_3\dot{\omega}_3(t) = (I_1 - I_2)\omega_1(t)\omega_2(t),$$

where $\omega := (\omega_1, \omega_2, \omega_3)$ is the vector of angular velocities around the three principal axes of the rigid body, and $I_1, I_2, I_3 > 0$ are the principal moments of inertia. This is depicted in Fig. B.13. The kinetic energy (due to rotation) is

$$\frac{1}{2}\left(I_1\omega_1^2 + I_2\omega_2^2 + I_3\omega_3^2\right).$$

(a) Prove that the origin $\omega = (0,0,0)$ is a stable equilibrium.

(b) Prove that the origin $\omega = (0,0,0)$ is not asymptotically stable.

Now assume that the moments of inertias are ordered as

$$0 < I_1 < I_2 < I_3.$$

(This implies a certain lack of symmetry of the rigid body, e.g., it is not a unit cube. An example where $0 < I_1 < I_2 < I_3$ is shown in Fig. B.13.)

(c) The origin $(0,0,0)$ is just one equilibrium. Determine all equilibria and explain what this implies about the stability properties.

(d) Determine the linearization around each of the equilibria.

(e) Use linearization to prove that steady spinning around the second principal axis $(0, \bar{\omega}_2, 0)$ is unstable if $\bar{\omega}_2 \neq 0$.

(f) *This is a tricky question.* Prove that both the kinetic energy

$$\tfrac{1}{2}\left(I_1\omega_1^2 + I_2\omega_2^2 + I_3\omega_3^2\right),$$

and the squared total angular momentum

$$I_1^2\omega_1^2 + I_2^2\omega_2^2 + I_3^2\omega_3^2$$

are constant over time, and use this to prove that steady spinning around the first and third principal axes is stable, but not asymptotically stable.
Remark: A spinning body spins stably both around the principal axis with the smallest moment of inertia and the principal axis with the largest moment of inertia. But around the other principal axis it is not stable. This can be demonstrated by (carefully) spinning this book in the air. You will see that you can get it to spin nicely around the axis with largest inertia – like a discus – and around the axis with smallest inertia – like a spear – but you will probably fail to make it spin around the other axis.

B.14 Consider the system $\dot{\mathrm{x}}(t) = f(\mathrm{x}(t))$ with equilibrium point $\bar{x}$, and let Ω be a neighborhood of $\bar{x}$. Suppose a Lyapunov function exists such that $\dot{V}(x) = 0$ for all $x \in \Omega$. Prove that this system with equilibrium $\bar{x}$ is not asymptotically stable.

B.15 Let $\mathrm{x}(t; x_0)$ be a solution of the differential equation $\dot{\mathrm{x}}(t) = f(\mathrm{x}(t)), \mathrm{x}(0) = x_0$. Prove that $\mathcal{O}(x_0) := \{\mathrm{x}(t; x_0) \mid t \geq 0\}$ is an invariant set for $\dot{\mathrm{x}}(t) = f(\mathrm{x}(t))$.

B.16 Consider a system $\dot{\mathrm{x}}(t) = f(\mathrm{x}(t))$, and assume that f is locally Lipschitz continuous. A trajectory $\mathrm{x}(t; x_0)$ is *closed* if $\mathrm{x}(t + s; x_0) = \mathrm{x}(t; x_0)$ for some t and some $s > 0$. Let $\mathrm{x}(t; x_0)$ be a closed trajectory of this system, and suppose that $V : \mathbb{R}^n \to \mathbb{R}$ is a C^1 Lyapunov function for this system on the entire state space $\mathbb{R}^n$. Prove that $\dot{V}(\mathrm{x}(t; x_0)) = 0$ for all $t \geq 0$.

B.17 In this exercise we look at variations of the system (B.8) from Example B.4.6. We investigate the system

$$\begin{aligned}\dot{\mathrm{x}}_1(t) &= \mathrm{x}_2(t) + \mathrm{x}_1(t)\left(\gamma - \mathrm{x}_1^2(t) - \mathrm{x}_2^2(t)\right),\\ \dot{\mathrm{x}}_2(t) &= -\mathrm{x}_1(t) + \mathrm{x}_2(t)\left(\gamma - \mathrm{x}_1^2(t) - \mathrm{x}_2^2(t)\right),\end{aligned}$$

with $\gamma \in \mathbb{R}$. Prove that the origin is an asymptotically stable equilibrium point if $\gamma \leq 0$, and that it is an unstable equilibrium point if $\gamma > 0$.

B.18 (Assumes knowledge of Appendix A.1.) Determine all $\alpha, \beta \in \mathbb{R}$ for which

$$\begin{bmatrix} \alpha & 0 & 0 \\ 0 & 1 & \beta \\ 0 & \beta & 4 \end{bmatrix}$$

(a) is positive definite.

(b) is positive semi-definite but not positive definite.

B.19 (Assumes knowledge of Appendix A.1.) Let the matrices A and Q be given by

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, \qquad Q = \begin{bmatrix} 4 & 6 \\ 6 & 10 \end{bmatrix},$$

(a) Determine a matrix P such that $A^{\mathrm{T}}P + PA = -Q$.

(b) Show that P and Q are positive definite and conclude that $\dot{\mathrm{x}}(t) = Ax(t)$ is asymptotically stable.

B.20 (Assumes knowledge of Appendix A.1.) Consider the matrix

$$A = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 1 & -2 \end{bmatrix}.$$

(a) Use a computer to determine the solution P of the Lyapunov equation

$$A^{\mathrm{T}}P + PA = -I.$$

(b) Check (without using a computer) that this solution P is positive definite.

B.21 Consider the linear differential equation

$$\begin{aligned} \dot{\mathrm{x}}_1(t) &= \mathrm{x}_1(t) + 2\mathrm{x}_2(t), \\ \dot{\mathrm{x}}_2(t) &= -\alpha \mathrm{x}_1(t) + (1-\alpha)\mathrm{x}_2(t). \end{aligned}$$

Determine all α's for which this differential equation is asymptotically stable around $\bar{x} = (0,0)$.

B.22 The *blue* phase portrait of Fig. B.14 is that of $\dot{\mathrm{x}}(t) = A\mathrm{x}(t)$ with

$$A = \begin{bmatrix} -1 & -\frac{\pi/2}{3} \\ 3\pi/2 & -1 \end{bmatrix}.$$

(a) Determine a diagonal positive definite matrix P of the form $P = \left[\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right]$ for which also $A^{\mathrm{T}}P + PA$ is diagonal.

(b) Show that $x^{\mathrm{T}}Px$ is a strong Lyapunov function for this system (with equilibrium $\bar{x} = 0$).

(c) Sketch in Fig. B.14 a couple of level sets $\{x \in \mathbb{R}^2 \mid x^{\mathrm{T}}Px = c\}$, and explain from this figure why indeed $\dot{V}(x) < 0$ for all nonzero x.

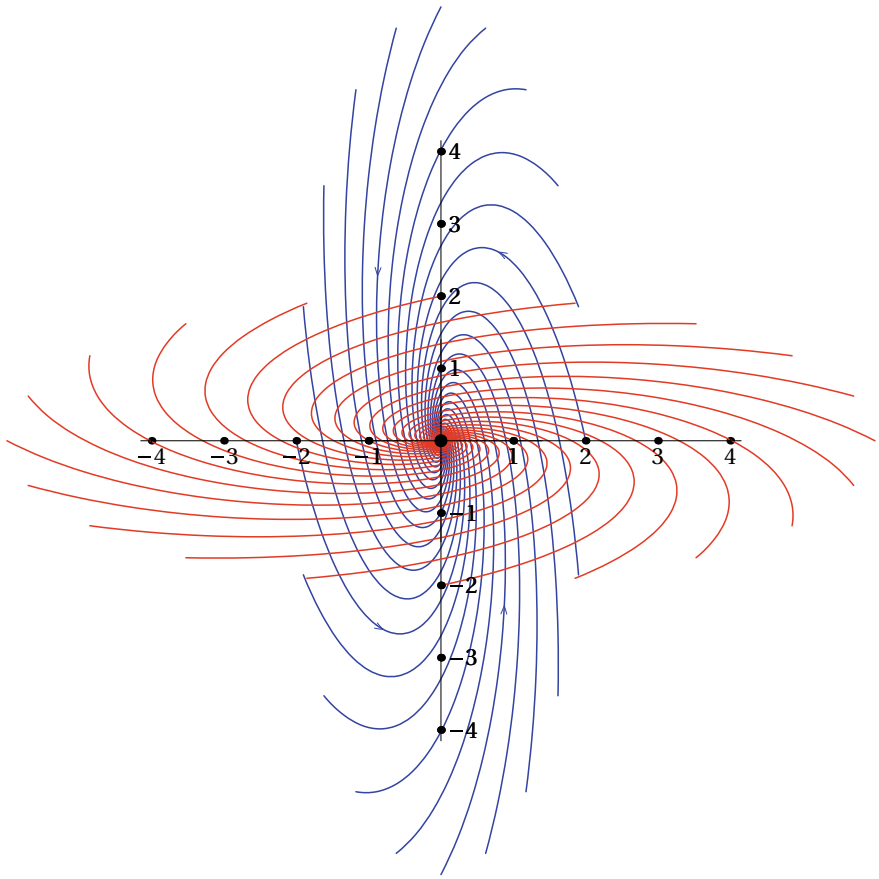


FIGURE B.14: The blue phase portrait is that of the system of Exercise B.22. This is also the phase portrait of the system $\dot{\mathbf{x}}(t) = A_{\text{even}}\mathbf{x}(t)$ of Exercise B.23. In red is the phase portrait of $\dot{\mathbf{x}}(t) = A_{\text{odd}}\mathbf{x}(t)$ of Exercise B.23. All trajectories (blue and red) converge to zero as $t \to \infty$.

B.23 Notice that the results we derived in this chapter are formulated only for *time-invariant* systems $\dot{x}(t) = f(x(t))$. For *time-varying* systems $\dot{x}(t) = f(x(t), t)$ the story is quite different, even if the system is linear of the form

$$\dot{x}(t) = A(t)x(t). \tag{B.29}$$

For such linear systems one might be tempted to conclude that it is asymptotically stable if for every t all eigenvalues of $A(t)$ have negative real part. In this exercise we will see that this is wrong. Consider the system (B.29) where

$$A(t) = \begin{cases} A_{\text{even}} & \text{if } \lfloor t \rfloor \text{ is even} \\ A_{\text{odd}} & \text{if } \lfloor t \rfloor \text{ is odd} \end{cases} \tag{B.30}$$

in which

$$A_{\text{even}} = \begin{bmatrix} -1 & -\frac{\pi/2}{3} \\ 3\pi/2 & -1 \end{bmatrix}, \qquad A_{\text{odd}} = \begin{bmatrix} -1 & -3\pi/2 \\ \frac{1}{3}\pi/2 & -1 \end{bmatrix}.$$

Here $\lfloor t \rfloor$ denotes the *floor* of t (the largest integer less than are equal to t). The system hence switches dynamics at every $t \in \mathbb{Z}$.

(a) Show that the eigenvalues of $A(t)$ at every t are $-1 \pm i\pi/2$.

In particular all eigenvalues of $A(t)$ have negative real part at every t. At this point it is interesting to have a look at the phase portraits of $\dot{x}(t) = A_{\text{even}}x(t)$ and $\dot{x}(t) = A_{\text{odd}}x(t)$, see Fig. B.14. The blue phase portrait is that of $\dot{x}(t) = A_{\text{even}}x(t)$, and the red phase portrait is that of $\dot{x}(t) = A_{\text{odd}}x(t)$. It can be shown that

$$\begin{aligned} e^{A_{\text{even}}t} &= e^{-t}\begin{bmatrix} \cos(\frac{\pi}{2}t) & -\frac{1}{3}\sin(\frac{\pi}{2}t) \\ 3\sin(\frac{\pi}{2}t) & \cos(\frac{\pi}{2}t) \end{bmatrix}, \\ e^{A_{\text{odd}}(t-1)} &= e^{-t}\begin{bmatrix} \cos(\frac{\pi}{2}t) & -3\sin(\frac{\pi}{2}t) \\ \frac{1}{3}\sin(\frac{\pi}{2}t) & \cos(\frac{\pi}{2}t) \end{bmatrix}. \end{aligned}$$

(b) Verify that $x(t) = e^{A_{\text{even}}t}x(0)$ for all $t \in [0,1]$, and $x(t) = e^{A_{\text{odd}}(t-1)}x(1)$ for all $t \in [1,2]$.

(c) Show that

$$x(2k+2) = \begin{bmatrix} -(3/e)^2 & 0 \\ 0 & -1/(3e)^2 \end{bmatrix} x(2k)$$

for all $k \in \mathbb{Z}$, and use it to conclude that the time-varying system (B.29) is not asymptotically stable.

(d) Use the above to sketch in Fig. B.14 the trajectory $x(t)$ for $t > 0$ with initial condition $x(0) = \left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right]$, and argue that this trajectory diverges as $t \to \infty$.

Remark: another "counterexample" can be described in words as follows: consider a mass attached to a spring where the positive spring constant k depends on time t. If the function $k(t)$ is constructed in such a way that $k(t)$ is small whenever the mass is far from the rest position but large whenever the mass passes through the rest position, then it is physically clear that the time-varying mass-spring system is unstable even in the presence of small damping. On the other hand, the eigenvalues at each time t have real part strictly less than zero.

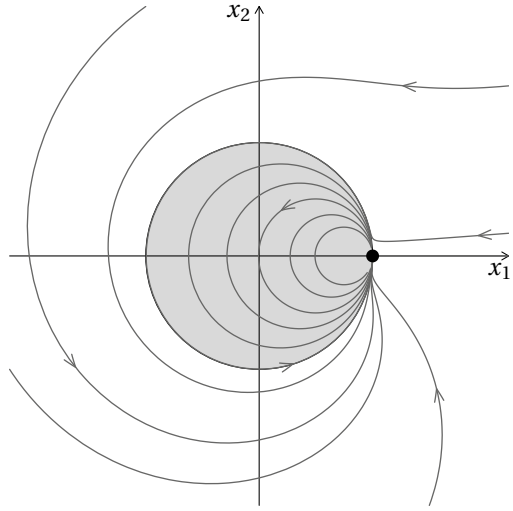


FIGURE B.15: Stable or not? Globally attractive or not? See Exercise B.7.

B.24 This exercise is based on an example from a paper by Ryan and Sontag[3]. It is about a system whose equilibrium is globally attractive yet not stable! Consider the system $\dot{x}(t) = f(x(t))$ with

$$f(x) = \begin{cases} \begin{bmatrix} -x_1(1-\frac{1}{\|x\|}) - 2x_2(1-\frac{x_1}{\|x\|}) \\ -x_2(1-\frac{1}{\|x\|}) + 2x_1(1-\frac{x_1}{\|x\|}) \end{bmatrix} & \text{if } \|x\| \geq 1, \\ \begin{bmatrix} 2(x_1-1)x_2 \\ -(x_1-1)^2 + x_2^2 \end{bmatrix} & \text{if } \|x\| < 1. \end{cases}$$

Notice that f inside the unit disc is defined differently than outside the unit disc. Nevertheless, $f(x)$ is locally Lipschitz continuous, also on the unit circle. Inside the unit circle, the orbits are arcs (parts of circles) that converge to $\bar{x} = (1,0)$, see Fig. B.15. Outside, $\|x\| \geq 1$, the system is easier to comprehend in polar coordinates $(x_1, x_2) = (r\cos(\theta), r\sin(\theta))$ with $r =$

[3] E.P. Ryan and E.D. Sontag. Well-defined steady-state response does not imply CICS. *System and Control Letters*, 55: 707–710, 2006.

$\sqrt{x_1^2 + x_2^2}$. This gives

$$\begin{aligned} \dot{\mathtt{r}}(t) &= 1 - \mathtt{r}(t), \\ \dot{\theta}(t) &= 4\sin^2(\theta(t)/2) = 2(1 - \cos(\theta(t))). \end{aligned} \tag{B.31}$$

(a) Derive (B.31).

(b) Show that $\bar{x} := (1,0)$ is the unique point of equilibrium.

(c) Argue that for $\|\mathtt{x}(0)\| > 1$ its phase portrait is as in Fig. B.15.

(d) Argue that the equilibrium $\bar{x}$ is globally attractive but not stable.

B.25 Let $A \in \mathbb{R}^{n\times n}$ and suppose that $A + A^{\mathrm{T}}$ is negative definite. Is the origin a stable equilibrium of $\dot{\mathtt{x}}(t) = A\mathtt{x}(t)$?

Solutions to Odd-Numbered Exercises

To avoid clutter in this appendix we often write $\mathrm{x}(t)$ simply as x.

Chapter 1

1.1 (a) $0 = \left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\right)(\dot{\mathrm{x}}^2 - \alpha \mathrm{x}^2) = -2\alpha^2 \mathrm{x} - \frac{\mathrm{d}}{\mathrm{d}t}(2\dot{\mathrm{x}}) = -2\alpha^2 \mathrm{x} - 2\ddot{\mathrm{x}}$. So $\ddot{\mathrm{x}} + \alpha^2 \mathrm{x} = 0$. Its solution (using characteristic polynomials) is $\mathrm{x}(t) = c\mathrm{e}^{\mathrm{i}\alpha t} + d\mathrm{e}^{-\mathrm{i}\alpha t}$ with c, d arbitrary constants. Equivalently, $\mathrm{x}(t) = a\cos(\alpha t) + b\sin(\alpha t)$ with a, b arbitrary constants.

(b) $0 = 2 - \frac{\mathrm{d}}{\mathrm{d}t}(2\dot{\mathrm{x}}) = 2 - 2\ddot{\mathrm{x}}$. So $\mathrm{x}(t) = \frac{1}{2}t^2 + at + b$.

(c) $0 = -\frac{\mathrm{d}}{\mathrm{d}t}(2\dot{\mathrm{x}} + 4t) = -(2\ddot{\mathrm{x}} + 4)$. So $\mathrm{x}(t) = -t^2 + at + b$.

(d) $0 = \dot{\mathrm{x}} + 2\mathrm{x} - \frac{\mathrm{d}}{\mathrm{d}t}(2\dot{\mathrm{x}} + \mathrm{x}) = 2\mathrm{x} - 2\ddot{\mathrm{x}}$. So $\mathrm{x}(t) = a\mathrm{e}^t + b\mathrm{e}^{-t}$.

(e) $0 = \left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\right)(\mathrm{x}^2 + 2t\mathrm{x}\dot{\mathrm{x}}) = 2\mathrm{x} + 2t\dot{\mathrm{x}} - \frac{\mathrm{d}}{\mathrm{d}t}(2t\mathrm{x}) = 2\mathrm{x} + 2t\dot{\mathrm{x}} - (2\mathrm{x} + 2t\dot{\mathrm{x}}) = 0$. The Euler-Lagrange equation hence is $0 = 0$. Every function x satisfies this, so every function is stationary. (Incidentally, it is not too hard to show that $J(\mathrm{x})$ equals $T\mathrm{x}^2(T)$, so J is constant if we specify the endpoint $\mathrm{x}(T) = x_T$. This explains why all x are stationary.)

1.3 (a) $0 = \frac{\partial}{\partial x}\left(\frac{\mathrm{d}}{\mathrm{d}t}G(t, \mathrm{x}(t))\right) - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial}{\partial \dot{x}}\left(\frac{\mathrm{d}}{\mathrm{d}t}G(t, \mathrm{x}(t))\right)\right)$.

(b) $\frac{\mathrm{d}}{\mathrm{d}t}G(t, \mathrm{x}(t))$ equals $\frac{\partial G(t, \mathrm{x}(t))}{\partial t} + \dot{\mathrm{x}}^{\mathrm{T}}(t)\frac{\partial G(t, \mathrm{x}(t))}{\partial x}$. So the Euler-Lagrange equation becomes

$$\begin{aligned} 0 &= \frac{\partial}{\partial x}\left(\frac{\mathrm{d}}{\mathrm{d}t}G(t, \mathrm{x}(t))\right) - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\left(\frac{\partial}{\partial t}G(t, \mathrm{x}(t)) + \dot{\mathrm{x}}^{\mathrm{T}}(t)\frac{\partial}{\partial x}G(t, \mathrm{x}(t))\right) \\ &= \frac{\partial}{\partial x}\left(\frac{\mathrm{d}}{\mathrm{d}t}G(t, \mathrm{x}(t))\right) - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial}{\partial x}G(t, \mathrm{x}(t))\right) \\ &= 0. \end{aligned}$$

This holds for all x. In the last equality we used that $\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial G(t, \mathrm{x}(t))}{\partial x}\right)$ equals $\frac{\partial}{\partial x}\left(\frac{\mathrm{d}}{\mathrm{d}t}G(t, \mathrm{x}(t))\right)$.

(c) $\int_0^T F(t, \mathrm{x}(t), \dot{\mathrm{x}}(t))\,\mathrm{d}t = \int_0^T \frac{\mathrm{d}}{\mathrm{d}t}(G(t, \mathrm{x}(t))\,\mathrm{d}t = G(T, x_T) - G(0, x_0)$ so the outcome is the same for all functions x that satisfy the given boundary conditions.

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G. Meinsma and A. van der Schaft, *A Course on Optimal Control*, Springer Undergraduate Texts in Mathematics and Technology,
https://doi.org/10.1007/978-3-031-36655-0

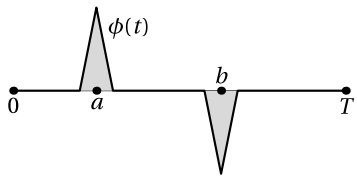
1.5 The constant $4\pi\rho v^2$ plays no role. So take $F(y,\dot y) = y\dot y^3$. Beltrami gives

$$C = \mathrm{y}\dot{\mathrm{y}}^3 - \dot{\mathrm{y}}\frac{\partial}{\partial \dot y}(\mathrm{y}\dot{\mathrm{y}}^3) = \mathrm{y}\dot{\mathrm{y}}^3 - 3\mathrm{y}\dot{\mathrm{y}}^3 = -2\mathrm{y}\dot{\mathrm{y}}^3.$$

Hence $\mathrm{y}\dot{\mathrm{y}}^3$ is constant. Now $\mathrm{y}(x) := y_1\,(x/x_1)^{3/4}$ satisfies this equation (verify this) and, in addition, then $\mathrm{y}(0) = 0$ and $\mathrm{y}(x_1) = y_1$ as required.

(By the way, the function $\mathrm{y}(x) = y_1(x/x_1)^{3/4}$ is not differentiable at $x = 0$.)

1.7 (a) By contradiction. If f is not constant then $f(a) \neq f(b)$ for some $a, b \in (0,T)$. Let ϕ be a continuous function with $\phi(a) = +1$ and $\phi(b) = -1$ of the form



This function satisfies $\int_0^T \phi(t)\,\mathrm{d}t = 0$ and for small enough "tents" around a, b the integral $\int_0^T f(t)\phi(t)\,\mathrm{d}t$ is nonzero because $f(t)$ is continuous and $f(a) \neq f(b)$. Contradiction. Hence f is constant.

(b) Momentarily denote $\frac{\partial F(t,\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))}{\partial x}$ simply as $F_x(t)$, and let $G_x(t)$ be an antiderivative of $F_x(t)$. Now $\int_0^T F_x^{\mathrm{T}}(t)\delta_x(t)\,\mathrm{d}t$ equals $[G_x^{\mathrm{T}}(t)\delta_x(t)]_0^T - \int_0^T G_x^{\mathrm{T}}(t)\dot\delta_x(t)\,\mathrm{d}t$. The term $[G_x^{\mathrm{T}}(t)\delta_x(t)]_0^T$ is zero because $\delta_x(0) = \delta_x(T) = 0$. Therefore (1.63) is the same as

$$\int_0^T \big(-G_x^{\mathrm{T}}(t) + \frac{\partial F(t,\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))}{\partial \dot x^{\mathrm{T}}}\big)\dot\delta_x(t)\,\mathrm{d}t.$$

Clearly, one possible antiderivative is $G_x(t) = \int_0^t \frac{\partial F(\tau,\mathrm{x}_*(\tau),\dot{\mathrm{x}}_*(\tau))}{\partial x}\,\mathrm{d}\tau$.

(c) Follows immediately from (a) and (b) [take $\phi = \dot\delta_x$ and realize that this ϕ is continuous and $\int_0^T \phi(t)\,\mathrm{d}t = [\delta_x(t)]_0^T = 0 - 0 = 0$.]

(d) From (c) and continuity of $\frac{\partial F(t,\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))}{\partial x}$ it follows that $\frac{\partial F(t,\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))}{\partial \dot x}$ is differentiable w.r.t. t and that $\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial F(t,\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))}{\partial \dot x} = -F(t,\mathrm{x}_*(t),\dot{\mathrm{x}}_*(t))$.

(e) Follows from the solution of (d). (Since F and x_* are C^1 it is guaranteed that the derivative exists and is continuous.)

1.9 (a)

$$\begin{aligned} 0 &= \left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot x}\right)(\dot{\mathrm{x}}^2 - 2\mathrm{x}\dot{\mathrm{x}} - \dot{\mathrm{x}}) \\ &= -2\dot{\mathrm{x}} - \frac{\mathrm{d}}{\mathrm{d}t}(2\dot{\mathrm{x}} - 2\mathrm{x} - 1) \\ &= -2\dot{\mathrm{x}} - (2\ddot{\mathrm{x}} - 2\dot{\mathrm{x}}) = -2\ddot{\mathrm{x}}. \end{aligned}$$

Thus $\mathrm{x}(t) = at + \mathrm{x}(0) = at + 1$. According to (1.43) we need that $2\dot{\mathrm{x}} - 2\mathrm{x} - 1 = 0$ at $T = 1$. For our $\mathrm{x}(t) = at + 1$ this means $2a - 2(a+1) - 1 = 0$, i.e., $-3 = 0$. Impossible.

(b) Follows from the previous part.

(c) For $\mathrm{x}(t) = at + 1$ we get $J(\mathrm{x}) = \int_0^1 a^2 - 2(at+1)a - a\,\mathrm{d}t = [a^2 t - (at+1)^2 - at]_0^1 = a^2 - (a+1)^2 - a + 1 = -3a$. This cost is unbounded from below (as a function of a), so does not have a minimum (not even a local minimum).

1.11 In Example 1.6.8 we saw that the 2nd derivative of $\sqrt{1+\dot{y}^2}$ w.r.t. $\dot{y}$ is $1/(1+\dot{y}^2)^{3/2}$. So here the Legendre condition is that $2\pi \mathrm{r}(x)/(1+\dot{\mathrm{r}}^2(x))^{3/2}$ has to be nonnegative for all x. That is the case because the hyperbolic cosine solution $\mathrm{r}_a(x)$ is ≥ 0.

1.13

$$\frac{\partial^2 F(t, \left[\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}\right], \left[\begin{smallmatrix} \dot{x}_1 \\ \dot{x}_2 \end{smallmatrix}\right])}{\partial \left[\begin{smallmatrix} \dot{x}_1 \\ \dot{x}_2 \end{smallmatrix}\right] \partial \left[\begin{smallmatrix} \dot{x}_1 \\ \dot{x}_2 \end{smallmatrix}\right]^{\mathrm{T}}} = \begin{bmatrix} \frac{\partial^2 F(\ldots)}{\partial \dot{x}_1^2} & \frac{\partial^2 F(\ldots)}{\partial \dot{x}_1 \partial \dot{x}_2} \\ \frac{\partial^2 F(\ldots)}{\partial \dot{x}_2 \partial \dot{x}_1} & \frac{\partial^2 F(\ldots)}{\partial \dot{x}_2^2} \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} > 0.$$

Hence the Legendre condition is satisfied.

1.15 (a) $0 = \left(\frac{\partial}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial \dot{x}}\right)(\dot{\mathrm{x}}^2 - \mathrm{x}) = -1 - \frac{\mathrm{d}}{\mathrm{d}t}(2\dot{\mathrm{x}}) = -1 - 2\ddot{\mathrm{x}}$.

(b) So $\mathrm{x}_*(t) = -t^2/4 + at + b$. Given $\mathrm{x}(0) = 0, \mathrm{x}(1) = 1$ it follows that $\mathrm{x}_*(t) = -t^2/4 + 5/4\,t$.

(c) $\frac{\partial^2 F(t,x,\dot{x})}{\partial \dot{x}^2} = 2 \geq 0$ so yes.

(d) The Hessian $H(t, x, y)$ is

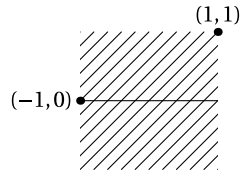
$$H(t, x, y) = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}.$$

It is positive semi-definite, so the condition is satisfied.

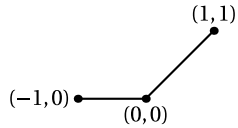
(e) Since $H \geq 0$ our x_* is globally optimal.

1.17 (a) Because $(1 - \dot{\mathrm{x}}(t))^2 \mathrm{x}^2(t) \geq 0$ for every x and t.

(b) $J(\mathrm{x}) = 0$ only if at every moment in time $\mathrm{x}(t) = 0$ or $\dot{\mathrm{x}}(t) = 1$. So either $\mathrm{x}(t) = 0$ are any of these:



Among these there is only one continuous function for which $\mathrm{x}(-1) = 0$ and $\mathrm{x}(1) = 1$:



(c) (Idea suffices.) Among the continuous functions no C^1 function is optimal because the unique optimal solution is not C^1. (One can also understand this question as: *among the C^1 functions* is there an optimal one? The answer is no: take a sequence of C^1 functions x_n that "converge" to the optimal continuous function. Then $\lim_{n\to\infty} J(\mathrm{x}_n) = 0$. So $\inf_{\mathrm{x}\,\mathrm{is}\,C^1} J(\mathrm{x}) = 0$ while no C^1 function x achieves $J(\mathrm{x}) = 0$.)

1.19 (a) The function $F + \mu M$ as used in Theorem 1.7.1 becomes

$$F(x, y, \dot{y}) + \mu M(x, y, \dot{y}) = (\rho g y + \mu)\sqrt{1 + \dot{y}^2}.$$

Since it does not depend on x we can apply Beltrami. This gives $\mathrm{y}(x) + \mu/(\rho g) = a\sqrt{1 + \dot{\mathrm{y}}^2(x)}$ for some integration constant a.

(b) The normal case is given in (a). For the abnormal case: Euler-Lagrange on $\sqrt{1 + \dot{\mathrm{y}}^2}$ gives $\ddot{\mathrm{y}}(x) = 0$ (see Example 1.2.5). Hence a straight line.

(c) We have the normal case if ℓ exceeds the distance between $(x_0, \mathrm{y}(x_0))$ and $(x_1, \mathrm{y}(x_1))$. The abnormal case if ℓ equals this distance (and if ℓ is less than this distance then no solution exists).

1.21 The function that *minimizes* $\int_0^1 \dot{\mathrm{x}}^2(t)\,\mathrm{d}t$ satisfies the Euler-Lagrange equation: $0 = -2\ddot{\mathrm{x}}(t)$, so is a function of the form $\mathrm{x}(t) = bt + c$. Given the boundary conditions $(\mathrm{x}(0) = 0, \mathrm{x}(1) = 1)$ this gives $b = 1, c = 0$, and, so, $\mathrm{x}(t) = t$. The Hessian is positive semi-definite. Therefore $\mathrm{x}_*(t) = t$ is an optimal solution, and, consequently, the minimal C is $C = \int_0^1 \dot{\mathrm{x}}_*^2(t)\,\mathrm{d}t = \int_0^1 1\,\mathrm{d}t = 1$.

Chapter 2

2.1 (a) $H(x, p, u) = pxu + x^2 + u^2$ and, hence, $\dot{\mathrm{p}} = -\mathrm{p}\mathrm{u} - 2\mathrm{x}, \mathrm{p}(T) = 2$.

(b) The u at any moment in time minimizes the Hamiltonian. Since the Hamiltonian is a convex parabola in u, the minimizing u follows from $0 = \frac{\partial H}{\partial u} = px + 2u$, i.e., $\mathrm{u}_* = -(\mathrm{p}\mathrm{x})/2$.

(c) The final condition on $\mathtt{p}$ is $\mathtt{p}(T) = 2$. Also, for $\mathtt{u}_* = -\mathtt{p}\mathtt{x}/2$ we have

$$H(\mathtt{x}, \mathtt{p}, \mathtt{u}_*) = \mathtt{p}\mathtt{x}\mathtt{u}_* + \mathtt{x}^2 + \mathtt{u}_*^2 = -\mathtt{p}\mathtt{x}(\mathtt{p}\mathtt{x}/2) + \mathtt{x}^2 + (\mathtt{p}\mathtt{x}/2)^2 = \mathtt{x}^2(1 - \mathtt{p}^2/4).$$

So $H(\mathtt{x}_*(T), \mathtt{p}_*(T), \mathtt{u}_*(T))$ is zero.

(d) For $\mathtt{u}_* = -\mathtt{p}\mathtt{x}/2$ the costate equation becomes $\dot{\mathtt{p}} = (\frac{1}{2}\mathtt{p}^2 - 2)\mathtt{x}$ with final condition $\mathtt{p}(T) = 2$. Clearly the constant $\mathtt{p}_*(t) = 2$ satisfies the final condition, and also the DE because then $\dot{\mathtt{p}}_* = 0 = (\frac{1}{2}\mathtt{p}_*^2 - 2)$.

(e) For $\mathtt{p} = 2$ (constant) we have $\mathtt{u}_* = -\mathtt{x}$ so $\dot{\mathtt{x}} = -\mathtt{x}^2$. See Example B.1.5: $\mathtt{x}(t) = 1/(t+1)$.

2.3 The Hamiltonian is $H = p(x+u) + \frac{1}{2}u^2 - 2u - 2x$. If u would have been free to choose, then the minimizing u would have been the one that achieves $0 = \partial H/\partial u = p + u - 2$. This gives $u = 2 - p$. Given that $\mathbb{U} = [0,4]$ and the fact that H is a convex parabola in u it is easy to see that the minimizing u is the element of $[0,4]$ that is closest to $\hat{u} := 2 - p$. The costate equation is $\dot{\mathtt{p}} = 2 - \mathtt{p}, \mathtt{p}(1) = 0$. Therefore $\mathtt{p}(t) = 2(1 - \mathrm{e}^{1-t})$. Thus $\hat{\mathtt{u}}(t) := 2 - \mathtt{p}(t) = 2\,\mathrm{e}^{1-t}$. We need $\mathtt{u}(t) \in [0,4]$. Now $\hat{\mathtt{u}}(t) = 4$ at $t = 1 - \ln(2) \approx 0.3069$, and $\hat{\mathtt{u}}(t) > 4$ if $t < 1 - \ln(2)$, and $\hat{\mathtt{u}}(t) \in [0,4]$ if $t \geq 1 - \ln(2)$. Therefore

$$\mathtt{u}_*(t) = \begin{cases} 4 & \text{if } t \in [0, 1 - \ln(2)) \\ 2\,\mathrm{e}^{1-t} & \text{if } t \in [1 - \ln(2), 1] \end{cases}.$$

The optimal control is continuous but not differentiable.

2.5 The Hamiltonian is $p(x+u) + \frac{1}{4}u^4$. So $\dot{\mathtt{p}} = -\mathtt{p}$ (without final condition), and the u minimizes iff $u^3 = -p$, so $\mathtt{u}_* = -\mathtt{p}^{1/3}$. The costate has the form $\mathtt{p}(t) = c\,\mathrm{e}^{-t}$. Hence $\mathtt{u}_*$ has the form $d\,\mathrm{e}^{-t/3}$. From $\dot{\mathtt{x}}(t) = \mathtt{x}(t) + d\,\mathrm{e}^{-t/3}$ it follows that $\mathtt{x}(t) = \alpha\,\mathrm{e}^t + \beta\,\mathrm{e}^{-t/3}$ for certain α, β. (The c, d, α, β are related but let us worry about that later.) The initial and final conditions become $1 = \mathtt{x}(0) = \alpha + \beta$ and $0 = \mathtt{x}(3) = \alpha\,\mathrm{e}^3 + \beta\,\mathrm{e}^{-1}$. So $\alpha = -\mathrm{e}^{-4}\beta$ and $1 = \beta(1 - \mathrm{e}^{-4})$. That is, $\beta = 1/(1 - \mathrm{e}^{-4})$ and $\alpha = -\mathrm{e}^{-4}/(1 - \mathrm{e}^{-4})$. Now that α, β are known the $\mathtt{x}_* = \alpha\,\mathrm{e}^t + \beta\,\mathrm{e}^{-t/3}$ follows, and also $\mathtt{u}_* = \dot{\mathtt{x}}_* - \mathtt{x}_* = -\frac{4}{3}\beta\,\mathrm{e}^{-t/3}$.

2.7 The cost to be minimized is $J(x_0, \mathtt{u}) = -\mathtt{x}_1(T)$, so $K(x) = -x_1$ and $L(x, \mathtt{u}) = 0$.

(a) $H(x, p, u) = p^{\mathrm{T}} f(x, u) + L(x, u) = p_1 x_2 + p_2 u$.

(b) $\dot{\mathtt{p}}_1(t) = 0, \dot{\mathtt{p}}_2(t) = -\mathtt{p}_1(t)$. Since $\mathtt{x}_2(t)$ has a final condition, the corresponding final condition on $\mathtt{p}_2(t)$ is absent. We just have $\mathtt{p}_1(T) = \partial K(\mathtt{x}(T))/\partial x_1 = -1$. The Hamiltonian equations in x are simply the given equations: $\dot{\mathtt{x}}_1(t) = \mathtt{x}_2(t), \dot{\mathtt{x}}_2(t) = \mathtt{u}(t)$ and the given initial and final condition.

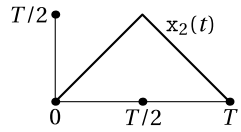
(c) $\mathtt{p}_1(t) = \mathtt{p}_1(T) = -1$. Then $\dot{\mathtt{p}}_2(t) = +1$, so $\mathtt{p}_2(t) = t + c$.

(d) Since $\mathtt{u}(t)$ minimizes the Hamiltonian we have $\mathtt{u}(t) = -\operatorname{sgn}(\mathtt{p}_2(t))$. Since $\mathtt{p}_2(t)$ is an increasing function, our $\mathtt{u}(t)$ switches sign at most once, from $\mathtt{u}(t) = 1$ to $\mathtt{u}(t) = -1$. As $\int_0^T \mathtt{u}(t)\,\mathrm{d}t = x_2(T) = 0$ it must be that $\mathtt{u}(t)$ switches sign half-way, at $t = T/2$:

$$\mathtt{u}(t) = \begin{cases} +1 & t < T/2, \\ -1 & t > T/2. \end{cases}$$

(By the way, this means that the constant c in $\mathtt{p}_2(t) = t + c$ is $c = -T/2$). Then the speed $\mathtt{x}_2(t)$ is this "tent" function

$$\mathtt{x}_2(t) = \int_0^t \mathtt{u}(\tau)\,\mathrm{d}\tau =$$



and the traveled distance $\mathtt{x}_1(T)$ is the area under the "tent": $\mathtt{x}_1(T) = T^2/4$.

By the way, we may also choose $J(x_0, \mathtt{u}) = \int_0^T -\mathtt{x}_2(t)\,\mathrm{d}t$ because that, too, equals $-\mathtt{x}_1(T)$. That choice works fine as well.

2.9 (a) $H(x, p, u) = p(-x + u) + (x - u)^2$, so $\dot{\mathtt{p}} = \mathtt{p} - 2(\mathtt{x} - \mathtt{u})$ (and no final condition on $\mathtt{p}$). The u that minimizes the Hamiltonian satisfies $0 = \frac{\partial H}{\partial u} = p - 2(x - u)$. So $\mathtt{p}_* = 2(\mathtt{x}_* - \mathtt{u}_*)$. Inserting this into the costate equations gives $\dot{\mathtt{p}}_* = 0$, hence the costate is constant, $\mathtt{p}_* = p_*$. The system equation then becomes $\dot{\mathtt{x}} = -\frac{1}{2}p_*$ indicating that $\mathtt{x}$ grows linearly with slope $-\frac{1}{2}p_*$. Given $\mathtt{x}(0) = 0, \mathtt{x}(T) = 1$ it follows that $\mathtt{x}(t) = t/T$. Hence $\mathtt{p}_*(t) = p_* = -2/T$ and $\mathtt{u}_*(t) = (t + 1)/T$.

(b) Yes because the Hessian of H w.r.t. x, u is $\left[\begin{smallmatrix} 2 & -2 \\ -2 & 2 \end{smallmatrix}\right] \geq 0$, so H is convex in x, u. (See Appendix A.7.) Also $\mathbb{U} := \mathbb{R}$ is a convex set.

(c) $\mathtt{x}_* - \mathtt{u}_* = -p_*/2 = 1/T$ so $\int_0^T (\mathtt{x} - \mathtt{u})^2\,\mathrm{d}t = 1/T$. The longer it takes the "cheaper" it is. This is to be expected because if for some final time $T_1 < T$ it would be cheaper then taking $\mathtt{u} = \mathtt{x}$ for $t > T_1$ would achieve this cheaper (lower) cost also for final time T. (Notice that taking $\mathtt{u} = \mathtt{x}$ "costs nothing".)

2.11 (a) $H(p, x, u) = px(1 - u) - \ln(xu)$.

(b) $\dot{\mathtt{x}} = \mathtt{x}(1 - \mathtt{u})$, $\mathtt{x}(0) = 1, \mathtt{x}(1) = \frac{1}{2}\mathrm{e}$, and $\dot{\mathtt{p}} = -\mathtt{p}(1 - \mathtt{u}) + 1/\mathtt{x}$. Setting $\partial H(x, p, u)/\partial u$ to zero gives $0 = -px - 1/u$.

(c) $0 = -px - 1/u$ gives $\mathtt{u}_* = -1/(\mathtt{p}_*\mathtt{x}_*)$.

(d) $\dot{\mathtt{p}} = -\mathtt{p}(1 + 1/(\mathtt{p}\mathtt{x})) + 1/\mathtt{x} = -\mathtt{p}$ so $\mathtt{p}(t) = c\,\mathrm{e}^{-t}$. Also, $\dot{\mathtt{x}} = \mathtt{x}(1 + 1/(\mathtt{p}\mathtt{x})) = \mathtt{x} + 1/\mathtt{p} = \mathtt{x} + c\,\mathrm{e}^t/c$. Hence $\mathtt{x}(t) = (t/c + d)\,\mathrm{e}^t$. The conditions $\mathtt{x}(0) = 1$ and $\mathtt{x}(1) = \frac{1}{2}\mathrm{e}$ determine $d = 1, c = -2$.

(e) Since $\mathtt{x}(0) > 0$ and $\dot{\mathtt{x}} \geq 0$ we have $\mathtt{x}(t) > 0$ for all $t > 0$. Now $\mathtt{u}_*(t) = -1/(\mathtt{p}(t)\mathtt{x}(t)) = 1/(2-t) > 0$. So yes.

2.13 (a) Once $\mathtt{x}(t) > 0$ it can only increase because $a > 0, \mathtt{u}(t) \geq 0$ so then $\dot{\mathtt{x}}(t) = a\mathtt{u}(t)\mathtt{x}(t) \geq 0$.

(b) Since $J(\mathtt{u}) = -\int_0^T \mathtt{x}_2(t)\,\mathrm{d}t$ we have $H = p_1 a u x_1 + p_2 a(1-u)x_1 - x_2$. The costate equations become

$$\dot{\mathtt{p}}_1 = -a\mathtt{u}\mathtt{p}_1 - a(1-\mathtt{u})\mathtt{p}_2, \qquad \mathtt{p}_1(T) = 0,$$
$$\dot{\mathtt{p}}_2 = 1, \qquad \mathtt{p}_2(T) = 0.$$

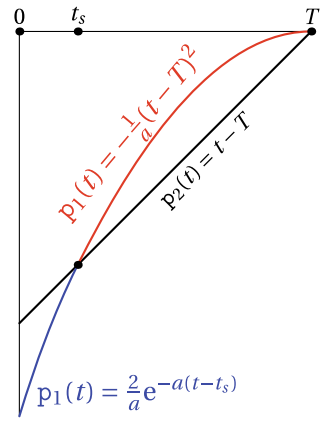
So $\mathtt{p}_2(t) = t - T$. Write H as $H = uax_1(p_1 - p_2) + \cdots$, thus the optimal u only depends on sign of $p_1 - p_2$. Now the clever trick: at the final time we have $\dot{\mathtt{p}}_1(T) = 0$, and since $\dot{\mathtt{p}}_2 = 1$ it follows that we have $\mathtt{p}_1(t) - \mathtt{p}_2(t) < 0$ near T. Therefore $\mathtt{u}(t) = 0$ near T. Now, as in Example 2.5.5, solve the equations backwards in time:

$$\mathtt{p}_1(t) = -\tfrac{1}{2}a(t-T)^2 \qquad \forall\, t \in [t_s, T].$$

Here t_s is the switching time (where $\mathtt{p}_1(t_s) = \mathtt{p}_2(t_s)$, i.e., $-\frac{1}{2}a(t-T)^2 = t - T$, so $t_s = T - 2/a$). We expect that $\mathtt{p}_1(t) < \mathtt{p}_2(t)$ for $t < t_s$ and so $\mathtt{u}(t) = 1$ for $t < t_s$. Again this requires some "ad hoc" argument: since at $t = t_s$ we have $\mathtt{p}_1(t) = \mathtt{p}_2(t)$, we see that $\dot{\mathtt{p}}_1(t_s) = -a\mathtt{p}_2(t_s) = 2 > 1$ (independent of $\mathtt{u}$). It shows that $\mathtt{p}_1$ increases faster than $\mathtt{p}_2$ around $t = t_s$, so for $t < t_s$ but close to t_s we have $\mathtt{p}_1(t) < \mathtt{p}_2(t)$. Then

$$\mathtt{u}_*(t) = \begin{cases} 1 & \text{if } t < T - 2/a \\ 0 & \text{if } t > T - 2/a \end{cases}.$$

For $T = 5, a = 0.5$ this gives $t_s = 1$ and the solution of $\mathtt{p}_1(t)$ and $\mathtt{p}_2(t)$ are as follows:



(The plot of $\mathtt{p}_1(t)$ is deceivingly smooth. We already saw that $\dot{\mathtt{p}}_1(t) \to 2$ both for $t \uparrow t_s$ and $t \downarrow t_s$. The behavior of $\mathtt{p}_1$ is, however, quite

different for $t < t_s$ and $t > t_s$. One is exponential in t, the other is polynomial in t.)

2.15 (We use $p_{1:n}$ to denote the first n entries of a vector p.)

(a) $f(z) = \left[\begin{smallmatrix} u \\ M(x,u) \end{smallmatrix}\right], L(z,u) = F(x,u), K(z) = \left[\begin{smallmatrix} 0 \\ c_0 \end{smallmatrix}\right]$,

(b)

$$H_\lambda(z,p,u) = p_{1:n}^{\mathrm{T}} u + p_{n+1} M(x,u) + \lambda F(x,u).$$

So

$$\dot{\mathrm{p}}_{n+1} = -\frac{\partial H}{\partial z_{n+1}} = 0.$$

(c) First note that the first n entries of the costate satisfy

$$\dot{\mathrm{p}}_{1:n} = -\frac{\partial H}{\partial x} = -\frac{\partial F}{\partial x} - p_{n+1}\frac{\partial M}{\partial x} \tag{B.32}$$

(if $\lambda = 1$). The optimal input satisfies $\frac{\partial H}{\partial u} = 0$. This means

$$\mathrm{p}_{1:n} + p_{n+1}\frac{\partial M}{\partial u} + \frac{\partial F}{\partial u} = 0. \tag{B.33}$$

Now (1.54) for $\mu_* := p_{n+1}$ (constant) is satisfied because

$$\begin{aligned}
&\frac{\partial(F + p_{n+1}M)}{\partial x} - \tfrac{\mathrm{d}}{\mathrm{d}t}\frac{\partial(F + p_{n+1}M)}{\partial u} \\
&= \frac{\partial F}{\partial x} + p_{n+1}\frac{\partial M}{\partial x} - \tfrac{\mathrm{d}}{\mathrm{d}t}(-\mathrm{p}_{1:n}) && \text{because of (B.33)} \\
&= \frac{\partial F}{\partial x} + p_{n+1}\frac{\partial M}{\partial x} - \left(\frac{\partial F}{\partial x} + p_{n+1}\frac{\partial M}{\partial x}\right) && \text{because of (B.32)} \\
&= 0.
\end{aligned}$$

(d) In the abnormal case we have $H_0 = p_{1:n}^{\mathrm{T}} u + p_{n+1} M(x,u)$. The costate equation then gives us $\dot{\mathrm{p}}_{1:n} = -\mathrm{p}_{n+1}\frac{\partial M}{\partial x}$, and the optimal u makes $\frac{\partial H}{\partial u}$ zero, so $p_{1:n} + p_{n+1}\frac{\partial M}{\partial u} = 0$. Then the abnormal equation (1.55) holds because $p_{n+1} \neq 0$ (which we were allowed to assume), and

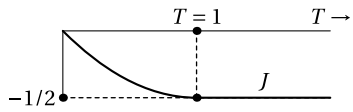
$$p_{n+1}\left(\frac{\partial M}{\partial x} - \tfrac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial M}{\partial u}\right)\right) = p_{n+1}\frac{\partial M}{\partial x} + \tfrac{\mathrm{d}}{\mathrm{d}t}\mathrm{p}_{1:n} = 0.$$

(Actually it can be shown that p_{n+1} is indeed nonzero, for if p_{n+1} would have been zero then the minimality property of the Hamiltonian would yield $\mathrm{p}_{1:n} = 0$, but Thm. 2.6.1 guarantees that a zero costate is impossible in the abnormal case, so $p_{n+1} \neq 0$.)

2.17 (a) $H(x,p,u) = pu + x$ so $\dot{\mathrm{p}} = -1, \mathrm{p}(T) = -1$. Therefore $\mathrm{p}(t) = T - 1 - t$. This costate swaps sign at $t = T-1$. Since u minimizes $pu + x$ we have $\mathrm{u}(t) = 0$ (minimal) for all $t < T-1$, and $\mathrm{u}(t) = 1$ for all $t > T-1$.

(b) $H = 0$ if $T \geq 1$, while $H = T - 1$ if $T < 1$.

(c) $J = -1/2$ if $T \geq 1$, while $J = T^2/2 - T$ if $T < 1$:



(d) Every $T \geq 1$ is optimal. It agrees with Thm. 2.7.1 because then $H = 0$.

2.19 (a) The same as the proof of Theorem 2.8.1 but with addition of the red parts:

$$
\begin{aligned}
&J(x_0, \mathrm{u}) - J(x_0, \mathrm{u}_*) \\
&= \int_0^T L(\mathrm{x}, \mathrm{u})\,\mathrm{d}t - L(\mathrm{x}_*, \mathrm{u}_*)\,\mathrm{d}t + K(\mathrm{x}(T)) - K(\mathrm{x}_*(T)) \\
&= \int_0^T (H(\mathrm{x}, \mathrm{p}_*, \mathrm{u}) - \mathrm{p}_*^{\mathrm{T}}\dot{\mathrm{x}}) - (H(\mathrm{x}_*, \mathrm{p}_*, \mathrm{u}_*) - \mathrm{p}_*^{\mathrm{T}}\dot{\mathrm{x}}_*)\,\mathrm{d}t + K(\mathrm{x}(T)) - K(\mathrm{x}_*(T)) \\
&= \int_0^T (H(\mathrm{x}, \mathrm{p}_*, \mathrm{u}) - H(\mathrm{x}_*, \mathrm{p}_*, \mathrm{u}_*)) - \mathrm{p}_*^{\mathrm{T}}(\dot{\mathrm{x}} - \dot{\mathrm{x}}_*)\,\mathrm{d}t + K(\mathrm{x}(T)) - K(\mathrm{x}_*(T)) \\
&\geq \int_0^T -\dot{\mathrm{p}}_*^{\mathrm{T}}(\mathrm{x} - \mathrm{x}_*) - \mathrm{p}_*^{T}(\dot{\mathrm{x}} - \dot{\mathrm{x}}_*)\,\mathrm{d}t + K(\mathrm{x}(T)) - K(\mathrm{x}_*(T)) \\
&= \left[-\mathrm{p}_*^{\mathrm{T}}(t)(\mathrm{x}(t) - \mathrm{x}_*(t))\right]_0^T + K(\mathrm{x}(T)) - K(\mathrm{x}_*(T)) \\
&\geq -\mathrm{p}_*^{\mathrm{T}}(T)(\mathrm{x}(T) - \mathrm{x}_*(T)) + \frac{\partial K(\mathrm{x}_*(T))}{\partial x^{\mathrm{T}}}(\mathrm{x}(T) - \mathrm{x}_*(T)) \qquad \text{(B.34)} \\
&= 0 \qquad \text{because } \mathrm{p}_*(T) = \frac{\partial K(\mathrm{x}_*(T))}{\partial x}.
\end{aligned}
$$

(Inequality (B.34) is because of convexity of K.)

(b) For every constrained state entry x_i we trivially have that (B.34) is zero (because $\mathrm{x}_i(T) = \mathrm{x}_{*i}(T)$ for every such entry).

2.21 Yes: $\mathbb{U} = \mathbb{R}$ is a convex set; $H(x, \mathrm{p}(t), u) = \mathrm{p}(t)u + x^2 + u^2$ is a convex parabola, so definitely convex in (x,u) [at every t]; $K(x) = 0$ so a convex function. Convexity of H can also be concluded from the fact that its Hessian w.r.t. (x,u) is positive (semi-)definite:

$$
\begin{bmatrix} \frac{\partial^2 H(x,\mathrm{p}(t),u)}{\partial x^2} & \frac{\partial^2 H(x,\mathrm{p}(t),u)}{\partial x \partial u} \\ \frac{\partial^2 H(x,\mathrm{p}(t),u)}{\partial u \partial x} & \frac{\partial^2 H(x,\mathrm{p}(t),u)}{\partial u^2} \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} > 0.
$$

Chapter 3

3.1 $K = -K_0, L = -L_0, J = -J_0$.

3.3 (a) For $V(x,t) = Q(x)$ the HJB equations become

$$0 + \min_{u\in\mathbb{R}}[Q'(x)xu + x^2 + u^2] = 0, \quad Q(x) = 2x.$$

So the final condition $V(x,T) = K(x) := 2x$ uniquely establishes $Q(x)$! Remarkable. For this $Q(x) = 2x$ the HJB equation simplifies to

$$0 + \min_{u\in\mathbb{R}}(2xu + x^2 + u^2) = 0.$$

It is indeed satisfied because $2xu + x^2 + u^2 = (x+u)^2$ so its minimum is zero (attained at $u = -x$.)

The $V(x,t) = 2x$ hence is our *candidate* value function, and $\mathtt{u}(t) = -\mathtt{x}(t)$ our *candidate* optimal control with candidate optimal cost $V(x_0,0) = 2x_0 = 2$.

(b) They are just *candidate* solutions because we still need to verify that the resulting closed loop $\dot{\mathtt{x}}_*(t) = \mathtt{x}_*(t)\mathtt{u}_*(t) = -\mathtt{x}_*^2(t)$ has a well defined solution. For $\mathtt{x}(0) = 1$ that is the case (see Example B.1.5): $\mathtt{x}_*(t) = 1/(t+1)$. *Now that* $\mathtt{x}_*(t)$ *is well defined*, Thm. 3.4.3 (Item 2) says that $\mathtt{u}_*(t) = -\mathtt{x}_*(t)$ *is* the optimal control and that $V(x_0,0) = 2x_0 = 2$ *is* the optimal cost.

(c) For $\mathtt{x}(0) = -1$ the *candidate* optimal input makes the closed-loop system satisfy $\dot{\mathtt{x}}_*(t) = -\mathtt{x}_*^2(t)$ with $\mathtt{x}_*(0) = -1$. In Example B.1.5 we saw that $\mathtt{x}_*(t) = -1/(-t+1)$ so this solution escapes at $t = 1 < T = 2$. Hence the candidate $\mathtt{u}_*(t) = -\mathtt{x}_*(t)$ is *not* optimal after all. (One can show that in this case the cost is unbounded from below.)

3.5 To turn maximization into minimization we need to swap the sign of the cost,

$$J_{[0,3]}(x_0,\mathtt{u}) := \mathtt{x}(3) + \int_0^3 (\mathtt{u}(t) - 1)\mathtt{x}(t)\,\mathrm{d}t.$$

(a) $\dot{Q}(t)x + \min_{u\in[0,1]}(Q(t)ux + (u-1)x) = 0$ and $Q(3)x = x$.

(b) Since $x > 0$ we may cancel x from the HJB equations to obtain $\dot{Q}(t) + \min_{u\in[0,1]}(Q(t)u + u - 1) = 0, Q(3) = 1$, which is

$$\dot{Q}(t) - 1 + \min_{u\in[0,1]}(Q(t)+1)u = 0, \qquad Q(3) = 1.$$

So

$$\mathtt{u}(t) = \begin{cases} 0 & \text{if } Q(t)+1 > 0 \\ 1 & \text{if } Q(t)+1 < 0 \end{cases}.$$

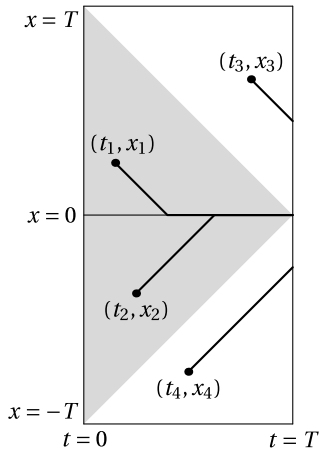
(c) As $Q(3) = 1$ we have $Q(t) + 1 > 0$ near the final time. So then $u = 0$ which turns the HJB equations into $\dot{Q}(t) - 1 = 0, Q(3) = 1$. Thus $Q(t) = t - 2$. This is the solution on $[1,3]$ for then we still have $Q(t) + 1 > 0$. On $[0,1]$ we have $\mathtt{u}(t) = 1$ so then the HJB equations become $\dot{Q}(t) + Q(t) = 0$ which, given $Q(1) = -1$, implies that $Q(t) = -\mathrm{e}^{1-t}$.

(d) $\mathtt{u}(t) = 1$ on $[0,1]$, and $\mathtt{u}(t) = 0$ on $[1,3]$. Then $\mathtt{x}(t)$ satisfies $\dot{\mathtt{x}}(t) = \mathtt{x}(t)\mathtt{u}(t)$ which is well defined for all $t \in [0,3]$. So the candidate optimal solution is truly optimal, and the candidate value function is the true value function.

(e) The optimal (minimal) cost is $V(x_0, 0) = Q(0)x_0 = -\mathrm{e}\, x_0$, so the maximal satisfaction is $+\mathrm{e}\, x_0$.

3.7 (a) We want the final state to be as close as possible to zero. So, whenever $\mathtt{x}(t)$ is nonzero, we steer optimally fast to zero: $\mathtt{u}(t) = -\operatorname{sgn}(\mathtt{x}(t))$. Once $\mathtt{x}(t)$ is zero, we take $\mathtt{u}(t) = 0$.

(b) The plot below shows a couple of state trajectories $\mathtt{x}(t)$ as a function of time:

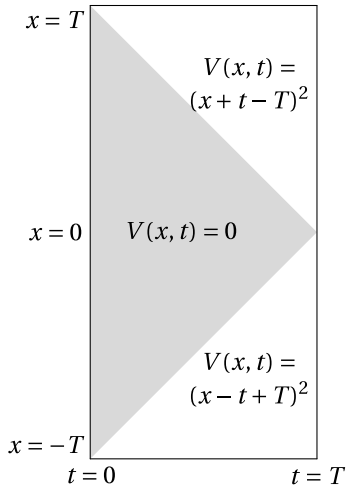


Clearly, the shaded triangle is determined by

$$|x| \le T - t$$

and for any (x,t) in this triangle the final state $\mathtt{x}(T)$ is zero, so then $V(x,t) = 0$. For any (x,t) above the triangle we have $\mathtt{x}(T) = x - (T - t) = x + t - T$, hence $V(x,t) = (x + t - T)^2$. Likewise for any (x,t) below the triangle we have $\mathtt{x}(T) = x + (T - t) = x - t + T$, hence $V(x,t) =$

$(x-t+T)^2$:



As a formula:

$$V(x,t) = \begin{cases} 0 & \text{if } |x| \le T-t \\ (x+t-T)^2 & \text{if } x > T-t \\ (x-t+T)^2 & \text{if } x < t-T \end{cases}.$$

Does it satisfy the HJB equations? The final condition $V(x,T) = K(x) = x^2$ is satisfied by construction. For (x,t) above the triangle, the HJB partial differential equation becomes

$$2(x+t-T) + \min_{u\in[-1,1]} (2(x+t-T)u) = 0.$$

The equality holds because in this region $x+t-T > 0$ so the minimizer is $u=-1$ and this renders the left-hand side of the HJB equation indeed equal to zero. On the triangle, the HJB equation is rather trivial,

$$0 + \min_{u\in[-1,1]} 0 = 0.$$

Below the triangle the HJB equation reads

$$-2(x-t+T) + \min_{u\in[-1,1]} (2(x-t+T)u) = 0,$$

which, too, is correct because now $x-t+T<0$ so $u=+1$ is the minimizer.

On the boundary $|x| = T-t$ of the triangle, the function $V(x,t)$ is continuously differentiable, and so the HJB equations hold for all x and all $t \in [0,T]$.

3.9 We momentarily use V_x to mean $\frac{\partial V(x,t)}{\partial x}$, and likewise for V_t.

(a) The u that minimizes $V_x x u + x^2 + \rho^2 u^2$ is $u = -V_x x/(2\rho^2)$. So the HJB equations (3.12) become

$$V_t + V_x x(-V_x x/(2\rho^2)) + x^2 + \rho^2 (V_x x)^2/(4\rho^4) = 0, \quad V(x,T) = 0.$$

Since $V(x,t) = x\rho G(z)$ with $z := (t-T)x/\rho$, we have

$$\begin{aligned} V_t &= x\rho G'(z)x/\rho && = x^2 G'(z), \\ V_x &= \rho G(z) + x\rho G'(z)(t-T)/\rho && = \rho(G(z) + zG'(z)). \end{aligned}$$

Thus the HJB equations become

$$x^2 G'(z) - (G(z) + zG'(z))^2 x^2/2 + x^2 + x^2 (G(z) + zG'(z))^2/4 = 0,$$

together with $G(0) = 0$. After cancelling the common term x^2 we get $G'(z) - \frac{1}{4}(G(z) + zG'(z))^2 + 1 = 0$.

(b) $u = -V_x x/(2\rho^2) = -\frac{x}{2\rho}(G(z) + zG'(z))$. Hence $\mathrm{u}_*(t) = -\frac{\mathrm{x}(t)}{2\rho}(G(z) + z'G(z))$ in which $z = (t-T)\mathrm{x}(t)/\rho$.

(c) Thus the closed-loop satisfies

$$\dot{\mathrm{x}}_*(t) = -\tfrac{1}{2\rho}\mathrm{x}_*^2(t)[G(z) + zG'(z)] \text{ in which } z = (t-T)\mathrm{x}_*(t)/\rho.$$

Recall that $\rho > 0$. If $\mathrm{x}(t) > 0$ then $z \le 0$ for every $t \in [0,T]$ so (see graphs) $G(z) + zG'(z) \ge 0$ and, thus, $\dot{\mathrm{x}}_*(t) \le 0$. Likewise, if $\mathrm{x}(t) < 0$ then $z \ge 0$ and $G(z) + zG'(z) \le 0$ and $\dot{\mathrm{x}}_*(t) \ge 0$. In both cases we see that $|\mathrm{x}_*(t)|$ decreases. Therefore, it does not escape on $[0,T]$, that is, the solution $\mathrm{x}_*(t)$ is well defined for all $t \in [0,T]$. Hence V is the value function, and the optimal cost is $V(x_0,0) = x_0\rho G(-Tx_0/\rho)$.

3.11 (a) The HJB equations become

$$\begin{aligned} (x-1)^2\dot{P}(t) + \min_{u\in\mathbb{R}}\left(2(x-1)P(t)(u-x) + (u-x)^2\right) = 0, \\ P(T)(x-1)^2 = \tfrac{1}{\beta}(x-1)^2. \end{aligned}$$

The minimizing u satisfies $2(x-1)P(t) + 2(u-x) = 0$, so $u - x = -(x-1)P$. Thus the HJB equations become

$$\begin{aligned} (x-1)^2\dot{P}(t) - 2(x-1)^2 P^2(t) + (x-1)^2 P^2(t) = 0, \\ P(T)(x-1)^2 = \tfrac{1}{\beta}(x-1)^2. \end{aligned}$$

Cancel the common factor $(x-1)^2$ and we find the simple ODE

$$\dot{P}(t) = P^2(t), \qquad P(T) = 1/\beta.$$

(b) For $P(t) := 1/(\beta+T-t)$ we have $P(T) = 1/\beta$ and $\dot{P}(t) = 1/(\beta+T-t)^2 = P^2(t)$. Correct. The previous part gives $\mathtt{u}_*(t) = \mathtt{x}(t) + P(t)(1-\mathtt{x}(t)) = \mathtt{x}(t) + \frac{1-\mathtt{x}(t)}{\beta+T-t}$. Thus the closed-loop system satisfies

$$\dot{\mathtt{x}}(t) = \frac{1-\mathtt{x}(t)}{\beta+T-t}, \qquad \mathtt{x}(0) = 0.$$

For $\mathtt{x}_*(t) := t/(\beta+T)$ we have $\dot{\mathtt{x}}_*(t) = 1/(\beta+T)$ and $(1-\mathtt{x}_*(t))/(\beta+T-t) = (1-t/(\beta+T))/(\beta+T-t) = 1/(\beta+T) = \dot{\mathtt{x}}_*(t)$. Correct. Consequently, $\mathtt{u}_*(t) = \dot{\mathtt{x}}_*(t) + \mathtt{x}_*(t) = \frac{1}{\beta+T} + \frac{t}{\beta+T} = \frac{t+1}{\beta+T}$. Correct. The optimal cost is $\frac{1}{\beta}(T/(\beta+T)-1)^2 + \int_0^T \dot{\mathtt{x}}^2(t)\,\mathrm{d}t = \frac{1}{\beta}(-\beta/(\beta+T))^2 + T(1/(\beta+T))^2 = 1/(\beta+T)$.

(c) $\lim_{\beta\downarrow 0} \mathtt{x}(T) = T/T = 1$. So it equals the desired voltage 1. This is to be expected because the quadratic term $(\mathtt{x}(T)-1)^2/\beta$ in the cost function blows up as $\beta \downarrow 0$ unless $\mathtt{x}(T) \to 1$.

3.13 (a) $H = -p\mathtt{x}u$ so $\dot{\mathtt{p}} = \mathtt{p}\mathtt{u}$. We also have $\mathtt{p}(T) = 1$.

If $x_0 > 0$ then $\mathtt{x}(t)$ is positive for all $t > 0$ so then H is minimal for $\mathtt{u}_*(t) = 1$ near $t = T$ (because $\mathtt{p}(t) \approx T > 0$). This gives $\dot{\mathtt{p}}_* = \mathtt{p}_*$. Since $\mathtt{p}_*(T) = 1$ we get

$$\mathtt{p}_*(t) = \mathrm{e}^{t-T} \qquad \text{if } x > 0.$$

If $x_0 < 0$ then $\mathtt{x}(t) < 0$ for all $t > 0$. Then near $t = T$ optimal is $\mathtt{u}_*(t) = 0$, so $\mathtt{p}_*(t)$ is constant,

$$\mathtt{p}_*(t) = 1 \qquad \text{if } x < 0.$$

(b) For $x_0 = 0$ the solution of $\dot{\mathtt{x}} = \mathtt{x}\mathtt{u}$ is zero for all time (not dependent on bounded $\mathtt{u}$). Thus every $\mathtt{u}$ results in the same cost J.

(c) If $\mathtt{x}_*(t) > 0$ then $\partial V(\mathtt{x}_*(t), t)/\partial x = \mathrm{e}^{t-T}$. It agrees with the above $\mathtt{p}_*(t)$.

If $\mathtt{x}_*(t) < 0$ then $\partial V(\mathtt{x}_*(t), t)/\partial x = 1$. It agrees with the above $\mathtt{p}_*(t)$.

3.15 The infinite horizon HJB equation (3.30) becomes

$$\min_{u\in\mathbb{R}} \big(V'(x)u + x^4 + u^2\big) = 0.$$

The minimizing u satisfies $V'(x) + 2u = 0$. Using this $V'(x) = -2u$ the HJB equation becomes $0 = -2u^2 + x^4 + u^2$. Hence $u = \mp x^2$ and, so, $V'(x) = \pm 2x^2$. Clearly, $V(x) = \frac{2}{3}|x|^3$ is the unique solution of the HJB equation that is nonnegative and such that $V(0) = 0$. The corresponding input is $\mathtt{u}_*(t) =$

$-\frac{1}{2}V'(\mathrm{x}(t)) = -\mathrm{x}^2(t)$ if $\mathrm{x}(t) > 0$, and $\mathrm{u}_*(t) = +\mathrm{x}^2(t)$ if $\mathrm{x}(t) < 0$. It is a stabilizing input. Exactly as in Example 3.6.1 we have for every stabilizing u that

$$\begin{aligned} J_{[0,\infty)}(x_0,\mathrm{u}) &= \int_0^\infty L(\mathrm{x}(t),\mathrm{u}(t))\,\mathrm{d}t \\ &\geq \int_0^\infty -\frac{V(\mathrm{x}(t))}{\partial x^{\mathrm{T}}} f(\mathrm{x}(t),\mathrm{u}(t))\,\mathrm{d}t \qquad \text{because of (3.30)} \\ &= \int_0^\infty -\dot{V}(\mathrm{x}(t))\,\mathrm{d}t = V(x_0) - \underbrace{V(\mathrm{x}(\infty))}_{0} = V(x_0). \end{aligned}$$

And since equality holds for $\mathrm{u} = \mathrm{u}_*$ (and u_* stabilizes) we see that u_* is the solution we seek.

3.17 (a) It is a variation of Example 3.6.1. The infinite horizon HJB equation (3.30) becomes

$$\min_{u\in\mathbb{R}} V'(x)(x+u) + u^4 = 0.$$

The minimizing u satisfies $V' + 4u^3 = 0$. Using this $V' = -4u^3$ the HJB equation (3.30) becomes $0 = -4u^3(x+u) + u^4 = -4u^3x - 3u^4$. Hence, either $0 = u = V'$ or $u = -\frac{4}{3}x$. For the latter, $V'(x) = 4(4/3x)^3$. So $V(x) = 3(4/3x)^4$ is a possible solution. Notice that $\mathrm{u}_* := -\frac{4}{3}\mathrm{x}$ stabilizes because then $\dot{\mathrm{x}} = \mathrm{x} + \mathrm{u} = -\frac{1}{3}\mathrm{x}$. Now, exactly as in Example 3.6.1, we have for every stabilizing u that

$$\begin{aligned} J_{[0,\infty)}(x_0,\mathrm{u}) &= \int_0^\infty L(\mathrm{x}(t),\mathrm{u}(t))\,\mathrm{d}t \\ &\geq \int_0^\infty -\frac{V(\mathrm{x}(t))}{\partial x^{\mathrm{T}}} f(\mathrm{x}(t),\mathrm{u}(t))\,\mathrm{d}t \qquad \text{because of (3.30)} \\ &= \int_0^\infty -\dot{V}(\mathrm{x}(t))\,\mathrm{d}t = V(x_0) - \underbrace{V(\mathrm{x}(\infty))}_{0} = V(x_0). \end{aligned}$$

And since equality holds for $\mathrm{u} = \mathrm{u}_*$ (and u_* stabilizes) we see that u_* is the solution we seek.

(b) Clearly $\mathrm{u}_*(t) = 0$ minimizes the cost. It renders the closed loop unstable: $\dot{\mathrm{x}}(t) = \mathrm{x}(t), \mathrm{x}(0) = 1$ (so $\mathrm{x}(t) = \mathrm{e}^t$).

3.19 (a) Practically the same as Exercise 3.14(a): let $t_1, \delta \in \mathbb{R}$. If $\mathrm{x}(t)$ satisfies

$$\dot{\mathrm{x}}(t) = f(\mathrm{x}(t),\mathrm{u}(t)), \qquad \mathrm{x}(t_1) = x$$

then, by time-invariance, the shifted $\tilde{\mathrm{x}}(t) := \mathrm{x}(t-\delta), \tilde{\mathrm{u}}(t) := \mathrm{u}(t-\delta)$ satisfy the same differential equation but with shifted-time initial condition

$$\dot{\tilde{\mathrm{x}}}(t) = f(\tilde{\mathrm{x}}(t),\tilde{\mathrm{u}}(t)), \qquad \tilde{\mathrm{x}}(t_1+\delta) = x.$$

Therefore

$$J_{[t_1,T]}(x,\mathtt{u}) = J_{[t_1+\delta,T+\delta]}(x,\tilde{\mathtt{u}}).$$

Hence, any cost that can be achieved over $[t_1,T]$ starting at $\mathtt{x}(t_1)=x$, can also be achieved over $[t_1+\delta,T']$ starting at $\mathtt{x}(t_1+\delta)=x$ for some T' (namely $T'=T+\delta$). So also the optimal cost-to-go (where we also optimize over the final time T) starting at x does not depend on the initial time.

(b) Notice that we cannot use (3.30) because we are given the value function $\mathcal{V}$ and not a solution V of (3.30). On the one hand we have by the chain rule that $\frac{\mathrm{d}\mathcal{V}(\mathtt{x}_*(t))}{\mathrm{d}t} = \frac{\partial\mathcal{V}(\mathtt{x}_*(t))}{\partial x^{\mathrm{T}}} f(\mathtt{x}_*(t),\mathtt{u}_*(t))$, and on the other hand we have exactly as in § B.5 that $\frac{\mathrm{d}\mathcal{V}(\mathtt{x}_*(t))}{\mathrm{d}t} = -L(\mathtt{x}_*(t),\mathtt{u}_*(t))$.

(c) The second identity in the displayed equation in (b) immediately yields $H(\mathtt{x}_*(t),\mathtt{p}_*(t),\mathtt{u}_*(t))=0$ for $\mathtt{p}_*(t)=\frac{\partial V(\mathtt{x}_*(t),t)}{\partial x}$. It is in line with Theorem 2.7.1 (at the final time) and the constancy of Hamiltonians (Theorem 2.5.6).

Chapter 4

4.1 So $A=3, B=2, Q=4, R=1, S=0$.

(a) $\mathscr{H} = \begin{bmatrix} 3 & -4 \\ -4 & -3 \end{bmatrix}$.

(b) Now

$$\begin{bmatrix} \mathtt{x}_*(t) \\ \mathtt{p}_*(t) \end{bmatrix} = \frac{1}{5}\begin{bmatrix} 4\mathrm{e}^{5t}+\mathrm{e}^{-5t} & -2\mathrm{e}^{5t}+2\mathrm{e}^{-5t} \\ -2\mathrm{e}^{5t}+2\mathrm{e}^{-5t} & \mathrm{e}^{5t}+4\mathrm{e}^{-5t} \end{bmatrix}\begin{bmatrix} x_0 \\ \mathtt{p}_*(0) \end{bmatrix}.$$

Given that $\mathtt{p}_*(T)=0$ we can determine $\mathtt{p}_*(0)$ (as a function of x_0):

$$\mathtt{p}_*(0) = \frac{2\mathrm{e}^{5T}-2\mathrm{e}^{-5T}}{\mathrm{e}^{5T}+4\mathrm{e}^{-5T}}x_0.$$

This determines $\mathtt{p}_*(0)$ and, therefore, determines $\mathtt{x}_*,\mathtt{p}_*$ for all time. Finally, $\mathtt{u}_* = -R^{-1}B^{\mathrm{T}}\mathtt{p}_* = -2\mathtt{p}_*$. So also $\mathtt{u}_*$ is determined for all time. The optimal cost is

$$\mathtt{p}_*(0)x_0 = \frac{2\mathrm{e}^{5T}-2\mathrm{e}^{-5T}}{\mathrm{e}^{5T}+4\mathrm{e}^{-5T}}x_0^2.$$

4.3 (a) $H(x,2p,u) = 2p^{\mathrm{T}}(Ax+Bu)+x^{\mathrm{T}}Qx+u^{\mathrm{T}}u$. So the Hessian of H w.r.t. (x,u) is $\begin{bmatrix} 2Q & 0 \\ 0 & 2I \end{bmatrix} \geq 0$.

(b)

$$\begin{aligned}
\dot{\mathtt{x}} &= A\mathtt{x} + B(\mathtt{u}_* + \mathtt{v}), & \mathtt{x}(0) &= x_0,\\
\dot{\mathtt{x}}_* &= A\mathtt{x}_* + B\mathtt{u}_*, & \mathtt{x}_*(0) &= x_0,\\
\Longrightarrow\quad \dot{\mathtt{z}} &= A\mathtt{z} + B\mathtt{v}, & \mathtt{z}(0) &= 0.
\end{aligned}$$

$$\begin{aligned}
J &= \int_0^T \mathtt{x}^{\mathrm{T}} Q\mathtt{x} + (\mathtt{u}_* + \mathtt{v})^{\mathrm{T}}(\mathtt{u}_* + \mathtt{v})\,\mathrm{d}t,\\
J_* &= \int_0^T \mathtt{x}_*^{\mathrm{T}} Q\mathtt{x}_* + \mathtt{u}_*^{\mathrm{T}}\mathtt{u}_*\,\mathrm{d}t,\\
\Longrightarrow\quad J - J_* &= \int_0^T (\mathtt{x}_* + \mathtt{z})^{\mathrm{T}} Q(\mathtt{x}_* + \mathtt{z}) - \mathtt{x}_*^{\mathrm{T}} Q\mathtt{x}_* + \mathtt{v}^{\mathrm{T}}\mathtt{v} + 2\mathtt{u}_*^{\mathrm{T}}\mathtt{v}\,\mathrm{d}t\\
&= \int_0^T (\mathtt{z}^{\mathrm{T}} Q\mathtt{z} + \mathtt{v}^{\mathrm{T}}\mathtt{v}) + 2\mathtt{z}^{\mathrm{T}} Q\mathtt{x}_* + 2\mathtt{u}_*^{\mathrm{T}}\mathtt{v}\,\mathrm{d}t.
\end{aligned}$$

$$\begin{aligned}
\tfrac{\mathrm{d}}{\mathrm{d}t}(\mathtt{p}_*^{\mathrm{T}}\mathtt{z}) &= \dot{\mathtt{p}}_*^{\mathrm{T}}\mathtt{z} + \mathtt{p}_*^{\mathrm{T}}\dot{\mathtt{z}}\\
&= (-Q\mathtt{x}_* - A^{\mathrm{T}}\mathtt{p}_*)^{\mathrm{T}}\mathtt{z} + \mathtt{p}_*^{\mathrm{T}}(A\mathtt{z} + B\mathtt{v})\\
&= -\mathtt{z}^{\mathrm{T}} Q\mathtt{x}_* + \mathtt{p}_*^{\mathrm{T}} B\mathtt{v}.\\
&= -\mathtt{z}^{\mathrm{T}} Q\mathtt{x}_* - \mathtt{u}_*^{\mathrm{T}}\mathtt{v}.
\end{aligned}$$

So $\int_0^T \mathtt{z}^{\mathrm{T}}(t)Q\mathtt{x}_*(t) + \mathtt{u}_*^{\mathrm{T}}(t)\mathtt{v}(t)\,\mathrm{d}t = [-\mathtt{p}_*^{\mathrm{T}}(t)\mathtt{z}(t)]_0^T = 0$ and, therefore, $J - J_* = \int_0^T \mathtt{z}^{\mathrm{T}}(t)Q\mathtt{z}(t) + \mathtt{v}^{\mathrm{T}}(t)\mathtt{v}(t)\,\mathrm{d}t \geq 0$.

4.5 The solution $\mathtt{x}$ of $\dot{\mathtt{x}} = A\mathtt{x} + B\mathtt{u}$ is linear in both x_0 and $\mathtt{u}$, so we can write $\mathtt{x} = \mathscr{L}(x_0, \mathtt{u})$ for some *linear* mapping $\mathscr{L}$.

(a) $J_{[t,T]}(\lambda x, \lambda \mathtt{u}) = \int_0^T (\lambda\mathscr{L}(x,\mathtt{u}))^{\mathrm{T}} Q(\lambda\mathscr{L}(x,\mathtt{u})) + \lambda^2\mathtt{u}^{\mathrm{T}} R\mathtt{u}\,\mathrm{d}t = \lambda^2 J_{[t,T]}(x,\mathtt{u})$.
As for the second equality, use that $J_{[t,T]}(x+z, \mathtt{u}+\mathtt{w})$ equals

$$\int_0^T (\mathscr{L}(x,\mathtt{u}) + \mathscr{L}(z,\mathtt{w}))^{\mathrm{T}} Q(\mathscr{L}(x,\mathtt{u}) + \mathscr{L}(z,\mathtt{w})) + (\mathtt{u}^{\mathrm{T}} + \mathtt{w}^{\mathrm{T}})R(\mathtt{u}+\mathtt{w})\,\mathrm{d}t$$

and that $J_{[t,T]}(x-z, \mathtt{u}-\mathtt{w})$ equals

$$\int_0^T (\mathscr{L}(x,\mathtt{u}) - \mathscr{L}(z,\mathtt{w}))^{\mathrm{T}} Q(\mathscr{L}(x,\mathtt{u}) - \mathscr{L}(z,\mathtt{w})) - (\mathtt{u}^{\mathrm{T}} - \mathtt{w}^{\mathrm{T}})R(\mathtt{u}-\mathtt{w})\,\mathrm{d}t.$$

The sum of these two cancels all cross terms and leaves

$$\int_0^T 2\mathscr{L}(x,\mathtt{u})^{\mathrm{T}} Q\mathscr{L}(x,\mathtt{u}) + 2\mathscr{L}(z,\mathtt{w})^{\mathrm{T}} Q\mathscr{L}(z,\mathtt{w}) + 2\mathtt{u}^{\mathrm{T}} R\mathtt{u} + 2\mathtt{w}^{\mathrm{T}} R\mathtt{w}\,\mathrm{d}t.$$

(b) $\mathscr{V}(\lambda x, t) = \min_{\mathtt{u}} J_{[t,T]}(\lambda x, \mathtt{u}) = \min_{\mathtt{u}} J_{[t,T]}(\lambda x, \lambda\mathtt{u}/\lambda)$ and because of (a) this equals $\lambda^2 \min_{\mathtt{u}} J_{[t,T]}(x, \mathtt{u}/\lambda) = \lambda^2\mathscr{V}(x,t)$.
We know that $J_{[t,T]}(\lambda x, \lambda\mathtt{u}) = \lambda^2 J_{[t,T]}(x,\mathtt{u})$. Since λ is constant we see that $\mathtt{u}$ minimizes $J_{[t,T]}(x,\mathtt{u})$ iff $\mathtt{u}$ minimizes $J_{[t,T]}(\lambda x, \lambda\mathtt{u})$, i.e., iff $\mathtt{w} := \lambda\mathtt{u}$ minimizes $J_{[t,T]}(\lambda x, \mathtt{w})$.

(c) Let $\mathtt{u}_x, \mathtt{w}_z$ be the minimizers of the right-hand side of (4.43). Then (4.43) by definition of value function becomes

$$J_{[t,T]}(x+z, \mathtt{u}_x + \mathtt{w}_z) + J_{[t,T]}(x-z, \mathtt{u}_x - \mathtt{w}_z) = 2\mathcal{V}(x,t) + 2\mathcal{V}(z,t).$$

The result follows because, again by definition of value function, $\mathcal{V}(x+z,t) \le J_{[t,T]}(x+z, \mathtt{u}_x + \mathtt{w}_z)$ and $\mathcal{V}(x-z,t) \le J_{[t,T]}(x-z, \mathtt{u}_x - \mathtt{w}_z)$.

(d) Essentially the same: define $\hat{\mathtt{u}} = \mathtt{u} + \mathtt{w}$, $\hat{\mathtt{w}} = \mathtt{u} - \mathtt{w}$. Then (4.43) becomes

$$J_{[t,T]}(x+z, \hat{\mathtt{u}}) + J_{[t,T]}(x-z, \hat{\mathtt{w}}) = 2J_{[t,T]}(x, (\hat{\mathtt{u}}+\hat{\mathtt{w}})/2) + 2J_{[t,T]}(z, (\hat{\mathtt{u}}-\hat{\mathtt{w}})/2).$$

Minimizing the left-hand side over all $\hat{\mathtt{u}}, \hat{\mathtt{w}}$ by definition of value function gives

$$\mathcal{V}(x+z,t) + \mathcal{V}(x-z,t) = 2J_{[t,T]}(x, (\hat{\mathtt{u}}_* + \hat{\mathtt{w}}_*)/2) + 2J_{[t,T]}(z, (\hat{\mathtt{u}}_* - \hat{\mathtt{w}}_*)/2),$$

and the right-hand side by definition is at most $2\mathcal{V}(x,t) + 2\mathcal{V}(z,t)$.

(e) We saw earlier that (4.43) for $\mathtt{u} = \mathtt{u}_z, \mathtt{w} = \mathtt{w}_z$ says

$$J_{[t,T]}(x+z, \mathtt{u}_x + \mathtt{w}_z) + J_{[t,T]}(x-z, \mathtt{u}_x - \mathtt{w}_z) = 2\mathcal{V}(x,t) + 2\mathcal{V}(z,t).$$

The previous two parts show that the above right-hand side equals $\mathcal{V}(x+z,t) + \mathcal{V}(x-z,t)$.

(f) $\mathcal{V}$ is the *minimal* cost, so the left-hand side of the equality of the previous part is nonnegative, while the right-hand side is non-positive. So they must both be zero! Hence $J_{[t,T]}(x+z, \mathtt{u}_x + \mathtt{w}_z) = \mathcal{V}(x+z,t)$. It shows that $\mathtt{u}_x + \mathtt{w}_z$ is optimal for $x+z$. Scaling z with a factor λ shows the result.

(g) Trivial: if $\mathtt{u}_*$ is linear in $\mathtt{x}(t)$ then so is $\mathtt{u}_*(t)$ at every t.

(h) Follows from (a) and the fact that $\mathcal{V}(x+\lambda z, t) = J_{[t,T]}(x+\lambda z, \mathtt{u}_x + \lambda \mathtt{w}_z)$.

4.7 (a) The RDE is $\dot{P} = -2P + P^2 - 3, P(0) = s$. We use Exercise 4.6: we have $\dot{P} = (P+1)(P-3)$, so $G := 1/(P+1)$ satisfies $\dot{G} = 4G - 1$. Hence $G(t) = \frac{1}{4} + c\,\mathrm{e}^{4t}$ for some c. Exercise 4.6 now says that

$$P(t) = -1 + \frac{1}{c\,\mathrm{e}^{4t} + 1/4} = \frac{3 - c\,\mathrm{e}^{4t}}{1 + c\,\mathrm{e}^{4t}}.$$

We need $P(T) = s$, so $c = \mathrm{e}^{-4T}\,\frac{3-s}{1+s}$. Write c as $c = \mathrm{e}^{-4T}\,d$ then the result follows.

(b) Since $\dot{P} = (P-3)(P+1)$ it is immediate that if $P(t) > 3$ then $\dot{P}(t) > 0$ so $P(t)$ increases. Conversely, if $-1 < P(t) < 3$ then $\dot{P}(t) < 0$ so $P(t)$ decreases.

(c) See the first sentence of the proof of Theorem 4.5.2.

(d) The larger s is the higher the penalty on the final state, so for both plots a "small" final value $\mathrm{x}(T)$ corresponds to a "large" s. A bit vague: if s is small and x is close to zero then the dominant term in the cost function is u^2. So (as long as x does not change too much over the rest of time) it is "optimal" to take u small, but then $\dot{\mathrm{x}} \approx \mathrm{x}$ so x starts to increase in magnitude.

4.9 (a) We have $B = R = I_2$ so it is clear that $\mathrm{u}_* = -R^{-1}B^{\mathrm{T}}P\mathrm{x} = -P\mathrm{x}$.

(b) The matrix $E = \left[\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right]$ achieves $\tilde{Q} = \left[\begin{smallmatrix} 18 & 0 \\ 0 & 10 \end{smallmatrix}\right]$, $\tilde{A} = \left[\begin{smallmatrix} 0 & 0 \\ 0 & -2 \end{smallmatrix}\right]$, $\tilde{B} = \frac{1}{2}\left[\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right]$. Then $\tilde{B}\tilde{B}^{\mathrm{T}} = \frac{1}{2}I_2$, so the ARE becomes

$$\left[\begin{smallmatrix} 0 & 0 \\ 0 & -2 \end{smallmatrix}\right]P + P\left[\begin{smallmatrix} 0 & 0 \\ 0 & -2 \end{smallmatrix}\right] - \tfrac{1}{2}P^2 + \left[\begin{smallmatrix} 18 & 0 \\ 0 & 10 \end{smallmatrix}\right] = 0.$$

Clearly a diagonal $\tilde{P}$ suffices. It is easy to see that $\tilde{P} = \left[\begin{smallmatrix} 6 & 0 \\ 0 & 2 \end{smallmatrix}\right]$ does the job, and since $P \geq 0$ it is the solution we need. Then $P = E^{-\mathrm{T}}\tilde{P}E^{\mathrm{T}} = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$.

4.11 (a) $\dot{\mathrm{z}} = -\alpha\,\mathrm{e}^{-\alpha t}\mathrm{x} + \mathrm{e}^{-\alpha t}\dot{\mathrm{x}} = -\alpha\mathrm{z} + \mathrm{e}^{-\alpha t}(A\mathrm{x} + B\mathrm{u}) = (-\alpha I + A)\mathrm{z} + B\mathrm{v}, \mathrm{z}(0) = x_0$. $J = \int_0^T \mathrm{z}^{\mathrm{T}}Q\mathrm{z} + \mathrm{v}^{\mathrm{T}}R\mathrm{v}\,\mathrm{d}t$.

(b) So $\alpha = 1$ and $\dot{\mathrm{z}} = \mathrm{v}$ and, hence, $A = 0, Q = R = B = 1$: $\dot{P} = P^2 - 1$ and $P(T) = 0$. Example 4.5.1 says $P(t) = \tanh(T - t)$. Now

$$\mathrm{u}_*(t) = \mathrm{e}^{+\alpha t}\mathrm{v}_*(t) = -\mathrm{e}^{+\alpha t}R^{-1}B^{\mathrm{T}}P(t)\mathrm{z}(t) = -P(t)\mathrm{x}(t).$$

4.13 (a) Note that PB for $B = \left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right]$ is the 2nd column of P, so $PB = \frac{1}{3}\left[\begin{smallmatrix} 1 \\ \sqrt{2} \end{smallmatrix}\right]$.

$$\begin{aligned}
&A^{\mathrm{T}}P + PA - (PB)R^{-1}(PB)^{\mathrm{T}} + Q \\
&= \tfrac{1}{3}\left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]\left[\begin{smallmatrix} 2\sqrt{2} & 1 \\ 1 & \sqrt{2} \end{smallmatrix}\right] + \tfrac{1}{3}\left[\begin{smallmatrix} 2\sqrt{2} & 1 \\ 1 & \sqrt{2} \end{smallmatrix}\right]\left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right] - \tfrac{1}{3}\left[\begin{smallmatrix} 1 \\ \sqrt{2} \end{smallmatrix}\right]3\left(\tfrac{1}{3}\left[\begin{smallmatrix} 1 & \sqrt{2} \end{smallmatrix}\right]\right) + \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right] \\
&= \underbrace{\tfrac{1}{3}\left[\begin{smallmatrix} -1 & -\sqrt{2} \\ 2\sqrt{2} & 1 \end{smallmatrix}\right] + \tfrac{1}{3}\left[\begin{smallmatrix} -1 & 2\sqrt{2} \\ -\sqrt{2} & 1 \end{smallmatrix}\right]} - \underbrace{\tfrac{1}{3}\left[\begin{smallmatrix} 1 \\ \sqrt{2} \end{smallmatrix}\right]\left[\begin{smallmatrix} 1 & \sqrt{2} \end{smallmatrix}\right]} + \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right] \\
&= \tfrac{1}{3}\left[\begin{smallmatrix} -2 & \sqrt{2} \\ \sqrt{2} & 2 \end{smallmatrix}\right] - \tfrac{1}{3}\left[\begin{smallmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 2 \end{smallmatrix}\right] + \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right].
\end{aligned}$$

(b) $A - BR^{-1}B^{\mathrm{T}}P = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right] - \left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right]3\left[\begin{smallmatrix} 0 & 1 \end{smallmatrix}\right]\frac{1}{3}\left[\begin{smallmatrix} ? & ? \\ 1 & \sqrt{2} \end{smallmatrix}\right] = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right] - \left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right]\left[\begin{smallmatrix} 1 & \sqrt{2} \end{smallmatrix}\right] = \left[\begin{smallmatrix} 0 & 1 \\ -2 & -\sqrt{2} \end{smallmatrix}\right]$.

The standard stability test uses eigenvalues. The eigenvalues are the zeros λ of $\det(\lambda I - (A - BR^{-1}B^{\mathrm{T}}P)) = \det\left[\begin{smallmatrix} \lambda & -1 \\ 2 & \lambda + \sqrt{2} \end{smallmatrix}\right] = \lambda(\lambda + \sqrt{2}) + 2$. So $\lambda_{1,2} = -\frac{1}{2}\sqrt{2} \pm \sqrt{3/2}\mathrm{i}$. They have negative real part, hence the closed loop is asymptotically stable.

(An easier test for stability is to verify that $V(x) := x^{\mathrm{T}}Px$ is a strong Lyapunov function for the closed-loop system. Indeed $P > 0$, and $\dot{V}(x) = -x^{\mathrm{T}}(Q + PBR^{-1}B^{\mathrm{T}}P)x$ and $-(Q + PBR^{-1}B^{\mathrm{T}}P) < 0$.)

4.15 (a) $S=0, Q=4, R=1, A=3, B=2$. The ARE is $-4P^2+6P+4=0$. The solutions P of the ARE are $P=(-6\pm\sqrt{6^2+4^3})/-8$ that is $P=-1/2$ and $P=2$. So we need $P=2$.

(b) (By the way, Lemma 4.5.6 *guarantees* that $A-BR^{-1}B^{\mathrm{T}}P$ is asymptotically stable because (A,B) is stabilizable and (Q,A) detectable.) We have $A-BR^{-1}B^{\mathrm{T}}P=3-4\times 2=3-8=-5$. Asymptotically stable indeed.

(c) $F=-R^{-1}B^{\mathrm{T}}P=-4$.

(d) $Px_0^2=2x_0^2$.

(e) The eigenvalues must be ± 5 because $A-BR^{-1}B^{\mathrm{T}}P=-5$.

(f) $\mathscr{H}=\left[\begin{smallmatrix}3&-4\\-4&-3\end{smallmatrix}\right]$, $P=2$. Then $\mathscr{H}\left[\begin{smallmatrix}1\\2\end{smallmatrix}\right]=\left[\begin{smallmatrix}-5\\-10\end{smallmatrix}\right]=-5\left[\begin{smallmatrix}1\\2\end{smallmatrix}\right]$.

4.17 (a) A needs to be asymptotically stable (only then is (A,B) stabilizable, and as a result (Q,A) is detectable).

(b) $PA+A^{\mathrm{T}}P+Q=0$.

(c) That $PA+A^{\mathrm{T}}P+Q=0$ has a unique positive semi-definite solution P if A is asymptotically stable and $Q\geq 0$. That is part 4 of Thm. B.5.2.

4.19 (a) Swapping n rows means the determinant gains a factor $(-1)^n$. Multiplying n rows with -1 means another $(-1)^n$. Hence, in total a factor $(-1)^{2n}=1$. So the sign of determinant does not change.

(b) Let $Z(\lambda)$ be the matrix of (a). Clearly, $(Z(-\lambda))^{\mathrm{T}}=Z(\lambda)$. Hence $r(\lambda)=\det(Z(\lambda))=\det(Z(-\lambda)^{\mathrm{T}})=r(-\lambda)$.

(c) For every zero $\lambda\neq 0$ also $-\lambda$ is a zero. Suppose it has $2m$ nonzero zeros $\lambda_1,\ldots,\lambda_{2m}$. Then $r(\lambda)=c\lambda^{2n-2m}\prod_{i=1}^{m}(\lambda-\lambda_i)(\lambda+\lambda_i)=c(\lambda^2)^{n-m}\prod_{i=1}^{m}(\lambda^2-\lambda_i^2)$. It is a function of λ^2.

Appendix B

B.1 (a) $\dot{V}(\bar{x})=\frac{\partial V(\bar{x})}{\partial x^{\mathrm{T}}}f(\bar{x})=\frac{\partial V(\bar{x})}{\partial x^{\mathrm{T}}}0=0$.

(b) Let $\bar{x}_1,\bar{x}_2$ be two equilibria. Global asymptotic stability of $\bar{x}_1$ means that also $\mathrm{x}(t;\bar{x}_2)$ would have to converge to $\bar{x}_1$, but it does not because, by definition of equilibrium, we have $\mathrm{x}(t;\bar{x}_2)=\bar{x}_2$ for all t.

(c) $\bar{x}$ is an equilibrium iff $A\bar{x}=0$. If A is nonsingular then $\bar{x}=A^{-1}0=0$. If A is singular then any element in the null space of A is an equilibrium (and the null space then has infinitely many elements).

B.3 (a) $\left[\begin{smallmatrix}\dot{\mathrm{x}}\\ \dot{\mathrm{k}}\end{smallmatrix}\right]=\left[\begin{smallmatrix}(a-\mathrm{k})\mathrm{x}\\ \mathrm{x}^2\end{smallmatrix}\right]$, $\left[\begin{smallmatrix}\mathrm{x}(0)\\ \mathrm{k}(0)\end{smallmatrix}\right]=\left[\begin{smallmatrix}x_0\\ 0\end{smallmatrix}\right]$. It is an equilibrium iff $(\bar{x},\bar{k})=(0,\bar{k})$ with $\bar{k}$ free to choose.

(b) $\dot{V}(x,k)=2x(a-k)x+2(k-a)x^2=0$. This V is C^1, is positive definite relative to $(x,k)=(0,a)$. So V is a Lyapunov function (on $\Omega=\mathbb{R}^2$) for equilibrium $(0,a)$.

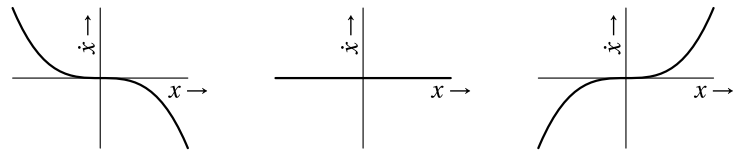
(c) Since $\dot{V}(x,k)=0$ we have that $\mathtt{x}^2+(\mathtt{k}-a)^2$ is constant. Hence $(\mathtt{k}(t)-a)^2 = C-\mathtt{x}^2(t) \le C$, which implies $\mathtt{k}(t)$ is bounded. (Notice that $C = x_0^2+a^2$.)

(d) Math question: since $\dot{\mathtt{k}}(t)\ge 0$ we have that $\mathtt{k}(t)$ is (besides bounded) also nondecreasing. This shows that $\mathtt{k}(t)$ converges as $t\to\infty$. Since $\mathtt{k}(t)$ converges and $\dot{\mathtt{k}}=\mathtt{x}^2$ we would expect $\mathtt{x}(t)\to 0$ as $t\to\infty$. Better: since $\dot{\mathtt{k}}=\mathtt{x}^2$ we have that $\dot{\mathtt{k}}(t)+(\mathtt{k}(t)-a)^2 = V(\dot{\mathtt{k}}(t),\mathtt{k}(t))$ which is constant. So as $\mathtt{k}$ converges, also $\dot{\mathtt{k}}$ converges. And then $\dot{\mathtt{k}}$ obviously converges to 0. Since $\dot{\mathtt{k}}=\mathtt{x}^2$ we thus conclude that $\mathtt{x}$ converges to zero as $t\to\infty$.

(e) The positive solution of $x_0^2+a^2=(k-a)^2$ is $k=a+\sqrt{a^2+x_0^2}$.

B.5 (a) If $\alpha\neq 0$ then $\bar{x}=0$, and if $\alpha=0$ then every $\bar{x}$ is an equilibrium. In any event, the linearization is $\dot{\delta}_x = A\delta_x$ with $A=\partial f(\bar{x})/\partial x = 3a\bar{x}^2$ which is zero, so $\dot{\delta}_x=0$.

(b)



ax^3 for some $a<0$ $\quad$ ax^3 for $a=0$ $\quad$ ax^3 for some $a>0$

If $a<0$ then the graph shows that $\mathtt{x}(t)>0$ implies $\dot{\mathtt{x}}(t)<0$, so $\mathtt{x}(t)$ then decreases (in the direction of 0). Likewise, if $\mathtt{x}(t)<0$ then $\dot{\mathtt{x}}(t)>0$ so $\mathtt{x}(t)$ then increases (again in the direction of 0). It keeps on moving until it reaches 0 (in the limit). It is asymptotically stable.

If $a=0$ then the graph obviously shows that $\dot{\mathtt{x}}(t)=0$. Hence $\mathtt{x}(t)$ is constant. It does not converge to 0 if $x_0\neq 0$ no matter how close x_0 is to zero.

If $a>0$ then the graph shows that $\mathtt{x}(t)>0$ implies $\dot{\mathtt{x}}(t)>0$, so $\mathtt{x}(t)$ increases (further away from 0). Likewise, $\mathtt{x}(t)<0$ then $\dot{\mathtt{x}}(t)<0$ so $\mathtt{x}(t)$ decreases (again further away from 0). The system is unstable.

(c) If $\alpha=0$ then every $\bar{x}$ is an equilibrium. Take $V(x)=(x-\bar{x})^2$. It is C^1 and positive definite (relative to $\bar{x}$) and $\dot{V}(x)=0$. So this V is a Lyapunov function and, consequently, every $\bar{x}\in\mathbb{R}$ is a stable equilibrium.

(d) If $\alpha<0$ then $\bar{x}=0$ asymptotically stable because $V(x)=x^2$ is a strong Lyapunov function: it is C^1, it is positive definite, and $\dot{V}(x)=2xf(x)=2\alpha x^4<0$ for all $x\neq 0$.

B.7 Let $\Omega=\mathbb{R}^2$.

(a) $V(x) > 0$ (relative to $\bar{x} = (0,0)$) and it is C^1, $\dot{V}(x) = 4x_1^3(x_2 - x_1) + 4x_2(-x_1^3) = -4x_1^4$. It is ≤ 0 so V is a Lyapunov function (on whatever neighborhood of $\bar{x}$).

(b) It is not a strong Lyapunov equation because $\dot{V}(x) = 0$ for all $x = (0, x_2)$.

(c) According to LaSalle there is a closed, bounded invariant neighborhood $\mathscr{K}$ of $\bar{x}$, and for every $x_0 \in \mathscr{K}$ the state $\mathrm{x}(t)$ for $t \to \infty$ converges to

$$\mathscr{G} = \{x \in \mathscr{K} \,|\, \dot{V}(\mathrm{x}(t;x)) = 0\} = \{x \in \mathscr{K} \,|\, \mathrm{x}_1(t;x) = 0, \dot{\mathrm{x}}(t;x) = f(\mathrm{x}(t;x))\}.$$

For $\mathrm{x}_1(t) = 0$ the system dynamics $\dot{\mathrm{x}}(t) = f(\mathrm{x}(t))$ become $0 = \mathrm{x}_2(t), \dot{\mathrm{x}}_2(t) = 0$. Which means $\mathrm{x}_2(t) = 0$. Hence $\mathscr{G} = \{\left[\begin{smallmatrix}0\\0\end{smallmatrix}\right]\}$ and we therefore have *asymptotic* stability.

B.9 (a) We need to solve $0 = x_1(-1 + x_2), 0 = x_2(1 - x_1)$. The first equation implies $x_1 = 0$ or $x_2 = 1$. If $x_1 = 0$ then the second equation holds iff $x_2 = 0$. If $x_2 = 1$ then the second equation holds iff $x_1 = 1$. So $(0,0)$ and $(1,1)$ are the only two equilibria.

(b) The Jacobian (for arbitrary x) is

$$\begin{bmatrix} -1 + x_2 & x_1 \\ -x_2 & 1 - x_1 \end{bmatrix}.$$

At $(0,0)$ and $(1,1)$ they respectively are

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

The first one is unstable (eigenvalue $+1$) so also the nonlinear system is unstable at $(0,0)$. The second one has imaginary eigenvalues only, so this says nothing about the stability of the nonlinear system.

(c) The function $y - \ln(y) - 1 \geq 0$ for all $y \geq 0$ (proof: its derivative is zero at $y = 1$ and its second derivative is > 0 so the function is minimal at $y = 1$ and then $y - \ln(y) - 1 = 0 \geq 0$). Likewise $V(\mathrm{x}_1, \mathrm{x}_2) = (\mathrm{x}_1 + \ln(\mathrm{x}_1) - 1) + (\mathrm{x}_2 - \ln(\mathrm{x}_2) - 1)$ is nonnegative and is minimal at $(x_1, x_2) = (1,1)$ (where $V = 0$). So V is positive definite relative to $(1,1)$. Clearly it is also C^1 for all $x_1, x_2 > 0$. (Yes, our state space is $x_1, x_2 > 0$.) Remains to analyze $\dot{V}(x)$:

$$\begin{aligned} \dot{V}(x) &= (1 - 1/x_1)x_1(-1 + x_2) + (1 - 1/x_2)x_2(1 - x_1) \\ &= 0. \end{aligned}$$

So $V(x)$ is preserved over time. Hence $(1,1)$ is stable, but not asymptotically stable.

B.11 (a) The first equation says $x_1 = 0$ or $x_1 = 1$ or $x_1 = 1/2$. The second equation says $x_2 = 0$. So three equilibria: $(0,0)$, $(1,0)$, and $(1/2,0)$.

(b) Jacobian (for general x) is

$$\begin{bmatrix} -2(x_1-1)(2x_1-1) - 2x_1(2x_1-1) - 2x_1(x_1-1) & 0 \\ 0 & -2 \end{bmatrix}.$$

At $(0,0)$, $(1,0)$, $(1/2,0)$ this becomes

$$\begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}, \quad \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}, \quad \begin{bmatrix} 1/2 & 0 \\ 0 & -2 \end{bmatrix}.$$

The first two have stable eigenvalues only (so the nonlinear system is asymptotically stable at $(0,0)$ and $(1,0)$).

(c) The third has a unstable eigenvalue ($1/2 > 0$) so the nonlinear system at $(1/2,0)$ is unstable.

B.13 (a) Since the model does not include damping we expect the kinetic energy to be constant, and $V(\omega) := \frac{1}{2}(I_1\omega_1^2 + I_2\omega_2^2 + I_3\omega_2^2)$ clearly is positive definite, and is C^1. Now

$$\begin{aligned} \dot{V}(\omega) &= \omega_1(I_2 - I_3)\omega_2\omega_3 + \omega_2(I_3 - I_1)\omega_1\omega_3 + \omega_3(I_1 - I_2)\omega_1\omega_2 \\ &= \omega_1\omega_2\omega_3[(I_2 - I_3) + (I_3 - I_1) + (I_1 - I_2)] = 0. \end{aligned}$$

So V is Lyapunov function: the origin is stable.

(b) In fact the origin is not asymptotically stable because the above V is constant over time.

(c) For every $\bar{\omega}_1, \bar{\omega}_2, \bar{\omega}_3 \in \mathbb{R}$ the three points $(\bar{\omega}_1, 0, 0)$, $(0, \bar{\omega}_2, 0)$, $(0, 0, \bar{\omega}_3)$ are equilibria.

(d) For these three types of equilibria the Jacobian's respectively are

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{I_3 - I_1}{I_2}\bar{\omega}_1 \\ 0 & \frac{I_1 - I_2}{I_3}\bar{\omega}_1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & \frac{I_2 - I_3}{I_1}\bar{\omega}_2 \\ 0 & 0 & 0 \\ \frac{I_1 - I_2}{I_3}\bar{\omega}_2 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & \frac{I_2 - I_3}{I_1}\bar{\omega}_3 & 0 \\ \frac{I_3 - I_1}{I_2}\bar{\omega}_3 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

(e) From the inequality $0 < I_1 < I_2 < I_3$ it follows that the Jacobian is of the form (with $\bar{\omega}_2 \neq 0$):

$$\begin{bmatrix} 0 & 0 & a\bar{\omega}_2 \\ 0 & 0 & 0 \\ b\bar{\omega}_2 & 0 & 0 \end{bmatrix}$$

for some $a, b < 0$. Its characteristic polynomial is $\lambda^3 - \lambda ab\bar{\omega}_2^2$. Since $ab\bar{\omega}_2^2 > 0$ we have an unstable real eigenvalue: $\lambda_1 = \sqrt{ab\bar{\omega}_2^2} > 0$. Conclusion: also the nonlinear system is unstable.

(f) In the first part we already showed that $V(\omega) := \frac{1}{2}(I_1\omega_1^2 + I_2\omega_2^2 + I_3\omega_2^2)$ is constant over time. Likewise we have for $W(\omega)$ defined as

$$W(\omega) = I_1^2\omega_1^2 + I_2^2\omega_2^2 + I_3^2\omega_3^2$$

that

$$\dot{W}(\omega) = (I_1(I_2 - I_3) + I_2(I_3 - I_1) + I_3(I_1 - I_2))2\omega_1\omega_2\omega_3 = 0.$$

Verifying stability is very technical. Consider an equilibrium $\bar{x} := (\bar{\omega}_1, 0, 0)$, and take an initial state $x_0 \in B(\bar{x}, \delta)$, that is, $x_0 = (\bar{\omega}_1 + \omega_1(0), \omega_2(0), \omega_3(0))$ with

$$\omega_1^2(0) + \omega_2^2(\omega) + \omega_3^2(0) < \delta^2.$$

Since both $V(\mathtt{x}(t; x_0))$ and $W(\mathtt{x}(t; x_0))$ are constant (over time) also

$$g(\omega_2, \omega_3) := W(\mathtt{x}) - 2I_1V(\mathtt{x}) = (I_2 - I_1)I_2\omega_2^2 + (I_3 - I_1)I_3\omega_3^2$$

is constant over time. (Notice that both $(I_2 - I_1)I_2$ and $(I_3 - I_1)I_3$ are positive.) Let $C = \min((I_2 - I_1)I_2, (I_3 - I_1)I_3) > 0$. Then

$$\omega_2^2(t) + \omega_3^2(t) \leq \frac{g(\omega_2(t), \omega_3(t))}{C} = \frac{g(\omega_2(0), \omega_3(0))}{C} \leq D\delta^2 \quad \forall\, t,$$

for some constant D (not depending on δ). Furthermore, since also $V(\mathtt{x}(t))$ is constant over time, we infer that $|\omega_1(t) - \bar{\omega}_1|^2$ is $\leq E\delta^2$ for all time for some constant E (not depending on δ). Given $\epsilon > 0$ we can thus choose $\delta > 0$ so small that $\|(\omega_1(t) - \bar{\omega}_1, \omega_2(t), \omega_3(t))\| < \epsilon$ for all $t > 0$ whenever $x_0 \in B(\bar{x}, \delta)$. So the equilibrium $\bar{x} := (\bar{\omega}_1, 0, 0)$ is stable. It is not *asymptotically* stable because the kinetic energy $V(\mathtt{x}(t))$ is constant over time, so the kinetic energy does not converge to zero as $t \to \infty$.

For the other equilibrium, $(0, 0, \bar{\omega}_3)$, a similar argument works.

B.15 Almost by definition: if $y \in \mathcal{O}(x_0)$ then $y = \mathtt{x}(t_1; x_0)$ for some t_1. Then $\mathtt{x}(t; y) = \mathtt{x}(t + t_1; x_0) \in \mathcal{O}(x_0)$.

B.17 For $\gamma > 0$ it is essentially the same as Example B.4.6. Anyway, for $r := x_1^2 + x_2^2$ one has $\dot{\mathtt{r}} = 2\mathtt{x}_1(\mathtt{x}_2 + \mathtt{x}_1(\gamma - \mathtt{x}_1^2 - \mathtt{x}_2^2)) + 2\mathtt{x}_2(-\mathtt{x}_1 + (\gamma - \mathtt{x}_1^2 - \mathtt{x}_2^2)) = 2(\mathtt{x}_1^2 + \mathtt{x}_2^2)(\gamma - \mathtt{x}_1^2 - \mathtt{x}_2^2) = 2\mathtt{r}(\gamma - \mathtt{r}^2)$. If $\gamma \leq 0$ then $\dot{\mathtt{r}} \leq -2\mathtt{r}^3$ so asymptotically stable (take, e.g., Lyapunov function $V(r) = r^2$.) If $\gamma > 0$ then the linearization around $r = 0$ is $\dot{\mathtt{r}} = (2\gamma)\mathtt{r}$, which is unstable (with eigenvalue $2\gamma > 0$). So also the nonlinear system is unstable in that case.

B.19 (a) $P = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right]$.

(b) $P > 0$ because $p_{11} = 1 > 0$ and $\det P = 1 > 0$. $Q > 0$ because $q_{11} = 4 > 0$ and $\det Q = 4 > 0$, so Thm. B.5.2 guarantees that $\dot{x}(t) = Ax(t)$ is asymptotically stable.

B.21 This is a linear system

$$\dot{x}(t) = \begin{bmatrix} 1 & 2 \\ -\alpha & (1-\alpha) \end{bmatrix} x(t).$$

Method 1: the characteristic polynomial of A is

$$\det(\lambda I - A) = \lambda^2 + (\alpha - 2)\lambda + 1 + \alpha.$$

The Routh-Hurwitz test says that a degree-two polynomial is asymptotically stable iff all its coefficients have the same sign, so asymptotically stable iff $\alpha > 2$. Ready.

Method 2: you might have forgotten about Routh-Hurwitz. Then compute its zeros (roots):

$$\lambda_{1,2} = \frac{2 - \alpha \pm \sqrt{(2-\alpha)^2 - 4(1+\alpha)}}{2}.$$

The sum of the $\lambda_{1,2}$ is $2 - \alpha$. Hence for stability we need $\alpha > 2$. If $\alpha > 2$ then $(2-\alpha)^2 - 4(\alpha+1) < (2-\alpha)^2$ so $\sqrt{(\alpha-2)^2 - 4(1+\alpha)} < |2-\alpha|$. Then $2 - \alpha \pm \sqrt{(2-\alpha)^2 - 4(1+\alpha)}$ both have negative real part.

Method 3 (really laborious): it is asymptotically stable iff $A^T P + PA = -I$ has a unique symmetric solution P, and $P > 0$. Write P as $P = \left[\begin{smallmatrix} p & q \\ q & r \end{smallmatrix}\right]$. The Lyapunov equation becomes

$$\begin{bmatrix} 1 & -\alpha \\ 2 & 1-\alpha \end{bmatrix} \begin{bmatrix} p & q \\ q & r \end{bmatrix} + \begin{bmatrix} p & q \\ q & r \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -\alpha & 1-\alpha \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

The three equations are

$$\begin{aligned} p - \alpha q &= 1, \\ 2p + (2-\alpha)q - \alpha r &= 0, \\ 2q + (1-\alpha)r &= 1. \end{aligned}$$

This turns out to give (yes, tricky):

$$p = \frac{2\alpha^2 - \alpha + 2}{\alpha^2 - \alpha - 2}, \qquad q = \frac{3\alpha - 2}{\alpha^2 - \alpha - 2}, \qquad r = \frac{6 + \alpha}{\alpha^2 - \alpha - 2}.$$

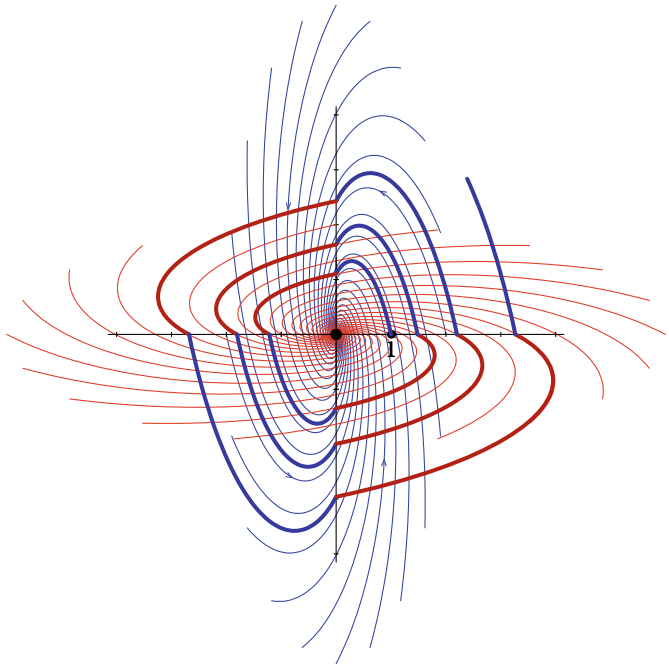
The $P = \left[\begin{smallmatrix} p & q \\ q & r \end{smallmatrix}\right]$ needs to exist, so we need $\alpha \neq -1$ and $\alpha \neq 2$. As $2\alpha^2 - \alpha + 2 > 0$ for all α, we have $p_{11} > 0$ iff $\alpha^2 - \alpha - 2 > 0$. This is the case iff $\alpha > 2$ or $\alpha < -1$. The determinant of P is $\frac{2\alpha^3 + 2\alpha^2 + 8\alpha + 8}{(\alpha^2 - \alpha - 2)^2}$. It is positive iff $\alpha > -1$. This combined with ($\alpha < -1$ or $\alpha > 2$) shows that P exists and is unique with $P > 0$ iff $\alpha > 2$.

B.23 (a) Both matrices have characteristic polynomial $\det(\lambda I - A) = (\lambda+1)^2 - \pi^2/2^2$. Its zeros are $-1 \pm i\pi/2$.

(b) Trivial.

(c) Follows from (b): $x(1) = e^{-1}\left[\begin{smallmatrix} 0 & -1/3 \\ 3 & 0 \end{smallmatrix}\right] x(0)$ and $x(2) = e^{-1}\left[\begin{smallmatrix} 0 & -3 \\ 1/3 & 0 \end{smallmatrix}\right] x(1)$, so $x(2) = e^{-2}\left[\begin{smallmatrix} -9 & 0 \\ 0 & -1/9 \end{smallmatrix}\right] x(0)$. Et cetera. Since $3/e > 1$ we have $\lim_{k\to\infty} \|x(2k)\| = \infty$ whenever $x_1(0) \neq 0$.

(d) Per quadrant follow the blue or red phase portraits (alternating):



B.25 So $Q := A^T + A < 0$. Then $V(x) := x_1^2 + x_2^2$ satisfies $\dot{V}(x) = x^T Q x < 0$, so yes.

Alternative proof: if $Av = \lambda v$ and $v \neq 0$ then $v^*(A + A^T)v < 0$ but $v^*(A + A^T)v = (2\,\mathrm{Re}(\lambda))\|v\|^2$. Hence $\mathrm{Re}(\lambda) < 0$.

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